BHAGAVATA COSMOLOGY

Vedic Alternative to Modern Cosmology

PAVANESHWAR DAS

Dedication

To His Divine Grace A. C. Bhaktivedanta Swami Prabhupāda
Founder-Ācārya of the International Society for Krishna Consciousness,
whose compassion, books and teachings have transformed me.

To His Holiness Rādhanath Swami, whose love, care and concern is
nourishing me.

To the teachers who have been guiding me all through my life.

To B. N. Chari who supported me during a crucial period in my life.

To my parents, well-wishers and friends, who have always supported me
in all my endeavours.

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Śrīla Śukadeva Gosvāmī says that the material creation of the Supreme Lord is unlimited and it is impossible to understand even with long lifespan, then what to speak of explaining the details of the structure and the functioning of this material universe. Nevertheless by the blessings of all the teachers of the past and present, in the form of their writings and teachings I am making a humble attempt to describe the same.

It is only by the tireless efforts and dedication of H.D.G. A.C. Bhaktivedanta Swami Śrīla Prabhupāda, the world is able to see the light of Bhāgavatam and appreciate the knowledge hidden in the Vedic literatures through his English translations and propagation. Words fall short to express my gratitude to Śrīla Prabhupāda for his selfless contribution.

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I express my gratitude to His Grace Sadāputa Prabhu (Richard L. Thompson) who presented this technical subject matter in his book "Vedic Cosmography and Astronomy" and attempted to bridge the gap between the Modern and the Vedic astronomy.

My deepest gratitude to His Holiness Danavir Goswami Mahārāja (Dr. Dane Holtzman) who tirelessly exploring the cosmology presented in Śrīmad-Bhāgavatam and other Puranic literatures and made it available through his various books." Taking inspiration from the works of these great scholars and taking references from their books, I am trying to describe the structure of the Universe as it is presented in the Śrīmad-Bhāgavatam and other Puranas.

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PREFACE

Śrīla Śukadeva Gosvāmī says that the expansion of the Supreme Lord's material creation is unlimited. It is not possible even for a person with the lifespan of a *devatā* to understand it by words or mind. But in order to answer the querry of King Parīkṣit that he wants to know the details of the structure of the universe, Śukadeva Gosvāmī presents a general description of the principal regions, such as Bhū-goloka (*Bhūloka*), with their names, forms, measurements and various symptoms (SB 5.16.4). Śrīla Śukadeva Gosvāmī himself admits that to give full details of this material universe would be impossible, and then what to speak of us with our insignificant limited understanding. Nevertheless, this is an attempt by a team of students at Bhaktivedānta Vidyāpītha, Govardhan Ecovillage, to understand and describe the structure of the universe, or *brahmāṇḍa*, as described in the fifth canto of *Śrīmad-Bhāgavatam*. Bhaktivedānta Vidyāpītha is a two-year, full time study course at Goverdhan Ecovillage. It covers the systematic study of *Śrīmad-Bhāgavatam and Śrī Caitanya Caritāmrta translated and commented by* Śrīla Prabhupāda.

As part of the two-year curriculum of this Bhaktivedānta Vidyāpītha course, we have undertaken this project to understand and present the Structure of the Universe, as described in the Vedic texts, in a pictorial way. This has started with an intention to come up with a 3-D working model, which presents the Structure of the Universe and the movement of various planets, such as the sun and the moon, in an understandable and simple way. We have tried to draw a parallel between the Vedic understanding of the Universe and the Modern understanding of the Universe. The commonalities and differences of the above two descriptions are presented for the consideration of the reader.

In doing this, we have drawn information from the following sources: (1) The commentaries of various Ācāryas on various Purānas, specifically the *Śrīmad-Bhāgavatam* (2) The commentary of Śrīla Visvanātha Chakravarti Thākura on the *Śrīmad-Bhāgavatam* (3) Other writings in the Vedic tradition, including the *Sūrya-siddhānta and Bhūgola varnanam* of Śrīla Vādirāja Tīrta Svāmi, and (4) Modern Western science about the description of the Solar System.

Our aim is to share an easy-to-understand presentation that will harmonize the descriptions of the Vedic texts and the descriptions of modern science related to the descriptions of the universe. This will help the reader to appreciate that the descriptions given in the scriptures are not mythical but factual.

This book primarily constitutes a preliminary study of the Vedic Cosmology as described in the Fifth Canto of the *Śrīmad-Bhāgavatam*, supported by the descriptions given in Śrīla Vādirāja Tīrtha swāmi's *"Bhūgola varnanam"*. The later part reconciles both the Vedic and Modern approaches in describing the Structure of the Universe.

The book includes the study of-
(1) The Structure of *Bhū-mandala* as presented in *Śrīmad-Bhāgavatam*
(2) The Structure of Fourteen-planetary System
(3) The movements of sun, moon and other planets (primarily the *nava-grahas*)
(4) The calculations of the Vedic Calendar (Traditional Indian Calendar)
(5) The Science behind the Eclipses and the change in the Seasons

According to ācāryas we should accept the cosmological statements in the Śrīmad-Bhāgavatam as authoritative and simply try to appreciate them. We will therefore adopt the working assumption that even though these statements may seem very hard to comprehend, they nonetheless do present an understandable and realistic description of the universe. To simplify these descriptions of the Śrīmad-Bhāgavatam, we have reffered to various ācāryas commentaries and quoted them all through the book, specifically Śrīla Vādirāja Tīrta Svāmi's *"Bhūgola varnanam"*. This will help the reader to appreciate the descriptions of the Śrīmad-Bhāgavatam.

It is my earnest desire, that this book can enable the reader to effectively appreciate the "Mystery behind the working of the Universe." It is also our hope that thise book along with the tiny 3-D Model of the Universe (Vedic Planetarium), can practically demonstrate the Working of the Universe. Thus, we can effectively present Kṛṣṇa Consciousness in the context of Vedic Cosmology. This, of course, was the plan of Śrīla Prabhupāda for the Temple of Vedic Planetarium (TOVP) in Śrīdhāma Māyāpura.

INTRODUCTION

Modern Western science is based on the idea that nature can be fully described by a single, rational world-model. However, the *Śrīmad-Bhāgavatam* points out that no person of this world is capable of fully describing the material universe "even in a lifetime as long as that of Brahmā" (SB 5.16.4). Thus the Vedic approach to the description of nature is based on the strategy of presenting many mutually compatible aspects of one humanly indescribable complete whole. This is the approach used in *Śrīmad-Bhāgavatam* while explaining the details of cosmology. The Universe described in the Vedic literature is higher-dimensional and beyond the range of our sense perception, thus it cannot be fully represented within three-dimensional space or within sense perceivable pictorial description. This requires the understanding of higher dimensional representation.

It was Śrīla Prabhupāda's earnest desire to explain the complex descriptions of the Fifth Canto of the *Śrīmad-Bhāgavatam* through the construction of a very large "*Vedic Planetarium*", a working 3-D Model. Śrīla Prabhupāda also desired that detaile information shall be presented on open verandas at different levels by means of dioramas, charts and films, etc. This is evident from Śrīla Prabhupāda's letter given below.

> "*Now our Ph.D.'s must collaborate and study the Fifth Canto to make a model for building the Vedic Planetarium. My final decision is that the universe is just like a tree, with root upwards. Just as a tree has branches and leaves, so the universe is also composed of planets which are fixed up in the tree like the leaves, flowers, fruits, etc....So now all you Ph.D.'s must carefully study the details of the Fifth Canto and make a working model of the universe. If we can explain the passing seasons, eclipses, phases of the moon, passing of day and night, etc., then it will be very powerful propaganda*" *(letter from Śrīla Prabhupāda to Svarūpa Dāmodara dāsa, April 27, 1976).*

Keeping all these viewpoints in mind and to fulfill the desire of His Divine Grace A.C. Bhaktivedanta Swami Śrīla Prabhupāda, many of his followers did lot of research to understand and explain the cosmology presented in the Fifth

Canto of the Śrīmad-Bhāgavatam in scientific way. In the initial days, His Grace Sadāputa Prabhu (Richard L. Thompson) presented his explanations in his book "Vedic Cosmography and Astronomy" and attempted to bridge the gap between the western and the Vedic science / astronomy. In the introduction of that book, he says "In this book we will discuss the cosmology presented in the Fifth Canto of the Śrīmad-Bhāgavatam and try to clarify its relationship with other prominent systems of cosmology, both ancient and modern." Later on His Holiness Danavir Goswami (Dr. Dane Holtzman) also did lot of research on the cosmology presented in Śrīmad-Bhāgavatam and other Puranic literatures. He presented the traditional Vedic astronomy through his various books such as "Vedic Cosmology, Puranic Cosmology and Bhugola Tattva, Acaryas Commentaries on Fifth Canto, etc..." Taking inspiration from the works of these great scholars and taking references from their books, we have tried to describe the structure of the Universe as it is presented in the Śrīmad-Bhāgavatam and other Puranas.

His Grace Drutakarmā Prabhu (Michael A. Cremo) is a research associate of the Bhaktivedanta Institute, specialized in history and philosophy of science. He authored 'Human Devolution' and co-authored 'Forbidden Archaeology - The Hidden History of the Human Race'. He lectures to academic and popular scientific audiences around the world in a continuing challenge to Darwinian evolution. He says that, "Most of the scientists and academics, though supporting the theory of evolution, are more open-minded and willing to consider alternatives. I see a new consciousness emerging that integrates science and religion into a cohesive paradigm of reality." When he was asked about his perspective on 'the change in the popular subjects such as human origins being taught in school textbooks', he responded saying that, there should be some diversity of ideas in the science curriculum and the views of the supporters of the alternatives should also be presented neutrally in a small minority of textbook pages and classroom time. This would reflect the actual realities in the world of science today and it would also respect the value of intellectual freedom. He further says "I find the university students very receptive to the idea that alternatives to the theory of evolution should have a place in the science curriculum. These young students will eventually become the new generation of leaders in the world of science and education. I ask them to work for change in the science curriculum."

Taking inspiration from his thoughts we are trying to present a Vedic alternative to the Modern Cosmology, the Bhāgavata Cosmology, based on the descriptions given in the Śrīmad-Bhāgavatam and other Vedic literatures. Let the reader carefully study these Vedic descriptions and compare it with the modern understanding of the Cosmology.

People in general see the '*words of the revealed scriptures*' through their '*experience in this world*'. One of the purposes of presenting this book is to enhance this perspective and develop a more harmonized way of approaching the revealed scriptures, especially when it comes to the complex description about the creation, cosmology and astronomy. Therefore, studying these Vedic descriptions will enable us to '*experience the world*' through the '*words of the revealed scriptures*'.

Śrīla Prabhupāda would often say that, only when one understands the greatness of the Supreme Lord Śrī Kṛṣṇa properly, then only one can invest ones firm faith in Him. Otherwise one may mistake Lord Kṛṣṇa to be one of the many demigods, or a commom man like us, or a powerful historical personality, or a myth only. Thus unless one understands Supreme Lord Śrī Kṛṣṇa's greatness properly one cannot appreciate His sweetness of Vṛndāvana. Otherwise there is a great chance that one will end up considering Lord Kṛṣṇa's sweet pastimes of Vṛndāvana to be simply some mundane activities or some fairy tales narrated to entertain the mind. Or one may end up developing the mentality of sahajiyas, who simply try to enter into Kṛṣṇa-līla artificially without considering one's current conditioning.

Śrīla Prabhupāda always recommended that a sincere reader of Śrīmad-Bhāgavatam must always start reading with the First Canto and then progressively go to the Tenth Canto, instead of directly jumping to Tenth Canto. Because without understanding the position of Lord Kṛṣṇa, which is systematically described and established in the first nine Cantos of Śrīmad-Bhāgavatam, if one directly enters into Tenth Canto pastimes, he / she is sure enough to be bewildered by seeing Lord Kṛṣṇa's human like pastimes, which appear to be very ordinary, or one may consider His Super Human feats to be simply an exaggeration. Thus there is a great danger in directly jumping to Tenth Canto simply to relish Vṛja-līla without firmly establishing oneself in the siddhānta properly.

Even Parīkṣit Mahārāja recomends the same by expressing his intention to Śrīla Śukadeva Gosvāmi to hear Lord's creation pastimes first, "You have described that I should hear and chant at the point of death. The subjects of hearing and chanting are Lord Kṛṣṇa's qualities, forms and pastimes. First I will ask about the pastimes of creation. Then I will ask about His sweet pastimes such as lifting Govardhana."

Even Lord Brahmā recommends his enlightened son, the great sage Nārada, to constantly hear the Creation Pastimes (*sṛṣṭi-līla* or the pastimes of the *puruṣāvatāras*) of the Supreme Lord, hearing which one is protected from Lord's illusory energy, *māyā*.

māyāṁ varṇayato 'muṣya īśvarasyānumodataḥ
śṛṇvataḥ śraddhayā nityaṁ māyayātmā na muhyati

> If the *jīva* constantly describes *māyā* (Lord's illusory energy) in relation to the Lord, remembers *māyā* or hears about *māya* with proper faith, he will not be bewildered by *māyā*. (SB 2.7.53)

Śrīla Visvanātha Chakravarti Thākura says that "The description of *māyā* or Lord's creation pastimes as an assistance to the processes of chanting (*varṇatyaḥ*), remembering (*anumodataḥ*) and hearing (*śṛnvataḥ*) is also *bhakti*. One should have faith that even *māyā-śakti* of the Lord, along with her expansion as *mahat-tattva* and other elements, is a devotee with the greatest devotion. Thus if one hears about *māyā* and *mahat-tattva* with this way of thinking, then one is not bewildered by *māyā*. Pastimes of the Lord related to *māyā* are not *māyā*. Rather they are transcendental."

Thus hearing, studying and describing the Creation Pastimes of the Lord and the Manifestation of His energies as this Gigantic Universe is in line with the desire of these great *ācāryas* coming in the line of *parampara* starting with Lord Kṛṣṇa Himself. This enables one to attain one's ultimate perfection, the life's purpose or goal, as Lord Kṛṣṇa says in BG 4.9 "One who understands in truth My spiritual birth and activities, attains Me in My eternal abode on giving up one's body, and does not take birth again in this Material World."

janma karma ca me divyam evaṁ yo vetti tattvataḥ
tyaktvā dehaṁ punar janma naiti mām eti so 'rjuna

Therefore, one should make it as a priority to systematically study the Creation Pastimes of the Lord and the descriptions of this Gigantic Universe, though they are highly technical and beyond the range of human imagination. But if one reads and describes them with proper faith and conviction, one is sure to understand them, which are difficult to understand, and develop unflinching faith and conviction at the Lotus Feet of the Supreme Personality of Godhead.

Generally any educational system should contain these five principle aspects, namely (1) skills or excellence, (2) origin or beginning, (3) values or morality, (4) character or qualities, and (5) purpose or goal. These five become subtler and subtler to perceive in progressive way. The Vedic Texts and the great sages in their presentations gave complete emphasis on the above five aspects of the knowledge, thus they properly guide the seeker in their progressive march. Unlikely, the Modern Educational System, which successfully neglected the rest of the four while giving more and more emphasis only to the first item, i.e., the development of the skills or making the student excellent in any specialized area. Thus the Modern Educational System deprives the student from learning the essential aspects of the life, which help one to understand the purpose of life and its existence and how to deal with the world around in an organic way, without any ulterior motives.

Among the commentaries of various ācāryas on the *Śrīmad-Bhāgavatam*, in addition to Śrīla Prabhupāda's commentary we primarily refer to the works of Śrīla Visvanātha Chakravarti Thākura (his commentary on the *Śrīmad-Bhāgavatam*) and Śrīla Vādirāja Tīrtha swāmi (his book known as *Bhūgola-Varnanam)*. In order to express our gratitude to these eminent Ācāryās, a brief sketch of their life and activities is presented in Appendix 1. This is to substantiate the authenticity of their works in this technical subject matter.

The wonderful activities and contribution of these ācāryas to the world are testimony to the authenticity of their writings, specifically their explanations in the subject of Cosmology or Astronomy. At present, science and traditional knowledge seem to be at loggerheads. Therefore, we shall endeavor to point out the limitations of scientific investigation and the authoritative declaration

of the puranas and Vedic literature in general as explained by these acaryas.

The Śrīmad-Bhāgavatam is the science of Lord Kṛṣṇa, the Absolute Personality of Godhead of whom we have preliminary information from the text of the Bhagavad-gītā. Śrīmad-Bhāgavatam is the transcendental science not only for knowing the ultimate source of everything but also for knowing our relation with Him and our duty toward perfection of the human society on the basis of this perfect knowledge. It is powerful reading matter and simply by a careful reading one will know God perfectly well.

This Bhāgavata Cosmology book describes some of the wonderful activities or pastimes of the Absolute Personality of Godhead as described in the Śrīmad-Bhāgavatam. Herein there is a brief summary description of the creation of unlimited number of Universes with their distinct features and divisions by the Supreme Lord, the first *puruṣa*, as it is described in the second and third Canto's of the Śrīmad-Bhāgavatam. It further describes how Lord Brahmā creates various living entities to populate the fourteen divisions manifested from the lotus originated from the navel of the Supreme Lord. The creation by Lord Brahmā of the immovable living entities, variety of animals, human beings and the devatās in four stages is briefly described herein as per discussion in the third Canto of Śrīmad-Bhāgavatam.

Chapter 1 of this book describes the Primary Creation (*Sarga* - the Creation of the totality of matter with elements manifesting from prakṛti by Lord Viṣṇu's glance) and the Sub-creation (*Visarga* - the creation of bodies for the jīvas by the resultant combination of the twenty-four elements from Lord Brahmā's mind and body).

The principle discussion of this book is "the Structure of the Universe, *Sthānaṁ*," which is elaborately discussed in the fifth Canto of Śrīmad Bhāgavatam. Śrīmad-Bhāgavatam Canto 5 describes *Sthāna*, the third of the ten subject matters of *Bhāgavata Purāṇa*. '*Sthānam*' defined as "*sthitir vaikuṇṭha-vijayaḥ*", the right situation for the living entities is to obey the laws of the Lord. *Sthānam*, the condition (*sthitiḥ*) of the *jīvas*, refers to the protection offered by the Lord to His devotees by destroying their suffering, shich shows the excellence of the Lord (*vaikuṇṭha-vijayaḥ*). *Sthāna* is basically the description of the geography of the Universe, where the demigods, men and

demons reside and its protection by the demigods. This points out to the maintenance of the inhabitants of the Bhūloka, Pātala and Svarga, who are situated there with their governing rules.

Chapter 2 describes "the Bhū-maṇḍala – the Horizontal Structure of the Universe". Herein there is a description how Priyavrata maharaja divides the Bhū-maṇḍala into seven islands (*dvīpas*) and seven Oceans. The innermost island Jambūdvīpa is divided into nine varṣas with nine boundary Mountains. The flow of river Ganga through various parts of Jambūdvīpa and the wonderful prayers offered by the residents of the nine varṣas of Jambūdvīpa are briefly described. Further there is a description of the other six islands of the Bhū-maṇḍala with their mountains, rivers and the places outside the seventh islands, such as svarna bhumi, Lokāloka Mountain and Āloka-varṣa.

Chapter 3 describes "the Fourteen Planetary Systems - the Vertical Structure of the Universe." Briefly describes the sun, moon and other planets, with their locations and characteristics. There is a description of Dhruva-loka, the Form of Dolphin and the Upper Planets, followed by the description of the Lower Planets (places below the sun), such as *bhuvar-loka* and *bila-svarga*. Finally after describing the glories of Lord Ananta, the holder of the Bhū-maṇḍala, this chapter ends with the description of the Hellish Planets, the places of punishment for sinful actions of sinful people.

Chapter 4 describes "the Vedic Planetarium", the movements of the sun, moon and other planets. Wherein it describes how the day and night are formed, how the lunar and solar months are calculated, and how the moon changes its phases over a period of a month. Finally gives a brief description of the movement of the other planets in the Zodiac.

Chapter 5 describes the vertical movements of the sun, moon and rahu, because of which there are changes in the seasons and the eclipses are formed once in a while. The Vedic understanding of the change in seasons and the change in day-night time over a period of one year is discussed by quoting various Puranas. Then the Vedic understanding of the eclipses and its occurance over a period of six months is discussed briefly as it is discussed in the Śrīmad-Bhāgavatam.

Chapter 6 briefly describes on how both the modern understanding of the Universe and Vedic understanding of the Universe are comparable to each other with computation and comparison of many of the parameters mathematically. Some of these calculations for comparison with modern scientific parameters are inspired by Prem Gauranga Prabhu and some of them are directly taken from his research work on this subject matter.

Definitions of terms used in this Book

Sarga: Primary Creation by Lord Maha-Vishnu, or the creation of the 24 elements, the formation of the Universal egg by their combination, manifestation of Virat-rupa and the birth of Lord Brahma.

Visarga: Secondary Creation by Lord Brahma or the creation of different living entities by Lord Brahma and his progeny, the Prajapatis and the Manus

Dvīpa: Dvi-apa "the land that contains water on both the sides"

Dvīpa is also known as that which drinks (sea water) by two sides, i.e., from East & West or from North & South. In the absence of such drinking, there would be deluges as sea-water would submerge all life forms on the island.

Dvīpa is also known as the place where the water entered twice or inside. Without it drinking, the heat of the sun's rays would cause the earth to lose its solidity. Due to such atomic disintegration, inhabitants of the island would be submerged (as the substrate would be lost).

Varṣa: the Land or the Valley between two Mountains

1 Yojana: A measurement of distance equal to 6.06 / 8 miles approximately

Saint Śrīla Vādirāja Tīrtha swāmi, a master of Vedic mathematics, gives its divisions and calculations:

7 Sarsapas (mustard seeds) - 1 Vrhi (length of a paddy seed)
3 Vrhis - 1 Angula (inch)
24 Angulas - 1 Hasta or 1 cubit (= 2 feet)
4 Hastas - 1 Danda, staff length (= 8 feet = 96 inches)
1000 Dandas - 1 Krosa (= 4000 hastas = 8000 feet or 96,000 inches)
4 Krosas - 1 Yojana (= 32000 feet or 16,000 hastas or 384,000 inches)
Considering 1 mile = 5278.87 feet or 1609 meters
1 Yojana = 6.06 miles (= 32000 feet / 5278.87 feet) approximately

Thus the scholars of Madhva accept 6.06 miles per Yojana for all the calculations referred in the book "*Bhugola Varnanam.*" Śrīla Visvanātha Chakravarti Thākura, Śrīla Prabhupāda and many other ācāryas used 8 miles per Yojana for all calculations in *Śrīmad-Bhāgavatam.* The Madhva

scholars reconcile this as below.

The fact that seeds and the like were bigger in the previous ages and as time progresses their sizes are reduced as indicated in SB 12.2.1. For the same reason, a man's cubit or Hasta measured 24 inches then as against 18 inches now obtained. Consequently, the average height of man in those days was 8 feet, of course, leaving margin for abnormalities. Incidentally, we may note that Acharya Sri Madhva is described as a personality of eight feet in height and that everyone could have darsan (or cite) of the bust of that personality as he moved in the midst of crowds in procession. By reconsidering the above change in Hasta (cubit) measurements, 1 Yojana will be:
1 Yojana = 16000 Hastas x 24/18 = 21333.333 Hastas = 42,666.667 feet
Considering 1 mile = 5278.87 feet or 1609 meters
1 Yojana = 8.08 miles (= 42,666.667 feet / 5278.87 feet) or 8 miles (approximately)

Richard L. Thompson (also known as Sadaputa das) mentions in his book 'Vedic Cosmography and Astronomy' about the calculation of Yojana.
One standard definition of a yojana is as follows "one yojana equals four krośas, where a krośa is the maximum distance over which a healthy man can shout and be heard by someone with good hearing." It is difficult to pin down this latter figure precisely, but it surely could not be much over two miles.

1 krośa = 2 miles
1 yojana = 4 krośas = 8 miles

Another definition is that a yojana equals 8,000 nṛ, or heights of a man. Using 8 miles per yojana and 5,280 feet per mile, we obtain 5.28 feet for the height of a man, which is not unreasonable.

1 yojana = 8,000 nṛ, or heights of a man
1 yojana = 8 miles = 8 x 5,280 feet per mile = 42,240 feet
Height of a man = 5.28 feet

To justify these reductions in scale, Vaṁśīdhara observes that the yojana is defined on the basis of the human body. Thus a yojana is 16,000 hastas or cubits (sometimes mentioned as 32,000 hastas), and a hasta is 24 finger-widths. Also, a hasta can be defined as one fifth the

height of a man standing with his arms stretched up. As the bodies of infants, children, and adults vary greatly in size, so the yojana also varies, and in this way one can explain differences between various estimates of distance.

Acarya Sri Vamsidhara in his commentary to SB 5.16.7 quoting Siva Purana gives the divisions and calculations of a Yojana.

1 Tryanuka = 6 Atoms
1 Vālāgra = 8 Tryanukas
1 Liksā = 8 Vālāgras
1 Yuka = 8 Liksās
1 Yava (Anguli) = 8 Yukas
1 Tāla = 8 Yava (Anguli)
1 Hasta = 2 Tāla (16 Angulis)
1 Danda = 4 Hastaṣ
1 Krośa = 2000 Dandas (~2 miles)
1 Yojana = 4 Krośas (8000 Dandas; 32000 Hastas; 512000 Angulis; 8.0825 miles)
Considering 1 mile = 1.609 km = 5278.8714 feet = 63,346.457 Angulis / Inches

1 MY: 1 Million Yojana (10 lakh yojanas)

1 AU: 1 Astronomical Unit, which is the radial distance between the earth globe and the sun, which is approximately equal to 150 million kilometres.

Interestingly Hanuman chalisa, written by Goswami Tulasidas in 15th century, gives the distance between sun and earth. This has been calculated much more accurately than the 17th century scientists even before 2 centuries. Here's a portion of Hanuman Chalisa, a two-line Prayer, computes this distance with great simplicity.

tumharo mantra vibhīshana māna, lankeshwara bhaya saba jaga jānā
juga sahastra yojana par bhānu, līlyo tāhi madhura phala jānu

This means that the sun (*bhānu*) is at a distance of "*Juga sahastra yojana.*" According to the traditional conversion practices that are in use as per Vedic Literature this comes to,

1 Juga = 12000 demigod years
1 Sahastra = 1000

1 Yojana = 8 miles

Thus Juga sahastra yojana = 12000x1000x8 = 96,000,000 miles, where 1 mile = 1.6 km

This further implies that the distance is 96,000,000 x 1.6 km = 153,600,000 km (~1AU)

Divya Yuga: Total time period of Satya, Treta, Dvapara & Kali Yugas
Manvantara: The rule of a Manu's time period (71 divya yugas)

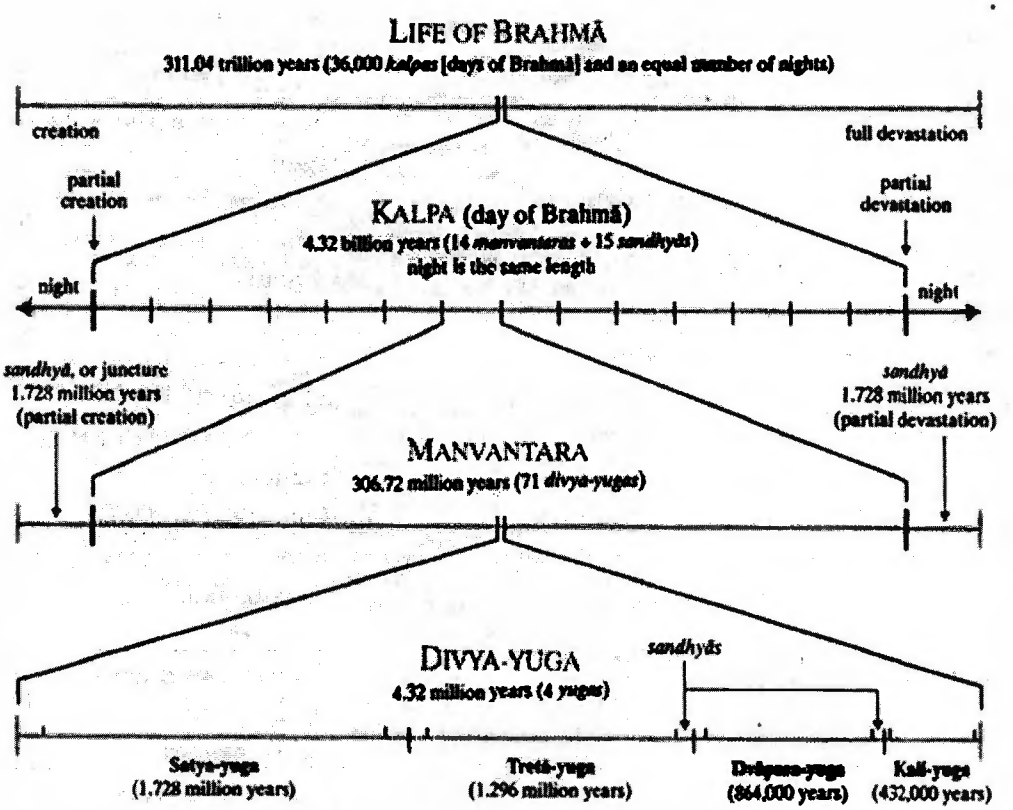
Kalpa or 1 Day of Brahma: 14 Manu periods (1000 divya yugas)
Maha Kalpa or Life of Brahma: 100 years according to his days and months, where 1 month is 30 days and 1 year is 12 months as per his calculations.

Calculation of Time, from the Atom to the Life of Lord Brahma[1]

Paramanu (Atom)	Ultimate indivisible particle of material manifestation.	
Param Mahan	Unmanifest aggregate of Atoms (Complete Manifestation).	
Atomic Time	Time required for the sun to pass an Atom or time taken to pierce a lotus through a needle.	
Greater Time (Brahma's life)	Time required to cover the unmanifest aggregate of Atoms.	
Trasareṇu	Two paramāṇus make an aṇu. Three aṇus make a trasareṇu, which is perceived by the eye, since if follow the air currents in the rays of the sun entering through a window lattice.	
One Truṭi	The time it takes for the sun to pass over three trasa-reṇus (3 hexatoms) = 8/13,500 sec	
1 truṭi	Time to the integration of 3 trasareṇus	8/13500 sec (1/1687.5 sec)
1 vedha	100 trutis	8/135 sec
1 lava	3 vedhas	8/45
1 nimeṣa	3 lavas	8/15 sec
1 kṣaṇa	3 nimeṣas	8/5 sec

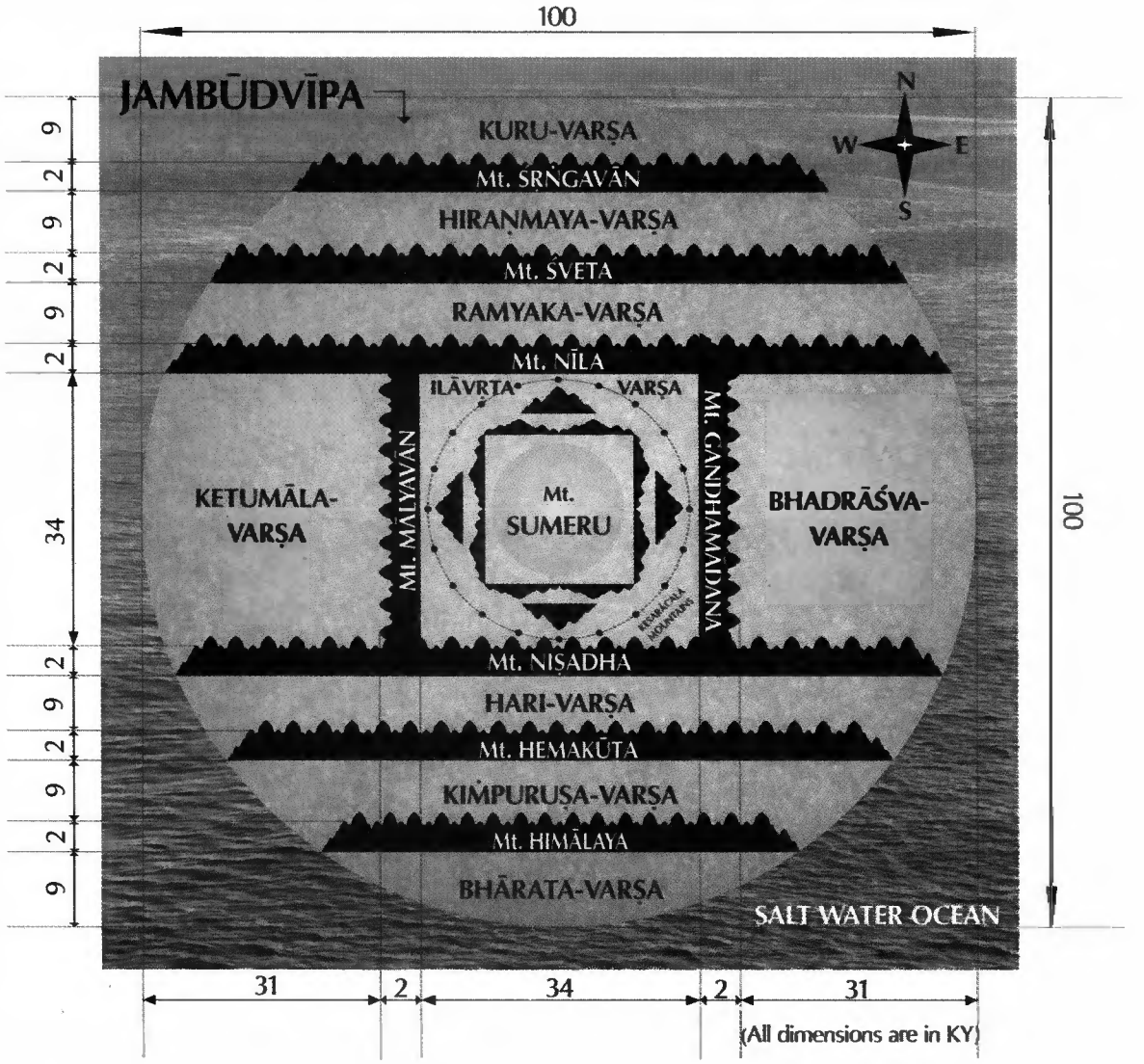
[1] SB 3.11: Calculation of divisions of time from divisions of distance

1 kāṣṭhā	5 kṣaṇas	8 sec
1 laghu	15 kāṣṭhās	2 min / 120 sec
1 nāḍikā / daṇḍa	15 laghus	30 min (sometimes 24 min)
1 muhūrta	2 daṇḍas	60 min (sometimes 48 min)
1 prahara - 1/4th of a day / night	6 or 7 daṇḍas	3 hours
yāma = 1 day or night	4 praharas	12 hours
1 Fortnight	15 days + nights	Half month
1 month	2 fortnights (white and black)	1 night and day of Pitā planets
1 season	2 months	One sixth of a year
1 movement of sun from south to north or north to south	1 day or night of the demigods	6 months (uttarayana or daksinayana)
1 calendar year	1 day and night of the demigods	2 solar movements (uttarayana & daksinayana)
Duration of human life	100 days of demigods	100 human years
1 year of demigods	360 days of demigods	360 human years
1 Satya Yuga	4,800 demigod years	1,728,000 human years
1 Treta Yuga	3,600 demigod years	1,296,000 human years
1 Dvapara Yuga	2,400 demigod years	864,000 human years
1 Kali Yuga	1,200 demigod years	432,000 human years
1 Divya Yuga (S+T+D+K)	12,000 demigod years	4,320,000 human years
1 Day or Night of Brahma	14 Manus & Demigods	1000 Divya Yugas
1 Manu's Life	71 Divya Yugas = 852,000 demogod years	306.72 million human years
1 Day of Brahma = 1 Night of Brahma	12 million demigod yr's	4.32 billion human years
1 Brahma's Life = His 100 years	864 billion demigod years	311.04 trillion human years
1 nimesa of Supreme Lord	Brahma's 100 years	311.04 trillion human years

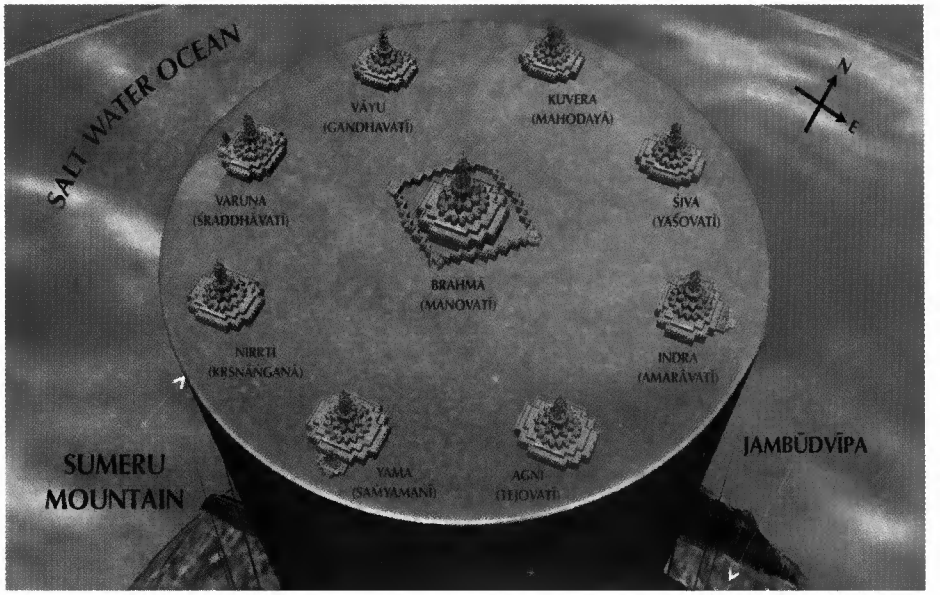


Expansions of the words used in this Book

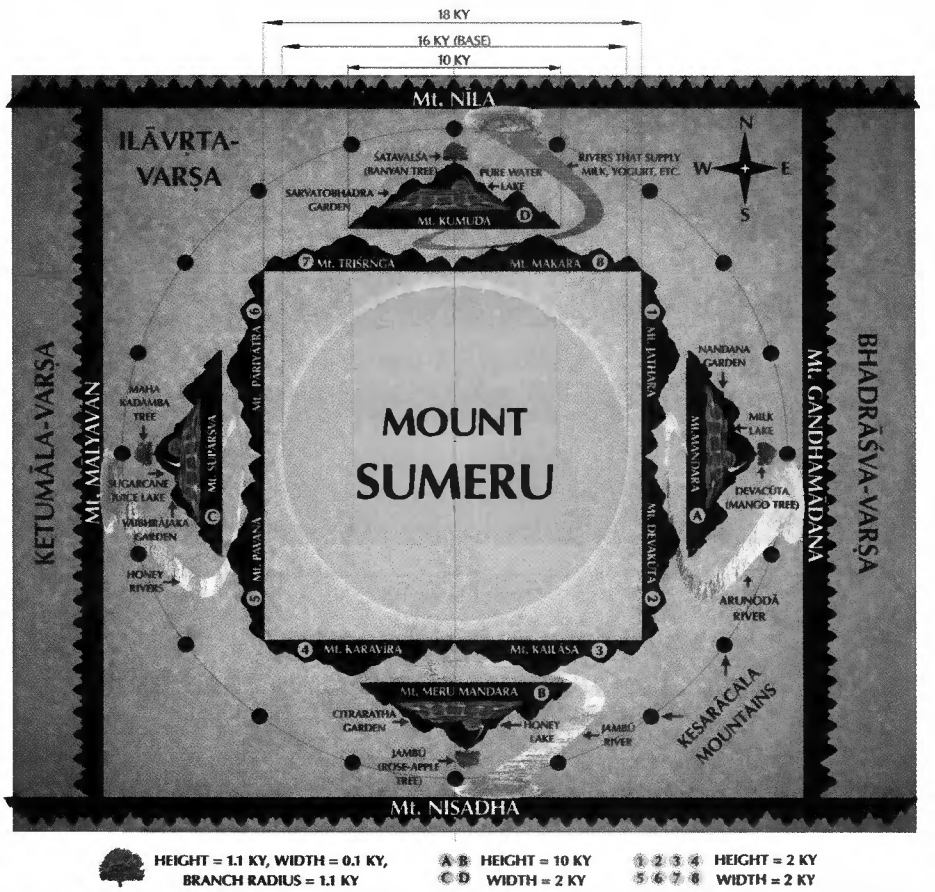
SB – Śrīmad-Bhāgavatam
BG – Bhagavad Gīta
VCT – Śrīla Visvanātha Chakravarti Thākura
MY – Million Yojanas
LxWxH – Length x Width x Height
1.2.3 – SB Canto One, Chapter Two, Verse Three



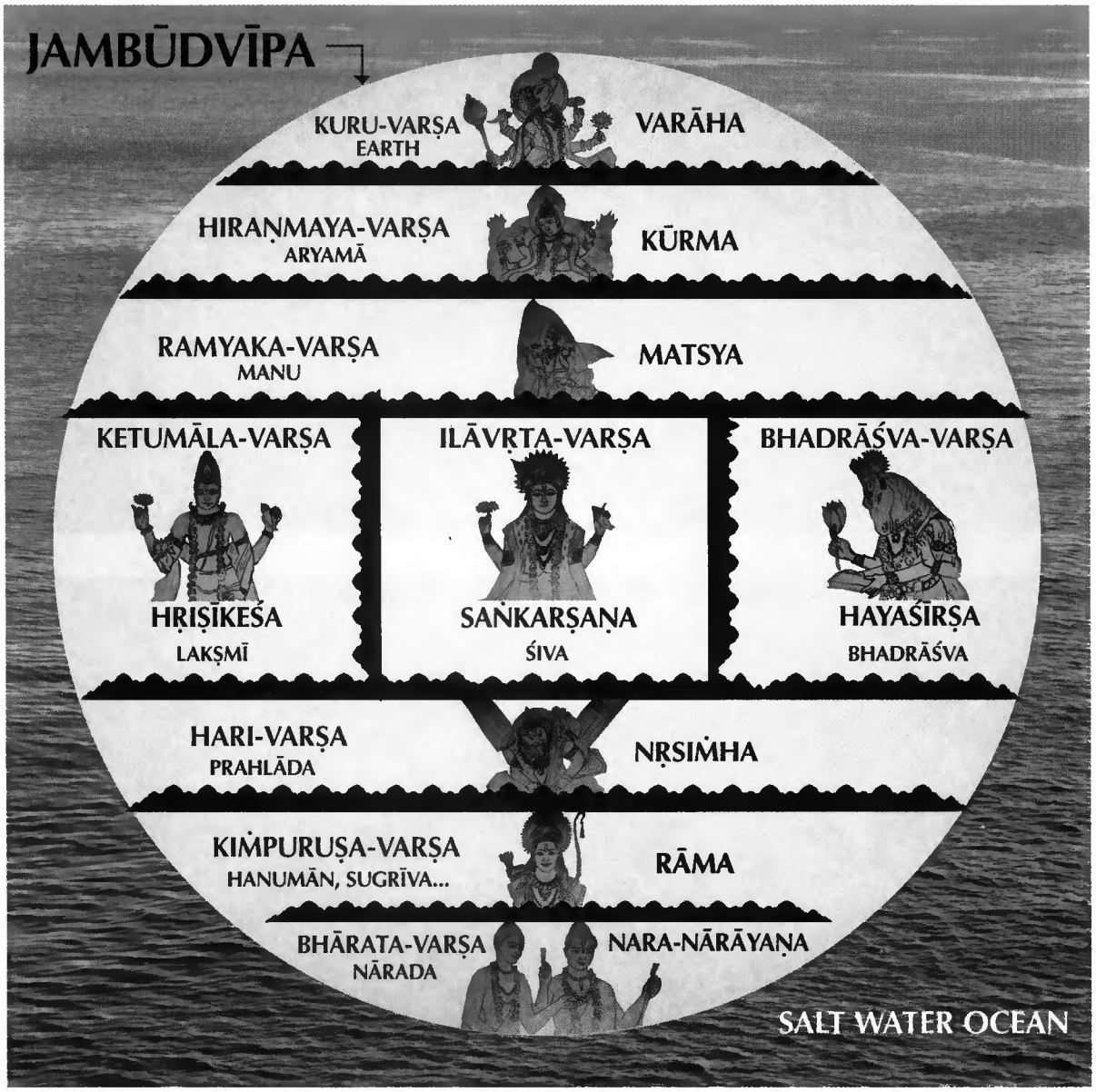
Top view of Jambūdvīpa – Jambūdvīpa and its varṣas



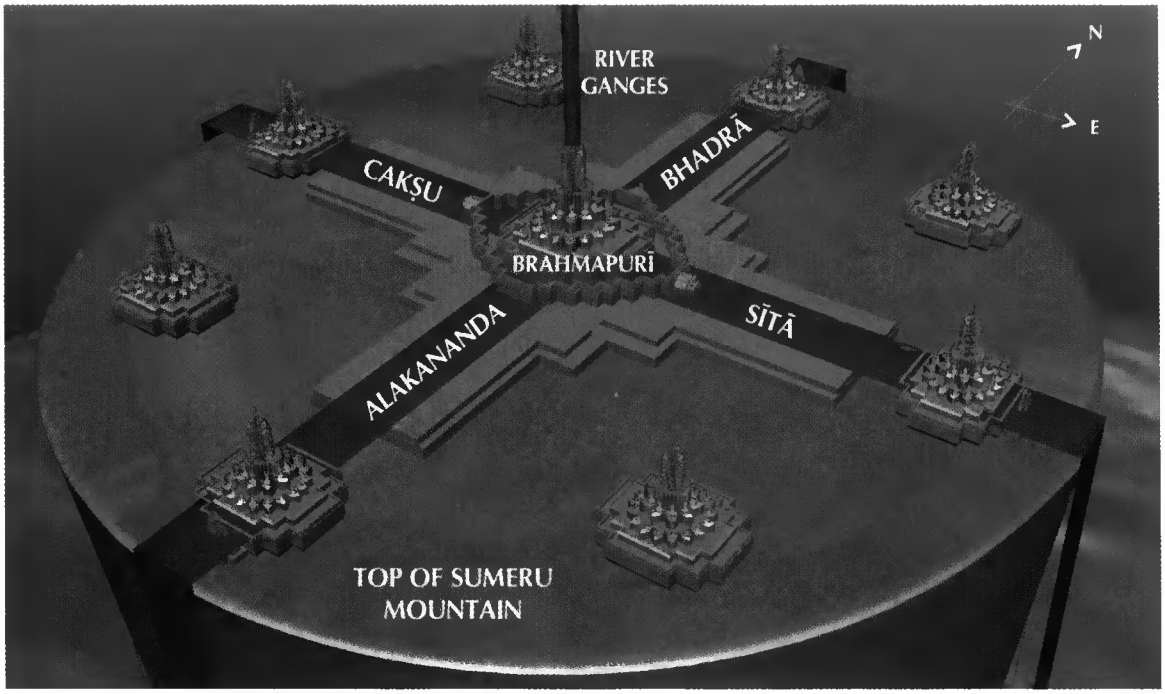
The Townships of Lokapālās on top of Mount Sumeru

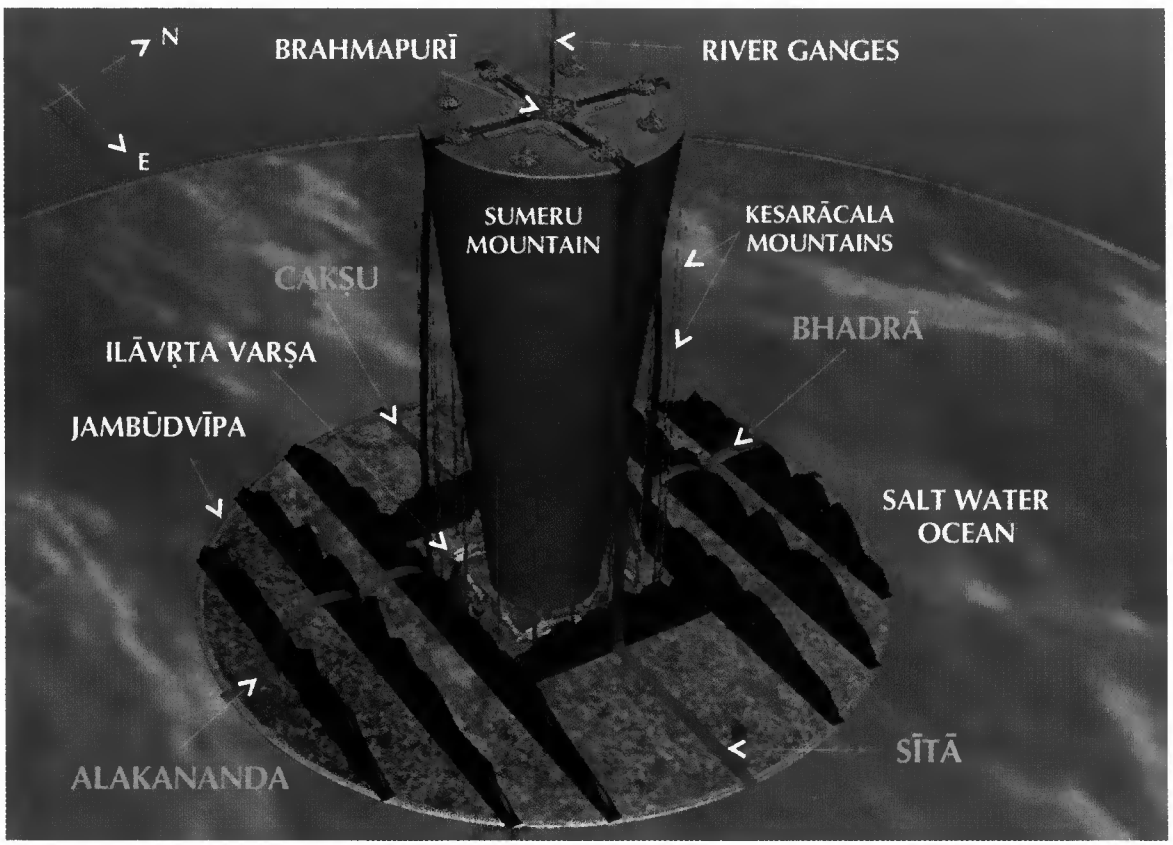


Top view of Ilāvṛta-varṣa

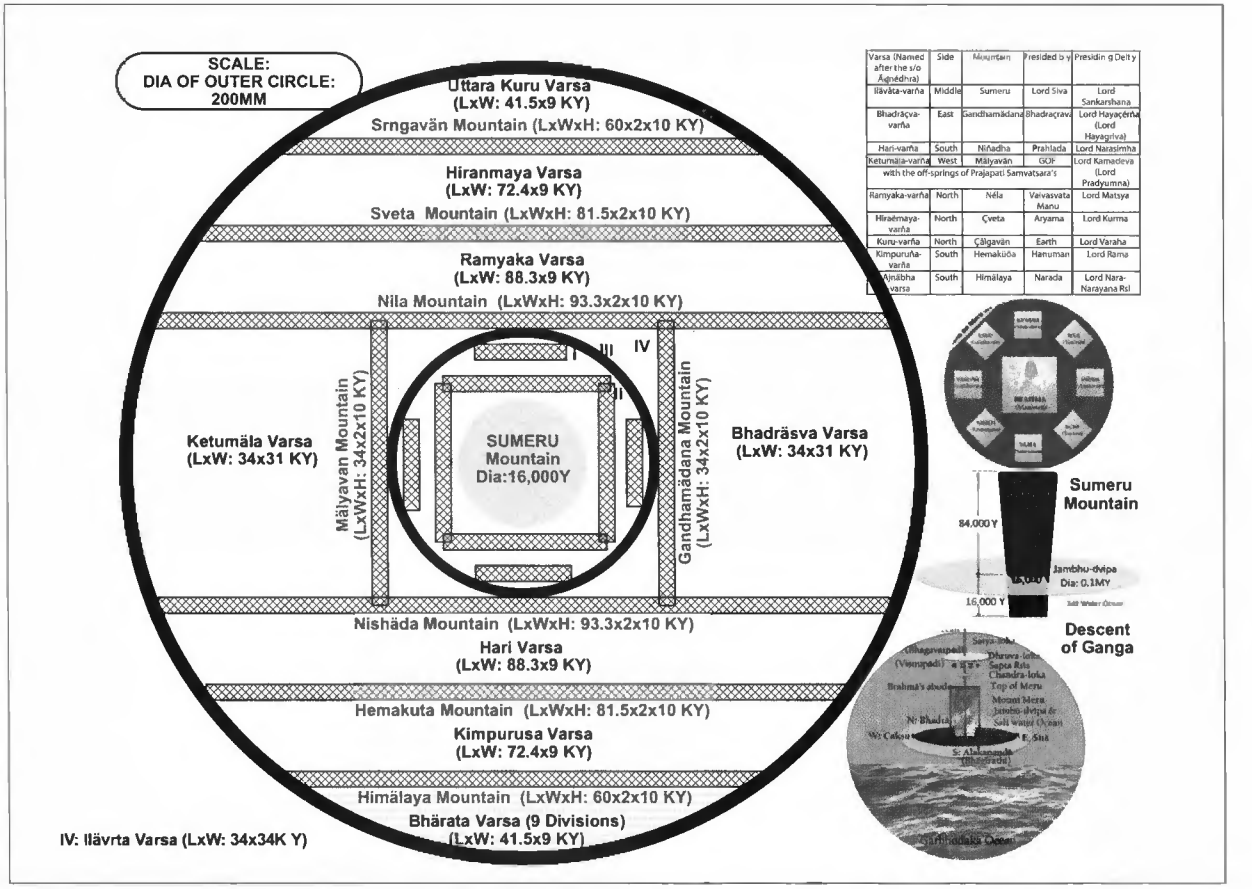


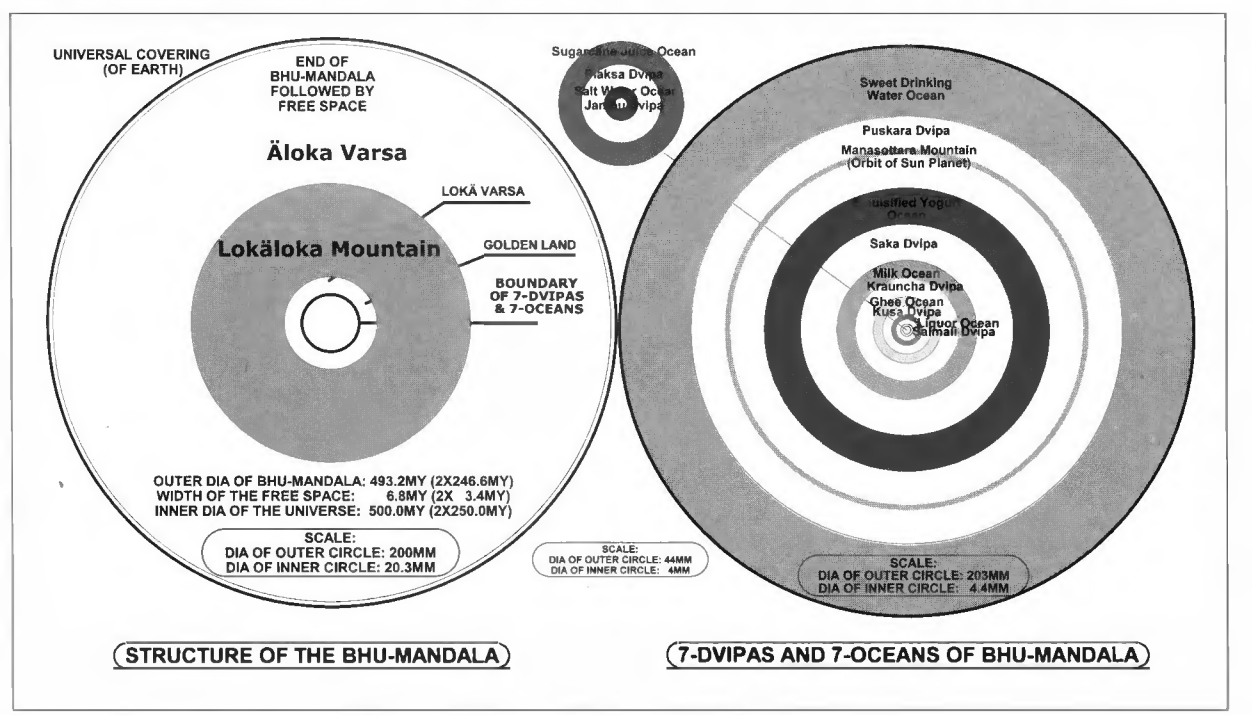
The presiding deities of the nine varṣas of Jambūdvīpa



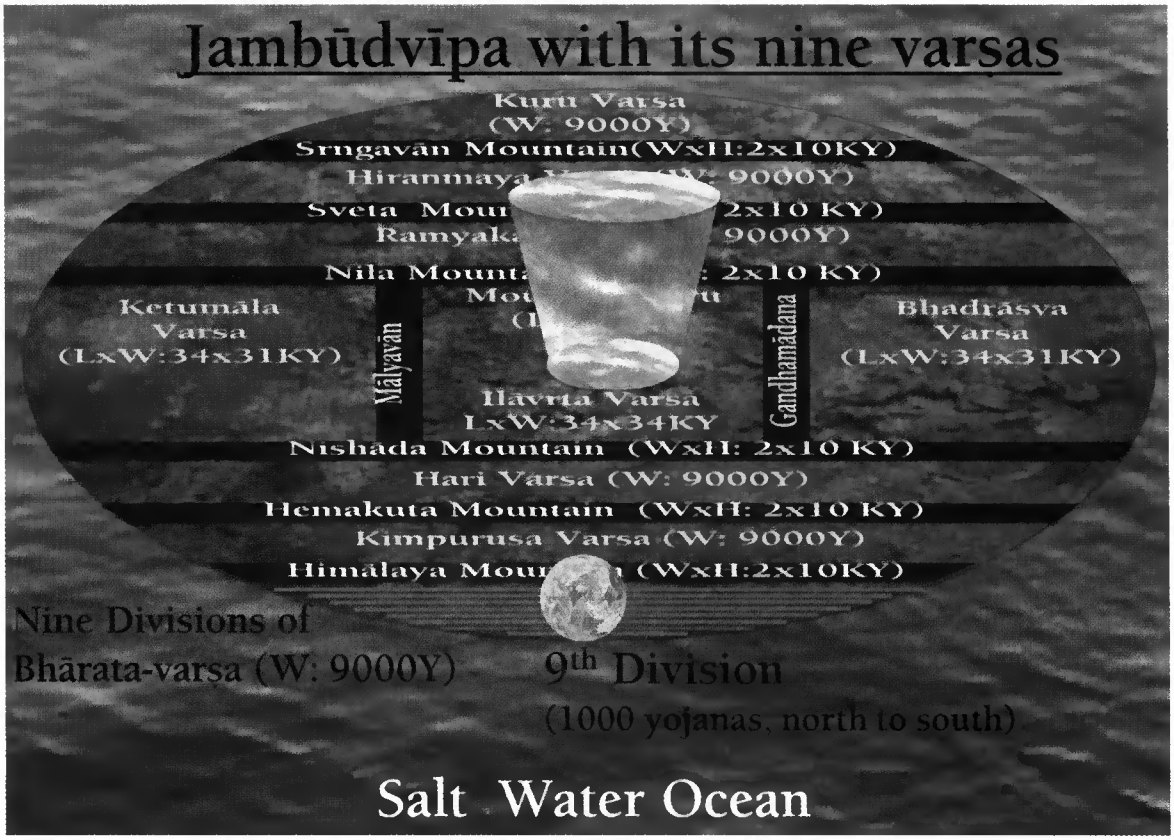


The Branches of River Gangā

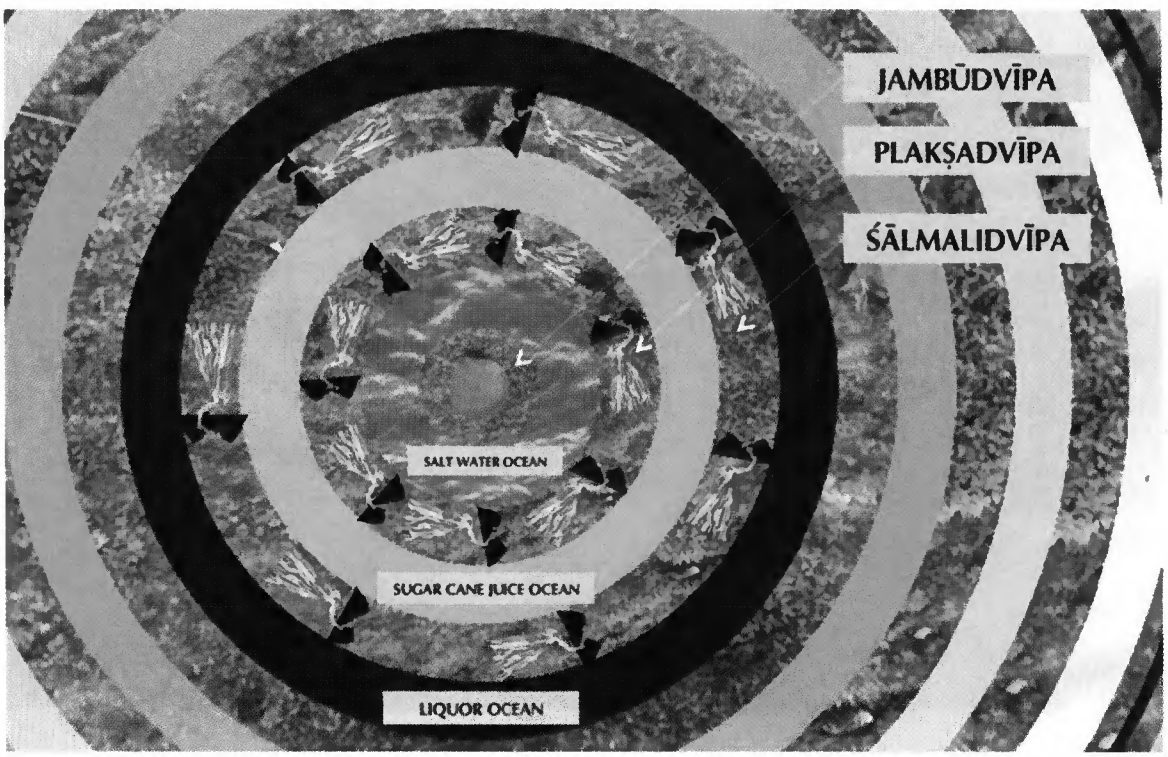




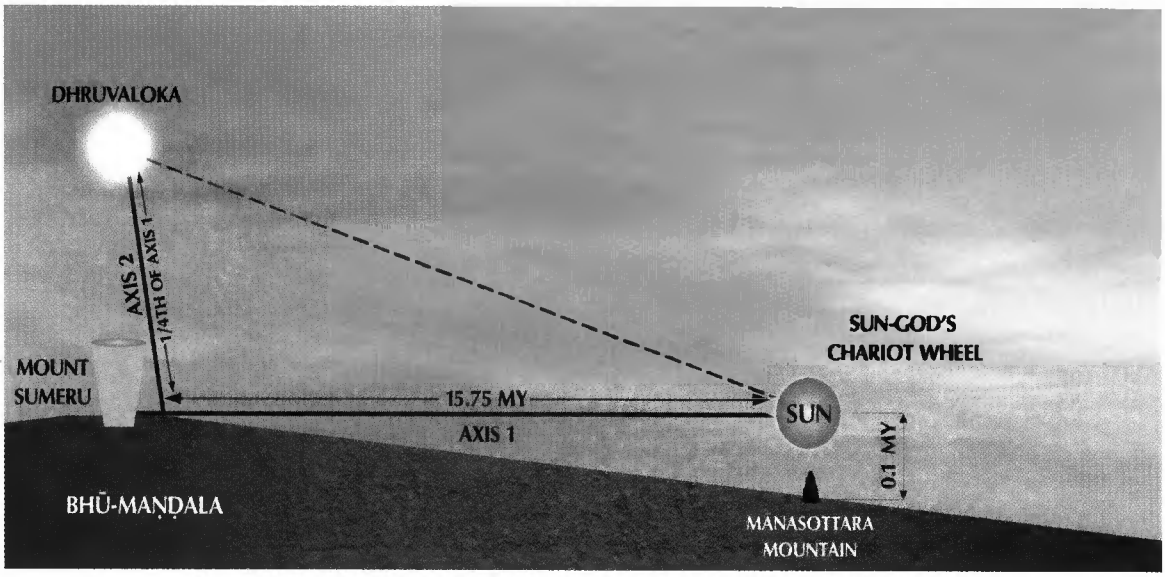
5.16-20: The horizontal cross-sectional view of the Bhū-mandala.



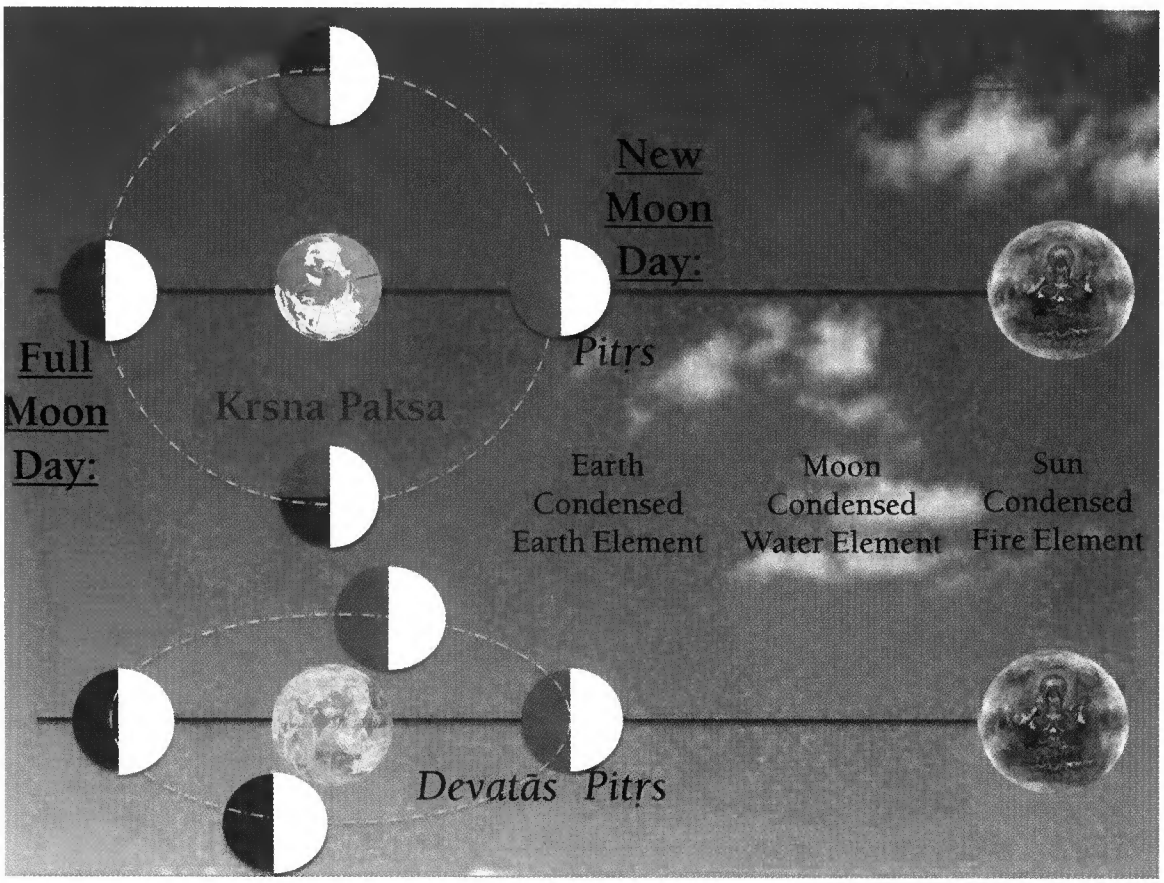
Isometric view of Jambūdvīpa – Divisions of Bhārata-varṣa



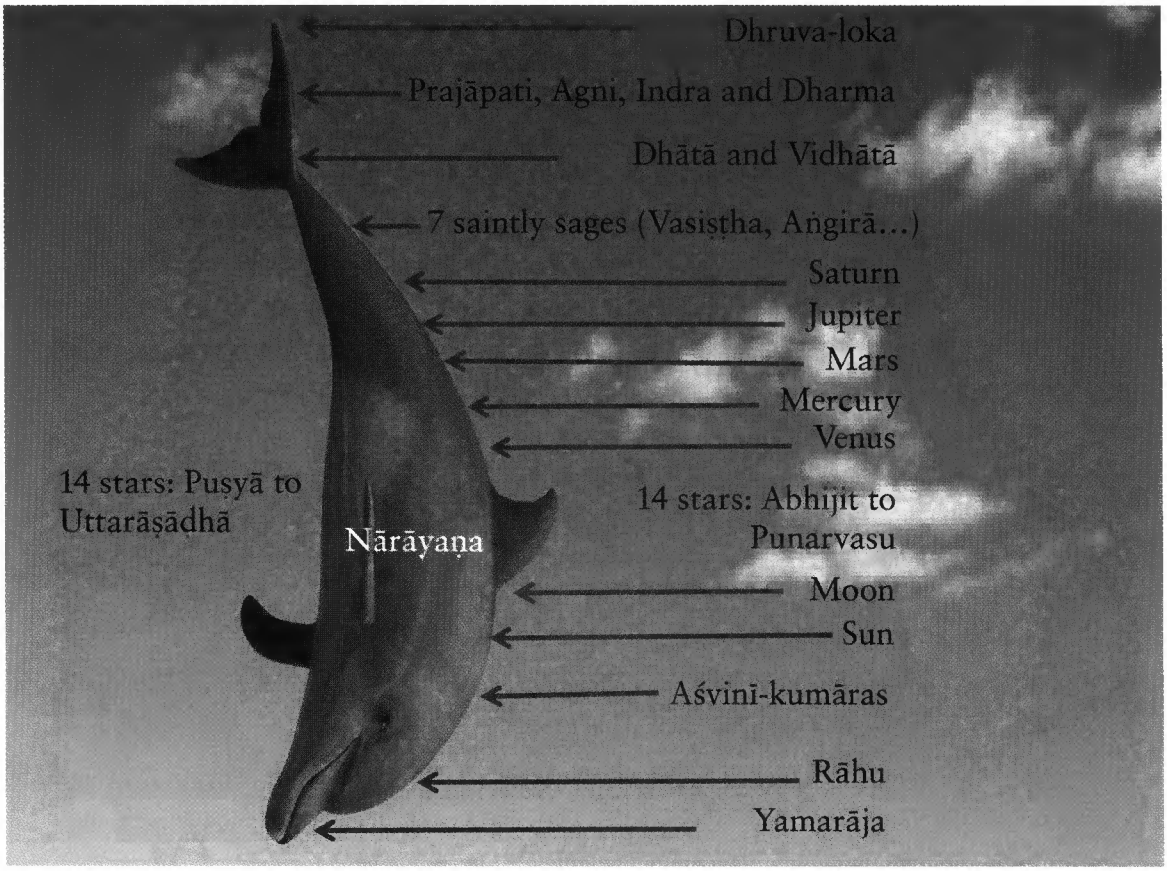
Mountain ranges and rivers of Plakṣadvīpa and Sālmalīdvīpa

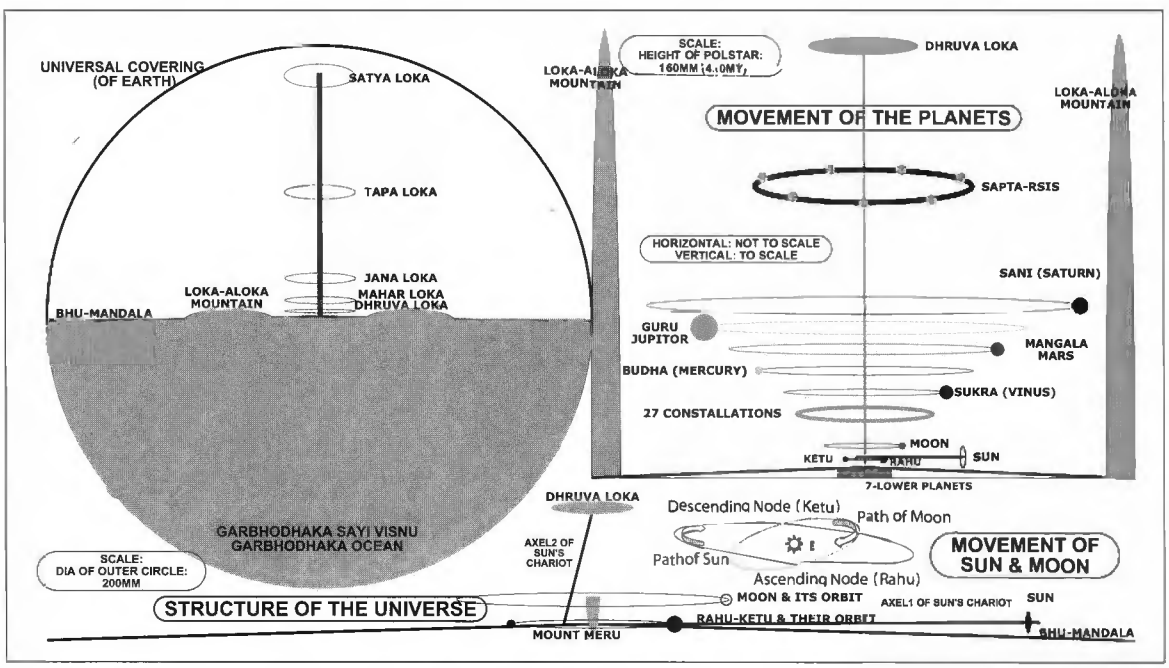


The Chariot of Sun-god

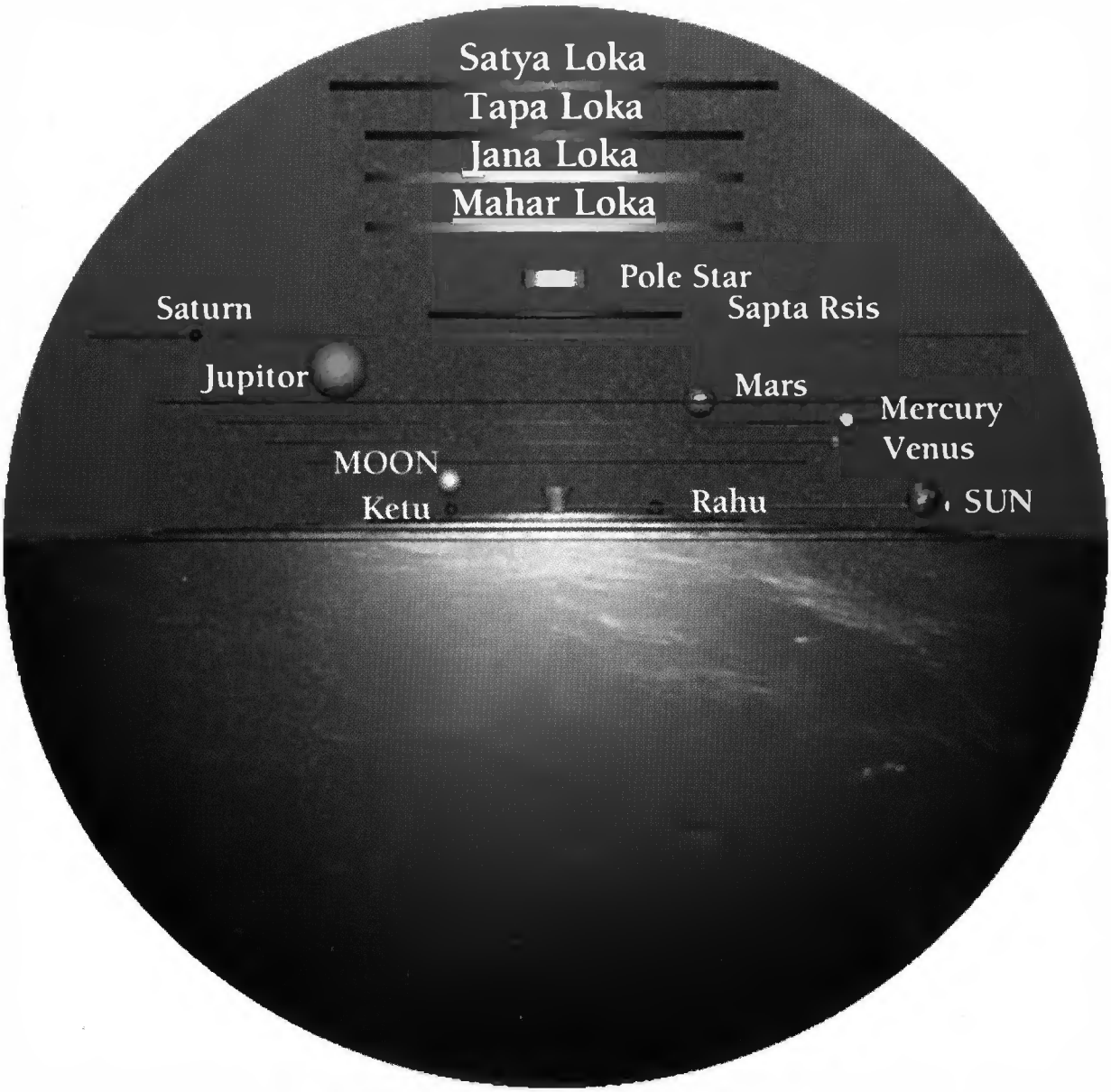


The Constitution of the Moon

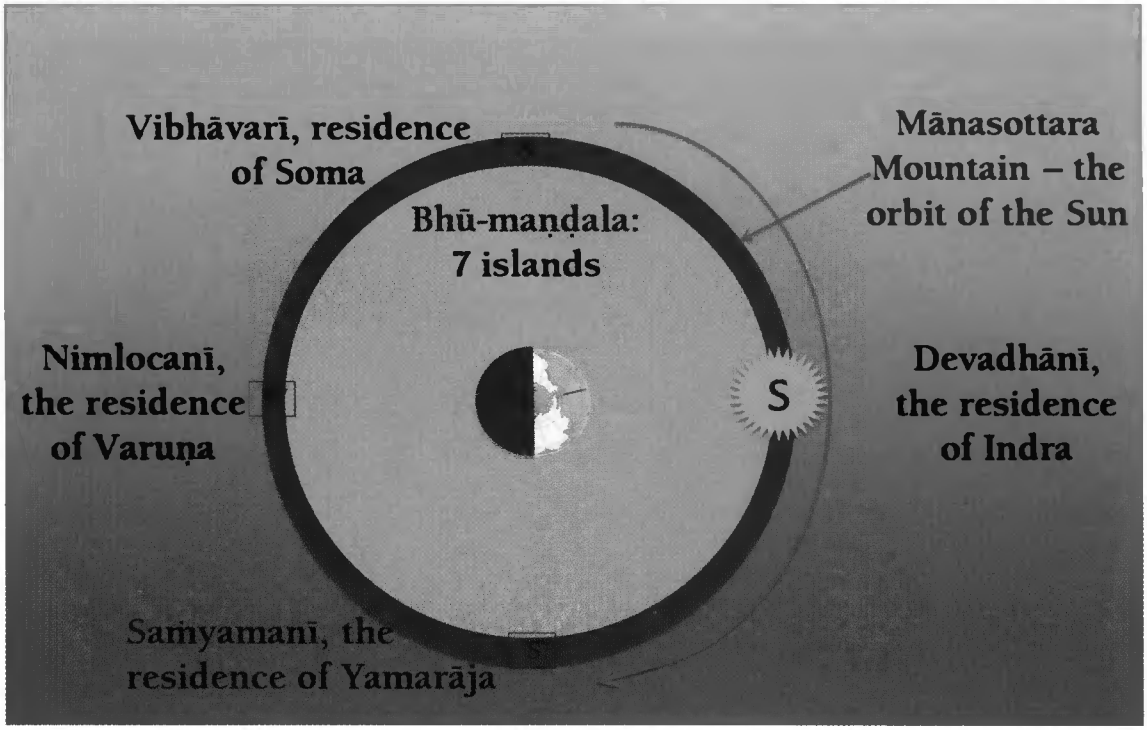




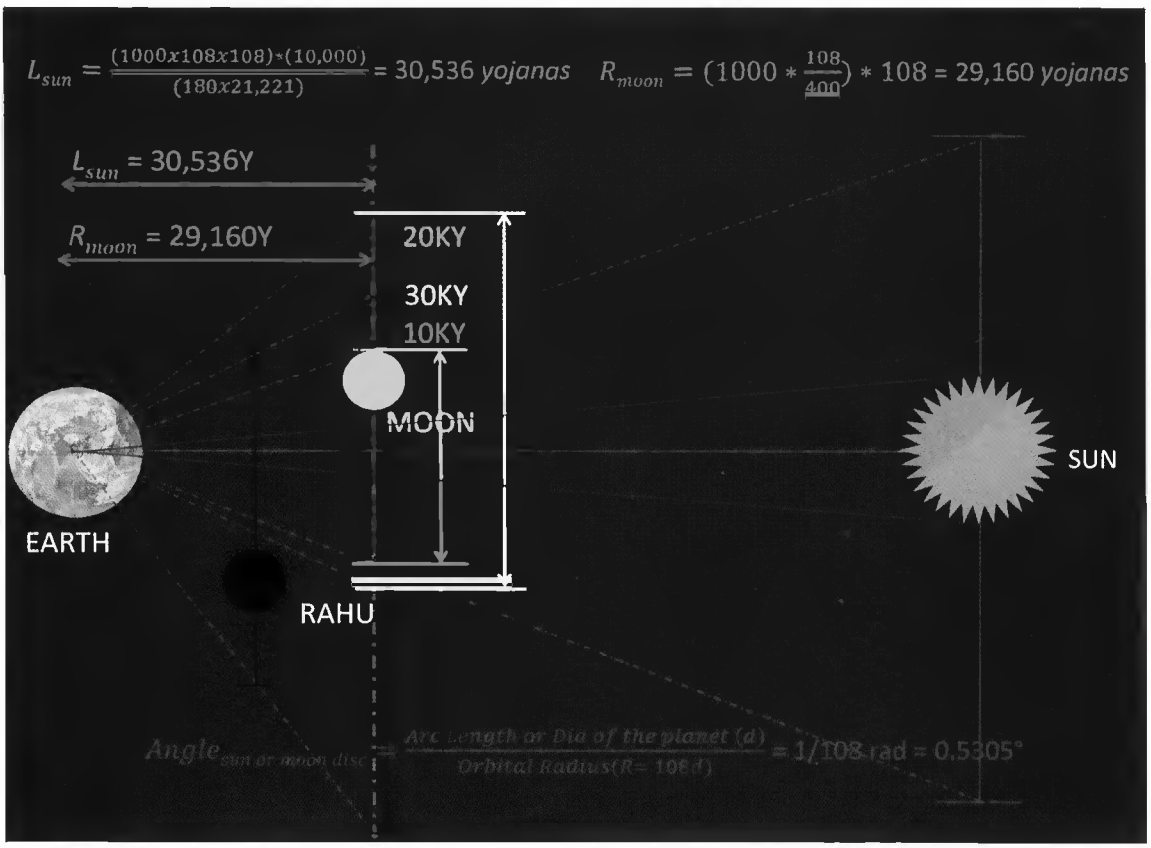
5.21-26: The vertical cross-sectional view of the Universe



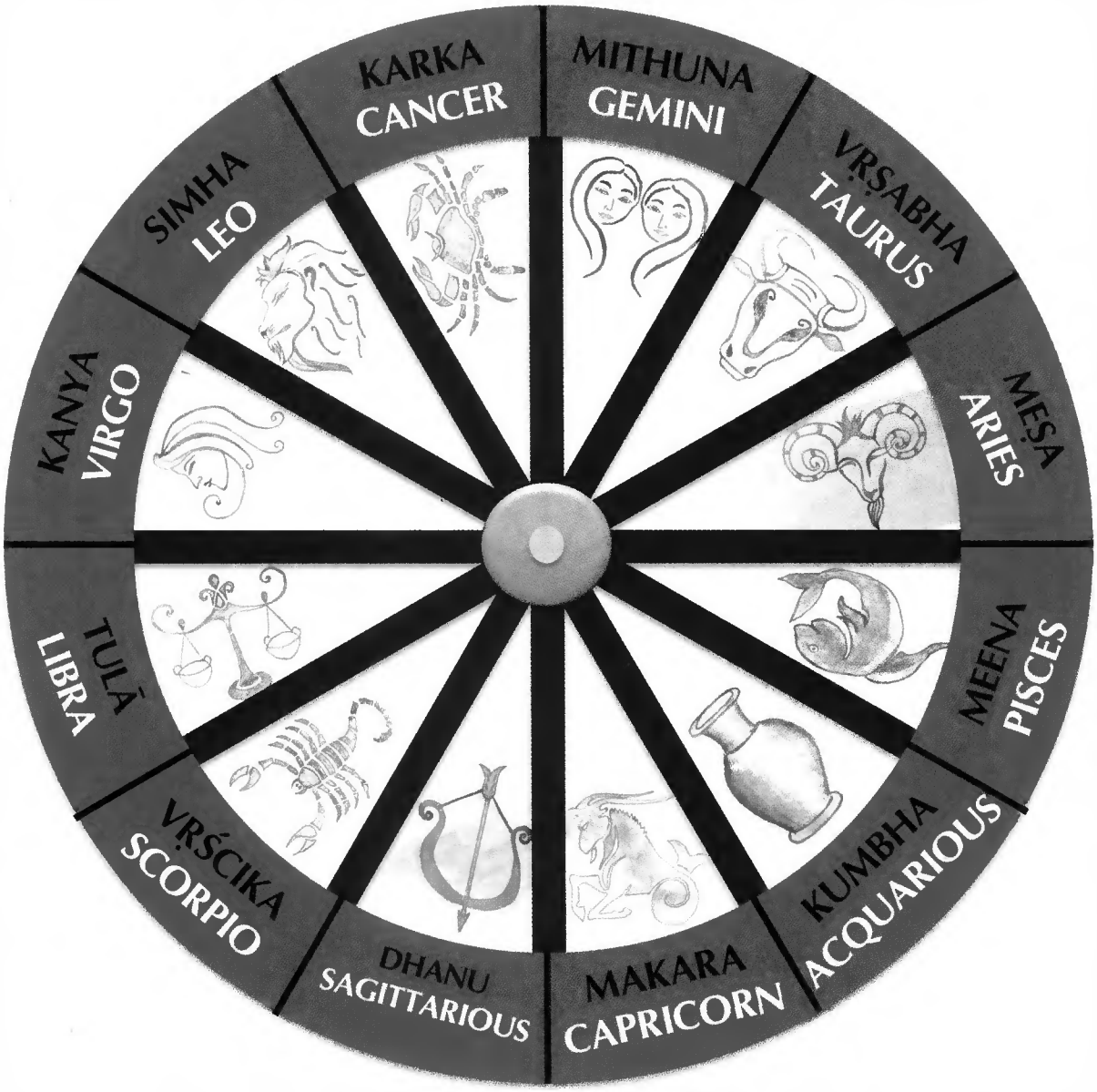
3D Structure of the Universe



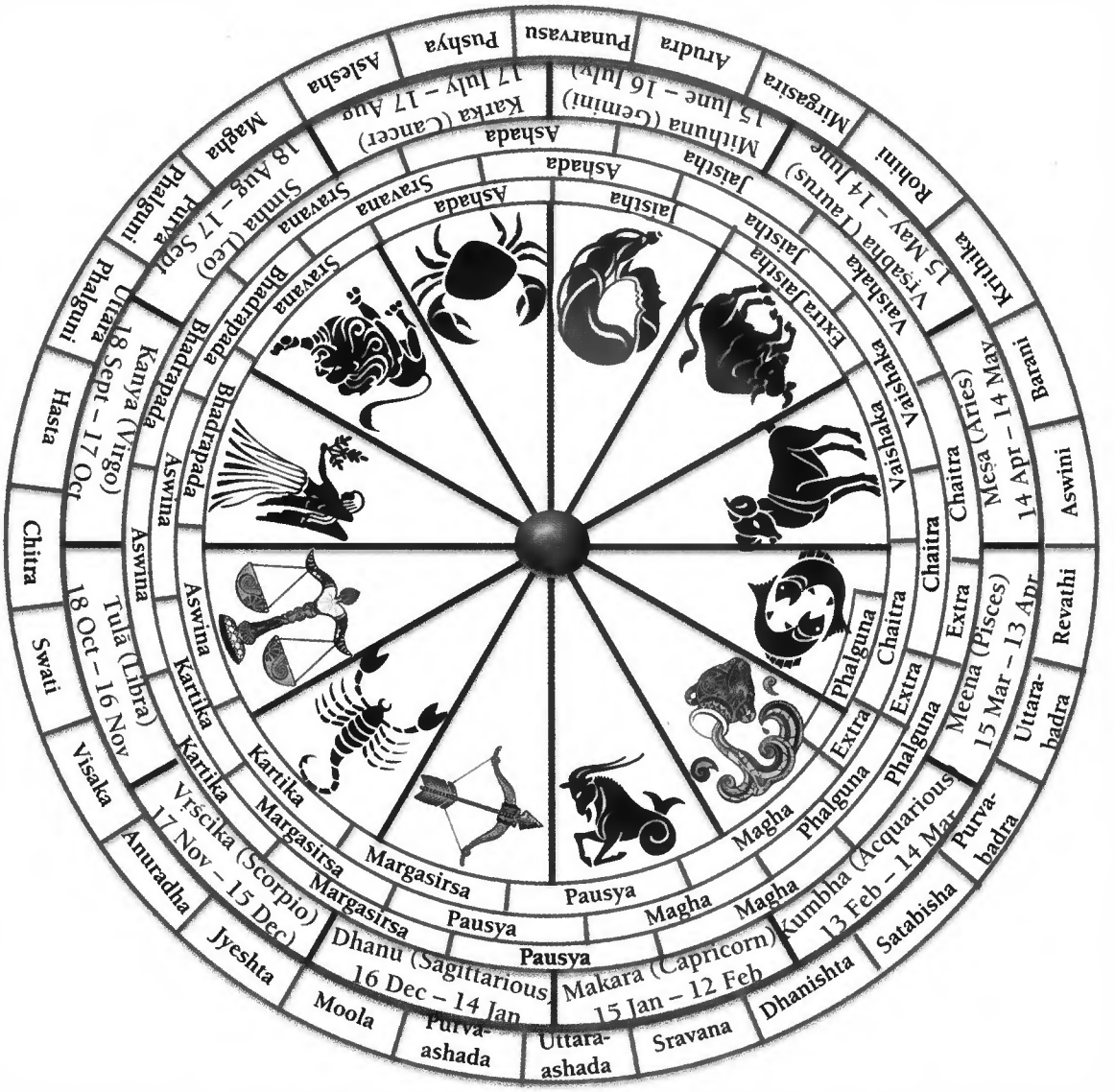
The Daily movement of Sun



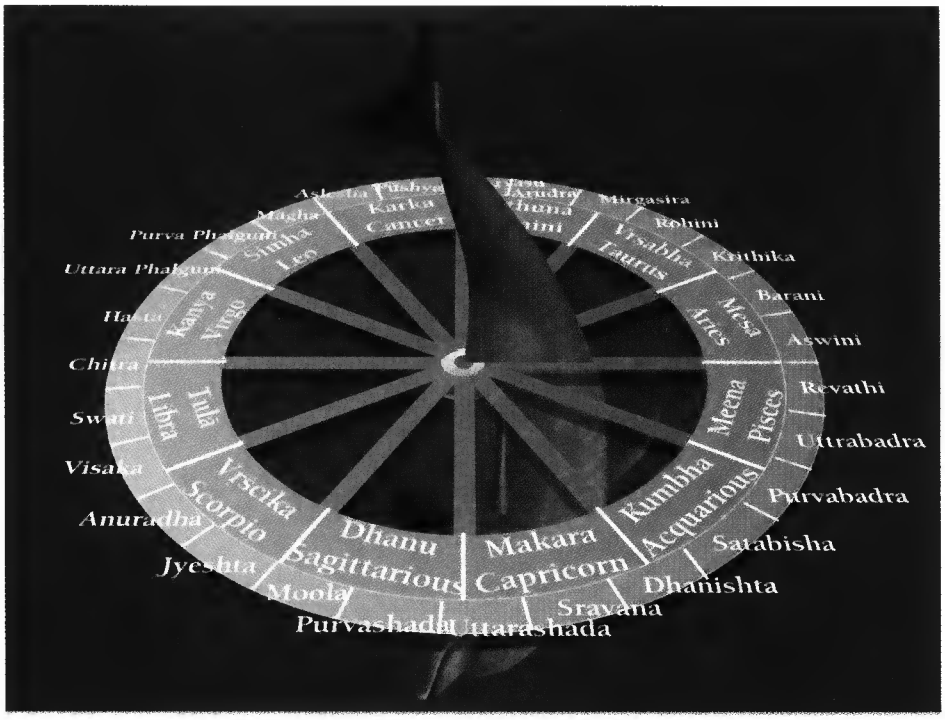
The Yearly movement of Sun, Moon and Rahu

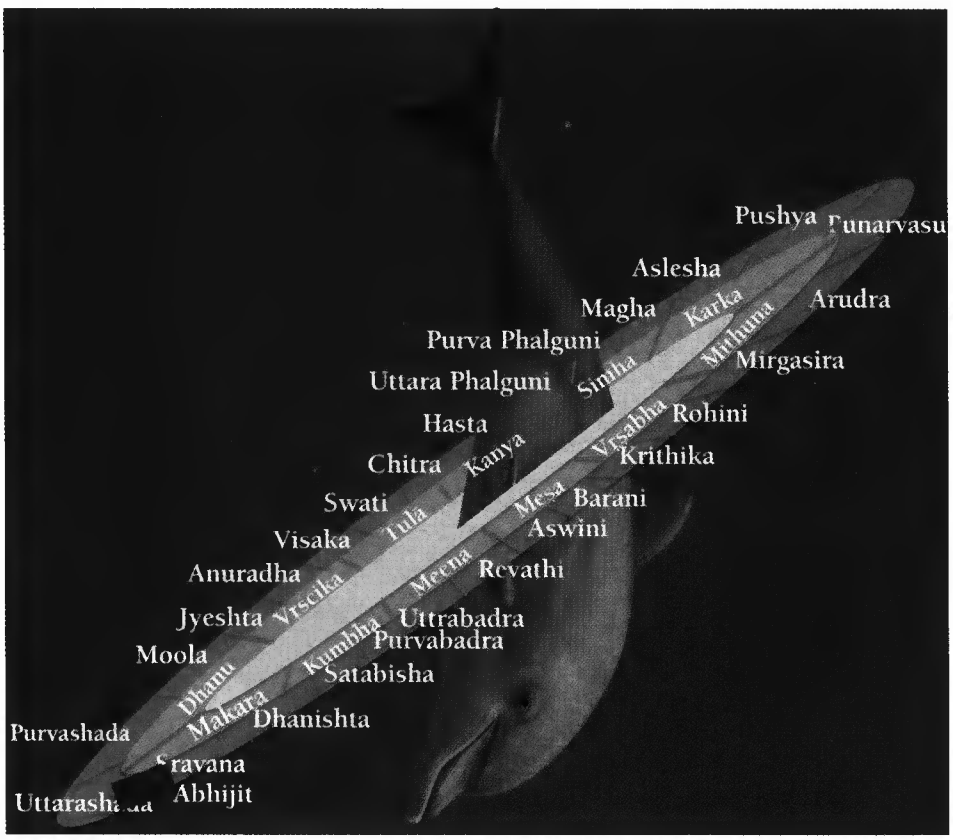


The Time-wheel or zodiac belt (kāla-cakra)

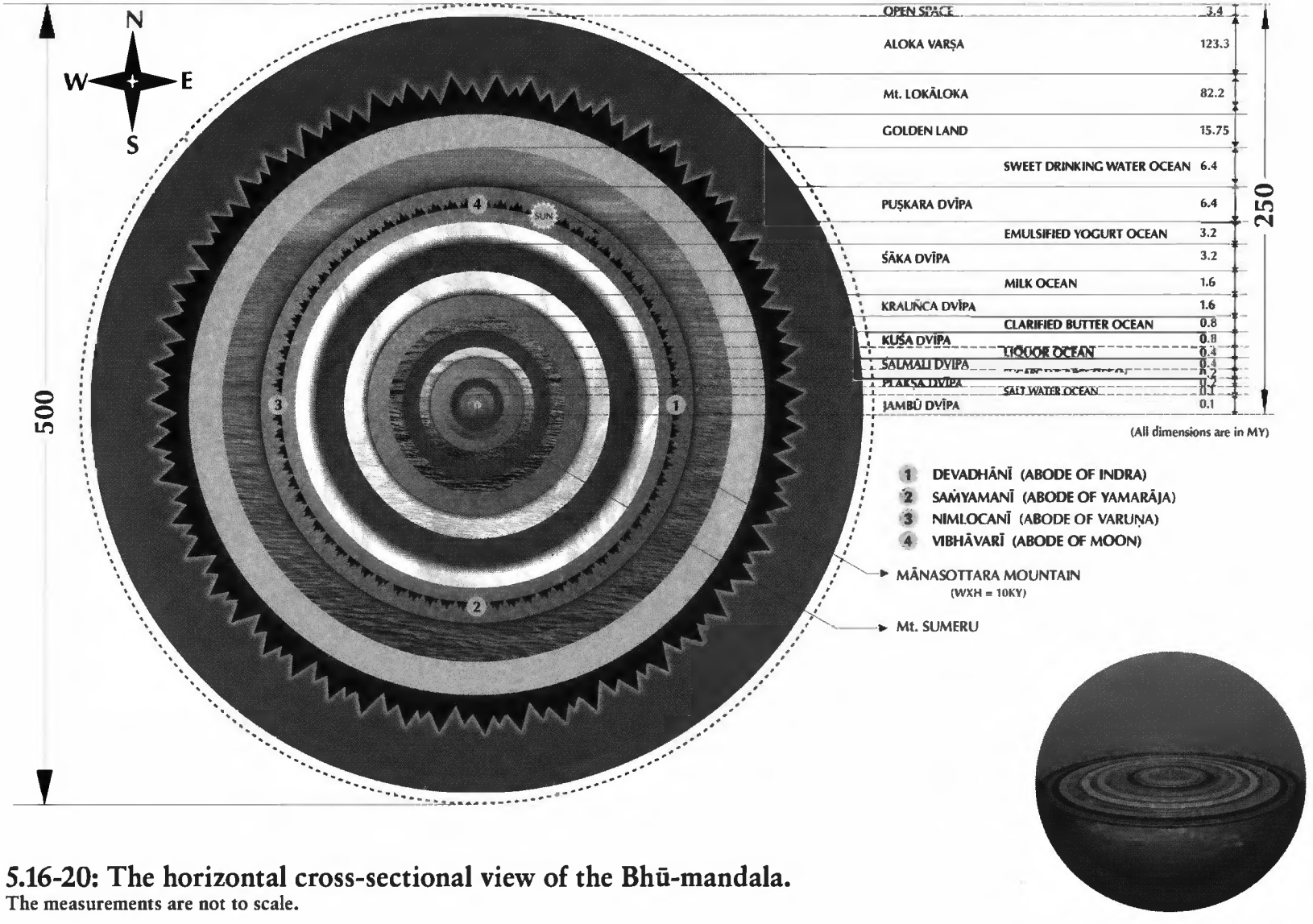


The Vedic Time-wheel or Zodiac-belt (Kāla-cakra)

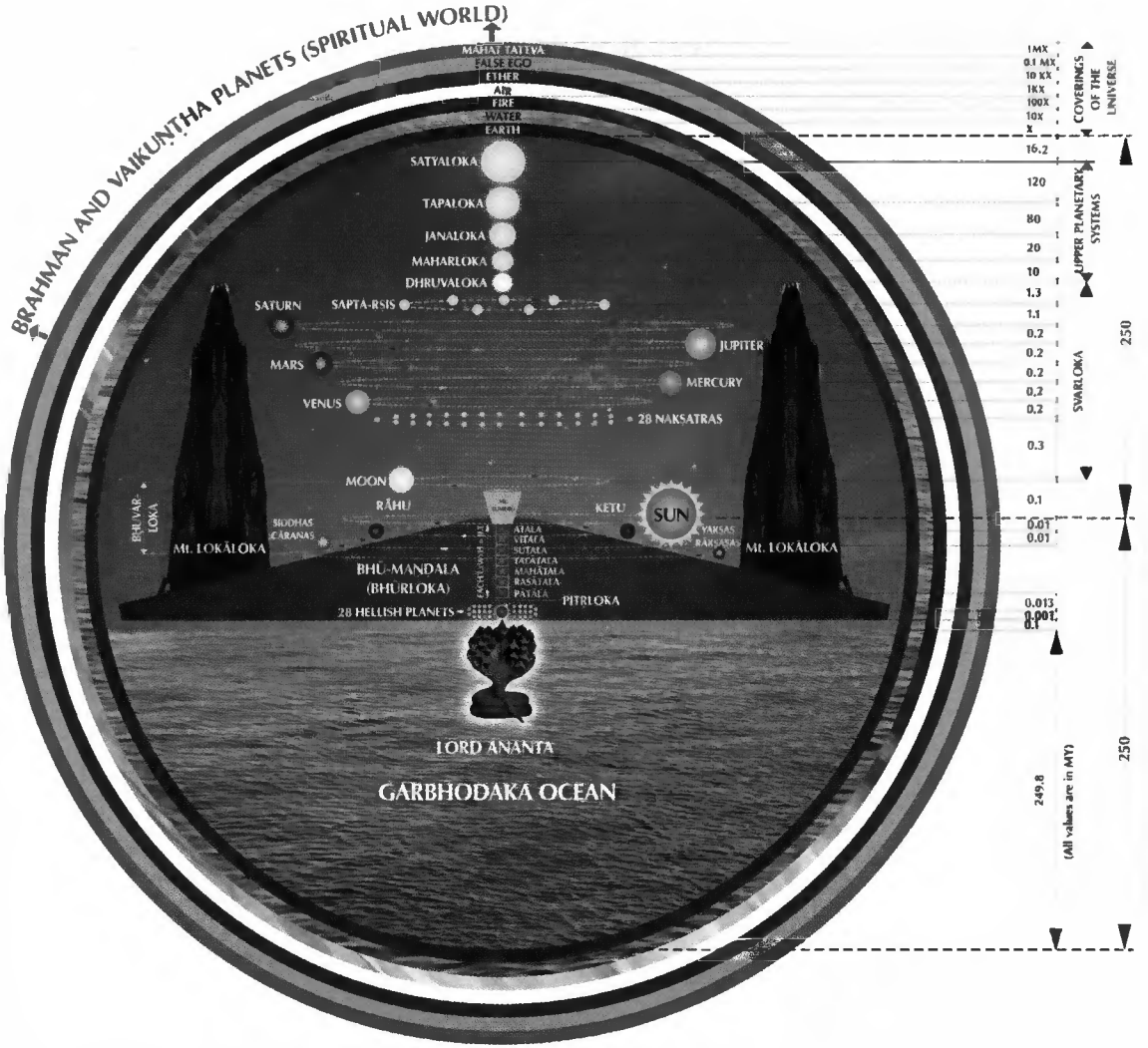




The Śiśumāra-cakra – The Inclined Kāla-cakra



5.16-20: The horizontal cross-sectional view of the Bhū-mandala.
The measurements are not to scale.



5.21-26: The vertical cross-sectional view of the universe - The Fourteen Planetary Systems
The measurements are not to scale.



3-D Model of the Universe

1

The Beginning

The Creation of the Universes and the Living Entities

Parīkṣit Mahārāja, reflecting that his act of garlanding the innocent sage Śamīka with a dead snake was reprehensible, became very depressed. Thinking that he had performed a deplorable act (*nīcam*), like an uncivilized person towards an innocent godly *brāhmaṇa* with great power, expected an unpreventable misfortune upon himself as a direct atonement for his sin in full force, so that he will not commit such an act again. And he desired that "Let the fire of an angry *brāhmaṇa* family burn up my kingdom, strength, and wealth today, since I am so sinful!
May I never again think sinfully of giving suffering to *brāhmaṇas, devatās* or cows!"

As he was thus thinking, a student Gauramukha, sent by Śamīka, revealed the fateful curse issued by Śṛṅgī, the sage's son that he would die from the bite of Takṣaka in seven days. He considered it beneficial that soon the fire of Takṣaka would be the cause of detachment from material life and thinking that service to Kṛṣṇa's lotus feet was the highest goal, he sat down on the bank of the Gaṅgā with a vow of not eating till death. When the noble-minded sages, the purifiers of the whole world, came along with their disciples there to see him, king Parīkṣit received them with proper respect and requested them to sing the spotless glories of the Supreme Lord (*gāyata viṣṇu-gāthāḥ).* As he requested them to determine the pure activity for person living and for a person about to die, the sages without agreement about the various processes of sacrifice, *yoga*, austerity, and charity, desired the appearance of a great personality in their minds. As they were thinking thus, the great son of Vyāsa, Śukadeva, wandering the earth as he pleased, arrived there as if it is the will of the Providence (*yadṛcchayā).*

Parīkṣit received him with proper respect, and after offering him a respectable seat enquired from him submissively about "What is perfection and the means to perfection, and what is to be heard, chanted remembered and worshipped, as well as what is not to be heard, chanted, remembered and worshipped?" Being pleased with Parīkṣit's inquiry, Śukadeva Gosvāmi answered him that "The ultimate perfection is attaining the feet of the Lord, and that is accomplished by hearing and chanting the names and pastimes of the Lord. This is the highest process."

Then the King thought "You have described that I should hear, chant and remember the qualities, forms and pastimes of Lord Kṛṣṇa at the point of death. First I will ask you about the pastimes of creation by His *māyā-śakti.* Then I will ask about His pastimes such as lifting Govardhana using His *cit-śakti.*" Contemplating thus the King requested Śukadeva Gosvāmi to speak about the activities of the *Puruṣāvatāras* such as creating the universe. After hearing the activities of the *Puruṣāvatāras* in relation to the creation of the universe, Parīkṣit then inquires from the sage on various topics related to Lord Kṛṣṇa. Being pleased with King's various questions about Lord Kṛṣṇa's pastimes, Śukadeva Gosvāmi then spoke the scripture called *Bhāgavata Purāṇa*, the essence of the Vedas, which was first spoken by the Lord to Brahmā in the first *kalpa*, consisting of four original seed verses known as *Catus-śloki-bhāgavatam.*

॥ SB 2.10.1-2 ॥

atra sargo visargaś ca sthānaṁ poṣaṇam ūtayaḥ ।
manvantareśānukathā nirodho muktir āśrayaḥ ॥

Śukadeva Gosvami says that "In this Purāṇa there are ten topics: creation, secondary creation, protection, mercy of the Lord, material activities, the conduct of the Manus, stories of the Lord, destruction of the universe, liberation and the ultimate shelter."

daśamasya viśuddhy-arthaṁ navānām iha lakṣaṇam ।
varṇayanti mahātmānaḥ śrutenārthena cāñjasā ॥

"The great devotees such as Vidura and Maitreya describe properly the nine topics in order to impart the highest knowledge of the tenth topic, through the words of the scripture and stories to illustrate their meaning."

Brief description of the ten subject matters of the *Śrīmad Bhāgavatam*,

Sarga: *bhūta-mātrendriya-dhiyām janma sarga udāhṛtaḥ*	*Sarga* refers to the manifestation of the gross material elements, *tan-mātrās*, senses, false ego and *mahat-tattva*, caused when the Lord transforms the *guṇas.*
Visargah: *brahmaṇo guṇa-vaiṣamyād visargaḥ pauruṣaḥ smṛtaḥ*	*Visarga* is the creation of bodies for the *jīvas* carried out by Brahmā. *Visarga* refers to the creation of moving and non-moving entities, means giving *jīvas* forms with intelligence, senses and body for engaging in material enjoyment, pursuing liberation or practicing *bhakti* according to their qualification.
Sthānam: *sthitir vaikuṇṭha-vijayaḥ*	*Sthānam* refers to the Lord protecting His devotee. This shows the excellence of the Lord. *Sthitiḥ* refers to the condition of the *jīvas* and *vaikuṇṭha-vijayaḥ* means that the Lord destroys the suffering of the *jīva.* After the creation, the condition of the *jīvas* is described by describing the different geographical locations in the Universe.
Poṣaṇam: *poṣaṇaṁ tad-anugrahaḥ*	*Poṣaṇam* means the Lord's mercy (in perfect peace of mind under Lord's protection). In protecting the *jīvas*, the Lord shows His mercy to some, the devotees performing *sādhana,* even if they are involved in sin by accident.
Ūtayaḥ: *ūtayaḥ karma-vāsanāḥ*	*Ūti* refers to the impressions created by the *jīva's* actions which lead them to future births. The impressions produced by actions either low or high. Pious or impious acts become the cause of future happiness and distress.
Manvantarani: *manvantarāṇi sad-dharma*	*Manvantara* refers to the conduct of the kings ruling the *manvantara* periods. The religious conduct of the kings reigning during the *manvantara*, revealed through narratives concerning them.
Iśānukathā:	The science of God - *Īśa-kathā* refers to

avatārānucaritaṁ hareś cāsyānuvartināṁ puṁsāṁ īśa-kathāḥ proktā nānākhyānopabṛṁhitāḥ	descriptions of the Lord's *avatāras* and their devotees, filled with various stories. The *avatāras* and their devotees should be heard about and glorified. Thus the devotional *aṅgas* of hearing & chanting of the *sādhaka* are described.
Nirodha: *nirodho 'syānuśayanam ātmanaḥ saha śaktibhiḥ*	*Nirodha* refers to the merging of the *jīva* along with his material identity into the Lord at the time of final devastation of the universe (winding up of the cosmic manifestation).
Muktih: *muktir hitvānyathā rūpaṁ sva-rūpeṇa vyavasthitiḥ*	*Mukti* means the *jīva's* attainment of his form as pure *ātmā* or as an associate of the Lord after giving up gross and subtle material bodies (permanent situation of the form of the living entity).
Aśrayaḥ: *ābhāsaś ca nirodhaś ca yato 'sty adhyavasīyate sa āśrayaḥ paraṁ brahma paramātmeti śabdyate*	The *āśraya*, Bhagavān, from whom arise creation, sub-creation, protection, mercy, *karma, dharma* of the kings, the appearance of *avatāras,* destruction and liberation is also called Paramātmā and *Brahman* (the Supreme Fountainhead, the Supreme Absolute Truth).

The third Topic '*Sthānaṁ*' is the principle discussion of this book, which is elaborately discussed in the Fifth Canto of Śrīmad Bhāgavatam. Prior to this, there is a brief description of the first two topics 'the primary and the secondary creation of the Universe (*Sargah & Visargah).* From the middle of Canto two to the middle of Canto five of Śrīmad-Bhāgavatam, elaborately describes the Primary Creation (*Sarga* - the Creation of the totality of matter with elements manifesting from prakṛti by Lord Viṣṇu's glance) and the Sub-creation (*Visarga* - the creation of bodies for the jīvas by the resultant combination of the 24 elements from Lord Brahmā's mind and body).

Prākṛtika-sṛṣṭi - Creation of the Universe consisting of 24 Elements (*Sarga*)

The prior state of the Creation: Before the creation of the universe (*idam agre*), Bhagavān, full of six powers, existed alone, since everything was merged

in Him. Out of His causeless mercy, the compassionate Lord felt imperfect without His plenary and separated parts and parcels. Thus to begin the material creation He impregnates the living entities into material nature and by the influence of eternal time all the living entities appear.

The process of Elementary Creation - Kāraṇa-sṛṣṭi[2]

- The Lord of *māyā,* the first puruśa, desiring to become many, by His own will, accepts time which was merged in Himself, as well as *karma* (the fate of the *jīvas)* and *svabhāva* which were merged in the *jīva*, through the agency of *māyā.*

  kālaṁ karma svabhāvaṁ ca māyeśo māyayā svayā |
  ātman yadṛcchayā prāptaṁ vibubhūṣur upādade ||

- Bhagavān, Lord of Mahā-vaikuṇṭha, in His *svāṁśa* expansion as the first *puruṣa* (also known as Kāraṇārṇavaśāyī Viṣṇu), at the beginning of the creation glances over *prakṛti.*

  Note: That energy composed of cause and effect is called *prakṛti* or *māyā.* This energy of the Lord is also known as *pradhāna,* in which the modes are in equilibrium state. By this energy the Lord creates the universe. This happens at the initial moment of the exhalation of the *mahā-puruṣa.*

- By His *svāṁśa* portion, the Lord placed the *jīva-śakti*, a semblance of consciousness (*vīryam*), in *māyā*, just by glancing from a distance (*puruṣeṇātma-bhūtena vīryam ādhatta vīryavān).* (Because the *jīva-śakti* enters *māyā-śakti*, it becomes dependent on *māyā-śakti.*)

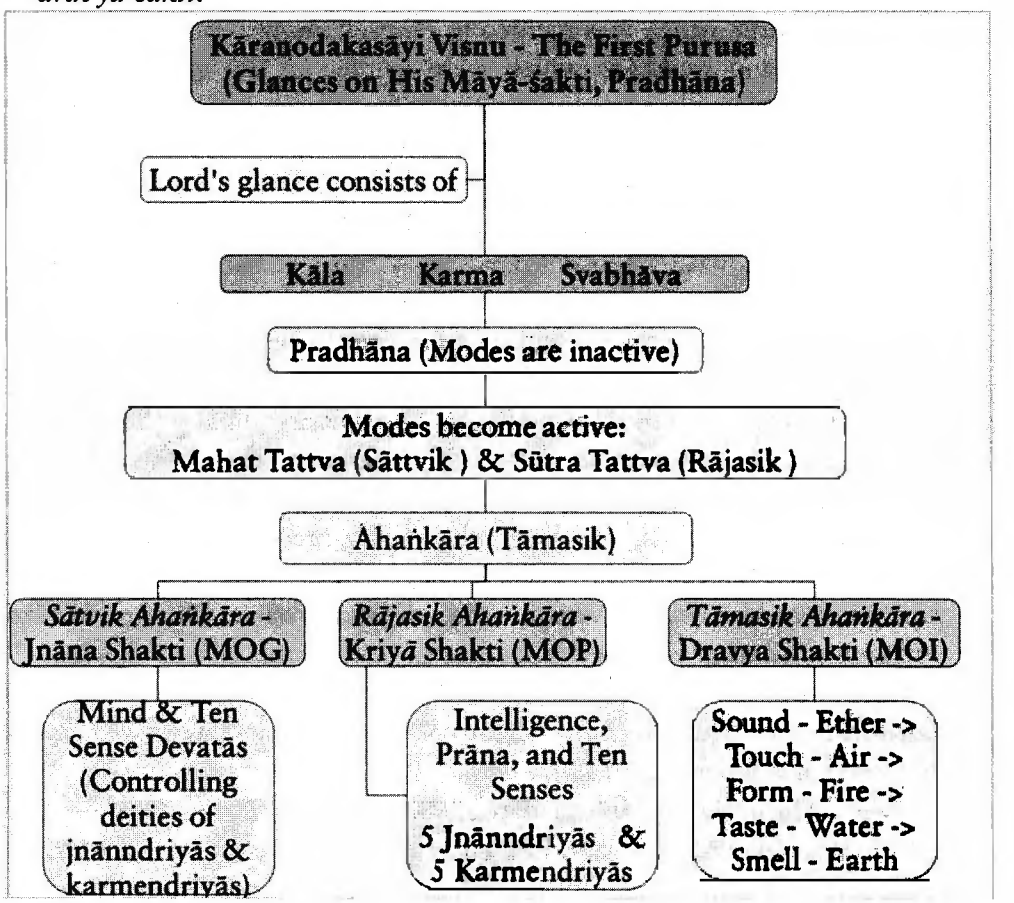
- Impelled by time, the invisible *prakṛti* (*pradhāna*) undergoing transformation necessary for creation and appears as *mahat-tattva*, composed of knowledge, in *sattva-guṇa.*

  Note: The *guṇas* are agitated from a state of equilibrium by time. They are transformed into another form by *svabhāva.* The *mahat-tattva* appears by the *karma* of the *jīvas.* It is composed of knowledge (*vijñānātmā*), by predominance of sattva. *Mahat-tattva* then becomes *sūtra-tattva* (a type of *mahat-tattva*), which is differentiated from *mahat-tattva* by the *kriya-śakti*, with a predominance of *rajas.* Thus it is said that mahat-tattva is predominated by sattva and rajas. All this is under the direction of the Lord.

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[2] SB 2.5.21-31 & SB 3.5.23-37: Description of Primary Creation or Sarga - Kāraṇa-sṛṣṭi

- From *mahat-tattva* predominated by *sattva* and *rajas,* a substance predominated by *tamas*, *ahaṅkāra,* arose, composed of *adhibhūta, adhyātma* and *adhidaiva.*
- *Ahaṅkāra* is the shelter of *adhibhūta, adhyātma* and *adhidaiva,* and transforms into gross elements, senses, mind and senses *devatās.* It is divided into *sattva, rajas* and *tamas*, called *jñāna-śakti, kriyā-śakti* and *dravya-śakti.*



- From the transformation of *ahaṅkāra* in *sattva* arose the mind and the sense *devatās* (controlling deities of the senses). This is adhidaiva aspect or kartrātmā. From the *devatās* of the senses, the sense objects become revealed.

  Note: From *ahaṅkāra* in *sattva* arose the mind and its deity as well as the ten presiding deities of the senses: the Direction deities, Vāyu, Sun, Varuṇa, two Aśvini-kumāras, Agni, Indra, Upendra, Mitra and

Prajāpati Dakṣa.

- From *ahaṅkāra* in *rajas* arose the knowledge and action senses (*adhyātma* aspect / *kāraṇa*).

  Note: From *ahaṅkāra* in *rajas* arose the intelligence (*śakti* of the five knowledge senses), *prāṇa* (*śakti* of the five action senses), and the ten senses. The ten senses are the ear, skin, nose, eye, tongue, voice, hands, feet, anus and penis.
- From *ahaṅkāra* in *tamas* arose the five sense objects (*tanmātrās)* starting with sound, and the five gross elements (*panca-mahābhūtas)* starting with ether (*adhibhūta* aspect / *kārya)*.

Formation of the Universe - Kārya-sṛṣṭi[3]

As long as the gross elements, senses and mind were not mixed together, it was not possible for them to produce material bodies. Coming together by the impulse of the Lord's energy (when the Lord entered them), accepting primary and secondary forms, they created the body of the whole universe and the individual bodies in it.

- The Supreme Lord, *bhagavān*, remembering the prayers of the personified elements, decided to create the Universal form by His *cit-śakti* so that the elements could perform their functions. Thinking thus, the Lord entered into it as *antaryāmī.*

  Note: How the Lord entered? Seeing the sleeping state in creating the universe because the elements were unmixed, the Lord, first by His energy of cohesion, appearing through time, and then as *antaryāmī,* entered the twenty-three elements (5 gross elements, 5 sense objects, 10 senses, mind, false ego and mahat-tattva) simultaneously.
- The Lord, entering by His *kriyā-śakti* (His energy of action) and awakening the dormant *karmas* of the *jīvas*, combined the elements together.
- The twenty-three elements, awakening to action by the *kriyā-śakti*, being stimulated by the Lord, produced the universal form (*adhipuruṣam*) by their portions. They formed a fetus or golden egg (*brahmānda* or *hiraṇyānda*) in which all the planets and living entities exist.
- This golden universal form (*hiraṇmayaḥ*), containing all the resting *jīvas (*totality of the universe - *samaṣṭi-virāṭ*), remained in the waters for a thousand years (*sahasra-parivatsarān*).

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[3] SB 2.5.32-35 & SB 3.6: Description of Primary Creation or Sarga - Kārya-sṛṣṭi

Note: This golden form was the totality of the universe (*samaṣṭi-virāṭ* or the Golden Egg). He remained in the Universe, which was in the Kāraṇa water for 1000 years.

- After a thousand years, the Lord (the *purusaḥ*, the soul of the mass of *jīvas* Hiraṇyagarbha), situated in time, *karma* and *svabhāva*, brought to life the non-living universe which was lying in the water of Kāraṇa ocean, by entering into it in His partial manifestation, known as Garbhodakaśāyī-Viṣṇu.
- That *puruṣa* pierced the universe and went outside. There He (as Kāraṇodakaśāyī-Viṣṇu or Mahā-Viṣṇu) resides with a thousand legs, feet, arms, eyes, faces and heads.

  Note: The Lord by His glance over prakṛti created the universe, and distinguishing it from Himself, He situated Himself outside.
- When this pure *puruṣa* (Garbhodakaśāyī-Viṣṇu) desired a sleeping place (*ayaṇam*) for Himself within the universe (*tasmin*), He created the waters of the Garbhodaka Ocean. The Lord, being naturally pure (*śuciḥ*), created water, which was pure (*śucīh*).
- He dwelled for many years (*sahasraṁ parivatsarān)* in that water created from Himself. Because of that, He is called Nārāyaṇa, since the waters *(nārā*) arose from the *puruṣa* (*nara*).
- That Lord, desiring to become many, rose from His bed after the sleep of universal destruction (the night of Lord Brahmā), and created the universal form shining brightly in three forms (*adhidaivam, adhyātmam* and *adhibhūtam*) by His energy.
- Various abodes for the *devatās* became differentiated in this universal form.
- Within the universe, the wise imagine the planets as His limbs. Buttocks and below - lower seven planets, hips and above - seven upper planets.
- Earth and the lower planets are imagined to extend from the feet (Bhūloka). Bhuvarloka is at the navel. Svarga-loka is imaged to be the head.
- **Conclusion:** The purpose of describing this wonderful Creation of the Universe is to understand that "the Supreme Lord is the basis of everything and engage all the created things in His worship without any ulterior motive. This enables one to attain the complete purification, the ultimate purpose of one's existence."
- This is further elaborated by Śrīla Śukadeva Gosvamī in SB 9.24.58 and SB 10.87.2

yan māyā-ceṣṭitaṁ puṁsaḥ
sthity-utpatty-apyayāya hi
anugrahas tan-nivṛtter
ātma-lābhāya ceṣyate

Even the actions of the Lord in the form of the *puruṣa* in relation to *māyā* for creation, maintenance and destruction of the universe, are His mercy alone. That mercy acts to produce detachment from the world and attainment of the Lord in the spiritual world.

- The action in the form of glance of the Lord (*puṁsaḥ*) upon *prakṛti* (*māyā-ceṣṭitam*) for creation, maintenance and destruction of the *jīvas*' bodies, is certainly because of His mercy.
- What then to speak of His activities such as lifting Govardhana, which are completely devoid of *māyā?*
- The Lord's mercy consists of glancing over *māyā,* thinking, "Let the *jīvas* get their material enjoyment after attaining material intelligence and senses!"
- "But how can this be mercy? The *jīvas*, by enjoying material objects and undergoing creation, maintenance and destruction of the universe, simply experience suffering."
- He acts so that the *jīvas* become detached from the world and its enjoyments, by *bhakti, jñāna* and *vairāgya*, and so that the *jīva* can attain the Lord (*ātmā-labhāya*).
- Without intelligence and senses, the *jīva* cannot attain *bhakti* or *jñāna.* The cause of the creation of the world is mercy to the *jīvas.*

buddhīndriya-manaḥ-prāṇān
janānām asṛjat prabhuḥ
mātrārthaṁ ca bhavārthaṁ ca
ātmane 'kalpanāya ca

The Supreme Lord manifested the material intelligence, senses, mind and vital air of the living entities so that they could indulge their desires for sense gratification, take repeated births to engage in fruitive activities, become elevated in future lives and ultimately attain liberation.

Note: Worship (*kalpanam*) of the Supreme Self (*ātmane*) is the means by which the living entities can fulfill the divine purpose for which they exist. Intelligence, mind and senses are meant to be used for worshiping the Supreme Lord.

Naimitika-sṛṣṭi - The Daily creation of Lord Brahmā (*Visarga*)

Birth of Lord Brahmā at the beginning of Brahmā's Day[4]

- The universe lay in the water of devastation when Garbhodakaśāyī Viṣṇu, full of His *cit-śakti,* with Śeṣa as His bed lay with His eyes closed during the night of Lord Brahmā.

  Note: Having destroyed all the gross bodies of the inhabitants of the three worlds by time, Garbhodakaśāyī Viṣṇu places their subtle bodies within Himself. This is called the aggregate subtle body of the *jīvas.* He then resides below Pātālaloka, in His own abode (*sve pade*), within the one expanse of water (Garbhodaka Ocean).
- After sleeping within that water for a thousand cycles of four *yugas* (the night of Lord Brahmā) with His *māyā-śakti,* He looked at the living entities from Brahmā to the non-moving beings (*lokān*) who had attained subtle forms within His body, in order to eject them from His body.
- After His glance entered into the subtle bodies of all the jīvas (*artha-sukṣma*) to pull them out, the material elements within Him with very subtle form (*tanīyān*) became agitated by rajas according to time, and were brought forth, pushed them upwards from His navel.
- The material elements, transforming into a lotus bud by time, stood above the waters of devastation, by the force of the awakened karmas of the jīvas. The lotus bud had Viṣṇu (*ātma*) as its source. It was like the sun, illuminating by its effulgence.
- Lord Viṣṇu as *antaryāmī* entered that lotus made of all the planets. That lotus held within itself the manifestation of the effects of all the *guṇas,* Svarga and Naraka, to be enjoyed by the *jīvas.* Brahmā appeared within that lotus. Thus he is called as self-born because he has no visible parentage.

Brahmā's Austerities and Lord's Instructions[5]

- Brahmā, situated in the center of the lotus, could not see the universe's planets. Turning his head around and glancing into space, he attained four heads to see in the four directions.
- Unable to find the source of the Lotus and himself, but hearing two words

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[4] SB 3.8.10-15: Birth of Lord Brahmā at the beginning of Brahmā's Day

[5] SB 3.8-9: Brahmā's austerities and Lord's Instructions

indicating 'ta-pa', Brahmā controlling his breath and mind, remained seated in complete concentration by meditation on the Lord.

- By his practice of meditation on the Lord for a hundred human years, he attained realization of the Lord. He saw the Lord, who manifested Himself in his heart, who could not be seen previously by searching.
- Desiring to create variety in the universe, Brahmā saw nothing except the lotus coming from the water in Viṣṇu's navel, himself, the three elements water, air & ether.
- Influenced by *rajas* and intent on creating, Brahmā concentrating his mind began to praise the Lord, worthy of worship.
- The Lord, understanding the mentality of Brahmā, concerning knowledge for constructing suitable bodies for the living beings, instructed him with profound words, dispelling his illusion, to carry out his task of creation and assured him of not being bound by such act of *rajoguna*.

Brahmā's Wonderful Creation[6]

- Brahmā then engaged in meditation for 100 celestial years more, as advised by the Lord, absorbing his mind in Nārāyaṇa.
- Becoming powerful with knowledge from increased meditation (matured knowledge & long penance) and mantra worship fixed on the Lord, Brahmā destroyed the wind along with the devastating water.
- Seeing the lotus on which he was situated spread up to Satyaloka, being inspired by the Lord, Brahmā entered into the whorl of the lotus, and divided it into 3 divisions and later expanded it into 14 divisions in order to produce varieties.
- As instructed by the Lord, Brahmā populated those 14 divisions with different varieties of living entities. As the universe exists now, it existed before devastation, and will exist after devastation.

  Note: "Because the universe has a beginning and end, the universe must be unreal. How can an unreal universe be divided?" This cannot be said. This universe existed before the great devastation (*agre*). After the devastation is over, it will exist. It is not a false object, but temporary.
- The primary creation (*Prākṛta* creations of Universes by Lord's *māyā-śakti, prakṛti*), which was previously described in detail, is summarized for easy understanding and it is grouped into six types. In addition, the subsequent

[6] SB 3.10, 12 & 20: Description of Lord Brahmā's Secondary Creation - Visarga

creation of Lord Brahmā (*Vaikṛta* - creations of the bodies for moving and non-moving entities) is grouped into four types, totaling ten types of creation.

1. Creation of mahat-tattva;
2. Creation of false ego;
3. Creation of the gross elements & tan-mātras;
4. Creation of knowledge and action senses;
5. Creation of sense devatās and mind;
6. Creation of ignorance (produced by Lord's *avidyā-śakti* to bewilder jīva)

   Note: The first five are related to transformations of *pradhāna*. The sixth is the creation of ignorance. This was created by *avidyā-śakti* of the Lord, which bewilders (*abuddhi*) the *jīva*. The meaning is this. *Māyā* has three conditions: *pradhāna, avidyā* and *vidyā*. By *pradhāna* the elements from *mahat-tattva* to earth are created (first five). These are real creations, by which the aggregate (*samaṣṭi*) and individual bodies (*vyaṣṭi*) of the *jīvas*, with gross and subtle bodies created. By *avidyā* (sixth) which bewilders the *jīva*, five types of ignorance— *avidyā, asmitā, raga, devṣa* and *abhiniveśa*-- are created. These are not real. The world, composed of real and unreal, is produced by *pradhāna* and *avidyā*. By *vidyā (karma, sānkya, yoga* and *bhakti),* knowledge which destroys the five types of ignorance is created.

- Lord Brahmā, the original creator, first produced the functions of ignorance: *tamas* (forgetfulness of one's real identity), *moha, mahāmoha, tāmisra* and *andha-tāmisra*. (SB 3.12.2 & 3.20.18)

  sasarjāgre 'ndha-tāmisram atha tāmisram ādi-kṛt
  mahāmohaṁ ca mohaṁ ca tamaś cājñāna-vṛttayaḥ

- Thus, Lord Bramā first created the ignorance of five types by his shadow (*tamas, moha, mahā-moha, tāmisram,* and *andha-tāmisram*), because false identity must arise in the *jīvas*. Without ignorance this is not possible.

  Note: Supreme Lord had already established the functions of ignorance at the end of the primary creation. However, at the beginning of the daily creation, they make their appearance in various forms starting with *tamas,* directly from Lord Brahmā.

- In creating the Yakṣas, Rākṣasas, *devatās* and humans, false identity must arise in the *jīvas*. Without ignorance this is not possible. Thus first the

functions of ignorance appeared on their own, through Brahmā.

1. *Tamas – avidyā (*ignorance of *svarūpa* of the *jīva*)
2. *Moha - asmitā (*egoism or false identification with the body*)*
3. *Mahā-moha - rāga (*attachment or applying possessiveness to objects of enjoyment*)*
4. *Tāmisram - dveṣa (*repulsion or hatred appears when one's enjoyment is obstructed*)*
5. *andha-tāmisram - abhiniveśa (*fear of death or absorption in material pleasure)

- Not present in the jīva, these were created by *avidyā*. Ignorance of *svarūpa*, false identity, hatred, fear and lamentation are actually only two functions of *avidyā, āvaraṇa* and *vikṣepa. Āvaraṇa* is called *avidyā* (*tamas*) or *ajñāna* and *vikṣepa i*s called *āsmitā (moha),* or *viparyāsa*. Attachment, hatred and fear of death (*rāga, dveṣa, abhiniveśa*), though they are qualities of the mind, are types of *vikṣepa*. Not being as prominent as *vikṣepa,* they are included in it.

  Note: In this regard, a conversation from sixth Canto is referred (SB 6.16.51VCT), wherein Lord Saṅkarṣaṇa instructs Citraketu maharāja about *jñāna,* in order to make him indifferent to questioning. Lord says, "Please listen. You should consider this. There are two types of real objects, permanent (or eternal) and temporary. There are three eternal objects: Brahman, Paramātmā, and Bhagavān with His associates. I am the Vedas (*śabda-brahma),* which are my breathing. I am the impersonal Brahman realized by *jñānīs*. These two are My forms and they are eternally existing (*śabda-brahma paraṁ brahma mamobhe śāśvatī tanū*). The insentient universe (consisting of the *jīvas* and their enjoyed objects, since I am non-different from my *māyā-śakti* and my *jīva-śakti*) made of matter is the temporary object. The *jīvas* are called temporary objects in this case because of their covering of ignorance (*ahaṁ vai sarva-bhūtāni bhūtātmā bhūta-bhāvanaḥ*). There are two types of unreal objects: experienced and unexperienced. The experienced unreal objects consist of dreams and magic. The unexperienced unreal objects consist of imaginary objects like a flower in the sky or a rabbit with horns." The First Canto of *Bhāgavatam* says that the permanent object is the best: *vedyam vāstavam atra vastu. (*SB 1.1.2)

- Then Brahmā created (as seventh creation) six types of immobile living

beings (*taṣṭhusām*) as the seventh creation, who distributes their nourishment upwards (six types of trees and plants: *vanaspaty-oṣadhi-latā-tvaksārā vīrudho drumāḥ).*

1. *Vanaspaty:* plants without flowers but having fruit
2. *Oṣadhi:* annual plants, perishing after a year
3. *Latā:* plants that need support to rise up, creepers
4. *Tvaksārā:* bamboo and other hollow plants
5. *Vīrudha:* plants which do not depend on support to rise because of stiffness of the trunk (bushes)
6. *Drumāḥ:* trees which give fruit through flowers

- The plants distribute their nourishment upwards. They are almost unconscious, but have a feeling of pain within on being cut. They are not fixed in size and have unlimited variety.

saptamo mukhya-sargas tu ṣaḍ-vidhas tasthuṣāṁ cayaḥ
vanaspaty-oṣadhi-latā-tvaksārā vīrudho drumāḥ
utsrotasas tamaḥ-prāyā antaḥ-sparśā viśeṣiṇaḥ

- Then he created (as eighth creation) different animals, who distribute their nourishment horizontally, of twenty-eight varieties. They have no concept of the future, are preoccupied only with food, understand their needs by smell, and are devoid of lengthy deliberation unable to remember anything within the heart.

tiraścām aṣṭamaḥ sargaḥ so 'ṣṭāviṁśad-vidho mataḥ
avido bhūri-tamaso ghrāṇa-jñā hṛdy avedinaḥ

- These moving entities include land dwelling animals, reptiles and snakes, water dwelling aquatics, land dwelling birds and water dwelling birds.
  - 1 to 25: Land dwelling animals (9+5+11 = 25)
    - 9 Animals of cloven hooves (cows, goats, buffalo, black deer, pigs, wild oxen, antelopes, sheep and camels)
    - 5 Animals without cloven hooves (donkeys, horses, mules, gaura deer, śarabhas and yaks)
    - 11 Animals with five nails (dogs, jackals, wolves, tigers, cats, rabbits, porcupines, lions, monkeys, elephants, and 25th the turtles)
  - 25th: reptiles (turtles, lizards) & snakes (3.20.47-48)
  - 26th: water dwelling aquatics (makara - alligators...)
  - 27th: land dwelling birds (herons, vultures, cranes, hawks, birds of prey, bhallukas, peacocks, crows, and owls) and

- 28th: water dwelling or aquatic birds (swans, Indian cranes, *cakravākas*...)

- Then Lord Brahmā created the human beings (as ninth creation), who are only one class and in whom the food is distributed downwards. Predominated by rajas, and they are dedicated to work and think themselves happy even in distress.

arvāk-srotas tu navamaḥ kṣattar eka-vidho nṛṇām
rajo 'dhikāḥ karma-parā duḥkhe ca sukha-māninaḥ

- Lord Brahmā then created the eight kinds of *devatās* (as tenth creation) such as the main *Devatās*; Pitrās; Asuras; Gandharvas and Apsaras; Yakṣas and Rāksasas; Siddhas, Cāraṇas and Vidhādhadras; Bhūtas, Pretas and Piśācas; and Kinnars, Kiṁpuruṣas and others.

N	Brahmā's Body	Transformed into	Creations, who accept the transformed forms
1	Ignorance (tamah)	Night, which gives rise to hunger & thirst	Yakṣas and Rākṣasas (*tamah*)
2	Brahmā's beauty, shining with effulgence	Day, in the MOG	The chief *devatās* (*sattva*)
3	Brahmā's buttocks or terrible body (rajas)	Evening twilight, which stimulates lust	The Daityas, who were lusty for women, imagined the twilight to be a woman of lusty conduct (*rajas*)
4	Beauty which he had relished	Moon light	Gandharvas and Apsarās headed by Viśvāvasu
5	Body of laziness	Sleep and contamination (yawning and sleep)	Bhūtas, Pretas and Piśācas, who were naked with loosened hair
6	Body of strength & invisibility	Power & Invisibility	Sādhyas (wise offer *havya*) & Pitṛs (wise offer *kavya*)
7	Body of disappearance	Power of disappearance	Siddhas, Cāraṇas & Vidyādharas
8	Beauty of Brahmā's reflection in a mirror	Displayed men and women together	Kinnara and Kimpuruṣas

The creation of *devatās* has eight divisions. The Supreme Lord performed these ten creations (*deva-sargaś cāṣṭa-vidho...sargās te viśva-sṛk-kṛtāḥ*).

- Then there was the creation of the four Kumāras, 11 Rudras, 10 Mānasa-putrās such as Marīci, and others such as Dharma, Adharma, Vāk and Kardhama through his various bodily limbs.

  The four Kumāras represent the four aspects of *vidyā (sāṅkhya, yoga, vairāgya* and *tapas*). The Mānasa-putrās headed by Marīci appeared as representatives of *karma-yoga.* Jñāna, vairāgya, tapas, yoga and karma only become successful by performance of bhakti, which gives results. Bhakti, the very form of all results, appeared as Nārada at the end.
- Then from his four mouths Lord Brahmā created various other manifestations such as Vedas, Upavedas or Vedic Supplements and Fire Sacrifices, Varnāśramas, Arts and Sciences, different Vedic Hymns headed by Pranava, Vedic Meters, Verse Writing, audible sounds in speaking – vaikhāri, and many similar things appeared.
- Understanding that the creation produced from the powerful sages (Marīci and others) was not sufficient, Brahmā again began to reflect in his heart: "How astonishing! Though I am engaged constantly in creation, the population does not increase. There is certainly some obstacle causing misfortune."
- While engaging himself properly and considering his bad luck, his body became two beings one with a beard & the other with breasts, who are known as the descendents of Brahmā.
- Thus thinking himself successful, finally Brahmā created by his mind Manu and his wife for increasing the population. Brahmā gave to Manu his own human form. Manu appeared and took the form of humans for completing the creation.
- Then those who were created previously glanced at Manu and began to praise Brahmā.

  aho etaj jagat-sraṣṭaḥ sukṛtaṁ bata te kṛtam ।
  pratiṣṭhitāḥ kriyā yasmin sākam annam adāma he ॥
- "O Creator of the universe! You have done excellent work in creating Manu. By this, the sacrificial actions will be established since Manu will protect everything. And we will be able to eat the remnants of sacrifice together." (SB 3.20.51)
- It is also stated in BG 3.10, "Lord Brahmā, having created the population

along with sacrifice, long ago spoke: 'May you prosper by this sacrifice. May it fulfill all your material desires'"

saha-yajñāḥ prajāḥ sṛṣṭvā purovāca prajāpatiḥ |
anena prasaviṣyadhvam eṣa vo 'stv iṣṭa-kāma-dhuk ||

Note: Long ago, along with *yajña*, Lord Brahmā created the creatures qualified for offering to Viṣṇu and said, "Increase more and more the population by this *dharma* or *yajña*. May this yajña bestow all desired enjoyment (*iṣṭa-kāma-dhuk*)."

- All the forms previously created by Lord Brahmā, increased through the forms of male and female, to perpetuate the creation till the end of Brahmā's day.
- Thus, by their union, they increased the population untill the end of Brahmā's day.
- This perpetual creation, which will continue till the end of Lord Brahmā's day is called as Nitya Creation or *Nitya-sṛṣṭi.*

The sequence of manifestations oof the creation

- Manifestation of five features of ignorance (*ajnana-vrittis)*
- First the creation of animals with five joints (aquatics)
- The creation of trees and plants (Immovable Trees)
- The creation of snakes (Reptiles)
- The creation of cows and buffalos (Animals or Birds and beasts)
- The creation of eight kinds of devatās (Yakṣas, Rākṣasas, Asuras, etc)
- The creation of the Four Kumāras
- The creation of Rudra or Lord Śiva
- The creation of Marīci and others (*Manasa-putras*)
- The creation of the humans (9th creation)
- Finally the Manus were created, who protect everything by establishing the sacrifices.

Brahmā's Instructions to his son Svāyambhuva Manu[7]

- To increase the population on the earthly plane, Lord Brahmā created two beings, a male Svāyambhuva Manu and the female Śatarūpā.
- Manu along with his wife approached Lord Brahmā with folding his hands

[7] SB 3.13.6-13: Lord Brahmā Instructs his son Svāyambhuva Manu to create progeny and rule the Earthly Planetary system (bhū-loka)

and offering respects enquired how they can serve him "O worshipable lord! Please instruct us how we can please you by performing actions according to our abilities, which after being performed, will give us fame in this life and a superior destination in the next life."

- Being pleased with his simplicity and surrender, Brahmā instructed him to serve the Supreme Lord, the guru of the entire Universe, diligently according to one's capacity with devotion and free from envy.
- Brahmā further instructed "O courageous son! After begetting in your wife, the off-springs with qualities similar to yourself protect the earth with *dharma* and worship the Lord with sacrifice. By protecting the population you will perform the highest service to me. The Supreme Lord will be satisfied with you as the protector of the population."
- Being instructed by Lord Brahmā, Svāyambhuva Manu became the ruler of the entire Bhū-mnadala and protected the people by providing all their necessities as a service to the Lord.
- Setting an ideal example, Svāyambhuva Manu retired from his responsibilities by handing over the ruler ship to his sons and grandsons. (SB 3.13 & 21-22)
- Then the lineage of Svāyambhuva Manu's second son Uttānapāda continued to rule the entire Bhū-mnadala until the time of Daksa, the son of Pracetās (SB 4.8-31)
- Later on the descendents of Priyavrata Mahaṙāja, first son of Svāyambhuva Manu, continued to rule the entire Bhū-mnadala until the end of the 1st Manu period. (SB 5.1-15)

Parīkṣit Maharāja's Eagerness to hear about the Bhū-maṇḍala[8]

Parīkṣit Maharāja, after hearing the lineage of Priyavrata Maharāja completely, inquires about the dimensions of the islands and oceans which he had heard about in the story of Priyavrata, which Śukadeva Gosvami discussed in SB 5.1.30-33. Now he expresses his eagerness to hear about them in detail.

- You have described the specific dimensions of Bhū-maṇḍala, which extends as far as the sun shines and within which the moon along with other luminaries are seen.

[8] SB 5.16.1-3: Questions by Parīkṣit Maharaja about the details of Jambūdvīpa.

- In Bhū-maṇḍala, seven oceans were created by the seven ditches made by the wheel of Priyavrata's chariot from which seven islands of Bhū-maṇḍala, each with distinctive construction appeared.
- I desire to know all about the oceans and islands, with their measurements and qualities.

Why Parīkṣit Mahārāja desires to hear this?

Parīkṣit Mahārāja says "The mind fixed on the Lord's gross form made of matter (*bhagavato guṇamaye sthūla-rūpe*) can absorb itself in the supreme Brahman, Bhagavān, Vāsudeva, light of the soul, who is beyond matter (*aguṇe 'pi sūkṣmatama ātma-jyotiṣi pare brahmaṇi*). Therefore, you should explain the structure of the universe (*guro 'rhasy anuvarṇayitum iti*)."

Note: Parīkṣit asks the question, not just for himself, but for the benefit of *bhakti-miśra-yogīs* present at the gathering who desired to concentrate their minds. The gross form of the universe made of *guṇas* belongs to the Supreme Lord. It is thus different from the Lord and indicates that the Lord is beyond the *guṇas*. Because you are the *guru* of all the sages you should explain this.

Śukadeva Gosvāmī Expresses his Humility[9]

"O King! It is not possible even for a person with the lifespan of a *devatā* (*vibudhāyuṣāpi puruṣas*) to understand by words or mind (*manasā vacasā*) the end of the transformations of *māyā's guṇas*, belonging to the Lord (*bhagavato māyā-guṇa-vibhūteḥ*). Therefore by giving a general description (*tasmāt prādhān-yenaiva*) I will explain the particulars of Bhū-loka (*bhū-golaka*), giving characteristics of name, shape and dimension (*viśeṣaṁ nāma-rūpa-māna-lakṣaṇato*)."

Note: Śukadeva Gosvāmī's purpose in describing the details of the structure of the universes is to show us that how the residents of different planets are primarily engaged in the worship of the Supreme Lord and thus to inspire us to take up the same. Canto 5 of Śrīmad-Bhāgavatam gives an overview of the geographical locations of different planetary systems but

[9] SB 5.16.4: Śukadeva Gosvāmī's Humility and immesurable manifestation of Lord's material Creation

not the full details, because the intention is to increase our appreciation for the Supreme Lord's wonderful Creation.

Division of Śrīmad-Bhāgavatam Canto 5 - Sthāna

Śrīmad-Bhāgavatam Canto 5 describes *Sthāna*, one of the ten subject matters of *Bhāgavata Purāṇa*. '*Sthānam*' defined as *"sthitir vaikuṇṭha-vijayaḥ"*, the right situation for the living entities is to obey the laws of the Lord. *Sthānam* refers to the Lord protection of His devotees. This shows the excellence of the Lord. *Sthitiḥ* refers to the condition of the *jīvas & vaikuṇṭha-vijayaḥ* means that the Lord destroys their suffering. *Sthāna* is the geography of the universe, where the demigods, men and demons reside and its protection by the demigods. The maintenance of the inhabitants of the Bhūloka, Pātala and Svarga, who are situated there with their governing rules is the main subject matter discussed in Canto 5, which is presented in this book with proper illustrations for easy understanding.

Birds eye view of Śrīmad-Bhāgavatam Canto 5

Part1: Description of "Bhū-maṇḍala – the Horizontal Structure of the Universe"

- Ch 1: Priyavrata divides the Bhū-maṇḍala into 7 Dvīpas & 7 Oceans
- Ch 1-15: Description of the descendents of Priyavrata maharaja, *visarga*
- Ch 16-19: Description of Jambūdvīpa, its mountains and trees
  - Ch 16-17: Jambūdvīpa, Its Divisions & Path of Ganges
  - Ch 17-19: Prayers offered by the residents of Jambūdvīpa (9 *varṣas*)
  - Ch 19: Glories of Bhārata Varṣa
- Ch 20: Description of the other six islands of the Bhū-maṇḍala with their mountains, rivers and the places outside the seventh island

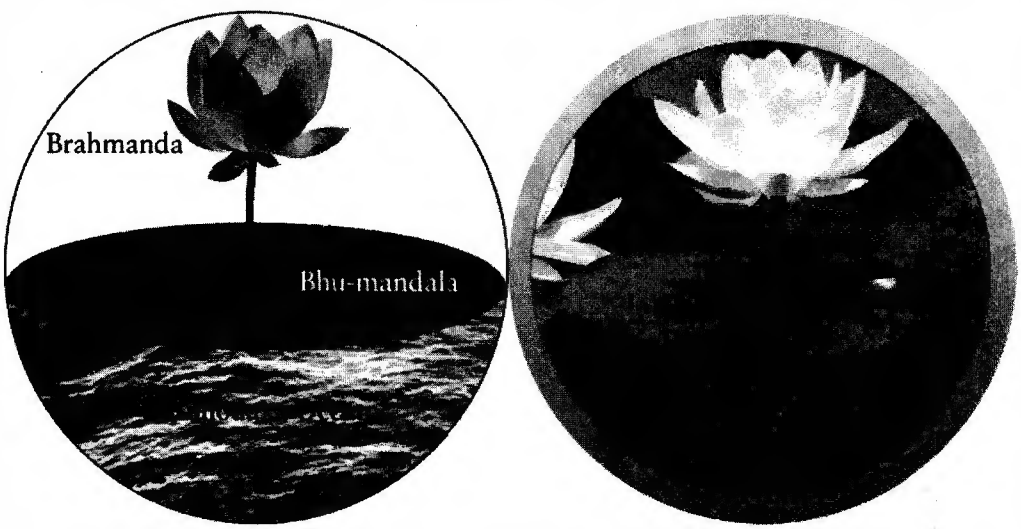
Part2: Fourteen Planetary Systems – The Vertical Structure of the Universe & the Movement of the Planets

- Ch 21-22: The Zodiac - Movement of Sun, Moon & other Grahas
- Ch 23: Description of Dhruva-loka, the Form of Dolphin and the Upper Planets
- Ch 24-25: Description of the Lower Planets (places below the sun)
- Ch 26: Description of Hellish Planets

2

The Bhū-Maṇḍala
The Horizantal Structure of the Universe

The Gigantic Universe, also known as the Brahmāṇḍa or the Hiranyāṇḍa or the Golden Egg, covered with seven layers starting with earth, is 500 Million Yojanas in diameter. This Brahmāṇḍa as described in the previous section appears like a lotus flower coming out of a lake, wherein the lake water is compared to the Garbhodaka Ocean, which fills the lower half of the Universe. The Lotus Leaf just above the water represents the Bhū-maṇḍala and the Lotus Stem and Flower represents the Fourteen Planetary System.



The earthly planetary system known as Bhū-maṇḍala resembles like a lotus flower. This primarily consists of seven islands which are situated as circular coverings in the lotus of Bhū-maṇḍala. Some say that these seven islands

resemble the whorl of that flower. The seven islands of the lotus like Bhū-maṇḍala are represented by seven circular coverings. The entire Bhū-maṇḍala, 500 Million Yojanas in diameter, situated in the middle of the Universe horizontally, was ruled solely by Svāyambhuva Manu, and the descendents of his second son Uttānapāda all the way up to Pracetās. When Daksa, the son of Pracetās, did not take up the rule, Priyavrata Maharāja, the first son of Svāyambhuva Manu, became the ruler of Bhū-maṇḍala being requested by his father and Lord Brahmā. During his time Priyavrata Maharāja divided the Bhū-maṇḍala into seven islands surrounded by seven oceans.

King Priyavrata divides the Bhū-maṇḍala into Seven Islands[10]

Knowing that Priyavrata, detached since childhood, would leave the house and go to the forest, Svāyambhuva Manu appointed his younger brother Uttānapāda to rule the kingdom. The descendents of Maharāja Uttānapāda, starting with Dhruva Maharāja all the way up to the Pracetās ruled the Bhū-maṇḍala. In this way, half of Svāyambhuva Manvantara period had passed. Dakṣa, who was born a second time as the son of the Pracetās, with a desire to attain his previous powers went to perform austerities, instead of taking up the ruler ship over Bhū-maṇḍala. Since there was no king, Svāyambhuva Manu, seeing no alternative, wanted to bring Priyavrata, his eldest son, back from the forest.

By serving Nārada Priyavrata quickly became fully knowledgeable of the highest goals and truths and attained initiation though meditation on *ātmā*. Then he became eager to limit himself with determination to experience only the Supreme Lord full of eternity, knowledge and bliss, and not any material object. At that moment, because he steadily manifested all the qualities mentioned in the scriptures for ruling the kingdom, his father Manu asked him to rule the world. "O Priyavrata! Now you must protect the world." He did not welcome that request. He did not desire to do so because he thought it was not beneficial for him. Why? He was constantly engaged in concentrating without break on the Lord with all his senses and activities. Though what his father said could not be rejected, he protested saying "I will not rule the kingdom." Why? He considered that he would be affected by improper

[10] SB 5.1: Priyavrata Maharaja's wonderful qualities and his surrender to Lord's will

qualities like lust and anger in ruling the kingdom. Then, Brahmā, always engaged in contemplating the prosperity of the creation of the universe, while determining the intentions of all the living beings, just as the king determines the intentions of all his subjects through his servants, surrounded by Marīci and others, as well as the personified Vedas, appeared on earth from Satyaloka to show proof of the *dharma* to be taught to Priyavrata. Being worshipped by Nārada, Manu and Priyavrata, Lord Brahmā looking at Priyavrata with mercy and a smile, spoke.

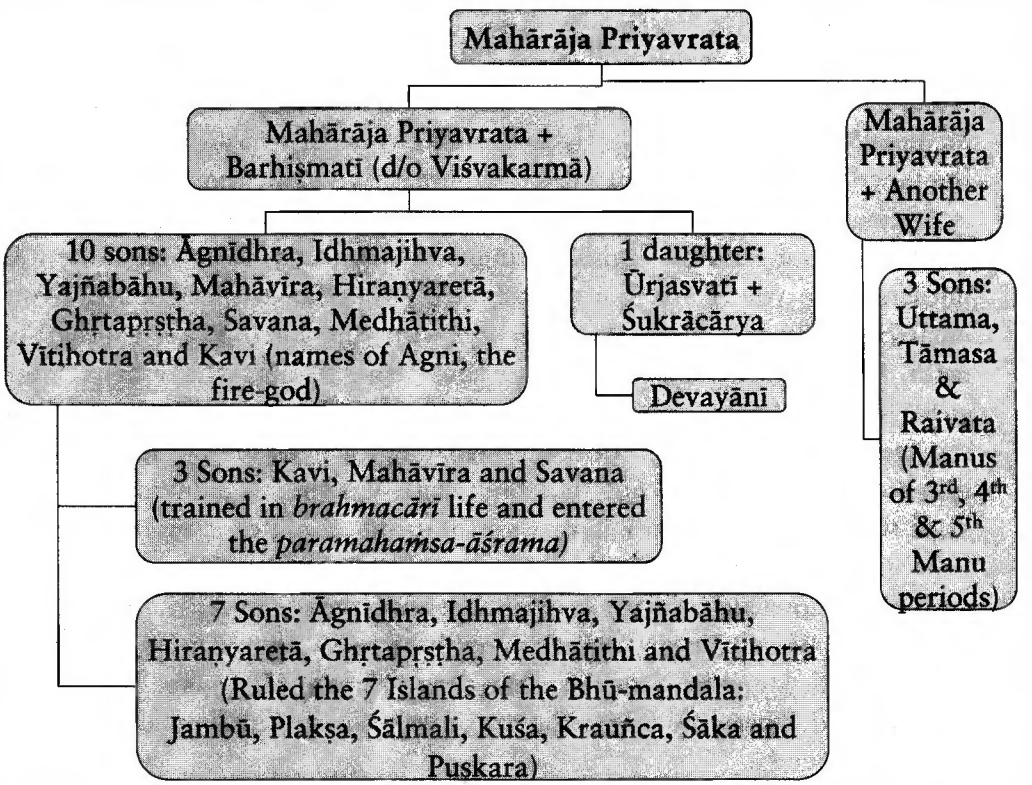
> Lord Brahmā had a merciful glance that "I have interrupted his *bhakti, jñāna* and *vairāgya,* but even in household life, may his *bhakti* should increase!" He looked at his grandson with a smile and thought - "To refuse the kingdom is your stubbornness. I will make you rule the kingdom. That is my stubbornness. I will see whose stubbornness will prevail." Then he spoke "Do not hate me, thinking that my words fixed on active life are intolerable to you. I am also dependent on your Lord. I will speak about what He wants you to do. Consider this with your intelligence. I speak the truth. What is the truth? I, Śiva and your father Manu, and Nārada, your *guru*, helplessly carry out the order of the Lord. Whatever He instructs in the heart of any of us, we must follow. By His order I am engaging you in ruling the kingdom. Do not find fault with your Lord, thinking that the Lord whom you worship has cast you into the ocean of *saṁsāra.* Do not hate him. One cannot avoid the order of the Lord by one's power of austerity, knowledge, or *yoga*, by independent intelligence, by wealth, by *dharma*, or by relying on others' strength or one's own. The Lord cannot be estimated. He quickly delivers one person whom He engages in actions in this world. Therefore following His order, rule the kingdom as requested by your father."

Being instructed, Priyavrata with great respect carried out the order of the great devotee Brahmā. Thus when Svāyambhuva Manu was not able to bring Priyavrata back, Brahmā came and made Priyavrata rule the kingdom.

Accepting Lord Brahma's order, Maharaja Priyavrata ruled over the Bhū-maṇḍala engaging in worldly affairs while thinking of the lotus feet of the Supreme Lord, free from material attachment. He married Barhiṣmatī, the d/o Prajāpati Viśvakarmā, and begot ten sons with abundant conduct, qualities, beauty, bravery, and magnanimity equal to his own, and a daughter, the

youngest of all, named Ūrjasvatī. Three among them, namely Kavi, Mahāvīra and Savana, lived in complete celibacy. In another wife, he begot three sons, named Uttama, Tāmasa and Raivata, who later took charge of 3rd, 4th & 5th Manvantaras.

Descendents of Priyavrata Mahārāja[11]



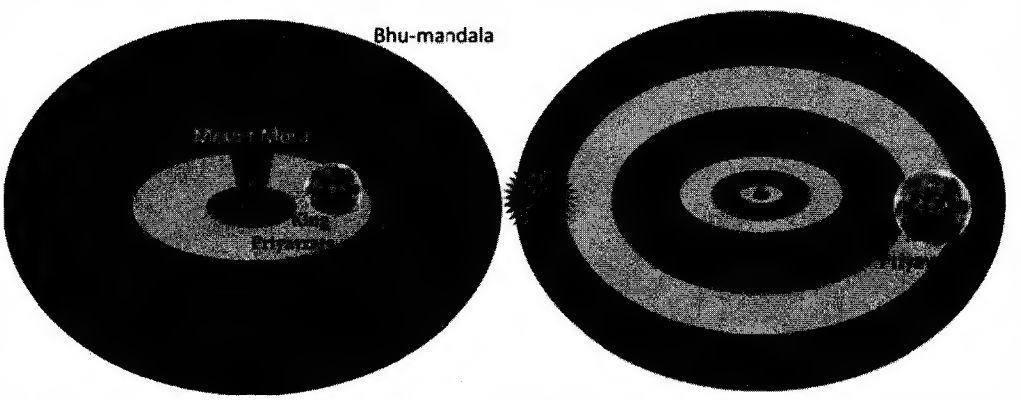
While these sons were engaged in the path of sense control, Maharaja Priyavrata, with powerful mind, ruled the Bhū-maṇḍala for 110,000,000 years (11 arbuda years[12] or 25 Divya-Yugas) and conquered his enemies by the mere sound of his bow drawn by two arms. Maharaja Priyavrata reigned while the sun, going around Meru, lit up half the surface of the Bhū-maṇḍala while the

[11] SB 5.1.23-29: Priyavrata Maharaja's rule and descendents

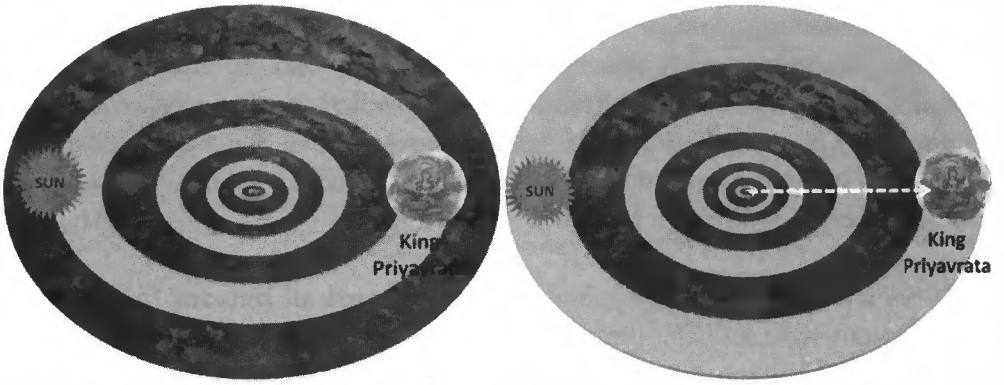
[12] Some say an arbuda is equl to 100,000,000 years. Sata – 100; sahasra – 1000; ayuta – 10,000; niyuta – 100,000; prayukta – 1,000,000; koti – 10,000,000; abudha – 100,000,000;

other half was dark. Attaining super-human power by worship of the Lord, Priyavrata, discontent with this condition, thinking that he would make the night into day by his brilliant chariot which traveled at the speed of the sun, becoming like a second sun, followed the movement of the sun seven times. The troughs created by the wheels of his chariot produced seven oceans, from which seven continents were formed.[13]

> By his worship of the Lord, Priyavrata surpassed all humans in influence. Discontent with half the Bhū-maṇḍala's surface being covered by darkness, he followed the movement of the sun (*taruṇim*), so that when the sun set, he would light up that area. He followed the sun, not to compete with the sun, but to give happiness to the inhabitants of Bhū-maṇḍala. This means that by the power of *yoga* of sages like Saubhari, Priyavrata had two bodies, one as the King and one as the sun. The movement of the wheel of his chariot produced troughs which became seven oceans. This means that he had seven chariots each with one wheel, by the power of his *yoga*. Their movement was higher than the sun's, and they circled around the plane of the Bhū-maṇḍala. The chariots were successively larger and larger in size, since the outer circles were larger and larger, each one twice the size of the previous one, traveling higher and higher so that the inhabitants of his city in Āryāvarta could see his chariot even at a great distance. Because the wheels of each chariot were broader than the previous ones, the oceans were successively broader.

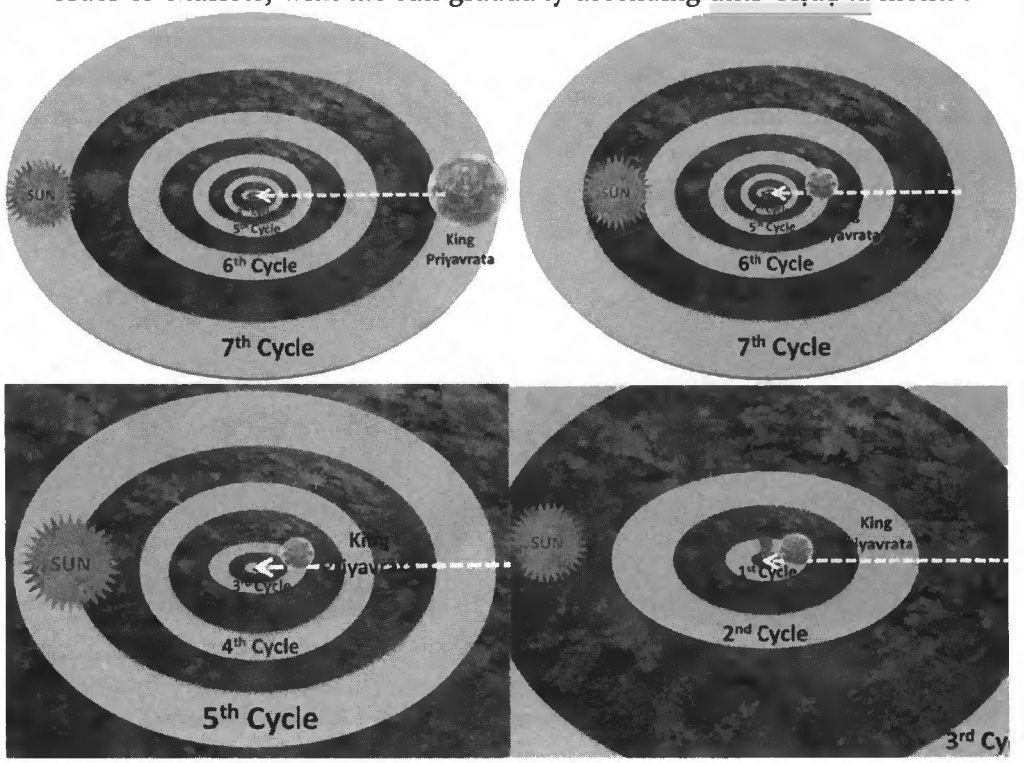


[13] SB 5.1.23-29: Priyavrata's rule and descendents & SB 5.1.30-35: Priyavrata's wonderful activities



He rode in each of these seven chariots for twenty-five days and forty-five and a half *ghaṭikas* each (totaling half a year, 25.758x7 = 180.3 days), starting with the summer equinox, with the sun gradually descending until Pauṣa month.

Again starting from the winter equinox, he would travel until the cycle was completed, with the same number of days fixed, but with the opposite order of chariots, with the sun gradually ascending until Aṣaḍha month.



Thus, "seven times" in the previous verse means that he traveled in seven chariots in the manner described above. There is no contradiction, since it is not proper to say that Priyavrata stopped pursuing the sun after only seven days, and some say he did not give up this activity. Because of these seven oceans (*yataḥ*), seven continents formed.

The troughs created by the wheels of his chariot produced seven oceans, from which seven continents were formed. These seven islands are known as Jambū, Plakṣa, Śālmali, Kuśa, Krauñca, Śāka and Puṣkara (*jambū-plakṣa-śālmali-kuśa-krauñca-śāka-puṣkara-saṁjñās*). Each island is twice as large as the one preceding it, and each is surrounded by an ocean on all sides. The seven oceans, respectively containing salt water, sugarcane juice, liquor, clarified butter, milk, yogurt, and drinking water are like trenches surrounding the seven continents of equal breath to its corresponding ocean (*kṣārodekṣu-rasoda-suroda-ghṛtoda-kṣīroda-dadhi-maṇḍoda-śuddhodāḥ sapta jaladhayaḥ*). Outside of each ocean is a continent, and outside each continent is another ocean. Such power is not surprising from persons who have conquered the six senses by the dust from the lotus feet of the Lord, since even an outcaste becomes immediately free of bondage of *karma* by chanting the Lord's name once.

The praise of Priyavrata, "Who could perform the actions of Priyavrata except the Supreme Lord himself? By the rim of his chariot wheel, he made seven oceans and destroyed the darkness of night. He arranged the situation of the Bhū-maṇḍala for the welfare of the people so that there were divisions and borders for each island by means of rivers, mountains and forests. (5.1.39-40)"

Note: He arranged the Bhū-maṇḍala with islands, and made boundaries by rivers and mountains for the happiness of the leaders of the people, so they could exist without quarrelling.

priyavrata-kṛtaṁ karma ko nu kuryād vineśvaram
yo nemi-nimnair akaroc chāyāṁ ghnan sapta vāridhīn
bhū-saṁsthānaṁ kṛtaṁ yena sarid-giri-vanādibhiḥ
sīmā ca bhūta-nirvṛtyai dvīpe dvīpe vibhāgaśaḥ

Mahārāja Priyavrata, the husband of Queen Barhiṣmatī, gave sovereignty over these seven islands to his seven respective sons, namely Āgnīdhra, Idhmajihva, Yajñabāhu, Hiraṇyaretā, Ghṛtapṛṣṭha, Medhātithi and Vītihotra.

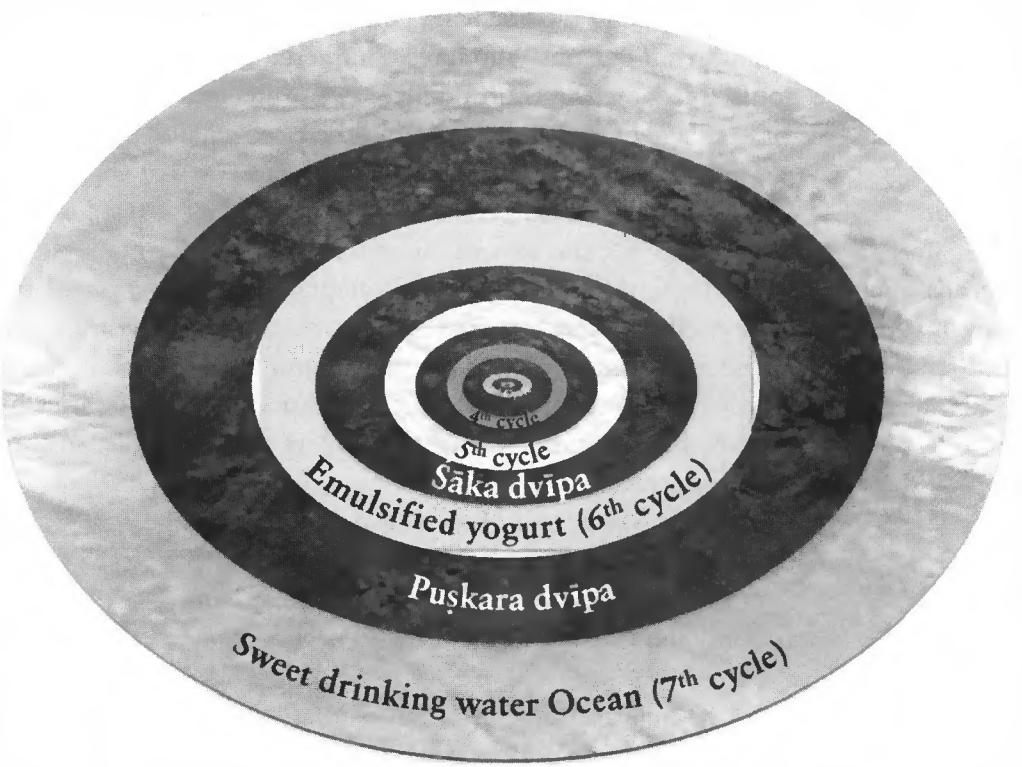
Distribution of the Seven Dvīpas among Priyavrata's Seven Sons

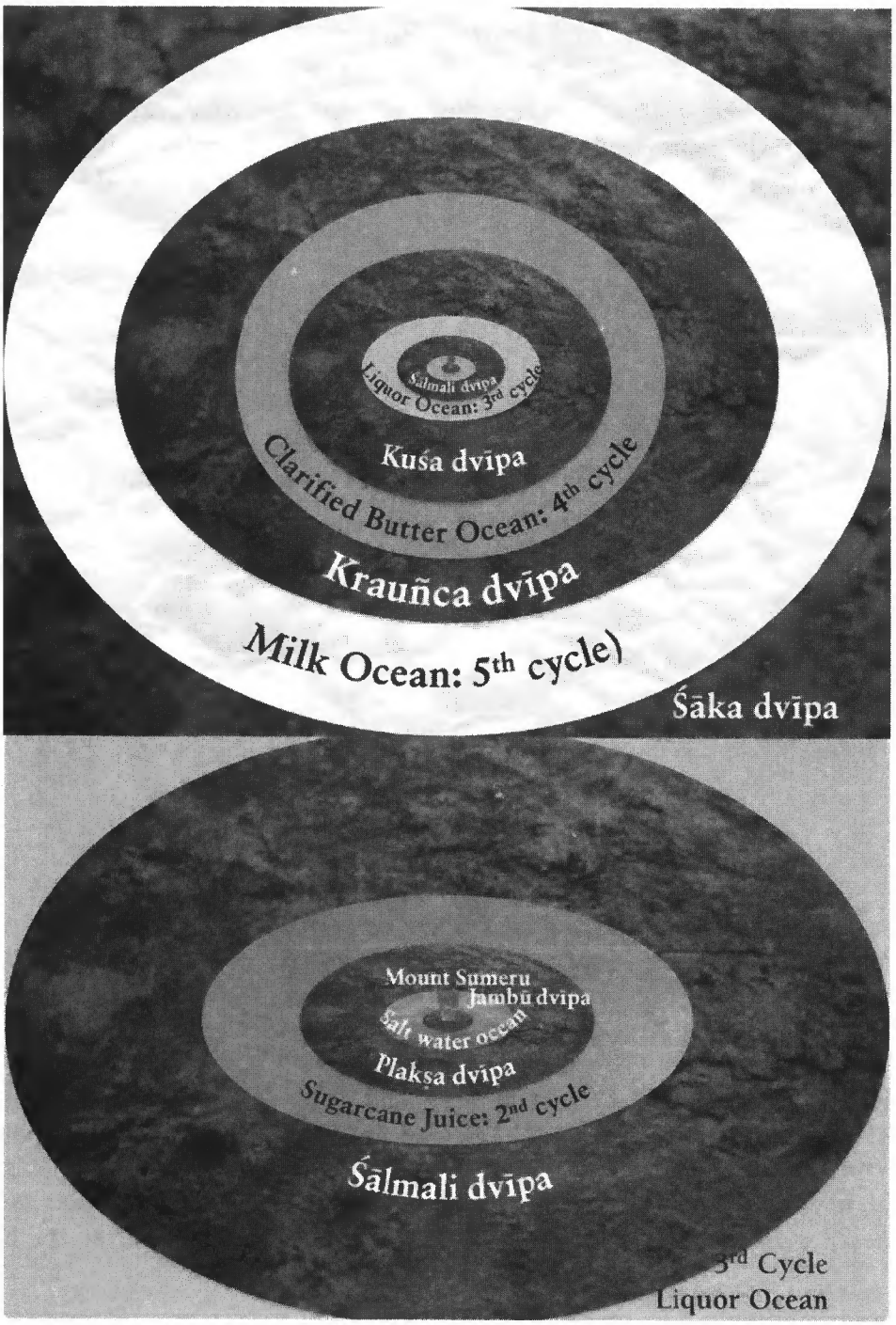
N	Island	Size	Ocean	Size	Priyavrata's Son
1	Jambūdvīpa	X	Salt water	X	Āgnīdhra
2	Plakṣadvīpa	2X	Sugarcane juice	2X	Idhmajihva
3	Śālmalidvīpa	4X	Liquor	4X	Yajñabāhu
4	Kuśadvīpa	8X	Clarified butter	8X	Hiraṇyaretā
5	Krauñcadvīpa	16X	Milk	16X	Ghṛtapṛṣṭha
6	Śākadvīpa	32X	Emulsified yogurt	32X	Medhātithi
7	Puṣkaradvīpa	64X	Sweet drinking water	64X	Vītihotra

- The breadth of the first island is (X=) 0.1 MY (also confirmed by Vīrarāghava Ācārya)
- The radius of the Seven Dvīpas with Seven Oceans ($R_{7D\text{-}7O}$)

  $R_{7D\text{-}7O} = 0.5X + X + 2*(2X+4X+8X+16X+32X+64X)$

  $= 1.5X + 2*126*X = 253.5X = 25.35$ MY (X = 0.1 MY)




Śālmali dvīpa
Liquor Ocean: 3rd cycle
Kuśa dvīpa
Clarified Butter Ocean: 4th cycle
Krauñca dvīpa
Milk Ocean: 5th cycle)
Śāka dvīpa
Mount Sumeru
Jambū dvīpa
Salt water ocean
Plakṣa dvīpa
Sugarcane Juice: 2nd cycle
Śālmali dvīpa
3rd Cycle
Liquor Ocean

Jambūdvīpa, Innermost Island of Bhū-maṇḍala[14]

The central portion (*kośa*) within the seven islands, which are situated as circular coverings (kośa) in the lotus of Bhū-maṇḍala (kuvalaya), is known as Jambūdvīpa. This Jambūdvīpa is 100,000 *yojanas* (*niyuta*) wide and is circular like a lotus leaf.

yo vāyaṁ dvīpaḥ kuvalaya-kamala-kośābhyantara-kośo
niyuta-yojana-viśālaḥ samavartulo yathā puṣkara-patram

Bhū-maṇḍala ~ Lotus Flower;
Seven Islands ~ Circular Coverings;
Jambūdvīpa ~ Whorl or Central Portion of the Lotus Flower

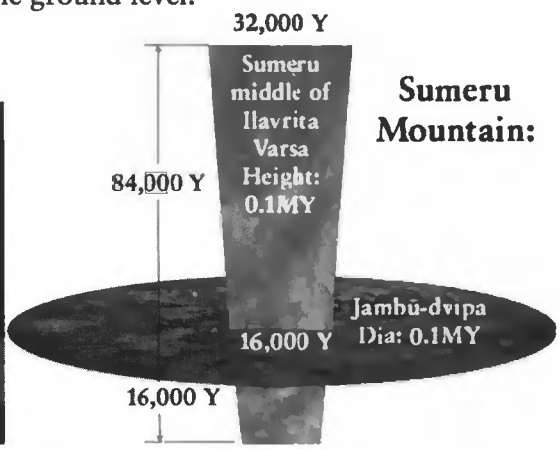


The Jambūdvīpa has been divided into nine divisions of land (*varṣa*[15]), each extending to a width of 9,000 yojanas. They are separated by eight mountain ranges (nine including the Mount Meru). In the center of these nine divisions is Ilāvṛta, and in its center (*nābhyām*) is Meru, the king of mountains, made of gold, whose height is equal to the length of Jambūdvīpa, and which acts as the pericarp of the lotus. At its top it measures 32,000 *yojanas* and at the base

[14] SB 5.16.4-10: Description of Jambūdvīpa, Mount Sumeru and Varsas with in Jambūdvīpa
[15] *Varṣa* means the Land between two Mountains (or the Vally)

16,000 *yojanas*. It extends 16,000 *yojanas* within the Jambūdvīpa. Thus 84,000 *yojanas* of Meru are visible above the ground level.





Divisions of Jambūdvīpa

North of Ilāvṛta-varṣa: Three mountains Nīla, Śveta and Śṛṅgavān mark the borders of three *varṣas* Ramyaka, Hiraṇmaya and Kuru and separate them from one another going further northward. The width and height of these mountains is 2,000 and 10,000 *yojanas* respectively. Lengthwise, they extend east and west to the ocean of salt water. The length of each mountain decreased by a little more than 10% compared to the previous mountain.

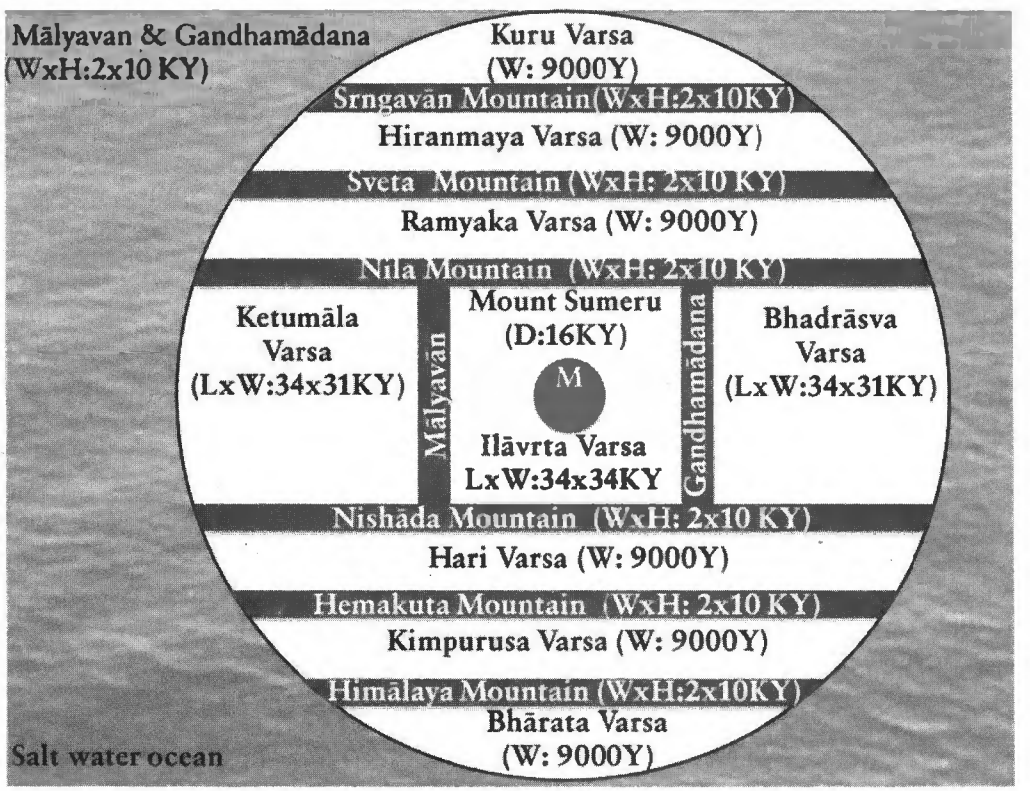
South of Ilāvṛta-varṣa: Three mountains Niṣadha, Hemakūṭa and Himālaya mark the boundaries of the three *varṣas* Hari-varṣa, Kimpuruṣa-varṣa and Bhārata-varṣa in the southern direction. Each of them is 2,000 *yojanas* wide and 10,000 *yojanas* high, similar to Nīla and the rest. Lengthwise, they extend east and west to the ocean of salt water.

West and East of Ilāvṛta-varṣa: Two mountains named Mālyavān and Gandhamādana mark the borders of the *varṣas* known as Ketumāla and Bhadrāśva in the west and east respectively. These two mountains extend as far as Nīla Mountain in the north and Niṣadha in the south. The width and height of these mountains is 2,000 and 10,000 yojanas respectively.

The *Vāyu Purāṇa* describes the form of Jambūdvīpa, "*dhanurvat saṁsthite*

jñeye dve varṣe dakṣiṇottare dīrghāṇi tatra catvāri caturasram ilāvṛtam." Two *varṣas* on the north and south are bow-shaped. Four other *varṣas* are long and Ilāvṛta is square. The *varṣa* in the south is Bhārata and that in the north is Uttarakuru. The four long *varṣas* are Kuṁpuruṣa, Harivarṣa, Ramyaka and Hiranmaya. Bhadrāśva and Ketumāla are also shaped like bows, bordered by Nīla and Niṣadha mountain ranges running horizontally and entering the sea.

Note: Their width is nine thousand *yojanas* except for Ketumāla and Bhadraśva in the middle. Ilāvṛta is nine thousand *yojanas* in all four directions from Meru (which has a 16,000 *yojanas* base). Ilāvrta, Bhadrāśva and Ketumālā are thus 34,000 *yojanas* in width (9000 + 16000 + 9000 = 34,000), thus they extend to Nīla in the north and Niṣadha in the south. Since Ilāvrta is square, it extends 34,000 *yojanas* in length, east to west. The width of Mālyavān and Gandhamādana is 2000 *yojanas* each, thus the length of Bhadrāśva and Ketumālā, from east to west, is 31,000 *yojanas* each (100 – 34 – 4 = 62; 62/2 = 31).



Measuring from south to north	
In a line from south to north, Bhārata-varṣa, Kiṁpuruṣa and Harivarṣa are 27,000 *yojanas* in total	9000 + 9000 + 9000 = 27,000
The mountains Himālaya, Hemakūṭa and Niṣadhā, each 2000 yojanas, make 6000 yojanas in total	2000 + 2000 + 2000 = 6,000
Sumeru and Ilāvṛta are 34,000 *yojanas* wide	9000 + 16,000 + 9000 = 34,000
Ramyaka, Hiraṇmaya and Kuru are 27,000 *yojanas* in total.	9000 + 9000 + 9000 = 27,000
Nīla, Śveta and Śṛṅgavān are at total of 6000 *yojanas*	2000 + 2000 + 2000 = 6,000
The total from south to north comes to	**100,000 *yojanas***
Measuring from east to west	
Bhadrāśva is	31,000 *yojanas*
Sumeru and Ilāvṛta are **(9 + 16 + 9 KY)**	34,000 *yojanas*
Ketumāla is	31,000 *yojanas*
The Gandhamādana and Mālyavān mountains are	4,000 *yojanas*
The total across is also	**100,000 *yojanas***

Note: The measurements such as 10,000 or 100,000 yojanas mentioned in the Śrīmad-Bhāgavatam are beyond our comprehension. Still they need to be considered authentic for they have been spoken by the greatest authorities like Śrīla Śukadeva Gosvāmī in the presence of the greatest personalities of the universe. The history has witnessed two greatest conversations, which were held to honour the passing away of two great souls, Śrīla Bhīṣmadev and Parīkṣit mahārāj. These conversations were witnessed by the greatest scholars of the Universe, such as Atri, Vasiṣṭha, Bhṛgu, Aṅgirās, Parāśara, Viśvamitra, Paraśurāma, Bhāradvāja, Gautama, Maitreya, Agastya, headed by Vyāsa and Nārada. In the first conversation, Śrīla Bhīṣmadev instructed Yudhiṣṭhira mahārāj on various dharmas suitable for men according to their natures, and according to ones varṇa and āśrama. These activities have qualities of renunciation and enjoyment, according to a person's degree of detachment or attachment. Śrīla Bhīṣmadev instructed Yudhiṣṭhira mahārāj while the sages were listening in the personal presence of Lord Kṛṣṇa Himself. Śukadeva Gosvāmī is the

personal witness to this conversation, whereas in the second conversation, he himself spoke to Parīkṣit maharāj for seven days about the complete perfection and the sādhana to be performed at all times for that perfection. During that seven-day discussion, in the presence of all the great sages, Śukadeva Gosvāmī briefly summarised the mysteries of the structure and the workings of this gigantic Universe. Due to lack of time and also not being the prime subject matter of that discussion, he just presented the essence within few chapters whit out missing any of the crucial details. Proper understanding and expansion of these few essential descriptions, reveals the complete mystery of the systematic Universal functioning. A brief glimpse of this has been presented herein.

In this regard, Madhvācārya quotes the following verses from the Brahmāṇḍa Purāṇa.

yathā bhāgavate tūktaṁ bhauvanaṁ kośa-lakṣaṇam
tasyāvirodhato yojyam anya-granthāntare sthitam
maṇḍode puraṇaṁ caiva vyatyāsaṁ kṣīra-sāgare
rāhu-soma-ravīṇāṁ ca maṇḍalād dvi-guṇoktitām
vinaiva sarvam unneyaṁ yojanābhedato 'tra tu

Śrīmad-Bhāgavatam is the authoritative text, which tells us about Cosmography; and information gathered from any other source should be in accordance with it. Except two corrections in reference to different *kalpas,* rest all should be considered accurate.

The lineage of King Āgnīdhra

In Pūrvacitti, the Apsarā sent by Lord Brahmā, King Āgnīdhra, the s/o King Priyavrata, begot nine sons in nine subsequent years. Because of drinking the breast milk of their mother, the nine sons of Āgnīdhra naturally had long limbs and strength. These sons of Āgnīdhra ruled the nine different part of Jambūdvīpa that they had received from their father. These kingdoms were named according to the names of the sons.

Ilāvṛta-varṣa and Sumeru Mountain[16]

[16] SB 5.16.11-29: Description of Ilavrta Varsa and Sumer Mountain

Meru, the king of mountains completely made of gold, is situated in the middle of Ilāvṛta-varṣa (*nābhyām*). Around this Meru there are four supporting mountains, known as Mandara, Merumandara, Supārśva and Kumuda, in four directions. Both the length and height of these mountains are 10,000 *yojanas* each. Since they act as the support of Meru in the center, they spread out in the four directions (*catur-diśam*). Their width is only 2000 yojanas.

Note: Commentary by Sri Giridhara Lāla ācārya – The four mountains in four directions from east such as Mandara, etc., are meant for the support of Meru, like a belt. Each of these is 10,000 *yojanas* extent and tall. These are created by the Supreme Lord. They are all golden, in the shape of a churning staff and are known from the reference in the 8th Canto as "the golden hill which crushed the demigods and the demons by its weight." They were using Mandara mountain for churning the milk-ocean.

On the summits of these four mountains starting with Mandara are a mango tree, a rose apple *(jambū)* tree, a *kadamba* tree and a banyan tree. They stand like flagstaffs. These trees are 100 *yojanas* wide and 1,100 *yojanas* in height. Their branches spread to a distance of 1,100 *yojanas*. On these four mountains are four huge lakes, filled with milk, honey, sugarcane juice, and pure water respectively. The celestial beings such as Siddhas, Cāraṇas and Gandharvas who use those waters naturally possess perfections of mystic yoga. There are also four celestial gardens named Nandana, Caitraratha, Vaibhrājaka and Sarvatobhadra. The *devatās*, along with their wives, enjoy within those gardens, while their glories are sung by the *upadevatās*.

Mango tree (Devacūta) on the summit of Mandara Mountain: When the Mangoes, as big as mountain peaks (861 *aratnis,* 861x1.5feet = 1291.5feet) and as sweet as nectar, fall from the top, they break and flow as a river called Aruṇodā. From drinking the water of the River Aruṇodā made mango juice with natural scent and other fragrances (*sugandhi*), the bodies of maidservants of Śiva's wife (the wives of Yakṣas), become fragrant. Touching their fragrant bodies, the wind perfumes the entire atmosphere for ten *yojanas* around.

Jambū tree on the summit of Merumandara Mountain: As the fruits of this Jambū tree fall from the top, the juice gliding from them (with the size of elephants) becomes Jambū-nadī River and flows to a distance of 10,000 yojanas in the southern side of Ilāvṛta flooding the entire land. The river flows,

increasing its length to the size of Ilāvṛta (*yāvat ilāvṛtam*), i.e. 9000 *yojanas*. The mud on both banks of the River Jambū-nadī, being moistened by the flowing juice, by a reaction with the air and the sunshine, produces huge quantities of gold called *jambū-nada,* which the *devatās* use for ornaments. All the *devatās* and their youthful wives wear this gold in the form of golden crowns, bangles and belts (*mukuṭa-kaṭaka-kaṭi-sūtrādy-ābharaṇa-rūpeṇa khalu dhārayanti*).

Mahā kadamba on Supārśva Mountain: Five rivers of honey, each about five *vyāmas*[17] (or 40 feet) wide, flow from the hollows of this tree. Flowing in western direction, these rivers give pleasure to itself and the whole of Ilāvṛta-varṣa. The air carrying the fragrance from the mouths of those who drink from those honey rivers, perfumes the land for a hundred *yojanas* around.

Banyan tree (Śatavalśa) on Kumuda Mountain: Many rivers flowing from the branches of this tree in the northern side of Ilāvṛta-varṣa gives happiness to Ilāvṛta and fulfills all the desires such as milk, yogurt, honey, ghee, molasses, food grains, clothes, bedding, seats and ornaments. By using the products of these flowing rivers, the residents have no suffering such as wrinkles, grey hair, fatigue, foul smelling perspiration, old age, disease, untimely death, cold, heat, loss of luster, or troubles. They all live their whole life with unlimited happiness[18].

Kesaracala Mountains: In addition to the four above mentioned mountains, there are another twenty mountains beautifully arranged around the foot of Mount Meru. They are situated like the filaments around the whorl of a lotus flower. The 20 mountains are Kuraṅga, Kurara, Kusumbha, Vaikaṅka, Trikūṭa, Śiśira, Pataṅga, Rucaka, Niṣadha, Sinīvāsa, Kapila, Śaṅkha, Vaidūrya, Jārudhi, Haṁsa, Ṛṣabha, Nāga, Kālañjara and Nārada.

Four Mountain Pairs: There are two more mountains on each side of Sumeru Mountain. In the East there are Jaṭhara and Devakūṭa, which extend to the north and south for 18,000 yojanas. In the West there are Pavana and Pāriyātra, extend to the north and south for 18,000 yojanas. In the South there

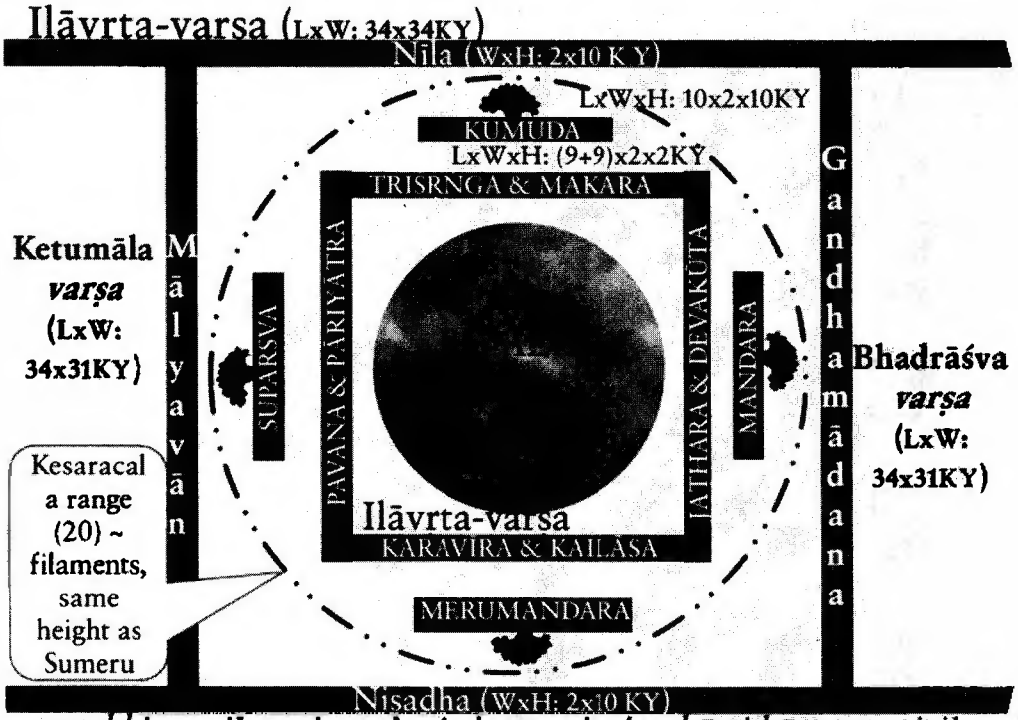
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[17] A *vyāma* is the space produced by extending the two arms and the fingers (8 feet).

[18] Refer to SB 5.16.25 for Śrīla Prabhupāda's explains the real cause of miseries in the world.

are Kailāsa and Karavīra, extend to the east and west for 18,000 yojanas. And in the North there are Triśṛṅga and Makara, extend to the east and west for 18,000 yojanas.

Note: Both mountains put together extend for 18,000 yojanas in length; it means each mountain is 9000 yojanas in length. The height and width of these mountains is 2000 yojanas (LxWxH: 18,000x2000x2000Y). Sumeru, a mountain of solid gold, surrounded by these eight mountains, shines like fire (*aṣṭabhir etaiḥ parisṛto 'gnir iva paritaś cakāsti kāñcana-giriḥ*).



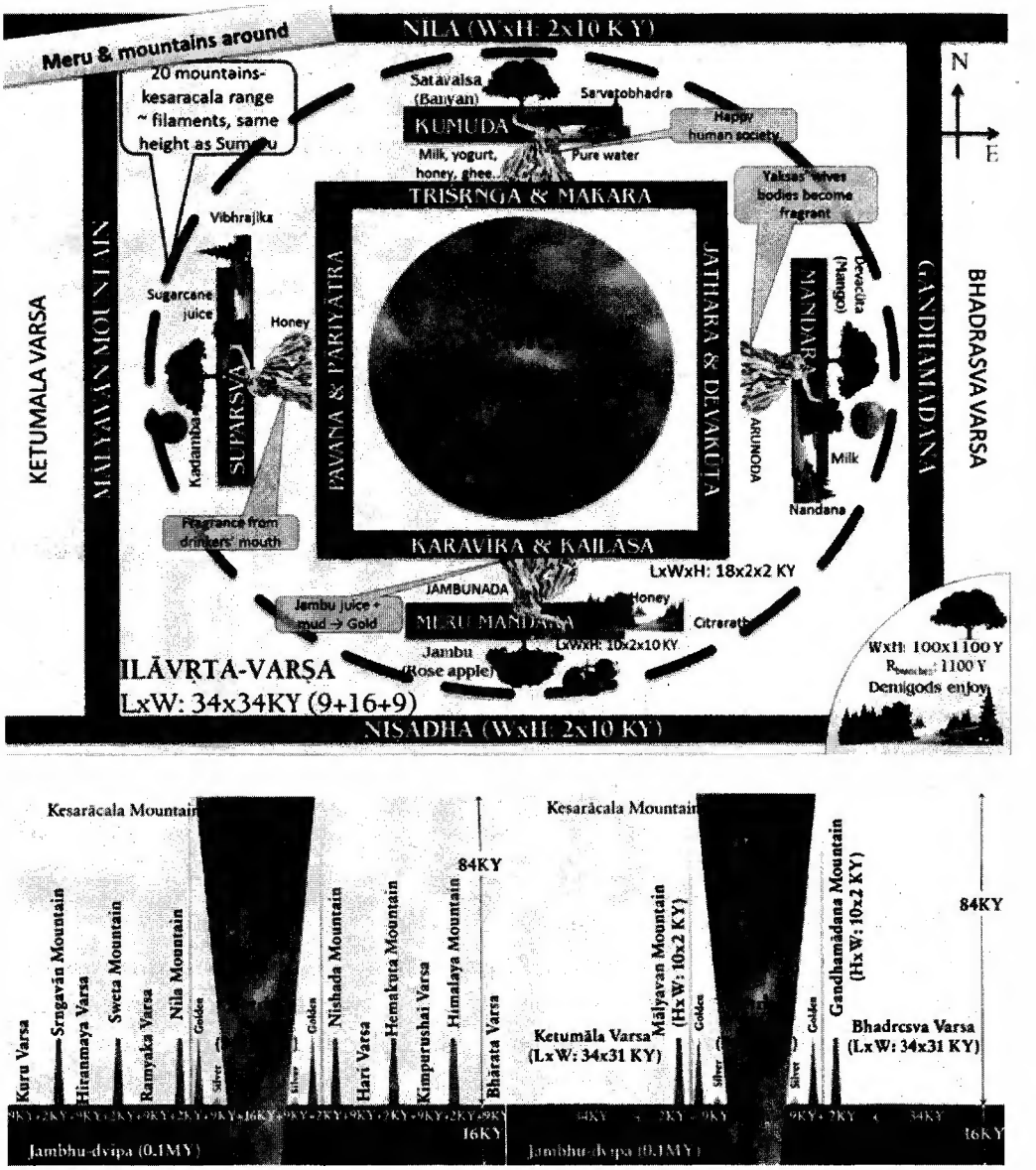
aṣṭabhir etaiḥ parisṛto 'gnir iva paritaś cakāsti kāṣcana-giriḥ.

Note: Commentary by Bhagavat Prasādācārya – On all four directions of Meru, leaving 1000 *yojanas* from its root, two mountains are located. Surrounded by these eight mountains, the golden mount Meru, glows like a sacred fire surrounded by dharba-grass on four sides (*aṣṭabhir etaiḥ parisṛto 'gnir iva paritaś cakāsti kāñcana-giriḥ*).

Note: Commentary by Srīdhara Svāmi[19] - In the four directions leaving 1000 yojanas from the root of Meru, like paridhi (squareperimeter) around

[19] Commentary on SB 5.16.27

a sacred fire Jaṭhara, Devakūṭa, etc., are situated and hence the size of 18,000 *yojanas* is mentioned. These mountains are to be known as silverish as Kailāsa is known to be so and these are analogous to it.

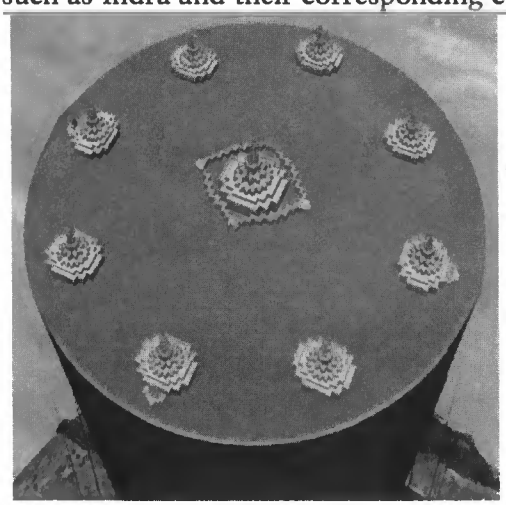


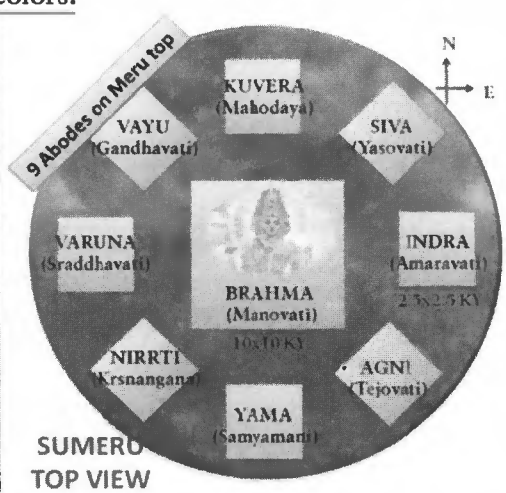
The Cross-section of Jambūdvīpa as seen from the South and West directions

The townships of Lord Brahmā & Eight Lokapalas on top of Mount Sumeru[20]

Residence of Brahma on the top of Sumeru: In the middle of the summit of mount Meru, which extends to a width of 32,000 yojanas, is the township of Lord Brahmā. It occupies an area of 10,000 yojanas in length and width (LxW: 10000x10000 *yojanas*). This township is made entirely of gold, and thus the learned scholars and sages call it as Śātakaumbhī or Manovatī.

Townships of 8 Lokapalas around Brahmapuri: Surrounding Brahmapurī in all directions atop Mount Meru are the residences of the eight principal governors of the directions (*loka-pālānām*), beginning with King Indra. These abodes are similar to Brahmapurī but are one fourth in size, i.e., 2,500 *yojanas* both in length and width (WxL: 2500x2500 *yojanas*). The devatās of the directions starting from east are Indra, Agni, Yama, Nirṛti, Varuṇa, Vāyu, Kubera, Śiva and their cities named as Amarāvatī, Tejovatī, Saṁyamanī, Kṛṣṇāṅganā, Śraddhāvatī, Gandhavatī, Mahodayā, and Yaśovatī. These cities are in the directions starting from the east, each with their particular deities such as Indra and their corresponding colors.



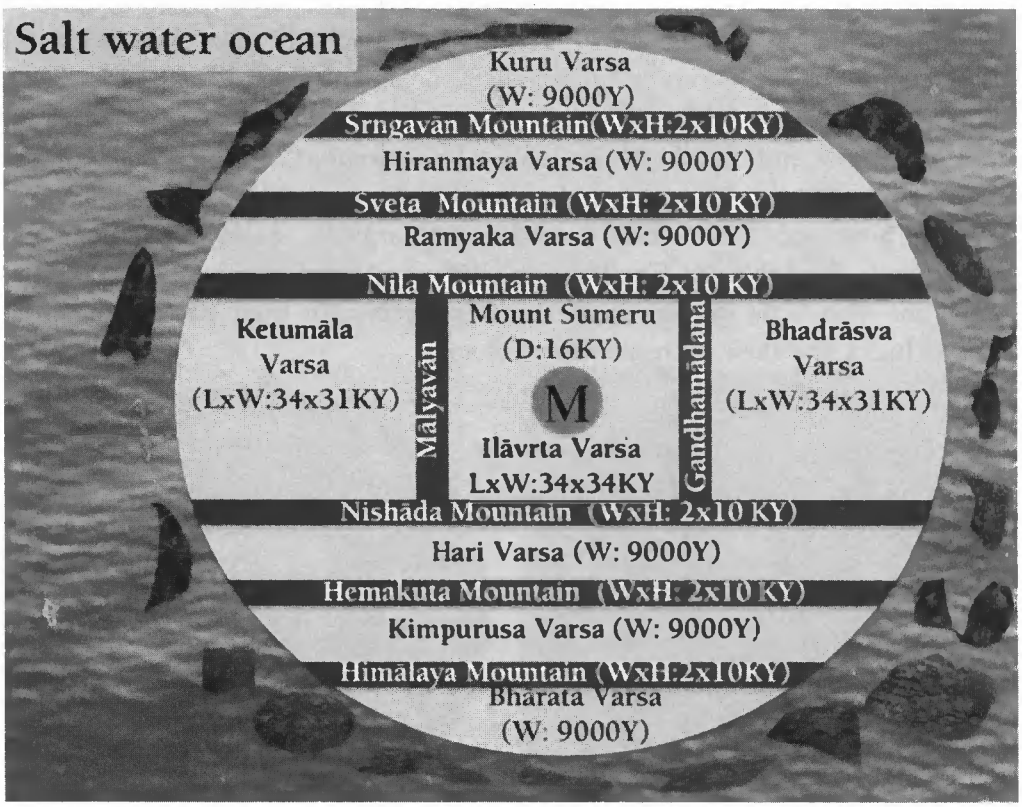


[20] SB 5.16.28-29: The townships of Lord Brahmā and eight lokapalas on the top of Mount Sumeru

Eight smaller islands around Jambūdvīpa[21]

In the opinion of some learned scholars, there are eight smaller islands surround Jambūdvīpa, which were created by the sons of Mahārāja Sagara, as they were by digging the earth while searching all over the world for their lost horse. The names of these islands are Svarṇaprastha, Candraśukla, Āvartana, Ramaṇaka, Mandara-hariṇa, Pāñcajanya, Siṁhala and Laṅkā.

jambūdvīpasya ca rājann upadvīpān aṣṭau haika upadiśanti sagarātmajair aśvānveṣaṇa imāṁ mahīṁ parito nikhanadbhir upakalpitān;



In SB 10.16.13 Lord Kṛṣṇa talks about Ramaṇaka Island, "O Kāliya, out of fear of Garuḍa, you left Ramaṇaka Island and came to take shelter of this lake. But

[21] SB 5.19.29-31: Islands around Jambūdvīpa

because you are now marked with My footprints, Garuḍa will no longer try to eat you." Kāliya went back to that island, the abode of snakes (*nāgālayaṁ*).

dvīpaṁ ramaṇakaṁ hitvā hradam etam upāśritaḥ
yad-bhayāt sa suparṇas tvāṁ nādyān mat-pāda-lāñchitam

Flow of the River Gaṅgā through Jambūdvīpa[22]

After describing the splendor of the mountains surrounding Sumeru like filaments, Śukadeva Gosvāmi now describes the very long stream of the Gaṅgā, the crest jewel of all splendors, by relating how Gaṅgā made her appearance. At the time of Vāmana *avatāra*, by stepping over the Bhū-maṇḍala with His right foot and from throwing His left foot in the air, He caused all places up to the top of the universe and even up to the layer of *prakṛti* (where *the guṇas* are pacified) to tremble. When He pierced the covering of the universe with the big toe of His left foot, the water of the Causal Ocean entered from outside (*vikramato vāma-pādāṅguṣṭha-nakha-nirbhinnordhvāṇḍa-kaṭāha-vivareṇa-antaḥ-praviṣṭā yā bāhya-jala-dhārā*). Thus the spiritual stream of water of the Kāraṇa Ocean, outside of the eight layers of the universe, entered within the Universe. The water was colored red from the *kuṁkuma* and the red color of the Lord's lotus feet, for it is arising from the water used to wash the Lord's feet (*tac-caraṇa-paṅkajāvanejanāruṇa-kiñjalkoparañjitākhila*). The touch of this water destroys the contamination of sin, but the water remains free of contamination (*jagad-agha-malāpahopa sparśanāmalā*). It was called by the name Bhagavatpadī, "water coming from the foot of the Lord." because at that time it was not known by other names. After a long time (a thousand *yugas*), the river descended to Dhruva-loka (*divaḥ mūrdhani*), which then became known as Viṣṇu-pada, the abode of Viṣṇu. Later on she is also known as Viṣṇupadī, Jāhnavī and Bhāgīrathī.

Note: Commentary by Srīdhara Svāmi[23] – Yuga here denotes the technical term used in astronomy, consisting of five years. And according to Sripati 'a yuga comprises five years'. That it took such a long time for the river to descend on heaven also suggests its other name "Mandākini Gaṅgā, the slow moving river."

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[22] SB 5.17.1-10: Descent, glories and the Path of Gaṅgā
[23] Commentary on SB 5.17.11

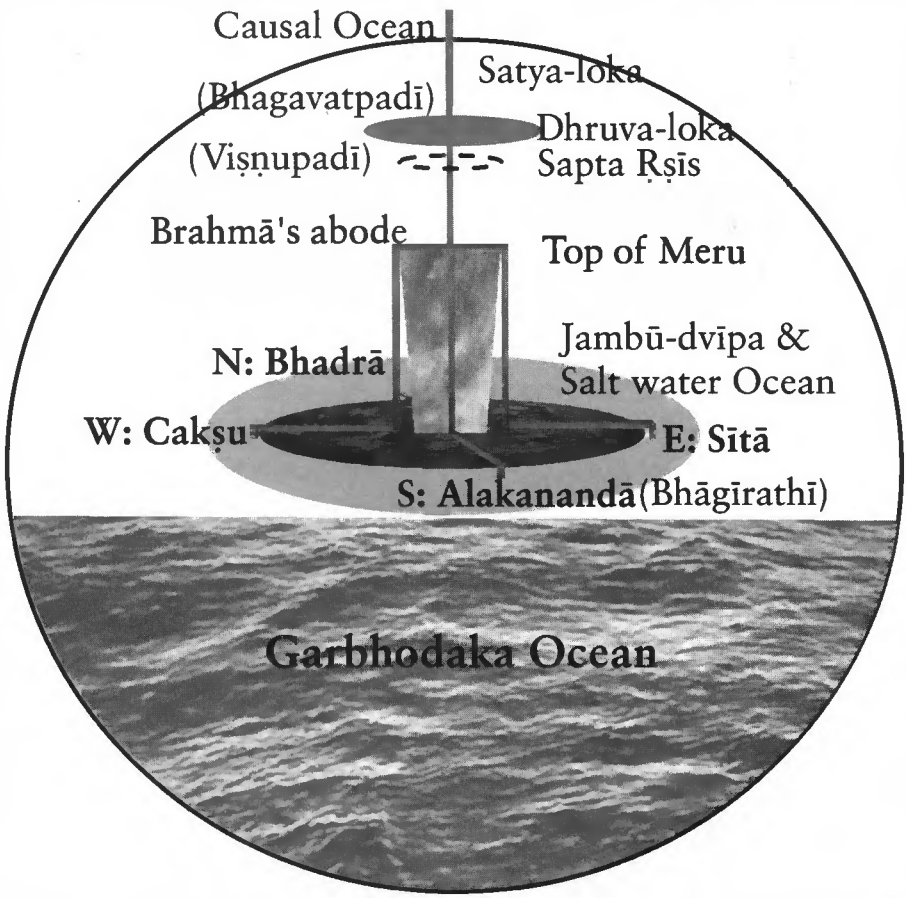
On Dhruva-loka, Dhruva Mahārāja with great respect even now holds on his head that water, thinking "This is the foot water from the deity of our family (*parama-bhāgavato 'smat-kula-devatā-caraṇāravindodakam iti*)." The seven great sages [Marīci, Vasiṣṭha, Atri and so on], who are situated just below Dhruva-loka, being aware of the influence of the water of the Gaṅgā, ignoring all other goals to this day respectfully keep the Gaṅgā's water in their matted locks, as much as persons desiring liberation respect liberation. After purifying the planet of the seven sages (*sapta ṛṣayah*) near Dhruva-loka, the Gaṅgā's water travels through the heavenly path (*deva-yānena*) carried by the throngs of celestial airplanes, inundates the moon (*indu maṇḍalam*) and finally reaches Lord Brahmā's abode atop Mount Meru (*āvārya brahma-sadane nipatati*). On top of mount Meru, the Gaṅgā divides into four branches and flow in four different directions. These branches are known as Sītā, Alakanandā, Cakṣu and Bhadrā, and they flow down to the ocean in East, South, West and North respectively.

- **Path of Sītā:** Starting from Brahmapurī atop Mount Meru, this branch runs down to the nearby peaks of the Kesarācala Mountains, and from there to the peak of Gandhamādana Mountain skipping over Ilāvṛta. It then flows into the land of Bhadrāśva-varṣa. Finally it reaches the ocean of salt water in the east.

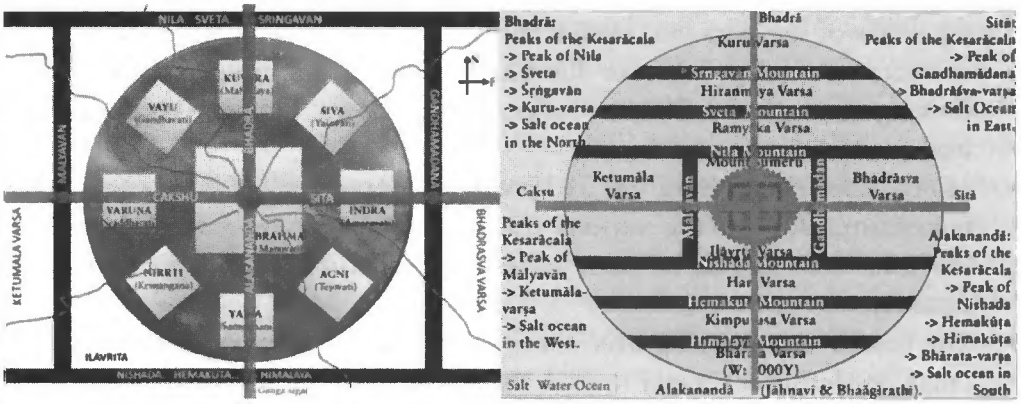
  Note: Commentary by Sri Giridhara Lāla ācārya – The Kesarācala (filament) Mountains are taller than the boundary hills Gandhamādana etc., and closer, thus the fall of Gaṅgā's eastern branch Sītā is first on them. The peak of the Kesara (Filament) Mountains are of the same height of the Meru (says Bhagavat Prasādācārya)
- **Path of Cakṣu:** Starting from Brahmapurī, touching the peaks of the Kesarācala Mountains, this branch falls on the summit of Mālyavān Mountain, then cascades into Ketumāla-varṣa with uninterrupted force and enters the ocean of salt water in the west.
- **Path of Bhadrā:** Starting from Brahmapurī atop Mount Meru, this branch fall onto the peaks of Kesarācala Mountain, then to Nīla, Śveta and Śṛṅgavān Mountains in succession (skipping over Ilāvṛta, Ramyaka and Hiraṇmaya *varṣas*). Then it runs down into the province of Kuru-varṣa and then finally enters into the saltwater ocean in the north.
- **Path of Alakanandā:** Starting from Brahmapurī, this branch fall onto the peak of Kesarācala Mountain, and then falls down with fierce force upon

the peaks of the Niṣadha, Hemakūṭa and Himakūṭa (Himālaya) mountains in succession (skipping over Ilāvṛta, Hari and Kimpuruṣa *varṣas*). Then flowing through the Bhārata-varṣa, it runs down into the ocean of salt water in the south.

- Many other rivers, both big and small, the daughters of Meru and other mountains, flow to the various tracts of land in hundreds of branches. (*anye ca nadā nadyaś ca varṣe varṣe santi bahuśo merv-ādi-giri-duhitaraḥ śataśaḥ*)
- The four rivers, skipping over many varṣas, flow down from the mountain range bordering the varṣa nearest the ocean onto the earth, and then with bowed head (so to speak), enter the ocean, their husband, giving up pride.

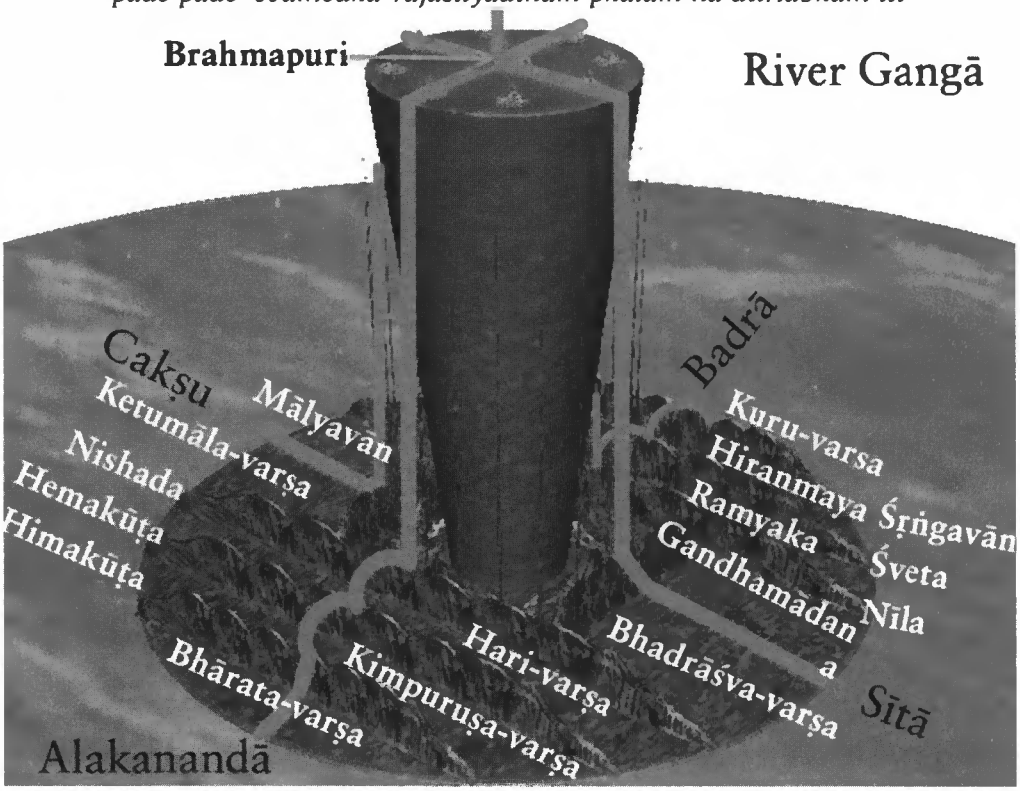


4 Branches of the Gaṅgā on the Top of Mount Meru and their passage through Jambūdvīpa



Benefits of Bathing in Ganges: It is not very difficult for persons who bathe in this river to achieve with every step the results of performing great sacrifices like the Rājasūya and Aśvamedha yajñas. Thus, the people who come to bathe in this river are very fortunate.

jaladhim abhipraviśati yasyāṁ snānārthaṁ cāgacchataḥ puṁsaḥ
pade pade 'śvamedha-rājasūyādīnāṁ phalaṁ na durlabham iti



Eight Heavenly Varṣas Vs. Bhārata-varṣa[24]

Among the nine divisions (*varṣas*) of Jambūdvīpa, the tract of land known as Bhārata-varṣa is understood to be the field of *karma* and the other eight *varṣas* are known as "heaven on earth," places to enjoy the remainder of pious acts after leaving Svarga. The other eight Varṣas are meant for very highly elevated pious persons, who after returning from the heavenly planets; enjoy the remaining results of their pious activities. There are three types of Svarga: heavenly (*divyaa-svarga*), earthly (*bhauma-svarga*) and subterranean (*bila-svarga*). These eight *varṣas* are the heavenly places on earth (*bhauma-svarga-pada-ni*).

tatrāpi bhāratam eva varṣaṁ karma-kṣetram anyāny aṣṭa varṣāṇi svargiṇāṁ puṇya-śeṣopabhoga-sthānāni bhaumāni svarga-padāni vyapadiśanti

Characteristics of the People in these Eight Varṣas

eṣu puruṣāṇām ayuta-puruṣāyur-varṣāṇāṁ deva-kalpānāṁ	Here like the *devatās* they live for ten thousand years according to earthly calculations.
nāgāyuta-prāṇānāṁ vajra-saṁhanana-bala	They have the bodily strength of ten thousand elephants, as sturdy as thunderbolts.
vayo-moda-pramudita-mahā-saurata-mithuna-vyavāyāpavarga-varṣa-dhṛtaika-garbha-kalatrāṇāṁ	The youthful duration of their lives is very pleasing and they have wives who conceive once in the last year of their lives.
tatra tu tretā-yuga-samaḥ kālo vartate	There the happiness is like that of the human beings who lived during Tretā-yuga.

Note: Because of the amount of material enjoyment, it is like that of Treta-yuga, since in Satya-yuga everyone is fixed in meditation and in Dvāpara and Kali there is more suffering.

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[24] SB 5.17.11–15: Description of 8 heavenly varsas (Bhārata-varṣa Vs Other Varṣas)

- SB 5.17.1-10: Descent, glories and Path of Ganga
- SB 5.17.11-15: Description of 8 heavenly varsas
- SB 5.17.16-24: Lord Siva's meditation of Lord Sankarshana, the deity of Ilāvṛta-varṣa

Mārkandeya Purāṇa says:

yāni kimpuruṣas tāni varṣanyaste dvijottama
te sumdijnani toyāni naiva vāryatra bhārate
vrksim svābhāviki deshyā toyottā manasi tathā

The waters in the eight varṣas, Kimpurusha and others come only from springs. Rain water appears only in Bhārata-varṣa. The eight varṣas are full of trees growing from these waters. The people living there are highly accomplished mentally (*manasi siddhis*). Their austerities also bring them accomplishment (*karmajā siddhi*). The trees there also bestow mystic perfections (*svābhāviki* or *deshya siddhi*) upon those who live under their shade. (Mar.Pur. 53.22-25)

na caitesu yugāvastā nādayo vyādayo na ca
punyāpunya samārambhā naiva tesu dvijottamā

There is no difference in the four yugas here. There are no physical or mental diseases here. There are no deeds which beget *punya* or *papa* (virtue or sin). (Mar.Pur. 53.26)

Opulences of these Eight Heavenly Varṣas:

- There they have hermitages and pleasant gardens splendid with trees and creepers whose branches are bent down by the abundance of fruits and flowers of all seasons.
- They sports in clear lakes resounding with the sounds of various groups of bees, swans, water hens, cranes and *kāraṇḍavas* excited by the fragrance of a variety of blooming lotuses growing in valleys between the mountains
- There, the chiefs of *devatās,* their glances and minds attracted by the playful glances and smiles of beautiful women, amuse themselves freely with ingredients supplied by their followers in water pastimes.
  *Though the heavenly residents enjoy superior facilities, but never forget their positions as eternal servants of Supreme Lord, but humans forget.

Expansions of the Supreme Lord in Nine Varṣas

In order to distribute His mercy to His devotees, the Supreme Lord, Nārāyaṇa, remains present in His various forms (*ātma-tattva-vyūhena*) in each of these nine tracts of land. It should be understood that these presiding deities of the various *varṣas* are deity forms of the Lord as described by Prāhlada in his prayers (ref SB 5.18.8).

Prayers offered by the Residents of different Varṣas of Jambūdvīpa:

Ilāvṛta-varṣa[25]: In this land the only male person is Lord Śiva. Due to Durgā Devi's curse no other male exists there. One who knows about Durgā's curse does not enter there. If any foolish man dares to do so, he turns into a woman[26]. Here, Lord Śiva resides with goddess Durgā, who is served by billion's of maidservants. Uttering different *mantras* in meditation, Lord Śiva worships Saṅkarṣaṇa, his source, the spiritual form in charge of ignorance, among the four forms of the Lord.

Note: the curse is effective only in the place between Kailāsa and Karavīra mountains in Ilāvṛta varṣa or continent (in the southern direction of Meru). Otherwise the existence of the demigods in Meru, earthly heavens and Lord's visit to various demigod abodes would be contradicted. (Commentary to SB 5.17.15)

oṁ namo bhagavate mahā-puruṣāya sarva-guṇa-saṅkhyānāyānantāyāvyaktāya nama iti

Lord Śiva prays: "I offer respects to Bhagavān, the great puruṣa, the revelation of all qualities, the unlimited, and the unfathomable."

Bhadrāśva-varṣa[27]**:** In Bhadrāśva, Bhadraśravā, the son of Dharmarāja, accompanied by his descendants and all the residents of the land, worships the dear form of Vāsudeva known as Hayaśīrṣa, the embodiment of *dharma,* approaching Him by intense concentration and by chanting different *mantras.*

oṁ namo bhagavate dharmāyātma-viśodhanāya nama iti.

Bhadraśravā with his followers pray: "I offer respects the Supreme Lord, to the embodiment of dharma, to the purifier of the ātmā."

Hari-varṣa[28]**:** In Hari-varṣa the Lord resides in the form of Narasiṁha. The great devotee Prahlāda the abode of qualities of a *mahā-puruṣa,* whose character purified the demons of his family, worships this pleasing form with uninterrupted, pure *bhakti* along with the inhabitants of this varṣa.

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[25] SB 5.17.16-24: Lord Siva's meditation of Lord Sankarshana

[26] Refer to SB 9.1 for further details

[27] SB 5.18.1-6: Prayer to Lord Hayaśīrṣa, deity of Bhadrāśva varṣa

[28] SB 5.18.7-14: Prayers to Deity of Hari- varṣa, Lord Narasimha

oṁ namo bhagavate narasiṁhāya namas tejas-tejase āvir-āvirbhava vajra-nakha vajra-daṁṣṭra karmāśayān randhaya randhaya tamo grasa grasa oṁ svāhā; abhayam abhayam ātmani bhūyiṣṭhā oṁ kṣraum

Prahlāda prays: I offer respects to the Supreme Lord Narasiṁha, the power among all powers. Please appear before me! O Lord with thunderbolt nails and teeth! Burn up the impressions of karma. Destroy ignorance! Please appear in my mind so that it becomes fearless.

Ketumāla-varṣa [29] : In Ketumāla, accompanied during the daytime by Prajāpati's sons (the predominating deities of the days: 36,000 = 100years * 360days) and accompanied during the night by Prajāpati's daughters (the predominating deities of the nights), Lakṣmīdevī worships the Lord in His most merciful form as Kāmadeva (Lord Pradyumna) by absorption in devotional service, and chants the following *mantras.*

oṁ hrāṁ hrīṁ hrūṁ oṁ namo bhagavate hṛṣīkeśāya sarva-guṇa-viśeṣair vilakṣitātmane ākūtīnāṁ cittīnāṁ cetasāṁ viśeṣāṇāṁ cādhipataye ṣoḍaśa-kalāya cchando-mayāyānna-mayāyāmṛta-mayāya sarva-mayāya sahase ojase balāya kāntāya kāmāya namas te ubhayatra bhūyāt

"I offer respects to Hṛṣīkeśa, to my husband, Kāmadeva, whose body, mind and intelligence have all excellent qualities, who is the master of the action senses, the knowledge senses and the *antaḥkaraṇa*, who has a perfect form like the full moon, who is the personified Vedas, personified food, personified nectar, the form of all forms, the energy of *citta*, the energy of the senses and energy of the body. I offer respects to him in this world and the next."

Ramyaka-varṣa[30]: In Ramyaka-varṣa, Vaivasvata Manu, the ruler of the land, even now worships in pure devotional service the dear form of Matsya who revealed Himself to Manu previously, by chanting various *mantras.* The Lord revealed Himself during Cākṣuṣa Manvantara when there was a flood.

oṁ namo bhagavate mukhyatamāya namaḥ sattvāya prāṇāyaujase sahase balāya mahā-matsyāya nama iti.

Manu prays: "I offer my respects to the incarnation made of pure sattva, the

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[29] SB 5.18.15-23: Prayer to the Deity of Ketumala varṣa, Lord Kamadeva/ Pradyumna

[30] SB 5.18.24-28: Prayers to Deity of Ramyaka varṣa, Lord Matsya

controller of the life air, the controller of the energy of the senses, mind, and body, the great fish."

Hiraṇmaya-varṣa[31]**:** In Hiraṇmaya-varṣa, the Supreme Lord Viṣṇu resides in the form of a tortoise (*bhagavān kūrma-tanuṁ*). Aryamā, the chief resident of Hiraṇmaya-varṣa, along with the other inhabitants of that land, worship this beloved form of the Lord by chanting different Vedic hymns.

oṁ namo bhagavate akūpārāya sarva-sattva-guṇa-viśeṣaṇāyānupalakṣita-sthānāya namo varṣmaṇe namo bhūmne namo namo 'vasthānāya namas te.

Aryamā prays: "I offer respects to the Supreme Lord, the tortoise (Lord Kūrma), a form of *śuddha-sattva*, having an invisible abode of Vaikuṇṭha, who has a huge, all-pervading body, and who acts as the support of all things."

Uttarakuru-varṣa[32]**:** In this tract of land the Supreme Lord lives as the Varāha incarnation (*bhagavān yajña-puruṣaḥ kṛta-varāha-rūpa*), the Lord of sacrifices. There, mother earth and all the other inhabitants worship him with unfailing devotional service and repeatedly chanting different Upaniṣad *mantras.*

oṁ namo bhagavate mantra-tattva-liṅgāya yajña-kratave mahā-dhvarāvayavāya mahā-puruṣāya namaḥ karma-śuklāya tri-yugāya namas te

I offer respects to the Supreme Lord, known in truth by mantras, who is the form of sacrifices with and without animals, who is the limbs of great sacrifices, who is the great puruṣa, who has a form of śuddha-sattva even in his actions, and who appears in three yugas visibly

Kimpuruṣa-varṣa[33]**:** In Kimpuruṣa-varṣa the great devotee Hanumān is always engaged along with the inhabitants of that land in devotional service to Lord Rāmacandra, the elder brother of Lakṣmaṇa and dear husband of Sītādevī. Hanumān, along with Arṣṭiṣeṇa and Sugriva, constantly hears with rapt attention the most auspicious glories of his master, sung by Gandharvas. Hanumān along with others worships the Lord by chanting various *mantras.*

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[31] SB 5.18.29-33: Prayer to Deity of Hiranmaya varṣa, Lord Kurma

[32] SB 5.18.34-39: Prayer to Deity of Uttarakuru varṣa, Lord Varāha (Boar incatnation)

[33] SB 5.19.1-8: Prayer to the Deity of Kimpurusa-varṣa, Lord Rama

oṁ namo bhagavate uttamaślokāya nama ārya-lakṣaṇa-śīla-vratāya nama upaśikṣitātmana upāsita-lokāya namaḥ sādhu-vāda-nikaṣaṇāya namo brahmaṇya-devāya mahā-puruṣāya mahā-rājāya nama iti

Hanumān prays: "I offer respects to the Supreme Lord, most praiseworthy, who is the personification of the best markings, qualities and vows. I offer respects to the Lord who accepted teachings from others, and who is worshipped by all types of people. I offer respects to the Lord who makes all good qualities the most excellent. I offer respects to the Lord worshipped by the brāhmaṇas, who is the great personality, the great king."

Bhārata-varṣa[34]**:** In Bhārata-varṣa, the Lord in the form of Nara-Nārāyaṇa, whose glories are inconceivable, to favor the *jñānīs,* mercifully performs austerities till the end of the *yuga*, which include realization of *ātmā,* abundant *dharma*, knowledge, renunciation, powers, sense control and freedom from false ego. The great sage Nārada muni, while teaching Sāvarṇī Manu (8th Manu) descriptions of realization of God by Sāṅkhya and *yoga* as spoken by the Lord, worships Nara-Nārāyaṇa with the greatest *bhakti* along with the inhabitants of Bhārata-varṣa, the followers of *varṇāśrama,* by chanting various Vedic *mantras.*

oṁ namo bhagavate upaśama-śīlāyoparatānātmyāya namo 'kiñcana-vittāya ṛṣi-ṛṣabhāya nara-nārāyaṇāya paramahaṁsa-parama-gurave ātmārāmādhipataye namo nama iti.

Nārada prays: "I offer respects to the Lord who has controlled senses, who has no attachment to persons not interested in self-realization (*ātmā*), who is the treasure for persons possessing nothing, who is the best of sages, the supreme guru of the paramahaṁsas, and the leader of the ātmārāmas."

Summary the prayers offered by the residents of 9-Varṣas of Jambūdvīpa

Varṣa	Ruler (9 sons of Āgnīdhra)	Side	Mountain	Presided by	Presiding Deity
Ilāvṛta-varṣa	Ilāvṛta	Middle	Sumeru	Lord Siva	Lord Sankarshana
Bhadrāśva-varṣa	Bhadrāśva	East	Gandhamādana	Bhadraśravā	Lord Hayaśīrṣa (Lord Hayagriva)
Hari-varṣa	Hari-varṣa	South	Niṣadha	Prahlada	Lord Narasimha
Ketumāla-	Ketumāla	West	Mālyavān	Goddess Lakṣmī	Lord Kamadeva (Lord

[34] SB 5.19.9-15: Prayers to the Deity of Bhārata-varṣa, Lord Nara-Narayana Rsi

varṣa					Pradyumna)
Ramyaka-varṣa	Ramyaka	North	Nīla	Vaivasvata Manu	Lord Matsya
Hiraṇmaya-varṣa	Hiraṇmaya	North	Śveta	Aryama	Lord Kurma
Kuru-varṣa	Kuru	North	Śṛṅgavān	Earth	Lord Varāha
Kimpuruṣa-varṣa	Kimpuruṣa	South	Hemakūṭa	Hanuman with other monkeys	Lord Rama
Ajnābha-varṣa	Nābhi	South	Himālaya	Narada	Lord Nara-Nārāyana Rsi

Qualities of Bhārata-varṣa[35]

After describing the prayers offered by the residents of the nine varṣas of Jambūdvīpa to their respective worshipable deities, Sukadeva Gosvami now describes the mountains and rivers of Bhārata-varṣa and then describes its specific divisions and characteristics.

Descendents of Āgnīdhra, the son of Priyavrata Mahārāja[36]

Āgnīdhra, the first son of Priyavata Maharaja's seven remaining sons, became the ruler of Jambūdvīpa as desired by his father. Then Āgnīdhra gave the ruler ship of the nine varṣas of Jambūdvīpa to his nine sons respectively, naming them after the names of his sons. Nābhi, his eldest son became the ruler of Ajnābha varṣa, the southernmost tract of land. Nābhi, son of King Āgnīdhra, desiring a son similar to the Lord, worshipped the Supreme Lord with great attention along with his wife Merudevī. Being pleased with King Nābhi's worship and to prove the words of *brāhmaṇas* should not be false, since exalted *brāhmaṇas* represent Lord's mouth, Lord agreed to appear in His partial form as the son of King Nābhi. Wanting to please King Nābhi, and desiring to show the path of the naked performers of austerity, who follow scripture and remain celibate, the Lord then appeared with a body of *śuddha-sattva* in the womb of Merudevī. Because of His excellent body, praised by many poets, His mental and physical strength, His beauty, fame, influence and courage, His father called Him as Ṛṣabha. He ruled the kingdom as instructed

[35] SB 5.19.16-20: Description of Bhārata-varṣa, the suthernmost island of Jambūdvīpa

[36] SB 5.2-5: Descendents of Āgnīdhra (5.2.17-23), Nabhi (5.3.1-3), Ṛṣabhadeva (5.4.8-13) & Bharata Maharaja (5.7.1-4)

by the *brāhmaṇas* and He begot in his wife Jayantī, given by Indra, a hundred sons similar to himself. Among His sons Bharata, the eldest, was a great *yogī*, having excellent qualities. Because of him, this Ajnābha-varṣa became known as Bhārata-varṣa. It is said in SB 11.2.17-18

teṣāṁ vai bharato jyeṣṭho nārāyaṇa-parāyaṇaḥ
vikhyātaṁ varṣam etad yan- nāmnā bhāratam adbhutam
sa bhukta-bhogāṁ tyaktvemāṁ nirgatas tapasā harim
upāsīnas tat-padavīṁ lebhe vai janṛnabhis tribhiḥ

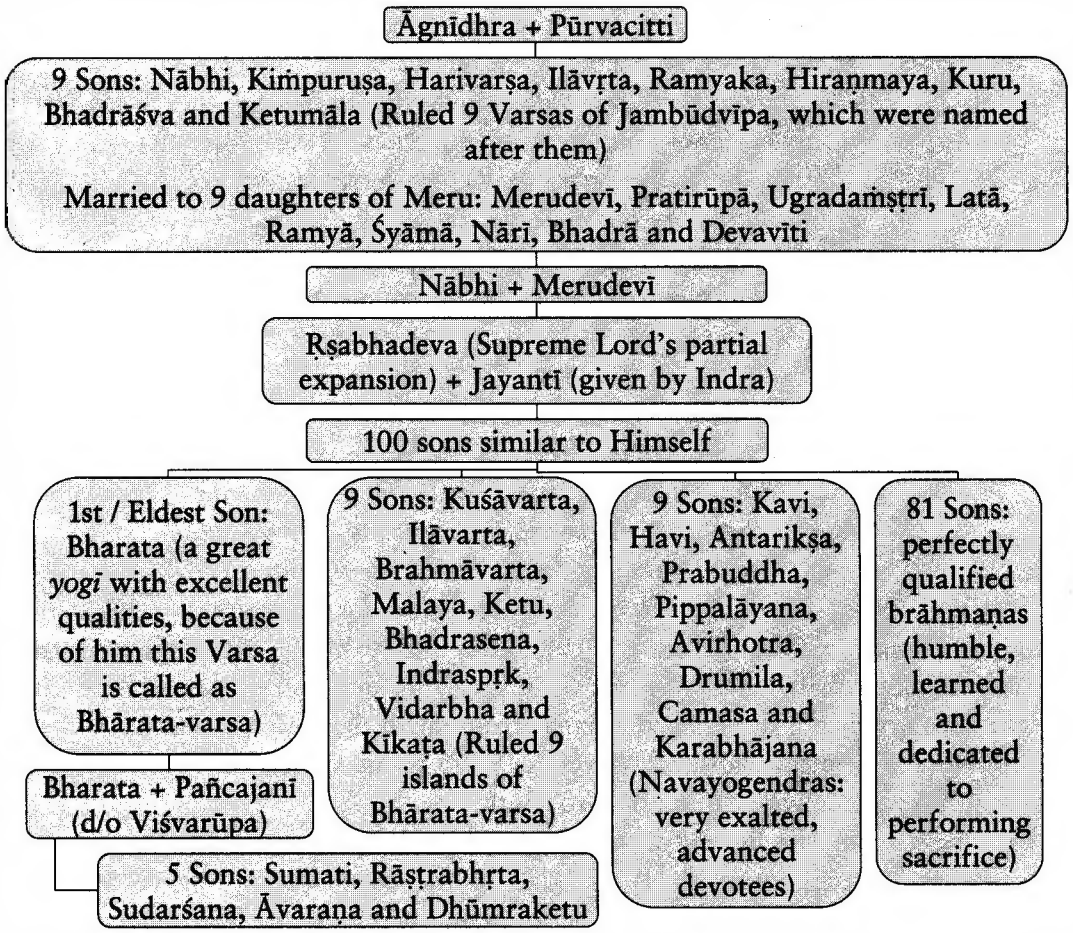
Of the one hundred sons of Ṛṣabha, the eldest, Bharata, was completely devoted to Nārāyaṇa. The famous varṣa became the astonishing Bhārata-varsā because of his name. King Bharata rejected this material world, considering all types of material pleasure temporary and useless. Leaving his house, he worshiped the Lord by severe austerities and attained the abode of the Lord after three lifetimes.

Younger than Bharata, but older than the other 90 sons (9: nava-mahā-bhāgavatāḥs, 81: karma-viśuddhā brāhmaṇās) were the sons named Kuśāvarta, Ilāvarta, Brahmāvarta, Malaya, Ketu, Bhadrasena, Indraspṛk, Vidarbha and Kīkaṭa became the rulers of the nine islands of Bhārata-varṣa. It is further stated in SB 11.2.19-22

teṣāṁ nava nava-dvīpa-patayo 'sya samantataḥ
karma-tantra-praṇetāra ekāśītir dvijātayaḥ
navābhavan mahā-bhāgā munayo hy artha-śaṁsinaḥ
śramaṇā vāta-rasanā ātma-vidyā-viśāradāḥ

Nine of the remaining sons of Ṛṣabhadeva became the rulers of the nine islands of Bhārata-varṣa and they exercised complete sovereignty over this. Eighty-one sons became brāhmaṇas and helped initiate the Vedic path of karma. The nine remaining sons of Ṛṣabha, greatly fortunate sages, discerning the highest truth (*artha-śaṁsinaḥ*), making great endeavor for realizing Paramātmā, and were learned concerning Paramātmā. They wandered about naked (*vāta-rasanā*).

Among the sons of Ṛṣabha, nine became rulers of nine islands such as Brahmāvarta within Bhārata-varṣa. Eighty-one sons became brāhmaṇas who started the path of karma. The nine sages *(nava-yogendras)* wandered the earth while seeing the entire universe, with all its gross and subtle objects, as a manifestation of the Supreme Lord and as non-different from Paramātmā.



In the tract of land known as Bhārata-varṣa, just as in Ilāvṛta-varṣa, there are many mountains and hills, with many large and small rivers flowing from their slopes. The inhabitants of Bhārata-varṣa contact these pure rivers by their minds, body and utterance of their names to get purified of themselves.

bhārate 'py asmin varṣe saric-chailāḥ santi bahavo; anye ca śata-sahasraśaḥ śailās teṣāṁ nitamba-prabhavā nadā nadyaś ca santy asaṅkhyātāḥ; etāsām apo bhāratyaḥ prajā nāmabhir eva punantīnām ātmanā copaspṛśanti;

Specific Qualities of Bhārata-varṣa: In Bhārata-varṣa, many destinations (heavenly, human and hellish) are prescribed for all people, because people take birth according to actions in *sattva, rajas* and *tamas*. All these destinations are

prescribed for the self, according to the quality of their actions, as indicated in the Vedas. Liberation is then achieved. (SB 5.19.19)

asminn eva varṣe puruṣair labdha-janmabhiḥ
śukla-lohita-kṛṣṇa-varṇena svārabdhena karmaṇā
divya-mānuṣa-nāraka-gatayo bahvya ātmana
ānupūrvyeṇa sarvā hy eva sarveṣāṁ vidhīyante
yathā-varṇa-vidhānam apavargaś cāpi bhavati

This takes place only in Bhārata-varṣa, and only in an area, whose dimension is 1000 yojanas.

Viṣṇu Purāṇa says

bhāratasyāsya varṣasya nava-bhedān niśāmaya
indra-dvīpaḥ kaśeruś ca tāmra-varṇo gabhastimān
nāga-dvīpas tathā saumyo gāndharvas tv atha vāruṇaḥ
ayaṁ tu navamas teṣāṁ dvīpaḥ sāgara-saṁbhṛtaḥ
yojanānāṁ sahasrantu dvīpo 'yaṁ dakṣiṇottarāt

Hear about the nine division of Bhārata: Indradvpia, Kaśeru, Tāmravarṇa, Babhasimān, Nagadvpiā, Saumya, Gandarva, Vāruṇa, and the ninth, bordering the ocean, which is a thousand yojanas from north to south. (Vi.Pu. 2.3.6-7)

Note: The total width of Bhārata is 9000 yojanas. Each division would be 1000 yojanas wide from north to south.

A person would be limited to only one of these divisions for his life. *Sāgara-saṁbhṛtaḥ* means situated at the edge of the ocean, according to Śrīdhara Svāmī. Though the ninth division is not mentioned by name it is understood to be called Navadvīpa (It is sometimes called as Bhārata-varṣa or Bhārata-khanda or Sudarsanadvīpa or Kumārakādvīpa). This place (ninth division, Bhārata-varṣa) is further described in the same work:

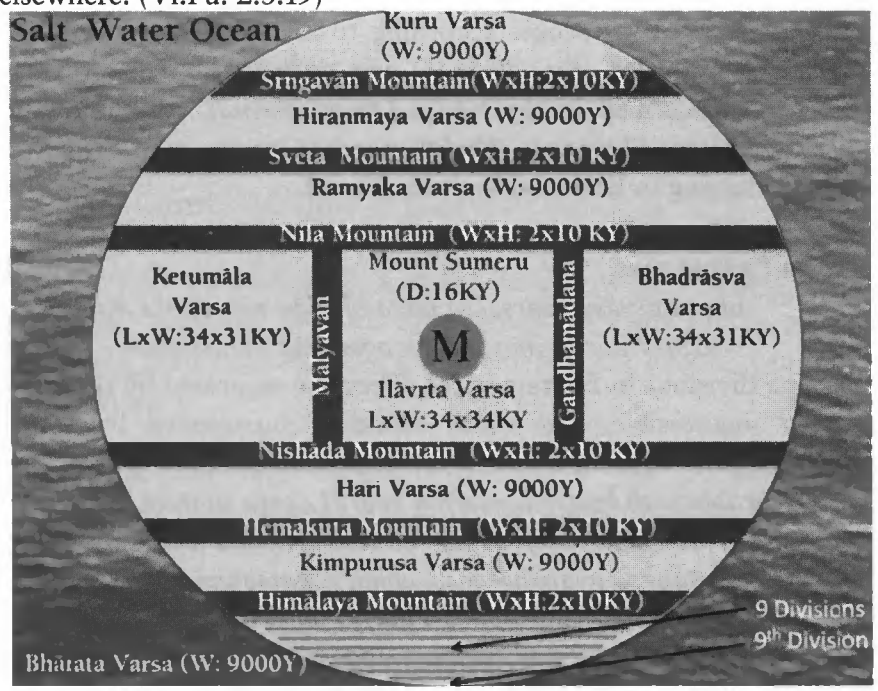
pūrve kirātā yasyānte paścime yavanāḥ smṛtāḥ
pūrva-deśādikāś caiva kāma-rūpa nivāsinaḥ
oṭrāḥ (puṇḍrāḥ) kaliṅgā magadhā dākṣiṇātyāś ca kṛtsnaśaḥ
mārukā mālavāś ca ity ādi

In the east are the Kirātas, in the west are the Yavanas. The inhabitants in the east are the residents of Kāmarūpa, Orissa, Kaliṅga, Magadha. The inhabitants in the south are Marukas and Mālavas. (Vi.Pu. 2.3.8 & 2.3.15)

catvāri bhārate varṣe yugānyatra mahāmune
kṛtaṁ tretādvāparañca kaliścānyatra na kvacit

In Bhārata-varṣa there are Satya, Treta, Dvāpara and Kali yugas, which do not

exist elsewhere. (Vi.Pu. 2.3.19)



In **Siddhānta-śiromaṇi**, Chapter One (Golādhyāya), in the Bhuvana-kośa section, the nine khaṇḍas are mentioned as follows: *(ref to CC Antya Lila 2.10)*

aindraṁ kaśeru sakalaṁ kila tāmraparṇam
anyad gabhastimad ataś ca kumārikākhyam
nāgaṁ ca saumyam iha vāruṇam antya-khaṇḍaṁ
gāndharva-saṁjñam iti bhārata-varṣa-madhye

"Within Bhārata-varṣa, there are nine khaṇḍas. They are known as (1) Aindra, (2) Kaśeru, (3) Tāmraparṇa, (4) Gabhastimat, (5) Kumārikā, (6) Nāga, (7) Saumya, (8) Vāruṇa and (9) Gāndharva."

Vāyu Purāṇa further says

bhāratasyāsya varṣasya nava bhedān nibodhata
sāgarāntaritā jñeyās te tv agamyāḥ parasparam

Hear about the nine divisions of Bhārata. They border the ocean and one cannot go from one to the other.

Many goals are undertaken by the individual according to prescribed actions in

sattva, rajas and *tamas (śukla-lohita-kṛṣṇa-varṇena)*, because (*hi*) all these goals will manifest for all people according to the qualities of action (*yathā varṇa-vidhānam*), caused by *dharma* and *adharma*. This takes place unavoidably, as prescribed in the Vedas (*ānupūrveyṇa*). And if one cannot accomplish liberation, liberation *(bhakti)* appears on its own. The word *ca* indicates that merging in Brahman is also achieved.

Mārkandeya Purāṇa says

bhāratasyāsya varṣasya nava bhedān nibodhata me
sāgarāntaritā jñeyās te tv agamyāḥ parasparam

There are nine divisions in Bhārata-varṣa. They are separated by the ocean and are mutually inaccessible. The ninth island is surrounded by water and measuring 1000 *yojanas* from south to north. (Mar.Pur. 54.5)

brāhmanah ksatriyā vaisyāh sudrāś canta sthitam dvija
ijjyādhyāya vānijyādyau karmabhih krtha pāvanah
teśam sa vyavahāraś ca ebhih karmābhirishyate
svargvpavarga prāptiś ca punya pāpo ca vai sadāh

There lives brahmanas, ksatriyas, vaisyas and sudras. Worship, studies, and trade are the activities here which cause people to interact with one another. One can attain svarga and moksa from here. There is the reckoning of punya and papa also. (Mar.Pur. 54.7-9)

te vimisram janapadā mlecchas cāryatra bhāgajah

People living here are a mixture of both *mlecchas* and *aryas*. They predominate in certain parts. (Mar.Pur. 54.15)

tad etat bhārata-varṣah sarva bijam dvijottamah
brahmatvam amres catvam devatvam martyatām tathā
prayānti karma bhumir iha na anya lokeshu vidyate
devānām 'pi viprayem sadā esa manoratah

This Bhārata-varṣa can bestow everything – the position of the creator (Brahmā), the position of the leader among the demigods, demigod hood, and even human birth also. As well it can bestow everything else as per *punya* and *pāpa* earned here. This is known as *karma-bhūmi,* for it can bestow everything according to *punya* and *pāpa.* Karma is not enjoined anywhere else other than Bhārata-varṣa. (Mar.Pur. 54.60-62)

What is the nature of this liberation and when does it occur?
That liberation, whose essential nature is unmotivated *bhakti-yoga* to the Lord full of qualities, takes place by destruction of the knot of ignorance, which causes various material goals, when there is association with devotees of the Lord.

The Demigods Glorify the Land of Bhārata-Varṣa[37]
The *devatās* glorify the land of Bhārata-varṣa, because attaining birth there is suitable for service to Mukunda. And the *devatās* want to know the kind of pious acts these residents perform or whether the Lord simply pleased with them, by which they attained birth there, whereas the *devatās* can only desire this, but not attain it.

The devatās further say: "What is the use of our performing difficult sacrifices, vows of austerity and acts of charity on earth, or attaining transient residence in heaven, where there is no remembrance of the lotus feet of Nārāyaṇa or where the memory is carried away by excessive enjoyment of the senses? (SB 5.19.22-24)

kiṁ duṣkarair naḥ kratubhis tapo-vratair dānādibhir vā dyujayena phalgunā na
yatra nārāyaṇa-pāda-paṅkaja-smṛtiḥ pramuṣṭātiśayendriyotsavāt

A moment's stay in Bhārata is better than living for a *kalpa* on Brahma-loka, since on Brahma-loka one must take birth again. Those who fix their minds on the Lord in Bhārata in a body subject to death, on giving up all pious and impious acts, attain Vaikuṇṭha.

kalpāyuṣāṁ sthānajayāt punar-bhavāt kṣaṇāyuṣāṁ bhārata-bhūjayo varam
kṣaṇena martyena kṛtaṁ manasvinaḥ sannyasya saṁyānty abhayaṁ padaṁ
hareḥ

Even Brahmaloka should not be accepted as a residence if there are no rivers of sweet topics about the Lord, no devotees who take shelter of those topics, and no great festivals or sacrifices of chanting the name of the Lord.

na yatra vaikuṇṭha-kathā-sudhāpagā na sādhavo bhāgavatās tadāśrayāḥ
na yatra yajñeśa-makhā mahotsavāḥ sureśa-loko 'pi na vai sa sevyatām

Those who attain birth as humans in Bhārata, who are fully endowed with

[37] SB 5.19.21-28: Demigods glorify Bhārata-varṣa

proper sense *devatās* (controlling deities of the senses), senses, and sense objects, but who do not endeavor for *bhakti-yoga* are again bound up, like birds trapped by a hunter. The Lord is one only, but called by different names. Though complete in himself, he is the master of blessings, and thus he happily accepts the pure offering of sacrifice along with materials, methods and *mantras* which are offered to various *devatās*, because of the *bhakti*. The Lord certainly gives desired objects to devotees who request them, but He does not give those objects in such a way that the devotee will ask again after finishing their enjoyment. In other words, the Lord gives His lotus feet, which include all desirables, to those worshippers who do not even desire them. If we have pious credits remaining from performing sacrifices, chanting the Vedas, or performing charitable acts, may those credits give us birth with remembrance of the Lord in Bhārata, since the Lord distributes happiness to His devotees in Bhārata-varṣa."

In the fourth Canto of *Śrīmad-Bhāgavatam* Nārada Muni speaks to the King Prācīna Barhiṣat[38]: "O King! There was a very famous king named Purañjana. Searching for a place to live he wandered the earth. Not finding a suitable place (*Śaraṇam* a place to live, a place for enjoyment, the body), he became depressed. Desiring objects for attaining enjoyment, he thought that not all the houses (the bodies of other species) on earth were suitable. Though attaining material happiness everywhere in all the houses, he did not consider it enough for satisfying his desires (since cow bodies and other were not suitable for enjoyment). Once, the King saw on the southern ridge of the Himalayas a city with nine gates, which he understood had good qualities."

sa ekadā himavato dakṣiṇeṣv atha sānuṣu
dadarśa navabhir dvārbhiḥ puraṁ lakṣita-lakṣaṇām

The meaning is this: "In Bhārata, in the land of karma (the 9th division of Bhārata-varṣa), as a result of performing actions to get human body, he saw a human body (*puram*) which had visibly good qualities. This means he was not blind or born in a low caste. Thus attainment of human body, with many good qualities, in Bhārata-varṣa is suitable for obtaining enjoyment and to attain liberation eventually."

In the seventh Canto of *Śrīmad-Bhāgavatam* Prahlāda Mahārāja speaks to his

[38] SB 4.25.10-13: Puranjana's search for a suitable house (Jiva's search for a suitable body)

classmates about the importance of practicing *bhāgavata-dharma.*

*kaumāra ācaret prājño dharmān bhāgavatān **iha***
durlabhaṁ mānuṣaṁ janma tad apy adhruvam arthadam

Prahlāda said: The wise person should practice *bhakti-yoga* in his life from youth. Human birth is rare and transient but effective for *bhakti.*

At kaumāra age (1-5 years) itself one should begin the practice of *bhāgavata dharma,* which is related to the Lord: hearing and chanting, having taken birth in the land of Bhārata (***iha***). This is the special prerogative of the land of Bhārata-varṣa. (SB 7.6.1)

Structure of Bhārata-varṣa

Śrī Thiruvenkata Rāmānuja Jīyar, a Śrī-Vaiṣṇava saint, while reading from the Bhīshma parva[39] section of Mahābhārata, came across the description of the shape of this Bhārata-khanda. Based on those slokas he drew the map of this tract of land. Wherein it says that, "As Dhritarashtra desired to hear about the different countries from which the warriors have assembled there in the battle field and also about the rivers, forests, mountains, provinces and dimensions of the whole earth, Sanjaya replies[40] that the earth is the foremost of all the elements for it contains all the properties of other elements, such as sound, touch, form, taste and fragrance. Living entities, whose physical body is made of these five elements, are seen throughout the universe. But the men on the earth try to understand their proportion by their reasoning ability.

tatra tatra hi dṛśyante dhātavaḥ pāñca bhautikāḥ ।
teṣāṁ manuṣyās tarkeṇa pramāṇāni pracakṣate ॥*6.10*॥

Those matters that are inconceivable can never be solved by ones reasoning, for what is above material nature is an indication that it is inconceivable.

acintyāḥ khalu ye bhāvā na tāṁs tarkeṇa sādhayet ।
prakṛtibhyaḥ paraṁ yattu tadacintyasya lakṣaṇam ॥*6.11*॥

Having stated its inconceivable nature, Sanjaya begins to describe the shape and different details of this earth. "This tract of land on which we are residing (the *Bharata-khanda,* 9th division of Bhārata-varṣa) is called *Sudarshanadvīpa,* since it looks beautiful to the eyes of the onlookers. Being circular it looks like

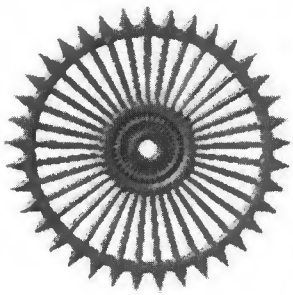
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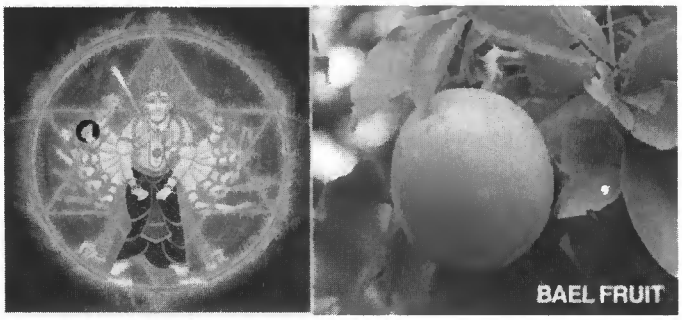
[39] Chapter 6, Jambu-khanda vinirmāna parva, Sri Bhishma-parva, Mahābhārata

[40] These verses can be seen from the 6th chapter of the Bhandarkar Oriental Research Institute (BORI) edition of Mahābhārat: BORI 6.12-17

the chariot wheel or the disc of the Supreme Lord for it is connected with the cyclic time (*satya-treta-dvārapara-kali*)."

sudarśanaṁ pravakṣyāmi dvīpaṁ te kurunandana ।
parimaṇḍalo mahārāja dvīpo'sau cakra saṁsthitaḥ ॥ *6.12* ॥





Note*: Parimaṇḍala* means a globe, a sphere, orbit or circumference as per Sanskrit dictionary. It also means round, circular or globular. *Sudarshana-dvīpa* is considered to be in the form of a globe like a Bael fruit.

It consists of many rivers associated with different water bodies, mountains that look like masses of clouds, different cities and many well established provinces.

nadījala praticchannaḥ parvataiś cābhra saṁnibhaiḥ ।
puraiś ca vividhākārai ramyair janapadais tathā ॥*12.13*॥

It is full of trees covered with flowers and fruits and with crops of various kinds and with many other rich things. It is surrounded on all sides with Salt Ocean.

vṛkṣaiḥ puṣpaphalopetaiḥ sampanna dhanadhānyavān।
lāvaṇena samudreṇa samantāt parivāritaḥ ॥*6.14*॥

As a person can see one's own face in a mirror, the form of this *Sudarshana-dvīpa* is seen from the mirror like moon.

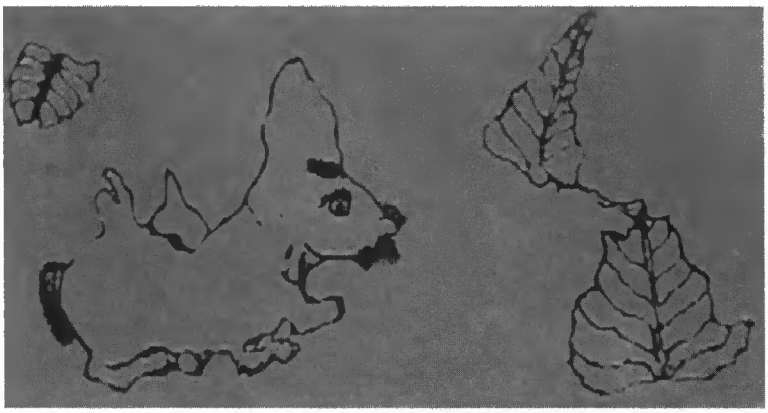
yathā ca puruṣaḥ paśyed ādarśe mukhamātmanaḥ ।
evaṁ sudarśana dvīpo dṛśyate candramaṇḍale ॥ *6.15* ॥

It is seen in two parts (*dviraṁśe*). First half of this island (*dvīpa*) appears as set of Peepal leaves (*pippalas,* sacred fig tree leaves) and the second half appears like a huge rabbit (*śaśo mahān*), and they are surrounded by different varieties

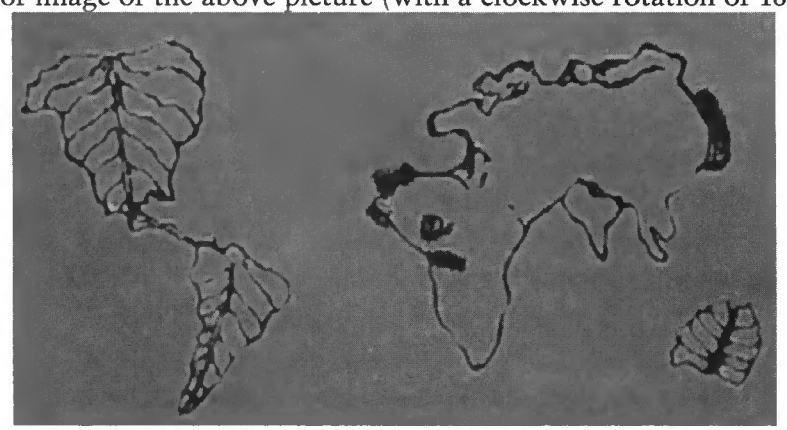
of plants and vegetation. Besides these portions, the rest of this island is covered with water.

dviraṁśe pippalas tatra dviraṁśe ca śaśo mahān |
sarvauṣadhi samāvāpaiḥ sarvataḥ paribṛṁhitaḥ || *6.16* ||
āpastato'nyā vijñeyā eṣa saṅkṣepa ucyate || *6.17* ||[41]

Śrī Thiruvenkata Rāmānuja Jīyar[42] sketched a drawing of one big Rabbit and Peepal tree leaves according to the description given above in the above Vedic texts.



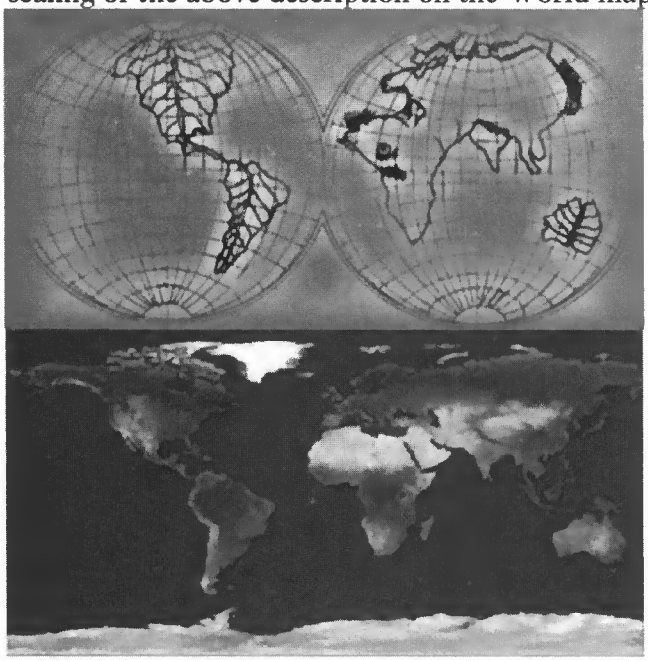
The mirror image of the above picture (with a clockwise rotation of 180°).



[41] In the next chapter Sanjaya begins to describe the different divisions of Jambūdvīpa.

[42] Born in Madhuramangalam, he was named Krishnan by his parents Ragavacharyar and Janaki ammal, since he was born on the same thirunakshathram as Lord Krsna and Periyavachan Pillai. He was born in the glorious Srivathsa lineage in the year 1805 CE. He lived for a short time (77 years) in this leela vibhuthi and ascended to paramapadham on Vishu varsham (in the year 1882 CE) thai month Krishna paksham thrayodhasi thithi.

Projection and scaling of the above description on the World map:



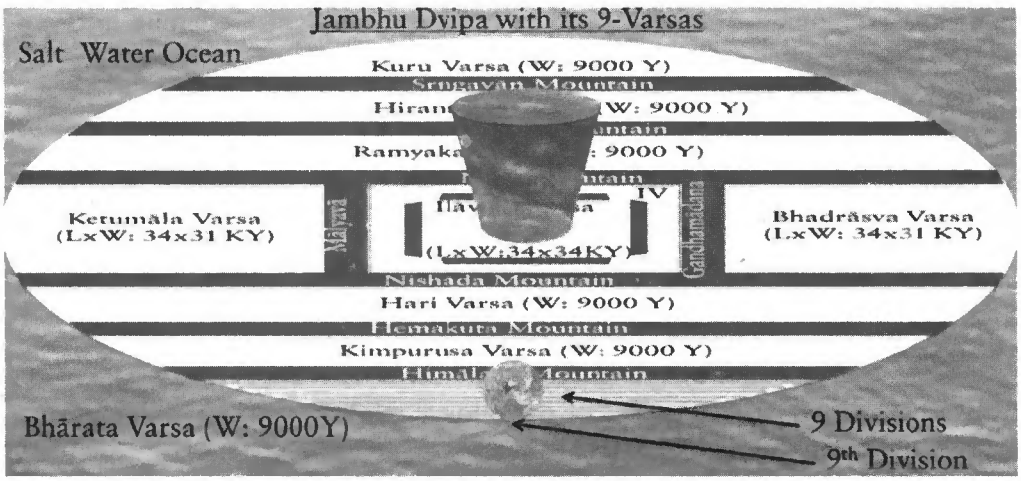
As per the above depiction it can be seen that the 9th division of Bhārata-varṣa, which is 1000 yojanas in diameter and called as Bhārata-varṣa or Bhārata-khanda or Sudarshana-dvīpa or Kumārakādvīpa, is the tract of land of our inhabitance and experience (our earth).

As per Surya Siddhanta 12.53-54 the shape of the earth can be understood to be spherical: "As the shape of earth is spherical (*vartula*), people in each place think themselves to be uppermost. But as the globe of earth is situated in ether, what then is its upper and lower side? Because of the smallness of their size, people see this earth as circular in all four directions."

sarvatraiva mahīgole svasthanam uparisthitam ।
manyante khe yato golastasya kvorddham kvavapy adhah ॥12.53॥
alpakāyatayā lokāh svasthānāt sarvatomukham ।
paśyanti vṛttāmapyetām cakrākāram vasundhaām ॥12.54॥

The diameter of the earth as per the scientific calculation:	
The earth's mean diameter	~12,742 km
Diameter of the earth at the equator	12,756.2 km
Diameter of the earth between the two poles	12,713.6 km

The diameter of the earth as described in SB 5.19.19:	
Diameter of Bhārata-khanda (the 9th division of Bhārata-varṣa)	1000 yojanas (9th part of 9000 yojanas)
Diameter in miles (1 yojana = 8 miles)	8000 miles
Diameter in km (1 mile = 1.609 km)	12,872 km (1000x8x1.609)



The diameter of the earth as per Surya Siddhanta chapter 1 verse 59: The radius of the earth at a point on the surface is the distance from the center of the earth to the mean sea level at that point. In chapter 1 verse 59 of Surya Siddhanta the diameter of the earth is given to be 1600 Yojanas. The diameter of a sphere as per Surya Siddhanta is equal to the distance measured from one end of the sphere to the other end over the surface. That is equal to half of the total circumference of a sphere or a circle ($C = 2\pi r$, $D = \frac{C}{\pi} = 2r$). Thus the diameter of earth's sphere is $D = \frac{C}{\pi} = \frac{1600*2}{\pi} = 1019$ yojanas.

Note: Comparing all the above numbers, it can be seen that they are closely matching to each other. This indicates that details presented by the bonafide scriptures are not mythical but can be verified by the modern scientific calculations.

Sankalpa mantra and its meaning

Just to show the connection of the above description of Jambūdvīpa, Bhārata-varṣa or Bhārata-khanda to the Vedic Hymns generally chanted at the

beginning of any auspicious ceremony or samskāra, a section of that is presented below. This Vedic hymn is commonly known as the Sankalpa Mantra. This sankalpa mantra is generally chanted before initiating any samskāra or auspicious ceremony, so that the controlling deities and the sages know the time and place at which the samskāra or auspicious ceremony is performed. This also indicates that the performer has taken an oath to righteously complete the ceremony to properly fulfill his religious duties as ordained. A small section of this sankalpa mantra, which is relevant to the context, is given below.

Sankalpa Mantra recited for January 14th, 2017, Saturday: (Durmukhi samvathsare, Daksināyane, Hemanta rithou, Pausa māse, Kṛṣṇa Pakshe, dvitiyām, Sani vāsare)

Srī govinda govinda govinda!
asya Srī-bhagavata mahā purushasya!
vishnor agjnayā pravartamānasya!

Adhya brahmana, (from the year of Brahma's birth)
Dvitīya-parārdhe, (in the 2nd half, 51st year)
Srī-Svetha-varāha-kalpe, (in the day of Brahmā known as Sveta-Varaha)
Vaivasvatha-manvantare, (in the reigning period of the current 7th Manu)
Ashta vimsati tame, (in the 28th maha-yuga of the manavantara)
Kaliyuge, prathame pāde, (in the first quarter of this kali-yuga)

Jambūdvīpe (in Jambūdvīpa, the central most island of the Bhū-maṇḍala)
Bhārata-varṣe, (the southernmost tract of land of Jambūdvīpa)
Bhārata-khande, (earth, 9th division of Bhārata varsha)
Meroh dakshine pārsve, (to the South of mount Meru)
Godhavari daksina tīre, (the land on the south of Godhavari River)
Mama svagruhe... (in our home)

Asmin vartamāne vyāvahārike, (in the current period now reigning)
Prabhavādi shashti samvatsarānām madhye, (among 60 years cycle starting from Prabhava)
Durmukhī nāma samvathsare, (Durmukhī – 8th April 2016 to 28th March 2017)
Daksināyane, (daksināyana period: 16th July 2016 to 14th Jan 2017, 7:50AM)
Hemanta rithou, (winter: Dhanu & Makara / Sagittarius & Capricorn or Margaśirśa & Pusya)

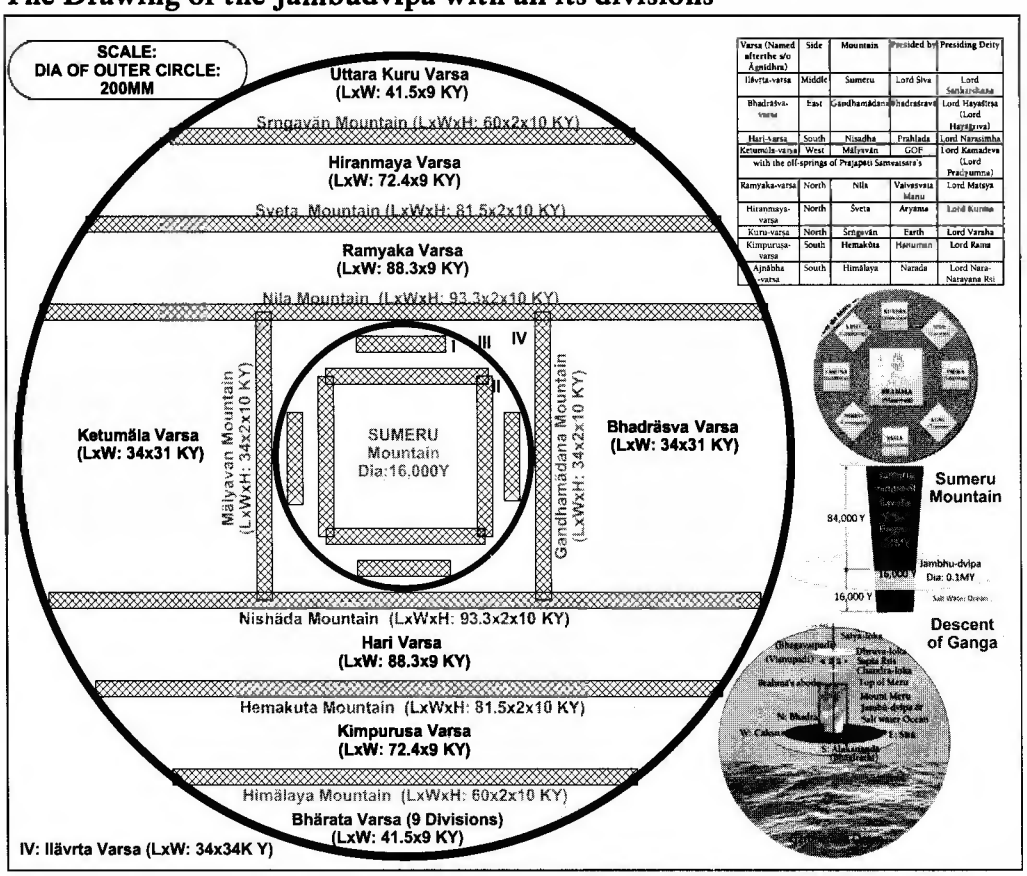
Pauśa māse (Dhanur māse), (Dhanu - Solar month 16th December 2016 - 14th January 2017; Pauśa – Lunar month, 30th Dec 2016 - 27th January 2017)
Kṛṣṇa Pakshe, (dark fortnight, purnima to amavasya)
Dvitiyām subha thithou, shani vāsara, (dvitiya or 2nd tithi and Saturday)
Ashlesha nakshathra (uttarāshāda) yukthaayaam, (ashlesha: sun star, uttarāshāda: moon star)

Srī-vishnu-yoga, Srī-vishnu-karana, Subha-yoga, Subha-karana,
Yevam guna, viseshana viśishthāyām, asyām dvādasyām Subha-tithou,
Srī-Bhagavad-ājnayā, Sri Bhagavat-kainkarya-rūpam ||

*Starting with "Jambūdvīpe" until "mama svagruhe" describe the location w.r.t the description of Jambūdvīpa discussed till now.

**Time details mentioned above in the Sankalpa mantra will be correlated in Chapter Four after describing the movements of the Planets.

The Drawing of the Jambūdvīpa with all its divisions



Remaining Six Islands of the Bhū-maṇḍala

After describing the divisions or varṣas in Jambūdvīpa, Sukadeva Gosvami now begins to describe the divisions of varṣas of the remaining six islands of the Bhū-maṇḍala with their mountains and rivers. This section will present the dimensions, specific characteristics and shapes of the six islands beginning with the island of Plakṣa. Also presents a brief description of the people, their prime occupations, their process of worship and their worshippable Deity.

Characteristics of the people & the common aspects of these Six Dvīpas[43]

- All the dvīpas have a tree or mountain, of special significance, after which the island gets its name.
- The first rulers of all these dvīpas are the sons of Priyavrata.
- 1st Dvīpa: 9 *varṣas*, 9 mountains and 4 rivers;
- 2-6 dvīpas: 7 *varṣas*, 7 mountains and 7 rivers;
- 7th dvīpa: 2 *varṣas*, 1 mountain and 2 rivers;
- They are named according to the names of their sons (9, 7 & 2).
- Giving the ruler ship to their respective sons, they retired from active life for full engagement in devotional service.
- In the middle five dvīpas there are seven mountains and seven rivers, with beautifully decorated hermitages, enormous lakes with flowers and birds.
- In the 7th dvīpa there is only one mountain in the middle of the island. This mountain divides the land into two parts, which are flooded by two rivers.
- One can immediately be free from material contamination by touching or bathing in those rivers. There are four castes of people live in those Dvīpas.
- They worship the Supreme Lord in His different forms (Sun, Moon, Agni, Varuna, Vāyu, Brahmā, etc...)
- They live for one thousand and more years with forms as beautiful as the *devatās* and produce offspring as the *devatās* do.
- Longevity, sensory prowess, physical and mental strength, intelligence and bravery are naturally and equally manifested in all the inhabitants there.

Jambūdvīpa

The innermost island, divided into nine tracts of lands or varṣas, with nine mountains (with Mount Sumeru) acting as their natural boundaries. Among

[43] SB 5.20.1 & 6: Characteristics of inhabitants of the Six Islands of the Bhū-maṇḍala

the nine varṣas, Ilāvrtā is in the middle consists of four huge trees, headed by Jambū tree, on its four directions. Because of this tree, the central island gets its name as Jambūdvīpa. Among the nine varṣas Bhārata-varṣa is considered to be the field of fruitive activities, and the other eight varṣas are considered as heavenly places on earth (*bhauma-svarga-pada-ni*). Dividing its nine varṣas among his nine sons, Āgnidrā retired from active life to engage in the devotional service of the Lord.

Plakṣadvīpa[44]

Jambūdvīpa is surrounded by an ocean of salt water, with the same breadth (100,000 yojanas). Just as a moat around a fort is surrounded by gardenlike forest, the saltwater ocean surrounding Jambūdvīpa is also surrounded by Plakṣadvīpa. Its breadth is twice that of the saltwater ocean (200,000 yojanas). Plakṣadvīpa gets its name from Plakṣa tree, which is shining like gold and as tall as the jambū tree (BxH: 100x1100 *yojanas*) and at its root is a fire with seven flames. Dividing its seven varṣas among his seven sons, Idhmajihva retired from active life to engage in the devotional service of the Lord.

There are seven varṣas, seven mountains and seven rivers in this land. The mountain ranges divide the *varṣas* and touch the ocean on both ends, running at right angles to the oceans. Haṁsas, Pataṅgas, Ūrdhvāyanas and Satyāṅgas are the four castes of people who live in Plakṣadvīpa. They represent the *brāhmaṇas, kṣatriyas, vaiśyas* and *śūdras.* They worship the Supreme Lord in the form of the sun-god, representing the Vedas, the gate to Svarga, by the path of *karma,* ritualistic ceremonies. "Let us take shelter of the sun-god, who is a reflection of Lord Viṣṇu, the all-expanding Supreme Lord, the oldest, the only worshipable, the Vedas and religion personified, and the origin of all auspicious and inauspicious results."

pratnasya viṣṇo rūpaṁ yat satyasyartasya brahmaṇaḥ
amṛtasya ca mṛtyoś ca sūryam ātmānam īmahīti

Sālmalīdvīpa[45]

Plakṣadvīpa is surrounded by an ocean of sugarcane juice, equal in breadth to the island itself. Similarly, the 3rd island Sālmalīdvīpa, twice as broad as Plakṣadvīpa (400,000 *yojanas*), is surrounded by an equally broad ocean of

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44 SB 5.20.2-5: Description of Plakṣadvīpa and the Prayers offered to Sun god by the residents
45 SB 5.20.7-12: Description of Sālmalīdvīpa and the Prayers offered to Soma by the residents

liquor. On Sālmalīdvīpa there is a śālmalī tree, which is as broad and tall as the plakṣa tree. Learned scholars say that this gigantic tree is the residence of Garuḍa, the king of all birds, where Garuḍa offers Lord Viṣṇu his Vedic prayers. *Chandaḥ-stutaḥ* refers to Garuḍa, who praises Viṣṇu by the hymns arising from his limbs. The *śalmalī* tree gives the island its name (*hūtaye*).
Yajñabāhu divided the Sālmalīdvīpa into seven tracts of land, gave them to his seven sons after naming them with names of his sons. In those tracts of land there are seven boundary mountains and seven holy rivers. Śrutidharas, Vīryadharas, Vasundharas and Iṣandharas by strictly following the cult of varṇāśrama-dharma worship the moon-god (soma), the expansion of the Supreme Lord, by chanting this mantra, "The moon-god by his own rays has divided the month into two fortnights, for the distribution of food grains to the *pitās* and the *devatās*. We pray to moon god that he may remain our king and guide, for he is the king of all the residents of the universe and divides time."

sva-gobhiḥ pitṛ-devebhyo vibhajan kṛṣṇa-śuklayoḥ
prajānāṁ sarvāsāṁ rājā-ndhaḥ somo na āstv iti

Kuśadvīpa[46]

Kuśadvīpa, surrounding the ocean of liquor, is twice as wide as the ocean of liquor (800,000 yojanas), is surrounded by an ocean of liquid ghee as broad as the island itself. On Kuśadvīpa there are clumps of kuśa grass, from which the island takes its name. Being created by the demigods, by Supreme Lord's will, they appear like a second form of fire with very mild and pleasing flames. Its young shoots illuminate all directions. Hiraṇyaretā, divided its seven parts among his seven sons according to the rights of inheritance. Then he retired from family life to engage in austerities. In those seven islands there are seven boundary mountains and seven holy rivers. The inhabitants known as Kuśalas, Kovidas, Abhiyuktas and Kulakas by bathing in the waters of those holy rivers will become purified. They worship the Lord in His aspect as the fire god by their expertise in ritualistic ceremonies (fire sacrifices) as per Vedic scriptures and chant this Mantra: "O fire-god, carry all the offerings of sacrifices, the yajñic ingredients, which we are offering the demigods, to the Supreme Lord, for He is the real enjoyer."

parasya brahmaṇaḥ sākṣāj jāta-vedo 'si havyavāṭ
devānāṁ puruṣāṅgānāṁ yajñena puruṣaṁ yajeti

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46 SB 5.20.13-17: Description of Kuśadvīpa and the Prayers offered to Agni by the residents

Note: This means, "Offer what is given in the name of the *devatās*, who are the limbs, to the Lord, who is the source of the limbs."

Krauñcadvīpa[47]

Krauñcadvīpa, surrounding the ocean of clarified butter, twice the width of the ocean of clarified butter (1,600,000 yojanas), is surrounded by an ocean of milk as broad as the island itself. By the presence of a great mountain, Krauñca, this island gets its name. Although the vegetables living on the slopes of Krauñca Mountain were attacked and devastated by the weapons of Kārttikeya, the mountain and the plants have become fearless because they are protected by Varuṇadeva and bathed on all sides by the milk ocean. The learned scholar Ghṛtapṛṣṭha divided the island among his seven sons, naming them according to the names of his sons. He completely retired from family life taking shelter at the lotus feet of the Lord and attained perfection.

In this island there are seven mountains, which indicate the boundaries of the seven tracts of land, and there are also seven rivers. The inhabitants known as Puruṣas, Ṛṣabhas, Draviṇas and Devakas using the waters of those sanctified rivers worship the Supreme Lord by offering a palm full of water at the lotus feet of Varuṇa, the demigod who has a form of water, while chanting the following mantra "O water of the rivers, with energy obtained from the Supreme Lord, you purify the three worlds (Bhūloka, Bhuvarloka and Svarloka), by taking away the sins. Kindly continue to purify us, who are always touching you, for this is your constitutional nature."

āpaḥ puruṣa-vīryāḥ stha punantīr bhūr-bhuvaḥ-suvaḥ
tā naḥ punītāmīva-ghnīḥ spṛśatām ātmanā bhuva iti

Śākadvīpa[48]

Śākadvīpa, surrounding the ocean of milk, twice the width of the ocean of milk (3,200,000 yojanas), is surrounded by an ocean of churned yogurt, as broad as the island itself. Due to the presence of a big and very fragrant śāka tree, which lends its scent or fragrance to the entire island, the island gets its name. Medhātithi divided the island among his seven sons, naming according to their names, then retired to fix his mind completely upon the lotus feet of the

47 SB 5.20.18-23: Description of Krauñcadvīpa and the Prayers offered to Varuna by the residents

48 SB 5.20.24-28: Description of Śākadvīpa and the Prayers offered to Vāyu by the residents

Supreme Lord, and entered into a forest, which is suitable for meditation. For these lands also, there are seven boundary mountains and seven rivers. The inhabitants known as Ṛtavrata, Satyavrata, Dānavrata and Anuvrata practice prāṇāyāma and mystic yoga, and in trance they worship the Supreme Lord in the form of Vāyu by chanting this mantra, "May the antaryāmī, the Supreme Lord who enters within and directly maintains the *jīvas* beings by the functions of the prāṇas, under whose control the universe is created, please protect us."

antaḥ-praviśya bhūtāni yo bibharty ātma-ketubhiḥ
antaryāmīśvaraḥ sākṣāt pātu no yad-vaśe sphuṭam

Puṣkaradvīpa[49]

Puṣkaradvīpa, surrounding the ocean of yogurt, twice as wide as the ocean of yogurt (6,400,000 yojanas), is surrounded by an ocean of very tasteful water as broad as the island itself. On Puṣkaradvīpa there is a great lotus flower with 100 million pure golden petals, as effulgent as the flames of fire, which is the sitting place of Lord Brahmā.

In the middle of that island is a great mountain named Mānasottara, which forms the boundary between the inner and outer varṣas. Its breadth and height are 10,000 yojanas. Because Mānasottara Mountain is circular, there are two circular varṣas, on either side of it. On that mountain, in the four directions, are the residential quarters of *devatās* such as Indra. Sun-god's chariot wheel, making up a year, going around Meru, travels on top of this mountain with northern and southern courses, causing day and night for the *devatās*.

Vītihotra divided the two parts of the island among his two sons named Ramaṇaka and Dhātaki and then personally engaged himself in activities for the sake of the Supreme Lord like his elder brother Medhātithi. The inhabitants of this tract of land worship the Lord as represented by Lord Brahmā, by actions which cause further actions, and offer prayers to the Lord as follows. "We offer respects to the Supreme Lord whom people, with the idea of servant and object of service, worship as Lord Brahmā, who is attained by karma (*karma-maya*), for he is fixed in devotion to the Lord. Thus he is non-different from the peaceful Lord, whom the Vedas are known."

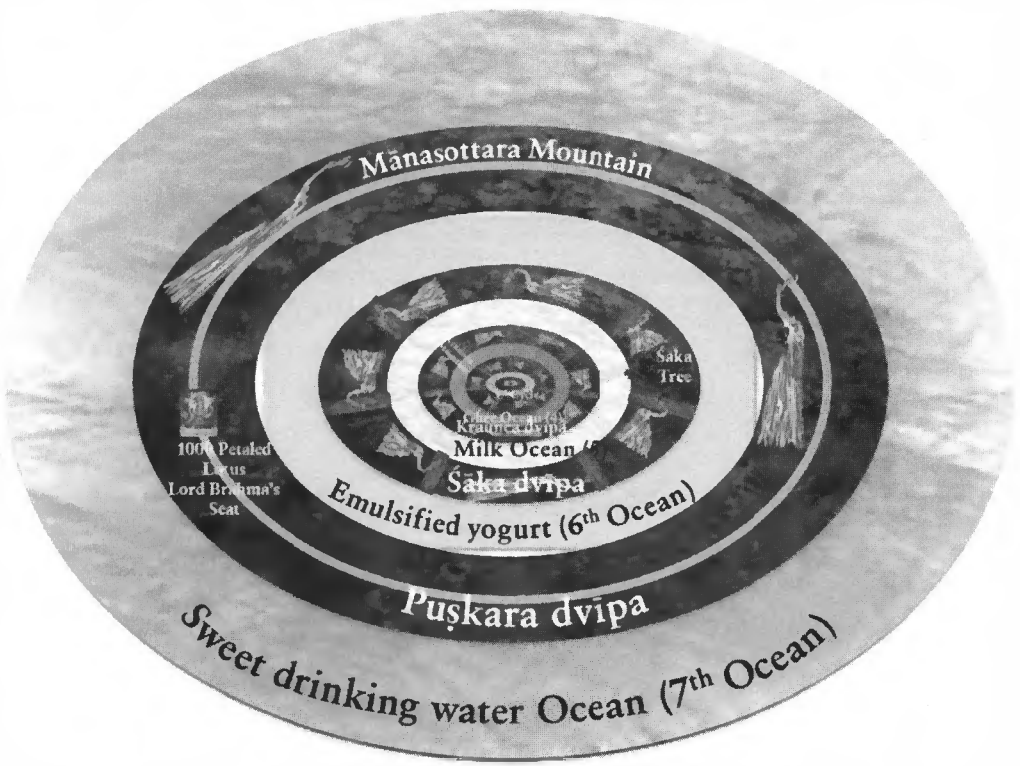
yat tat karmamayaṁ liṅgaṁ brahma-liṅgaṁ jano 'rcayet
ekāntam advayaṁ śāntaṁ tasmai bhagavate nama iti

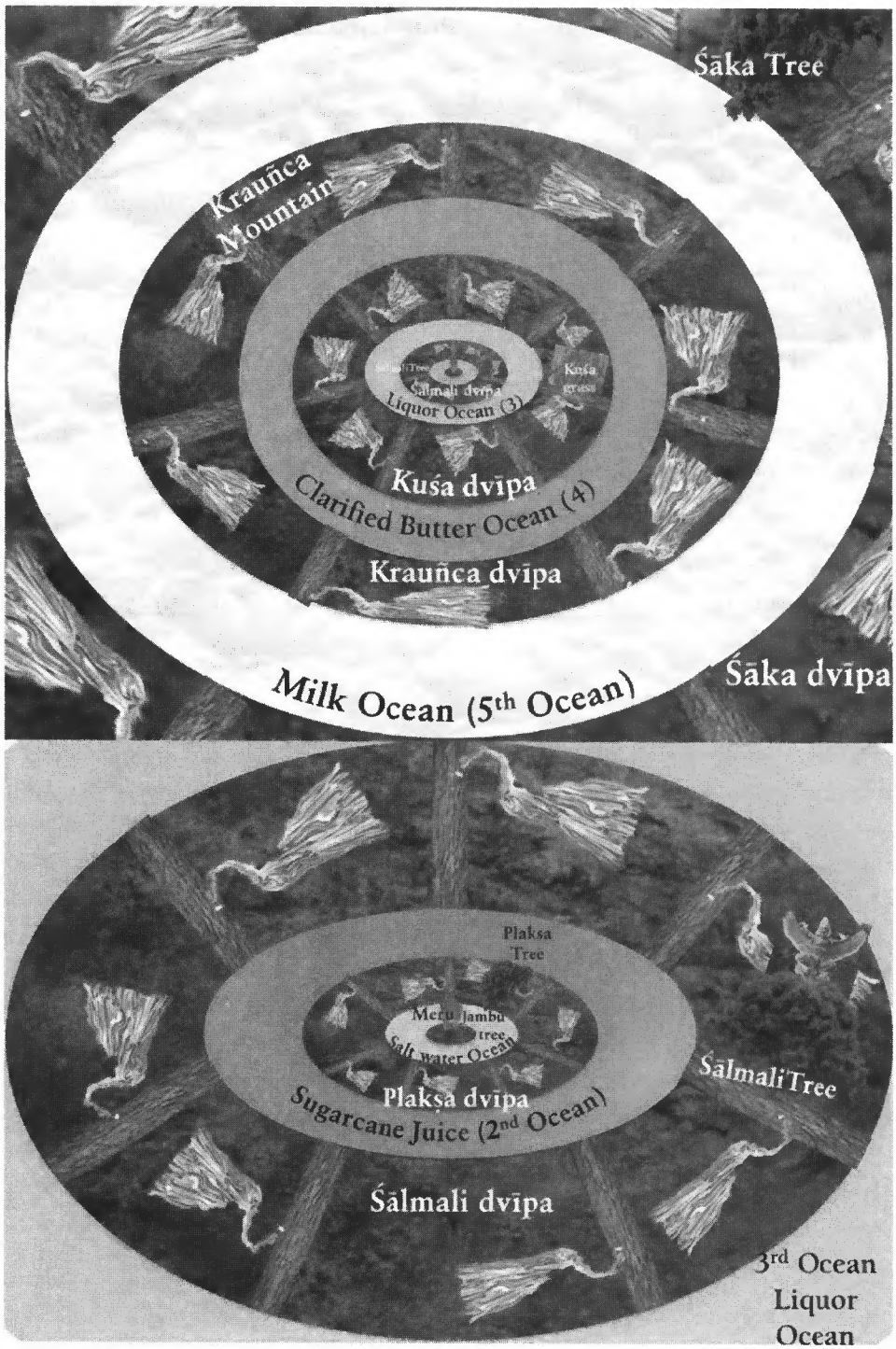
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49 SB 5.20.29-33: Description of Puṣkaradvīpa and the Prayers offered to Lord Brahmā by the residents

Summary of the Seven Islands of the Bhū-mandala

	Island	Ocean	Width (MY)	R_{Outer} (MY)	Ruler (s/o Priyavrata)	Deity	Specialty
1	Jambū dvīpa	Salt water	0.05+0.1	0.15	Āgnīdhra -> 9 Sons	9 deities 9 varṣas	Jambū tree
2	Plakṣa dvīpa	Sugarcane juice	0.2+0.2	0.55	Idhmajihva -> 7 Sons	Sun	Plakṣa tree
3	Śālmali dvīpa	Liquor	0.4+0.4	1.35	Yajñabāhu -> 7 Sons	Moon	Śālmalī tree
4	Kuśa dvīpa	Clarified butter	0.8+0.8	2.95	Hiraṇyaretā -> 7 Sons	Agni	Kuśa grass
5	Krauñca dvīpa	Milk	1.6+1.6	6.15	Ghṛtapṛṣṭha -> 7 Sons	Varuna	Krauñca mountain
6	Śāka dvīpa	Emulsified yogurt	3.2+3.2	12.55	Medhātithi -> 7 Sons	Vayu	Śāka tree
7	Puṣkara dvīpa	Sweet drinking	6.4+6.4	25.35	Vītihotra -> 2 Sons	Brahma	Lotus flower
	Mānasottara		0.01	15.75 (Radius: to its middle point)			




Śāka Tree
Krauñca Mountain
Śālmali dvīpa
Liquor Ocean (3)
Kuśa dvīpa
Clarified Butter Ocean (4)
Krauñca dvīpa
Milk Ocean (5th Ocean)
Śāka dvīpa
Plakṣa Tree
Meru Jambū tree
Salt water Ocean
Plakṣa dvīpa
Sugarcane Juice (2nd Ocean)
Śālmali dvīpa
Śālmali Tree
3rd Ocean
Liquor
Ocean

Lokāloka Mountain[50]

Sukadeva Gosvami, after describing the specific divisions, mountains and rivers of the seven dvīpas, surrounded by seven oceans, now he continuous to describe the remaining land of Bhū-maṇḍala outside the seven dvīpas area.

Land of Gold

Lokāloka Mountain, surrounding the area beyond the ocean of sweet water, divides the countries that are full of sunlight from those not lit by the sun. What is that area in between the sweet water ocean and Lokāloka Mountain? It is a tract of land surrounding the ocean of sweet water, as broad as the area between the middle of Meru and the middle of Mānasottara Mountain (15,750,000 yojanas). Since its surface is like a mirror, any object dropped there cannot be seen again, this land is called as Gold Land. Therefore all living entities have abandoned this place.

Note: Srila Madhvācārya says "One and half times the previous one is the extent of the Golden Land and it has a diamond boarder at the end, which is 150,000 yojanas.
Width of Puskara-dvīpa: 6.4 MY;
Width of Golden land: 1.5 x 6.4 = 9.6 MY;
Previous width: 0.1 + 0.2 + 0.4 + 0.8 + 1.6 + 3.2 = 6.3;
Width of Golden land: 1.5 x 6.3 = 9.45 MY;
Width of Diamond boarder: 0.15 MY;
Total Width of Golden land and Diamond boarder: 9.45 + 0.15 = 9.6 MY;

Nature or the Significance of the Lokāloka Mountain

The great mountain stands in between the lands inhabited by living entities (*loka*) and those that are uninhabited (a*loka*), separating them into two, is therefore celebrated as Lokāloka.

Height of the Lokāloka Mountain

Being created by the Lord in all eight directions, it extends up to the end of the three worlds (*Bhūrloka*, *Bhuvarloka* and *Svarloka*) in height, and acts like the outer border. The rays of the sun, of other planets, up to Dhruvaloka, and of the stars and luminaries spread throughout the three worlds, but cannot spread beyond the boundaries of this mountain. Because it extends as high as

[50] SB 5.20.34-41: Description of Lokāloka Mountain

Dhruvaloka, it defines the limit of the three worlds. Dhruvaloka is 3,900,000 *yojanas* above the earth. It is above Svarga and below Maharloka.

Far-end of Lokāloka Mountain

Learned scholars have thus described the locations of the planets, their particular symptoms, measurements and shapes. The distance between the centre of Sumeru and the far end of Lokāloka mountain is one fourth of the diameter of the universe, or 125,000,000 yojanas.

Lokāloka's distance from Meru is one quarter of the measurement of universal shell (*bhū-golasya*). From the middle of Meru to the middle of Mānasottara is 15,750,000 *yojanas*. From the middle of Mānasottara to the other shore of fresh water ocean is 9,600,000 *yojanas*. The golden land measures 15,750,000 *yojanas*. The total measurement from the middle of Meru to the beginning of Lokāloka Mountain is then 41,100,000 *yojanas* (15.75 + 9.6 + 15.75 = 41.1). The distance from start of Lokāloka to the end Lokāloka is 82,200,000 *yojanas* (twice to the inner width). Total is 123,300,000 *yojanas* from Meru to the far end of Lokāloka. Thus it is said that 125,000,000 *yojanas* (approximately) is the measurement from Meru to the end of Lokāloka. It is the same measurement on the other side of Meru.

Note: some say the height and width of the Lokāloka Mountain are (HxW): 5.7x5.05 MY. Some others say that the width of the Loka varṣa, outside the sweet drinking water ocean, is same as the radius of Mānasottara Mountain. Thus, the far end of Loka varṣa is 41.1MY (25.35+15.75). Beyond that is Golden land. Visnu Purana says that the two fold Golden Land (82.2MY) is avoided by all creatures. Beyond that, there is Lokāloka Mountain, the gap filler. Then there is another 125MY wide region of darkness all around.

Four Gaja-patis on top of Lokāloka Mountain

Lord Brahmā, the supreme spiritual master of the entire universe, established the four Gaja-patis (Ṛṣabha, Puṣkaracūḍa, Vāmana and Aparājita) on the top of Lokāloka Mountain in four directions, for they are responsible for maintaining the planetary systems of the universe.

Bhū-maṇḍala or Bhūloka is 493,200,000 (4 times to 123,300,000) *yojanas* in diameter with the oceans, islands and mountains. There is thus a space of 3,400,000 *yojanas* from the universal shell all around, for the inner

diameter of the Universal shell is 500 MY. Because of this Śeṣa holds up the earth and the elephants of the directions steady it. If the Bhū-maṇḍala touched the edges of the shell of the universe, it would not move downwards at all. To explain the dimensions in any other way would make the descriptions mentioned here useless, since it has been mentioned that the diameter to the shell of the universe is 500,000,000 *yojanas*. It would also be impossible for Vāraha to lift up the earth (*bhū-maṇḍala*) which has sunk in the Garbodhaka Ocean during Svāyambhuva and Cākṣuṣa Manvantara, if *bhū-maṇḍala* was 500,000,000 *yojanas* in diameter.

Note: In the Śveta-varāha-kalpa (first day in the first month of fifty-first day of Brahmā's life) at the beginning of Svāyambhuva-manvantara, Śveta-varāha (white in color) appeared from the nostril of Brahmā and lifted up the earth. Then He disappeared. In the sixth Cākṣusa- manvantara, when there was a sudden deluge, Nīla-varāha (black in color) appeared in the water, lifted the earth and killed Hiraṇyakṣa. In SB 3.13 the great sage Maitreya combines both appearances in his narration, which can be understood from the explanatory verses of *Laghu-bhāgavatāmṛta* 1.3.10-19. Few verses from SB 3.13 indicate that Lord Varāha lifted the entire *Bhū-maṇḍala,* during both the instances.

Understanding that the boar of black color, who raised the earth on the tips of His tusks while playing like an elephant was the Supreme Lord, Brahmā and others praised Him with prayers while folding their hands. (SB 3.13.33)

tamāla-nīlaṁ sita-danta-koṭyā kṣmām utkṣipantaṁ gaja-līlayāṅga
prajñāya baddhāñjalayo 'nuvākair viriñci-mukhyā upatasthur īśam

The sages headed by Brahmā said: O Lord! O lifter of the earth (*bhū-dhara*)! The earth with its mountains (*sa-bhū-ḍharā*) held by You on the tips of Your tusks (*daṁṣṭrāgra-koṭyā*) appears beautiful. It is like a lotus with its leaves held on the tusk of mad elephant rising from the water (*vanāt*). (SB 3.13.40)

daṁṣṭrāgra-koṭyā bhagavaṁs tvayā dhṛtā
virājate bhūdhara bhūḥ sa-bhūdharā
yathā vanān niḥsarato datā dhṛtā
mataṅ-gajendrasya sa-patra-padminī

Your form of a boar (*saukaraṁ*), personifying the Vedas (*trayīmayaṁ*), appears beautiful with the earth (*bhū-maṇḍalena*) held by Your tusks, just

as a huge mountain (*kulācalendrasya*) appears beautiful with a huge cloud held on its peak (*śṛṅgoḍha-ghanena*). (SB 3.13.41)

trayīmayaṁ rūpam idaṁ ca saukaraṁ
bhū-maṇḍalenātha datā dhṛtena te
cakāsti śṛṅgoḍha-ghanena bhūyasā
kulācalendrasya yathaiva vibhramaḥ

How these elephants support the universe?
The Supreme Lord, surrounded by His associates such as Viṣvaksena, remains on Lokāloka for the welfare of all people in all the varied planets and for increasing various powers of the elephants, who are protectors of the planets and His own expansions.

Why does the Lord remain there? The Lord remains there until the end of the *kalpa* to maintain all the varied planets that are created by His *yoga-māyā*. He remains there in order to protect the various affairs of the world produced by His own (*ātma*) *yoga-māyā*. If the Lord does not protect what He has created from His own energy, then who else can protect them?

Aloka-varṣa beyond Lokāloka Mountain[51]
Lokāloka Mountain extends for the same distance as the total diameter of the mountain range (82.2MY) and distance outside the mountain range is the same as the distance from Meru to the far edge of Lokāloka Mountain. Aloka-varṣa, situated outside the range of Lokāloka Mountain, extends to a distance of 123.3MY (~125MY). Beyond Aloka-varṣa is the destination of those who aspire for liberation from the material world, which is beyond the control of modes and completely pure. (The place indicated here is beyond the Universal covering. The *yogīs* pierce the shell of the universe and go there, desiring liberation. It is beyond the jurisdiction of the modes of material nature, and therefore it is completely pure.) Lord Kṛṣṇa took Arjuna through this place to bring back the sons of the brāhmaṇa.

Aloka-parināmāṁ means 82,200,000 *yojanas*. This is the total distance within the mountain circle, or the diameter of the circle, and also the width of the Lokāloka Mountain. The distance from the center of Meru to the far edge of Lokāloka is 123,300,000 *yojanas*. From the far edge of Lokāloka to

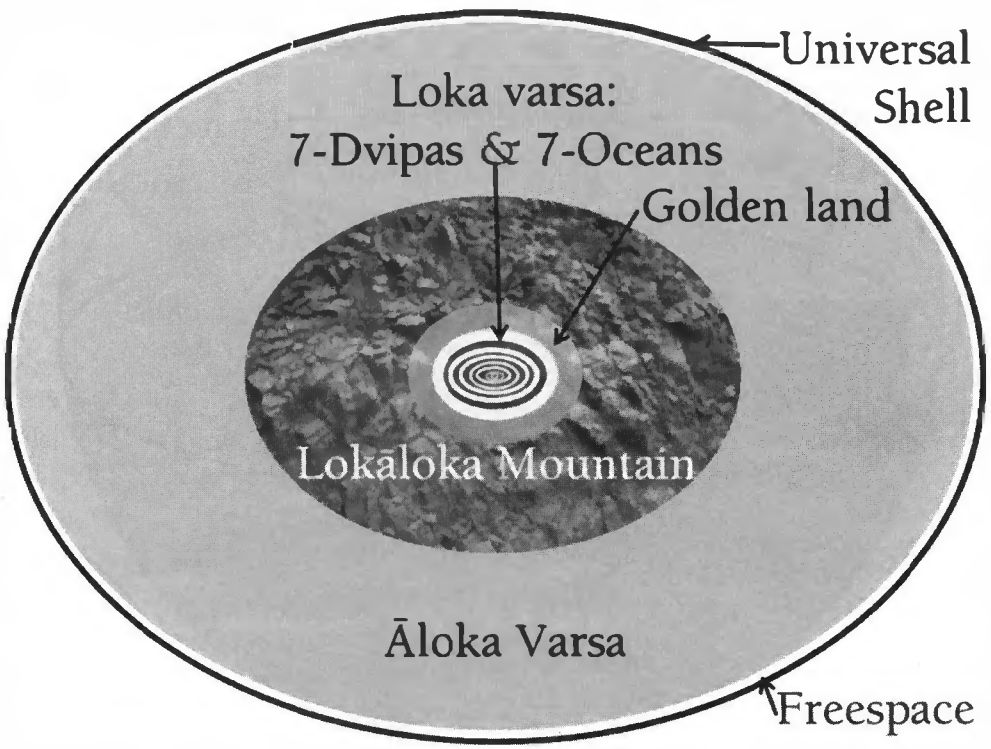
[51] SB 5.20.42: Description of Aloka-varṣa beyond Lokāloka Mountain

the (outer) edge of Bhū-maṇḍala is another 123,300,000 yojanas, which is equal to its inner width. This is the width of the Aloka-varṣa. Thus, the radius of Bhū-maṇḍala is 246,600,000 *yojanas* and its diameter is 493,200,000 *yojanas*.

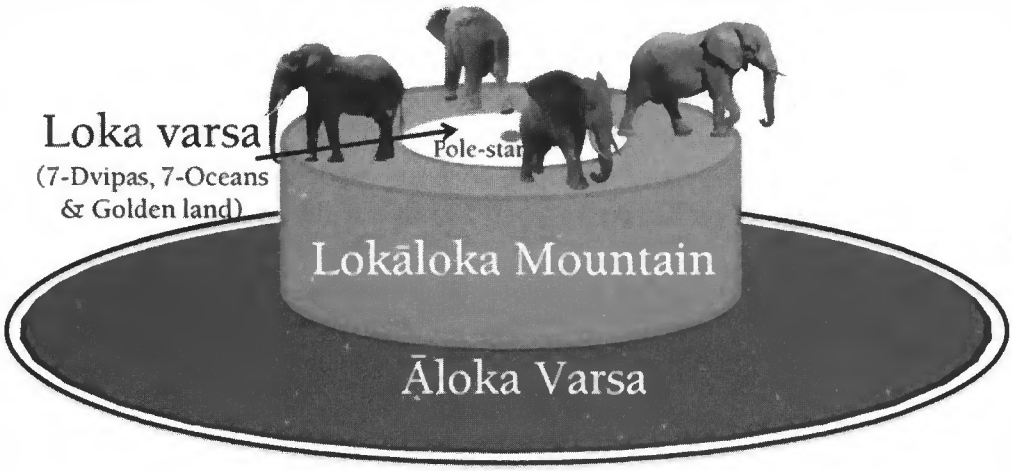
Then there are seven layers of universal covering made up of earth, water, fire, air, ether, false ego & mahat-tattva, each 10 times wider than the previous one.

Summary of the Land outside the Seven Islands of the Bhū-maṇḍala

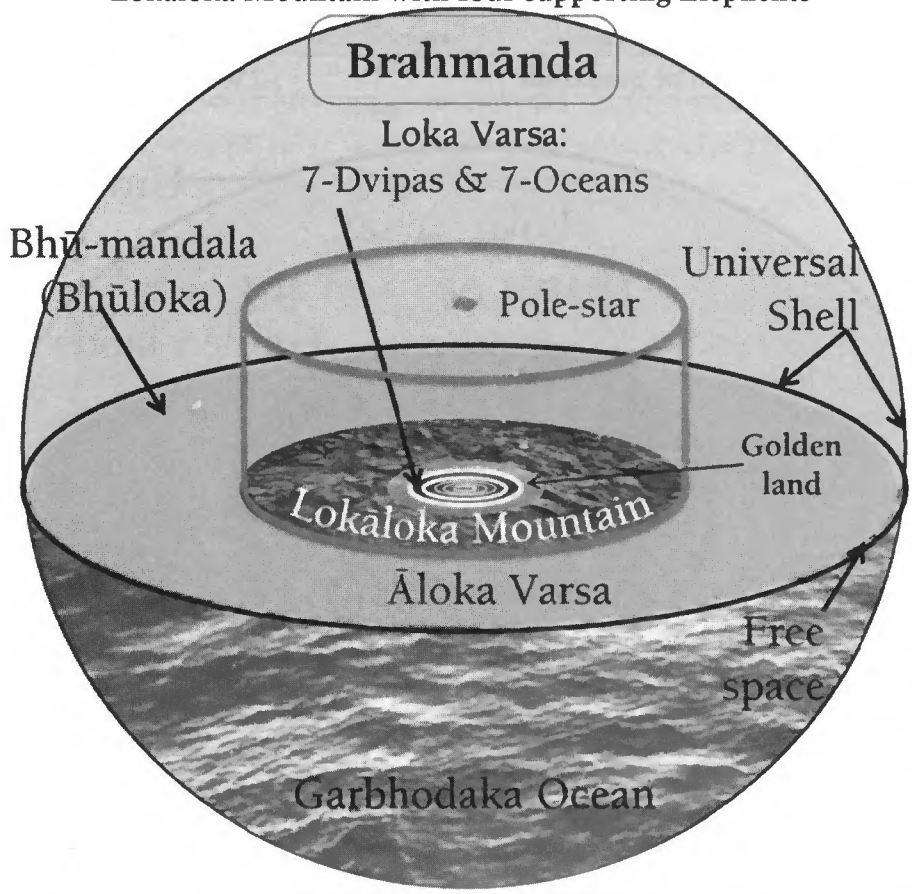
SN	Island	Width (MY)	R_{Outer} (MY)
8	Golden Land	15.75	41.1
9	Lokāloka Mountain	82.2	123.3
10	Āloka Varṣa	123.3	246.6
11	Free space	3.4	250
12	Universal coverings	7 layers	>250 (250 + 7 layers)
13	Destination of pure souls	-	>250+ Universal covering



Summary description of Bhū-maṇḍala - the Land outside the Seven Islands



Lokāloka Mountain with four supporting Elephents

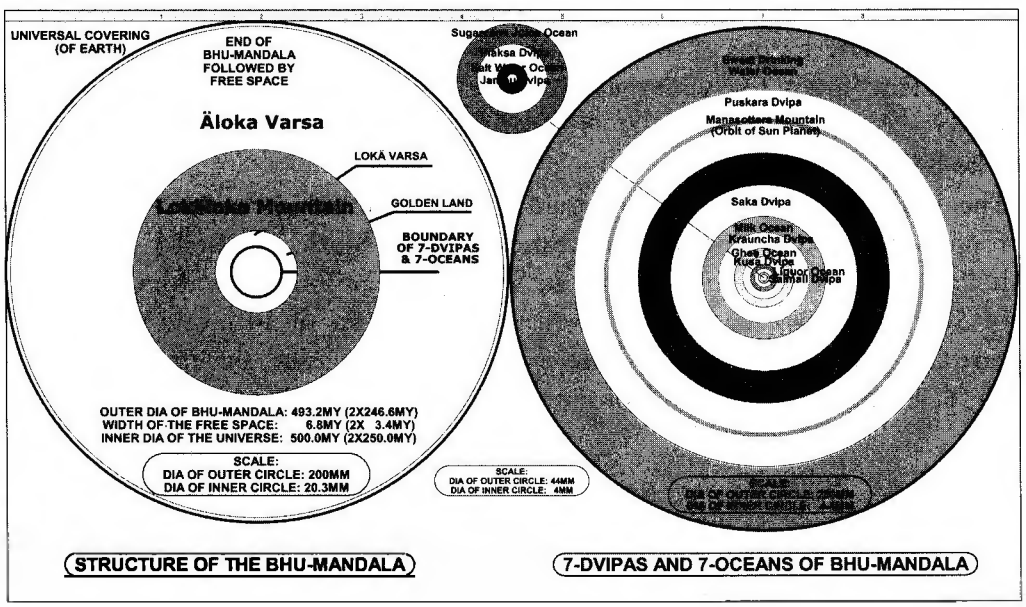


Summary depiction of the Divisions of the Bhū-maṇḍala

Some basic Calculations

Radius of Mānasottara Mountain Peak	15.75 MY (orbit of sun)
Outer radius of Svarna bhumi (inner radius of Lokāloka Mountain)	15.75+9.6+15.75 = 41.1 MY
Width of Lokāloka mountain = Diameter within the mountain range	2x41.1 = 82.2 MY
Radius of far end of Lokāloka mountain (1/4th of the Diameter of the Universe)	41.1 + 82.2 = 123.3MY ~=125MY (1 billion miles)
Radius of the Bhū-mandala (Bhūloka)	123.3 + 123.3 = 246.6 MY
Diameter of the Bhū-mandala (Bhūloka)	2x246.6 = 493.2 MY
Free space to the universal shell all around	3.4MY

The pictorial representation of the Bhū-maṇḍala with all its divisions according to the dimensions



The Structure of the Universe as per "Bhūgola Varnanam"

"Bhūgola Varnanam" is wonderful firsthand information about the structure of this Universe by Srila Vādiraja Tīrtha, one of the greatest scholars in the line of Madhva. He is well known for his scholarly writings and his commentaries on the various works of Srila Madvācārya. Here are few notes taken from this book "Bhūgola Varnanam", that will help us appreciate the wonderful creation of the Supreme Lord, whose artistic creation is fully embedded with many many inconceivable functionalities consisting of at most precession.

Śrīmad-Bhāgavatam is the authoritative text, which tells us about Cosmography; and information gathered from any other source should be in accordance with it. That is what ācārya Madhva says quoting the following verses from the Brahmāṇḍa Purāṇa.

yathā bhāgavate tūktaṁ bhauvanaṁ kośa-lakṣaṇam
tasyāvirodhato yojyam anya-granthāntare sthitam

Thus, the descriptions of the Cosmography presented in the book "Bhūgola Varnanam" are in line with the descriptions given in Śrīmad-Bhāgavatam.

The Brahmanda[52]

The Brahmanda or the hollow golden shell containing the universe is fully round in every direction, east to west or north to south. The Brahmanda or the hollow golden shell is in spherical structure with an internal width or diameter of 500 million yojanas. This spherical shell consists of 10 layers made up of earth, water, fire, air, ether, false ego, mahat-tattva and the remaining three layers are the layers made up of the three modes of Pradhana.

The Bhū-mandala

Within this golden shell just above the Garbhodaka Ocean, which fills the lower half of the shell, there is a vast land spread across the diameter of the shell. This vast land, consisting of three parts, is known as Bhū-mandala. The innermost part is generally known as the circular earth, inhabited by different living entities. Surrounding it, the middle part, twice as wide as the circular earth, is known as the region of darkness or *andhatamas* and it is circular in

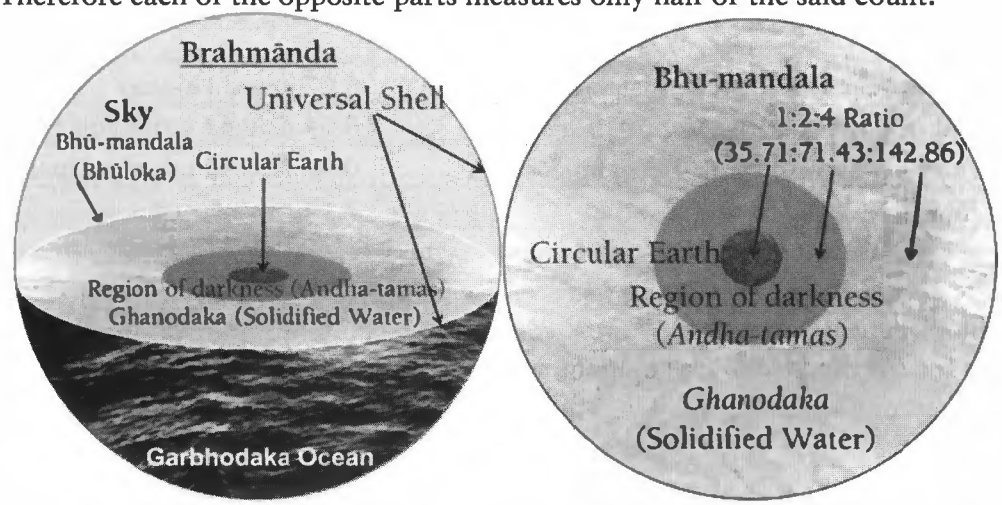
[52] Refer to the section "The Brahmanda (14 - 17)"

shape. Surrounding this middle part is the circular Ghanodaka stretch (solidified water), which is twice its inner complement, 'the region of darkness.'

Upanishads describe the circular earth as "Prithvi" that which is spacious with abundance of wealth for living beings. The circular earth is said to stretch from sea to sea as the brahmana seers conceive it. It has the Meru Mountain at its very centre and it contains seven islands and seven oceans stretching around the Meru centre. These three divisions are circular in form. The three together – circular earth, dark region and Ghanodaka - measure fifty crore[53] yojanas from end to end. This is factual and not fictitious.

Measurements of the three Regions (in the ratio of 1:2:4)

Of the three regions mentioned, the circular earth occupies a circular stretch having a diameter of little more than seven crore yojanas. Being double to circular earth in stretch, the region of utter darkness measures little more than fourteen crore yojanas. Considering the stretch of Ghanodaka as being twice that of the dark region, it measures little more than twenty eight crore yojanas. From the Meru centre, the circular earth measures 3 1/2 crore yojanas in all directions. The region of darkness and the Ghanodaka region measure 14 crore and 28 crore yojanas respectively putting the opposite parts together. Therefore each of the opposite parts measures only half of the said count.



[53] 1 crore is equal to 100 lakh or 10 million

Area	Total Width (MY)	Width (MY)	$Radius_{outer}$ (MY)
Central earth	71.43	35.715	35.715
Dark Region	142.86	71.430	107.14
Ghandodaka	285.71	142.86	250.00

Circular earth[54]

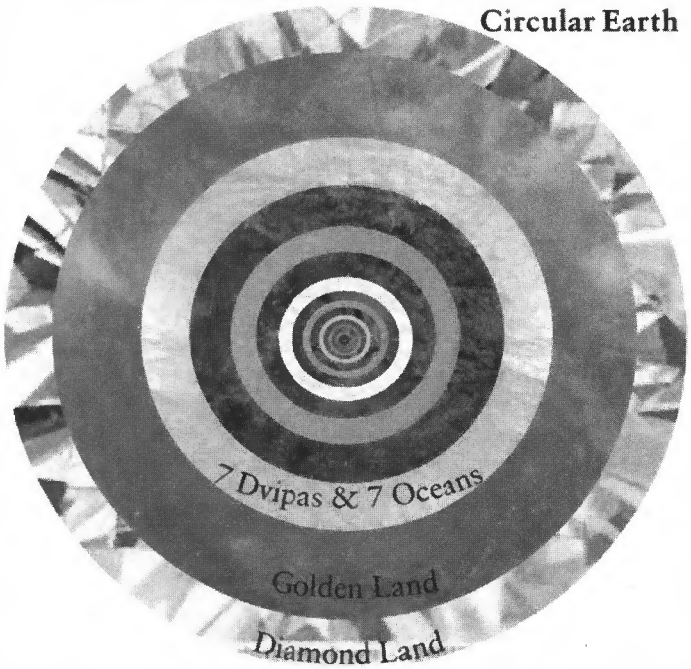
The circular earth is understood to be consisting of the seven islands and the seven oceans, the middle most island Jambūdvīpa measuring one lakh yojanas, and each of the outer islands being twice as large as their inner complements. Similarly, the seven surrounding oceans too should be considered. Thus, the diameter of the circular earth measures 5 crore and 7 lakh yojanas up to the encircling land of gold, which itself is surrounded by the land of diamond. The last two are known as "Antyadhyardha sthala, or the terrestrial region." Lokaloka parvata surrounds this entire terrestrial region or the land of gold and land of diamond.

The land of gold, the region around Suddhodaka Ocean, measures 64 lakh yojanas on each side (total 1.28 crore yojanas). "*Adhyardha*" does not actually mean half; but it means 'with a little addition to half.' Therefore, a little more than half of 64, that is 42 lakh yojanas (instead of 32) is the width of the land of diamond on either side. The total measurements of the Circular earth

Circular earth with seven islands and oceans	**5.07 crore yojanas**
Land of gold (*svarna-bhumi*), both sides	0.64 & 0.64 crore yojanas
Land of diamond (*vajralepa*), both sides	0.42 & 0.42 crore yojanas
The total width	7.19 crore yojanas

The excess of 5 lakh yojanas (7.19 - 7.14 = 0.05) should be assigned to Brahma-bhumi extending beyond '*Vajralepa*' (Land of diamond) and forming part of the foot of the Lokāloka mountain just a fringe of it.

[54] Refer to the section "The Terrestrial Region (55 - 58)"



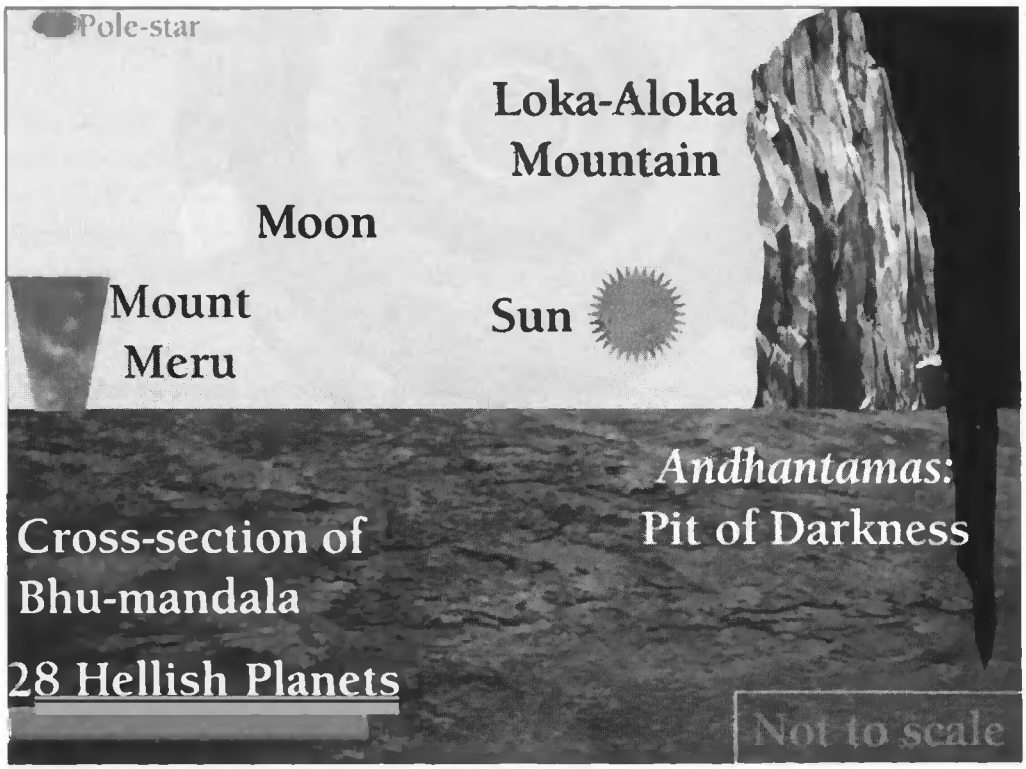
Andhantamas – The region of complete darkness[55]

The pit of darkness known as "*Andhantamas*" is situated below the earth level, sunk in Ghanodaka. This region surrounds the Lokāloka parvata and is termed as "*Adhogati* or the downward path." Because *Andnantamas* is below the level of *Ghanodaka*, it is possible for sinners to go down into it to some extent and to come up again. The pit of darkness is very deep. Sinners, who deserve greater punishment than what they may suffer in the ordinary hells, are made to go down into the pit of darkness to some extent and they come up. But those, who are thrown down into the lower parts known as "*tamas, maha tamas and andhantamas*", never come up. They are the third kind of souls who are doomed to suffer in the eternal hells (for long duration).

The principle of darkness extends even higher than the Lokāloka parvati, which is 51 lakh yojanas high. It is petrified darkness like water turned into ice. If it were not so, where was the need for Sri Kṛṣṇa, Lord of the Yadus, to

[55] Refer to the section "Andhantamas, the region of complete darkness (79 -83)"

create a tunnel through it by means of His Cakra (disc weapon)? When Sri Kṛṣṇa had to drive Arjuna's chariot towards "Anantasana" in the Ghanodaka far beyond the petrified circle of darkness, He had to cut through it and make a tunnel. It was easy for the disc of light (Cakra) to cut through the darkness. This was on the occasion of bringing back the dead sons of the brahmana.



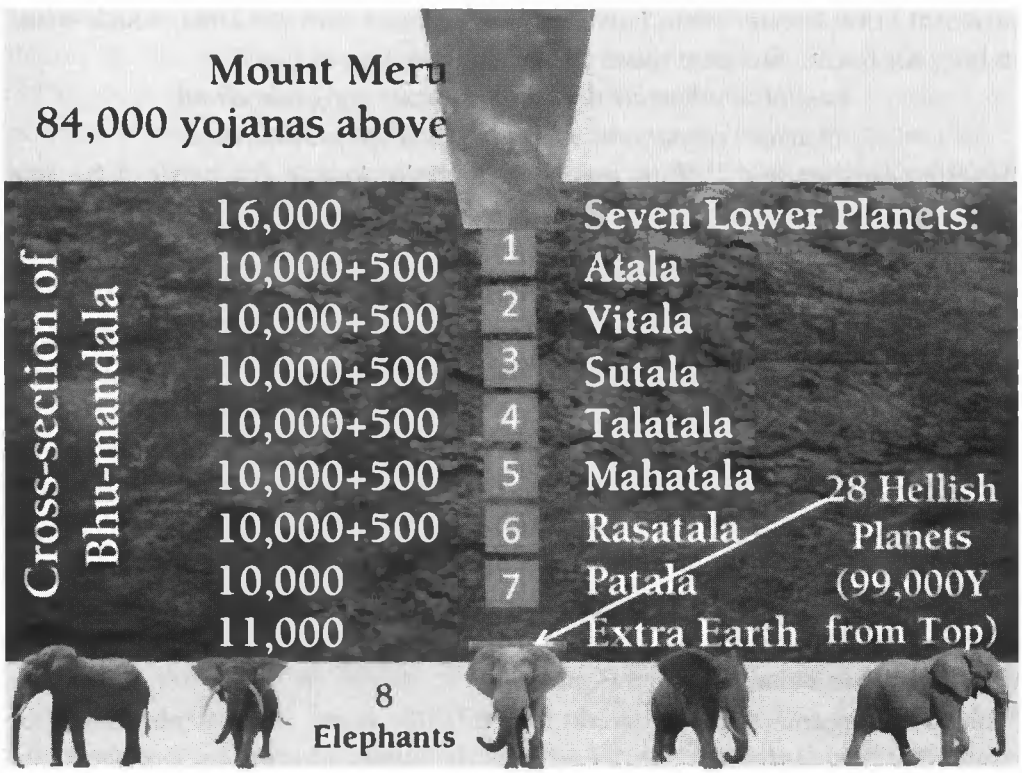
Lower Planetary System[56]

Within the hollows of the earth down below are seven worlds arranged one below the other. They are Atala, Vitala, Sutala, Talatala, Mahatala Rasatala and Patala. Each of these is 10,000 yojanas in extent (width and height). Thus they occupy a total area of 70,000 yojanas. The Meru Mountain has its base into the earth to the extent of 16,000 yojanas. The gaps between the seven nether worlds make a total of 3000 yojanas (500 yojanas in each case). The

[56] Refer to the section "The Nether Worlds (93 - 98)"

earth extends still further below to a depth of 11,000 yojanas; it is all rocky in nature. Thus 16+70+3+11 = 100 thousand yojanas in total is the thickness of the circular earth.

Height of the seven Lower Planets	70,000 yojanas (7x10,000)
Base of Mount Meru into the earth	16,000 yojanas
The gaps between the seven nether worlds	3000 yojanas (6x500Y each)
Earth extends further below to a depth of	11,000 yojanas
Total thickness of the circular earth	100,000 yojanas



Note: In Śrīmad-Bhāgavatam, the eight directional elephants (*Asta-diggajas*) are said to be standing beneath Pātalaloka, supporting the worlds above them. Evidently, their location should be within the range of the last mentioned 11,000 yojanas.

It is said in SB 8.8.4-5 that "Then, the king of elephants, named Airāvata, with four tusks, was generated. It defied the glories of white Kailāsa Mountain, the

abode of Lord Śiva. Thereafter, O King, eight great elephants of the directions (*aṣṭau dig-gajā*), headed by Airāvata, were generated. Eight she-elephants, headed by Abhramu, were also generated."

airāvaṇādayas tv aṣṭau dig-gajā abhavaṁs tataḥ
abhramu-prabhṛtayo 'ṣṭau ca kariṇyas tv abhavan nṛpa

It is further said in SB 10.62.4-7, where Banasura is speaking to Lord Siva, that "O Lord Mahādeva, I bow down to you who are just like a desire tree fulfilling all desires. These one thousand arms you bestowed upon me have become merely a heavy burden. Besides you, I find no one in the three worlds worthy to fight. Eager to fight with the elephants who rule the directions, O primeval Lord; I went forth, pulverizing mountains with my arms, which were itching for battle. But even those great elephants fled in fear."

kaṇḍūtyā nibhṛtair dorbhir, yuyutsur dig-gajān aham
ādyāyām cūrnayann adrīn, bhitās te 'pi pradudruvuh

Banasura further said, "With my arms intensely itching for battle, I became eager to fight against the elephants of the directions that hold up the bottom of the universe. Seeing me pounding the mountains into powder in order to relieve the itching of my arms, the elephants of the directions became afraid."

Reference from Vālmīki Rāmāyaṇa, Bala-kāṇḍa, Chapters 39 and 40

Once Mahārāja Sagara inaugurated the performance of an *aśvamedha-yajña*, with Aṁśumān (the son of Asamañjasa) protecting the challenge horse as it wandered over the earth. However, during the performance of the yajña, King Indra appeared in the guise of a Rākṣasa and stole the horse from the sacrificial arena. At this, the priest fervently urged Mahārāja Sagara to recover the horse at all costs. He implored that if the sacrifice remained incomplete, it would prove to be most inauspicious.

Mahārāja Sagara then ordered his 60,000 sons (the stepbrothers of Asamañjasa) to search the entire earth for the horse, allotting each of them one square yojana. He further instructed his sons that if the horse could not be found elsewhere, to look below the earth (Jambūdvīpa).

Thus, as Mahārāja Sagara and his grandson Aṁśumān waited at the sacrificial arena, the 60,000 princes searched the entire earth. At last, when they were unable to find the horse anywhere, the sons of Sagara began digging into the earth. This, however, caused the inhabitants of the lower worlds, such as the

Nagās, to become greatly distressed. Indeed, even the earth itself became extremely disturbed. Even after digging up the earth 60,000 yojanas (1 yojana each prince) below up to Rasātala and killing many living beings in the process, the sons of Sagara could not find the sacrificial horse, and thus returned to their father disappointed. When the 60,000 princes inquired from their father what they should do next, the King ordered them to continue digging further down, until the horse was found.

The sons of Sagara resumed their excavating of all four quarters of the earth by making tunnels from Rasātala, and soon came to see the four dig-gajās (Virupāksa in the East, Mahāpadma in the South, Saumanasa in the West, and Bhadra in the North), the colossal elephants that protect the four directions and support the earth. These gigantic elephants support the entire weight of the earth upon their backs, and when, due to exhaustion, they sometimes shake their heads for relief, it creates the phenomena of earthquakes. Finally, after returning to Rasātala and then penetrating the earth in the northeastern direction all the way down to Pātala (at around 70,000 yojanas in depth), the sons of Sagara came upon Lord Kapila sitting in meditation, while nearby, the sacrificial horse stood grazing peacefully. Hastily assessing the situation, Sagara's sons assumed Kapiladeva to be the thief. Thus, they angrily took up their digging implements and rushed to kill Him. Upon seeing this act of aggression, Kapiladeva became transcendentally angry. Then, simply by uttering the mantra "hum," He reduced the 60,000 sons of Sagara to ashes.

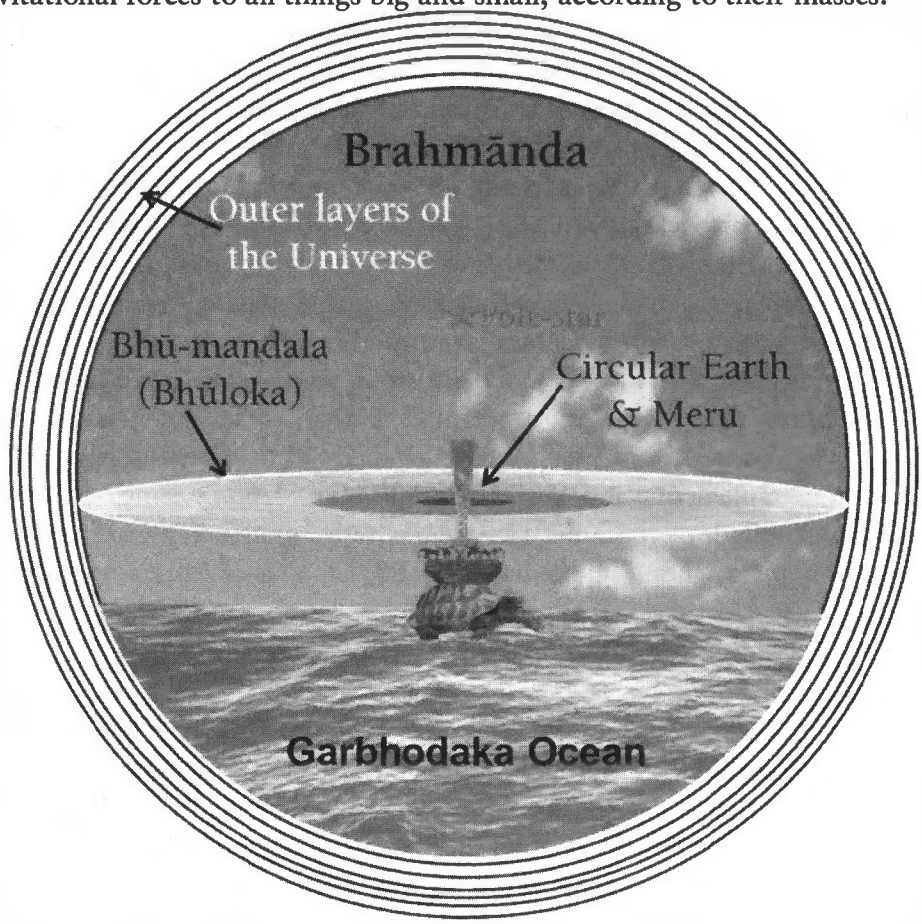
Ādi Śeṣa - The holder of circular earth[57]

All the serpents of Patala loka worship Ādi Śeṣa. He has his situation in the Ghanodaka down below; and he is immortal. He finds himself supported by 'Vayu Kurma' Mukhya Prana in tortoise form; Ādi Śeṣa's tail rests thereon. He is quite strong. Lifting one of his thousand hoods, he supports the worlds (Bhū-maṇḍala) above him so that the earth does not sink into Ghanodaka. Hence, he is hailed as "*Sarva loka dhurandharah.*"

The circular earth is not just a globe of 8000 miles in diameter, going round the sun. In fact, it does not move at all. It has a stretch of 30 crore miles

[57] Refer to the section "Ādi Śeṣa (99 - 102)"

(50MY - 7 Dvīpas & 7-Oceans), with the Meru mountain at the centre and the nether worlds beneath. The gravitational force is towards 'Visnu Kurma' who supports the Brahmanda with the whole universe within it. Gravitational force is nothing but the will of the Omnipotent Lord working through material bodies. God's ways are wonderful, not easily understood just by human intellect. He has huge forms as well as forms minuter than the minutest. The 'Visnu Kurma' form enters into the various 'Vayu Kurma' forms giving gravitational forces to all things big and small, according to their masses.



Sri Laghu-Bhāgavatāmrta, chapter 3, Text 67, describes that Lord Kūrma acted as the support of the bhū-mandala, from the very beginning of the day of Lord Brahmā.

pādme proktaṁ dadhe kṣauṇi-mayam evārthitaḥ suraiḥ

śāstrāntare tu bhū-dhārī kalpādau prakaṭo 'bhavat

In the Padma Purāṇa, it is said that on the request of the ruling devatās, Lord Kūrma lifted the bhū-mandala (earth). In another scripture, also it is said that He appeared in the beginning of the kalpa and lifted the bhū-mandala (earth). Aryamā, the chief resident of Hiraṇmaya-varṣa, worship their beloved Lord Kūrma by chanting the following hymns (SB 5.18.30).

oṁ namo bhagavate akūpārāya sarva-sattva-guṇa-viśeṣaṇāyānupalakṣita-sthānāya namo varṣmaṇe namo bhūmne namo namo 'vasthānāya namas te.

I offer respects to the Supreme Lord, the tortoise (akūpārāya), a form of śuddha-sattva (*sarva-sattva-guṇa-viśeṣaṇāyā*), having an invisible abode of Vaikuṇṭha, who has a huge, all-pervading body (*varṣmaṇe*), and who acts as the support of all things (*avasthānāya).*

Ādi Śeṣa by himself is not competent to support the worlds. His strength is derived from the Supreme Lord Sankarsana, whom he worships in his heart. The space occupied by Ādi Śeṣa on Ghanodaka is 99,000 yojanas.
[In fact, the Serpent form supporting the worlds is Lord Sankarsana Himself; and He is known as "Ananta." Ādi Śeṣa is only like a tool in the hands of a master.]

Location of Hellish Planets[58]

On the left hand side (south[59], in this instance) lower than the visible earth level, there is a big moat, many yojanas wide and deep. At its edge is 'Samyamanipura', god Yamadharma's city, where he is always attending to his duties. There are thousands of his servants who are very cruel by appearance. The 'Vaitarani' river is here; and it is a terror to sinners. After crossing this river with a foretaste of the sufferings in store, they are taken down into hells like 'Raurava', which are horrible. Thousands of such hells are there for punishing sinners of different kinds. These hells are in the southern direction, at the level of the *'Asta-diggajas'* beneath Pātala loka. Another capital city of Yamadharma's kingdom is also in Heaven, which is the harbinger of happiness

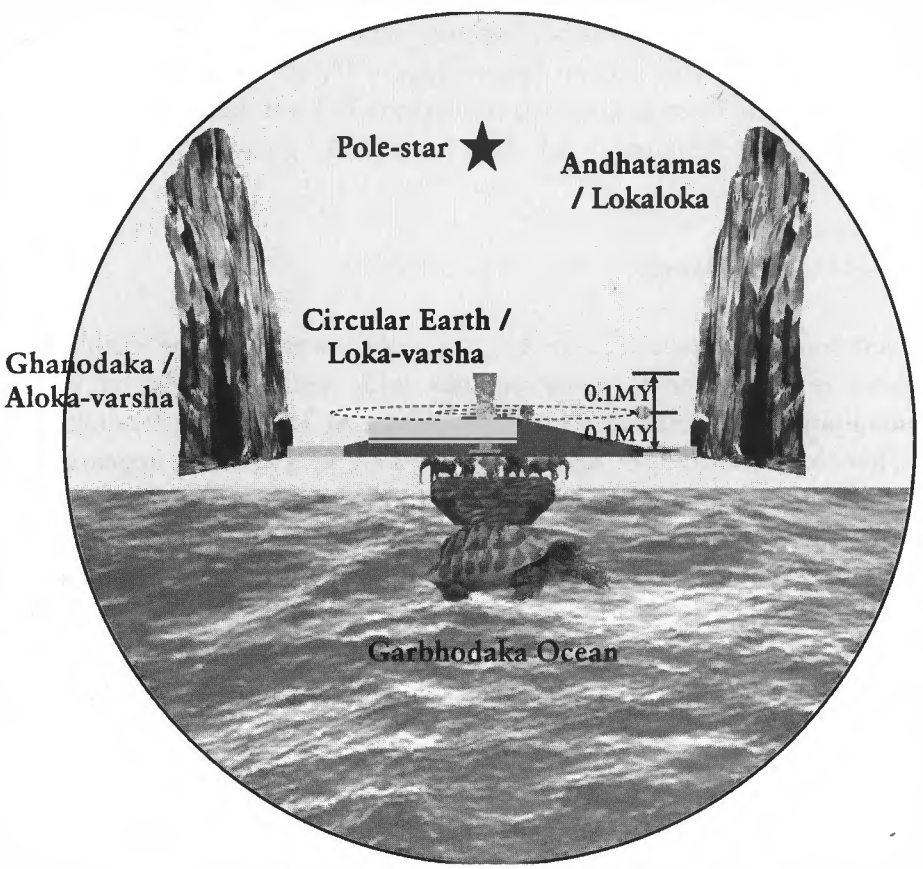
[58] Refer to the section "Location of Hells (103 - 107)"

[59] This south is not the south as discussed while describing the structure of Jambūdvīpa, instead it is True South, referring to the direction beneath the Bhū-maṇḍala or downword direction of the Universe

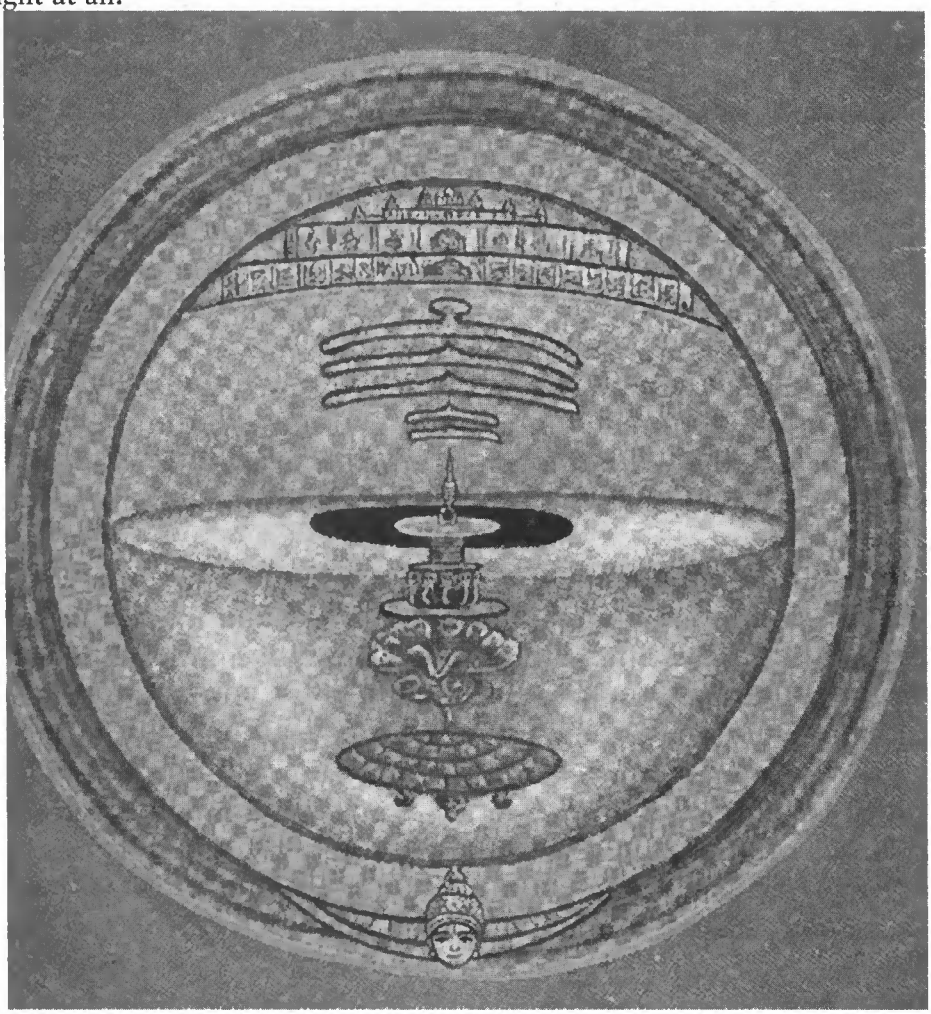
to those who have done meritorious deeds here on earth.

In Śrīmad-Bhāgavatam, the hells are said to be located beneath the earth, but above Ghanodaka. Since the hard earth extends downward, upto where it finds support on Ādi Śeṣa's head, the hells, which are in line with the directional elephants, are naturally above Ghanodaka and below the surface of the visible circular earth.

> "Here is substantial ground for us to conceive that the seven islands with their surrounding oceans are not flat like a single rupee coin; but they take a terraced type of arrangement. Thus, the Puṣkaradvīpa is two lakh yojanas lower than the upper edge of the Basket-like Jambūdvīpa; and the surface of Ghanodaka, beyond the region of darkness, is in line with Puṣkaradvīpa. The location of hells is slightly above this line."



Underneath the earth borne by Ādi Śeṣa, Ghanodaka extends to a depth of 24 crore, 99 lakh yojanas (249.9MY), where it touches the golden shell of Brahmanda. That shell itself is 100 crore yojanas in thickness; and it rests on the back of Visnu Kurma[60], almost like an atom. In the case of Ādi Śeṣa, the weight of the vast earth is compared to that of a mustard seed; in the case of Visnu Kurma, the weight of the Brahmanda, with the entire universe in it, is compared to that of an atom. Really speaking, this is only to make us to see the difference between Ādi Śeṣa and Maha Visnu In the case of the latter it is no weight at all.



[60] Refer to the section "Vishnu Kurma (106 - 109)"

Jambūdvīpa

The island in the middle of the Circular earth, consisting of seven islands and extending upto *svarna-bhumi* followed by *vajralepa*, is known as Jambūdvīpa. This Jambūdvīpa consists of nine varṣas, wherein the middle varṣa is Ilavrta varṣa, the residence of Lord Siva with his wife Bhavani, which has Mount Meru at its centre. The Meru is at the centre of Jambūdvīpa and its nine varṣas are Ilavrta (middle), Bhadrasva (east), Ketumala (west), Ramyaka, Hiranmaya and Uttara Kuru (north); Harivarṣa, Kimpurasa varsha and Bharatha varṣa (south).

The Sumeru Mountain[61]

The golden Meru Mountain is one lakh yojanas high. It is surrounded by the nine Khandas (Varṣas of Jambūdvīpa). It is the abode of Siddhas (souls who have attained a high degree of perfection). For all measurements concerning the universe, the Meru is the central point of reference. This mountain is thin at the base and goes on broadening gradually towards its top which is quite broad. The mountain resembles the "Unmatta" flower. At the base it has a width of 16,000 yojanas and it goes down into the circular earth to the extent of 16,000 yojanas. From the base up to the top, the Siddhas and yogis dwell.
At the top, the mountain has a width of 32,000 yojanas. It is round at the top and three sided at its base (like the cornered base of a nail). From the base to the top the mountain measures 84,000 yojanas. The circumference at the top is 96,000 yojanas (in round figures). There, in the middle, is the city of Brahma, known as "*Stakaumbhi*". Its palatial mansions are beautified with rare gems. The city is 10,000 yojanas long and equally broad. There, the Gandharva chiefs, the prominent ones among the Siddhas, sages like Vasista and gods headed by Indra worship Brahma seated on the throne of gems. At the edges of the Meru top there are eight cities of the directional gods, one in each of the eight directions. The Meru Mountain supports them all.

The Descent of River Ganga[62]

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[61] Refer to the section "The Meru Mountain (153 - 163)"
[62] Refer to the section "The River Ganga (164 - 171)"

River Ganga pours down constantly in the open courtyard of Brahma's palace such that he might use the waters for his *'sandhya vandana'* (worshipping Surya Narayana thrice a day morning, afternoon and evening). The other gods there also use it. By Brahma's behest the river Ganga takes four courses to come down to earth from the Meru top and to purify the denizens here. Even now we can see the southern branch. The branch 'Sita' joins the sea (Lavana Samudra) on the east. *'Alakananda'* pours into the southern sea and is quite famous as the purifier of the world. The branch 'Caksu' reaches the western sea and 'Bhadra' joins the northern sea. This river purifies those who use its waters in worshipful manner where ever they may be in the four quarters (of Jambūdvīpa). The river was again brought to our earth in another form by the efforts of Bhagiratha, and known as Bhagirati. This river is called as *'Mandakini'* in the Himalayas. When it began to pour down in the Himalayan region, Lord Siva locked it up in the meshes of his knotted hair. (This was to mitigate the force with which Ganga came down; hence the name *'Mandakini'*). Being let down from the knotted hair and following Bhagiratha the fresh flow of waters joined the original course of the river at 'Devaprayaga'. After crossing the point at 'Devaprayaga', the river went down into the bowels of the earth to reach 'Kapila asrama' in the nether world (specifically Patala loka) to sanctify the ashes of the 60,000 sons of Sagara cakravarti, burnt down by the angry look of Sage Kapila.

The Ganga River flows both in visible and invisible forms (*rupantarena vahati*), it is also in the form of the stars of the Milky Way known as "Akasa Ganga". It is said that if there is rain in the day time without any clouds in the sky, that rain water is directly dropped down from Akasa ganga. Whoever takes bath in that rain water is said to be completely purified, as a result of that bathing, and becomes eligible candidate for liberation.

Ilavrta Varṣa[63]

Ilavrta Khanda (or Varṣa) spreads around the Meru base, and it measures 9000 yojanas from its Meru base, in every direction. There are four supporting mountains on each of the four sides around the base. These mountains are each 2000 yojanas wide and 10,000 yojanas high.

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[63] Refer to the section "Ilavrta Khanda (171 - 183)"

The four supporting mountains are Mandara, Merumandara, Kumuda and Suparsva, in the four directions commencing from the east. There are four gigantic trees on the four mountains Cuta (mango), Jambu (rose apple tree), Plaksa (Indian fig) and Nyagrodha (banyan) respectively. Each of these trees is 100 yojanas high and the spread of the branches is to the extent of 1100 yojanas. The fruits of the trees resemble mountain peaks. The ripe jambu fruits of the tree upon Merumandara (south) drop down and the juice flows as the Jambu River. The juice spreading on its banks, when dried up, becomes the finest gold which is known as *'Jambunada'*. The divine beings, who visit this region, drink the juice and make different kinds of ornaments for themselves out of *'Jambunada'* (gold).

According to Śrīmad-Bhāgavatam, there are eight mountains, each 9000 yojanas long; two of them placed on each side. Since they are compared to the 'Kusa' grass bundles put around the sacrificial altar, each of the two mountains should be in a continuous line, with some gap or no gap between them.

Śrīmad-Bhāgavatam is the authoritative text, which tells us about Cosmography; and information gathered from any other source should be in accordance with it. That is what Acarya Madhva says:

yathā bhāgavate tūktaṁ bhauvanaṁ kośa-lakṣaṇam
tasyāvirodhato yojyam anya-granthāntare sthitam

In this Varṣa (specifically around the area of the two mountains Kailasa and Karavira, stretching from east to west, in the southern side of Mount Meru) there is only the fair sex moving about day and night. No male person enters it for fear of Bhavani Devi's curse. If anyone trespasses, he is transformed into female form like Ila, the son of Manu. The female attendants of Gauridevi are one hundred million (arbuda) in number. Lord Siva, along with Gauridevi, worships the form of Lord Samkarsana, one of the "*caturmurtis*", Vasudeva, Samkarsana, Pradyumna and Aniruddha. Lord Samkarsana is the sun brightening the hearts of yogis.

This Ilavrta Varṣa is bounded by four mountains on the four sides. On the eastern side is the Gandhamadana Mountain stretching between Nila and Nisadha mountains. It is 2000 yojanas in width. Of the same width, on the

southern side, is the Nisadha Mountain. It stretches from the eastern ocean upto the western ocean. This refers to the Lavana Samudra around Jambūdvīpa. On the western side is the Malyavan Mountain, lying between Nlla and Nisadha even like the Gandhamadana. On the northern side of Ilavrta is the Nila mountain, as wide and as long as Nisadha. Siddhas inhabit these mountains.

Bhadrasva Varṣa[64]

The eastern part of Jambūdvīpa is occupied by only one Varṣa, the Bhadrasva. There are two other boundary mountains within the Khanda, Jatara and Devakuta, to the east of Gandhamadana. The two mountains extend between Nila and Nisadha, each being 2000 yojanas wide. Barring the space occupied by the three mountain ranges (Gandhamadana, Jatara and Devakuta) the Bhadrasva Varṣa measures 27 thousand yojanas from west to east. Although Bhadrasva is a single Varṣa, the two extra mountain barriers mentioned, have been created by the Omniscient Lord to maintain symmetry on all the four sides. A similar arrangement is there in the western Varṣa. With the two additional mountain ranges the Bhadrasva Khanda is divided into three stretches, each nine thousand yojanas in width. It should be thus construed because it is said in the texts that each of the nine Khandas of Jambūdvīpa is 9000 yojanas in width. Symmetry is thus maintained.

Three varṣas in the South of Meru[65]

To the south of the mighty Nisadha Mountain is the Hari Varṣa. Its boundary mountain on the south is Hemakuta, 2000 yojanas wide and 10,000 yojanas high. Its length is 10% less than that of Nisadha. Therefore, it is 90,000 yojanas in length. To the south is Kimpurusa Varṣa, 9000 yojanas in width. It is bounded on its south by the Himavan Mountain, similar to Hemakuta in breadth and height. The length is less by 10,000 yojanas (in round figures) on account of the circular shape of Jambūdvīpa and the ocean surrounding it. Then comes Bhārata-varṣa (to the south of the Himavan mountain), having the same width as that of the neighboring Varṣa. It stretches up to the ocean on

[64] Refer to the section "Bhadrasva Khanda (183 - 193)"
[65] Refer to the section "The three Southern Varsas (193 - 208)"

the southern side. This is the Varṣa that provides opportunities for human beings to earn merit.

On a peak of the Himavan Mountain, the sacred "Badarikasrama" is situated. Here, Lord Narayana with Nara (Lord Nara-Narayana) is worshipped by expert yogis. Sage Narada and the seven rsis too worship Nara-Narayana, day after day. It is here too that the Lord in the form of Vedavyasa is served by Acarya Sri Madhva, the third '*avatara*' of Sri Vayudeva. There are many other '*munis*' following his example reside here.

The Himavan Mountain referred to is the boundary mountain between Kimpurusa Varṣa and Bhārata-varṣa - 2000 yojanas wide, 10,000 yojanas high and 80,000 yojanas long. But, the Himalaya Mountain with which we are familiar is only 50 yojanas wide, 1 yojana (5.5 miles, w.r.t sea level) high and 330 yojanas long. Its location is in 'Dandakaranya' of Bhārata-khanda (our globe), which forms just a big lump at the southern edge of Bhārata-varṣa, which has a width of 9000 yojanas. The Badarikasrama and the Kilasa in our Himalaya are only mini prototypes. Then, when it is said that Acarya Sri Madhva flew off from Anu Badari to Maha Badari, it means that he took a jump from our Himalaya to the bigger Himalaya covering a distance of 10,000 yojanas (60,600 miles) in height, this is the height of bigger Himalayas. This is nothing to be marveled at in the case of Sri Madhvacarya who is the avatar of Vayudeva, who enters into every one of the activities of this universe.

Ketumala Varṣa[66]

On the western part of Jambūdvīpa there is only one varṣa, the Ketumala resembling the Bhadrasva on the east. This extends upto the sea on its west. Here too, as in Bhadrasva Varṣa, there are two extra mountains in proper positions, maintaining the 9000 yojanas measure regarding the widths of the Varṣas. Pavana and Pariyatra are the two mountains. Lord Sri Hari has created them to make the island look symmetrically beautiful. "The last sentence of the above Paragraph speaks of the artistic sense of Saint Śrīla Vādirāja Tīrtha swāmi; and it also pays a humble tribute to Sri Maha Vishnu, the Artist of artists."

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[66] Refer to the section "Ketumala Varsa (208 -213)"

Three Varṣas in the North of Meru[67]

To the north of the Nila mountain is the Ramyaka Varṣa. The boundary mountain on the north is Sveta, which extends between the eastern and the western oceans with length 10% less than Nila. To its north is the Hiranmaya Varṣa bounded by Sringavan in the north. Its length is 10% less than that of Sveta. There are many Siddhas and rsis reside here. Beyond Hiranmaya is the Uttara Kuru Varṣa, inhabited by divine beings and rsis. This varṣa, 9000 yojanas wide, extends up to the ocean on the north. It is like its counter-part Bhārata-varṣa on the south. Here, Lord Sri Narayana is always worshipped in His Yajna Varaha (Bhu-Varaha) form by Bhudevi and the occupants of the Varṣa. Hari varṣa and the other two to the south of Ilavrta varṣa are similar to Ramyaka, Hiranmaya and Uttara kuru on the northern side. Each of these six varṣas extends from ocean to ocean, lengthwise.

When it is said of a few mountains that they are inhabited by Siddhas and rsis, it is indicative of the fact that all other mountain regions are similarly the abodes of sagely persons. In fact, that there is not an inch of space in this universe which is not made use of for some purpose or the other. Every mountain, every valley, every river has its own presiding deity with assistants to provide amenities to the devotees of the Lord. Thus the entire universe is populated by different living entities.

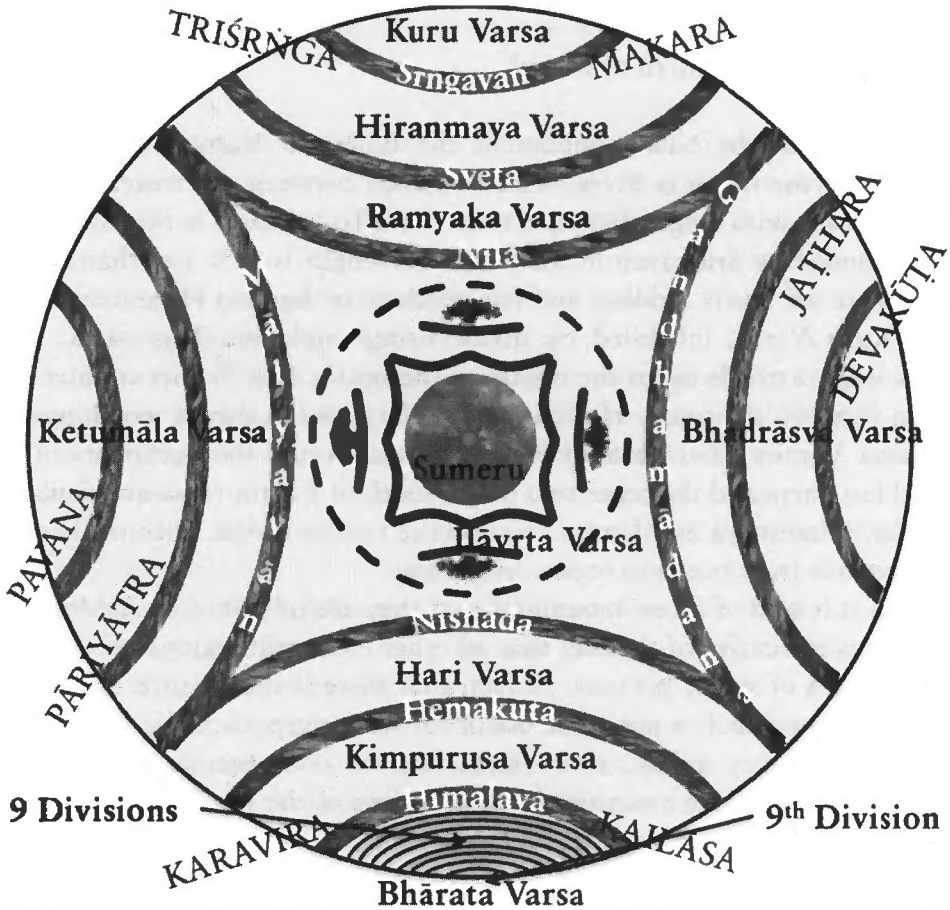
Four additional Mountains on Jambūdvīpa[68]

On the eastern side of Himadri, there is another Kailasa mountain wherein the three-eyed Lord Siva, the spouse of Umadevi, resides. On the western side of Himavan, there is the Karavira Mountain. In the same way, there are two mountains on each side of Srngavan (in the north of Meru) to the east and to the west. They are Trsrnga and Makara. Here too there are siddhas and rsis in large numbers. With these four mountains, the Meru appears to be strong with four arms.

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[67] Refer to the section "The three Northern Varsas (214 - 224)"

[68] Refer to the section "Four other Minor Mountains (224-227)"



"Ground plan of Jambudvip – Bhārata-varṣa with its 9 Divisions"

Mention of Kailasa as a separate mountain (in the east of Himavan, the bigger Himalayas) where Lord Siva resides with Umadevi reminds that Lord Siva's abode is in several parts of the universe, discharging various functions to bless the living beings and to earn merit for himself to a very high degree. He also resides at the Kailasa peak of the Himalayas of Bhārata-desa (present day India) in addition to the Kailasa Mountain mentioned above. His other two residences are at the foot of the Meru Mountain and in Tapoloka above. It means that he resides in all these places simultaneously and not frequently. Divine personalities are capable of being at different places at the same time with their fractional spiritual parts, having the fullness of the original for all purposes.

Lavana Samudra, the Salt Water Ocean[69]

The Jambūdvīpa, the vast land containing of nine varṣas with golden Meru at its centre, is the midmost island, with a diameter of one lakh yojanas. It is surrounded by the '*Lavana Sarnudra*' (ocean of blackish salt water), the width of which is also one lakh yojanas.

Before proceeding further to learn about the other six islands surrounding Jambūdvīpa, it is necessary to have a clear concept of Bhārata-varṣa and Bhārata-khanda which form parts of that island. Bhārata-varṣa is the southernmost division with the bigger Himalaya as its northern boundary and having a width of 9000 yojanas. Therefore, "Jambūdvīpa should be considered as being basketlike in shape for obvious reasons. It has an altitude of one lakh yojanas." This statement may be supported by the following argument.

The island Jambūdvīpa is circular. The orbits of the sun and the moon are almost in line with the Equator of Bhārata-khanda, our globe[70]. This is corroborated by the words of Śrīmad-Bhāgavatam which speaks of the sun's Udagayana or Uttarayana (northern course), Daksinayana (southern course) and Vaisuvadayana (course corresponding to the celestial equator). The orbit of Rahu is said to be 10,000 yojanas lower than that of the sun. Birds like Garuda (white-breasted eagle) and swan and syena (hawk) are said to fly in regions still farther down below. All these are not possible if the entire earth (with seven oceans and seven dvīpas), 5 crore yojanas wide, should be entirely flat. It must therefore have higher and lower parts, Jambūdvīpa being at the highest level.

Bhārata-khanda (our globe) is perked[71] at the southern edge of the basket-like Jambūdvīpa. It is not a planet going around the sun. This concept is substantiated by the statement of one, Sri Anantharamu of Bangalore, who had joined the crew of the first Indian Antarctic expedition. He says that "the south-pole region projects downwards like a cone. He also says that

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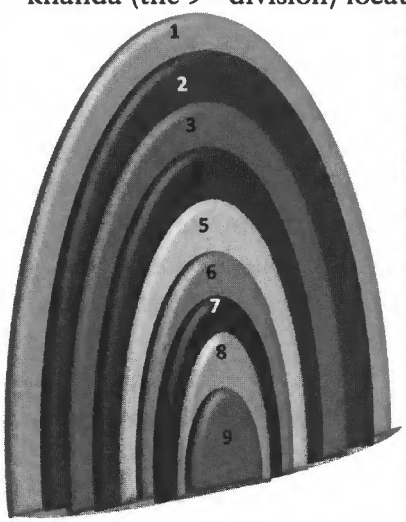
[69] Refer to the section "Lavana Samudra (227-229)"

[70] Refer to above section "Location of Hellish Planets"

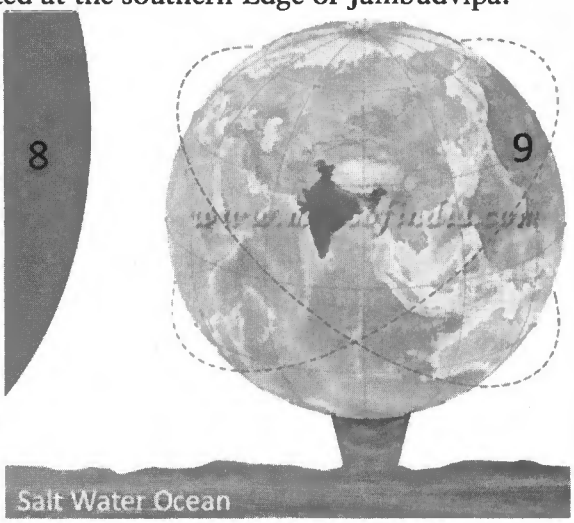
[71] Definition of Perk by Merriam-Webster: to stick up or out jauntily; to thrust up the head, stretch out the neck

this region has not been properly surveyed. Therefore, it is indeed a fiat lie to say that the south-pole region is slightly flat like north-pole region."

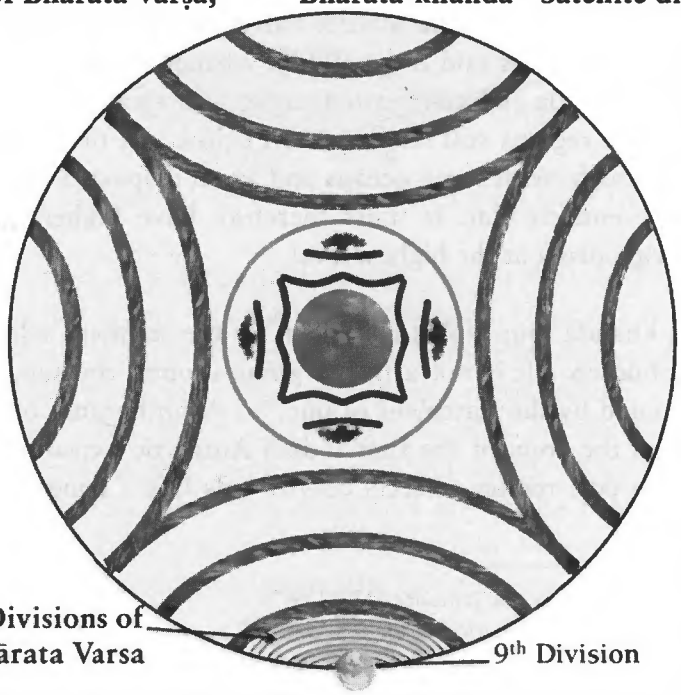
The below image shows the 9 Divisions of Bhārata-varṣa with Bhārata-khanda (the 9th division) located at the southern Edge of Jambūdvīpa.

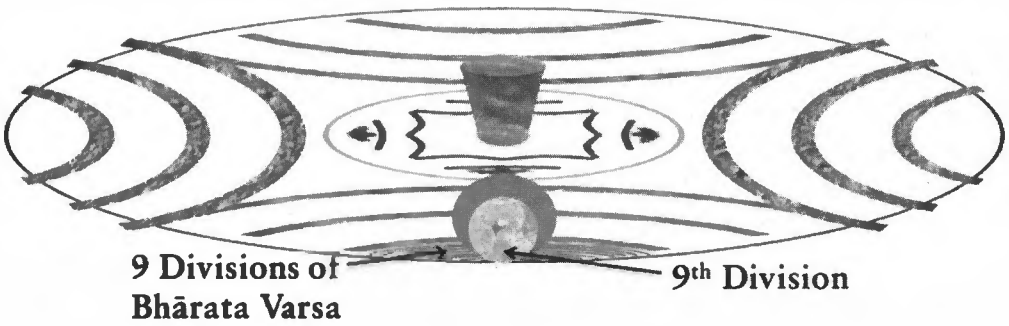


9 Divisions of Bhārata-varṣa;



Bhārata-khanda - Satellite directions





Remaining Six Islands of the Bhū-maṇḍala[72]

Surrounding the Lavana Samudra there is the Plakṣadvīpa, which is so called because it abounds with Plakṣa trees (Indian fig trees) of gigantic size. It is double the width of the ocean of brackish water or the ocean of salt water (i.e., two lakh yojanas). Of similar width is the 'Iksu Samudra' (ocean of waters with the taste of sugarcane juice) around Plakṣadvīpa, which is populated by Siddhas (the highly qualified souls). Around it is the Śālmalidvīpa, four lakh yojanas in width and the Sura Samudra (of wine-like waters), of the same width, spreads around it. After that there is Kuśadvīpa, where kuśa grass clusters grow very very tall like blazing pillars. It is double the width of Śālmalidvīpa. It is surrounded by Sarpi Samudra, ocean of waters with the qualities of ghee, having the same width. Then there is Krauñcadvīpa, which derives its name from the mountain of the same name. This island and the Dadhi Samudra (with waters of yogurt quality) have twice the widths of their inner complements. Each of these is 16 lakh yojanas wide. The next is Śākadvīpa surrounded by Ksira Samudra (ocean of milk-like waters), each 32 lakh yojanas in width. The last of the seven islands is Puṣkaradvīpa (island of huge lotus flowers of gold), 64 lakh yojanas wide. It is surrounded by the ocean of pure waters Subhodaka Samudra having the same width.

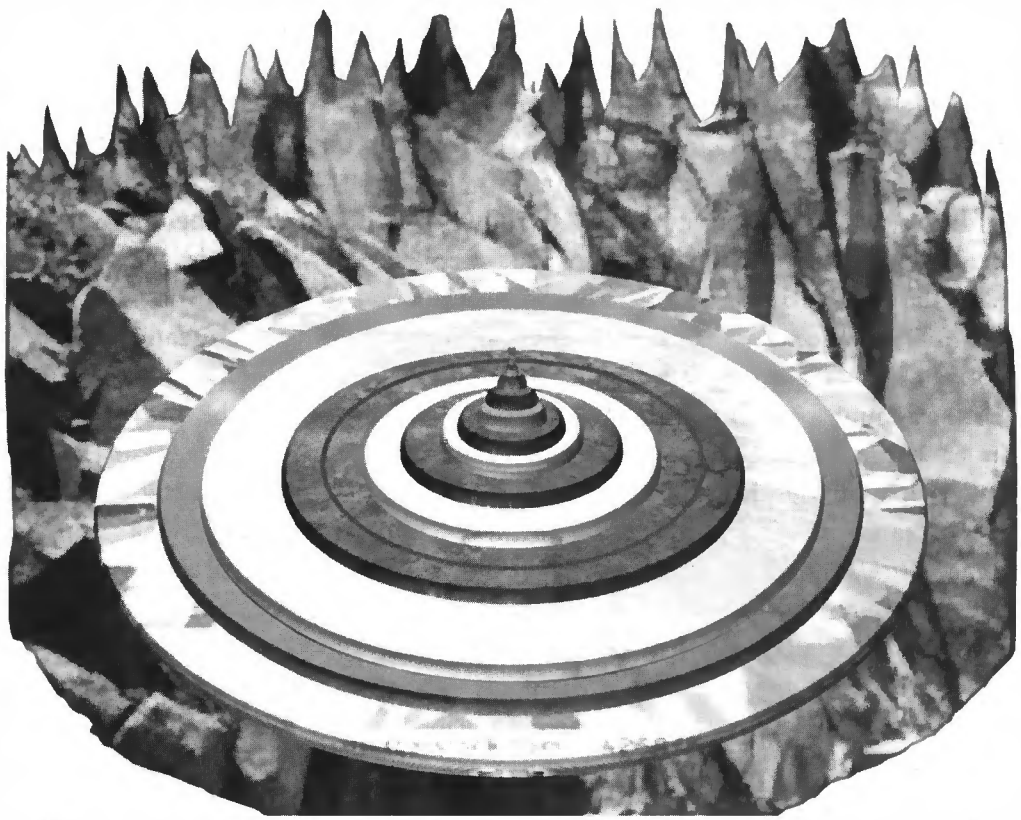
The entire circular earth is surrounded by the land of gold '*Hema bhumi*', along with the adjoining land of diamond '*Vajralepa*'. The Lokaloka Mountain goes round the whole stretch of earth described so far. The limit of the earth is marked by this mountain ring.

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[72] Refer to the section "The other Six Islands (230 -241)"

In this way, the earth spreads round the Meru centre on all sides and it is there to provide all the needs of living beings including the divine class, the sages, the pitas, the sovereigns as well as human beings. This earth provides happy conditions to one and all. The presiding goddess, "Bhudevi" is most dear to Lord Sri Narayana. The earth is plentiful with all kinds of plant life and holy forests, crowded with the hermitages of '*munis*' or sages. Rivers with holy waters are also there. Mother earth keeps its living beings happy and contented. There are rich mines containing precious and useful metals and the rarest gems. It is for this reason that the earth gets its name 'Vasundhara'. She is indeed the mother of the world displaying proverbial quality of forbearance. Being seated on the lap of Lord Varaha, she looks quite pretty. (3.13.40)

daṁṣṭrāgra-koṭyā bhagavaṁs tvayā dhṛtā
virājate bhūdhara bhūḥ sa-bhūdharā ।



"Seven Islands and Sevan Oceans, Surrounded by Loka-aloka Parvata"

Summary of Bhū-mandala - Būgola Varnanam Vs. Śrīmad-Bhāgavatam

	Iceland	Ocean	Width (MY)	R_{Outer} (MY)	Iceland	Ocean	Width (MY)	R_{Outer} (MY)
1	Jambūdvīpa	Salt water	0.05+0.1	0.15	Jambūdvīpa	Salt water	0.05+0.1	0.15
2	Plakṣadvīpa	Sugarcane juice	0.2+0.2	0.55	Plakṣadvīpa	Sugarcane juice	0.2+0.2	0.55
3	Śālmalidvīpa	Liquor	0.4+0.4	1.35	Śālmalidvīpa	Liquor	0.4+0.4	1.35
4	Kuśadvīpa	Clarified butter	0.8+0.8	2.95	Kuśadvīpa	Clarified butter	0.8+0.8	2.95
5	Krauñcadvīpa	Emulsified yogurt*	1.6+1.6	6.15	Krauñcadvīpa	Milk*	1.6+1.6	6.15
6	Śākadvīpa	Milk	3.2+3.2	12.55	Śākadvīpa	Emulsified yogurt	3.2+3.2	12.55
7	Puṣkaradvīpa	Sweet drinking water	6.4+6.4	25.35	Puṣkaradvīpa	Sweet drinking water	6.4+6.4	25.35
8	*Hema bhumi* *Vajralepa'* *Brahmabhumi*		6.4 4.2 -0.235	31.75 35.95 35.715	Golden Land		15.75	41.10
9	Dark Region with Lokāloka Mountain		71.430	107.14	Lokāloka Mountain		82.2	123.3
10	Ghandodaka		142.86	250.0	Āloka-Varṣa Free space		123.3 3.4	**246.6** 250

***Note1:** Considering the comparison of the descriptions of the 'Būgola Varnanam' with 'Śrīmad-Bhāgavatam', the location of the Emulsified Yogurt Ocean and Milk Ocean is interchanged. The descriptions given in 'Būgola Varnanam' are supported by Vāyu Purāṇa and also by Śrīla Madvācārya in his commentary to Śrīmad-Bhāgavatam and Mahābhārata.' Thus, we can also consider the descriptions of the 'Būgola Varnanam.'

Note2: Commentary by Bhagavat Prasādācārya on SB 5.20.25 – In Viṣṇu Purāṇa the position of the Milk Ocean and the Curd Ocean (Emulsified Yogurt Ocean) is interchanged. It should be interpreted as the position of a different Brahma-kalpa (day of Brahma).

Saint Śrīla Vādirāja Tīrtha swāmi has given a brief but precise description of the features of Brahmanda, not with a view to duplicate the details found elsewhere; but it is to say things without leaving room for doubts regarding locations, measurements and such other things. The way in which the text is prepared indicates that there must have been a good deal of confusion in the minds of the learned men of his days. He has removed misapprehensions in the clearest manner. In a concluding verse he himself says that it is so: "*vakyaartha kathana vyaajaat bhugolam iti varnitam*".

The Million Dollar Question

After hearing the descriptions of all these dvīpas or islands and their divisions known as varṣas, there arises a doubt that "Why Mount Meru, the other varṣas of Jambūdvīpa, the Bigger Himalayas and the other dvīpas of Bhārata-varṣa, which are very close to our earth sphere, are not seen by us?" But the people on the earth can see the moon, sun and many other planets, which are very far compared to Mount Meru or other parts of Jambūdvīpa.

The typical answer to this million-dollar question from the Madhva scholars is that "The upper worlds and even the neighbouring regions of our globe (perched at the southern edge of Jambūdvīpa) are all invisible to our eyes and to our instruments because they differ in composition although the entire universe is composed of the same five elements. It is said in the puranas *'adrśya dhātavah sarve samutpanna jala ādayah.'*"

They further emphasize on the importance of relying on the *sabdha pramana* over *pratyaksa pramana,* when it comes to such topics which are beyond our sense perception. They say that "The question of questions is

- Will our scientists ever get over the human tendency of succumbing to pride and prejudice in order to know the truth?
- Will they use a little introspection to shift from the mere physical to the spiritual?

Let the wise hold the torch of spiritual light against the darkness of materialism and let there be a Renaissance of religious culture based on Revelations.

How to understand this mystery?[73]

The creation of Brahmanda, the golden egg enveloping the universe, is out of the *gandha tanmatra* or *prithvi.* Inside the shell of Brahmanda *'ganda tanmatra'* principle is concentrated. The contents of the brahmanda are the five subtle material principles, the *tanmatras,* dissolved in the watery principle entering into Brahmanda. Watery principle concentrated as hardened water

[73] Refer to the section Annexure 5 – Pancikarana "Creation of Gross matter by *Alodana* or Chemical combination"

(Ghanodaka, Garbhodaka) to fill the lower half of the hallow shell, the upper half remaining as Subtle Sky. Earth is the formation of mineral salts, etc., settling down to occupy the middle part of the universe and above the surface of the hardened subtle waters (Ghanodaka). The formation of earth (5 crore yojanas in diameter) in geologistic and artistic manner is the hand of God.

All this is Vedic information, not scientific guess. We speak of suns and solar systems within the millions of galaxies. Have the scientists found at least one more solar system within our own galaxy? We also speak of galaxies of different types. Are there any galaxies seen in the southern celestial hemisphere except the two Magellanic Clouds[74]? If not, what is beyond there to the south? For the information of scientists it may said

- What looks like a galaxy is just a cluster of millions of divine vehicles in the form of stars, racing at unimaginable velocities around the upper worlds.
- There are many starlike planets, the abodes of Yaksas, Guhyas and the like, in the southern sky up to a certain extent.

Excepting the few conservatives among scientists, who have built up prejudicial complexes in themselves, all other great scientists have admitted the limitations of human knowledge and have declared that the mystery about the "beginning' is ever a mystery for them.

The following are the words of Einstein himself; one of the most leading scientists of our days. "The most beautiful and most profound emotion we can experience is the sensation of the mystical. It is the power of all true science. He to whom this emotion is a stranger, who can no longer wonder and stand rapt in awe, is as good as dead. To know that what is impenetrable to us really exists, manifesting itself as the highest wisdom and the most radiant beauty which our dull faculties can comprehend only in their most primitive forms this knowledge, this feeling is at the centre of true religiousness. My religion consists of a humble admiration of the illimitable superior spirit who reveals himself in the slight details we are able to perceive with our frail and feeble minds. That deeply emotional conviction of the presence of a superior

[74] The Magellanic Clouds are two irregular dwarf galaxies visible in the Southern Celestial Hemisphere; they are members of the Local Group and are orbiting the Milky Way galaxy. Because both show signs of a bar structure, they are often reclassified as Magellanic spiral galaxies.

reasoning power, which is revealed in the incomprehensible universe, forms my idea of God."

Vedic literature, therefore, comes to our aid where modern science fails and ends. Here, the main point to note is that the universe is not governed only by physical laws throughout as our scientists would fondly believe without actual experimentation; there are also divine laws working through natural and supernatural laws.

For example, the calculation of time is the same for all, depending upon the "Kala Cakra" (Time Wheel); but its application differs. A day of Brahma = 4,320,000 x 2000 human years; of other divine beings = 360 human days; of the pitas = 30 human days. Thus there is relativity as well as commonness of time in the three cases cited. Without putting to test this valid explanation how is it right to stick on to the Theory of Relativity (of Einstein and his recommended book 'The universe and Einstein' by Lincoln Barnet) or any other theory? Another case study of the measurement of stellar distances to examine further is given below. "However minutely precise may be our geometrical or other methods based on parallax, spectroscope, classification of stars according to luminosity and the like, how can we be sure of stellar distances and distances between stars and between galaxies. As some scientists have confessed, if it may be about 50% wrong, look at the difference that two stars said to be 100 trillion miles apart, will be only 10 million miles apart.[75]"

The fundamental error that is being committed today by philosophers, scientists and the common folk is their blind dependence on the theory of evolution and rejecting the truth about God and His creation. But the Vedic description of the creation is systematic and convincing. Therefore, in conclusion instead of simply depending on sensual perception, as most of the scientists usually do, let the mankind depend on the revelations of the Vedic Literature, which are more scientific than modern science. The Vedic Literatures are far beyond the human mistakes, precise in their explanations and are the word of God.[76]

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[75] Refer to "Vertical Structure of the Universe as per Bhūgola Varnanam" for more details

[76] Refer to SB 5.16.10 purport for Śrīla Prabhupāda's viewpoints in this regard

Understanding this concept through various Analogies

Analogy1: The water which appears in three different states behaves differently. The water as solid ice block is stationary, does not move, and held tightly within the grip of the hand. But in its liquid state, the water always tries to move towards the valley, thus it is called as '*nimna-ga*', that which flows downwards. Water requires a container to be held at one place. And when the water is in gaseous state, known as moisture, it cannot be seen, but perceived by touch, for it is spread all over. Thus, though it is the same water, when it changes its state and composition, it behaves differently and becomes invisible. Similarly, the other parts of the universe, although made up of same five elements, some appear and some does not appear because of the change in their composition.

Analogy2: Similarly the dust particles of earth, in solid stare can be held within the grip of the hand, when mixed with the water still can be held, though it is flowing down. But when it mixes with the air and moving along with the flowing air, it cannot be held in the hand though it is slightly perceived. Thus the other parts of the universe, though made up of the earth element, may not appear due to their different state of composition.

Analogy 3: The gold in our experience in this material world is '*jada*' or non-conscious. It just follows the will of the goldsmith and transforms into any shape that he wants. But the gold in the Spiritual world is *cinmaya*, full of consciousness and emotions. The gold or the golden ornaments of the Lord converses with the Lord and experiences the exchange of emotions. This principle applies to any other object or person in the Spiritual world, for there everything is *cinmaya* or conscious, and nothing is *jada* or unconscious. Trying to extrapolate the gap between the above two, the gold in the celestial realm is neither completely *jada* nor completely *cinmaya*, for the heavenly realm lies somewhere in the middle as per its constitution. When a devata makes his appearance and disappearance to a human vision according to his own will, his golden ornaments also make their appearance and disappearance simultaneously. Thus there the gold attains the constitution somewhat close to that of the demigods. Thus it is subtle in general and does not appear to the common person's vision.

Mahabhārata, Vana Parva, Nalopākhyāna Parva (Ch 50-77), Chapter 54: the Svayamvara of Damayanti (the Bhandarkar Oriental Research Institute edition) states some of the interesting details on the nature and the special characteristics of the *devatās* in contrast to the human beings.

sā paśyad vibudhān sarvān asvedān stabdha locanān |
hṛṣita sragrajohīnān sthitān aspṛśataḥ kṣitim ||23||

Damayanti then beheld all the gods (who assembles in her svayamvara ceremony assuming the same bodily appearance of Nala, the king of Niṣadhas) not perspiring, with fixed eyes and spotless garlands around their necks and seated without touching the earth.

chāyā dvitīyo malān asragrajaḥ sveda samanvitaḥ |
bhūmiṣṭho naiṣadhaścaiva nimeṣeṇa ca sūcitaḥ ||24||

She, on the other hand, saw Nala standing preeminent with his own shadow (unlike the devatas) full of perspiration and with floral wreaths stained with dust, and was also seated on the ground with blinking eyes.

The above two verses indicate that the devatās, whose garlands never fade, do not perspire, do not blink their eyes, do not touch the ground and specifically do not produce any shadow in front of the sun other lighting sources. Similarly the golden Mount Meru, other varṣas of Jambūdvīpa, the Bigger Himalayas and the other dvīpas of Bhu-mndala, though they are very close to the earth (on which we are living), cannot be seen because they are subtle to our vision, for they are all celestial abodes reserved for pure pious beings. Only with required purification or by the attainment of mystic siddhis all these places can become visible to one's eyes. Otherwise, they are inaccessible in this self-same body, for they are called "*agatam*", not approachable.

Then how can we accept their existence? Only through the scriptural references proving their existence we can believe their existence, for the authenticity of the pastimes establishes the authenticity of the places connected to those pastimes.

Below references in relation to Jambūdvīpa and Bhārata-varṣa establish the authenticity of the existence of these places.

SB 1.12.5-6: "... *jambūdvīpādhipatyaṁ ca ...*" – Yudhiṣṭhira Maharaja & Pāndavas

sampadaḥ kratavo lokā mahiṣī bhrātaro mahī |
jambūdvīpādhipatyaṁ ca yaśaś ca tri-divaṁ gatam ||
kiṁ te kāmāḥ sura-spārhā mukunda-manaso dvijāḥ |
adhijahrur mudaṁ rājñaḥ kṣudhitasya yathetare ||

O *brāhmaṇa*! Yudhiṣṭhira maharāja had all wealth, sacrifices, planets, queens, brothers, the earth, lordship over Jambūdvīpa, and fame in Svarga, desirable even for the *devatās*. Did these things give joy to the King, whose mind was only fixed on Mukunda without deviation and nothing else?

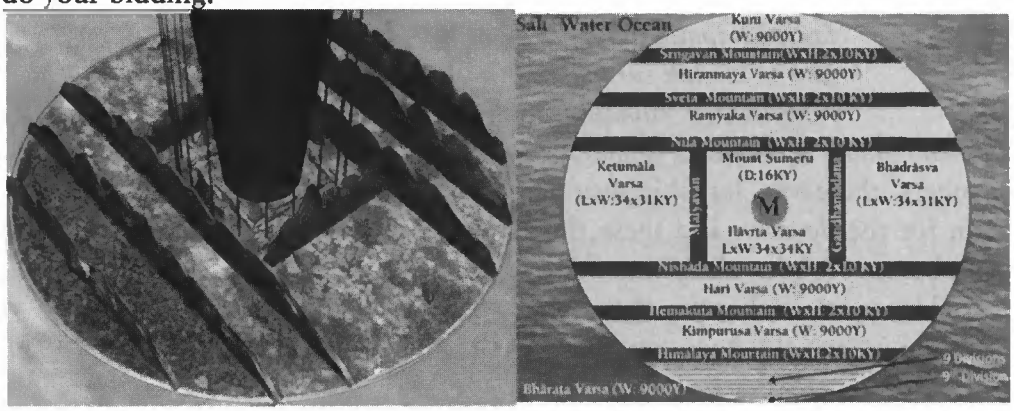
This verse indicates that Yudhiṣṭhira maharāja had rularship over Jambūdvīpa and he is considered as the monarch of whole Jambūdvīpa

The Rājasūya Sacrifice of Yudhiṣṭhira Maharaja (Mahābhārata, Sabha Parva)

Arjuna was becoming successful in his conquests in the northern direction. He first overpowered the king of the Kulindas, then the Anastas, then the Kalkuttas. He defeated the many kings on the island continent of Sakala, then encountered King Bhagadatta, a powerful warrior and defeated him after an eight days battle. Arjuna continued toward the north. He reached the mountains and vanquished the many barbarian tribes. Arriving in Himalayas, he subdued the region's kṣatriyas, and then passed over the Himālayas. To the north of the mountains lay the land inhabited by the Kimpuruṣas, semi-celestial beings born of Yakṣa women. Incensed at the approach of a human army, they fought a fierce battle, but finally Arjuna subdued Kimpurushas ruled by Durmaputra. The Pāṇḍava then conquered the land of the Guhakas called Harataka, another tribe of celestial fighters, followed by a region inhabited by Gandharvas.

At last the son of Pandu arrived in the country north of Harivarṣa and desired to conquer it. Thereupon certain frontier guards (from Northern Kurus province) endowed with huge bodies approached him with gallant hearts and advised him, "O son of Prtha, this tract of land can never be conquered by you. If you value your wellbeing, please return to your kingdom, for any human who enters this region, will surely perish. We are satisfied with you prowess; O hero, your conquest must stop here. This is where the northern Kurus live, and war does not exist here. Even if you were to enter this region,

you could not see anything for human eyes cannot perceive the subtle forms that exist here. If, however, you desire something, please tell us so that we may do your bidding."



"If your land cannot be seen by human beings," Arjuna said, "then I shall not enter it. However, I desire that some offering be paid to King Yudhiṣṭhira as tribute." Immediately, guards brought celestial cloths made of the finest silk, and they also brought dazzling jewelled ornaments and skins. After conquering all countries of the north and extracting tribute and subordination from those kings, Arjuna brought back the enormous wealth to King Yudhisthira. Thus Arjuna got the name Dhananjaya, the one who conquered the unlimited wealth.

The Forest dwelling days of the Pāndavas (Vana parva of Mahābhārata)

After Arjuna had left for the Himālayas to appease Lord Siva for obtaining Pasupatasra and the Devine weapons of various demigods, his brothers and Draupadī continued to live in the forest known as Kāmyaka vana. They missed him sorely, and they had no idea how long he might be away or whether or not he would ever return. But they knew they would only be successful in their fight to regain the kingdom with Arjuna's assistance. Thus they prayed for his safe passage to the mountains and for his success in gaining the celestial weapons.

One day the great sage Nārada visited the Pāndavs and instructed them to visit various holy places. Just after Nārada departed Lomaśa Ṛṣi arrived and informed them about Arjuna's accomplishments. "O heroes, while journeying throughout the worlds I went to Amarāvatī, Indra's great city. There I saw the exalted king of the celestials and, to my astonishment, your brother Arjuna

sharing his throne. Indra asked me to come to you and to assure you of Arjuna's welfare. Thus I have come with all speed." Lomaśa told the Pāṇḍavas how Arjuna had acquired the celestial weapons, including Siva's famous weapon. He had also learned to sing and dance from the Gandharvas. He would be returning soon. "Your brother asked me to lead you to the tirthas so that you may gain pious merits. Indra also made this request, and it is my desire as well. Although I have already been to all the tirthas twice, I shall go for a third time with you."

After visiting many sacred tirthas and bathing in numerous holy rivers and lakes they reached the Himālayas, where Lomaśa warned the party to proceed with caution. "There are great Himālayas and the peaks of Kailāsa, Gandhamādana, Triśṛṅga, Makaragiri, and Mount Meru" he said, pointing ahead to the splendid mountain rising into the clouds. "Here reside thousands of invisible celestial beings. The Kimpuruṣas, Yakṣas, Kinnaras, Suparṇas, Nāgas and Rākṣasas, all as swift as the wind and as powerful as a thousand mighty elephants, move about on these mountain ranges. Sinful men cannot gain access here, for they will be destroyed by the celestials." Lomaśa had told the brothers that they would meet Arjuna in the mountains as he descended from heaven onto the summit of the Gandhamādana Mountain. They decided to go there and await Arjuna's arrival. As they felt fatigued, Bhīma summoned his son Ghaṭotkaca, who along with his raksasa friends carried them through skyways keeping them on their back. The Rākṣasas flew swiftly through the heavens along the Siddha's path. As the party travelled they saw beneath them the beautiful lands inhabited by Vidhyadharas, Kinnaras, Kimpuruṣas, Gandharvas, and other divine beings. Finally, they all arrived at the great Himālayan Mountain. The Rākṣasas took them high onto the mountain to Badarīkā Ashram, the ancient hermitage of Nara and Nārāyaṇa Ṛṣis.

The Pāṇḍavas remained in Badarīkā Ashram for six days and nights. On the seventh day, a wind blew up from the northeast, carrying a single celestial lotus. The thousand-pedalled golden lotus flower fell at Draupadī's feet, who desiring more requested mighty Bhīma to fetch some of them. When Bhīma climbed up the mountain, he had an encounter with an effulgent monkey, who questioned Bhīma that "Who are you? Where are you going? Do you not know that you cannot proceed further? This path leads to heaven and men cannot access it. Only those who are successful in ascetic practices can go to the

celestial regions. Therefore, give up your quest and turn back. O foremost of men, do not try to force your way past and thus die for nothing." Then after a brief pleasing conversation, being requested by Bhīma, Hanumān showed him the way to the forest where the thousand-petalled lotuses grew. "There is the path which leads to the Saugandhika forest, and there you will see Kuvera's gardens, which are guarded by the Yakṣas and Rākṣasas. In a great lake lie the flowers which you seek for your wife."

As Bhīma approached Saugandhika to fetch some flowers, the thousands of Yakṣa and Rākṣasa guards of Kuvera stopped him for they are the guardians of that lake. Bhīma fought with the guards and killed them in thousands. Yudhiṣṭhira understanding that Bhīma had disturbed the celestials by attempting to take the lotuses from them, quickly went there and apologized for his brother's acts and consoled them with gentle words. With Kuvera's permission they stayed there for some time.

SB 4.6.23: At the foot of Kailāsa, demigods saw Alakāpuri & Saugandhika forest

Lord Brahmā pacified the *devatās*, who visited him being mutilated by Virabadra and his followers for siding Daksa, in his offence to Lord Śiva and the death of his dear daughter Sati. After instructing all the *devatās,* Lord Brahmā along with the Pitās, *devatās* and the lords of the progeny, departed from his place and went to the abode of Lord Śiva, known as the Kailāsa, the best among mountains, which is very dear to Lord Śiva. Kailāsa is always filled with persons perfected by *yoga, mantra,* austerity, herbs and birth, and with *devatās*, Kinnaras, Gandharvas and Apsaras. The mountain is surrounded by the Gaṅgā waters, made more sacred by the bathing of Satī. The *devatās,* seeing that mountain of Śiva with wonder, went there. They saw at that mountain a pleasant city called Alakā and the forest called Saugandhika where there were lotuses called *saugandhika.*

dadṛśus tatra te ramyām alakāṁ nāma vai purīm
vanaṁ saugandhikaṁ cāpi yatra tan-nāma paṅkajam

Outside the city were the rivers Nandā and Alakanandā, purified by the dust of the lotus feet of the Lord. Alakā was filled with hundreds of airplanes made of gold, pearls and precious jewels and with women. It was like the sky filled with clouds and lightning. The *devatās* passed through Alakā, the city of Kubera and the Yakṣas, and the Saugandhika forest, pleasant with desire trees

having colorful flowers, fruits and leaves. Seeing that forest used by the Kimpuruṣas and the pools which were decorated with lotus garlands, furnished with steps of cat's eye jewel, the *devatās* then saw from far off the banyan tree of Śiva.

vaidūrya-kṛta-sopānā vāpya utpala-mālinīḥ
prāptaṁ kimpuruṣair dṛṣṭvā ta ārād dadṛśur vaṭam

The tree was a hundred *yojanas* high and its branches extended seventy-five *yojanas.* Its shade was uninterrupted. It had no bird nests, and was cool. Under that tree which was the shelter of aspirants for liberation, and which bestowed great powers, the *devatās* saw Śiva sitting, appearing like death, but seeming to have given up his anger. (SB 4.6.8-33)

SB 4.10.4-5: "rudrānucara-sevitām dadarśa himavad-droṇyāṁ" – Dhruva Maharaj

When Dhruva Mahārāja heard of the killing of his brother Uttama, overwhelmed with lamentation and anger, mounting his victorious chariot, he went to the city of the Yakṣas, Alakāpurī. Going north, the King saw the city filled with Yakṣas and followers of Śiva in a valley of the Himalayas.

gatvodīcīṁ diśaṁ raja rudrānucara-sevitām
dadarśa himavad-droṇyāṁ purīṁ guhyaka-saṅkulām

SB 1.16.13: "*bhadrāśvaṁ...jagṛhe*" - Parīkṣit Mahārāja the king of Jambūdvīpa

When Parīkṣit Mahārāja, expert in battle, was staying in Kuru-jāṅgala inspecting his kingdom, hearing the rather unpleasant news that Kali had entered, he took up his bow. He had gone out from his city to gather tribute with his troops along with chariots, horses, and elephants on a chariot with lion flag, pulled by black horses and well decorated. Conquering Bhadrāśva, Ketumāla, Bhārata, the northern Kuru provinces, and other countries such as Kimpuruṣa, he received taxes.

bhadrāśvaṁ ketumālaṁ ca bhāratam cottarān kurūn |
kimpuruṣādīni varṣāṇi vijitya jagṛhe balim ||

Madhvacharya travels to Upper Badarīkāshram of the Greater Himālayas

It is widely accepted that Madhva was an incarnation of the demigod Vayu,

said to be in charge of controlling the wind. The idea that he descended for a particular mission and was thus greatly empowered to perform many uncommon deeds is offered as an explanation for the extraordinary events that surrounded his life. Uncommon feats of strength, flying in the sky, causing on two separate occasions plants to sprout from seeds by singing a particular, eating four thousand bananas and one hundred pots of milk at one sitting, using small amounts of food to feed hundreds, meeting Vyasadeva at Badrinath, etc., are only a few of the wonderful feats of Madhvacarya. Being called by Srila Vyasadeva to meet him in Upper Badarīkāshram, Madhvacarya easily jumped from lower Badarīkāshram of our Himalaya to upper Badarīkāshram of the bigger Himalaya covering a distance of 10,000 yojanas in height, this is the height of bigger Himalayas. This is nothing to be marveled at in the case of Sri Madhvacarya who is the avatar of Vayudeva, who enters into every one of the activities of this universe.

SB 9.1.25-26: At Meru's foot in Sukumāra forest, Sudyumna become a woman.

That hero Sudyumna, accompanied by a few ministers and associates and riding on a horse brought from Sindhupradeśa, wearing armor and holding an attractive bow and astonishing arrows, went in the north pursuing animals while hunting in the forest. There in the north, at the foot of Mount Meru, he entered a forest known as Sukumāra where Lord Śiva always enjoys with Umā.

sukumāra-vanaṁ meror adhastāt praviveśa ha
yatrāste bhagavāñ charvo ramamāṇaḥ sahomayā

When Sudyumna, who was expert in subduing enemies, entered the forest, he saw that he had become a woman and his horse had become a mare. When his followers also saw their gender reversed, they looked at each other in confusion.

Why was this place so empowered, and who made it so powerful? Some sages of fixed vows and whose own effulgence dissipated the darkness of all directions once came to see Lord Śiva in that forest. When the goddess Ambikā saw the great saintly persons, she was ashamed because at that time she was naked. She immediately got up from the lap of her husband and put on her cloth. Seeing Lord Śiva and Pārvatī engaged in intimacy, all the sages refrained from approaching and departed for the hermitage of Nara-Nārāyaṇa. Thereupon, just to please his wife, Lord Śiva said, "Any male entering this

place shall immediately become a female!" Since that time, no male had entered that forest. But now King Sudyumna, having been transformed into a female, began to walk with his associates from one forest to another. This pastime is related to Ilāvrta Varṣa.

SB 3.23.39-: Kardama travels along with his wife Devahūti in the arial mansion (airplane)

Using that airplane, praised by the Siddhas, Kardama, surrounded by damsels like Kuvera, enjoyed for a long time in the valleys of Meru Mountain (*Kulacalendra*), the pleasure ground of the deities of the eight directions, with its pleasant breezes, the companions of Cupid, and which reverberated with the sound of auspicious water walls of the Gaṅgā (Dyudhuni). Pleased with his wife, he enjoyed in Vaiśrambhaka, Surasana, Nandana, Puṣpabhadraka, Mānasa Sarovara and Caitrarathya, which are the gardens of the *devatās*. He surpassed the *devatās* with their airplanes as he traveled like the air to other planets on his giant, glowing vehicle propelled by his desire. After showing his wife the greatly amazing territory (*maṇḍala*) of the earth (*bhuvo golaṁ*) with its structure (*sva-saṁsthayā* means with its composition of islands and continents), the great yogī returned to his hermitage.

prekṣayitvā bhuvo golaṁ patnyai yāvān sva-saṁsthayā
bahv-āścaryaṁ mahā-yogī svāśramāya nyavartata

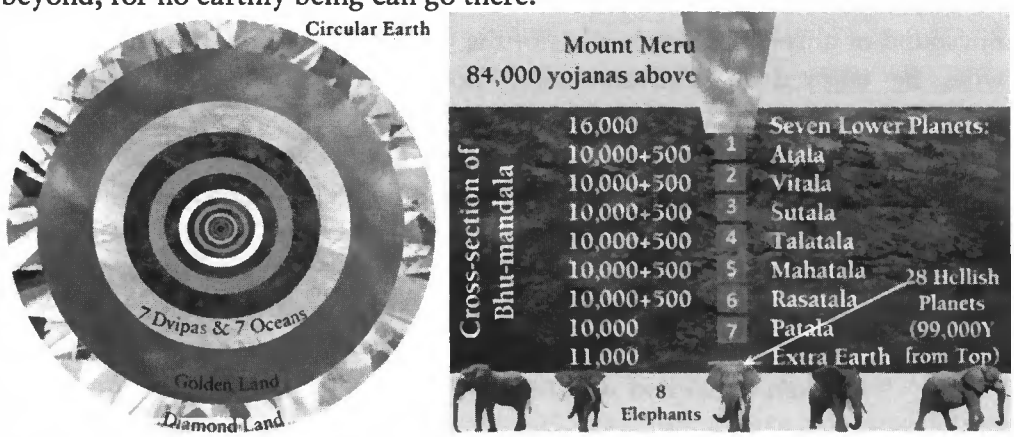
References in relation to the Seven Dvīpas of Bhū-maṇḍala

Kiṣkindhyā-kāṇḍa of Vālmīki Rāmāyaṇa: Monkeys on four directions

Rāma then said, "Our first mission is to locate the abode of Rāvaṇa and ascertain whether Sītā is still alive. My dear Sugrīva, you should organize the search parties and then, when Sītā is found, I will give the necessary orders."

Being so ordered, Sugrīva immediately called for the monkey king, Vinda, and ordered, "I want you and your soldiers to search everywhere in the Eastern direction, including the seven oceans and seven islands. Beyond the ocean of milk there is an ocean of fresh water, wherein lies the subterranean fire. Thirteen yojanas from the Eastern shore of the fresh water ocean is the mountain called Jatarupasila, which is made entirely out of golden rocks. It is there that Lord Ananta Śeṣa resides, supporting the entire earth. This marks the outer boundary of the Eastern Quarter. I am giving you one month to

search through this entire area. Anyone who reports back later than that will be executed for neglecting the royal order."
Next, Sugrīva dispatched Aṅgada, Nīla, Hanumān, Jāmbavān, Mainda, Dvivida and others to the South, making the son of Vāli their leader. After describing the Southern region, Sugrīva instructed the monkeys, "Eight hundred miles beyond the Northern shore of the salt ocean is an island that I think must surely be the residence of Rāvaṇa. Further to the South is Bhogavatī, the capital of Rasātala, which is presided over by Vāsuki. Then, beyond the Ṛṣabha mountain, which marks the furthest boundary of Bhū-maṇḍala, is the abode of Yamarajā. Do not search the Pitrlokā or anywhere beyond, for no earthly being can go there."



Sugrīva then dispatched Susena and his followers to the West. Instructing them, he said, "In the middle of the salt ocean is Prāgjyotiṣapura, the City of Narakāsura, and the Western limit is where the sun sets. Beyond that there is no information as to what exists, and so you should not go there." Finally, Satabali was dispatched to the North, and Sugrīva informed him, "You will first come to the land of the Mlecchas, and then the Himalayan Mountains. Beyond these mountains is 800 miles of desolate land. After that there is Mount Kailāsa, where Kuvera resides. Beyond Kailāsa is Krauncha mountain, then Uttara-Kuru province, and finally, the northern salt-water sea, in the middle of which is a golden mountain called Somagiri. Therein, Lord Viṣṇu, Brahmā and Śiva reside. When you see that mountain, you should turn back, for it is not possible to go beyond that."
After the monkeys had departed, Rāma inquired from Sugrīva, "How is it that you have acquired such extensive knowledge of the earth's geography?"

Sugrīva replied, "After Vāli returned to Kiṣkindhyā and banished me, even though I voluntarily surrendered the kingdom to him, he remained angry and continued to chase after me. Thus, out of fear for my brother, I continually wandered over the surface of the entire earth seeking shelter. Finally, after some time, Hanumān informed me of Matanga's curse."

Below references indicate that these kings have ruled the seven dvīpas of the Bhū-mandala during their reign:

- SB 9.2.26: *"tasyāvikṣit suto yasya maruttaś cakravarty abhūt"* – King Marutta
- SB 9.4.15-16: *"ambarīṣo mahā-bhāgaḥ sapta-dvīpavatīṁ maḥīm"* – Ambarish Mah
- SB 9.6.34: *"sapta-dvīpavatīm ekaḥ śaśāsācyuta-tejasā"* – King Mandata
- SB 9.8.4: *"sagaraś cakravarty āsīt sāgaro yat-sutaiḥ kṛtaḥ"* – King Sagara
- SB 9.11.6: *"aprattaṁ nas tvayā kiṁ nu bhagavan bhuvaneśvara"* – Lord Rāma
- SB 9.19.23: *"bhū-maṇḍalasya sarvasya pūrum arhattamaṁ viśām"* – King Pūru

  Note: Ambarīṣa was the emperor of the seven islands and then Pūru, s/o Yayati, became the next emperor. When the members of moon dynasty become emperors of 7-islands, then the members of sun dynasty become the kings of Bhārata-varṣa and vice-varsa.
- SB 9.20.23: *"pitary uparate so 'pi cakravartī mahā-yaśāḥ"* King Bharatha, s/o Duṣmanta and Shakuntala
- SB 9.23.24: *"arjunaḥ kṛtavīryasya sapta-dvīpeśvaro 'bhavat"* – Kārtavīrya Arjuna

Below reference describes the travel to Bila-svarga (Pātāla) by the sons of King Sagara and the descent of the River Gangā[77]

Sagara Mahārāja worshipped the Lord who is the supreme controller, who is the soul of the Vedas, the soul of the *devatās,* and the soul of all beings by horse sacrifices under the instructions of his spiritual master, the great sage Aurva. But, Indra stole the sacrificial horse and kept it in the Ashrama of Lord

[77] SB Canto 9 Chapters 8 & 9; Ramayana Bala Kanda, Chapters 39 to 43

Kapila. King Sagara had two wives, Sumati and Keśinī. The sons of Sumati, who were very proud of their prowess, following the order of their father Sagara, searched for the lost horse. While doing so, they dug up the earth (Jambūdvīpa) everywhere[78].

sumatyās tanayā dṛptāḥ pitur ādeśa-kāriṇaḥ
hayam anveṣamāṇās te samantān nyakhanan mahīm

"Oh, sons, you may therefore proceed to search for the horse and its thief on entire earth as far as it is garlanded by the ocean, and safety betides you all. Oh, sons, let each prince advance searching one square yojana of earth, by my order you dig up the earth until the appearance of the horse, while searching for the stealer of that horse." on going round the earth in its entirety those great-mighty princes did not find that horse, then as said by their father they mapped the earth to a unit of one square yojana for each of them, and each of them hollowed out each and every area on the surface of earth with their arms that have the touch of the diamonds. Mother earth bewailed while she is ruptured with spears also that are similar to thunderbolts, besides with very grueling ploughs. there emerged an unbearable rumpus from the serpents, asuras, demons, and other beings living underneath the surface of earth, while they are killed during hollowing the earth. Thus sixty thousand square yojanas of the earth is dug over, so as to make the earth's outermost plane as the unsurpassed rasātala, the sixth subterranean and the nethermost plane.

The sons of Sagara have endeavored everywhere digging over the Jambūdvīpa, which is verily congested with mountains in this way. Then on hollowing the entire earth and on making trips around her without finding the horse, all of the sons of Sagara have come together and ongoing to their father they spoke to him. To which, he ordered them 'Let the earth be dug further, may safety be with you, and verily splitting open the surface of the earth you catch hold of the thief of the horse, and on achieving your purpose you may return to me.' On catching up the words of their father, the great-souled Sagara, his sixty thousand sons rushed towards the surface of the rasātala, the netherworld. While digging the earth there they beheld a mountain similar easterly elephant named Viruupaaksha, which is bearing the eastern side of earth's surface on its head. That great easterly elephant Viruupaaksha is bearing the entire eastern earth along with its mountains and forests on its head. On certain occasions,

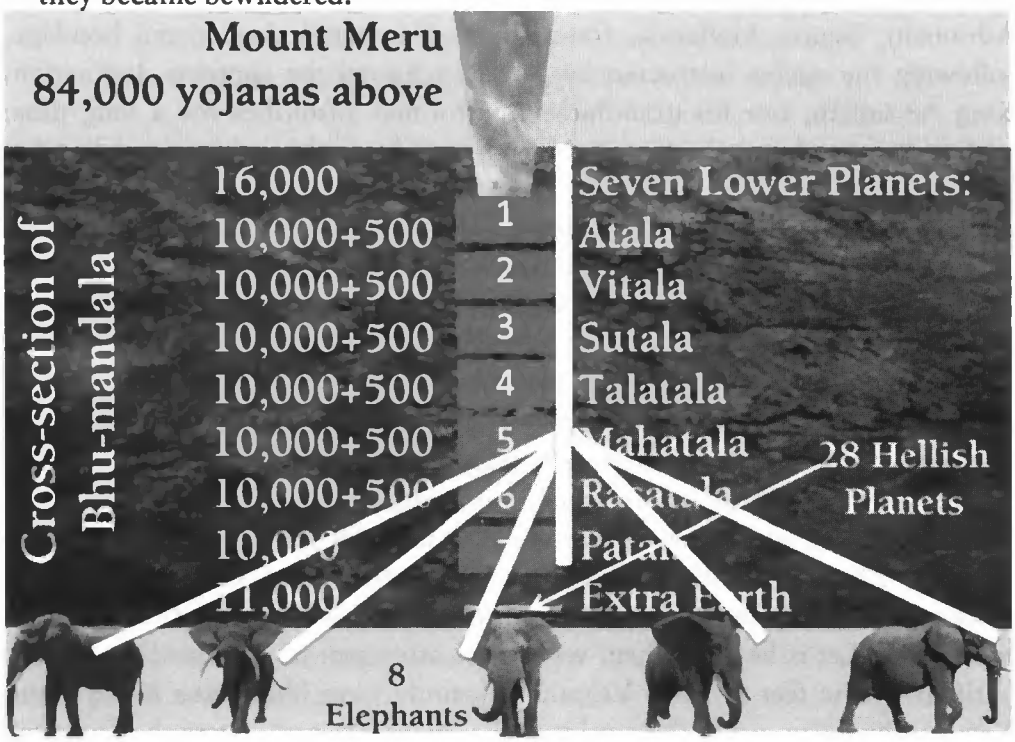
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[78] SB 9.8-9: 60,000 Sons of Sagara Mahārāja dug the Jambūdvīpa all the way upto Pātala-loka

when that great-elephant moves its head desiring respite in tiresomeness then earthquakes will occur on earth. Then they also met three other directional elephants (Mahapadma, Sumanasa, and Bhadra) after splitting the earth.

The sixty thousands sons of Sagara further burrowed to the surface of the earth, and then they dug towards the auspicious northeast direction. Thereafter, in the northeastern direction, they saw the horse near the hermitage of Kapila. "Here is the man who has stolen the horse," they said. "He is staying there with closed eyes. This sinful man should be killed!" The sixty thousand sons of Sagara, raising their weapons, attacked the sage. The sage opened his eyes. The sons of Sagara, losing their intelligence because of Indra, disrespected a great person. Consequently, by fire emanating from their own bodies, they were immediately burned to ashes.

Note: The third material element fire increased in their bodies because of their offense and burned them. Because they thought that Kapila was Indra, they became bewildered.



60,000 sons of Sagara searching the lost horse dug up the earth by 70,000 yojanas

Another son of Sagara Mahārāja named Asamañjasa was born from the King's second wife, Keśinī. The son of Asamañjasa was known as Aṁśumān, and he worked for the benefit of Sagara Mahārāja, his grandfather. Aṁśumān, ordered by the King to search for the horse, followed the same path dug up by his uncles. He saw the horse near a pile of ashes. The great Aṁśumān, seeing the sage Kapila, who was Viṣṇu himself, offered respects with folded hands and recited prayers with attention. The great sage Kapila, the powerful incarnation of Viṣṇu, being very merciful to Aṁśumān, spoke to him with intelligence, "O son! Take this sacrificial horse belonging to your grandfather! Your ancestors, who have been burnt to ashes, can be delivered only by Ganges water, and not by any other means."

Aṁśumān circumambulated Kapila and pleased him by bowing his head. Aṁśumān brought back the horse, and with this horse, Mahārāja Sagara performed the rest of the sacrifice. After delivering charge of his kingdom to Aṁśumān, Sagara Maharaja, freed from all material desires and bondage, following the means instructed by Aurva, achieved the supreme destination. King Aṁśumān, like his grandfather, performed austerities for a long time, with a desire to bring the Gaṅgā. Nonetheless, he could not do so, and in due course of time, he died. Like Aṁśumān, Dilīpa, his son, was unable to bring the Gaṅgā to this material world, and he also died in due course of time. Then Dilīpa's son, Bhagīratha, performed very severe austerities to bring the Gaṅgā to this material world. Gaṅgā appeared before King Bhagīratha and said, "I am pleased. I will grant you a boon." The King bowed his head before her and explained his desire to deliver his ancestors. To this Gaṅgā replied, "When I fall from the sky to the earth (*mahī-tale*), who will bear the force? If I am not sustained, I shall break the earth (*bhū-talaṁ*) and end up in Rasātala." To this Bhagīratha said, "Lord Śiva, the soul of all beings, in whom the universe exists like a cloth with vertical and horizontal threads, will sustain the force of your current." After saying this, king Bhagīratha satisfied Lord Śiva by performing austerities. Quickly satisfied with Bhagīratha, Lord Śiva accepted the proposal by saying, "Let it be so." Then, with great attention, he sustained the Gaṅgā purified by the feet of Lord Viṣṇu. The saintly king Bhagīratha brought the Gaṅgā, which can purify the world, to that place on earth where the bodies of his forefathers lay burnt to ashes. She flowed behind Bhagīratha mounted on his chariot moving as fast as the wind and purified many places. She then

sprinkled water on the sons of Sagara who had been burned to ashes. By their ashes being sprinkled with that water, the sons of Sagara who had been destroyed as punishment for offending a *brāhmaṇa* attained Svarga.

Conclusion: Considering all the above references it is clearly seen that the great Kings of Satya Yuga and Treta Yuga ruled the entire seven islands of the *bhū-maṇḍala,* as their single monarch, but as time progressed during Dvapara Yuga their monarchy is limited to Jambūdvīpa only, as in the case of Yudhiṣṭhira maharaja and Parīkṣit Mahārāja. And in Kali Yuga after Parīkṣit Mahārāja, the rule of the kings is limited to only one island of Bhārata varṣa eventually only a part of it is ruled by those kings. This is all because as time progresses a person's purity and strength is reduced, thus his influence and control over the Land is restricted. Śukadeva Gosvāmī mentions in SB 12.2.1 that "By the powerful influence of the age of Kali religion, truthfulness, cleanliness, tolerance, mercy, duration of life, physical strength and memory will all diminish day by day."

tataś cānu-dinaṁ dharmaḥ satyaṁ śaucaṁ kṣamā dayā
kālena balinā rājan naṅkṣyaty āyur balaṁ smṛtiḥ

Above all these reasons lies the main reason that the will of the Lord and His purpose in creating the world in such a way that different kind of people are endowed by different level of powers and control. This is all meant for the purification of the individual, as indicated in SB 5.21.7 & 5.22.3 that "The wise teach about the sunrise, noon and sunset and midnight in different cities for the purpose of engaging and disengaging people in work according to particular times. And the original cause of the cosmic manifestation, the Supreme Lord, Nārāyaṇa, made of the Vedas worshipped by the sages by recitation of the Vedas, divided Himself into twelve forms and six seasons for purifying the activities of the people (*karma-viśuddhi-nimittaṁ*)." This means through time the Lord regulates the actions, which are meant for purification.

Similarly for the sake of one's purification Lord has given different level of access and power to different living beings. Thus, though we, as part of the ninth division of Bhārata varṣa, cannot be able to see its other divisions or other varṣas of Jambūdvīpa or even the huge Mount Meru. Instead we are given the access to see the various planets (specifically the nava-grahas, starting with sun, moon, etc.), stars and constellations in the sky, though they are very

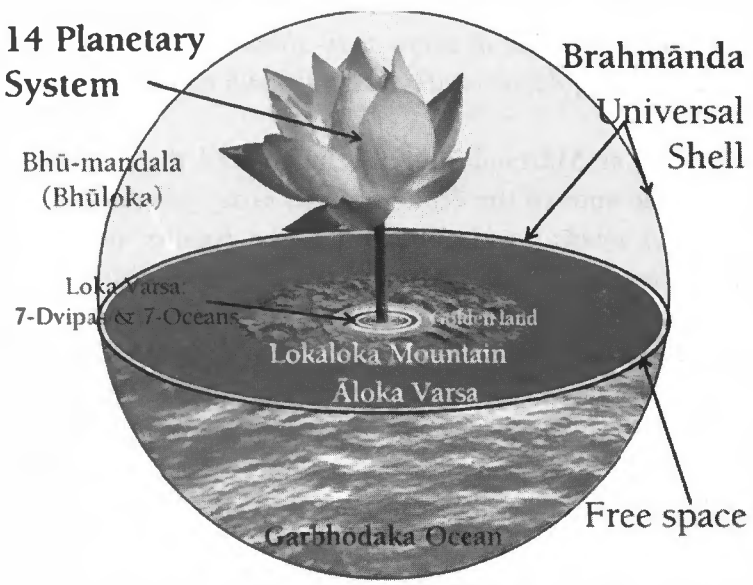
far. One of the reasons for this is by knowing the position of various planets (*nava-grahas*), stars and constellations in the sky, the Panchānga is formed, wherein many predictions such as the divisions of the day, thiti calculations, and variations in the seasons, time of eclipses and the auspicious times to perform various household *samskaras* are present. Even the forecast of an individual is seen based on their position. All these things will help an individual to perform some activities and to desist from other activities (*pravṛtti-nivṛtti-nimittāni samaya-viśeṣeṇa);* as a result one can focus on the goal of life, i.e., the purification of one's existence.

All these statements are in line with the purpose of the compilation and teachings of the great transcendental literature, the Śrīmad Bhāgavatam. This is explicitly indicated by Śrīla Sūta Gosvāmi while speaking to the Sages of Naimiśāraṇya headed by Śaunaka Ṛṣi in SB 1.3.40 and SB 1.7.4-6 about its compilation by Śrīla Vyāsadeva for the ultimate benefit of the world.

3

Fourteen Planetary Systems
Vertical Structure of the Universe

After describing the breadth of the universe (horizontal structure of the universe) in the last five chapters (from 16 to 20) with its form and characteristics, now Sukadeva Gosvami begins to describe the height or the vertical structure of the Universe in the next six chapters (from 21 to 26).



By the dimensions of Bhū-maṇḍala (500MY in diameter) experts indicate the dimension of the sky above the earth (*divo maṇḍala-mānaṁ*). They are like two halves of a bean seed (*Niṣpāva*). *Antarikṣa* joins the two together.[79] Just as by seeing the dimension of one of the halves of the seed one can understand the dimension of the other half, one should consider the dimension of the earthly

[79] SB 5.21.1-2: Diameter and divisions of Universe

sphere (*golaka*) and the heavenly sphere to be the same. The word *golaka* is used. This means that the universe is compared to a globe. Here it should mean a semi-sphere, since the comparison is made to half a seed. Between them is the *antarīkṣa*, measuring 200,000 *yojanas*. It is joined to them on both its sides.

The Sun[80]

The sun is situated (vertically) in the middle of the universe, in between Bhūrloka and Svargaloka, which is called Bhuvarloka (*antarikṣa* or outer space). Thus situated in the middle of the *antarīkṣa*, the sun, master of heat, heats up and lights up the three worlds by its light (*yan-madhya-gato bhagavāṁs tapatāṁ patis tapana ātapena tri-lokīṁ pratapaty avabhāsayaty ātma-bhāsā*). The distance between the sun and the circumference of the universe (above and below) is 250 Million *Yojanas*. The sun is in the middle of the universal shell. It means that the distance between the sun and the universal shell upwards and downwards is the same (250,000,000 *yojanas*).

aṇḍa-madhya-gataḥ sūryo dyāv-ābhūmyor yad antaram
sūryāṇḍa-golayor madhye koṭyaḥ syuḥ pañca-viṁśatiḥ

The sun is known as Mārtaṇḍa because he entered the unconscious universe (Mārtaṇḍa, he who entered the dead universe, *mṛte 'ṇḍa eṣa etasmin yad abhūt tato mārtaṇḍa iti vyapadeśaḥ*). The sun is the totality of *jīvas* with subtle covering Hiraṇyagarbha. Thus, he is called as Hiraṇyagarbha, because he is the very existence of the gross body of the universe called the *hiraṇyāṇḍa*, the golden egg (*hiraṇyagarbha iti yad dhiraṇyāṇḍa-samudbhavaḥ*). The name of the sun is used to explain the reason for it being situated in the middle of the universe. The directions, sky, the heavens, and earth and other divisions, as well the places of enjoyment, liberation and hell, such as Atala, are all differentiated by the sun. All living entities depend upon the heat and light of the sun-god.

sūryeṇa hi vibhajyante diśaḥ khaṁ dyaur mahī bhidā
svargāpavargau narakā rasaukāṁsi ca sarvaśaḥ

He is called dṛg-īśvara, for he enables everyone to see. Thus, he is the eye and life of all living beings, of *devatās*, animals, humans, reptiles and plants.

deva-tiryaṅ-manuṣyāṇāṁ sarīsṛpa-savīrudhām
sarva-jīva-nikāyānāṁ sūrya ātmā dṛg-īśvaraḥ

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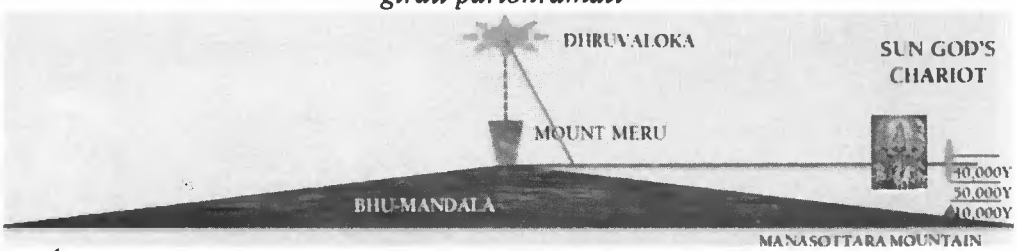
[80] SB 5.20.43-46: Glories of the Sun god

Situated in the middle of the antarīkṣa, the sun, master of heat, heats up and lights up the three worlds by its light. Situated in the middle of the antarīkṣa means vertically the sun is at a height of 100,000 yojanas above Puṣkaradvīpa (seventh dvīpa of Bhū-maṇḍala), because antarīkṣa extends to 200,000 yojanas. Horizontally the sun is at a distance of 15.75MY from the centre of Mount Meru.

Chariot of the Sun-god[81]

The chariot of the sun-god has only one wheel. One side of the axle (Length: 15,750,000 *yojanas*) carrying the wheel rests upon the summit (*mūrdhani*) of Mount Sumeru[82] (being fixed on the lower portion of Meru), and the other rests upon Mānasottara Mountain. Affixed to the outer end of the axle, the wheel continuously rotates on Mānasottara Mountain like the wheel of an oil-pressing machine.

yasyaikaṁ cakraṁ tasyākṣo meror mūrdhani kṛto mānasottare kṛtetara-bhāgo yatra protaṁ ravi-ratha-cakraṁ taila-yantra-cakravad bhraman mānasottara-girau paribhramati



Śrīdhara Svāmī explains that the chariot wheel rotates on Mānasottara at a height of 50,000 *yojanas* on a surface made flat by the wind. Because Mānasottara is only 10,000 *yojanas* high, the total height at which the wheel rotates is 60,000 *yojanas*. The sun god is situated in the enclosure at a place 40,000 *yojanas* above the wheel. Thus, the sun god is situated at a height of 100,000 *yojanas* above Puskara-Dvīpa, as indicated above.

Brahma-saṁhitā 5.52 says that, 'I worship Govinda, the primeval Lord, under

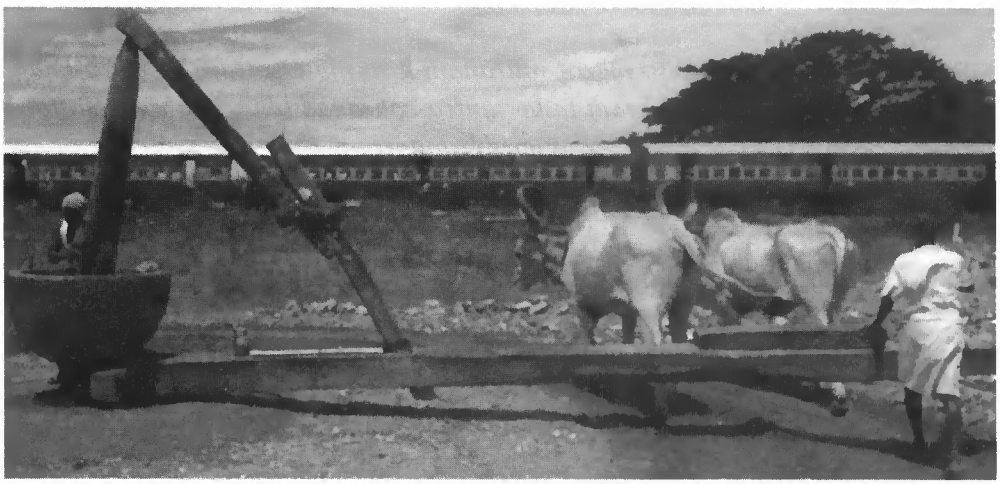
[81] SB 5.21.13-18: Description of the Sun-god's chariot

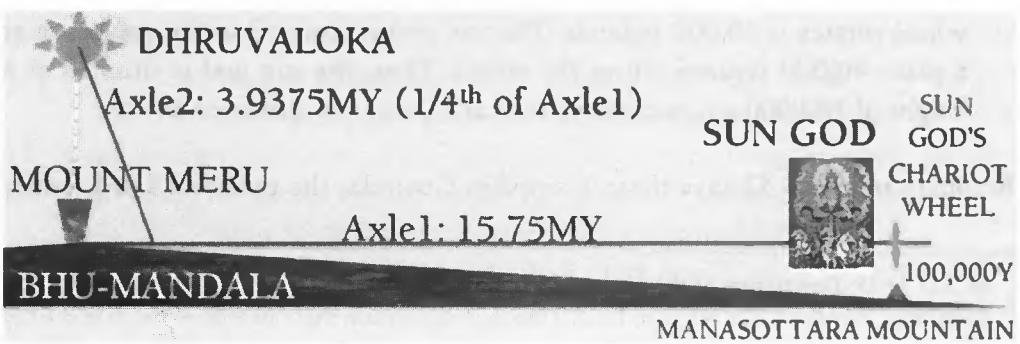
[82] Sometimes *mūrdhani* also refers to Bace. Thus acaryas explain that one side of the axle is fixed on the lower portion (at the bace) of Meru.

whose control even the sun, who is the eye of the Lord and king among all the planets, rotates in the time-wheel (*kāla-cakrah*).'

yac cakṣur eṣa savitā sakala-grahāṇāṁ
rājā samasta-sura-mūrtir aśeṣa-tejāḥ
yasyājñayā bhramati saṁbhṛta-kāla-cakro
govindam ādi-puruṣaṁ tam ahaṁ bhajāmi

As in an oil-pressing machine, the first part of the first axle *(kṛtamūlaḥ)* with the wheel at its end is attached to a second axle, which is one fourth as long (3,937,500 *yojanas* = 15.75 / 4 MY). The second axle is bound to the first exactly opposite to the salt water ocean. The upper end of the second axle is connected to Dhruvaloka by a rope of wind (*tasminn akṣe kṛtamūlo dvitīyo 'kṣas turyamānena sammitas taila-yantrākṣavad dhruve kṛtopari-bhāgaḥ*). According to SB 5.23.1, Dhruvaloka is 3.9MY above the Bhū-mandala.





Size of the chariot: The carriage (or the enclosure, *niḍāḥ*) of the sun-god's chariot is estimated to be 3,600,000 *yojanas* long and one-fourth as wide, 900,000 *yojanas (ratha-nīḍas tu ṣaṭ-triṁśal-lakṣa-yojanāyatas tat-turīya-bhāga-viśālas tāvān)*. The chariot's horses, named after Vedic meters such as Gāyatrī etc, are harnessed by Aruṇadeva to a yoke that is also 900,000 *yojanas* wide. Controlling these horses Aruṇa drives the chariot continuously carrying the sun-god (*ravi-ratha-yugo yatra hayāś chando-nāmānaḥ saptāruṇa-yojitā vahanti devam ādityam*). Although Aruṇa sits in front, he looks backward toward the sun-god, being situated at sun-god's left-side (*purastāt savitur aruṇaḥ paścāc ca niyuktaḥ sautye karmaṇi kilāste*).

Vāyu Purāṇa describes the horses: the seven horses are the seven Vedic meters. They carry the sun from the left and are bound to the side of the wheel. The axel is connected with the wheel.

saptāśva-rūpa-cchandāṁsi vahante vāmato ravim
cakra-pakṣa-nibaddhāni cakre vākṣaḥ samāhitaḥ

In the Viṣṇu Purāṇa it is stated that "The seven horses yoked to the sun-god's chariot are named Gāyatrī, Bṛhati, Uṣṇik, Jagatī, Triṣṭup, Anuṣṭup and Paṅkti. These names of various Vedic meters designate the seven horses that carry the sun-god's chariot."

gāyatrī ca bṛhaty uṣṇig jagatī triṣṭup eva ca
anuṣṭup paṅktir ity uktāś chandāṁsi harayo raveḥ

Facing the chariot, the sixty thousand brāhmaṇa sages known as Vālakhilyas, each the size of a thumb, travel in front and offer prayers to the almighty sun-god with Vedic mantras (*tathā vālikhilyā ṛṣayo 'ṅguṣṭha-parva-mātrāḥ ṣaṣṭi-sahasrāṇi purataḥ sūryaṁ sūkta-vākāya niyuktāḥ saṁstuvanti*). Similarly, other Sages, Gandharvas, Apsarās, Nāgas, Yakṣas, Rākṣasas and *Devatās,* in fourteen groups in some months and pairing up to make seven groups in other months, continuously perform different ritualistic ceremonies to worship the Supreme Lord as Sūrya.

While the sages glorify the sun-god with the hymns of the Sāma, Ṛg and Yajur Vedas, which reveal his identity, the Gandharvas also sing his praises and the Apsarās dance before his chariot. The Nāgas bind the chariot firmly and the Yakṣas harness the horses to the chariot, while the powerful Rākṣasas push from behind. (SB 12.11.47-48)

sāmarg-yajurbhis tal-liṅgair ṛṣayaḥ saṁstuvanty amum
gandharvās taṁ pragāyanti nṛtyanty apsaraso 'grataḥ
unnahyanti rathaṁ nāgā grāmaṇyo ratha-yojakāḥ
codayanti rathaṁ pṛṣṭhe nairṛtā bala-śālinaḥ

Viṣṇu Purāṇa says that these seven groups of fourteen associates regulate the proper times for snow, heat and rain.

so 'yaṁ sapta-gaṇaḥ sūrya-maṇḍale muni-sattama
himoṣṇa vāri-vṛṣṭīṇāṁ hetutve samayaṁ gataḥ

People, by following *varṇāśrama* conduct and worshipping the sun with faith by *aṣṭāṅga-yoga* (*yoga-vitānaiḥ*) and by regulated activities authorized by the Vedas (*trayyā*), easily attain the highest goal. The sun, soul of worlds, situated in the zodiac belt in the *antarikṣa* between heaven and earth, enjoys twelve months, indicated by zodiac signs, limbs of a year (*saṁvatsarāvayavān*).

The Moon[83]

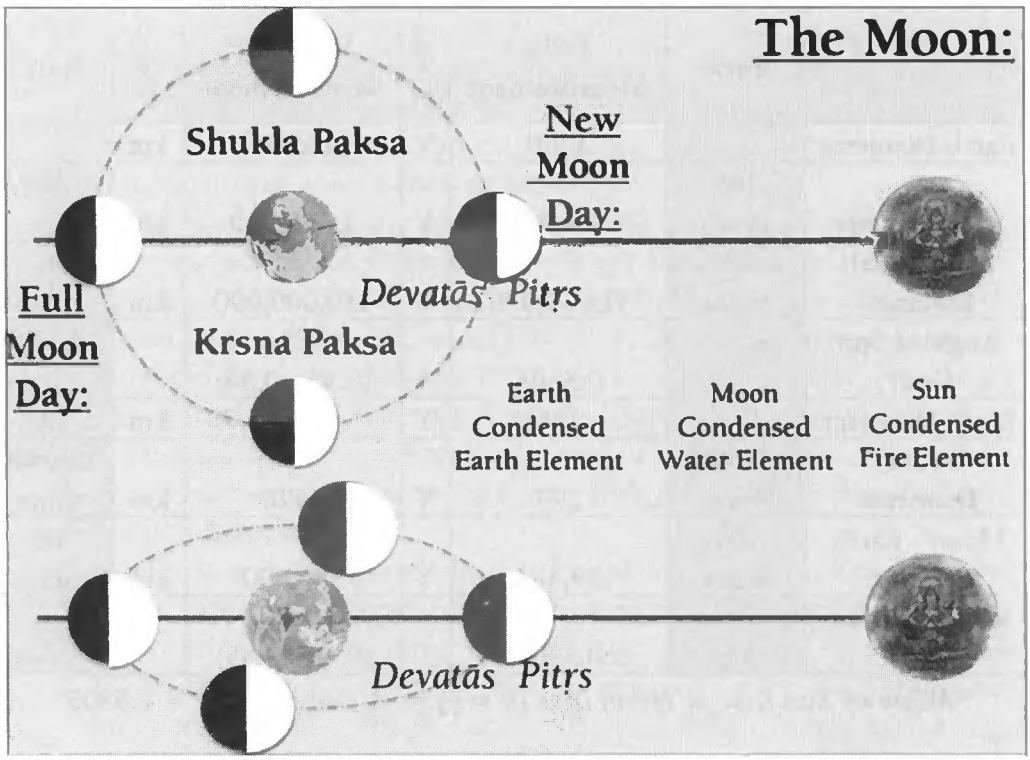
The moon, situated 100,000 *yojanas* above the rays of the sun (*evaṁ candramā arka-gabhastibhya upariṣṭāl lakṣa-yojanata upalabhyamāno*), moving with quicker speed, covers the sun's year in two fortnights and one constellation in thirty *muhūrtas* (one day).



The moon with its waxing and waning phases, creates day for the *devatās* during the waxing phases, and creates day for the Pitṛs during the waning phases. The waxing and waning phases are day and night for the *devatās,* and night and day for the Pitṛs. This means the waxing phase (*sukla paksa*) is suitable for worship of *devatās* and the waning phase (*Kṛṣṇa paksa*) is suitable for worship of Pitṛs.

atha cāpūryamāṇābhiś ca kalābhir amarāṇāṁ kṣīyamāṇābhiś ca kalābhiḥ
pitṝṇām aho-rātrāṇi pūrva-pakṣāpara-pakṣābhyāṁ vitanvānaḥ

[83] SB 5.22.8-10: Movements and glories of the Moon-god



Note: The new moon day (*amāvāsya*, end of waning phase) can be considered as mid-day for the Pitṛs and mid-night for the *devatās*. In addition, the full moon day (*paurnami*, end of waxing phase) can be considered as the mid-night for the Pitṛs and mid-day for the *devatās*.

Inter-relation between Earth-Sun-Moon

The Vedic literatures describe a special relationship between sun & earth and moon & earth w.r.t, their dimensions. The diameter of the sun is 108 times the diameter of the earth. The distance from the sun to the earth is 108 times the diameter of the sun. The diameter of the moon is 108/400 times the diameter of the earth. The average distance of the moon from the earth is 108 times the diameter of the moon.

	Ratio	Vedic Measurement	Units	Scientific Measurement	Units	Ratio
Earth Diameter		1000	Y	12,756	km	
Sun Diameter	108 times	108,000	Y	1,390,000	km	109 times
Sun – Earth Distance	108 times	11,664,000	Y	150,000,000	km	108 times
Angle of Sun Disc*		0.5305	°	0.53 (0.52 - 0.54)	°	
Earth Diameter		1000	Y	12,756	km	
Moon Diameter	108/400 times	273	Y	3476	km	109/400 times
Moon - Earth Distance	108 times	29,484	Y	384,400	km	110 times
Angle of Moon Disc*		0.5305	°	0.52 (0.49 - 0.55)	°	

$*Angle\ of\ Sun\ Disc\ or\ Moon\ Disc\ (\theta = \frac{r}{l}) = \frac{1}{108} rad = \frac{180}{\pi * 108}° = 0.5305°$

Considering all the above detail, that one part of the moon is inhabited by the *devatās* and the other part is inhabited by the pitas, it is very difficult for a common person to go to moon planet and access it in its original constitution, for it is heavenly abode. Then, how is it possible for an ordinary person without obtaining proper qualification to enter into moon planet, a heavenly abode, though the modern scientists can say that they have gone to moon with the aid of Rocket Science? Though they may be successful in reaching the moon planet, with the use of all advanced technologies, they cannot see or access the moon planet as it is. It is like, if someone without any knowledge of different varieties of diamonds and jewels enters into a diamonds shop, he only sees different stones with different colors and shapes. Thus, one cannot appreciate them and their value. To appreciate the value of different jewels one need proper knowledge about them. Alternatively, it is like, if someone enters into Vrndāvan without any knowledge of it being the divine abode of the Supreme Lord Sri Kṛṣṇa, that person simply ends up seeing different roads, buildings, people and animals. One cannot access the devotional mood of

the vrajavāsis and the mercy of the Supreme Lord simply by one's material senses and material qualifications. One actually needs spiritual qualification to receive Lord's mercy and access the mood of Vrndāvan. Similarly, one cannot go and access the moon without proper knowledge about it and proper qualification to access it.

As per the Modern scientific developments, the first human-made object to reach the surface of the moon was the Soviet Union's Luna 2 mission, on 13 September 1959. The United States' Apollo 11 was the first manned mission to land on the moon, on 20 July 1969. There have been six manned U.S. landings (between 1969 and 1972) and numerous unmanned landings, with no soft landings happening from 22 August 1976 until 14 December 2013. To date, the United States is the only country to have successfully conducted manned missions to the moon, with the last departing the lunar surface in December 1972.

- **Apollo 11** was the spaceflight that landed the first two humans on the moon. Mission commander Neil Armstrong and pilot Buzz Aldrin, both American, landed the lunar module *Eagle* on July 20, 1969, at 20:18 UTC. Armstrong became the first to step onto the lunar surface six hours after landing on July 21 at 02:56:15 UTC; Aldrin joined him about 20 minutes later.
- They spent about two and quarter hours together outside the spacecraft, and collected 47.5 pounds (21.5 kg) of lunar material to bring back to earth.
- Michael Collins piloted the command module *Columbia* alone in lunar orbit while they were on the moon's surface. Armstrong and Aldrin spent just under a day on the lunar surface before re-joining *Columbia* in lunar orbit.

Indian Space Research Organisation (ISRO) performed a controlled crash impact with its Moon Impact Probe (MIP) on 14 November 2008. The MIP is an ejected probe from the Indian Chandrayaan-1 lunar orbiteer and they performed remote sensing experiments during its descent to the lunar surface. The Chinese lunar orbiteer Change 1 executed a controlled crash onto the surface of the moon on 1 March 2009. The rover mission Change 3 was launched on 1 December 2013 and soft-landed on 14 December.

But, as per the Vedic understanding as it is mentioned in *Vayu Purana,*

Adhyaya 53.56-57[84], "The discs of sun (surya) and moon (Chandra) are devine spheres of Burning luster. They resemble a circular pot. The moon planet is made up of condensed water element and the sun planet is made up of condensed fire element"

surya-chandramasor divyeh mandale bhasvare khage
jwala thejomaye shukle vrtta kumbha nibhe shubhe
ghana toyatmakam tatra mandalam shashina smrutam |
ghana tejomayam shukla mandalam bhaskarastha tu ||

It is stated in SB 8.11.3-6, where in Indra rebukes Bali, who was trying to conquer the Heavens, "I send ignorant thieves, who want to ascend to Svarga or beyond and want to control Maharloka by creating illusions, to planets lower than their previous residence."

ārurukṣanti māyābhir utsisṛpsanti ye divam |
tān dasyūn vidhunomy ajñān pūrvasmāc ca padād adhaḥ ||

Another appropriate example as mentioned in SB 9.7.5-6, wherein Triśaṅku tries to go to heaven in the self-same body without having necessary qualification, "By the influence of Viśvāmitra, he went to Svarga in the same body, but because of the prowess of the *devatās* he fell downward. But, by the power of Viśvāmitra, he did not fall all the way down; even today he can still be seen hanging in the sky, head downward (Triśaṅku-svarga)."

Also in SB 5.22.9, it is further stated that "The moon with its waxing and waning phases, creates day for the devatās during the waxing phases, and creates day for the Pitṛs during the waning phases. It indicates that half of the moon is occupied by the Devatas and the other half is occupied by the Pitṛs (fore fathers). Thus, Svargaloka starts from moon planet and goes all the way upto Dhruva Planet. Living entity can go to Svarga or Pitrloka only after gaining the necessary qualification and after leaving this body. This may not possible in the self-same body for the people in general; but there may be some exceptions like Arjuna, Yudhisthir maharaj, King Raivata, Nahusha, etc..., who are so exalted that they were able to go and also come back in their self-same bodies.

The moon is called as life itself, supporting the life of all beings (*sarva-jīva-nivaha-prāṇo*). The moon is the life of all beings (*jīvah*) because it produces

[84] Refer to Puranic Cosmology chapter 19

food in the form of plants by his nectarine coolness that influences the growth of food grains. *Śruti says annaṁ hi prāṇināṁ prāṇāḥ:* food is the life of all beings. The moon is called *jīvaḥ* because it is made of immortal nectar necessary for living and he is the chief living being within the universe.

The moon composed of sixteen phases described as the Supreme Lord, the mind, food, nectar, and the life of *devatās,* Pitṛs, humans, ghosts, animals, birds and reptiles and to be all-pervading, since it is the cause of all life.

- The moon represents the Supreme Lord for he is full of potencies (*vibhūti*)
- Manomaya: because he is the presiding deity of the mind
- Annamaya: because he gives food to all in the form of herbs and plants
- Amṛtamaya: because he nourishes the devatas by providing nectar.
- Sarvamaya [all-pervading]: Everyone is satisfied by his presence.

As stated in BG 15.12-13, Lord Kṛṣṇa says, "Know that the light of the sun, moon and fire, which illuminates this universe, is My light. Entering the earth by My power, I support all beings. I nourish all plants, becoming the moon, full of sweet juice."

yad āditya-gataṁ tejo jagad bhāsayate 'khilam
yac candramasi yac cāgnau tat tejo viddhi māmakam
gām āviśya ca bhūtāni dhārayāmy aham ojasā
puṣṇāmi cauṣadhīḥ sarvāḥ somo bhūtvā rasātmakaḥ

Note: The Lord reasons, "know that the light situated in the sun, in the moon and in fire, which illuminates the whole universe, belongs to Me. Actions for attaining jīva's enjoyment according to his karma are accomplished by the light of the sun which has risen and the fire which is blazing. These actions are causes of happiness and destroy darkness and lethargy for the jīva. When the moon rises, there is nourishment of herbs, dissipation of heat and pain, and relaxation in the moonlight. The light which produces these effects in the sun, moon and fire is My vibhūti of light. Entering the earth, which is like a lump of dust, I make it firm and support all moving and non-moving things and entities by My śakti. The sacred Ṛk Veda 8.7.3.1 says '*yena dyaur ugrā pṛthivī ca dṛḍhā*: by whom the heavens and powerful earth are made firm.' If not for that power, the earth would perish, dissolving like a fistful of sand. Becoming the moon, made of nectar (*rasātmakaḥ somaḥ*), I nourish all plants, such as rice. I

make things full of various tasty flavors. For the *jīvas* situated on the earth and enjoying various tastes, I alone accomplish their experience of various tastes, after making the places of enjoyment such as palaces, gardens, tanks and playgrounds (from earth)."

Vasudeva speaks to Lord Kṛṣṇa in SB 10.85.7 "The glow of the moon, the brilliance of fire, the radiance of the sun, the twinkling of the stars, the flash of lightning, the permanence of mountains and the aroma and sustaining power of the earth—all these are actually You."

kāntis tejaḥ prabhā sattā candrāgny-arkarkṣa-vidyutām
yat sthairyaṁ bhū-bhṛtāṁ bhūmer vṛttir gandho 'rthato bhavān

The Constellations or Nakṣatrās[85]

The constellations (*nakṣatrāṇi*), twenty-eight in number with Abhijit, are being inspired by the Lord move in clockwise direction around the Meru Mountain on the wheel of time (*kālāyane*). They are situated at a distance of 300,000 *yojanas* above the moon. Thus the constellations are 400,000 *yojanas* above sun.

tata upariṣṭād tri-lakṣa-yojanato nakṣatrāṇi meruṁ dakṣiṇenaiva kālāyana
īśvara-yojitāni sahābhijitāṣṭā-viṁśatiḥ

By the supreme will of the Supreme Lord (*īśvara-yojitāni*), they are fixed to the wheel of time. Thus, they do not have a different motion, but move clockwise along with the Time wheel. *Śruti* says *abhijin nāma-nakṣātram upariṣṭād āṣāḍhānām adhastāc chroṇāyāḥ*: Abhjit constellation is composed of the last part of Āṣāḍhā and the first part of Śravaṇā constellations. The astronomical works also confirm this *uttarāṣāḍhāśeṣārdhāt śravaṇādau liptakā-catuṣke ca abhijit tatstha khecare vijñeya rohiṇī-viddha*: Abhijit consists of the last quarter of Uttarāṣāḍha constellation and the first quarter of Śravaṇa constellation. Each constellation is consisting of four *daṇḍas* or four *padas*.

Muhūrte means "at a suitably auspicious time" and *abhijite muhūrte* means "with an auspicious ascendant which give complete victory." Abhijit-muhūrta generally occurs at midday. The auspicious period known as *abhijit* is most opportune for victory and it commences at midday (*abhijin nāma yogo*

[85] SB 5.22.11-17: Description of the movements and position of the Stars and Planets above Moon

mauhūrtika). Lord Kṛṣṇa says in SB 11.16.27 "Among months I am Mārgaśīrṣa, and among lunar houses I am the auspicious Abhijit (*māsānāṁ mārgaśīrṣo 'haṁ nakṣatrāṇāṁ tathābhijit*)." Each day the sun passes once through the location of the lunar constellation Abhijit (*sūrye cābhijiti sthite*), marking the period most auspicious for victory. Thus, the *muhūrta* Abhijit coincides with the high noon.

Other Planets below Drhuva-Loka

The Venus, Śukra-graha or Uśanā-graha

Venus is observed 200,000 *yojanas* above the constellations. It moves like the sun, with fast, slow and medium speed, sometimes behind and sometimes in front of the sun (*vārkasya śaighrya-māndya-sāmyābhir gatibhir arkavac carati*). It is always considered favorable for all people, generally causing rain by its movements and removing evil influence of other planets, which are obstacles to rain (*vṛṣṭi-viṣṭambha-grahopaśamanaḥ*). Venus is 600,000 *yojanas* above sun. By its movement (*cāreṇa*), it causes rain. It corrects the obstacles to rain caused by other planets.

The Mercury or Budha-graha

Mercury, the son of the moon (*budhaḥ soma-suta*), is observed 200,000 *yojanas* above Venus. It moves in a similar fashion (*uśanasā budho vyākhyātas tata*). It is generally auspicious. When it moves away from the sun (*arkād vyatiricyeta*) then it produces a fearful situation with drought, cyclones, and continual clouds (*ativātābhra-prāyānāvṛṣty-ādi-bhayam āśaṁsate*). Mercury is 800,000 *yojanas* above sun. It moves similar to Venus. It moves sometimes in front and sometimes behind the sun.

The Mars, Mangala-graha or Aṅgāraka-graha

Mars is observed 200,000 *yojanas* above Mercury. If not traveling retrograde (*na vakreṇābhivartate*), it passes through one of the twelve zodiac sign every three fortnights (*tribhis tribhiḥ pakṣair ekaikaśo rāśīn*). Generally it is inauspicious, indicating grief and distress (*prāyeṇāśubha-graho 'gha-śaṁsaḥ*). Mars will be 1,000,000 *yojanas* above sun. If it does not move retrograde (*yadi na vakreṇābhivartate*) it traverses one zodiac sign every three fortnights.

The Jupiter, Bṛhaspati-graha or Guru-graha
Situated above Mars by 200,000 *yojanas,* If not traveling retrograde (*yadi na vakraḥ syāt*), Jupiter moves through each zodiac sign in one *prativatsara*[86] (*ekaikasmin rāśau parivatsaraṁ parivatsaraṁ carati*). It is generally auspicious for the *brāhmaṇas.* Jupiter is 1,200,000 above sun.

The Saturn, Śanaiścara or Śani-graha
Above Jupiter by 200,000 *yojanas* is Saturn, which travels slowly through one zodiac sign in thirty months (*ekaikasmin rāśau triṁśan māsān vilambamānaḥ*). In this way it travels through all twelve signs in thirty *anuvatsaras. Anuvatsara* sometimes means the lunar year, but here probably it means simply a solar year. It is generally inauspicious for all people. From the sun, the Saturn is at a height of 1,400,000 *yojanas.*

The Sapta- Ṛṣis or the Seven Sages
The Seven Sages are situated above Saturn by 1,100,000 *yojanas.* Thinking of the welfare of the people, they rotate clockwise around Dhruvaloka, the abode of Viṣṇu (*śam anubhāvayanto bhagavato viṣṇor yat paramaṁ padaṁ pradakṣiṇaṁ prakramanti*). The sages are 2,500,000 *yojanas* above sun planet.

The Dhruva Loka (Polestar)[87]
1,300,000 *yojanas* above the planets of the seven sages is the place that learned scholars describe as the abode of Lord Viṣṇu (*viṣṇoḥ paramaṁ padam*). There, the son of Mahārāja Uttānapāda, the great devotee Mahārāja Dhruva, remains until the end of the *kalpa* (*kalpa-jīvinām*), around whom Agni, Indra, Prajāpati, Kaśyapa and Dharma rotate with respect.

agninendreṇa prajāpatinā kaśyapena dharmeṇa ca
samakāla-yugbhiḥ sabahu-mānaṁ dakṣiṇataḥ kriyamāṇa

Dhruvaloka is 3,800,000 *yojanas* above sun. All of them together at the same time in the form of the constellations (*samakāla-yugbhiḥ*) rotate around the Polestar, Dhruva loka. This Polestar, being established by the Lord, shines constantly and acts as the pivot (*sthāṇur ivāvaṣṭambha*) of all the luminaries, planets and constellations (*jyotir-gaṇānāṁ graha-nakṣatrādīnām*).

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[86] *Parivatsara* can refer to the solar year or to the Jupiter year which is 360 days.
[87] SB 5.23.1-3: Glories of Dhruva planet (Polestar)

After hearing Himself praised by the intelligent boy Dhruva having great determination, the Supreme Lord (Pṛṣnigarbha or Dhruva-priya), attracted to His servant who approached Him, desiring the best position in the three worlds (*padaṁ tri-bhuvanotkṛṣṭaṁ*), which has not been attained by any of his forefathers or others (SB 4.8.37), accepted the praise and then spoke, "O prince, determined in vows! I know what you resolved in your heart. All auspiciousness to you! I will award that place rarely achieved, and also *prema*." (SB 4.9.19)

vedāhaṁ te vyavasitaṁ hṛdi rājanya-bālaka
tat prayacchāmi bhadraṁ te durāpam api suvrata

"Please accept that place now. I give you that radiant place which is eternal (*dhruva-kṣiti*), not attained by anyone else. The zodiac of planets, constellations and stars (*grahaṛkṣa-tārāṇāṁ*) revolve around it like oxen walking around a central pillar to grind grains. That places remains till the end of Brahmā's life (*kalpa-vāsinām*). Bharaṇi, Kṛttika, Cancer, Venus (*dharmo 'gniḥ kaśyapaḥ śukr*) and the Big Dipper (*vanaukasaḥ*, the seven sages) revolve around Dhruva-loka in clockwise direction (*caranti dakṣiṇī-kṛtya*) along with the stars (*satārakāḥ*)." (SB 4.9.20-21)

nānyair adhiṣṭhitaṁ bhadra yad bhrājiṣṇu dhruva-kṣiti
yatra graharkṣa-tārāṇāṁ jyotiṣāṁ cakram āhitam
medhyāṁ go-cakravat sthāsnu parastāt kalpa-vāsinām
dharmo 'gniḥ kaśyapaḥ śukro munayo ye vanaukasaḥ
caranti dakṣiṇī-kṛtya bhramanto yat satārakāḥ

This place remains fixed for the duration of Brahmā's life for those living for that duration (*kalpa-vāsinām*). At the time of final destruction, some say that Dhruva ascends to Mahā-Vaikuṇṭha. Even though existing within the material universe, Dhruva-loka has the qualities of Śvetadvīpa, Mathurā and Dvārakā.

"You will go to My abode (*mat-sthānaṁ*), worshipped by all planets (*sarva-loka-namaskṛtam*), higher than the Big Dipper (*upariṣṭād ṛṣibhyas*), from which you will not return (*yato nāvartate*), having gone there (*gataḥ*)."

tato gantāsi mat-sthānaṁ sarva-loka-namaskṛtam
upariṣṭād ṛṣibhyas tvaṁ yato nāvartate gataḥ

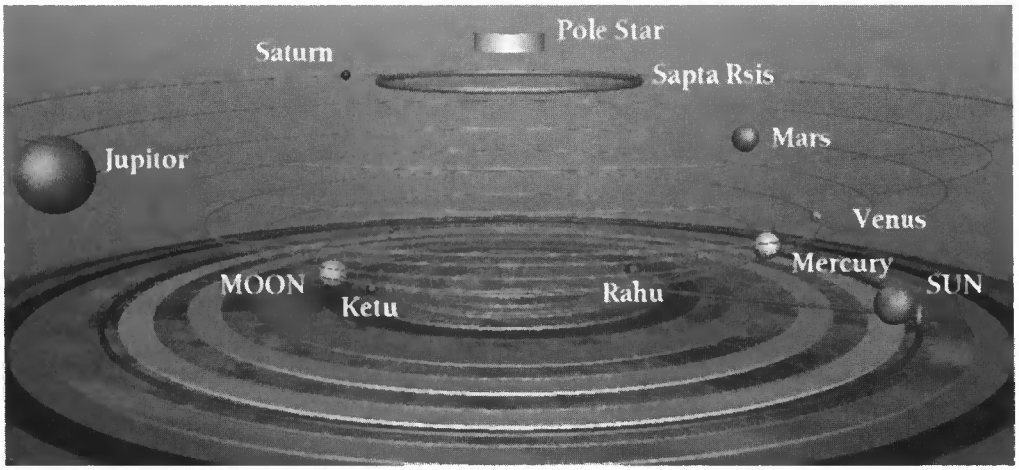
As stated in SB 4.12.22-23 – The esteemed servants of the Lord, Sunanda and Nanda, smiling, approached and spoke to Dhruva, who stood with folded hands and bowed head, absorbed in the feet of Kṛṣṇa.

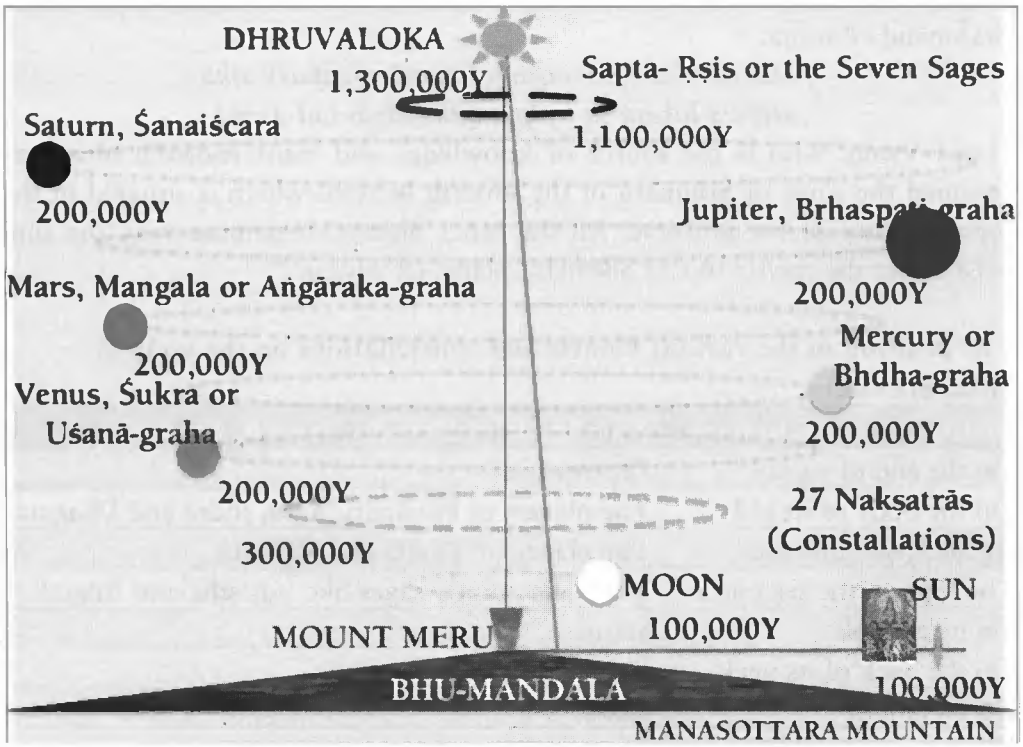
> "O King! All good fortune to you, who pleased the Lord by austerity when you were five years old! We, the associates of the Supreme Lord, creator of all the universes, have come to take you in the same body to the abode of the Lord. You have attained the planet of Viṣṇu (*viṣṇu-padaṁ*), difficult to achieve, which the *devatās* (the seven sages) see but cannot attain. Stay there. The moon, sun (*candra-divākarāda*) and other planets, the constellations and stars (*graharkṣa-tārāḥ*) circumambulate that supreme place clockwise (*pariyanti dakṣiṇam*). O immortal (*āyuṣman*) Dhruva! Stay in that supreme abode of Viṣṇu (*tad viṣṇoḥ paramaṁ padam*), a place not achieved by your forefathers or any others, which is worshipped by the whole universe (*jagatāṁ vandyaṁ*)." (SB 4.12.24-26)

Summary of the Vertical Position of different Planets

Planet	Delta Hight (MY)[88]	Hight wrt Sun (MY)
Dhruva Loka (Pole-star)	1.3	3.8
Saptarsi (7 sages)	1.1	2.5
Śanaiścara (Saturn)	0.2	1.4
Bṛhaspati-graha (Jupiter)	0.2	1.2
Aṅgāraka (Mars)	0.2	1.0
Budha, s/o Moon (Mercury)	0.2	0.8
Śukra-graha (Venus)	0.2	0.6
Constellations	0.3	0.4
Moon	0.1	0.1
Sun	0	0

[88] Delta Hight is the Hight of the Planet w.r.t. the previous Planet (planet just below it)





Note: The horizontal distance between Mount Meru to sun is four times to the vertical height between Mount Meru and Dhruvaloka (or the polestar).

The Śiśumāra Planetary Systems[89]

Some describe this zodiac belt (*jyotir*-anīkaṁ) to be the form of a śiśumāra [dolphin], used for worshipping Vāsudeva in meditation. (*kecanaitaj jyotir-anīkaṁ śiśumāra-saṁsthānena bhagavato vāsudevasya yoga-dhāraṇāyām anuvarṇayanti*). They meditate upon it as the virāṭ-rūpa. This form of the śiśumāra or dolphin (*kuṇḍalī-bhūta-dehasya*) has its head downward and its body coiled. The coiled body of the dolphin turns toward the right side. Its body is balanced because its two sides are occupied by an equal number of stars, i.e. 14 on either side. The body of the śiśumāra should be considered as the external form of Lord Viṣṇu, the Supreme Lord, and not the eternal form of the Lord. This Śiśumāra-cakra is the resting place of all the demigods and all the stars and planets. Śrīla Madhvācārya quotes the following verse from the Brahmāṇḍa Purāṇa.

jñānānandātmano viṣṇuḥ śiśumāra-vapuṣy atha
ūrdhva-lokeṣu sa vyāpta ādityādyās tad-āśritā

"Lord Viṣṇu, who is the source of knowledge and transcendental bliss, has assumed the form of Śiśumāra in the seventh heaven, which is situated in the topmost level of the universe. All the other planets, beginning with the sun, exist under the shelter of this Śiśumāra planetary system."

The position of the various Planets and constellations on the body of Śiśumāra-cakra

Bodily limb of Śiśumāra	The Abode of Devatās, Stars & Planets
On the end of its tail	Dhruva-loka
On the body of its tail	The planets of Prajāpati, Agni, Indra and Dharma
At the base of its tail	The planets of Dhātā and Vidhātā
The hips on the śiśumāra	The seven saintly sages like Vasiṣṭha and Aṅgirā
On its genitals	Saturn
On the back of its neck	Jupiter
On its mouth	Mars
Within its life air (*prāṇāpāna*)	Mercury
On its navel	Venus

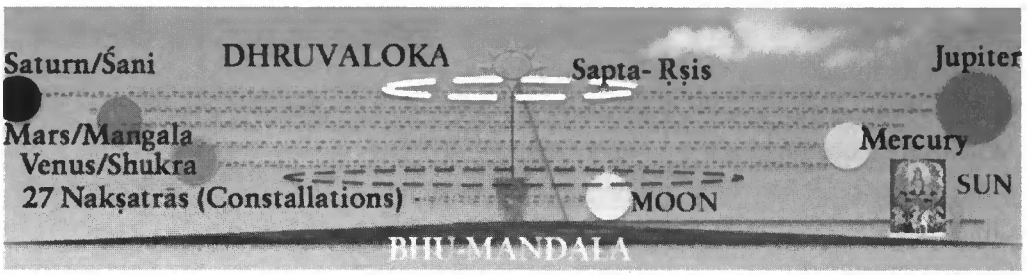
[89] SB 5.23.4-7: Description of body of Śiśumāra (Machine of stars, planets ~ Dolphin)

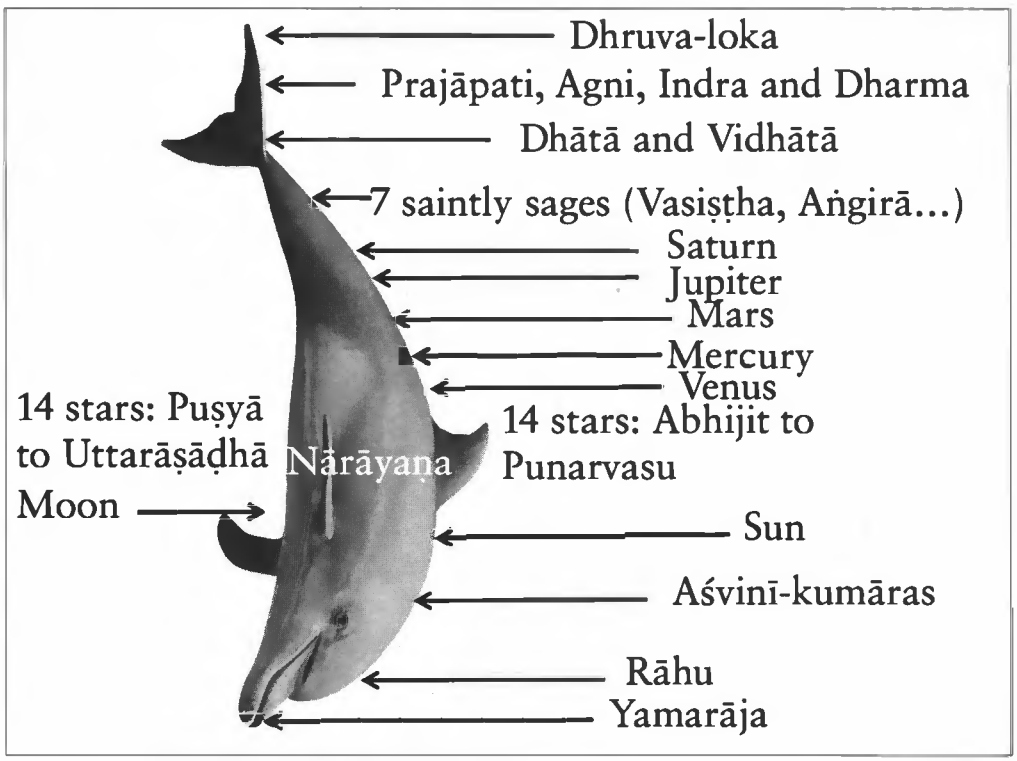
Right side of the coiled body	The 14 stars from Abhijit to Punarvasu (*Udgayanāni*)
Left side of the coiled body	The 14 stars from Puṣyā to Uttarāṣāḍhā (*Dakṣiṇāyanāni)*
On the back of the śiśumāra	Ajavīthī stars (first portion of the constellations on the southern path)
On its abdomen	The Ganges that flows in the sky (the MilkyWay)
Within its mind	The moon
On its chest	The Sun
Within the core of heart	Nārāyaṇa
On its breasts	The Aśvinī-kumāras
On its neck	Rāhu
On its upper chin	Agasti
On its lower chin	Yamarāja
All over its body	Comets
In its pores	the numerous stars

The position of the 28 Constellations on the body of Śiśumāra-cakra

Bodily limb of the Śiśumāra-cakra	The abode of demigods, all the stars & planets
On the right side of the coiled body of the Śiśumāra-cakra, which turns toward its right	the 14 constellations from Abhijit to Punarvasu (*Udgayanāni*)
On its left side (its body is balanced because its two sides are occupied by an equal number of stars)	The 14 stars from Puṣyā to Uttarāṣāḍhā (*Dakṣiṇāyanāni)*
On the right & left sides of the loins	The stars named Punarvasu and Puṣyā
On its right and left feet	Ārdrā and Aśleṣā
On its right and left nostrils	Abhijit and Uttarāṣāḍhā
At its right and left eyes	Śravaṇā and Pūrvāṣāḍhā
On its right and left ears	Dhaniṣṭhā and Mūlā

On the ribs of the left of its body	The 8 stars from Maghā to Anurādhā[90] (southern course)
On the ribs on the right side.	The 8 stars from Mṛgaśīrṣā to Pūrvabhādra (northern course)
On the right and left shoulders.	Śatabhiṣā and Jyeṣṭhā





[90] The constellations Mṛgaśiras, Rohiṇī, Kṛttika, Bhāraṇī, Āśvinī, Revatī, Uttarabhadrā and Pūrvabhadrā situated on the right side are in reverse order.

Meditating on the form of Śiśumāra and its Benefits[91]

Process: Morning, noon and evening, daily, while controlling speech, gazing at the form of Viṣṇu composed of all the *devatās (sarva-devatāmayaṁ rūpam)*, one should worship it using this *mantra:* "I offer my respectful obeisance's unto You, the shelter of all planets, the form of time, the master of the *devatās*, the Supreme Lord. I meditate on You."

"namo jyotir-lokāya kālāyanāyānimiṣāṁ pataye mahā-puruṣāyābhidhīmahīti"

Benefits: Persons who offer respects three times a day to this form and remember this form of the Lord (*adhidaivikam*) which is composed of the planets, constellations and stars (*graha-rkṣa-tārā-mayam*), and which destroys all sins (*pāpāpahaṁ*), while chanting this *mantra,* quickly destroy all sins born of time (*naśyeta tat-kālajam āśu pāpam*).

Distinction between the terms like *graha, rkṣa, tārā* etc

- **Nakṣatra:** The stars that never fade (1 ray). These are the stars associated with Chandra: "*nakṣatrāṇām ahaṁ śaśī*"
- **Rksha:** Group of stars that are situated in the Zodiac (Ashwini, Bharani, Krtika etc...). They are 27 in number and bigger in size like Budha.
- **Taraka:** Other small stars. Those who do good deeds cross this world (*tr* - to cross) and attain these placess. "Tara" also means they are white and bright. Their size starts from half yojana as big as 100 yojanas also.
- **Graha:** Planets like sun (the king of all), moon, etc... "*jyotiṣāṁ ravir aṁśumān*".

  ādityānām ahaṁ viṣṇur jyotiṣāṁ ravir aṁśumān |
  marīcir marutām asmi nakṣatrāṇām ahaṁ śaśī ||

  Among Ādityas I am Viṣṇu, and among lights, I am the many-rayed sun. Among the winds, I am Marīci. I am the moon amidst the stars.

  Note: Among the twelve Adityas I am Vāmana (Lord Viṣṇu). Among illuminators, I am the sun whose rays spread throughout the universe (*aṁśumān* - having rays). Among the forty-nine Maruts, I am Marīci. I am the moon, who showers nectar, the lord of the constellations. In this verse the genitive case is used to indicate an outstanding member of a group or a relationship (eg. lord of the constellations).

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[91] SB 5.23.8-9: Mantra to worship Sisumara and the Result

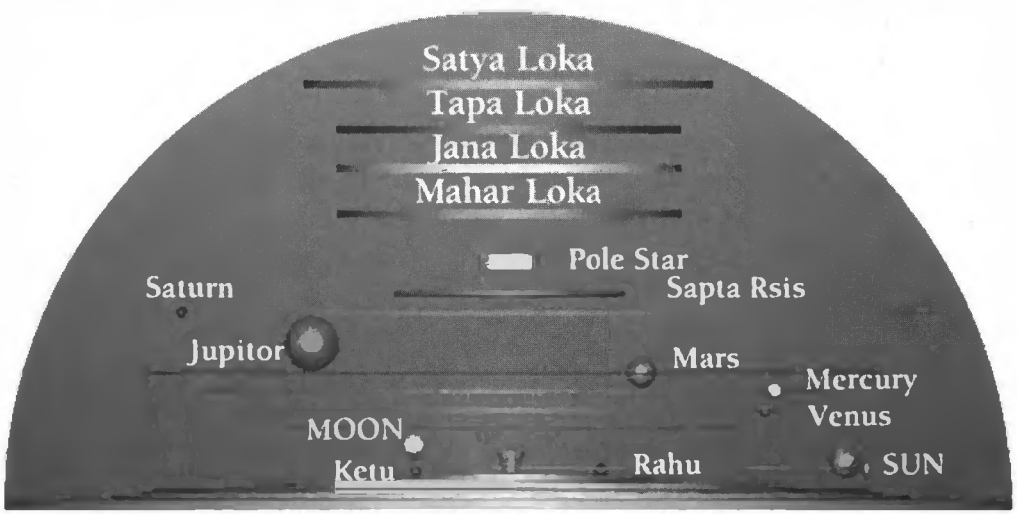
Planets above Dhruva-Loka

As per the details given in the Purport of SB 5.23.9 referring to Viṣṇu Purāṇa, Dhruvaloka is situated 3,800,000 *yojanas* away from the sun. Maharloka is 10,000,000 *yojanas* from Dhruvaloka. Janaloka is 20,000,000 *yojanas* from Maharloka. Tapaloka is 80,000,000 *yojanas* from Janaloka. Satyaloka is 120,000,000 *yojanas* from Tapaloka. From the sun to Satyaloka is 233,800,000 *yojanas*. Vaikuṇṭha is 26,200,000 *yojanas* above Satyaloka. That is the covering of the universe. This is mentioned in the *Viṣṇu Purāṇa*. Thus, it is seen that 250,000,000 *yojanas* from the sun is the shell of the universe.
Similarly the dimensions of the lower half of the Universe are mentioned in the *Viṣṇu Purāṇa*. The earth (Bhū-maṇḍala) is 100,000 *yojanas* from the sun. The seven lower planets continue to Pātālaloka for 70,000 *yojanas* below the surface of the earth (Bhū-maṇḍala). From Pātālaloka to Śeṣa on the Garbodhaka Ocean is 30,000 *yojanas*. The Garbodhaka Ocean is 200,000 *yojanas* from the sun. The ocean is 249,800,000 *yojanas* deep. Thus from top to bottom, the universe is 500,000,000 *yojanas*.

Vertical Structure of the Universe - The Position of the Upper Planets

Planet	Delta Hight (MY)	Hight w.r.t. Sun	Characteristics of the Planets and People over there
Start of Vaikuṇṭha planets	-	>260 + shell	According to Viṣṇu Purāṇa the covering of the universe is 250 MY away from the sun. The Vaikuṇṭha planets begin 26.2 MY above Satyaloka or 10 MY above the covering of the universe.
Region of Liberated souls	10	260 + Shell	This is the area between the covering of the universe & the Vaikuṇṭha planets.
Universal covering	7 Layers	>250	Universal covering consists of seven layers of made up of Earth, water, fire, air, ether, false ego and mahat-tattva, each ten times wider than the previous one. **Refer Note2 below

Free-space	16.2	250	According to Viṣṇu Purāṇa the covering of the universe starts 16.2MY above Satyaloka and 250MY (2 billion miles) away from the sun. (In this free space above Satyaloka, there is Visnuloka, which the seers call Vikuṇṭhā. *Refer Note1 for further details)
Upper Planetary System			
Satyaloka	120	233.8	Abode of Lord Brahma (he resides along with his associates)
Tapoloka	80	113.8	Abode of great Sages & Rsis (4-Kumaras headed by Sanath Kumara resides here. This loka is fully populated with Siddhas and Rsis; and provides abundance of amenities of all kinds)
Janaloka	20	33.8	Abode of great Sages & Rsis (Sages headed by Brghu Muni)
Maharloka	10	13.8	Abode of great Sages & Rsis (It is populated by Siddhas and Rsis in large numbers)



***Note1: References of the Vaikuṇṭha-loka from Bāgavatam**

From the combination of Śubhra and his wife named Vikuṇṭhā, there appeared the Supreme Lord, Vaikuṇṭha, along with *devatās* called the Vaikuṇṭhas, who

were His expansions. Just to please the goddess of fortune, Lord Vaikuṇṭha, at her request, created another Vaikuṇṭha planet, which is worshiped by everyone. Just as the Lord's appearance is called a birth, so the appearance of the Lord's abode is called "creating" a planet. It is not a material creation, since the Lord and His abode are both eternal. They are described similarly, with the intention of saying that they are eternal. *Krama-sandarbha* says "This Vaikuṇṭha planet of the son of Vikuṇṭha is the same as the original Vaikuṇṭha beyond the eight coverings of the universe seen by Brahmā before He created the living entities, as described in the Second Canto."

> The Lord, gratified by his worship, showed Brahmā Mahā-vaikuṇṭha, to which nothing is superior, which is free of suffering, confusion and fear of offenses and which is praised by the *devatās* who constantly see the self. In Vaikuṇṭha there is no *rajas* or *tamas*, and no *sattva* mixed with *rajas* and *tamas*. There is no influence of time. There is no influence of *māyā* at all, what to speak of its products such as material elements. In Vaikuṇṭha the inhabitants are fully dedicated to the Lord and are worshippable by the *devas, asuras* and devotees. That place is filled with shining air vehicles belonging to the inhabitants, and shines with the forms of divine women. The planet appears to be a sky filled with clouds and flashing lightning. (SB 2.9.9-10,13)

The activities and most glorious qualities of the Vaikuṇṭha abode have already been described. A mortal who succeeds in describing the qualities of Viṣṇu can count the particles of dust on earth. Vaikuṇṭha was described in the Third Canto Chapter 15 and 16. Its qualities are such things as fondness for the Vedas and *brāhmaṇas*.

> Brahmā said: My mental sons born before you, the Kumāras, having given up all desires in this world, wandering to various planets through the sky went to the abode of the Supreme Lord (*vaikuṇṭha-nilayaṁ*) who purifies all hearts. This abode is worshipped by all people. In that place all men reside with forms like the Lord and they worship the Lord with unmotivated service since ancient times. There, the Supreme Lord, the original person, the form of the highest *dharma*, attainable by the Vedas, established in pure *sattva*, gives pleasure to us, His devotees. (SB 3.15.12-15)

Other references about the the Vaikuṇṭha-loka

Canto 2: From the feet to the hips, the planets from Pātala to earth are imagined. Bhuvarloka extends from the navel. Svarga is situated at the heart, and Maharloka is at the chest. Janaloka is on the neck, Tapaloka is on the breast and Satyaloka is on the heads of that form. Above this is the Lord's planet which is eternal (*mūrdhabhiḥ satyalokas tu brahmalokaḥ sanātanaḥ*). (2.5.38-39) Above Satyaloka is the planet of the Lord, Vaikuṇṭha. But this is not the object of meditation as a limb of the universal form because it is eternal. Though it exists within the universe it is like the Lord, eternal.

Canto 9: Thereafter, being disappointed, for he was refused by Brahmā and Śiva of protection from Sudarśana cakra, Durvāsā Muni went to Vaikuṇṭha, where the Supreme Lord resides with his consort Lakṣmī. (SB 9.4.60)

tato nirāśo durvāsāḥ padaṁ bhagavato yayau
vaikuṇṭhākhyaṁ yad adhyāste śrīnivāsaḥ śriyā saha

Canto 10: Bhṛgu then left the abode of Lord Śiva, Mount Kailāsa, and went to Vaikuṇṭha, where Lord Janārdana resides (*atho jagāma vaikuṇṭhaṁ yatra devo janārdanaḥ*). (SB 10.89.6) He then went to Vaikuṇṭha in the material world, made by the son of Vikuṇṭha (Vaikuṇṭha). This was a special Viṣṇu-loka, where Lord Vaikuṇṭha resided.

**** Note2: The Lord is situated within and outside the Universe (*Puruṣa-sūkta* Conformed)**

That *puruṣa* pierced the universe and went outside. There He resides with a thousand legs, feet, arms, eyes, faces and heads. (SB 2.5.35)

sa eva puruṣas tasmād aṇḍaṁ nirbhidya nirgataḥ ।
sahasrorv-aṅghri-bāhv-akṣaḥ sahasrānana-śīrṣavān ॥

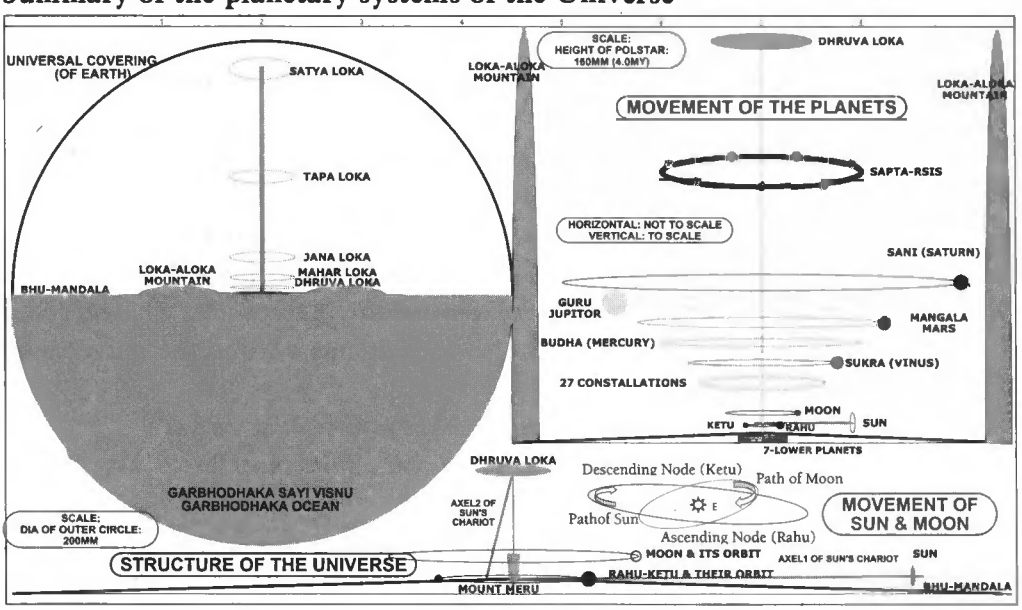
Whatever existed in the past, whatever exists in the present and whatever will exist in the future, is covered by the Lord. He extends beyond the universe by ten fingers. (SB 2.6.16)

sarvaṁ puruṣa evedaṁ bhūtaṁ bhavyaṁ bhavac ca yat ।
tenedam āvṛtaṁ viśvaṁ vitastim adhitiṣṭhati ॥

Because the Supreme Lord is the possessor of the *māyā-śakti*, everything in this world is non-different from Him. This confirms the meaning of the *Puruṣa-*

sūkta. The line starting *sahasra-śīrṣā* and three verses starting with *brāhmaṇo 'sya mukham āsīt* in the *Puruṣa-sūkta* were described in SB 2.5.35. Now *puruṣa evedaṁ sarvam* (the Lord is everything in the universe) from the *Puruṣa-sūkta* is illustrated in three and half verses (SB 2.6.13-16). After showing that the Supreme Lord is the cause of the material world, Brahmā states that the Lord is beyond the material world, and limits the whole world: this world is covered by the Supreme Lord, because He exists (*tiṣṭhati*) greater than (*adhi*) this world. He remains, exceeding it by ten fingers (*vitastim*). This is stated to express His superiority to the universe, not to express His dimension, since He cannot be measured, being unlimited. This illustrates the *Puruṣa-sūkta* line *sa bhūmiṁ sarvataḥ spṛṣṭvā atyatiṣṭhad daśāṅgulam*: touching the earth (universe), He extended Himself beyond it by ten fingers. *Bhūmin* means *viśvam* in the context, and *daśāṅgulam* means *vitastim*. *Atyatiṣṭhad* means *adhitiṣṭhati*. He extended Himself beyond the universe by ten fingers means He is eternally situated outside the ten layers of the Univesre, in His eternal abode, Mahā-vaikuṇṭha. Ten layers of the Univesre are the layers made up of earth, water, fire, air, ether, *ahankāra*, *mahat-tattva* and the three layers of *prakṛti* (layer of *tamo-guna*, *rajo-guna* and *sattva-guna*). Thus the Lord eternally resides outside the ten layers of the universal shell.

Summary of the planetary systems of the Universe



Planet	Delta Hight (MY)	Hight wrt Sun (MY)
Beginning of Vaikuntha planets	-	>260 + shell
Region of Liberated souls	10	260 + Shell
Universal covering	10 / 7 Layers (shell)	>250 (250 + shell)
Free-space	16.2	250
Upper Planetary System		
Satyaloka	120	233.8
Tapoloka	80	113.8
Janaloka	20	33.8
Maharloka	10	13.8
Svarloka		
Dhruvaloka	1.3	3.8
Saptarsi (7 Saintly sages)	1.1	2.5
Śanaiścara (Saturn)	0.2	1.4
Bṛhaspati-graha (Jupiter)	0.2	1.2
Aṅgāraka (Mars)	0.2	1
Budha, s/o Moon (Mercury)	0.2	0.8
Śukra-graha (Venus)	0.2	0.6
Constellations	0.3	0.4
Moon	0.1	0.1
Sun	0	0
Bhūloka / Bhūrloka		
Bhū-mandala (earthly plane)	-0.1	-0.1

Vertical Structure of the Universe as per "Bhūgola Varnanam"

The Heavenly Abodes[92]

The sun's route lies on the wheel of the Kala Cakra, one lakh yojanas above the earth. He is there to bless the world with his bright auspicious rays. On another wheel, one lakh yojanas still higher, the moon moves upon its own path. Above the moon's orbit, the star world extends up to 30 lakh yojanas. In the middle areas the merited souls, such as Gandharva and Siddha class of divine personalities, along with their consorts, enjoy life, moving as they do in

[92] Refer to the section "The Star World (320 - 345)"

their divine vehicles. They eat nice food, drink sweet juices and they sing and dance. They drink the sweet waters of the Ganga pervading in the sky (Akāsa Ganga). Their vehicles are crores and crores in number and they are in the form of stars. (They are stars of their own kind). Even now men can see them. Because of the brilliance of these star-like vehicles and of the stars and planets, the Time Wheel is called as *Jyotiscakra* the 'Wheel of Light'.

The location of the constellations (the 27 star groups) is in between the moon and the stars. The planet Sukra (Venus) is two lakh yojanas higher than the stars. Budha (Mercury) is two lakh still higher; Kuja (Mars) another two lakh higher; another two lakh higher is Brhaspati (Jupiter), the divine priest. Two lakh yojanas above Jupiter is Sanaiscara (Saturn). The Saptarsi mandala is 11 lakh yojanas above Saturn; and Dhruva-mandala (polestar) is 14 lakh yojanas above the orbit of the Saptarsis. Thus, the polestar is at an altitude of 37 lakh yojanas above the earth.

Planet	ΔHight (MY)	Hight wrt Earth (MY)
Earth	0	0
Rahu & Ketu	0.09	0.09
Sun chariot	0.01	0.1
Moon (above sun rays, exactly above salt water ocean)	0.1	0.2
Gandarvas, siddhas & other divine personalities live, moving in their airplanes (In between moon and stars)		
27 Constellations are situated In between Moon and Stars		
Sukra (Venus)	0.2	0.4
Budha (Mercury)	0.2	0.6
Kuja (Mars)	0.2	0.8
Brhaspati (Jupiter)	0.2	1.0
Sani (Saturn)	0.2	1.2
Saptarsi	1.1	2.3
Dhruva mandala	1.4	3.7

Some critical comments from the translator of Bhugola Varnanam: "Here we are certainly at loggerheads with modern scientists and

astronomers, who say that the nearest star 'Alpha' is at least 4 light years (4.22 light years precisely) away and that the polestar is still farther away (about 433 light-years). Having seen how Vedic information is fully correct, we should not be afraid of finding fault with human calculations. There must be something wrong somewhere with their instruments or their mode of calculation. The polestar is only 2,22,00,000 miles (3.7 million yojanas) away from us, and not more than 200000000000000 miles away (433 light-years). Just as there are star-like divine vehicles below the polestar, there are also divine vehicles of greater brilliance and velocities at higher altitudes near the upper worlds, moving in different directions. Perhaps, these are mistaken as Pulsars and Quasars and Galaxies. As the planets go along their courses, Gandharvas and Apsaras sing and dance and go with them in front. They make themselves merry again and again with sweet scents, flowers, delicious food and nice drinks. The learned brahmanas on earth chant hymns in their praise and offer oblations and salutations. These divine beings bless the good souls with auspicious things and curse the bad ones with disastrous effects. (Here is the truth about the satellites or moons of the planets.)

The Puṣkaradvīpa surrounds the Milk Ocean (Ksirabdhi). In its middle part runs the sun above Manasottara mountain ring brightening all directions. Some say, out of ignorance, that the sun sinks into the western ocean and enters the Nether worlds and then rises in the east. This is against Vedic information. It should not be accepted. On account of the terraced formation of the seven islands, Puṣkaradvīpa is the lowest; and therefore the setting point and the rising point of the sun are at lower levels in comparison with the location of our part of the earth. Hence it looks as though the Kala Cakra goes down and comes up.

The ignorant say that the earth is round like a rupee coin, surrounded by a deep ocean. This is how the early Greeks and the Europeans of the medieval period thought. Indians were no exception, influenced as they were by the Greeks after Alexander's invasion of India. It is sad to note that the great Aryabhatta unwittingly accepted the Heliocentric theory proposed by the later Greeks.

The Upper Planetary System[93]

Including the visible terrestrial world, the upper worlds are seven in number, the seventh and the uppermost being 'Satya loka'. The 'Antariksa loka' begins one lakh yojanas above the earth. The Svarga loka is at an altitude of 50 lakh yojanas. It is beautiful with divine palatial buildings. Antariksa loka and Svarga loka are one crore yojanas breadth wise. Above Svarga loka is Mahar loka, at an altitude of 75 lakh yojanas above the former. It is populated by Siddhas and rsis in large numbers. Above it is Jana loka at an altitude of 1.125 crore yojanas. Then comes Tapa loka, 1.6875 crore yojanas higher up. This loka is fully populated with Siddhas and rsis; and provides abundance of amenities of all kinds. The three lokas now spoken of Mahar loka, Janaloka and Tapoloka, have abodes beautified with varieties of gems in plenty. The extent of these three lokas is similar, each measuring a third fraction of 50 crore yojanas and 2 krosas.

Putting together the altitudes of all the lokas spoken of and adding the gaps in between the adjacent lokas, the total altitude will come to 4.0625 crore yojanas. Having a similar distance in between, the Satya loka begins. Therefore, it is at an altitude of 8.125 crore yojanas above the earth. That is where Brahma's abode is situated. Its extent is similar to that of the three lokas beneath it. The Satya loka itself has an altitude of 8.125 core yojanas. Thus the total altitude of all the upper worlds will be 16.25 crore yojanas. The Satya loka of Brahma is snapped like a tuber at its base. It is somewhat thin in the middle like a stalk and, at the top, it spreads out all round like the petals of a lotus flower. It is very very wide. The lokas of the universe, being all shaped to resemble lotus flowers, the whole thing looks like a huge lotus blooming upon the Ghanodaka. Considering the earth itself, we should take into account, the pedestal-like Jambūdvīpa, the terraced arrangement of the other six islands and the mountain ring of Puṣkaradvīpa; and also the other mountain ranges of the other islands. Then, Bhuloka verily looks like a huge lotus flower.

The abode of Lord Visnu, Vaikuntha[94]

At an altitude, twice that of Satya loka, there is Visnu loka, which the seers

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[93] Refer to the section "The Upper Worlds (110 - 123)"

[94] Refer to the section "Vaikuntha Loka (129 - 133)"

call Vaikunta. It is at an altitude of 16.25 crore yojanas above the earth. Its extent is twice that of Brahma's loka. It is neither created, nor destroyed. It is there that the released souls move about in celestial planes of the non-material world.

The Measurements of the Brahmanda[95]

Measuring from the base of the Meru mountain, up to the shell of Brahmanda above, it is 25 crore yojanas. All the lokas are contained within this range. Upward, eastward, westward, north or south, or in any other direction it measures 25 crore yojanas. This is with reference to the centre at the base of the Meru Mountain. But, putting together the measurement from the Meru centre towards the south up to the shell of Brahmanda, the Brahmanda as a whole measures fifty crore yojanas every way from end to end. It is fully round and made of gold. Its thickness is 100 crore yojanas.

Position of the Upper World	ΔHight (MY)	Hight w.r.t Earth (MY)
The Larger Earth	0	0
Antariksa, Above earth (1 crore yojanas breadth wise)	0.1	0.1
Svarga Loka, Above antariksa (1 crore yojanas breadth wise)	4.9	5.0
Mahar loka, abode of Siddas and Rsis (Extend for a distance of 50/3 crore yojanas in width)	7.5	12.5
Jana loka, abode of Siddas and Rsis (Extend for a distance of 50/3 crore yojanas in width)	11.25	23.75
Tapo loka, abode of Siddas and Rsis (Extend for a distance of 50/3 crore yojanas in width)	16.875	40.625
Gap between Tapa loka and Satya loka	40.625	81.25
Satya loka extends to a height of 81.25MY (extends for a distance of 50/3 crore yojanas in width)	81.25	162.5
Vaikuntha, Released souls move in celestial planes (Extend for a distance of 2*50/3 crore yojanas)	81.25	243.75
Gap to universal shell	6.25	250
Total	250	250

[95] Refer to the section "The Brahmanda (129 - 133)"

Three abodes of Lord Visnu[96]

Upon the Ghanodaka on the northern side is the Abode of Visnu, known as '*Anantasana*'. It is Sridevi who is in the form of that loka. The learned say that it is at a height of one lakh yojanas. A similar Abode is in the milk ocean (Ksira Samudra) at an altitude of one and a half lakh yojanas with reference to its neighboring earth. This Abode is called '*Svetadvīpa*'. The three Abodes Vaikunta, Svetadvīpa and Anantasana are only the forms of Sridevi.

Universal Coverings[97]

The golden Brahmanda is the earthly envelope surrounding the universe. It is 100 crore yojanas thick. Surrounding it are nine other envelopes, each ten times bigger than its inner complement. They are the envelopes made of the subtle material principles water, fire, air, sky, ahamkara, mahat-tatva, tamas, rajas and satva. The fourteen worlds of the universe are all contained at the core of these envelopes. Such is this wonderful universe. But, with all its envelopes, it is so tiny that it is not enough to occupy fully a single hair-pit on the body of Lord Sriman Narayana. Then, who else among the gods can stand comparison with Him? Let the learned ones find out. The universe (Brahmanda) with its ten envelopes starting with earth measures 100000000050 crore yojanas.

No	Universal Covering	MY
	Barahmanda	500
1	Earth layer	1000
2	Water layer	10000
3	Fire layer	100000
4	Air layer	1000000
5	Ether layer	10000000
6	Ahamkara layer	100000000
7	Mahat tattva layer	1000000000
8	Tamas layer	10000000000
9	Rajas layer	100000000000
10	Satva layer	1000000000000
	Total width	1111111111500 or 1000000000500

[96] Refer to the section "The Three Abodes of Visnu (133 - 149)"

[97] Refer to the section "Nine outer Envelopes (150 - 152)"

Bhuvar-loka - Planets below Sun and above Earth[98]

After describing the planets above sun, Śukadeva Gosvāmī now describes the Residences of the Siddhas and others, as well as the seven lower planets starting with Atala. Bhūvarloka or Antarikṣa described as the place of enjoyment for the Yakṣas, Rākṣasas, Piśācas, ghosts and so on (*yakṣa-rakṣaḥ-piśāca-preta-bhūta-gaṇānāṁ vihārājiram antarikṣaṁ*). Antarikṣa extends (downward below) as far as the wind blows and the clouds float in the sky (*yāvad vāyuḥ pravāti yāvan meghā upalabhyante*). Above this limit there is no more air. The sky between the heaven and earth is called antarikṣa, or outer space (*dyāv-āpṛthivyor antareṇa nabho-valayasya*). The *antarīkṣa*, measuring 200,000 yojanas, is joined to them on both its sides. Antarikṣa adjoins the top of the earth and the bottom of heaven (*te antareṇāntarikṣaṁ tad-ubhaya-sandhitam*).

The Planet Rāhu

The speakers of the Purāṇas (*ity eke yo 'sāv*) say that 10,000 *yojanas* below the sun is the planet known as Rāhu, which moves like one of the constellations (*adhastāt savitur yojanāyute svarbhānur nakṣatravac caratīty*). This movement is in the direction opposite to the sun and the other planets. Rāhu, the son of Siṁhikā and its presiding deity, is the most abominable of all *asuras,* and he has attained the position of a *devatā* by the mercy of the Lord (*amaratvaṁ grahatvaṁ cālabhata bhagavad-anukampayā*), though he is not qualified to be a *devatā* (*a-tat-arhaḥ*).

> SB 6.6.37 says, "In his wife Siṁhikā, Vipracitti (son of Danu and Kaśyapa) begot one hundred and one sons, of whom the eldest is Rāhu and the others are one hundred Ketus. All of them attained positions as planets."

He created an obstacle previously by coming between the sun and moon to drink nectar. He created enmity with them when they understood what he was doing and stopped him from drinking nectar. Thus, obstructing the sun and moon previously, he has become their enemy, and attacks them on the full and dark moon days (*rāhor yaḥ parvaṇi tad-vyavadhāna-kṛd vairānubandhaḥ sūryā-candramasāv abhidhāvati*). *Parvaṇi* means the full moon and dark

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[98] SB 5.24.1–6: Description of Bhuvar-loka - Planets below Sun and above Earth

moon. On these days it creates an obstacle to the full moon and dark moon for those looking from the earth. Hearing of Rāhu's attack, the Lord engages His dear Sudarśana weapon to protect the sun and moon both (*ubhayatra*). Seeing the intolerable heat of the weapon rotating in front of him for forty-eight minutes (*muhūrtam*), Rāhu, frightened and worried, flees to a distant place. People say that (*iti vadanti lokāḥ*) this is the eclipse (*uparāgam*). Because of either precise or imprecise alignment, there are full and partial eclipses. Rāhu does not actually devour the sun or moon, since he is situated 10,000 *yojanas* away from the sun, as per vertical measurement.

It is said in Bhugola Varnanam[99] that "The Rahumandala (Rahu's orbit) is situated upon the lowest wheel of the Kala Cakra, at an altitude of 90,000 yojanas above the earth and thus 10,000 yojanas below the level of the sun's path. The planet Ketu, with a hundred tail-like projections, is also at the same spot, is as if the two planets are twins whose bodies are joined together, back-to-back. The glorious Lord cut off Rahu's head (as also the head of Ketu) when he attempted to share nectar stealthily at the time of 'Amrta-manthan'. The heads of the two demons have occupied the status of planets having just tasted a few drops of nectar. They are there only to torment the sun and the moon. However, their spiritual presence in the planets is negligible. The oblations offered to Rahu and Ketu are accepted by divine presences dwelling in those two planets; and the Lord Himself accepts the offerings through those divine presences and blesses the worshippers.

Planet of Siddhas, Cāraṇas and Vidyādharas

The planets known as Siddhaloka, Cāraṇaloka and Vidyādhara-loka are situated 10,000 *yojanas* below Rāhu. They possess many mystic powers, and travel to different places at will. SB 6.17.1-5 says, "After offering obeisances to the direction from which Lord Ananta had disappeared, Citraketu began traveling in outer space as the head of the Vidyādharas on a brilliant airplane given to him by Lord Viṣṇu."

Planet of Yaksas, Raksasas and Pisacas

Within the outer sky called *antarikṣa* beneath Vidyādhara-loka, Cāraṇaloka and Siddhaloka, are the places of enjoyment for the Yakṣas, Rākṣasas, Piśācas,

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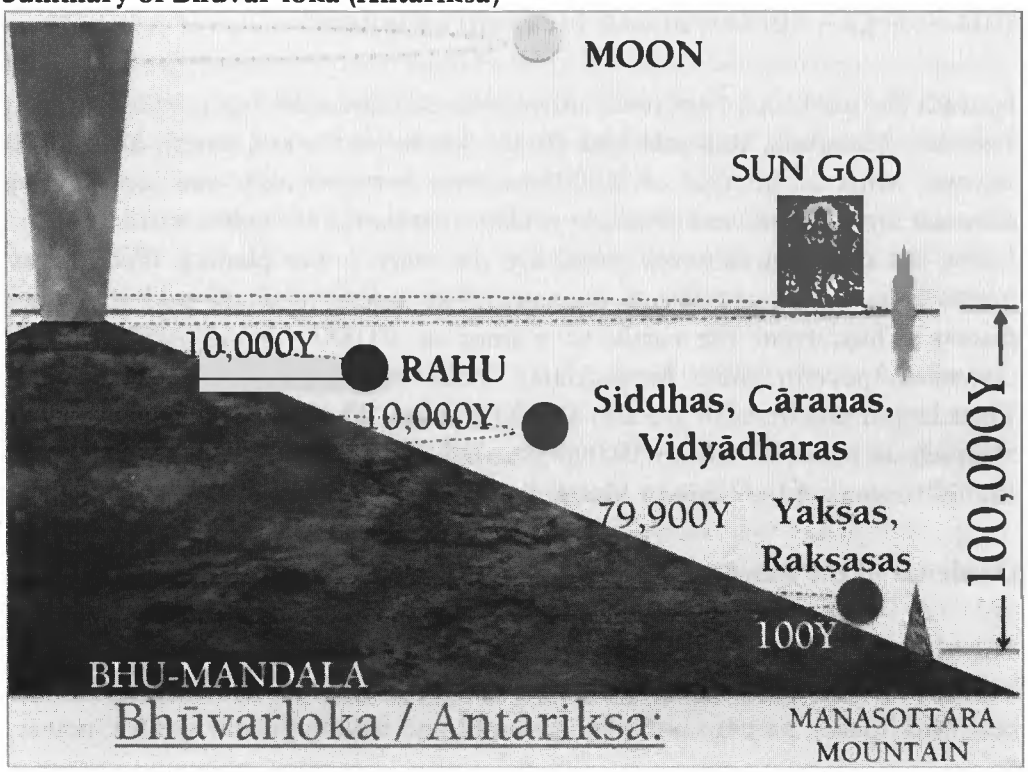
[99] Refer to the section "Planets Rahu and Ketu (309 - 320)"

ghosts and so on (*yakṣa-rakṣaḥ-piśāca-preta-bhūta-gaṇānāṁ vihārājiram antarikṣaṁ*). *Antarikṣa* extends downward as far as the wind blows and the clouds float in the sky (*yāvad vāyuḥ pravāti yāvan meghā upalabhyante*).

Limits of Bhū-mandala

Below the abodes of the Yakṣas and Rākṣasas by a distance of 100 *yojanas* is the Bhū-mandala (*tato 'dhastāc chata-yojanāntara iyaṁ pṛthivī*), whose upper limits extend as high as swans, hawks, eagles and similar large birds, who are transformations of earth, can fly (*yāvad dhaṁsa-bhāsa-śyena-suparṇādayaḥ patattri-pravarā utpatantīti*). This describes the upper limit of Bhūrloka above the earth plane (Bhū-mandala), which approximately can be considered as around 50 yojanas. All our space expiditions in the Northern side could be possible all the way upto this extent, may not be beyond this, for that comes under the preview of higher beings.

Summary of Bhuvar-loka (Antariksa)



Planet	ΔDepth (MY)	Depth (MY)	Characteristics of the Planets and People over there
Bhūvarloka or Antarikṣa: the places of enjoyment for the Yakṣas, Rākṣasas, Piśācas, ghosts and so on. Antarikṣa or outer space is the sky b/w the earthly sphere and heavenly sphere.			
Sun	0	0	Situated in the middle of the Universe
Rāhu	0.01	0.01	Moves like one of the stars.
Siddha, Cāraṇa & Vidyādhara-loka	0.01	0.02	Naturally endowed with mystic powers, can go from one planet to another easily.
Yakṣa-Rakṣaloka	0.0799	0.0999	The places of enjoyment for the Yakṣas, Rākṣasas, Piśācas, ghosts and so on.
Bhūloka / Bhūrloka			
Puṣkaradvīpa (Bhū-mandala)	0.0001	0.1	Its upper limits extend as high as swans, hawks, eagles & similar large birds can fly.

Bila Svarga - Subterranean Heavenly Planets[100]

Beneath the earth there are seven other planets, known as Atala, Vitala, Sutala, Talātala, Mahātala, Rasātala and Pātāla whose width and length are 10,000 *yojanas,* with an interval of 10,000 *yojanas* between each one (*avaner apy adhastāt sapta bhū-vivarā ekaikaśo yojana-ayuta-antareṇ-ayāma-vistāreṇa*).
Below the earth, in different parts, are the seven lower planets. *Bhū-vivaraḥ* means "in hollow cavities of the earth." It refers to Atala and the other planets. Thus, from the earth, to a limit of 10,000 *yojanas* Atala extends (*antaram*, space or limit). From Atala, Vitala extends another 10,000 *yojanas.* Their length and breadth are also 10,000 *yojanas.* The heights of the places are as much as possible (within 10,000 *yojanas*). Atala starts after the remaining 16,000 *yojanas* of the Sumeru Mountain.

Opulence of the Residents of Bila-svarga

The Daityas, Dānavas and Nāgas (*daitya-dānava-kādraveyā*), who are householders always enamored by their wives, children, friends and society in beautiful houses, gardens and play grounds, and having greater wealth, power,

[100] SB 5.24.7-15: Description of Bila svarga - Planets below Earth and their opulences

bliss and sense enjoyment than the higher planets (*bila-svargeṣu svargād apy adhika-kāma-bhogaiśvaryānanda*), live there in illusory enjoyment, with desires more unimpeded than in Svarga.

The demon named Maya, who is an expert artist and architect, has constructed cities, decorated with the finest houses of the leaders, having artificial areas crowded with couples of pigeons, parrots, mynas, Nāgas and demons, with wondrous houses, walls, gates, assembly houses, temples, yards and resting places made of the finest jewels. In these planets there are beautiful gardens, surpassing the beauty of heaven (*amara-loka-śriyam atiśayitāni*), resplendent with the beauty of trees having branches bent down with beautiful shoots, and clumps of fruits and flowers which give the greatest joy to the senses and mind, and with the sensual pleasure of various sweet cries of flocks of frolicking birds which have made their nests in forests of lotuses, white water lilies, night blooming lotuses, white lotuses, blue lotuses, pink lotuses and hundred-petalled lotuses, in ponds disturbed by jumping fish and frequented by pairs of birds.

In these planets there is no fear because there are no divisions of time with day and night (*aho-rātrādibhiḥ na kāla-vibhāgair*). *How do they get light?* The gems on the heads of the best of the great snakes drive away all darkness (*mahāhi-pravara-śiro-maṇayaḥ sarvaṁ tamaḥ prabādhante*). The inhabitants do not have anxiety or sickness from food or drink because of consuming the juice of divine herbs and elixirs (*divyauṣadhi-rasa-rasāyana-anna-pāna-snānādibhir ādhayo vyādhayo*). They do not experience various effect of age (*valī-palita-jarādayaś ca*).

> First, they do not have wrinkles or grey hair or old age. Second, they do not experience change of bodily complexion, bad odor, and perspiration. Finally they do not experience fatigue (*klama*) or lack of pleasure (*glāniḥ*). Thus they are free from the effects of aging, caused by bathing, drinking and eating (*na deha-vaivarṇya-daurgandhya-sveda-klama-glānir iti vayo 'vasthāś*). Living comfortably, they do not experience the influence of death except from the energy of the Lord called the Sudarśana-cakra. When the Sudarśana disc enters those provinces, the pregnant wives of the demons all have miscarriages (*garbha-srāva*[101]) due to fear of its effulgence.

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[101] If the child is lost up to the fourth month after pregnancy it is called *garbha-srāva* and if the child is lost in the fifth or sixth month it is called *garbha-pāta.*

Seven Lower Planetary Systems[102]

Atala

A demon named Bala, the son of Maya Dānava, had created ninety-six kinds of mystic powers in Atala. Some people hold some of these powers even today. Simply by yawning, the demon Bala further created three kinds of women, known as *svairiṇī, kāmiṇī* and *puṁścalī*. Those women who enjoy in their own caste are called *svariṇī*. Those who enjoy outside their caste are called *kāminī*. Those who are very fickle in enjoying are called *puṁścalī*. When a man enters their houses (*bilāyanam*, which are like holes), these women make him fit with *hāṭaka*, and enjoy with him by embraces, talking, attractive smiles and playful glances. Drinking this beverage, the man, thinking he has the strength of ten thousand elephants, as if blind with intoxication, boasts "I am God, and am perfect."

Vitala

Below Atala, on Vitala, Lord Śiva, the master of gold mines, lives with his wife, surrounded by his personal associates, the ghosts, for producing progeny. From Vitala a river called Hāṭakī made from their vital energy flows. It is heated by Agni (*pibati*) and hardened by Vāyu. The men and their wives wear that gold called Hāṭaka along with their jewel ornaments with great admiration in the houses of the great demons.

Sutala

Below Vitala, on Sutala, the son of Mahārāja Virocana, Bali Mahārāja, who is famous and praised by verses, resides even now. He was cheated of the three worlds by the Lord, who, desiring to please Indra, appeared in the form of Vāmana, born from Aditi. Reinstated by the Lord's mercy on Sutala and blessed with wealth not possessed even by Indra, he worships with fearless heart the Lord who is most worthy of worship.

Talātala

Below Sutala, in Talātala resides Maya, king of the Dānavas, lord of three cities. His three cities were burned by Śiva, who desired the good of the three

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[102] SB 5.24.16-31: Description of Seven Lower Planets

worlds. By Śiva's grace Maya, master of the magicians, regained his position. Protected by Śiva, without fear of Sudarśana, he is worshipped.

Mahātala

Below Talātala, in Mahātala, a group of snakes with many hoods, descendents of Kadrū, headed by Kuhaka, Takṣaka, Kāliya and Suseṇa, who are controlled by anger, reside. Addicted to material enjoyment and bewildered by the association of wives, children, friends and relatives, they enjoy themselves but constantly live in fear of Garuḍa (*puruṣa-vāhāt,* the carrier of the Lord), king of the birds.

Rasātala

Beneath Mahātala, in Rasātala, live the Paṇis, descendents of Diti and Danu, as well as the Nivāta-kavacas, Kāleyas and Hiraṇya-puravāsīs. These are all different types of demons. They are all enemies of the *devatās,* and reside in holes like snakes. From birth they are extremely powerful and cruel, but their strength and pride are always defeated by the auspicious Sudarśana-cakra of the Lord.

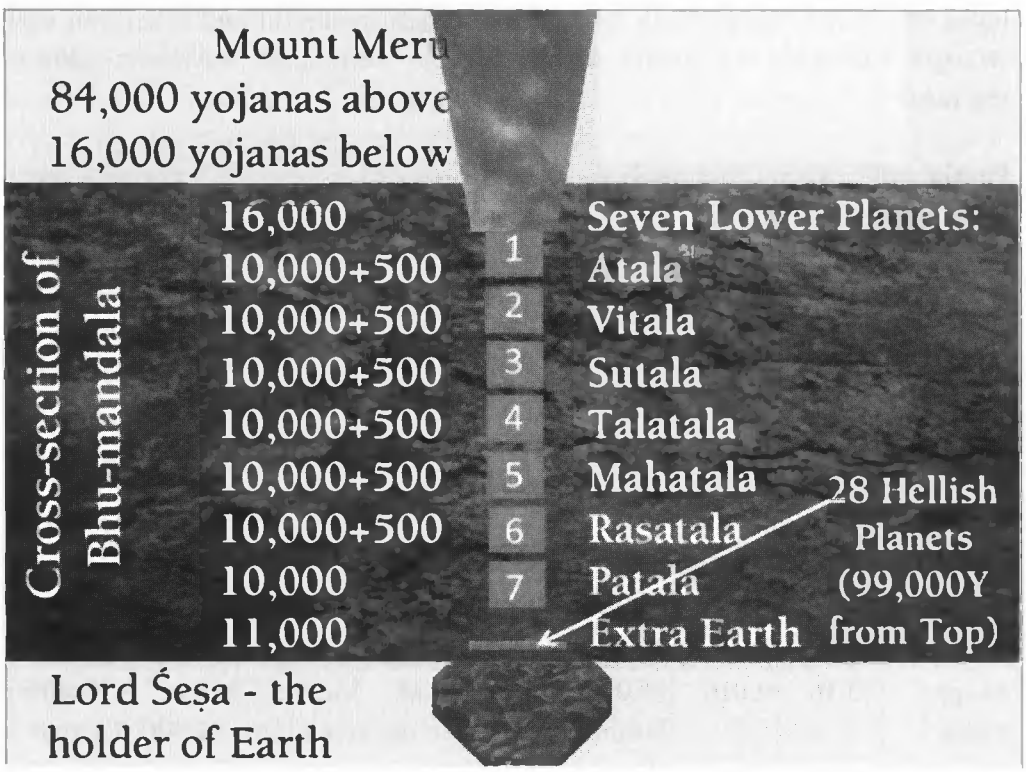
Pātāla

Below Rasātala, in Pātāla, reside the leaders of Nāga-loka, such as Śaṅkha, Kulika, Mahāśaṅkha, Śveta, Dhanañjaya, Dhṛtarāṣṭra, Śaṅkhacūḍa, Kambala, Aśvatara and Devadatta, headed by Vāsuki. They have many hoods and are very envious. The shining jewels fixed on five, seven, ten, a hundred or a thousand hoods, light up the darkness of the caves of Pātāla with their effulgence.

Summary of the Seven Lower Planetary systems

Planet	ΔHight (MY)	Hight (MY)	Characteristics of the Planets and People over there
Lower planetary systems - ***Bhū-vivaraḥ*** "in hollow cavities of the earth."			
Mount Meru	0.016	0.016	16,000 Yojana of Mount Meru is within Jambūdvīpa, while the remaining 84,000 Yojanas is above Jambūdvīpa.
Atala	0.01	0.026	Ruled by the demon Bala, the s/o Maya Dānava.

Vitala	0.01	0.036	Lord Śiva, lives with his wife, surrounded by his associates.
Sutala	0.01	0.046	Bali Mahārāja, s/o Virocana, resides here worshipping the Lord.
Talātala	0.01	0.056	Ruled by Maya, the king of Dānavas, lord of three cities.
Mahātala	0.01	0.066	Abode of many hooded snakes headed by Takṣaka, descendents of Kadrū.
Rasātala	0.01	0.076	Abode of Paṇis, descendents of Diti & Danu, Nivāta-kavacas, Kāleyas etc.
Pātāla or Nāgaloka	0.01	0.086	The residence of the leaders of many hooded Nāgas, headed by Vāsuki.
Gap b/w them	0.003	0.089	The total gap among the seven nether worlds (500 yojanas in each case) make 3000 yojanas.



Places below Pātāla - Lord Ananta and Hellish Planets

Lord Ananta[103]

Lord Ananta, an expansion of the Supreme Lord, holder of the earth, lives 30,000 *yojanas* below the planet Pātāla. The form of Ananta which holds up the universe is a *śaktyāveśa* form according to *Laghu-bhāgavatāmṛta.* Anantadeva, the source of Rudra who destroys the universe, an expansion of the Supreme Lord, sometimes called *tamasī.* Ananta is called *tamasī* because he is the supervisor of destruction, an activity of *tamas,* but he is not a form composed of *tamas.* The Lord is called Saṅkarṣaṇa because he draws the enjoyer and the object of enjoyment (*draṣṭṛ-dṛśayoḥ*) completely together. This attraction is characterized by the false identity of "I am the enjoyer and this is my object of enjoyment." The devotees meditate upon Saṅkarṣaṇa, as the presiding deity of *ahaṅkāra* in order to become purified of the false conception of I and mine. This universe (*kṣiti-maṇḍalaṁ*), situated on one of Lord Ananta's thousands of hoods, appears just like a white mustard seed (SB 5.25.2).

yasyedaṁ kṣiti-maṇḍalaṁ bhagavato 'nanta-mūrteḥ sahasra-śirasa
ekasminn eva śīrṣaṇi dhriyamāṇaṁ siddhārtha iva lakṣyate.

And In Ilāvṛta-varṣa, Lord Śiva always worships Lord Saṅkarṣaṇa, his source and the spiritual form in charge of ignorance, by uttering various mantras in meditation indicating Lord's greatness in continuously supporting the *bhū-maṇḍala* for the welfare of all the living entities (SB 5.17.21) - "The sages say that He is the cause of maintenance, creation and destruction of the universe, that He is without material guṇas, and He is unlimited. He does not even feel the universe (*bhū-maṇḍala*) situated on His thousands of hoods, since it is like a mustard seed."

yam āhur asya sthiti-janma-saṁyamaṁ
tribhir vihīnaṁ yam anantam ṛṣayaḥ
na veda siddhārtham iva kvacit sthitaṁ
bhū-maṇḍalaṁ mūrdha-sahasra-dhāmasu

He is called unlimited because of his knowledge of the *tattvas.* He does

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[103] SB 5.25.1-15: Abode, Activities and Glories of Lord Ananta

> not know the universe (*bhū-maṇḍala*), which is situated somewhere on one of his hoods. Because He is situated below *bhū-maṇḍala* measuring 493,400,000 *yojanas,* His hoods are not wide enough to hold it. However, persons situated there can see that by His inconceivable energy, His hoods become unlimited in dimension, and thus *Bhū-maṇḍala* appears like a mustard seed. That is actual, not an illusion. This is an example of His astonishing unlimited nature just mentioned.

When Ananta desires to destroy the entire creation, from between his two moving eyebrows appears three-eyed Rudra, composed of anger, with eleven expansions, carrying a raised trident. Because *māyā* is controlled by his desire, and when Lord desired to destroy this universe (*idam*), the angry Rudra (an effect of *tamas*) appears. Just as a lotus composed of *rajo-guṇa* appears from the navel of the second *puruṣa* who desires to create, Rudra composed of *tamas* appears from the brow of Ananta when Ananta desires to destroy. Lord Ananta is an ocean of unlimited spiritual qualities. For the welfare of all living entities within this material world, he resides in his abode, restraining his anger and intolerance until the proper time. He restrains his anger because at the time of maintenance, it is not suitable to display anger.

Since the Lord is unlimited, what person, even having a thousand tongues, can count the powers of the great Lord of immeasurable powers, Ananta with a thousand heads, who has placed on just one of his heads the small particle known as the universe or Bhū-mandala (*bhū-golaṁ*) with its mountains, rivers oceans and living beings? (5.25.12) On one head, the universe is situated. Who even with a thousand tongues can count the Lord's powers?

mūrdhany arpitam aṇuvat sahasra-mūrdhno
bhū-golaṁ sagiri-sarit-samudra-sattvam
ānantyād animita-vikramasya bhūmnaḥ
ko vīryāṇy adhi gaṇayet sahasra-jihvaḥ

The independent Supreme Lord Ananta, possessing unlimited powers and great qualities, situated below the lower planets, easily supports the Bhū-mandala (*kṣmām*) for its continued existence (SB 5.25.13).

evam-prabhāvo bhagavān ananto duranta-vīryoru-guṇānubhāvaḥ
mūle rasāyāḥ sthita ātma-tantro yo līlayā kṣmāṁ sthitaye bibharti

May I offer respects to the Supreme Lord with a thousand hoods, after whose actions creators of the universe can use their action senses, after whose

perception, the knowledge senses can gather knowledge, and on whose head Bhū-mandala rests like a mustard seed (SB 6.16.48)!

yaṁ vai śvasantam anu viśva-sṛjaḥ śvasanti
yaṁ cekitānam anu cittaya uccakanti
bhū-maṇḍalaṁ sarṣapāyati yasya mūrdhni
tasmai namo bhagavate 'stu sahasra-mūrdhne

Mahabharata, Adiparva, chapter 36 states that Ādi Śeṣa, first son of Kadru and Kashyapa, performed severe austerities not to be with his other envious snake brothers and to be friendly with with his cousine brother Garuda. Pleased with his austerities for self-denial and love of peace, lord Brahma requested him to perform a difficult act for the welfare of all creatures.

prīto'smyanena te śeṣa damena praśamena ca |
tvayā tvidaṁ vacaḥ kāryaṁ manniyogātprajāhitam ||18||

Brahma says: You must bear this unsteady earth, with all its mountains, forests, seas, mineral reservoirs, and cities. Arise! Hold her in place so that the life of the living beings is not disturbed!

imāṁ mahīṁ śailavanopapannāṁ; sasāgarāṁ sākarapattanāṁ ca |
tvaṁ śeṣa samyakcalitāṁ yathāvat; saṅgṛhya tiṣṭhasva yathācalā syāt ||19||

To which Ādi Śeṣa agreed to follow his command, "O divine lord of all creatures, O granter of boons, O lord of the earth, O lord of every created thing, O lord of the universe, I shall hold the earth steady as you command. O Prajapati kindly place it upon My head."

yathāha devo varadaḥ prajāpatir; mahīpatir bhūtapatir jagatpatiḥ |
tathā mahīṁ dhārayitāsmi niścalāṁ; prayaccha tāṁ me śirasi prajāpate ||20||

In obedience to Brahma's command Ādi Śeṣa, the powerful elder brother of the snakes, entering into a hole, went to the other side of the earth. He held that goddess (earth) upon his head with her belt of seas all around her.

tatheti kṛtvā vivaraṁ praviśya sa; prabhurbhuvo bhujagavarāgrajaḥ sthitaḥ |
bibharti devīṁ śirasā mahīmimāṁ; samudranemiṁ parigṛhya sarvataḥ ||22||

The abode of Yamarāja, king of Pitṛs[104]

A class of Pitṛs headed by Agiṣvāttās reside beneath the seven lower planets (*bhūmeḥ*), and slightly above the Garbhodaka Ocean (*jalāt*), desiring virtuous

[104] SB 5.26.4-6: The location and activities of Yamarāja in Pitrloka

blessings for their family members by great concentration on the Lord.

adhastād bhūmer upariṣṭāc ca jalād yasyām agniṣvāttādayaḥ pitṛ-gaṇā diśi svānāṁ gotrāṇāṁ parameṇa samādhinā satyā evāśiṣa āśāsānā nivasanti

There, the king of the Pitṛs, the son of Vaivasvata named, Yamarāja (*bhagavān pitṛ-rājo vaivasvataḥ*), along with his followers, in obedience to the orders of the Lord, awards unavoidable punishment for sin according to the action to the living beings who have been brought to his place after death by his servants. Yama awards punishment according to the crime, which cannot be avoided, allotted according to the faulty action. By that frightening experience he makes the *jīvas* turn towards the Lord.

The Hellish Planets[105]

All the hellish planets are situated within an intermediate space within the three worlds, on the southern side of the universe, beneath the seven lower planets (*bhūmeḥ*), and slightly above the Garbhodaka Ocean *(antarāla eva tri-jagatyās tu diśi dakṣiṇasyām adhastād bhūmer upariṣṭāc ca jalād)*. As stated in SB 3.30.24, the Hells are situated 99,000 yojanas below the earth: "Brought along the path measuring ninety-nine thousand *yojanas* in two or three moments, he then receives punishment."

yojanānāṁ sahasrāṇi navatiṁ nava cādhvanaḥ
tribhir muhūrtair dvābhyāṁ vā nītaḥ prāpnoti yātanāḥ

> Note: The path is ninety-nine thousand *yojanas*. Because of greater sin, it takes only two moments to traverse the path. Muhūrta can mean a moment or a period of 48 minutes. Vijaya-dhvaja-tīrtha explains that the time of travel can be longer, such as ten days, a fortnight or a month.

Pulling him out of his gross body and putting him in a body suitable for punishment, binding him with ropes by the neck, like a king's servants, they forcibly take the prisoner to be punished on a long road. (SB 3.30.20)

yātanā-deha āvṛtya pāśair baddhvā gale balāt
nayato dīrgham adhvānaṁ daṇḍyaṁ rāja-bhaṭā yathā

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[105] SB 5.26.1-7: The science and the descriptions of 28 hellish planets

His heart is pierced by their scolding. Trembling, bitten by dogs on the path, in great pain, he remembers that he is receiving the results of his sins. Afflicted by hunger and thirst, burned by the winds made hot as a forest fire by the sun, beaten on his back by a whip, with difficulty, and though not capable, he moves on a path covered with hot sand, without a resting place with water (*nirāśramodake*). Falling down here and there with fatigue, fainting and being pulled up repeatedly, he is quickly brought along the path of sinners to the abode of Yama. Some authorities say that there are a total of twenty-one hellish planets. According to others, in addition to the twenty-one hells, seven more exist. Thus, there are twenty-eight hells, places of pain for the living entities.

Why do these hellish planets exist?

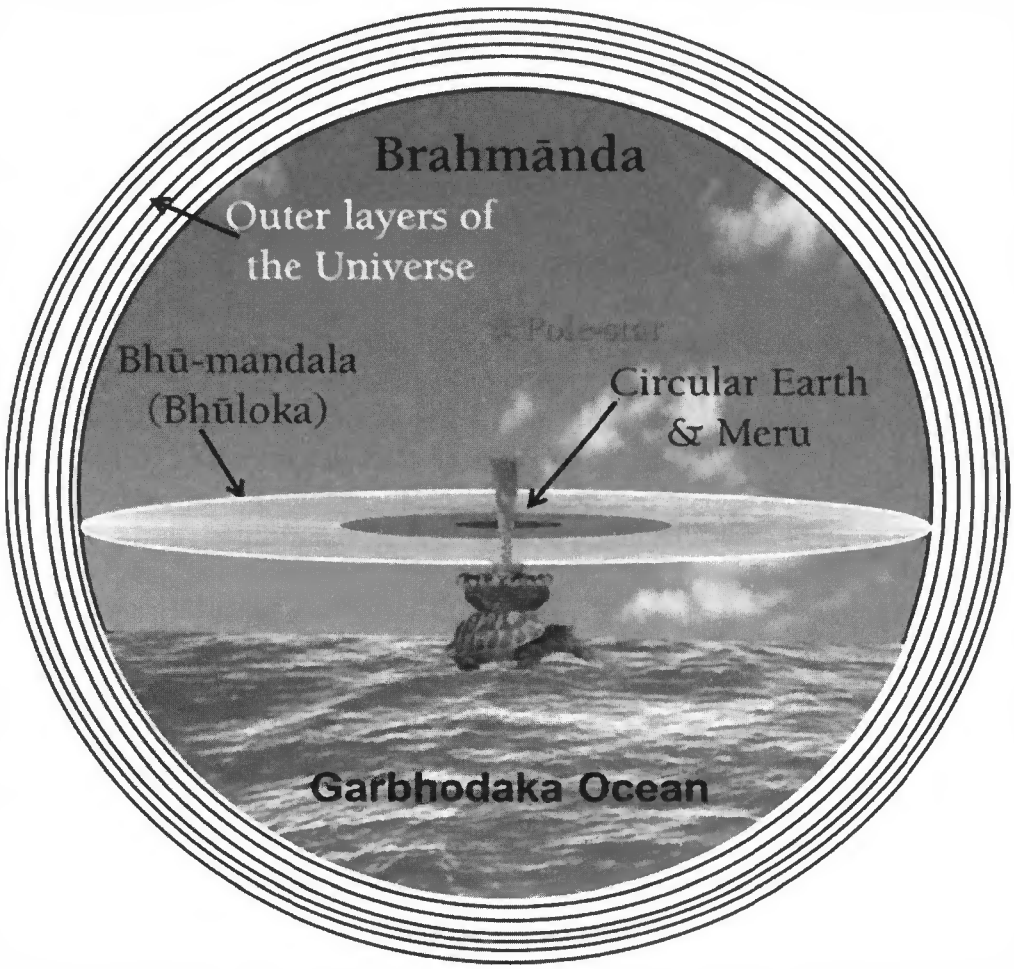
Because of the faith of the doer arising from the three *guṇas,* different varieties of destinations for action appear. All these destinations (heaven, earth or hell) appear for all people in proportion to the *guṇas* (*tri-guṇatvāt kartuḥ śraddhayā karma-gatayaḥ pṛthag-vidhāḥ sarvā eva sarvasya tāratamyena bhavanti*). By faith in *sattva-guṇa, dharma* arises, and from that, happiness for the performer. By faith influenced by rajas a mixture of *dharma* and *adharma* arise, and by them, there arises a mixture of happiness and distress. By faith influenced by *tamas, adharma* only arises, and by that completely distress. By degrees of those types of faith, there are degrees of happiness and distress. No one is always situated in *sattva,* nor is anyone always in *rajas* or *tamas.* Thus, all variety of destinations for all types of *jīvas* arise, but at different times.

The destinations for performing acts out of desire caused by beginningless ignorance are merely transformations of those desires (forbidden acts of *adharma*), with thousands of varieties (*visadṛśaṁ bhavati yā hy anādy-avidyayā kṛta-kāmānāṁ tat-pariṇāma-lakṣaṇāḥ sṛtayaḥ sahasraśaḥ pravṛttās*). The results of performing *dharma* according to scriptures have been described already as happiness on celestial heaven, heaven on earth, and heaven below the earth (*divya-svarga, bhauma-svarga* & *bila-svarga*).

Just as for the performer of *dharma* (*tathā*), for the performer of *adharma* characterized by forbidden acts, by his faith with different degrees of *tamas,* results appear. By performing *adharma* due to inattention, there is a small

amount of *tamas*. By performing sin with knowledge there is greater amount of *tamas*. By performing sin with complete rejection of God there is complete *tamas*. The *jīva* has ignorance without beginning. It is impossible to say when or how the *jīva* developed a relationship with ignorance. The results of desires arising by the *jīva's* beginningless ignorance of three types are transformations of those desires, which are known as various hells. In the kingdom of Yamarāja there are hundreds and thousands of hellish planets (places of pain for the living entities), among them twenty-eight hells are prominent.

Summary of the Placess below Pātāla



Planet	Δdepth (MY)	Depth (MY)	Characteristics of the Planets and People over there
Pitrloka	0.010	0.099	Agiṣvāttās, a class of Pitṛs, dwell here & Yamarāja awards punishment to the sinful.
Hellish planets	0.000	0.099	Situated 99,000Y below the earth, in between the three-worlds and Garbhodaka Ocean
Remaining earth	0.001	0.100	Earth extends still further below Pātāla to a depth of 11,000 yojanas, and it is all rocky in nature.
Below Lower planetary systems			
Śeṣa Nāga*	0.1	0.2	Holds the *kṣiti-maṇḍalaṁ* on one of his thousands of hoods, and feels it like a mustard seed.
Garbhodaka Ocean	249.8	250	Made up of the sweat of Lord Garbhodakasāyi Visnu. Fills the lower half of the universe. Lord sleeps in it on the bed of Ananta.
End of Universe	0	250	The lower end of the Universe, which is 250 MY below the sun planet.

*Lord Ananta, an expansion of the Supreme Lord, holder of the earth, lives 30,000 yojanas below the planet Pātāla. 30,000 yojanas = 16,000 yojanas of Mount Meru base + 3000 yojanas of gaps in between seven lower planets + 11,000 yojanas of solid rocky earth below Pātāla.

Śrīla Śukadeva Gosvāmi's concluding Words

Śukadeva Gosvāmi concludes his descriptions of the Structure of the Universe, by clearly pointing out the purpose of the Creation and the purpose of our existence. He says that "The impious people must enter the various hellish planets according to the degree of their sin. And the pious people enter Svarga according to the degree of their piety. Nevertheless, both the pious and impious are again brought to earth, Bhārata-varṣa, by the remaining results of *dharma* and *adharma*." This means again they return to earth by the remnants of *dharma* and *adharma*, since this is the place where *dharma* and *adharma* accrue (*iha tu punar-bhave ta ubhaya-śeṣābhyāṁ niviśanti*). Because from here (earth, Bhārata-varṣa), one can attain liberation easily.

In the beginning Śrīmad-Bhāgavatam described about how one can progress on the path of liberation: "The *yogī*, having gone by the path of Brahma-loka,

goes to Vaiśvānara by means of the ether, through the luminaries of the *suṣumna-nāḍī*. Being free of all desires, he then approaches the Śiśumāra constellation above, which is related to the Lord." (SB 2.2.24)

vaiśvānaraṁ yāti vihāyasā gataḥ suṣumṇayā brahma-pathena śociṣā
vidhūta-kalko 'tha harer udastāt prayāti cakraṁ nṛpa śaiśumāram

The same is elaborately described by the descriptions of the fourteen planetary systems and the path of sun during uttarāyana and daksināyana. In the Purāṇas the universe divided into fourteen parts is described as the gross body of the Supreme Lord Nārāyaṇa, made of His material *māyā*. If one reads the description of this external form of the Lord with great faith, or if one hears about it or explains it to others, develops faith, *bhakti,* and pure intelligence, and he will understand the topic of the Supreme Lord, which is like an Upaniṣad (*upageyam*), difficult to understand.[106] This is the highest benefits of hearing about the Universal Structure, though it is difficult to understand. Because this amazing gross body of the Lord is the shelter of all the *jīvas.*

"anuvarṇitam ādṛtaḥ paṭhati śṛṇoti śrāvayati sa upageyaṁ bhagavataḥ
paramātmano 'grāhyam api śraddhā-bhakti-viśuddha-buddhir veda"

Although developing this consciousness is very difficult, by this process one can purify oneself and gradually come to an awareness of the Supreme Absolute Truth. One should hear of the universal position of the virāṭ form of the Lord as described in Śrīmad-Bhāgavatam. That will help one save himself from material conditional life and gradually elevate him to the path of liberation so that he can go back home, back to Godhead. After hearing of the gross and subtle forms of the Lord, the renunciate, just as he leads the controlled mind by the intelligence to the gross form of the Lord, leads the mind to subtle form of the Lord. (*śrutvā sthūlaṁ tathā sūkṣmaṁ rūpaṁ bhagavato yatiḥ; sthūle nirjitam ātmānaṁ śanaiḥ sūkṣmaṁ dhiyā nayed iti*). This amazing gross body of the Lord, consisting earth, islands, varṣas, rivers, mountains, sky and oceans, the lower planetary systems, the directions, the hellish planetary systems and luminaries, is the shelter of all living entities. Thus, Śukadeva Gosvāmi concludes.

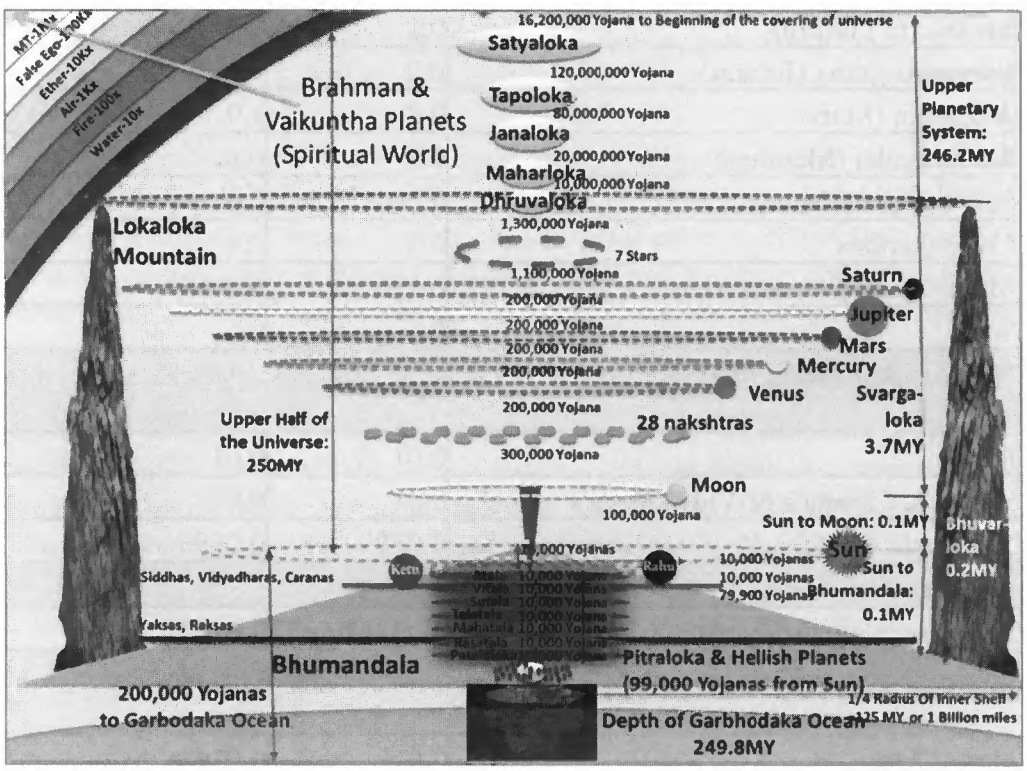
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[106] SB 5.26.38-40: Result of hearing and speaking about Universal form

Summary of the Vertical Structure of the Universe

Planet	Δ Hight (MY)	Hight wrt Sun (MY)
Region of Liberated souls & Vaikuṇṭha planets		> 250 + Shell
Free-space & Outer covering	16.2	250
Upper Planetary System		
Satyaloka	120	233.8
Tapoloka	80	113.8
Janaloka	20	33.8
Maharloka	10	13.8
Svarloka		
Dhruvaloka	1.3	3.8
7 Saintly Stars	1.1	2.5
Śanaiścara (Saturn)	0.2	1.4
Bṛhaspati-graha (Jupiter)	0.2	1.2
Aṅgāraka (Mars)	0.2	1.0
Budha-graha (Mercury)	0.2	0.8
Śukra-graha (Venus)	0.2	0.6
Constellations	0.3	0.4
Moon	0.1	0.1
Sun	0	0
Bhūvarloka: Places of enjoyment for the Yakṣas, Rākṣasas, Piśācas, ghosts and so on.		
Rāhu	0.01	0.01
Siddha's, Cāraṇa's & Vidyādhara's	0.01	0.02
Yakṣa's & Rakṣa's	0.0799	0.0999
Bhūloka / Bhūrloka		
Bhū-mandala (earthly plane)	0.0001 (0.0)	0.1 (0.0)
Lower Planets: Placesses of daityas, dānavas & nāgas in hollow cavities of earth, ***bhū-vivaraḥ***		
Mount Meru	0.016	0.016
Atala	0.010	0.026

Vitala	0.010	0.036
Sutala	0.010	0.046
Talātala	0.010	0.056
Mahātala	0.010	0.066
Rasātala	0.010	0.076
Pātāla	0.010	0.086
Gap between planets (500Yx6)	0.003	0.089
Pitrloka & Hellish planets	0.010	0.099
Below Lower planetary systems		
Remaining earth	0.001	0.10
Śeṣa Nāga	0.10	0.20
Garbhodaka Ocean	249.8	250
Universe's end	0	250



4

The Vedic Planetarium

Daily Movements of the Sun, Moon and other Planets

The Polestar (Dhruva-loka)[107], the place that learned scholars describe as the abode of Lord Viṣṇu (*viṣṇoḥ paramaṁ padam*), around whom Agni, Indra, Prajāpati, Kaśyapa and Dharma rotate with respect together at the same time in the form of the constellations (*samakāla-yugbhiḥ sabahu-mānaṁ dakṣiṇataḥ kriyamāṇa idānīm*). This star, established by the Lord, which is the pivot (*sthāṇur ivāvaṣṭambha*) of all the luminaries, planets and constellations (*sa hi sarveṣāṁ jyotir-gaṇānāṁ graha-nakṣatrādīnām*), which are made to rotate constantly around it by the inconceivable force of the Lord's time energy (*animiṣeṇāvyakta-raṁhasā bhagavatā kālena bhrāmyamāṇānāṁ*), shines constantly (*śaśvad avabhāsate*).

> *Sthānuḥ* means a pillar or stationary, since it remains without being moved by the wheel of time, the rotation of the zodiac. Rather, it is the pivot for the rotating zodiac. All others are dependent upon the rotation of the zodiac, but Dhruvaloka controls the zodiac rotation. There, the son of Mahārāja Uttānapāda, the great devotee Mahārāja Dhruva, remains until the end of the *kalpa* (*kalpa-jīvinām ājīvya upāste*). "How did it get this ability?" It has been allotted this power by the Lord, who can cause to do, not do or do differently, in order to show the excellence of His devotee.

Just as bulls yoked together and tied to a central post to thresh rice tread around that pivot without deviating from their proper positions (*yathā-sthānaṁ maṇḍalāni caranty*) one bull being closest to the post, another in the

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[107] SB 5.23.1-3: Glories of Dhruva planet (Polestar)

middle, and a third on the outside, so all the planets and luminaries, fixed in the zodiac (*kāla-cakra*), revolve around the polestar in their respective orbits (*paricaṅ kramanti*), some higher and some lower, till the end of the *kalpa* (*ākalpāntaṁ*), propelled by wind (*vāyunodīryamāṇā* or *dakṣiṇāvarta pravāha*). Just as clouds or eagles, controlled by the wind (*vāyu-vaśāḥ*), pushed by *karma* (*karma-sārathayaḥ*), move about in the sky, so the planets, supported by matter and the Lord (*prakṛti-puruṣa-saṁyogānugṛhītāḥ*), and moving (*parivartante*) by their past *karmas* (*karma-nirmita-gatayo*), do not fall to the ground.

> In terms of the planets, *tri-savanaiḥ* refers to the summer solstice, equinox and winter solstice. The planets are fixed at lower and higher (*antar-bahir-yogena*) positions in the zodiac belt. Thus, Saturn and other planets are fixed by the Supreme Lord below the seven sages, and the moon and other planets are fixed above the sun. They move in clockwise direction keeping Mount Meru and Polestar to their right, being turned by the parvartaka-pravaha wind. "Why do they not fall from the sky?" Just as clouds float in the sky, the planets also float in the sky being assisted by karma.

The Zodiac Belt or the Kālacakra-gata[108]

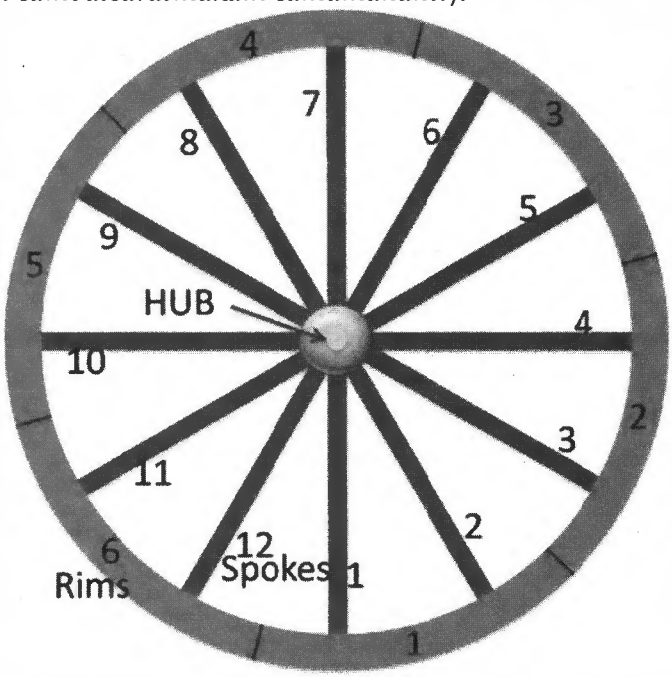
The original cause of the cosmic manifestation, the Supreme Lord (*bhagavān ādi-puruṣa*), Nārāyaṇa, made of the Vedas (*trayīmayaṁ*) worshipped by the sages by recitation of the Vedas (*trayīmayaṁ*), divided Himself into twelve forms (*dvādaśadhā vibhajya*) and six seasons (*ṣaṭsu vasantādiṣv ṛtuṣu*) for purifying the activities of the people (*karma-viśuddhi-nimittaṁ*), and He arranged the qualities of the seasons for enjoyment according to their *karma.*

> The Lord famous as time, divided Himself into twelve forms and into six seasons for purifying the activities of the people. This means through time He regulates their actions. He arranged for the qualities of the seasons such as heat and cold, which are experienced according to *karma (yathopajoṣam).* Thus the Lord manifests Himself as the Time-wheel to regulate and purify our activities. This is for the purpose of engaging and disengaging people in work according to particular times (*bhūtānāṁ pravṛtti-nivṛtti-nimittāni samaya-viśeṣeṇa*).

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[108] SB 5.22.3-5: Glories of the Supreme Lord, who descended as the Time-wheel

This form of the Lord is known as kālacakra-gata or Time-wheel, in which the chariot of sun-god travels in an orbit known as samvatsara. This wheel has twelve spokes, six rims, and three hubs (*yasyaikaṁ cakraṁ dvādaśāraṁ ṣaṇ-nemi tri-ṇābhi saṁvatsarātmakaṁ samāmananti*).



The spokes of the wheel are the twelve months. The six rims are the six seasons. The three hubs[109] are the three sets of four months or the seasons corresponding to Uttarāyana, Visuvāyana and Daksināyana. The sun, soul of worlds, situated in the zodiac belt (kālacakra-gato) in the *antarikṣa* between heaven and earth (*antareṇa nabho-valayasya*), enjoys twelve months (*dvādaśa māsān*), indicated by the names of zodiac signs (*rāśi-saṁjñān*), which are limbs of the year (*saṁvatsarāvayavān*). Thus, the sun, as time or the month, enjoys a twelfth portion of the celestial circle, which moves on its own. People, by following *varṇāśrama* conduct and worshipping the sun with faith by *aṣṭāṅga-yoga* (*yoga-vitānaiḥ*) and by regulated activities high and low, which are authorized by the Vedas (*trayyā*), easily attain the highest goal.

Some describe the great machine or the zodiac belt (*jyotir-anīkaṁ*), consisting

[109] The three hubs are three cātur-māsya periods or the position of the wheel at the crossing of *uttarāyaṇa, dakṣiṇāyana* and *viṣuvāyana* respectively.

of the stars and planets, resembles the form of a *śiśumāra* [dolphin] in the water. It is sometimes considered an incarnation of Kṛṣṇa, Vāsudeva. Therefore, some yogīs contemplate this imaginary *śiśumāra* to be swimming in the sky the way a dolphin swims in water. They meditate upon it as the virāṭ-rūpa, the gigantic form of the Supreme Lord, because it is actually visible. Thus, they describe this zodiac belt (*jyotir-anīkam*) to be the form of a dolphin and worship it as Vāsudeva in meditation. This form of the *śiśumāra* has its head downward and its body coiled. The coiled body of the dolphin turns toward the right side.

Notes from Bhugola Varnanam on Kala-Chakra[110]

The wonder-working Lord has formed this Kala Cakra in order to create facility for the courses of the nine planets, the Rasis (constellations of stars), the Taras (stars), the seven sages (Saptarsis) and the divine, dazzling star-like vehicles of Gandharvas, Siddhas and the merited human souls. The courses are located differently in space and the Lord holds them all in their courses at all times. The courses are all circular at different distances and altitudes, and they look quite brilliant with the luster of rare gems. The courses are thousands and thousands of yojanas apart. The wheels of the Kala Cakra are twelve (nine for the nine Planets, one for the Saptarsis, one for the stars and one for the Rasis or constellations). The knower's of scriptures call this as Kala Cakra. Since both Hemanta and Sisira seasons are considered together, the Cakra is considered as having five spokes (*Pancara*); and, with reference to the twelve months, it is said to be having twelve spokes (*Dvadasara*).

Since the mechanism helps the counting of time divisions, it is known as Kala Cakra. This Kala Cakra extends over crores of yojanas in the wide sky. Above the Meru mountain, the Supreme Lord, in the form of Simsumara (the scorpion form) stands to a Height of 36 lakh yojanas, with his pretty face turned downward and his tail end held up to touch the Dhruva Mandala (polestar orbit). The body of this form is ringed in the middle and the tail part goes up. The brilliance of this form of Lord "Simsumara is much more than that of a thousand suns. It is He who makes the Time Wheel to go round and round as if it is a part of Himself. The main shaft-like power line has its starting point

[110] Refer to the Section "Kala chakra, The Time-wheel (242-256)"

atop the Meru Mountain and extends outward to the extent of crores of yojanas.

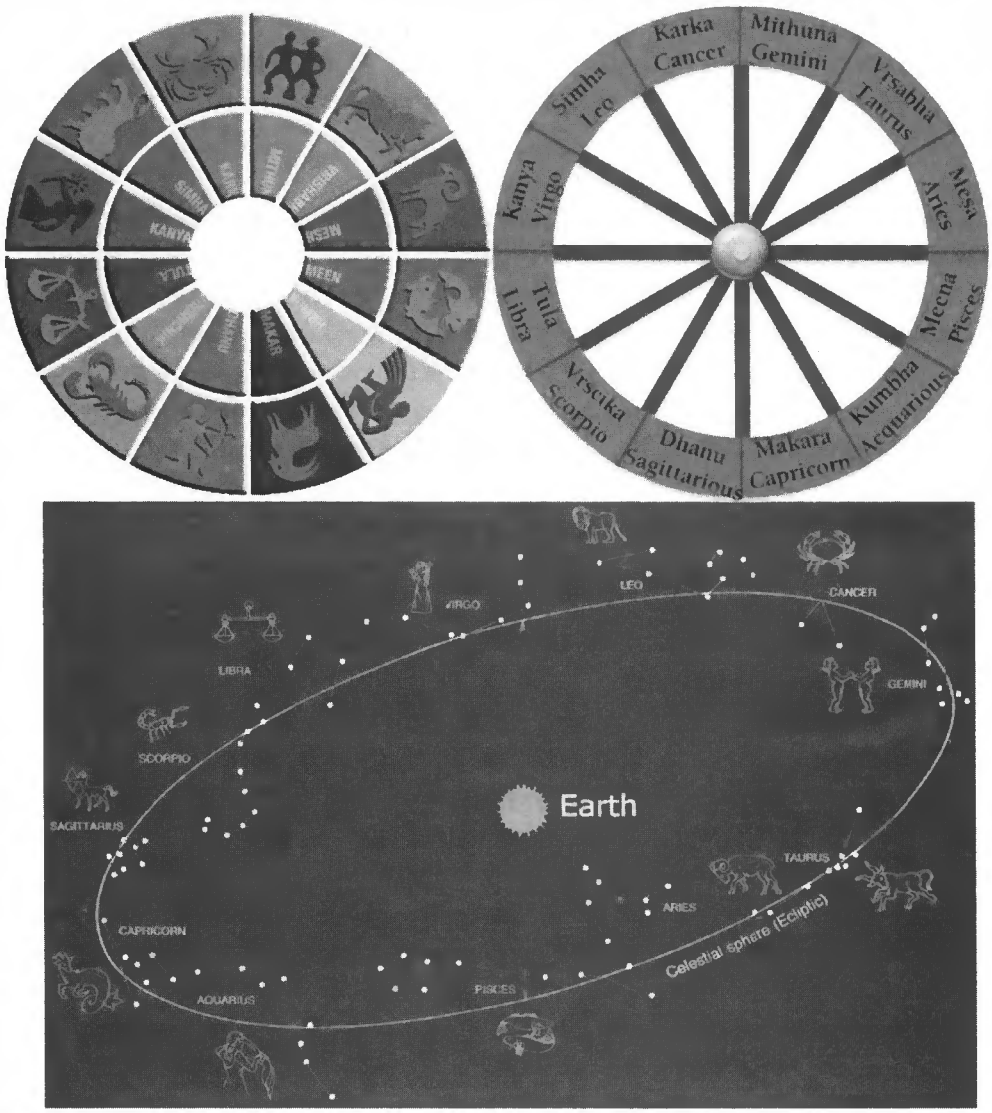
Although Time is an abstract concept, yet it is something. Time is something although it cannot be perceived by our external senses, as material objects are perceived. The order of events, parts and wholes, continue to be and they have their pasts, presents and futures. Therefore, the sense of Time is eternal. Therefore eternality, in its abstract sense (apart from events), is undivided and inconceivable; but yet it is there. It is known as Maha Kala, which cannot be recognized as such by us; but the Omniscient eternal coordinator of all things knows it as something that is.

The Maha Kala, running like a continuous stream, presents itself to be recognized in the form of the Time Wheel giving room for the movements of the planets and enabling human beings to take note of the passing of time in its fractional parts like moments and its wholes like hours, days, months and years. Time is not dependent on its relation to the occurrence of events. In fact, in a way, everything else depends upon Time; or Time is almost synonymous with the Will of God. Thus, the 'Will of God' orders the events of the universe.

> In the description of the Kala Cakra given by Saint Śrīla Vādirāja Tīrtha swami, we see that there are star-like divine vehicles distinct from the 27 star groups and the constellations. They are said to be crores and crores in number and crores and crores of yojanas apart. Certainly, our Saint did not use a modern telescope, nor did he use electronic methods for computing numbers and distances. Yet, here is the truth about heavenly bodies. Vedic information is correct. Modern findings, though approximating Vedic truth, are nevertheless faulty to an extent.
> Suppose a star-like vehicle of a Gandharva king's servant is following the star like vehicle of the Master. The two being different in brilliance, how can the scientist distinguish them precisely? Evidently, his calculations will be erroneous. The mention of *Dhruva mandala* suggests that the seemingly motionless polestar has its own very small local orbit. The word *Mandala* is suggestive. Scientists do not know how the celestial bodies acquired their motions at unimaginable velocities. They simply call it 'constant velocity'. Why and how, they cannot tell.

The pictorial representation of the Zodiac Belt (*kālacakra-gata*):

The 12 Zodiac signs, such as Simha, Tula etc., get their names based on the form manifested by the cluster of stars representing that sign. For example, if a set of stars are arranged in a Lion shape, then that particular sign is called as Leo or Simha Rāśi. This is shown in a pictorial form below.

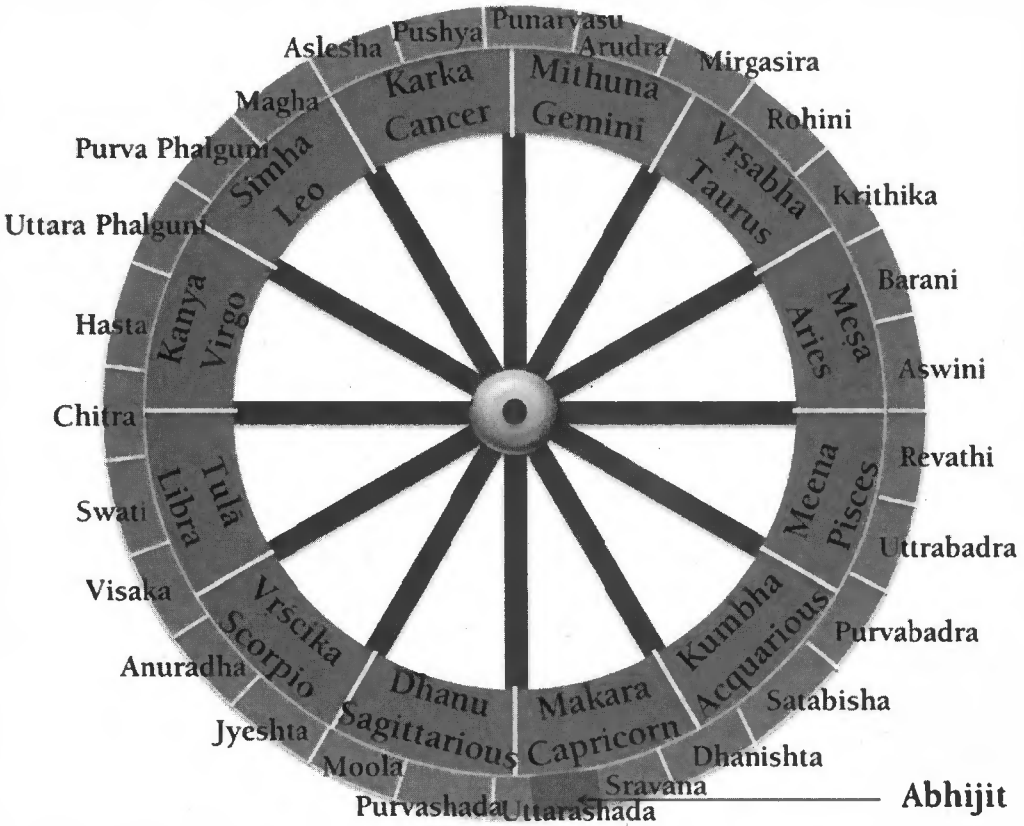


The Constellations or Nakṣatrās

The constellations, twenty-eight in number with Abhijit, are being inspired by the Lord move in clockwise direction around Mount Meru on the wheel of time (*nakṣatrāṇi merum dakṣiṇenaiva kālāyana īśvara-yojitāni sahābhijitāṣṭā-viṁśatiḥ*). They do not have a different motion, but move clockwise along with the Time wheel (*kālāyane*). *Śruti* says *abhijin nāma-nakṣātram upariṣṭād āṣāḍhānām adhastāc chroṇāyāḥ:* Abhijit constellation is composed of the last part of Āṣāḍhā and the first part of Śravaṇā constellations. The astronomical works also confirm this: Abhijit is the last part of Āṣādha and the first part of Śravaṇā. Each constellation is consisting of four daṇḍas or four padas.

Superimposition of the Constellations on the Zodiac Belt, Śiśumāra chakra:

<table>
<tr><td>Leo / Simha
Magha 1-2-3-4
Purva Phalg 1-2-3-4
Uttara Phalguni 1</td><td colspan="2">Cancer / Karka
Aslesha 1-2-3-4
Pushya 1-2-3-4
Punarvasu 4</td><td>Gemini / Mithuna
Mirgasira 3-4
Arudra 1-2-3-4
Punarvasu 1-2-3</td><td>Taurus / Rishabha
Mirgasira 1-2
Rohini 1-2-3-4
Krithika 2-3-4</td></tr>
<tr><td>Virgo / Kanya
Uttara phalg 2-3-4
Hasta 1-2-3-4
Chitra 1-2</td><td colspan="3" rowspan="2">Numbers (1-2-3-4) are pada or part.
Each part is 3 1/3 degree
Each Constellatiom is 13 1/3 degree
Each Zodiac sizn has 2.25 Constellations
Each Zodiac sizn has 9 padas of 3 constelltions</td><td>Aries / Mesha
Krithika 1
Barani 1-2-3-4
Aswini 1-2-3-4</td></tr>
<tr><td>Libra / Tula
Chitra 3-4
Swati 1-2-3-4
Visaka 1-2-3</td><td>Pisces / Meena
Revathi 1-2-3-4
Uttrabadra 1-2-3-4
Purvabadra 4</td></tr>
<tr><td>Scorpio / Rischika
Visaka 4
Anuradha 1-2-3-4
Jyeshta 1-2-3-4</td><td>Sagittarius / Dhanu
Uttarashada 1
Purvashada 1-2-3-4
Moola 1-2-3-4</td><td>Capricorn / Makara
Uttarashada 2-3-4
Sravana 1-2-3-4
Dhanishta 1-2</td><td>Aquarius / Kumbha
Purvabadra 1-2-3
Satabisha 1-2-3-4
Dhanista 3-4</td></tr>
</table>



Notes from Bhugola Varnanam on Kala-Chakra[111]

There is one horizontally placed wheel of gold (not the gross material gold with which we are familiar), located just above the Mansottara mountain in Puṣkaradvīpa, the outermost of the seven islands. This wheel always goes round and round and it is meant to take the entire Kala Cakra with it.

> Every heavenly body has its own wheel forming part of the Kala Cakra to go round along with it, but with varying velocity. The entire Kala Cakra may be compared to an open umbrella with its apex at the polestar and its base at the main wheel. The main wheel of the Kala Cakra, which is meant to take the entire mechanism with it, is studded with stars and star-like bodies. Thus, it is known as Jyotiscakra. This wheel moves because Lord Simsumara moves it.

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[111] Refer to the Section "Main Wheel (Dhara) of Kala Cakra (257 - 265) and Sun's Chariot (266 - 295)"

The movement of the sun's chariot is similar to the movements of the other celestial chariots those of the planets, the stars and the like. Every one of these chariots has its own circular golden path befitting it. Upon the main wheel of the Kala Cakra runs the sun's chariot drawn by seven horses. This chariot has only one wheel and its course upon the Kala Cakra is right above the Mansottara mountain ring.

The Kala Cakra or Jyotiscakra moves with great velocity day after day, having the Meru Mountain at the centre. This Kala Cakra covers a distance of nine crore and fifty-one lakh yojanas every day, taking the stars and planets along with it. The changes in season's cold, warm and hot occur on account of this Jyotiscakra. This is due to the northern and the southern courses manipulated by the lifting and lowering of the Time Wheel by the tail end of Lord Simsumara. Because of the three seasonal changes effected, this Cakra is said to consist of three sections of its hub (Trinabhi). This Cakra (main wheel) itself is made to run without the need of horses to draw it; hence it is called '*anarva*, horseless.'

The Kala Cakra extends from the Meru centre to the mid part of the Maṁasottara Mountain and reaches right upto the polestar above. The Svetadvīpa Abode of Lord Sri Narayana is located in the Milk Ocean, within the bounds of the Cakra.

Daily Movement of the Sun through the Zodiac Belt[112]

Situated in the middle of the antarīkṣa, the sun, master of heat, heats up and lights up the three worlds by its light (*yan-madhya-gato bhagavāṁs tapatāṁ patis tapana ātapena tri-lokīṁ pratapaty avabhāsayaty ātma-bhāsā*). As per the learned, the sun travels over all sides of Mānasottara Mountain in a circle whose length is 95.1 million yojanas (*evaṁ nava koṭaya eka-pañcāśal-lakṣāṇi yojanānāṁ mānasottara-giri-parivartanasyopadiśanti*).

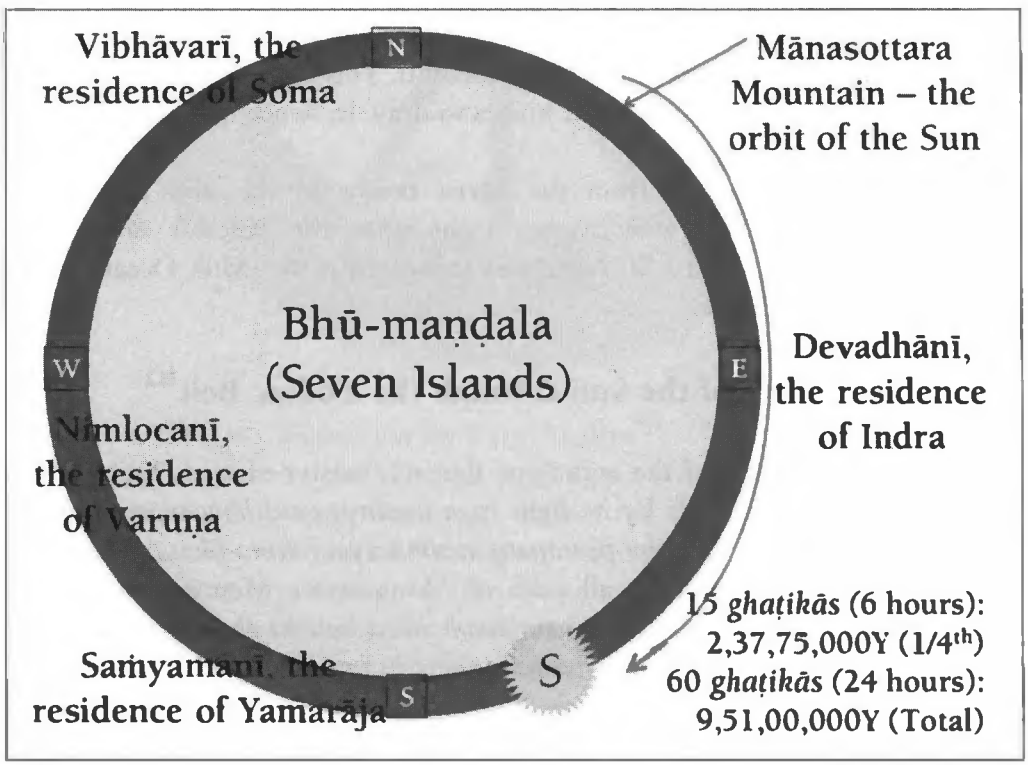
2*pi*R ~= 2*3*15.85 MY = 95.1 MY or 760.8 million miles

The distance from Mount Meru to Mānasottara Mountain (Radius of Mānasottara) is 15.75MY. The path of the sun's chariot is 0.2MY wide. Thus, the radius till the middle of sun's path is 15.85MY

[112] SB 5.21.7–12: Description of the daily movement of the Sun

The wise teach about the city called Devadhānī, east of Mount Sumeru, possessed by King Indra; a place known as Saṁyamanī, possessed by Yamarāja, to the south of Meru; a place known as Nimlocanī, possessed by Varuṇa, to the west of Meru; and a place named Vibhāvarī, possessed by the moon-god to the north of Meru are on the path of the suns revolution on Mānasottara Mountain. They teach about the sunrise, noon and sunset and midnight in these cities for the purpose of engaging and disengaging people in work according to particular times. They teach about the four directions of Meru.

tāsūdaya-madhyāhnāstamaya-niśīthānīti bhūtānāṁ
pravṛtti-nivṛtti-nimittāni samaya-viśeṣeṇa meroś catur-diśam



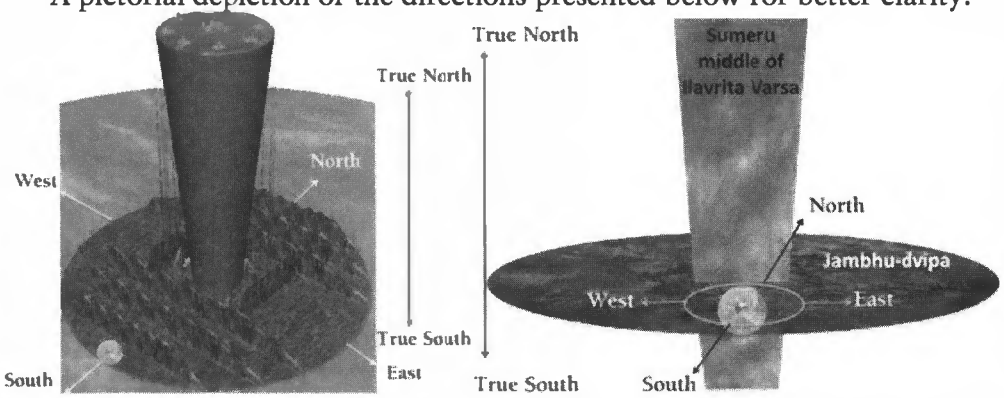
They teach about the sunrise, noon and sunset in these cities, being in the four directions from Meru. This means the following.

- Those who reside to the south of Meru perceive sunrise, when the sun is starting from the east (or eastern city of Indra).

- Those residing in the west perceive sunrise when the sun is starting from the south
- Those residing in the north perceive sunrise, when the sun is starting from the west, and
- Those residing to the east perceive sunrise, when the sun is starting from the north.

As per *Viṣṇu Purāṇa* all the inhabitants have Meru in the north (relatively speaking). Because whichever side they see the sun is east for them, and to their left hand side is north and to their right hand side is south respectively. Thus, all the inhabitants of Bhū-maṇḍala perceive that Mount Meru is on their northern side, and they perceive the remaining three directions accordingly.

Note: These north and south directions as mentioned above and also as mentioned in chapter 16 are only for the sake of explaining the geography of Bhū-maṇḍala and the Movement of the sun. They are not the True North and True South directions, which will be discussed in future sections while discussing the sun's movement during *uttarāyaṇa* and *dakṣiṇāyana*. A pictorial depiction of the directions presented below for better clarity.



Daily Motion of the Sun and Day-Night perception of by different people

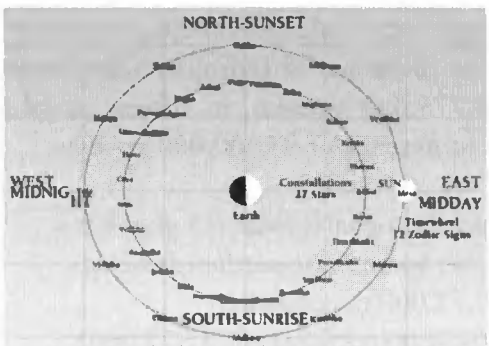
From Devadhānī, the residence of Indra, in the east, the sun travels to Saṁyamanī, the residence of Yamarāja, (in the south) then to Nimlocanī, the residence of Varuṇa (in the west), from there to Vibhāvarī, the residence of the moon-god (in the north), and from there again to the residence of Indra (in the East). When the sun rises in one city, it sets at the city diametrically opposite

(*yatrodeti tasya ha samāna-sūtra-nipāte nimlocati*). When it shines with perspiration at noon, it is midnight at the city diametrically opposite (*yatra syandenābhitapati tasya haiṣa samāna-sūtra-nipāte prasvāpayati*). Those who see the sun set and rise do not see where it has gone in the interim.

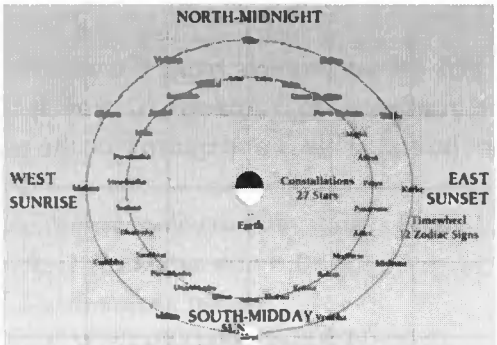
> The sunrise means seeing from far off apparently in contact with the earth, because of the sun's rotational movement. (*Viṣṇu Purāṇa* says *yair yatra dṛśyate bhāsvān sa teṣām udayaḥ smṛtaḥ:* wherever the sun is seen by particular persons, they call that the sunrise.) First sight of the sun is called sunrise. Seeing the sun apparently in the middle of the sky is noon (*syandena*). Sunset (*nimlocati*) is not seeing the sun, because it apparently enters the earth. Midnight (*prasvāpayati*) is the sun in a very far position from the observer or in the opposite side of the observer. *Śruti* states that "*adbhyā vā eṣa prātar udety apaḥ sāyaṁ praviśati*" the sun rises from the ocean in the morning, and enters it in the evening. This *śruti* statement when one is standing on the ocean shore is conventional, not actual.

The arrangements for rising, setting, midday and midnight according to the different *varṣas* are next described.

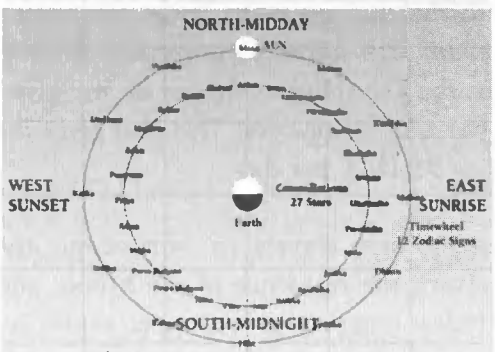
- When the sun is in the Eastern direction of Meru, then those who reside on the South of Meru perceive sunrise, on West midnight, on the East midday and on North sunset.
- When it rises (first sight of the sun) in one city, it sets at the city diametrically opposite.
- When it shines with perspiration at noon, it is midnight at the city diametrically opposite.
- Amongst the four directions around Meru, wherever the sun is seen to rise, it is noon in the *varṣa* to the east, midnight to the *varṣa* in the west, & sunset in the *varṣa* to the north.
- When it is noon in southern *varṣa*, it is sunset in the eastern *varṣa*, sunrise in the western *varṣa* and midnight in the northern *varṣa*.
- When the sun is situated in the eastern varṣa at noon, there is sunrise in the southern *varṣa*, sunset in the northern *varṣa* and midnight in the western *varṣa*.
- For the people on top of the Meru, the sun give heat at all times (around 6 months), being situated at midday.



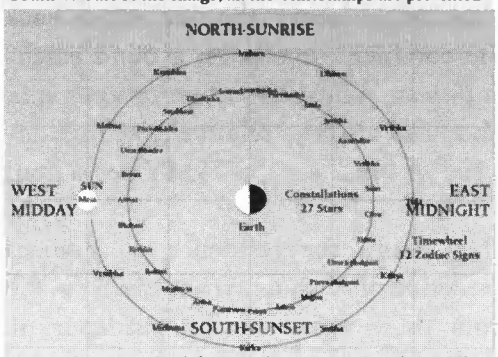
Considering Earth Globe (i.e. Bharata Varsa) in the Middle and the Place under consideration (i.e. INDIA) in the Southern Part of the Image, all the Timestamps are presented



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Considering Earth Globe (i.e. Bharata Varsa) in the Middle and the Place under consideration (i.e. INDIA) in the Southern Part of the Image, all the Timestamps are presented

Note: See the above images in a clockwise manner to perceive sunrise, noon, sunset and midnight of a person residing in southern *varṣa* (*Bhārata-varṣa*) after every 6 hours of a day. Same can be seen for the other *varṣas* respectively.

- Thus, all the people situated in all the *varṣas* consider themselves situated to the south of Meru and simply see sunrise, noon and sunset in their own *varṣa,* and know the events of the sun in other *varṣas* by the previously mentioned conception.
- In summary, the first sight of the sun is called sunrise (*dṛśyate*), where the sun rises, simultaneously it sets (*Nimlocati*) at a point exactly opposite.
- After a period of fifteen *ghaṭikās* (15 *ghaṭikās* = 15*24 minutes = 6 hours) after sunrise, the sun is in the middle of the sky, causing perspiration (*syandena*), and at the opposite point it causes midnight (*prasvāpayati*).
- Thus, after every fifteen *ghaṭikās* (6 hours) the people residing in a *varṣa* experience sunrise, noon, sunset and midnight in sequence respectively.

When the sun travels from Devadhānī, the residence of Indra, to Saṁyamanī, the residence of Yamarāja, it travels 23,775,000 *yojanas*, in fifteen *ghaṭikās* (six hours). This is one quarter of the circumference of 95,100,000 yojanas.

yadā caindryāḥ puryāḥ pracalate pañcadaśa-ghaṭikābhir (15 ghaṭikās = 6 hours*) yāmyāṁ sapāda-koṭi-dvayaṁ yojanānāṁ sārdha-dvādaśa-lakṣāṇi sādhikāni (2,37,75,000Y) copayāti*
(2,00,00,000+25,00,000+12,00,00,000+50,000+25,000 = 2,37,75,000Y)

Thus, in 24 hours, the sun comes back to the starting point, after completing one complete revolution around earth (*evaṁ tato vāruṇīṁ saumyām aindrīṁ ca punas*). Thus, this observation explains the Day-Night changes of any given place and also the relative time in a day w.r.t, other placess. This also gives the speed of sun, i.e., 23.775MY per 6 hours or 95.1MY per day.

Thus, from the residence of Yamarāja the sun travels to Nimlocanī, the residence of Varuṇa, from there to Vibhāvarī, the residence of the Moon, and from there again to the residence of Indra (*evaṁ tato vāruṇīṁ saumyām aindrīṁ ca punas*). The chariot of the sun-god, meant for worship, travels through the four residences mentioned above at a speed of 3,400,800 *yojanas* in a *muhūrta.*

evaṁ muhūrtena catus-triṁśal-lakṣa-yojanāny (34,00,000) aṣṭa-śatādhikāni (800) sauro rathas trayīmayo 'sau catasṛṣu parivartate purīṣu

There are 30 *muhūrtas* in a day, Thus, the speed of the sun is 30x3.4008 = 102.024MY per Day. This is the second speed of sun.

In his orbit through a circumference of 95,100,000 *yojanas* on the circle of the earth, the sun travels 2,000 *yojanas* and two *krośas*[113] in a moment. This is the Third Speed of the sun, which is 2000.5 yojanas per moment (8/5 seconds) = 108.027MY per Day.

lakṣottaraṁ sārdha-nava-koṭi-yojana-parimaṇḍalaṁ(9,51,00,000) bhū-valayasya kṣaṇena sagavyūty-uttaraṁ dvi-sahasra-yojanāni sa bhuṅkte.
((9,00,00,000+50,00,000+1,00,000 = 9,51,00,000Y)

---

[113] *Gavyūti* is two *krośas* (half a *yojana*). 1 *yojana* = 2 *gavyūtis* = 4 *krośas* = 8 *miles*

The three different speeds of the chariot of sun-god

Speed of the Sun	S (MY /hour)	C: S*24 (MY)	R: C/6 (MY)
Speed1 in SB 5.21.10: Distance traveled by sun in 15 ghaṭikās (6 hours) is 23,775,000 yojanas	3.9625	95.1	15.85 (15.136)
Speed2 in SB 5.21.12: Distance that sun travels through four cities of Demigods in a muhūrta (48 minutes) is 3,400,800 yojanas.	4.251	102.024	17.004 (16.38)
Speed3 in SB 5.21.19: Distance covered by sun in the orbit through Bhū-maṇḍala in a moment or kṣaṇa (1.6 sec) is 2000.5 yojanas	4.501125	108.027	18.0045 (17.193)

* 1 Ghatika = 24 min's (30 Ghatikas in a day or night or 12 hours)
S: Speed of the chariot of sun-god (MY/hour);
C: Distance travelled in 24 hours (C = S*24 MY)
R: Radius of sun chariot's path (R = C/(2*pi) = C/6 MY, where pi = 3 approx.)
The R value in the bracket is with pi = 3.1416
Thus, the chariot of the sun-god travels through the four residences in his orbit through a circumference of 95.1MY on the circle of the earth above Mānasottara Mountain continuously.

Yearly Movement of the Sun w.r.t. the Time-wheel[114]

For the people on Meru, the sun give heat at all times, being situated at midday. The sun keeps Meru to the left and to the right (*tatratyānāṁ divasa-madhyaṅgata eva sadādityas tapati savyenācalaṁ dakṣiṇena karoti*). From the residence of Yamarāja the sun travels to Nimlocanī, the residence of Varuṇa, from there to Vibhāvarī, the residence of the moon-god, and from there again to the residence of Indra. In a similar way, the moon, along with the other stars and planets, becomes visible in the zodiac belt and then sets.

evaṁ tato vāruṇīṁ saumyām aindrīṁ ca punas tathānye ca grahāḥ somādayo nakṣatraiḥ saha jyotiś-cakre samabhyudyanti saha vā nimlo-canti (5.21.11)

Hearing that the powerful sun-god travels around Dhruvaloka with both Dhruvaloka and Mount Sumeru on his right, yet at the same time he faces the

[114] SB 5.22.1-2: Clockwie and Counterclockwise Movements of Sun

zodiac signs and keeps Sumeru and Dhruvaloka on his left in SB 5.21.8, Parīkṣit asks for clarification on "*how can sun keep Sumeru and Dhruvaloka on both his left and right simultaneously?*"

> Daily the sun appears to be moving in a clockwise direction. However, you have said that, facing the constellations, the sun goes counterclockwise. How can this be conceived, since it is contradictory?
>
> *yad etad bhagavata ādityasya meruṁ dhruvaṁ ca pradakṣiṇena parikrāmato rāśīnām abhimukhaṁ pracalitaṁ cāpradakṣiṇaṁ bhagavato pavarṇitam amuṣya vayaṁ katham anumimīmahīti*

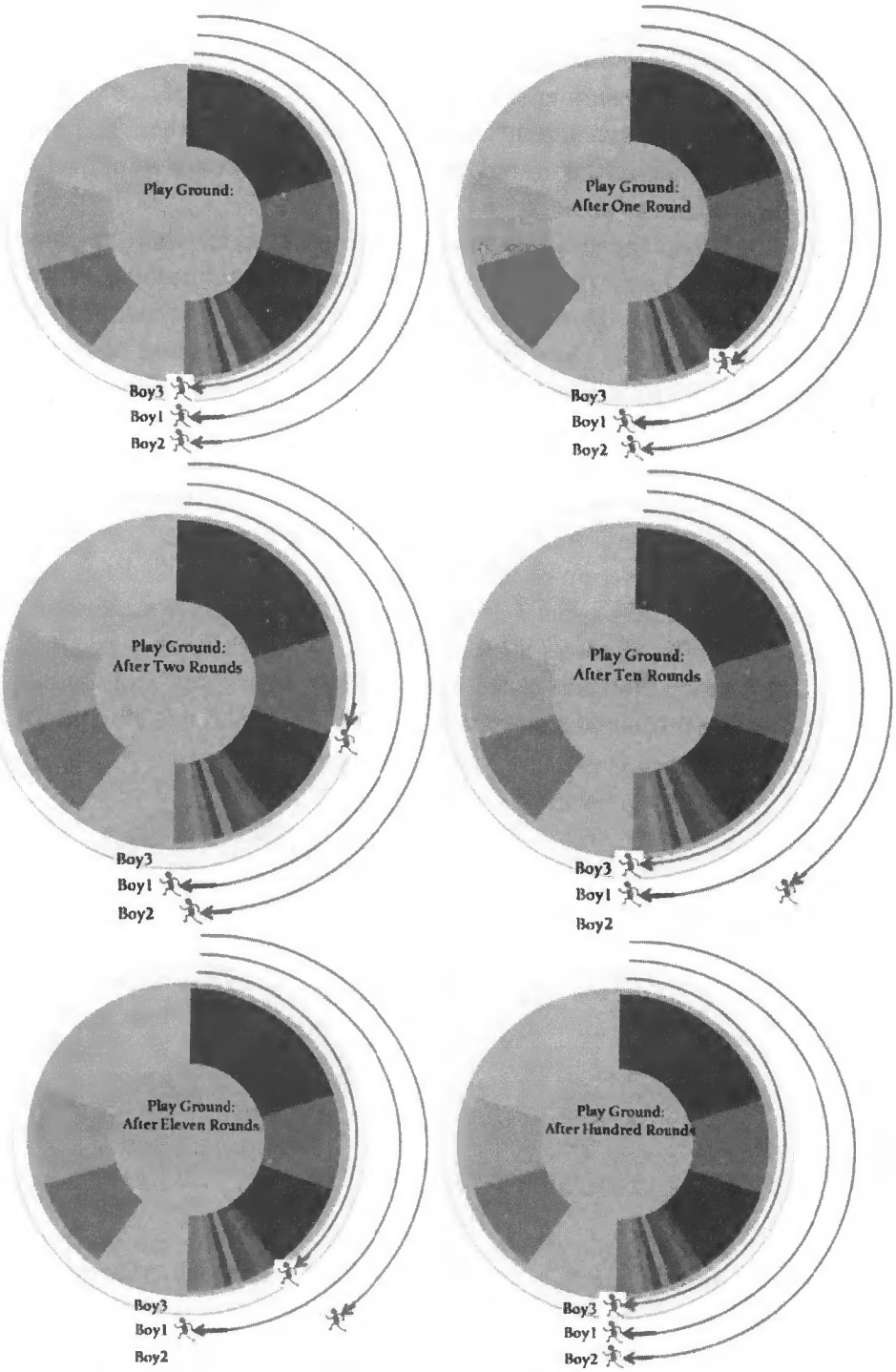
Śukadeva Gosvāmī answers with Potter's wheel analogy: "Just as the motion of ants (*pipīlika-ādīnām*) on a potter's wheel (*kulāla-cakreṇa*) moving in the opposite direction (*bhramatā saha bhramatām*) from the rotation of the wheel is different from that of the wheel, because they can be observed moving from place to place on the wheel.

yathā kulāla-cakreṇa bhramatā saha bhramatāṁ tad-āśrayāṇāṁ pipīlikādīnāṁ gatir anyaiva pradeśāntareṣv

Similarly, the motion of the planets and sun moving with the wheel of time which is moving clockwise around Meru and Dhruva as seen by the stars is different from the movement of the wheel of time (*kāla-cakreṇa*), since they can be observed moving in different constellations and zodiac signs (*nakṣatra-rāśibhir*) in the opposite direction." Thus, Śukadeva Gosvāmī clarifies his statement. They are moving in the opposite direction from the wheel (*bhramatā saha bhramatām*). There is no contradiction since there are two separate movements: the movement of the wheel and the individual movement.

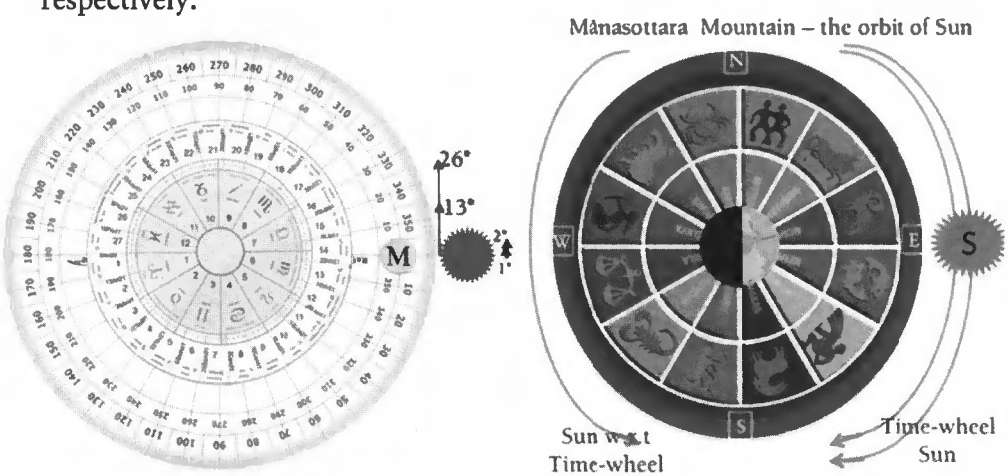
> Though the sun keeps Meru on its left (*savyena*) in relation to the moving constellations, but being turned by the *parvartaka-pravaha* wind (from *purva* to *daksina*), sun keeps the Meru to the right on a daily basis. Thus the sun moves from east to west in the sky on daily basis, but in relation to the zodiac and other stars, the sun appears to move from Aries to Taurus, and to other signs in the opposite direction. Thus, the sun can be said to be moving counterclockwise as well as clockwise.
>
> *evaṁ nakṣatra-rāśibhir upalakṣitena kāla-cakreṇa dhruvaṁ meruṁ ca pradakṣiṇena paridhāvatā saha paridhāvamānānāṁ tad-āśrayāṇāṁ sūryādīnāṁ grahāṇāṁ gatir anyaiva nakṣatrāntare rāśy-antare*

The Movement of the Sun and Moon in the Zodiac Belt


Play Ground:
Boy3
Boy1
Boy2
Play Ground:
After One Round
Boy3
Boy1
Boy2
Play Ground:
After Two Rounds
Boy3
Boy1
Boy2
Play Ground:
After Ten Rounds
Boy3
Boy1
Boy2
Play Ground:
After Eleven Rounds
Boy3
Boy1
Boy2
Play Ground:
After Hundred Rounds
Boy3
Boy1
Boy2

The movements of the sun and moon w.r.t the Zodiac Belt explained through the analogy of three boys moving around a Playground

- When three boys running in a playground with different speeds in clockwise direction (*savyena*), after one round we see their positions are much different than the initial position. After second round, they are entirely in a different position.
- As shown above, the Boy2 is moving with 1% lesser than the speed of Boy1, then after every round, he lags behind by 1% of total distance w.r.t the Boy1. Thus, after 10 rounds he is behind by 10% and after 100 rounds he is behind by 100%. Then Boy2 meets Boy1 at the starting point with the completion of 99 rounds only.
- Similarly, the Boy3 is moving with 10% lesser than the speed of Boy1. Thus, after every round he lags behind by 10% of total distance w.r.t the Boy1. Then after 10 rounds, he meets Boy1 at the starting point with the completion of 9 rounds and after 100 rounds he meets Boy1 at the starting point with the completion of 90 rounds.
- Thus, we can see that when Boy1 completes 100 rounds Boy2 completes 99 rounds and Boy3 completes 90 rounds only.
- Thus, we can see different speed among Time wheel, sun and moon as per the above illustration when we compare them with Boy1, Boy2 and Boy3 respectively.



- Therefore, though the sun and moon moves from east to west on daily basis in the sky from our point of view, but in relation to the zodiac and other stars, the sun, moon and other planets appears to move from Aries to

Taurus, and to other signs in the opposite direction. Thus, the sun is said to be moving in clockwise direction as well as in counter-clockwise direction.

- Therefore, sun, moon and other luminaries move in clockwise direction on daily basis, but w.r.t. Time-wheel they move in counter-clockwise direction and seem to be entering into different zodiac signs at different intervals.

Actual Movement of Sun and Moon w.r.t the Time wheel

Following the above analogy, consider that in a circle of 360° after every round the sun falls behind by 1° w.r.t Kala-chakra, and similarly the moon falls behind by 13° w.r.t Kala-chakra. The computation of angle lags for both sun and moon w.r.t., the Time-wheel, after different number of rounds, are tabulated below and compared with the above illustrations.

Boy1	**Boy2**	**Boy3**	**Boy2 - lag**	**Boy3 – lag**
Start	Start	Start	0	0
1 round	0.99 round	0.9 rounds	1%	10%
2 rounds	1.98 rounds	1.8 rounds	2%	20%
10 rounds	9.9 rounds	9 rounds	10% (0.1 round)	100% (1 round)
11 rounds	10.89 rounds	9.9 rounds	11% (0.11 round)	110%(1.1round)
100 rounds	99 rounds	90 rounds	1 round	10 rounds
		On a circle of 360°		
Time-wheel	**Sun**	**Moon**	**Sun - lag**	**Moon – lag**
Start	Start	Start	0	0
1 round (360°)	359°	347°	1°	13°
2 rounds (2x360°)	2x359°	2x347°	2°	26°
30 rounds (30x360°)	30x359°	30x347°	30° (1/12th round)	390° (13/12 rounds)
360 rounds (360x360°)	360x359°	360x347°	360° (1 round)	360x13° (13 rounds)

However, considering the actual motion of Time-wheel, sun and moon as explained in Śrīmad-Bhāgavatam, when Time-wheel completes 366 rounds in clockwise direction, the sun completes 365 rounds, indicating the completion

of one year with 365 days as per solar calendar. In addition, moon only completes 352.667 rounds, making 13.333 reverse revolutions, indicating 12.333 months as per lunar calendar. Thus, after the completion of every round the sun falls behind by 0.9836° w.r.t Kala-chakra, and the moon falls behind by 13.11475° respectively.

$$Sun\ daily\ lag = 360° - \frac{365\ rounds\ x\ 360°}{366\ rounds} = \frac{1\ round\ x\ 360°}{366\ rounds} = 0.9836°$$

$$Moon\ daily\ lag = 360° - \frac{352.667\ rounds\ x\ 360°}{366\ rounds}$$
$$= \frac{13.333\ rounds\ x\ 360°}{366\ rounds} = 13.11475°$$

These calculations are based on the consideration of 365 days in a year instead of 365.25 days and appearance of an extra month once in every three years instead of actual appearance of extra month after every 32nd or 33rd month approximately. These approximations considered only to provide the simple explanation of all the calculations. Therefore, in every three years, the sun completes three reverse revolutions and the moon completes 40 reverse revolutions w.r.t the Time-wheel

$$Sun\ daily\ lag = \frac{3\ rounds\ x\ 360°}{3\ years\ x\ 366\ rounds} = 0.9836°$$

$$Moon\ daily\ lag = \frac{40\ rounds\ x\ 360°}{3\ years\ x\ 366\ rounds} = 13.11475°$$

Thus, moving around like this for 366 revolutions of Kala chakra, the chariot of sun meets Kala chakra at the starting point with the completion of 365 revolutions, i.e. 1 reverse revolution covering all the 12 Zodiac signs and 27 Constellations on the Time wheel. The sun's chariot, yoked with seven horses and driven by Aruna, moving slower than the Kala Cakra, touches the constellations and Zodiac Signs gradually, one after the other.

Similarly, the moon, being much slower than that of the sun, remains far behind the sun every day by 12.13115° and the Time wheel by 13.11475°. Thus, by its quick retrogression, the moon covers each zodiac sign in 2.3 days and all the 12 zodiac signs in one month (27 to 28 days). The velocity of the moon's chariot, therefore, exceeds that of the sun as far as retrograde motion is

concerned.

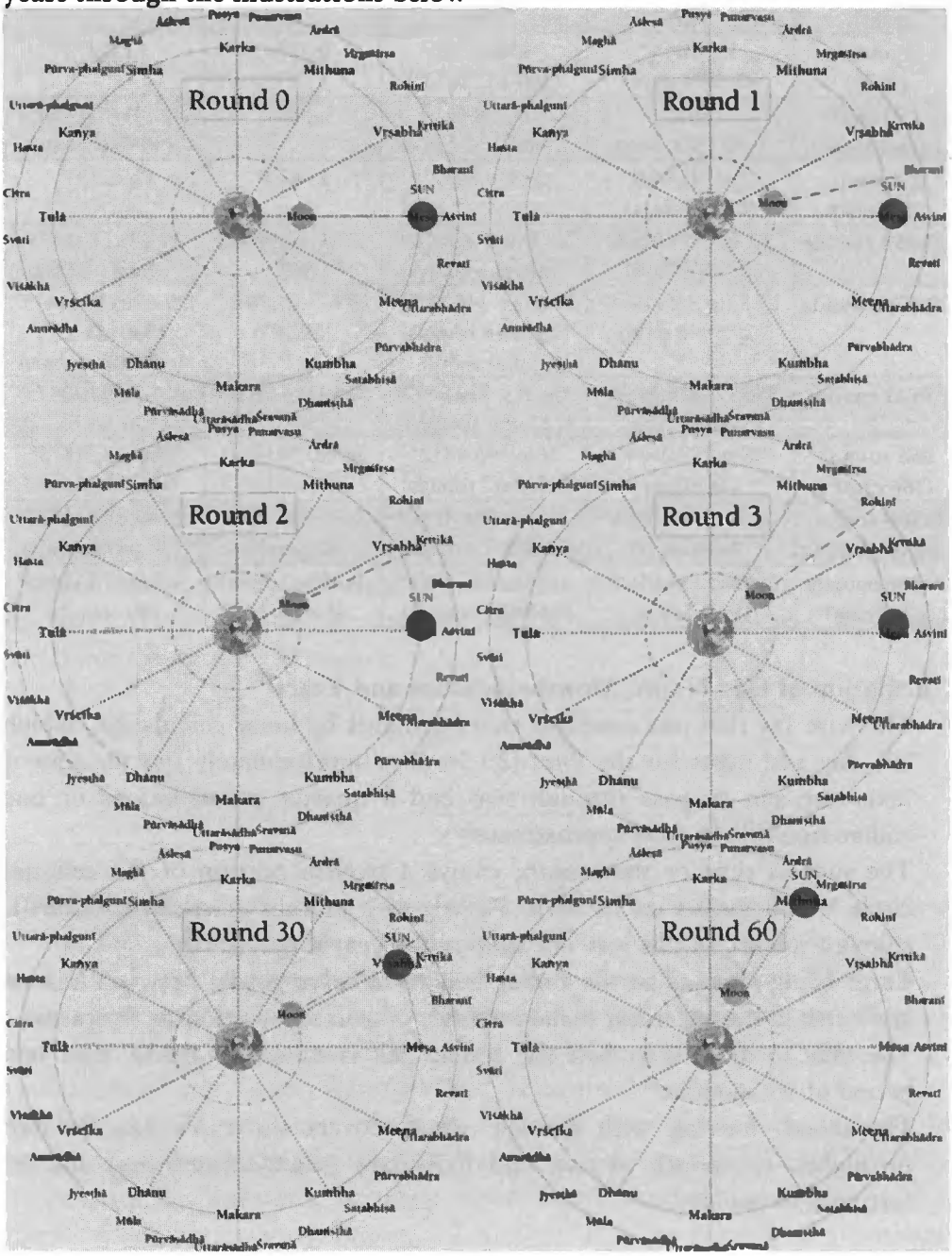
Time-wheel	Sun	Moon	Sun - lag	Moon – lag
Start	Start	Start	0	0
1 round (360°)	359.0164° (0.9973 days)	346.8852° (360-13.11475°)	0.9836°	13.11475°
1.016667 round (366°)	365° (1.013889 days)	352.6667° (366°-13.3333°)	1°	13.333° (1 star) (360°/27 stars)
2 rounds (2x360°)	2x359.0164° (1.9945 days)	2x346.8852°	1.9672°	26.2295°
27.450 rounds	27.45x359.0164° (27.375 days)	27.45x 346.8852° (26.45 rounds)	27.45x0.9836° (27°)	27.45x13.11475° (1 round / 27 stars)
29.676 rounds	29.676x359.0164° (29.5946 days)	29.676x 346.8852° (28.5946 rounds) (1 lunar month)	29.676x0.9836° (29.19°)	29.676x13.11475° (360+29.19°) (~1+1/12 rounds)
356.11 rounds	356.11x359.0164° (355.135 days)	356.11x 346.8852° (12 lunar months)	356.11x0.9836° (350.27°)	356.11x13.11475° (12x360+350.27°)
366 rounds (366x360°)	366x359.0164° (365days)	366x346.8852° (352.667 rounds)	366x0.9836° (1 round)	366x13.11475° (13.333 rounds)
2x366 rounds (2x366x360°)	2x366x359.0164° (2x365days)	2x366x346.8852° (352.667 rounds)	2x366x0.9836° (2 rounds)	2x366x13.11475° (26.667 rounds)
3x366 rounds (3x366x360°)	3x366x359.0164° (3x365days)	3x366x346.8852° (352.667 rounds)	3x366x0.9836° (3 rounds)	3x366x13.11475° (40 rounds)

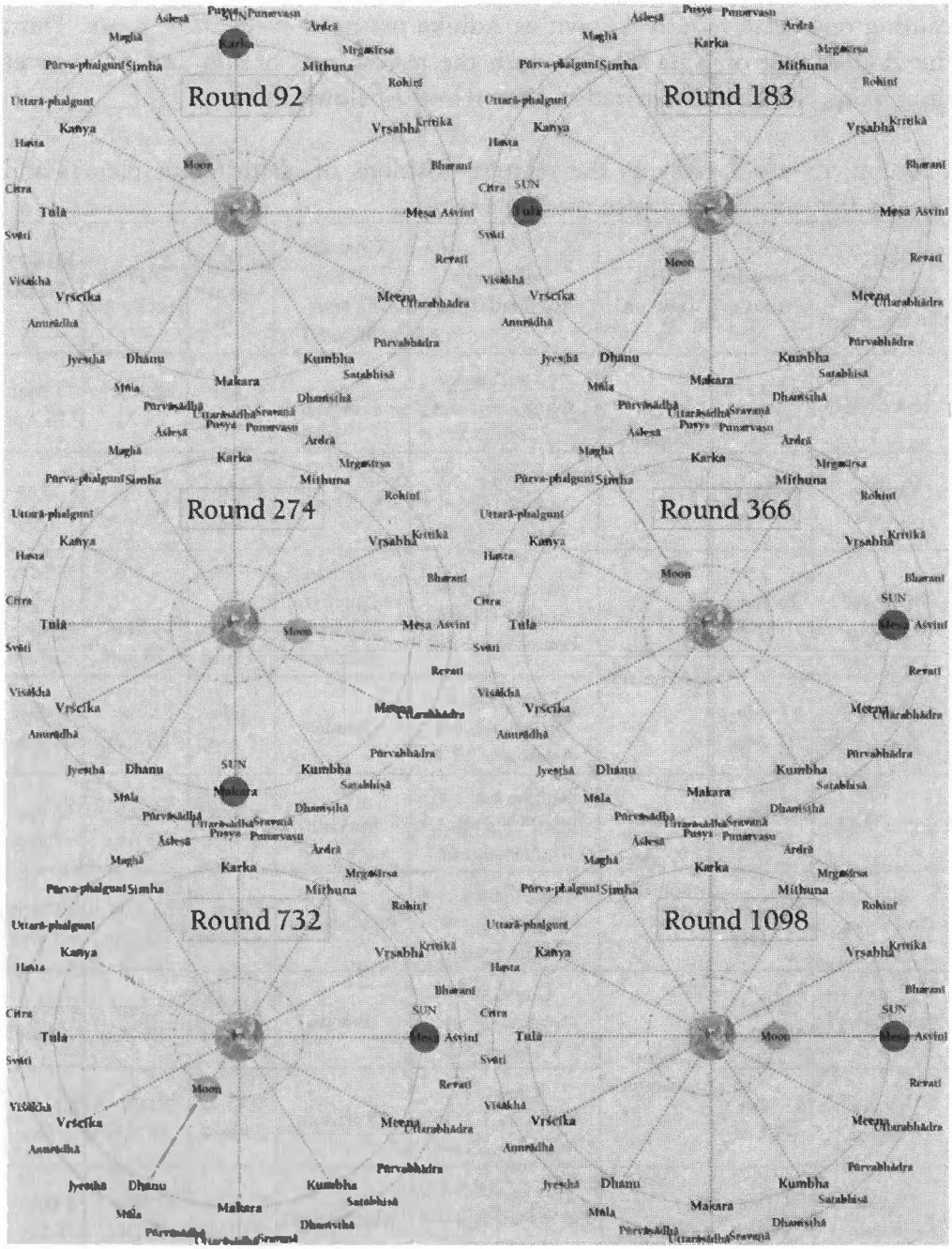
Calculation of Day-Night, Months, Seasons and Years[115]

- The wise say that one month is two fortnights by lunar calculation, which is a day and night for the Pitṛs (29.56 days approximately), or the time it takes the sun to pass through two and a quarter constellations or one zodiac sign (30.46 days approximately).
- The sun, as time or the month, enjoys a twelfth portion of the celestial circle which moves on its own. Passing two months *(ṣāstham āṁśam)* it enjoys a season. In this way one limb of the year is described.
- Thus, being situated in the zodiac belt (*kāla-cakra-gataḥ*) between heaven and earth in the antarikṣa (*nabho-valayasya*) sun enjoys months & seasons.
- The time to move over half the zodiac belt is called an *ayana*. This is a period of six months.
- The moon, moving with quicker speed, covers the sun's year in two fortnights, its month in two and half days (*sapādarkṣābhyām*) and its fortnight in one day.

[115] SB 5.22.5-8: Calculation of Day-Night, Months, Seasons and Years

The movements of sun and moon can be seen on the Kalachakra over 3 years through the illustrations below





From the above illustrations, we can observe that moon completes 37 reverse revolutions while sun covered only 3 reverse revolutions in 36 months, Thus,

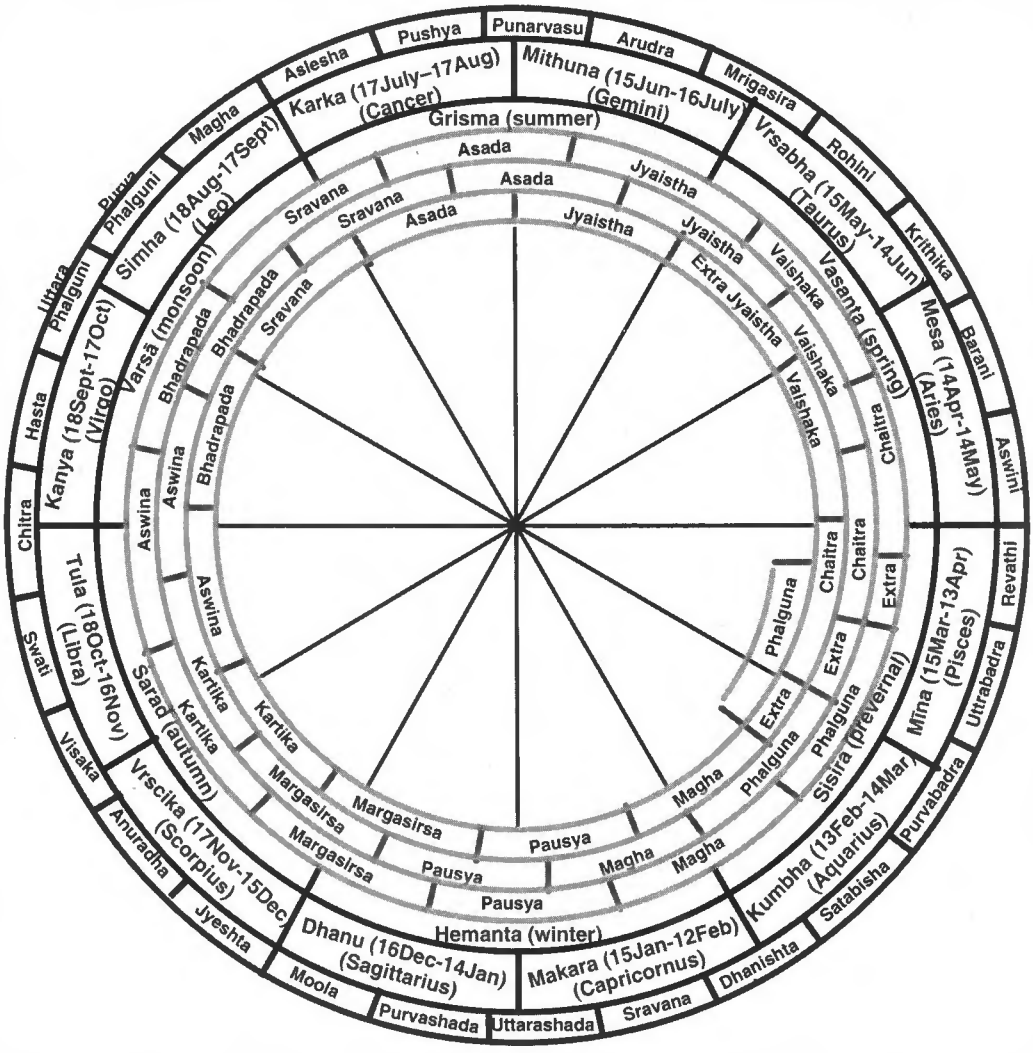
causing one extra month (known as Adhika masa) in every three years. Thus, the Zodiac Belt or Kala Chakra with the movements of sun and moon over three years can be summarized as shown in the below table

The Zodiac-wheel gives us the relative positions of all different planets and stars as they move along with the Time wheel.

Rāśi Saura Māsa (Zodiac Sign)	**Calender Months**	**Rtu (Season)**	**Nakshatra Constellation**	**Chandra Māsa (Lunar Month)**	**2016-17 Durmukhi**	**2017-18 Hevilambi**	**2018-19 Vilambi**
Meṣa (Aries)	14 Apr - 14 May	Vasanta (spring)	Aswini 1-2-3-4 Barani 1-2-3-4 Krithika 1	Chaitra	8 Apr - 6 May	29 Mar - 26 Apr	18 Mar - 16 Apr
Vṛṣabha (Taurus)	15 May - 14 Jun		Krithika 2-3-4 Rohini 1-2-3-4 Mirgasira 1-2	Vaishaka	7 May - 5 June	27 Apr - 25 May	17 Apr - 15 May
Mithuna (Gemini)	15 Jun - 16 July	Grīṣma (summer)	Mirgasira 3-4 Arudra 1-2-3-4 Punarvasu 1-2-3	Extra Jyaistha Normal Jyaistha	6 June - 4 July	26 May - 24 Jun	16 May -13 Jun 14 Jun - 13 July
Karka (Cancer)	17 July - 17 Aug		Punarvasu 4 Pushya 1-2-3-4 Aslesha 1-2-3-4	Asada	5 July - 2 Aug	25 Jun - 23 July	14 July -11 Aug
Simha (Leo)	18 Aug - 17 Sept	Varśā (monsoon)	Magha 1-2-3-4 PurvaPhalg1-2-3-4 UttaraPhalguni 1	Sravana	3 Aug - 1 Sept	24 July - 21 Aug	12 Aug - 9 Sept
Kanya (Virgo)	18 Sept - 17 Oct		UttaraPhalg 2-3-4 Hasta 1-2-3-4 Chitra 1-2	Bhadrapada	2 Sept - 30 Sept	22 Aug - 20 Sept	10 Sept - 9 Oct
Tulā (Libra)	18 Oct - 16 Nov	Śarad (autumn)	Chitra 3-4 Swati 1-2-3-4 Visaka 1-2-3	Aswina	1 Oct - 30 Oct	21 Sept - 19 Oct	10 Oct - 7 Nov
Vṛścika (Scorpius)	17 Nov - 15 Dec		Visaka 4 Anuradha1-2-3-4 Jyeshta 1-2-3-4	Kartika	31 Oct - 29 Nov	20 Oct - 18 Nov	8 Nov - 7 Dec
Dhanu (Sagittarius)	16 Dec - 14 Jan	Hemanta (winter)	Moola 1-2-3-4 Purvashada1-2-3-4 Uttarashada 1	Margasirsa	30 Nov - 29 Dec	19 Nov - 18 Dec	8 Dec - 5 Jan
Makara (Capricornus)	15 Jan - 12 Feb		Uttarashada 2-3-4 Sravana 1-2-3-4	Pausya	30 Dec - 27 Jan	19 Dec - 16 Jan	6 Jan - 4 Feb

			Dhanishta 1-2				
Kumbha (Aquarius)	13 Feb - 14 Mar	Śiśira (prevernal)	Dhanista 3-4 Satabisha 1-2-3-4 Purvabadra 1-2-3	Magha	28 Jan - 26 Feb	17 Jan - 15 Feb	5 Feb - 6 Mar
Mīna (Pisces)	15 Mar - 13 Apr		Purvabadra 4 Uttrabadra 1-2-3-4 Revathi 1-2-3-4	Phalguna	27 Feb - 28 Mar	16 Feb - 17 Mar	7 Mar - 5 Apr

Pictorial representation of the above Table - Vedic Time-wheel (Zodiac)



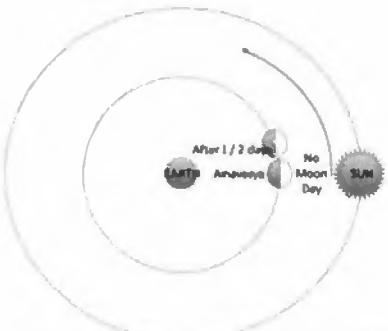
- Outer Most Circle: 27 Constellations with 2.25 of them in each Zodiac sign (Column4)
- First Inner Circle: 12 Zodiac Signs of the Zodiac Belt (Column 1-2)
- Second Inner Circle: Six Seasons of a Year (Column 3)
- Third Inner Circle: Lunar Months for the year 2016-17 (Durmukhi Lunar year- Column6)
- Fourth Inner Circle: Lunar Months for the year 2017-18 (Hevilambi Lunar year - Column7)
- Inner Most Circle: Lunar Months for the year 2018-19 (Vilambi Lunar year - Column 8)
- 3 Inner most Circles: 3 Lunar Years to indicate the formation of Extra Lunar Month (Column 5-8)

Details of different divisions (angles) of the above Time-wheel

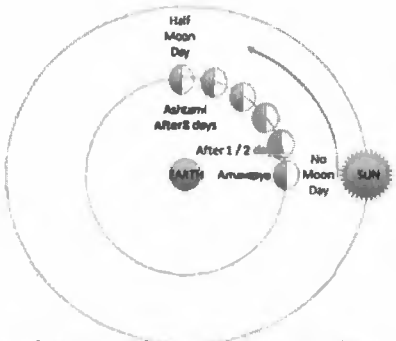
Rāśi Saura Māsa (Zodiac Sign)	**Rtu (Season) Each 60°**	**Nakshatra / Constellations Each 13.333°**	**2016-17 Durmukhi Each 29.189°**	**2017-18 Hevilambi Each 29.189°**	**2018-19 Vilambi Each 29.189°**
0°	0°	0°, 13.333°, 26.667°	354°	344.311°	332.62°
30°		40°, 53.333°	23.109°	12.42°	1.73°
60°	60°	67.667°, 80°	52.218°	41.529°	30.84°/59.949°
90°		93.333°, 106.667°	81.328°	70.638°	89.058°
120°	120°	120°, 133.333°, 146.667°	110.437°	99.748°	118.167°
150°		160°, 173.333°	139.546°	128.857°	147.277°
180°	180°	186.667°, 200°	168.655°	157.966°	176.386°
210°		213.333°, 226.667°	197.765°	187.075°	205.495°
240°	240°	240°, 253.333°, 266.667°	226.874°	216.184°	234.604°
270°		280°, 293.333°	255.983°	245.294°	263.713°
300°	300°	306.667°, 320°	285.092°	274.403°	292.823°
330°		333.333°, 346.667°	314.201°	303.512°	321.932°
360°	360°	360°		END	351.041°

Change in the Phases of the Moon[116]

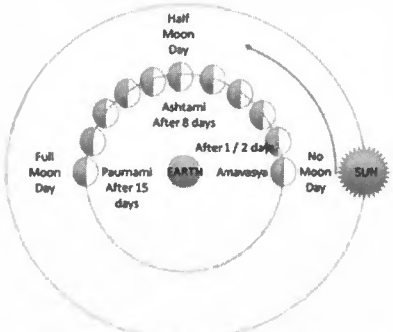
Due to the change in the relative position of the moon and the sun on the Zodiac or Kala-chakra after every day, the portion of the moon seen from the people on the earth varies daily. The portion of the moon seen from the people on the earth gradually increases from new-moon day to full-moon day and then gradually decreases to none from full-moon day to new-moon day. This can be seen from the below illustrations



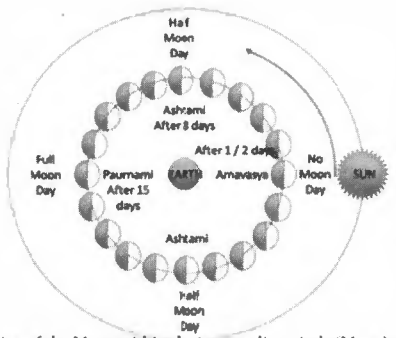
The portion of the Moon within the intermediate circle (Moon's orbit) is seen by the People on the Earth.
Shaded portion: Dark Moon; Non-shaded Portion: Bright Moon.



The portion of the Moon within the intermediate circle (Moon's orbit) is seen by the People on the Earth.
Shaded portion: Dark Moon; Non-shaded Portion: Bright Moon.

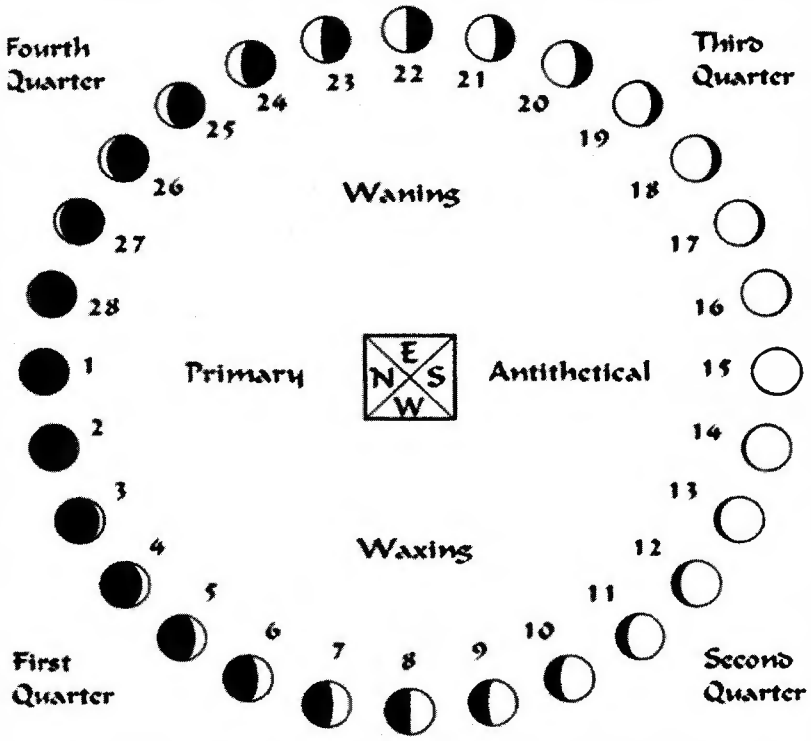


The portion of the Moon within the intermediate circle (Moon's orbit) is seen by the People on the Earth.
Shaded portion: Dark Moon; Non-shaded Portion: Bright Moon.



The portion of the Moon within the intermediate circle (Moon's orbit) is seen by the People on the Earth.
Shaded portion: Dark Moon; Non-shaded Portion: Bright Moon.

[116] SB 5.22.8-10: Understanding the formation of the Phases of The Moon



Movements of the Other Planets[117]

From the residence of Yamarāja the sun travels to Nimlocanī, the residence of Varuṇa, from there to Vibhāvarī, the residence of the moon-god, and from there again to the residence of Indra. In a similar way, the moon, along with the other stars and planets, becomes visible in the zodiac belt and then sets *(evaṁ tato vāruṇīṁ saumyām aindrīṁ ca punas tathānye ca grahāḥ somādayo nakṣatraiḥ saha jyotiś-cakre samabhyudyanti saha vā nimlo-canti, SB 5.21.11)*

The time it takes for the sun to pass completely through the *antarikṣa* along with earth plane and heavenly planes, with slow, quick and medium speeds, is called *saṁvatsara, parivatsara, iḍāvatsa, anuvatsara,* and *vatsara.* It should be understood that the *antariṣka* is equal in size to the earthly and heavenly planes surrounding on either side. The different names of the year according to use

[117] SB 5.22.11-17: The movement of the Stars & and Other Planets above the Moon in the Zodiac Belt

based on different calculations has been described in the Third Canto (SB 3.11.13-14).

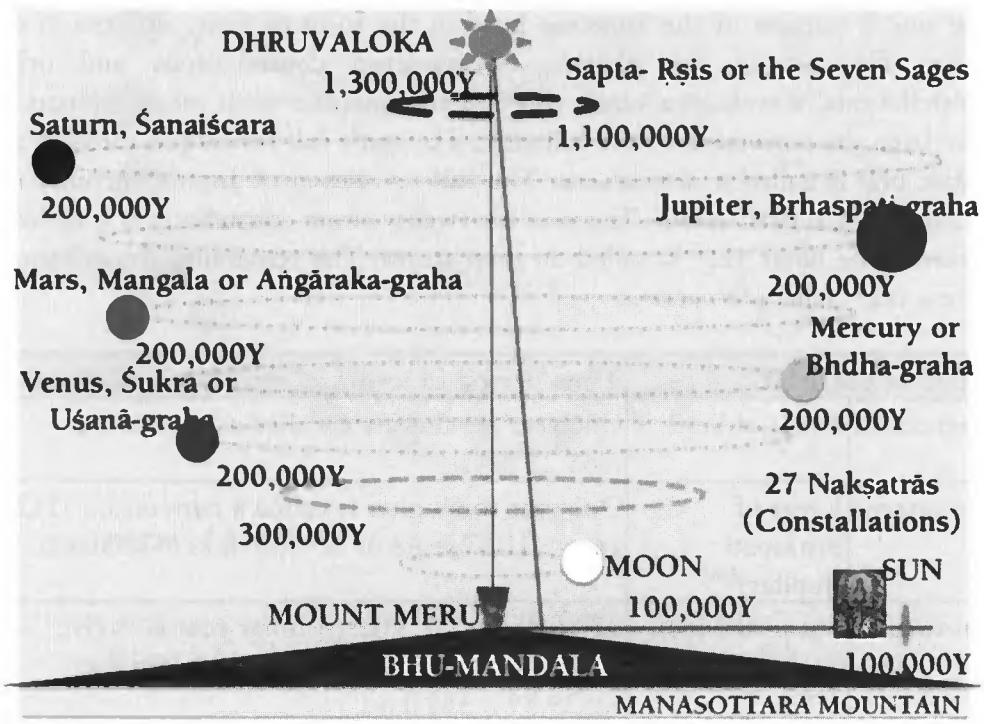
graharkṣa-tārā-cakra-sthaḥ paramāṇv-ādinā jagat
saṁvatsarāvasānena paryety animiṣo vibhuḥ
saṁvatsaraḥ parivatsara iḍā-vatsara eva ca
anuvatsaro vatsaraś ca viduraivaṁ prabhāṣyate

The sun, a portion of the Supreme Lord in the form of time, situated in the zodiac belt among the planets, twenty-seven constellations and other constellations, travels in a circle through the universe with measurements of time from the *paramāṇu* to the full year. The sun's full revolution through the zodiac belt is called a *saṁvatsara.* The full revolution of Jupiter through the zodiac is called *parivatsara.* The year for twenty-seven constellations is called a *vatsara.* The lunar year is called an *anuvatsara.* The remaining constellations have a year called i*dā-vatsara.*

Name of the orbit		Time period in terms of earthly calculations
Samvatsara	1 year of Sun	One year for the sun is called a *saṁvatsara* (365.25 days)
Parivatsara	1 year of Brhaspati (Jupiter)[118]	One year for Jupiter is called a *parivatsara* (11.87 years): 11.87 years or 12*360 days (4320days).
Anuvatsara	1 year of Moon	327.6 days = 27.3*12 (A lunar year is twelve lunar months from full moon to full moon: 12*29.59 = 355 days)
Vatsara	12 months of 27 Constellations	324 days: Since there is no separate orbital motion, their time is measured using the moon's movements (12*27 days)
Idā-vatsara	1 year of Stars or Other Constellations	Because of there is no measuring system in relation to the remaining constellations, they have a year called *idā-vatsara* with solar months of thirty days (total of 12*30 = 360 days)

[118] Sometimes Parivastara is considered as the time taken by Jupitor to cross one Zodiac sign, i.e., 360 days.

This gives the different names of the years for the orbiting sun and other heavenly bodies. All the planets and all the hundreds and thousands of stars are driven around the polestar in clockwise direction by the *parvartaka-pravāha* wind and will continue to be so until the end of creation (*ākalpam*). All these planets and stars float in the air within the vast sky being connected to polestar with the ropes of wind.



The below Table gives the details of the Angular Speeds and the time to complete the Zodiac-belt of all the Nine Planets (*nava-grahas*) on the Zodiac Belt in counter clockwise direction.

Nava-Grahas	Angular Speed (degree/day)	Ratio wrt Sun	Time for one Zodiac-sign	Time to complete 12 Zodiac-signs
Shani - Saturn[119]	1/30°	0.0333°	900 days (2.5 years)	10800 days (30 years)

[119] Slowest moving of the 9 planets

Guru - Jupiter	1/12°	0.0833°	360 days (1 year)	4320 days (12 years)
Mangala - Mars	2/3°	0.667°	45 days (1.5 month)	540 days (1.5 years)
Budha[120] - Mercury	1°	1°	30 days (1 month)	360 days (1 year)
Shukra[121] - Venus	1°	1°	30 days (1 month)	360 days (1 year)
Chandra - Moon[122]	13°	13°	2.25 days	27 days (1/12.3 years)
Surya - Sun	1°	1°	1 month	1 year
Rahu - Dragonhead	-1/18°	-0.0556°	540 days (1.5 years)	6480 days (18 years)
Ketu* - Dragon Tail	-1/18°	-0.0556°	540 days (1.5 years)	6480 days (18 years)

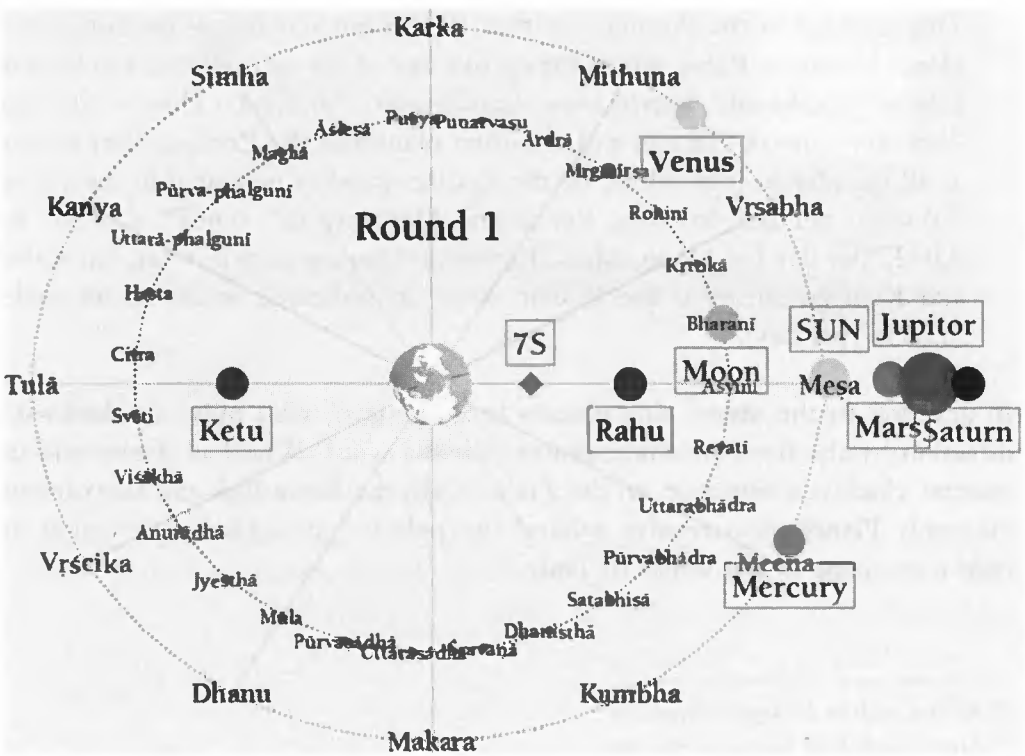
*Ketu (South Node) is always in the opposite side to Rahu (North Node). The speakers of the Purāṇas say that 10,000 *yojanas* below the sun is the planet known as Rāhu, which moves like one of the constellations (*adhastāt savitur yojanāyute svarbhānur nakṣatravac caratīty*). This is in the direction opposite the sun and the other planets in the Zodiac. That means as all the planets lags behind by the Zodiac wheel as indicated in the above Table (1° per day for Sun, Venus and Mercury, 13°, 0.667°, 0.0833° & 0.0333°per day for Moon, Mars, Jupiter and Saturn respectively), but Rahu and Ketu go ahead of the Zodiac wheel as indicated in the above table (0.0556° per day).

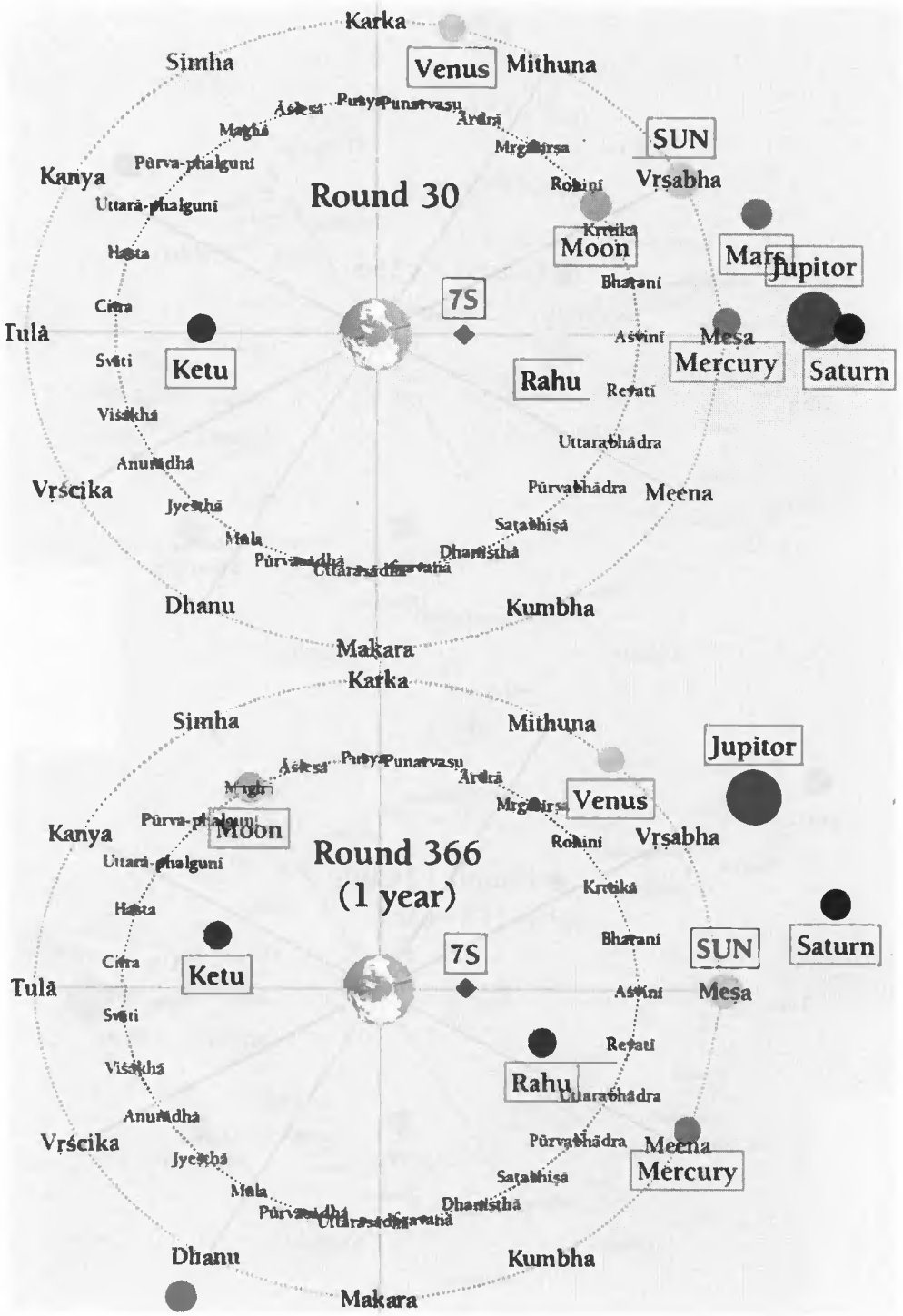
In addition to the above nine planets (nava-grahas), who move in clockwise direction by the force of dakṣiṇāvarta pravāha wind (Wheel of Time) and in counter clockwise direction on the Zodiac-belt, the Sapta-Rsis and the various Heavenly Planets also revolve around the polestar in clockwise direction at their own speed on the Wheel of Time.

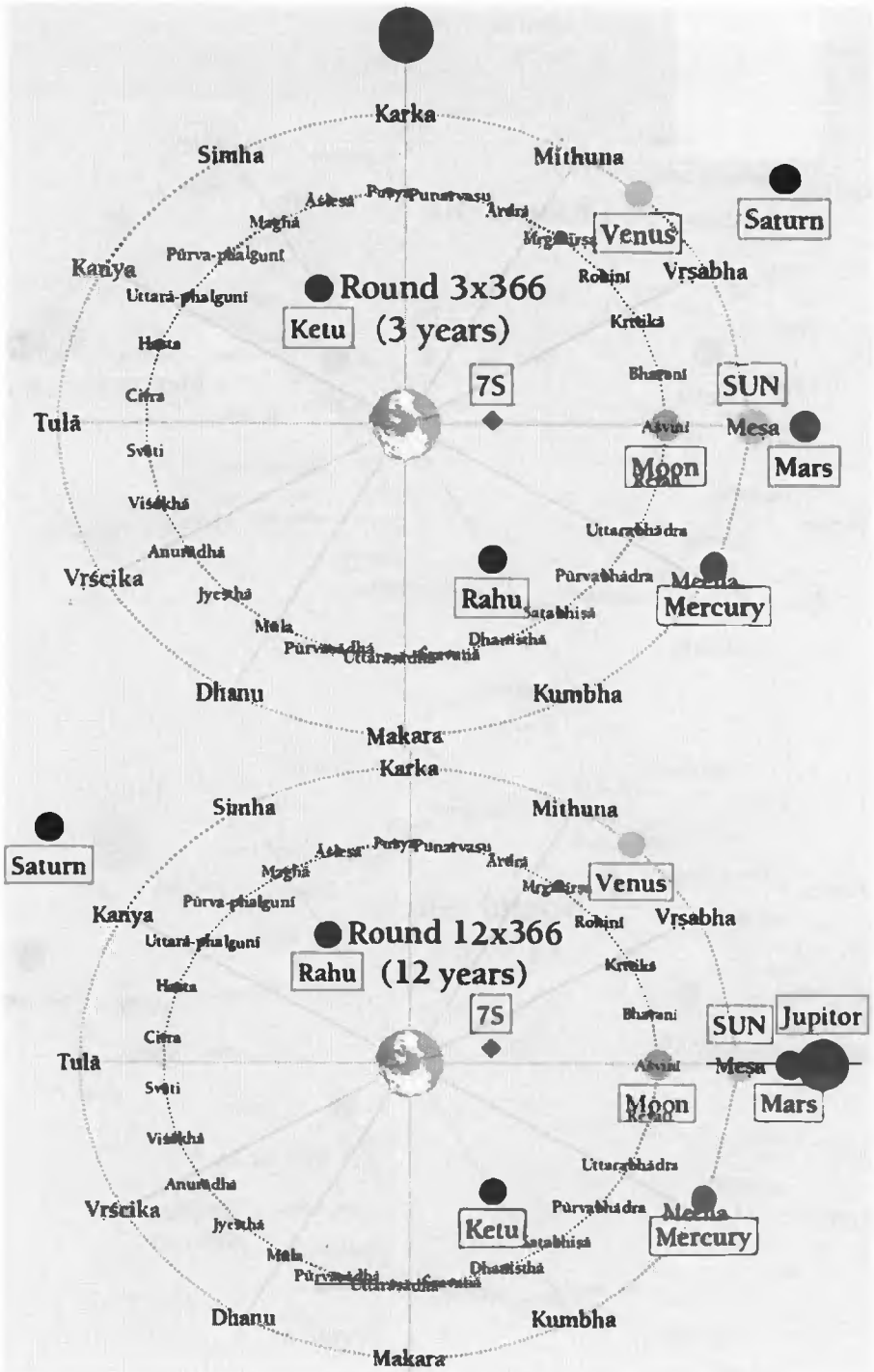
[120] Always with in 27 degrees from sun
[121] Always within 48 degrees of the sun
[122] Fastest moving of the 9 planets

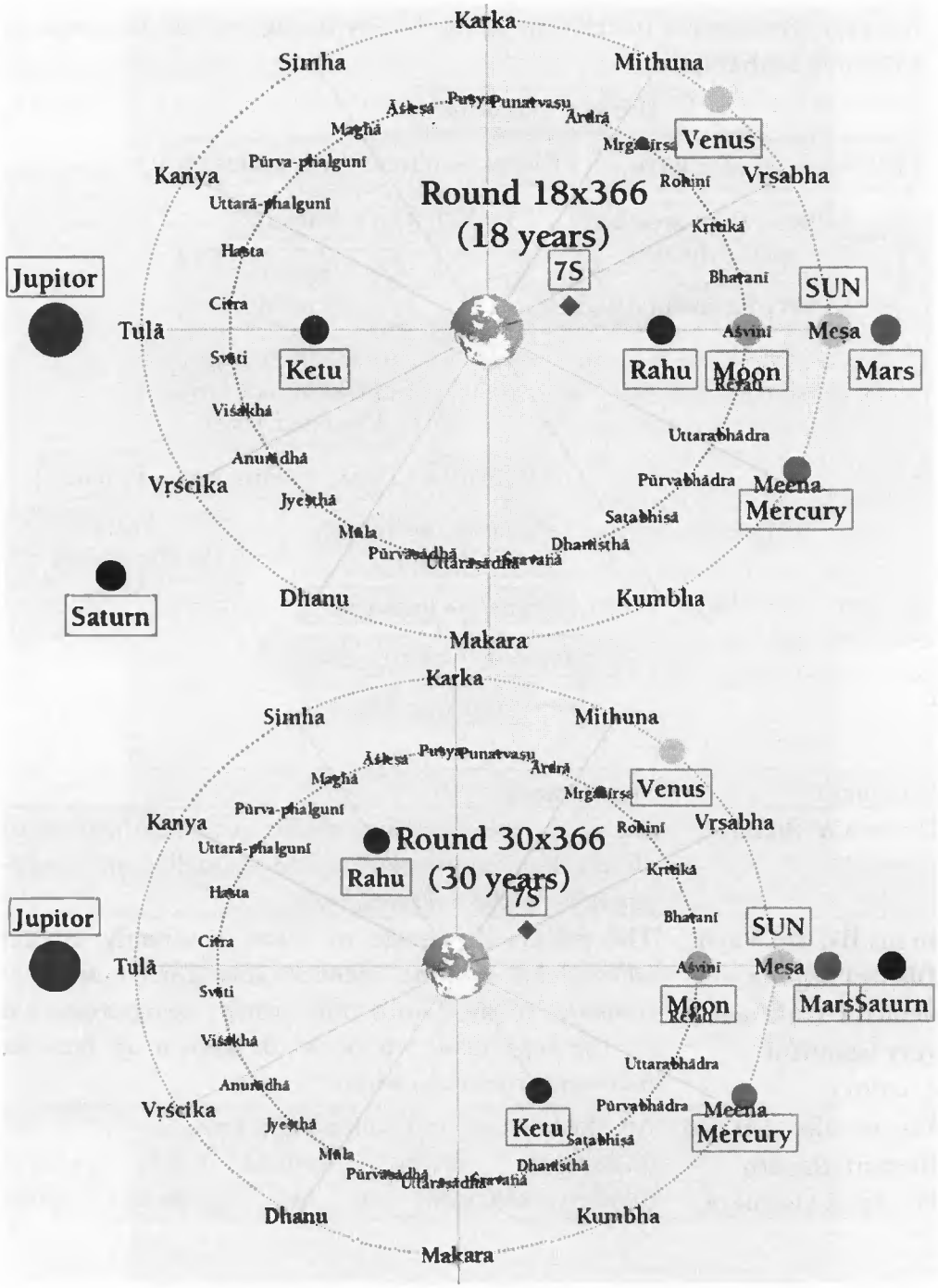
Approximate position of the planets on the Time-wheel

Planet	Years/ zodiac	(°) Initial Position	Speed (°/Day)	1 round	30 rounds	1 year	3 years	12 years	30 years
Time wheel		0	0.000	0.000	0.000	0	0.000	0.000	0
Rahu	-18	0	-0.055	-0.055	-1.639	-20	-60	-240	-600
Ketu	-18	180	-0.055	179.9	178.36	160	120	-60	-420
Sun	1	0	0.984	0.986	29.508	360	1080	4320	10800
Moon	0.081	0	13.115	13.115	393.51	4800	14400	57600	144K
Constellations		0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Venus	1	48	0.984	48.984	77.508	408	1128	4368	10848
Mercury	1	-27	0.984	-26.01	2.508	333	1053	4293	10773
Mars	1.5	0	0.656	0.656	19.672	240	720	2880	7200
Jupiter	12	0	0.082	0.082	2.459	30	90	360	900
Saturn	30	0	0.033	0.033	0.984	12	36	144	360
7sages/heaven	360	0	0.003	0.003	0.082	1	3	12	30
Polestar		0	0.000	0.000	0.000	0.000	0.000	0.000	0.000

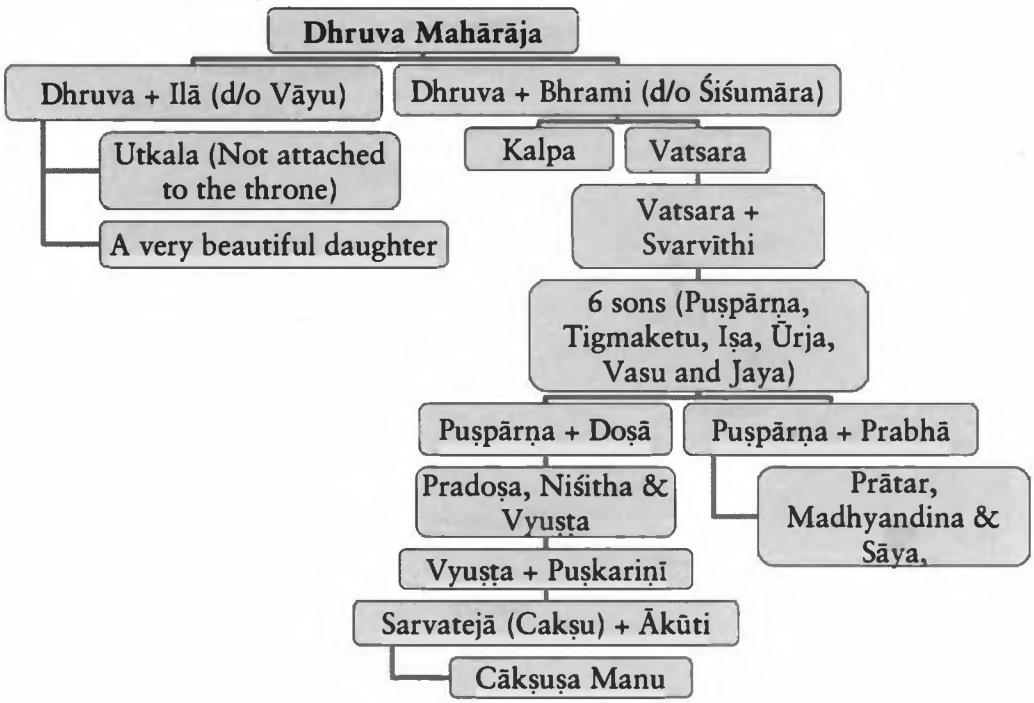



Karka
Simha
Venus
Mithuna
Āśleṣā
Puṣya
Punarvasu
Ārdrā
Maghā
Mṛgśirṣa
SUN
Pūrva-phalguni
Kanya
Round 30
Rohiṇī
Vṛṣabha
Uttarā-phalguni
Kṛttikā
Moon
Mars
Jupitor
Hasta
Bharaṇī
7S
Citra
Tulā
Aśvinī
Meṣa
Svāti
Ketu
Rahu
Mercury
Saturn
Revatī
Viśākhā
Uttarabhādra
Anurādhā
Vṛścika
Pūrvabhādra
Meena
Jyeṣṭhā
Śatabhiṣā
Mūla
Dhaniṣṭhā
Pūrvāṣāḍhā
Uttarāṣāḍhā
Śravaṇā
Dhanu
Kumbha
Makara
Karka
Simha
Mithuna
Jupitor
Āśleṣā
Puṣya
Punarvasu
Ārdrā
Mṛgśirṣa
Venus
Kanya
Moon
Pūrva-phalguni
Round 366
(1 year)
Rohiṇī
Vṛṣabha
Uttarā-phalguni
Kṛttikā
Hasta
Saturn
Bharaṇī
SUN
7S
Citra
Ketu
Tulā
Aśvinī
Meṣa
Svāti
Revatī
Rahu
Viśākhā
Uttarabhādra
Anurādhā
Vṛścika
Pūrvabhādra
Meena
Mercury
Jyeṣṭhā
Śatabhiṣā
Mūla
Dhaniṣṭhā
Pūrvāṣāḍhā
Uttarāṣāḍhā
Śravaṇā
Dhanu
Kumbha
Makara


Karka
Simha
Mithuna
Saturn
Venus
Kanya
Vṛsabha
Round 3x366
Ketu
(3 years)
7S
SUN
Tulā
Meṣa
Mars
Moon
Vṛścika
Rahu
Mercury
Dhanu
Kumbha
Makara
Karka
Simha
Mithuna
Saturn
Venus
Kanya
Vṛsabha
Round 12x366
Rahu
(12 years)
7S
SUN
Jupitor
Tulā
Meṣa
Mars
Moon
Vṛścika
Ketu
Mercury
Dhanu
Kumbha
Makara


Karka
Simha
Mithuna
Āśleṣā
Puṣya
Punarvasu
Ārdrā
Maghā
Mṛgaśirsa
Venus
Pūrva-phalgunī
Kanya
Rohiṇī
Vṛsabha
Uttarā-phalgunī
Round 18x366
(18 years)
Kṛttikā
Hasta
7S
Jupitor
Bharanī
SUN
Citra
Tulā
Aśvinī
Meṣa
Svāti
Ketu
Rahu
Moon
Mars
Revatī
Viśākhā
Uttarabhādra
Anurādhā
Vṛścika
Pūrvabhādra
Meena
Jyeṣṭhā
Mercury
Śatabhiṣā
Mūla
Dhaniṣṭhā
Pūrvāṣāḍhā
Uttarāṣāḍhā
Śravaṇā
Saturn
Dhanu
Kumbha
Makara
Karka
Simha
Mithuna
Āśleṣā
Puṣya
Punarvasu
Ārdrā
Maghā
Venus
Mṛgaśirsa
Pūrva-phalgunī
Kanya
Rohiṇī
Vṛsabha
Uttarā-phalgunī
Round 30x366
Rahu
(30 years)
Kṛttikā
Hasta
7S
Jupitor
Bharanī
SUN
Citra
Tulā
Aśvinī
Meṣa
Svāti
Moon
Mars
Saturn
Revatī
Viśākhā
Uttarabhādra
Anurādhā
Vṛścika
Ketu
Pūrvabhādra
Meena
Jyeṣṭhā
Śatabhiṣā
Mercury
Mūla
Dhaniṣṭhā
Pūrvāṣāḍhā
Uttarāṣāḍhā
Śravaṇā
Dhanu
Kumbha
Makara

Planetary Movements in relation to the family lineage or the Descendents of Dhruva Mahārāja[123]



Personality	Significance
Dhruva Mahārāja	Resides in polestar (Dhruvaloka), which is the pivot of all the luminaries, planets and constellations *(jyotir-ganānāṁ graha-nakṣatrādīnām)*
In his Ilā, d/o Vāyu, Dhruva begot a son named Utkala and a very beautiful daughter	The planets are made to rotate constantly around Dhruvaloka by the inconceivable force (*avyakta-raṁhasā*) of the Lord's time energy, being connected by the rope made up of winds known as *Pravaha (parvartaka-pravaha* wind).
Dhruva also married Bhrami, the d/o Prajāpati Śiśumāra,	All the planets and luminaries, fixed in the zodiac (Śiśumāra), revolve around the polestar (*bhrāmyamāṇānāṁ*) in their respective orbits

[123] SB 4.10.1-2: Descendents of Svayambhuva Manu & SB 4.13.12-20: The descendents of King Vatsara (s/o Dhruva Maharaja)

and begot two sons Kalpa and Vatsara	(*saṁvatsara* – sun's orbit), till the end of the *kalpa (ākalpāntaṁ)*, propelled by wind (*vāyunodīryamāṇā*)
Kalpa	Mahārāja Dhruva remains in Dhruvaloka, the abode of Lord Viṣṇu, until the end of the *kalpa (ākalpāntaṁ)*, around whom the zodiac belt with all the planets and stars rotate with respect.
Vatsara	The wheel of the chariot of the sun god, making up a year (*saṁvatsarātmakaṁ*), going around Meru (*paribhramataḥ*), travels on the top of the mānasottara-mountain. The chariot of the sun-god has only one wheel known as the year (*saṁvatsarātmakaṁ*). The time it takes for the sun to pass through the *antarikṣa* along with earth plane and heavenly planes, with slow-quick-medium speeds, is called *saṁvatsara, parivatsara, iḍāvatsara, anuvatsara,* and *vatsara.*
Svarvīthi, Vatsara's dear wife, gave birth to six sons	The time it takes for the sun to pass completely through the *antarikṣa* (*nabho-maṇḍalaṁ*) along with earth plane and heavenly planes (*saha dyāv-āpṛthivyor maṇḍalābhyāṁ*) is called *saṁvatsara, parivatsara, iḍāvatsa, anuvatsara,* and *vatsara.*
The six sons are Puṣpārṇa, Tigmaketu, Iṣa, Ūrja, Vasu and Jaya	The Supreme Lord, Nārāyaṇa, divided Himself into twelve forms and six seasons for purifying the activities of the people, by arranging the qualities of the six seasons for enjoyment according to their *karma.* This wheel known as year (*saṁvatsarātmakaṁ*) on which the chariot of sun god travels has twelve spokes (*dvādaśāraṁ* - twelve months), six rims (*ṣaṇ-nemi* - six seasons), and three hubs (*tri-ṇābhi* - three sets of four months). The six seasons Vasanta (spring), Grīṣma (summer), Varśā (monsoon), Śarad (autumn), Hemanta (winter) & Śiśira (prevernal) represent the six sons of Vatsara and Svarvīthi.
Puṣpārṇa	Vasanta Ṛtu (Spring season), best of all the seasons, consisting of 60 to 61 days approximately.
Puṣpārṇa's two wives, Prabhā and Dosā.	The two wives were referring to the two parts of a day, i.e., the day time (Prabhā) and the night time (Doṣā).

Prabhā's three sons: Prātar, Madhyandina and Sāyam	The day time (Prabhā) has three divisions known as morning (Prātar), noon (Madhyandina) and evening (Sāyam)
Doṣa's three sons: Pradoṣa, Niśitha, and Vyuṣṭa.	The night time (Doṣā) also has three divisions known as early night (Pradoṣa), mid-night (Niśitha), and last part of the night or dawn (Vyuṣṭa)
Vyuṣṭa begot a son named Sarvatejā in his wife Puṣkariṇī	Early in the morning the bright sun (Sarvatejā), the destroyer of the darkness of all directions, appears from the Eastern Ocean (Puṣkariṇī)
Sarvatejā, also called Cakṣu	By the presence of sun we get external vision (Cakṣu) to see the world, being uncovered by the darkness.[124] Only by the presence of the sun of knowledge (Cakṣu) the darkness within the heart is destroyed and one can distinguish the right from the wrong.
Sarvatejā in his wife Ākūti begot a son named Cākṣuṣa Manu	One should take the advantage of the early morning hours to perform *sādhana* (Cākṣuṣa Manu), under the guidance of a bonafide spiritual master.
Cākṣuṣa Manu in his wife Naḍvalā begot sons without passion	By such spiritual practices, one will be freed from all *anarthās* (headed by passion or lust) and develops all divine qualities.

Note: These are explanations given by Śrīla Vīrarāgav ācārya, a respectable Śrī Vaishnava Ācārya, in his commentary to Śrīmad-Bhāgavatam.

Transitions of the nine planets into the twelve zodiac signs in sequence as per the astrological calculations (Jyotir-veda or Vedic Astrology) are presented below for better understanding of the Planetary Movements

	Sun	Moon	Venus	Mercury	Mars	Jupiter	Rahu	Ketu	Saturn
Capricorn Makara	14/1/15,22:52	20/1/15, 8:57	30-12-14	01-01-15	18-12-12	09-12-2008 30-07-2009	05-12-26	17-08-17	20-03-1990 14-12-1990
Aquarius Kumbha	13/2/15,11:49	22/1/15, 8:49	23/1/15, 2:00	09-03-15	25-01-13	01-05-2009 19-12-2009	18-05-25	29-01-16	05-03-1993 09-11-1993
Pisces Meena	15/3/15,8:33	24/1/15, 9:14	16/2/15, 7:53	28-03-15	04-03-13	01-05-2010 05-12-2010	30-10-23	12-07-14	01-06-1995 16-02-1996
Aries Meṣa	14/4/15,16:41	26/1/15, 11:48	12/3/15, 19:38	12-04-15	12-04-13	08-05-11	12-04-22	23-12-12	17-04-98

[124] The sun is the eye (*dṛg-īśvaraḥ*) and life (ātmā) of all living beings, of *devatās,* animals, humans, reptiles and plants – SB 5.20.46

Taurus Vṛsabha	15/5/15,12:44	28/1/15, 17:12	6/4/15, 19:03	27-04-15	23-05-13	16-05-12	22-09-20	06-06-11	06-06-00
Gemini Mithuna	15/6/15,18:01	31/1/15, 1:17	2/5/15, 17:40	05-07-15	05-07-13	30-05-13	06-03-19	16-11-09	22-07-2002 07-04-2003
Cancer Karka	17/7/15,3:24	2/2/15, 11:28	30/5/15, 16:16 13/8/15, 10:24	20-07-15	19-08-13	18-06-14	17-08-17	29-04-08	05-09-2004 25-05-2005
Leo Simha	17/8/15,10:37	4/2/15, 23:17	5/7/15, 9:46 1/10/15, 13:35	04-08-15	05-10-13	13-07-15	29-01-16	11-10-06	31-10-2006 15-07-2007
Virgo Kanya	17/9/15,9:49	7/2/15, 12:09	3/11/15, 10:48	23-08-15	29-11-13 25-03-14	11-08-16	12-07-14	24-03-05	09-09-09
Libra Tulā	17/10/15,21:26	10/2/15, 00:47	30/11/15, 10:55	29-10-15	04-02-14 14-07-14	11-09-17	23-12-12	05-09-03	14-11-2011 03-08-2012
Scorpio Vṛścika	16/9/15,21:07	12/2/15, 11:08	25/12/15, 16:49	17-11-15	05-09-14	11-10-2018 22-04-2019	06-06-11	16-02-02	02-11-14
Sagittarious Dhanu	16/12/15,11:46	14/2/15, 17:37	19-01-2016, 6:10	6/12/15 14/1/16,15:1	18-10-14	29-03-2019 04-11-2019	16-11-09	30-07-00	26-01-2017 26-10-2017
Capricorn Makara	15/1/2016,1:17	16/2/15, 20:05	12-02-2016, 14:39	26/12/15 9/2/16, 2:59	27-11-14	29-03-2020 19-11-2020	29-04-08	11-01-99	23-01-20
Total Years	**1 year**	**27 days**	**404 days**	**360 days**	**23 months**	**11 years 4 months**	**18 years 7months**	**18 years 7months**	**29 years 10months**
Years/Sign	**1 month**	**2.25 days**	**34 days**	**30 days**	**57 days**	**11.3month**	**1.55 years**	**1.55 years**	**2.5 years**

Another set of details:

	Sun	Moon	Venus	Mercury	Mars	Jupiter	Rahu	Ketu	Saturn
Capricorn Makara	15/1/16,1:17	16/2/15, 20:05	12-02-16, 14:39	26/12/15 9-2-16, 2:59	27-11-14	29-3-20 19-11-20	29-04-08	11-01-99	23-01-20
Aquarius Kumbha	13-02-16, 14:26	18/2/15, 19:57	07-03-16, 20:55	02-03-16,0hr	5/1/15,00:21	20-11-21	11-10-06	24-06-97	28-04-22 17-01-23
Pisces Meena	14-03-16, 11:08	20/2/15, 19:15	01-04-16, 3:10	19-03-2016, 4:27	12/2/15,15:04	13-04-22	24-03-05	05-12-95	29-03-25
Aries Meṣa	13-04-16, 19:37	22/2/15, 20:01	25-04-16, 10:36	03-04-2016, 3:33	23/3/15,22:01	21-04-23	05-09-03	18-05-94	02-06-27 23-02-28
Taurus Vrsabha	14-05-16, 16:29	24/2/15, 22:46	19-05-16, 19:30	08-06-16, 9:40	3/5/15,23:20	01-05-24	16-02-02	29-10-92	08-08-29 16-04-30
Gemini Mithuna	14-06-16, 23:04	27/2/15, 07:01	13-06-16, 5:20	27/6/16,7:54	16/6/15,1:28	14-05-25	30-07-00	12-04-91	30-05-32
Cancer Karka	16-07-16, 9:56	01/03/15, 7:37	07-07-16, 15:17	11/7/16, 9:34	31/7/15,2:30	18-10-25	11-01-99	23-09-89	12-07-34
Leo Simha	16-08-16, 18:20	04/03/15, 5:26	01-08-16, 1:08	27/7/16,6:56 9/9/16, 18:12	15/9/15,19:35	30-10-26 28-2-28	24-06-97	06-03-88	27-08-36
Virgo Kanya	16-09-16, 18:16	6/3/15, 18:20	25-08-16, 11:33	19-8-16, 16:52 3-10-16, 17:46	3/11/15,10:06	26-11-27 29-3-29	05-12-95	18-08-86	22-10-38 12-07-39
Libra Tulā	17-10-16, 6:14	9/3/15, 6:50	18-09-16, 23:49	21-10-16, 14:06	24/12/15,8:34 18/6/16,11:12	26-12-28 01-05-30	18-05-94	29-01-85	27-01-41 25-09-41
Scorpio Vṛścika	16-11-16, 6:03	11/3/15, 17:41	13-10-16, 15:16	09-11-16, 00:28	20/2/16,14:54 12-7-16,4:21	24-01-30 13-06-31	29-10-92	13-07-83	11-12-43 29-08-44
Sagittarious Dhanu	15-12-16, 20:42	14/3/15, 1:31	07-11-16, 11:49	28-11-16, 21:59	18-9-16,6:16	17-02-31 15-10-31	12-04-91	23-12-81	07-12-46
Capricorn Makara	15-01-16, 01:17	16/3/15, 5:47	02-12-16, 18:34		1-11-16,7:44	05-03-32	23-09-89	05-06-80	06-03-49 03-12-49
Total Years	1 year	27.3 days	294 days	360 days	23 months	12 years	18 years 7months	18 years 10months	29 years 10months
Years/Sign	1 month	2.25 days	24.5 days	30 days	57 days	12month	1.55 years	1.55 years	2.5 years

A brief note on Sade Sati of Saturn (Sani graham)

Saturn does his rounds of the twelve Rāśi's of the Zodiac once in about 30 years. In the process, he is supposed to stay for about 2 ½ years, in each sign of the Zodiac, before moving into the next. Saturn is expected to cause troubles

and turmoil when he transits the 12th, 1st and 2nd from the *Janma Rāśi* (1st Rāśi for a person). This period of 7 ½ years, is known as Sade Sati. Everyone will have to experience this period once in 30 years. Of these periods, the middle 2 ½ years when Saturn transits the first or the Janma Rāśi is called Janmasani. The Sade Sati cycle can normally occur thrice, or even four times, though rarely, in one's lifetime. Those born with the Sade Sati at birth will have it recurring for the second time after 30 years; then for the third time after 60 years; and thereafter after 90 years, provided one is destined to live that long. Most of those, for whom the Sade Sati occurs for the first time many years after birth, may not have to undergo the fourth cycle, unless they are destined to live over ninety years. The other two ominous places for Saturn are the 4th called Ardhashtama sani and the 8th called Ashtama Sani.

A brief note on Vimsottari Mahādasha

According to Vedic astrology, planets have a fixed number of years, or Mahadasha (Great period) assigned to them. When one is born, one's birth could be under the Mahadasha of any of the nine planets. Vimshottari dasha system has a fix cyclic order of the Mahadasha of planets. This dasha system works based on Nakshatras (constellation). Vimshottari dasha starts right after the birth of the native and runs until the end of his life. Vimshottari in Sanskrit stands for the number 120. Vimshottari Dasha assumes that the maximum duration of life of an individual human being is 120 years, which is the aggregate duration of all nine planetary periods i.e., Ketu 7, Venus 20, Sun 6, Moon 10, Mars 7, Rahu 18, Jupitor 16, Saturn 19 and Mercury 17, in the order of their operation. Generally, the lunar months are named after corresponding solar months in which the initial new-moon falls. Twelve lunar months make one Samvastsara, each Samvastsara bearing a specific name commencing from Prabhava ending with Akshaya completing a cycle of 60 years. Vimsottari cycle of 120 years corresponds to two of lunar-year cycles.

Description of Indian Traditional Calendar

Indian traditional calendar is a collective name for most of the luni-sidereal (moon + stars) calendars and sidereal (with respect to the distant stars) calendars used in India since ancient times. In the Indian traditional calendar, the day starts with the sunrise. It is allotted five (pancha) "*properties*" or "*limbs*", called *angas*. They are,

1. The Tithi (one of 30 divisions of a synodic month) active at sunrise
2. The Vāsara (ancient nomenclature), vāra (modern nomenclature), or weekday
3. The Nakśatra (one of 27 divisions of the celestial ecliptic) in which the moon resides at sunrise
4. The Yoga (one of 27 divisions based on the ecliptic longitude of the sun and moon) active at sunrise time
5. The Karana (divisions based on tithis) active at sunrise.

Together 5 limbs or properties are labeled under as the *pancāṅga* (Sanskrit: *panca* = five, *anga* = limbs). Thus, *pancāṅga* basically refers to a system of calculating and keeping a date into five parts.

Tithi

A tithi is the time required for the angular distance between the sun and the moon to increase in steps of 12° starting from 0°. The tithi at sunrise of a particular day shall be the prevailing tithi for the whole day. The day change at every sunrise, i.e. from one sunrise to next sunrise is considered as a Vedic day. As per Vaishnava calendar the *tithi* during *brahma-muhurtha,* one and half hour to two hours before sunrise (2 muhurthas: 2x48min = 96min), of a particular day shall be the prevailing *tithi* for the whole day.
The calculation of a *tithi* based on the position of sun and moon in transit.

$$Tithi = \frac{(Moon\ Longitude - Sun\ Longitude)}{12} = Value\ ranges\ from\ 0\ to\ 30$$

If the calculated result is greater than 30, we have to subtract 30 from the result. The net result is Tithi. Names of tithis starting with new-moon day (Amavasya) as per the number obtained above.

N	Value	Phase	Tithi	No	Value	Phase	Tithi
1	1-1.99	Shukla	Pratipada	16	16-16.99	Krishna	Pratipada
2	2-2.99	Shukla	Dvitiya	17	17-17.99	Krishna	Dvitiya
3	3-3.99	Shukla	Tritiya	18	18-18.99	Krishna	Tritiya
4	4-4.99	Shukla	Chaturthi	19	19-19.99	Krishna	Chaturthi
5	5-5.99	Shukla	Panchami	20	20-20.99	Krishna	Panchami

6	6-6.99	Shukla	Shashthi	21	21-21.99	Krishna	Shashthi
7	7-7.99	Shukla	Saptami	22	22-22.99	Krishna	Saptami
8	8-8.99	Shukla	Ashtami	23	23-23.99	Krishna	Ashtami
9	9-9.99	Shukla	Navami	24	24-24.99	Krishna	Navami
10	10-10.99	Shukla	Dasami	25	25-25.99	Krishna	Dasami
11	11-11.99	Shukla	Ekadasi	26	26-26.99	Krishna	Ekadasi
12	12-12.99	Shukla	Dwadasi	27	27-27.99	Krishna	Dwadasi
13	13-13.99	Shukla	Trayodasi	28	28-28.99	Krishna	Trayodasi
14	14-14.99	Shukla	Chaturdashi	29	29-29.99	Krishna	Chaturdashi
15	15-15.99	Shukla	Purnima	30	30-0.99	Krishna	Amavasva

Vāsara

Vāsara refers to the weekdays and the names of the week in many western cultures bear striking similarities with the Vāsara: Sanskrit name and modern names of the day are given below

N	Sanskrit name*	English Name*	Celestial object
1	Ravi vāram / Bhānu-vāsara	Sunday	Ravi, Aditya = Sun
2	Soma vāram / Indu-vāsara	Monday	Soma = Moon
3	Maṅgala vāram / Bhauma-vāsara	Tuesday	Maṅgala = Mars
4	Budha vāram / Saumya-vāsara	Wednesday	Budha = Mercury
5	Guru vāram / Brhaspati vāsara	Thursday	Brhaspati = Jupiter
6	Śukra vāram / Bhrigu-vāsara	Friday	Śukra = Venus
7	Śani vāram / Sthira-vāsara	Saturday	Śani = Saturn

*Sanskrit calendar day begins at sunrise & English calendar day begins at 00:00 Hrs

Nakṣatra

The ecliptic is divided into 27 Nakṣatrās, which are variously called lunar houses or asterisms. These reflect the moon's cycle against the fixed stars, 27 days and 7¾ hours, the fractional part being compensated by an intercalary 28th nakṣatra titled Abhijit.

The ecliptic divided into the nakṣatras eastwards starting from a reference point, which is traditionally a point on the ecliptic directly opposite the

star Spica called *Citra* in Sanskrit. It is generally known as Meṣādi - "start of Aries"; this is when the equinox, where the ecliptic meets the equator, was in Aries (today it is in Pisces, 28 degrees before Aries starts – as per the scientists).

N*akṣatras* are (in this context) not just single stars but are segments on the ecliptic characterized by one or more stars. Hence, there is more than one star for each *nakṣatra*. Nakshatras are divisions of ecliptic, each 13°20', starting from 0° Aries. The *purnima* of each month is synchronized with a *nakshatra*.

Calculation of Nakṣatra

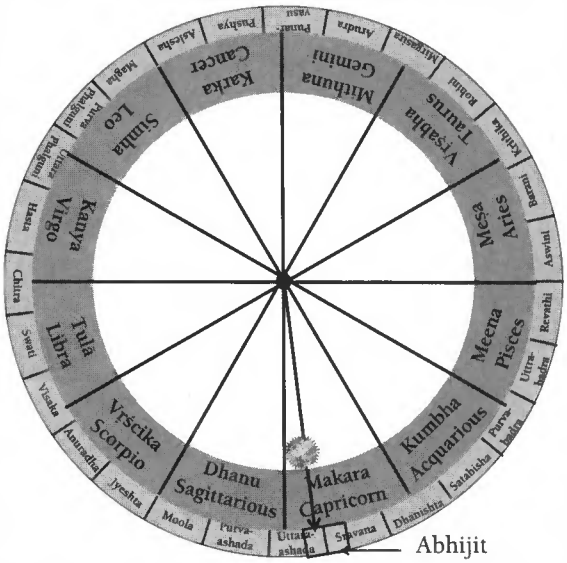
$$Naksatra = \frac{(Moon\ Longitude\ from\ Aries)}{13\ degree\ 20\ minutes} = Value\ ranges\ from\ 0\ to\ 27$$

Division of the path of Moon in the sky

No	Starname	No	Starname	No	Starname
1	Ashvini	10	Magha	19	Mula
2	Bharani	11	Purva Falguni	20	Purva Ashadha
3	Krittika	12	Uttara Falguni	21	Uttara Ashadha
4	Rohini	13	Hasta	22	Shravana
5	Mrugashira	14	Chitra	23	Dhanishta
6	Ardra	15	Swati	24	Shatataraka
7	Punarasu	16	Vishakha	25	Purva Bhadrapada
8	Pushya	17	Anuradha	26	Uttara Bhadrapada
9	Aslesha	18	Jyeshtha	27	Revati

Abhijit Nakṣatra:

Abhijit constellation is composed of the last part of Uttarāṣāḍhā and the first part of Śravaṇā constellations. The astronomical works also confirm this. Thus, Abhijit is at the beginning of auspicious northern course of sun (*Uttarāyana punya kāla*). In addition, as per its position in the ecliptic it indicates the noontime muhurtha of the day, which lasts for about 48 minutes. Thus, the noon muhurtha of a day is considered as the Abhijit muhurtha, or auspicious time, during which Lord Rāma and Lord Vāmana appeared.



Yoga

Yoga means "union", but in astronomical calculations, it is used in the sense of "alignment". The angular distance along the ecliptic of each object, w.r.t. Aries is called the longitude of that object. The sum of the longitude of the sun and the longitude of the moon are normalized to a value ranging between 0° to 360°. When this sum is divided into 27 parts (800' = 13°20'), these parts are called as the *yoga*. The *yoga* that is active during sunrise of a day is the prevailing *yoga* for the day.

Calculation of Yoga

$$Yoga = \frac{(Moon\ Longitude + Sun\ Longitude\ from\ Aries)}{13\ degree\ 20\ minutes}$$
$$= Value\ ranges\ from\ 0\ to\ 27$$

Names of the Yoga starting with Aries as per the number obtained above.

No	Starname	No	Starname	No	Starname
1	Vishkumbh	10	Ghand	19	Parigh
2	Preeti	11	Vridhi	20	Shiv
3	Aayushman	12	Dhruv	21	Sidh
4	Saubhagya	13	Vyaghat	22	Sadhya
5	Shobhan	14	Harshan	23	Shubh
6	Atigand	15	Vajra	24	Shukla
7	Sukarma	16	Sidhi	25	Brahm
8	Dhriti	17	Vyatipat	26	Ainder
9	Shool	18	Variyaan	27	Vaidhriti

Karana

A karana is half of a tithi. To be precise, a karanas is the time required for the angular distance between the sun and the moon to increase in steps of 6° starting from 0°. The karana at sunrise of a particular day shall be the prevailing karana for the whole day. The day changes at every sunrise i.e. from sunrise 1 to sunrise 2 - is one Vedic day. One tithi consists of two karanas. There are total 11 Karanas, among them seven are chara or repeating karans and four are fixed Karans.

No	Fixed Karans	No	Cara or Repeating Karans
1	Śhakuni	1	Vava or Bava
2	Chatushpada	2	Valava or Bālava
3	Nāga	3	Kaulava
4	Kimstughana	4	Taitila or Taitula
		5	Gara or Garaja
		6	Vanija
		7	Vishti or Bhadra

The first half of the 1st *tithi* (of Śukla Paksa) is always Kinstughna karana. Hence this karana is "fixed". Next, the 7-repeating karanas repeat eight times to cover the next 56 half-*tithis*. Thus, these are the "repeating" (Cara) karanas. The 3 remaining half-*tithi*s take the remaining three "fixed" karanas in order. Thus, these are also "fixed" (sthira) karanas. Thus, one gets 60 karanas of a month from those 11 preset karanas.

Calculation of Karana: $Karana = \frac{Tithi}{2} = Value\ ranges\ from\ 0\ to\ 60$

Note: Out of the five limbs of *pancāṅga*, the first three are in use and the last two are not so much in use now. Generally these calculations are made based on 77.11E Longitude and 28.39N Latitude.

Another way to describe the five limbs of the Vedic Calendar

The Hindu calendar is called the *Panchanga (paṣcāṅga or paṣcāṅgam)* which means the five limbs because it deals with five divisions of time (Kāla), which are:

1. Māsam or Lunar Month
2. Pakṣa or Lunar Fortnight
3. Tithi or Lunar day
4. Vāra or Weekday
5. Nakṣatra or Lunar Mansion

The Panchāṅga or "almanac" is of the utmost importance as it tells us all the relevant data that we need for any astrological consideration. A working knowledge of astrology is considered the sine-qua-non of every educated Hindu.

There is a verse which says,

caturaṅga balo rājā jagatim vaśam ānayet |
aham paṣcāṅga balavān ākāśaṃ vaśam ānayet ||

'With the strength of the four-fold army the king conquers the earth, but I, armed with the paṣcāṅgam conquer the regions of space !'

Description of Lunar Calendar

Māsam or Lunar Month

When a new moon occurs before sunrise on a day, that day is said to be the first day of the lunar month. Therefore, it is evident that the end of the lunar month will coincide with a new moon. A lunar month has 29 or 30 days (according to the movement of the moon). The tithi at sunrise of a day is the only label of the day. Sometimes two successive days have the same tithi. In such a case, the latter is called an *adhika* tithi where *adhika* means "extra". Sometimes, one *tithi* may never touch a sunrise, and hence no day will be labeled by that tithi. It is then said to be a *Tithi-Ksaya* where *Ksaya* means "loss".

Month names

There are 12 months in Indian traditional Lunar Calendar. Determining, which name a lunar month takes is somewhat indirect. It is based on the *rāshi* (Zodiac sign) into which the **sun** transits within a lunar month, i.e. before the new moon ending the month. There are 12 *rāśi* names; there are twelve lunar month names. Commonly used names of the 12 Lunar Months are: Chaitra, Vaishak, Jyeshtha, Ashadha, Shravana, Bhadrapada, Ashvina, Kartika, Mragshirsha, Pausha, Māgha and Phalguna. When the sun transits

into the *Meṣa rāśi* in a lunar month, then the name of the lunar month is *Caitra*. When the sun transits into *Vṛṣabha*, then the lunar month is *Vaiśākha*. So on.

No	Sun's Transition into	Month Name
1	Transition into Meṣa (Aries)	Chaitra
2	Transition into Vṛṣabha (Taurus)	Vaishaka
3	Transition into Mithuna (Gemini)	Jyaistha
4	Transition into Karka (Cancer)	Asada
5	Transition into Simha (Leo)	Sravana
6	Transition into Kanya (Virgo)	Bhadrapada
7	Transition into Tulā (Libra)	Aswina
8	Transition into Vṛścika (Scorpius)	Kartika
9	Transition into Dhanu (Sagittarius)	Margasirsa
10	Transition into Makara (Capricornus)	Pausya
11	Transition into Kumbha (Aquarius)	Magha
12	Transition into Mīna (Pisces)	Phalguna

The Sanskrit grammatical derivation of the name of a lunar month is based on its central full moon occurring at or near a particular *nakṣatra* or constellation. For example, when Pūrnimā occurs in or near *Viśākha nakṣatra*, that lunar month is titled as *Vaiśākha Māsa*. The same is for other lunar months. The lunar months are split into two *Pakśās* of 15 days. The waxing phase is called *Śukla Pakśa*, light half, and the waning phase is called *Kṛṣṇa Pakśa*, dark half. There are two different systems for making the lunar calendar,

- *Amāvāsyanta* or *mukhya mana system* – a month begins with a new moon and ends at new moon, mostly followed in the southern states (traditional Indian calendar follows this system)
- *Pūrnimānta* or *gauna mana system* – a month begins with a full moon and ends at full moon, followed more in the North (Vaisnava calendar follows this system)

A month is defined by traditionally in four different ways:—

1. **Nakṣatra māsa - stellar month:** *60 Ghaṭis are said to equal one Stellar day, 30 stellar* days make one Nakṣatra Māsam (month).

2. **Sāvana māsa:** The time between two consecutive sunrises is considered as one sāvana day and 30 such days constitute Sāvana month.
3. **Cāndra māsa:** The time between two consecutive new moon (*amavāsya)* days is called lunar month and it consists of 30 lunar tithis. Some calculate the month from one Full moon (*pūrṇima)* to the next. The names of the lunar months are those of the lunar mansion (*nakṣatra)* of the full moon day of that particular month. The moon is a powerful transmitter of solar energy as it is the closest of the heavenly bodies to the earth, and as such the lunar month is the most precise and is used for the religious calender.
4. **Saura māsa:** The zodiac is treated as a circle having 360 degrees. The time taken by the sun to travel 1 degree of this circular zodiac is called one solar day. 30 such solar days combine to make one solar month. The zodiac is divided into 12 (Rasi) signs, namely Mesha, Vrishaba... and so on. Each sign occupies 30 degrees. The sun's movement through one Rasi (sign) is completed on one solar month which is also referred to as one Sankramana. Thus sun's passage through Meṣa is called Meṣa Sankramana or Sankrānti.

Pakṣa or Lunar Fortnight

The lunar month is made up of two fortnights called Pakṣa. The śukla-pakṣa is the period in which the moon is growing stronger or "waxing" — reaching its peak strength when it is opposite the sun at full-moon. This is considered as a good period to start new undertakings and to do all auspicious work. The kṛṣṇa-pakṣa is the period when the moon is "waning" or growing weaker and reaches its weakest point when in conjunction with the sun at new-moon (*Amavāsya). The kṛṣṇa-pakṣa* is considered an inauspicious time in general although there are a few good days.

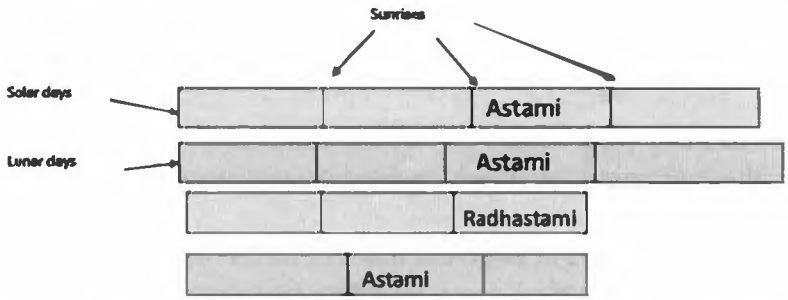
In the spiritual realm of the ancestors, one of our months equals one day and so the Bright fort-night is considered to be a day for the ancestors and the Dark fortnight is the night. Amavāsya is the dawn in the realm of the departed spirits and thus is a good time for worshipping the ancestors.

Tithi or Lunar day

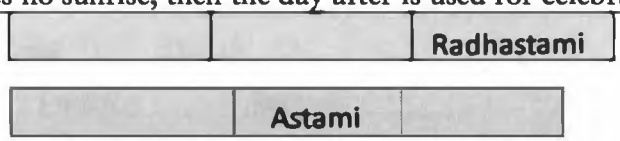
A tithi is the time required for the angular distance between the sun and the moon to increase in steps of 12° starting from 0°. As per traditional Indian Calendars, the tithi at sunrise of a particular day shall be the prevailing tithi for the whole day. The day change at every sunrise, i.e. from one sunrise to

next sunrise is considered as a Vedic day. But, as per Vaishnava calendar the *tithi* during *brahma-muhurtha,* one and half hour to two hours before sunrise, of a particular day shall be the prevailing *tithi* for the whole day.

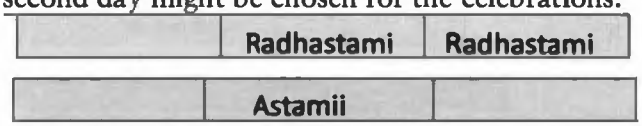
Since the lunar tithi is not equal to 24 hours, the beginning of a tithi may be any time of day or night.



If a tithi touches no sunrise, then the day after is used for celebrations.

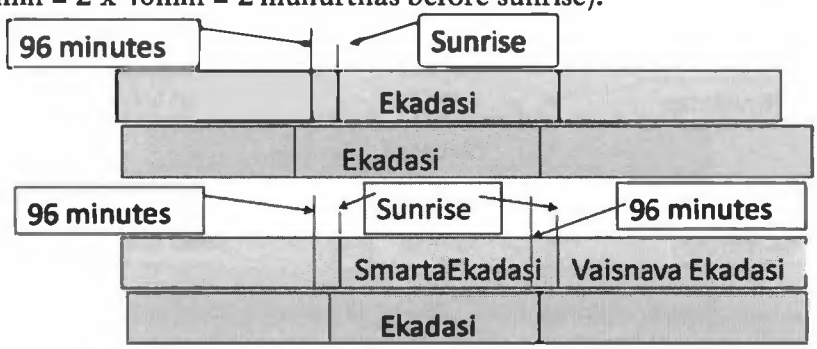


If a tithi touches two sunrises, then two days are labeled with the same tithi name. First or second day might be chosen for the celebrations.

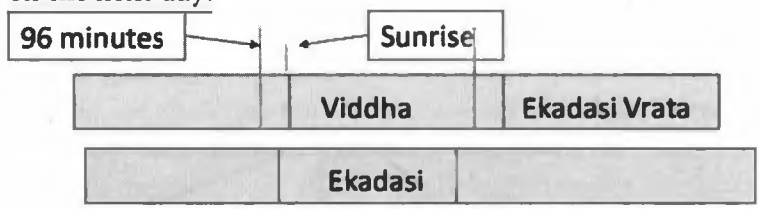


Ekadasi Rule

As per Vaishnava calendar, if a day to be considered as Ekadasi, then the Ekadasi tithi must start 1 hour 36 minutes before the sunrise of that given day (i.e., 96min = 2 x 48min = 2 muhurthas before sunrise).

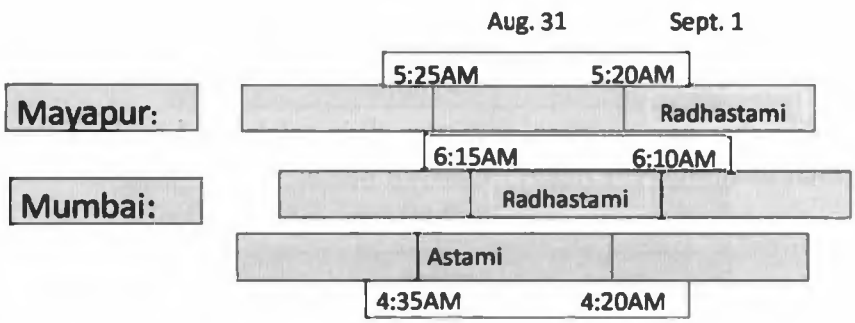


If an Ekadasi tithi does not touch any of the Brahma Muhurtha, then that Ekadasi tithi is called as Viddha ekadasi. And the Ekadasi tithi vow is celebrated on the next day.



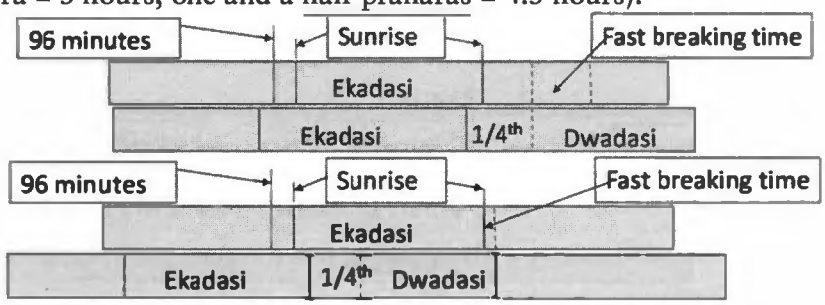
Sometimes Ekadasi vrata occurs on different days at different locations.

- Since the beginning of a tithi occurs simultaneously everywhere in the world,
- but the sunrise occurs at different times at different planes,
- Thus, sometimes Ekadasi vrata or other vratas are observed on different days in different places, even within one country.



Parana Rule - Breaking fast

Ekadasi fast should be broken after sunrise, of the area under consideration, and after one quarter of dvadasi tithi is completed (i.e., approximately after 6 hours), but before the end of dvadasi tithi. Max fast breaking time can be one and a half praharas from sun rise of a given location (i.e., 4 hours 30 minutes; 1 prahara = 3 hours; one and a half praharas = 4.5 hours).



Observing Vratas and Festivals

- Ekadasi: fasting till parana time next day, after sunrise.
- Age: eight to eighty
- Anukalpa or permitted foods may be taken without breaking the fast during the vrata.
- Anukalpa: roots, fruits, milk, water, medicine, order of guru
- Avatara Appearance Days: Gaudiya Panjika recommends fasting on these days, breaking the next day:
- Rama, Narasimha, Balarama, Varaha, Vamana, Kṛṣṇa, Gauranga, Nityananda and Advaita.
- In ISKCON generally appearance of Kṛṣṇa and Gauranga are observed as fasts.
- Anukalpa is taken after celebration at the appearance time (midnight, sunset). Fast is broken the next day.
- The other appearance days are optional as fast days.
- Celebration takes place at the time of appearance if known, with abhisheka and feast.

Vamana Dvadasi and Varaha Dvadasi:

- In the case of Vamana and Varaha dvadasi, since ekadasi fast is followed by dvadasi fast, the two are combined on the ekadasi, and there is no fasting on the dvadasi,
- On the dvadasi there is performance of worship of Vamana and Varaha, followed by a feast.

Appearance times:

- Lord Kṛṣṇa - midnight
- Lord Gauranga - sunset
- Lord Rama - noon
- Lord Narasimha - evening (probably)
- Lord Vamana - noon
- Lord Varaha, Lord Balarama, Lord Nityananda and Advaita-acarya - appearance time is unknown.

Appearance days of Acaryas, gurus

- Worship and feasting is usually done at noon time since the main meal is supposed to be at noon. However, the worship could be done in the morning and the feast observed after that.

- If the appearance day of the acarya is on ekadasi, the worship is done on the ekadasi and the feasting is done the next day.

Muhurta

A muhūrta technically means a period of 48 minutes [2 ghatis]. There are 30 muhūrtas in a day. They are used for deciding auspicious times for performing important activities like marriage, moving into a new home etc.

	Diurnal	Muhurta	Effect		Nocturnal	Muhurta	Effect
1	0600 - 0648	Rudra	–	16	1800 - 1848	Girisa	–
2	0648 - 0736	Ahi	–	17	1848 - 1936	Ajapāda	–
3	0736 - 0824	Mitra	+	18	1936 - 2024	Ahirbudnya	+
4	0824 - 0912	Pitru	–	19	2024 - 2112	Pūṣā	+
5	0912 - 1000	Vasu	+	20	2112 - 2200	Aśvini	+
6	1000 - 1048	Vara	+	21	2200 - 2248	Yama	–
7	1048 - 1136	Viśvedeva	+	22	2248 - 2336	Agni	–
8	1136 - 1224	Vidhi	+	23	2336 - 2424	Vidhātṛ	+
9	1224 - 1312	śatamukhi	+	24	2424 - 0112	Chanda	+
10	1312 - 1400	Puruhuta	–	25	0112 - 0200	Aditi	+
11	1400 - 1448	Vahni	–	26	0200 - 0248	Jīva	+
12	1448 - 1536	Naktancara	–	27	0248 - 0336	Viṣṇu	+
13	1536 - 1624	Varuṇa	+	28	0336 - 0424	Yumigadyuti	+
14	1624 - 1712	Aryama	+	29	0424 - 0512	Brahma	+
15	1712 - 1800	Bhaga	–	30	0512 - 0600	Samudram	+

Extra months (Adhika Māsa)

When the sun does not at all transit into any *rāśi* but simply keeps moving within a *rāśi* in a lunar month (i.e. before a new moon), then that lunar month will be named according to the first upcoming transit. It will also take the epithet of *adhika* or "extra". For example, if a lunar month elapsed without a solar transit and the next transit is into *Meṣa*, then this month without transit is labeled *Adhika Caitra Māsa*. The next month will be labeled according to its transit as usual and will get the epithet *nija* ("original") or *Śuddha* ("unmixed"). Thus, it is called *Śuddha Caitra Māsa*.

Extra Month, or *adhika māsa* (māsa = lunar month in this context) falls every 32.5 months. It is also known as *puruśottama māsa*, so as to give it a devotional name. Thus, 12 lunar months (*māsa*) is equal to approximately 355 to 356 days, while in a solar year there are 365 or 366 days (366 in a leap year) which create difference of 9 to 11 days, which is offset every 3rd year by

adding one extra month of 30 days. No *adhika māsa* falls during Kārtika to Māgha (Kartika, Mragshirsha, Pausha, and Māgha).

List of Extra Months appeared in the last few decades

Month Name	Period of Extra Month	Period of Normal Month
Ashada	17/06/1996 - 15/07/1996	16/07/1996 - 14/08/1996
Jyaistha	15/05/1999 - 13/06/1999	14/06/1999 - 13/07/1999
Ashwina	18/09/2001 - 16/10/2001	17/10/2001 - 15/11/2001
Shravana	18/07/2004 - 16/08/2004	17/08/2004 - 14/09/2004
Jyaistha	17/05/2007 - 15/06/2007	16/06/2007 - 14/07/2007
Vaishaka	15/04/2010 - 14/05/2010	15/05/2010 - 12/06/2010
Bhadrapada	18/08/2012 - 16/09/2012	17/09/2012 - 15/10/1012
Asada	17/06/2015 - 16/07/2015	17/07/2015 - 14/08/2012
Jyaistha	16/05/2018 – 13/06/2018	14/06/2018 – 13/07/2018
Ahwina	18/09/2020 – 16/10/2020	17/10/2020 - 15/11/2020
Shravana	18/07/2023 – 16/08/2023	17/08/2023 – 14/09/2023
Jyaistha	17/05/2026 – 15/06/2026	16/06/2026 – 14/07/2026
Caitra	16/03/2029 – 13/04/2029	14/04/2029 – 13/05/2029
Bhadrapada	19/08/2031 – 16/09/2031	17/09/2031 – 16/10/2031

Lost months (Kṣaya Māsa)

If the sun transits into two *rāshis* within a lunar month, then the month will have to be labeled by both transits and will take the epithet *kṣaya* or "loss". This is considered to be a "loss" because in this case, there is only one month labeled by both transits.

For example, if the sun transits into *Meṣa* and *Vṛṣabha* in a lunar month, then it will be called *Caitra-Vaiśākha kṣaya-māsa*. There will be no separate months labeled *Caitra* and *Vaiśākha*. A Kṣaya-Māsa occurs very rarely. Known gaps between occurrences of Kṣaya- Māsas are 19 and 141 years. The last was in 1983. January 15 through February 12 was *Pauṣa-Māgha kṣaya-māsa*. February 13 onwards was (Adhika) Phālguna.

Religious observances in case of extra and lost months

Among normal months, adhika months, and kshaya months, the earlier are considered "better" for religious purposes. That means, if a festival should

fall on the 10th *tithi* of the *Āshvayuja* month (this is called Vijaya-dasami) and there are two Āśvayuja (Āśvina) months caused by the existence of an adhika Āśvayuja, the first adhika month will not see the festival, and the festival will be observed only in the second *nija* month. However, if the second month is *āshvayuja kshaya* then the festival will be observed in the first adhika month itself.

When two months are rolled into one in the case of a *kshaya māsa*, the festivals of both months will also be rolled into this Kṣaya Māsa. For example, the festival of Mahā-shivarātri which is to be observed on the fourteenth tithi of the Māgha Kṛṣṇa-Pakṣa was, in 1983, observed on the corresponding *tithi* of Pauṣa-Magha Kṣaya Kṛṣṇa-Pakṣa, since in that year, Pauṣa and Māgha were rolled into one, as mentioned above. When two months are rolled into one in the case of a Kṣaya Māsa, the festivals of both months will also be rolled into this Kṣaya Māsa.

Solar Months and Seasons

The transits of the sun through various constellations of the zodiac (*rāśi*) are used, then we get Solar months, which do not shift with reference to the Western or Gregorian calendar. The Solar months along with the corresponding Indian traditional seasons and Western or Gregorian months are

Rāśi - Saura Māsa (solar months)	Western or Tropical months	Rtu (season)	Lunar Month	Western or Tropical months
Meṣa (Aries)	Apr-May	Vasanta (spring)	Chaitra	Mar-Apr
Vṛṣabha (Taurus)	May–June		Vaishaka	Apr-May
Mithuna (Gemini)	June–July	Grīṣma (summer)	Jyaistha	May–June
Karka (Cancer)	July-Aug		Asada	June–July
Simha (Leo)	Aug-Sept	Varśā (monsoon)	Sravana	July-Aug
Kanya (Virgo)	Sept-Oct		Bhadrapada	Aug-Sept
Tulā (Libra)	Oct-Nov	Śarad (autumn)	Aswina	Sept-Oct
Vṛścika (Scorpius)	Nov-Dec		Kartika	Oct-Nov

Dhanu (Sagittarius)	Dec-Jan	Hemanta (winter)	Margasirsa	Nov-Dec
Makara (Capricornus)	Jan-Feb		Pausya	Dec-Jan
Kumbha (Aquarius)	Feb-Mar	Śiśira (prevernal)	Magha	Jan-Feb
Mīna (Pisces)	Mar-Apr		Phalguna	Feb-Mar

Year names

The New Year day is the first day of the *shukla paksha* of *Chaitra-māsa*. The epoch (starting point or first day of the 0th year) of the current era of Indian traditional calendar (both solar and luni-solar) is February 18, 3102 BCE in the proleptic Julian calendar or January 23, 3102 BCE in the proleptic Western or Gregorian calendar. According to the Purānās this was the moment when Lord Śrī Kṛṣṇa returned to His eternal abode. Both the solar and lunisolar calendars started on this date. Since then, each year is labeled by the number of years elapsed since the epoch.

Apart from the numbering system outlined above, there is also a cycle of 60 calendar year names, called Samvatsaras, which started at the first year (at elapsed years zero) and runs continuously

1. Prabhava (1987-88)	21. Sarvajeeth (2007-08)	41. Plavanga (2027-28)
2. Vibhava	22. Sarvadhāri (2008-09)	42. Kīlaka
3. Shukla	23. Virodhi (2009-10)	43. Saumya
4. Pramoda	24. Vikrita (2010-11)	44. Sādhārana
5. Prajāpati	25. Khara (2011-12)	45. Virodhikruthi
6. Āngirasa	26. Nandana (2012-13)	46. Paridhāvi
7. Shrīmukha	27. Vijaya (2013-14)	47. Pramādicha
8. Bhāva	28. Jaya (2014-15)	48. Ānanda
9. Yuva	29. Manmadha (2015-16)	49. Rākshasa
10. Dhātri	30. Durmukhi (2016-17)	50. Anala
11. Īshvara	31. Hevilambi (2017-18)	51. Pingala
12. Bahudhānya	32. Vilambi (2018-19)	52. Kālayukthi
13. Pramādhi	33. Vikāri (2019-20)	53. Siddhārthi
14. Vikrama (2000-2001)	34. Shārvari (2020-21)	54. Raudra

15. Vrisha (2001-02)	35. Plava (2021-22)	55. Durmathi
16. Chitrabhānu (2002-03)	36. Shubhakruti	56. Dundubhi
17. Svabhānu (2003-04)	37. Sobhakruthi	57. Rudhirodgāri
18. Tārana (2004-05)	38. Krodhi (2024-25)	58. Raktākshi
19. Pārthiva (2005-06)	39. Vishvāvasu (2025-26)	59. Krodhana
20. Vyaya (2006-2007)	40. Parābhava (2026-27)	60. Akshaya (2046-47)

This system contains the concept of leap year also. Every 4th year will have 366 days and the others only 365. The starting point is Meshadi or Mesha Sankranti, (1st of the month Meṣa or the Indian traditional lunar New Year). It is also calculated a day by day mode.

Time cycles in Indian Traditional calendar system

Year	365.25 days or 6 seasons
Season	about 60 days or 2 lunar months
Lunar Month	2 pakshas, shukla (waxing) and krishna (waning)
Paksha	15 tithis (1-14, 15th is purnima or amavasya)
Ahoratra or day and night	24-hour period or 60 ghatikas or 30 muhurtas or 8 praharas
Muhurta	48 minutes or 30 kalas (approx)
Kala	1.6 minutes or 96 secons or 30 kasthas
Kastha	3.2 seconds or 15 Nimishas (0.213 seconds)

The lunar years are synchronized with the solar sidereal year by adding a month every three years. The extra month is termed as "Adhik Masa" (extra month). This extra month is also called as Mala Masa (impure month) in Eastern India.

Date conversion

Converting a date from an Indian calendar to the Common Era can require a complex computation. To obtain the approximate year in the Common Era (CE)

- Shaka: add 78-79
- Vira Nirvana Samvat: subtract 527-526
- Yudhishthira Samvat: Subtract 3101 (Ascension of Lord Krishna at age 125) from common era
- Sri Krishna Samvat: Subtract 3226 (Birth of Lord Sri Krishna) from

common era

Correspondence between calendars

The Western or Gregorian year 2000 CE corresponds, respectively with

1. Year 5102 in the Kaliyuga calendar; (3102 BCE)
2. Year 2544 in the Buddha Nirvana calendar; (544 BCE)
3. Year 1922 in the Saka calendar; (78 CE)
4. Year 514 in the Gaurabda Gaudiya calendar (1486 CE)

This is a brief explanation of the calculations of the Indian Traditional calendar system. For detailed explanations on the same, the reader can refer to the relevant books.

Considering all the above details about the various planetary movements and the calculations of various parameters of the calendar based on their movements, we will examine the Sankalpa mantra, which is generally chanted before initiating any samskāra or auspicious ceremony to indicate the time and place at which the samskara is performed

Sankalpa Mantra with its meaning in Connection to the Movements of the Planets

Sankalpa mantra recited for March 29, 2017, Wednesday: Hevilambi samvathsare, Uttaraayane, Vasantha rithou, Mīna mase, Shukla Pakshe, Dwitiyasyam, Budha vāsare)

srī govinda govinda govinda!
asya srī-bhagavata mahā purushasya!
vishnor agjnayā pravartamānasya!

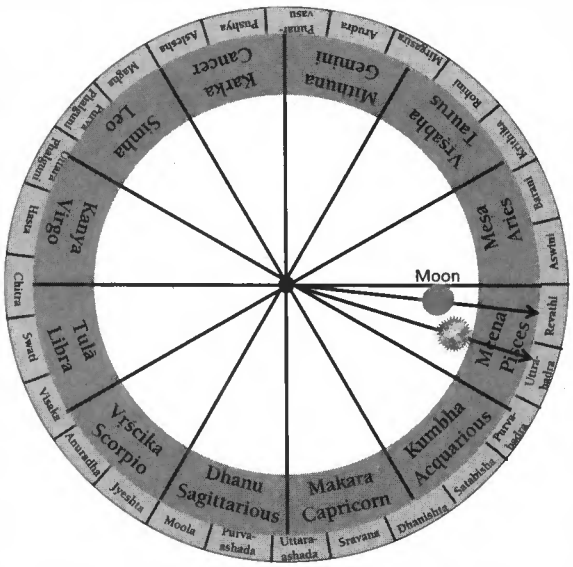
Jambhu dvīpe, Bhārata-varṣe, (in Jambhu dvīpe, the southern tract of land)
Bhārata-khande, (the earth, 9th division of Bhārata varsha)
Meroh dakshine pārsve (bhāge), (to the South of mount Meru)
Godhāvari daksina tīre, (on the southern side of the holy river Godhavari)
Sahyadri ksetra samīpe, (close to the Sahyadri hills)
Mama svagruhe, devatā sannidau (In the presence of the God dwelling in my house)

Adhya brahmana, (from the year of Brahma's birth)
Dvitīya-parārdhe, (in the 2nd half, 51st year)
Srī-Svetha-varāha-kalpe, (in the kalpa of Sveta-Varaha)
Vaivasvatha-manvantare, (in the reining period of the current 7th Manu Vaivaswatha)
Ashta vimsati tame, (in the 28th maha-yuga of the manavantara)
Kaliyuge, prathame pāde, (in the first quarter of this kali-yuga)

Asmin vartamāne vyāvahārike, (in the current period now reigning)
Prabhavādi shashti samvatsarānām madhye, (among 60 years cycle starting from Prabhava)
Hevilambi nāma samvathsare, (Hevialambi – March 28, 2017 to March 17, 2018)
Uttarāyane, (Uttarāyana – January 15, 2017 to July 16, 2018)
Vasanta rithou, (spring season)
    Spring: Meṣa & Vṛṣabha - solar months or Chaitra & Vaishaka - lunar months;
Mīna māse (Chaitra māse),
    Mīna or Pisces - Solar month: March-14, 05:45PM – April-14, 02:17AM;
    Chaitra māsa – Lunar month: March-28, 08:26AM – April-26, 05:45PM;
    Sun sign: Mīna (Mar-14, 05:45PM to April-14, 02:17AM);
    Moon sign: Mīna (Mar-27, 09:41AM to Mar-29, 11:39AM)
Shukla Pakshe, (bright fortnight, Pratipada to Purnima; 28 March – 11 April)
Dwitiyasyam subha thithou, (dwitiya - second tithi)
    Pratipada - first tithi: Mar-28, 08:26AM to Mar-29, 05:44AM
    Dwitiya - second tithi: Mar-29, 05:44AM to Mar-30, 02:47AM
    Sunrise: 6:35AM; Moonrise: 7:30AM
Budha vāsara, (Wednesday weekday; March 29, 2017)
Uttara Bhadrapada (Revati) nakshathra yukthaayaam,
    Uttara Bhadrapada - sun star (Mar-18, 02:06AM to Mar-31, 12:53PM)
    Revati nakshathra - moon star (Mar-28, 01:41PM to Mar-29, 11:39AM)

Srī-vishnu-yoga, Srī-vishnu-karana, Subha-yoga, Subha-karana,
Yevam guna, viseshana viṣishthāyām, asyām dvādasyām Subha-tithou,
Srī-Bhagavad-ājnayā, Sri Bhagavat-kainkarya-rūpam ||

This sankalpa mantra indicates that on this particular day (March 29th, 2017), sun is in Uttara Bhadrapada nakshathra and Mīna rāsi (14 Mar - 14 April), while moon is moving in Revati nakshathra, and Mīna rāsi (27 March - 29 April). The lunar month is Chaitra māsa (28 March – 26 April) and solar months is Mīna or Pisces (14 March – 13 April).



As per the traditional Indian calendar, the tithi at sunrise of a particular day shall be the prevailing tithi for the whole day, because a Vedic day is from one sunrise to next sunrise. But as per Vaishnava calendar the tithi during *brahma-muhurtha* (1.5 to 2 hours before sunrise), of a particular day shall be considered as the prevailing tithi for the whole day. As per traditional Indian calendar, the tithi of March 29th, 2017 is dwitiya, but as per Vaishnava calendar it is Pratipada. This indicates that the tithi transition took place between *brahma-muhurtha* (04:00AM) and sunrise (06:35AM), Thus, there is a difference in both the tithis. Thus, the auspicious beginning of the traditional New Year, Hevilambi (Gudi padwa festival) is celebrated in all over Indian states on 28th March, while it is on 29th March as per Vaisnava calendar. See the picture for the position of the sun and the moon in the Zodiac

Thus, w.r.t the movements of sun and moon, for a given date there will be a nakshathra and rāsi for sun and there will be anther nakshathra and rāsi for

moon respectively. The sun rāsi acts as its month, but moon has its own month with different name as per sun's transition into a rāsi (sun sign). Sun's position is generally indicated by its movement in a rāsi, and it is called as sun-sign. Similarly moon's position is indicated by its movement in a nakshathra, and it is called as moon-star or nakshathra. There are few case studies presented below for clear understanding

1) Appearance of Lord Kṛṣṇa

At the auspicious time for the appearance of the Lord, the entire universe was surcharged with all the qualities of goodness, beauty and peace. The constellation Rohini appeared, as did stars like Aśvini. The sun, the moon and the other stars and planets were very peaceful. Lord Kṛṣṇa appeared on the midnight of Śrāvana-kṛṣṇa-aṣthami, the eighth day of the waning fourtnight in the month of Śrāvana, under Rohini nakshatra. Sun is moving in Simha rāsi while the moon is moving in rohini nakshatra.

SB 10.3.8

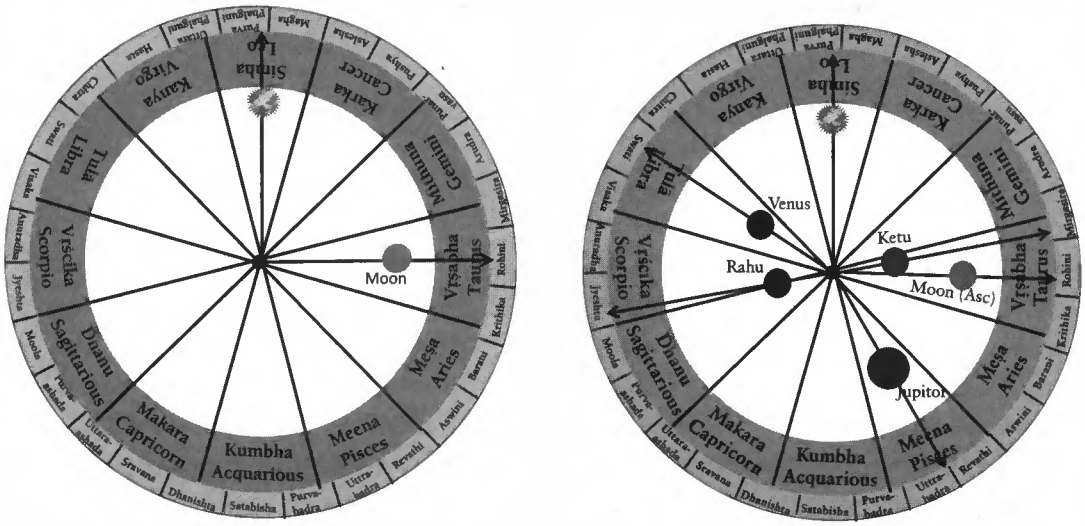
devakyāṁ deva-rūpiṇyāṁ viṣṇuḥ sarva-guhā-śayaḥ
āvirāsīd yathā prācyāṁ diśīndur iva puṣkalaḥ

The Supreme Lord, Viṣṇu, who is situated in the core of everyone's heart, appeared from the heart of Devakī in the dense darkness of night, like the full moon rising on the eastern horizon.

- *induḥ iva:* full moon; Kṛṣṇa appeared like the full moon rising in the eastern sky. Though it was the eighth day of waning at the time, the moon thought in bliss, "The Lord is ornamenting our dynasty with His birth in this world." Then it became round like the full moon.
- According to *Hari-vaṁśa*, the birth took place in the eighth month of pregnancy, before the expected time, in order to bewilder Kaṁsa.
- In the *Kha Manikya,* an astrological work, Lord Kṛṣṇa's birth is described: The Moon, Mars, Mercury and Saturn were exalted and Taurus was ascendant. Jupiter was in Pisces, Sun was in Leo, Venus in Libra and Rāhu in Scorpio. At midnight, on Wednesday with the moon in the Rohiṇī nakṣatra, Lord Śrī Kṛṣṇa appeared.

Midnight of Śrāvana-kṛṣṇa-aṣthami, rohinyām	
Moon, Mars, Mercury and Saturn	In exalted Posion

Ascendant rāsi (Lagna)	Taurus (Vṛṣabha rāsi)
Jupiter	Pisces (Mīna rāsi)
Sun	Leo (Simha rāsi)
Venus	Libra (Tulā rāsi)
Rāhu	Scorpio (Vṛścika rāsi)
Moon	Rohiṇī nakṣatra
Year	Srimukha nama samvatsara
Month	Śrāvana-māsa
Fortnight	kṛṣṇa-paksa
Tithi	Aṣthami
Muhūrta	23rd Muhūrta (Vidhātṛ: 23:36-24:24)
Day and Time	Wednesday midnight



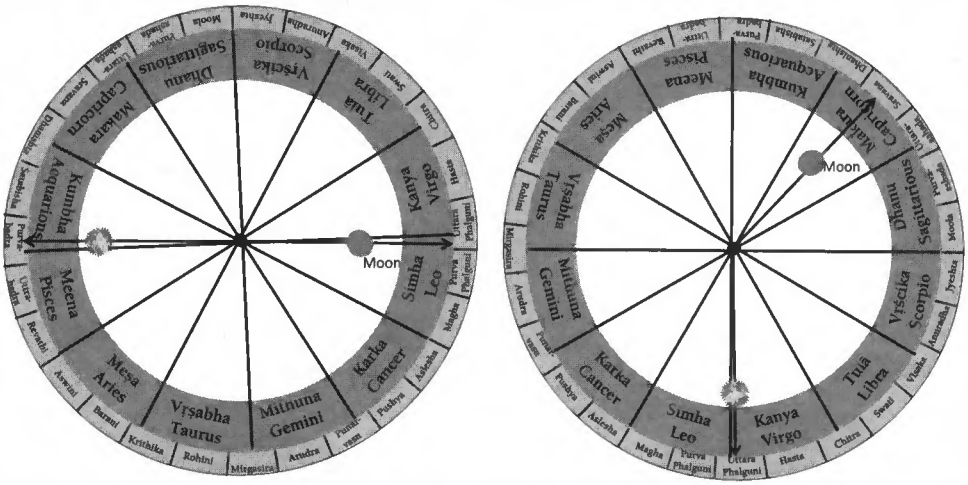
Appearance of Lord Kṛṣṇa in the Mid-night

2) Appearance of Arjuna

Arjuna appeared on Srimukha Phalguna full-moon day during the day in Uttara Star. As per MB Virata Parva, Goharana Parva, and Chapter 44 (Arjuna speaks to price Uttara, just before the Virata battle, describing his 10 names): "I was born on the lower peaks of the Himalayas when the constellation Uttara Phalguna was on the ascendant and hence they call me Phalguni."

- Year: Srimukha nama samvatsara;

- Month: Phalguna & Fortnight: Shukla paksa
- Tithi: Full-moon day (Paurnami) & Nakshatra: Uttara Phalguna



Appearance of Arjuna at Sun-set & Appearance of Lord Vāmana at Noon

3) Appearance of Lord Vāmanadeva

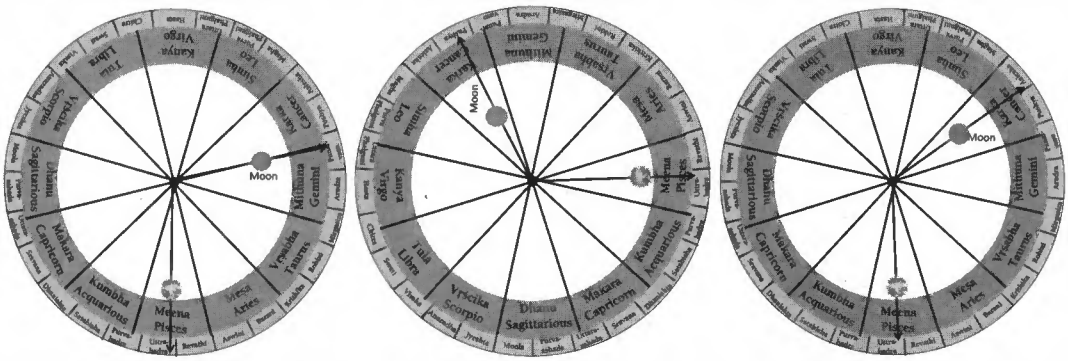
On the day of Dvādaśī of Bhādra month, when the moon was in the constellation Śravaṇa (*śroṇāyāṁ śravaṇa-dvādaśyāṁ*), at the auspicious moment of Abhijit (*muhūrte 'bhijiti*), Lord Vāmanadeva appeared in this universe. All the constellations and planets (*nakṣatra-tārādyāś*) were generous. *Śravaṇa-dvādaśī* means the bright fortnight *dvādaśī* of Bhādra month. The moon was in Śravaṇa constellation. The moon was also in the first portion of Śravaṇa constellation called Abhijit. The constellations such as Aśvinī and the planets (*tāra)* such as Jupiter and Venus, as well as the luminaries consisting of the sun and moon (*ādyāḥ*) were very generous *(dakṣiṇam*) at His birth. (SB 8.18.5)

4) Appearance of Lord Rāma, Lakṣmaṇa, Bharata and Śatrughna

Lord Rāma appeared on caitra-śukla-navami noon time, during Abhijit muhurtha, under Punarvasu nakshatra. Bharata appeared early next morning, on Caitra-śukla-dasami under Pusyami nakshatra. Lakṣmaṇa and Śatrughna appeared on the same day noon, Caitra-śukla-dasami under Aslesa nakshatra.

	Tithi	Nakshathra
Lord Rāma	Caitra-śukla-navami	Punarvasu
Bharata	Caitra-śukla-dasami	Pusyami
Lakṣmaṇa and Śatrughna	Caitra-śukla-dasami	Aslesa

Moon takes to transit one nakshatra approximately 24 hours and 20 minutes. Thus, moon can transit from Punarvasu to Pusyami, and then to Aslesa in little over a day, i.e. from previous day's noon to next day's noon.



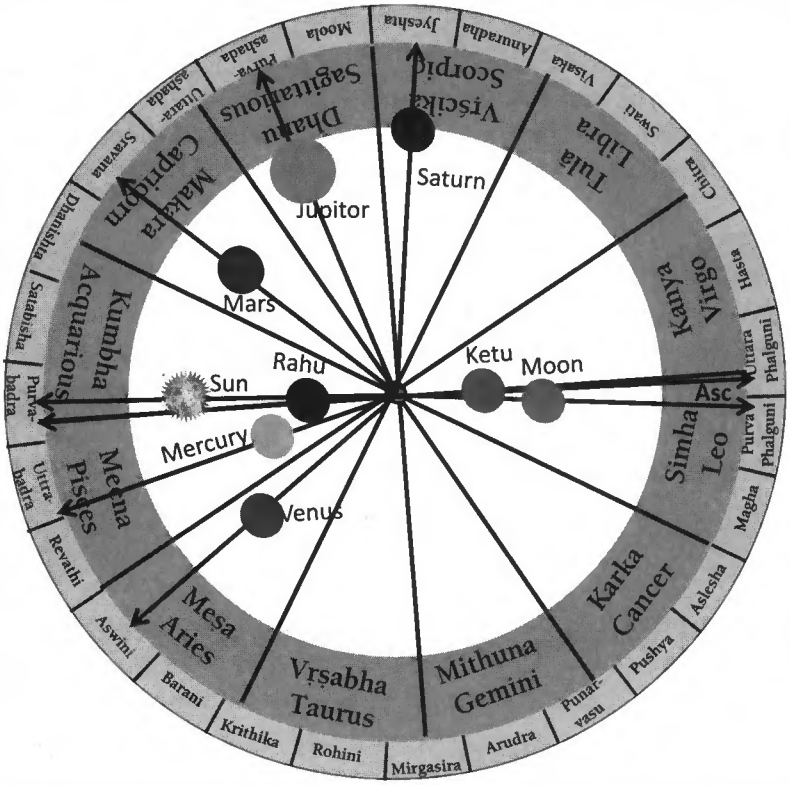
Appearance of Lord Rāma: at Noon; Bharata: at Sun-rise, & Lakṣmaṇa-Śatrughna: Noon

5) Appearance of Lord Caitanya

On the full-moon evening of the month of Phālguna when coincidentally there was a lunar eclipse, Lord Caitanya took birth (*phālguna-pūrṇimā-sandhyāya prabhura janmodaya*). According to the Jyotir-veda, or Vedic astrology, the auspicious birth moment is described as follows: The moon was in Leo [the figure of the lion in the zodiac], Leo was the ascendant (it means it that the sign Leo is ascending in the eastern direction), several planets were strongly positioned, and the ṣaḍ-varga and aṣṭa-varga showed all-auspicious influences. (*siṁha-rāśi, siṁha-lagna, ucca graha-gaṇa ṣaḍ-varga, aṣṭa-varga, sarva sulakṣaṇa*). Thus, in the year 1407 of the Śaka Era [A.D. 1486], in the month of Phālguna [February-March], in the evening of the full-moon day, the desired auspicious moment arrived. The explanation of the horoscope given by Bhaktivinoda Ṭhākura is that at the time of the birth of Lord Caitanya

Mahāprabhu the planets were situated as follows:

Appearance of Lord Caitanya at moon-rise or at sun-set on full-moon day



Planet	Rāśi or Sun-sign	Nakṣatra or Lunar Mansion
Lagna (Ascendent)	Siṁha-rāśi (Leo)	
Śukra (Venus)	Meṣa-rāśi (Aries)	Aśvinī
Ketu (the 9th planet)	Siṁha-rāśi (Leo)	Uttaraphalgunī
Candra (the moon)	Siṁha-rāśi (Leo)	Pūrvaphalgunī (11th mansion)
Śani (Saturn)	Vṛścika-rāśi (Scorpio)	Jyeṣṭhā
Bṛhaspati (Jupiter)	Dhanu-rāśi (Sagittarius)	Pūrvāṣāḍhā
Maṅgala (Mars)	Makara-rāśi (Capricorn)	Śravaṇā
Ravi (the sun)	Kumbha-rāśi (Aquarius)	Pūrvabhādrapāda
Rāhu	Kumbha-rāśi (Aquarius)	Pūrvabhādrapāda
Budha (Mercury)	Mīna-rāśi (Pisces)	Uttarabhādrapāda

6) Case study from Mahābhārata - Conflict between the Pāndavas and Kauravas

Srila Madhvācārya analyses a conflict that arises between Pāndavas and Kauravas, due the variations in the calculation of a year based on the movement of the sun and the moon in the Zodiac, in his book known as Mahābhārata Tātparya Nirnaya. Śrīla Vādirāja Tīrtha gave elaborate explanations analyzing these intricate calculations.

Having completed the twelve-year exile (Vanavās) and one year incognito period (Agnātavāsa), the Pāndavās were engaged in great festivities along with the Viratās, Pānchalas and the Yādhavas. Abhimanyu's wedding and most of all reunion with Sri Krishna – all these gave immense joy to the Pāndavas, whose limits knew no bounds. On other side, after having been terrorized in the war immensely by Arjuna, Duryodhana and others along with Karna and Shakuni, and Bhishma went back to Hastināpura and organized a meeting. They started devising yet another conspiracy to see how they can send the Pāndavas back to the forest again. Finally the Kauravas said that 'we have spotted Arjuna who is still in Agnatavasa today. Therefore, Pāndavas should return to Vanavasa.'

Bhishma who did not agree to the evil thought of Kauravas said that since Agnatavasa was over, Arjuna had not violated the rules and it is not right for them to return to Vanavasa. Even Drona supported him. Ignoring both their words, evil Kauravas sent messengers to Pandavas telling "You should return to forest again". Dharmarāja rejected Duryodhana's argument telling that, Drona and other knowledgeable Brāhmanas know that they have completed Agnatavāsa completely. Therefore Kauravas argument cannot be accepted.

Importance of Chandramana: The Kauravas told that Pāndavas had not completed Agnatavāsa based on Sauramāna calculation. But Chandramāna was the method of calculation used during that time. Because, sun is the lord of days, since different divisions of a day like morning, noon, night, etc., are distinguished based on sun's movements. Chandra is the lord of fortnight, month and year, since they are calculated based on moon's movements. Therefore, knowledgeable people consider saumya year as the prime in calculating the months.

There are five famous ways of calculating time, namely – saura, chāndra, savana, nakshatra and brhaspatya. Calculation based on chaitra etc (moon months) is called chandramāna. Calculation based on mesha etc (sun months) is called sauramāna. A calculation based on stars where 27 days equals one month is nakshatramāna. Calculation based on when planet Guru moving from one zodiac to other is called brhaspatya. According to savana calculation there is exactly 360 days in a year.
Since sun is the Lord of the days, counting of days should be done using the method of sauramāna. But since moon is the Lord of fortnights, month and year, chandramāna method should be employed to calculate them. Therefore it can be summarized that, since Pāndavas Vanavāsa and Agnatavāsa went by years, Thus, they should be decided based on chandramāna calculation.

Therefore even in the performance of different Yagnās (Sacrificial rites) or different samskārās candramāna is considered and not sauramāna. Without considering all this, Kauravas out of greed, did not give the kingdom back to Pāndavas. Pāndavas got Brāhmanas to announce that they had completed Agnatavasa correctly and left to Upaplavya city.

Therefore, since chandramāna is the correct method for calculating the years, and since Pāndavas had correctly completed Agnatavasa according to it, there was no need for them to return back to Vanavasa. If Kauravas being ignorant of such methods of calculations was one reason, the other reason was their greed to retain the kingdom without giving it to Pandavas was the other reason. It can also be understood that during the time of deciding the time of Vanavasa they never gave clear rules about which calculations should be used so that in the future they could come up with such contradictory opinions. Since Pāndavas opined that chandramāna was universally accepted calculation by knowledgeable people, they did not clearly point it out and also their opinion was devoid of any mischief. Only Kauravas did so out of evil intention. This makes it clear that Kauravas, only out of evil intent made false allegations on Pandavas.

Below are the recorded dates of Vanavāsa and Agnatavāsa that Pāndavas went through

- Forest Life (Vanavāsa) started: Sarvadhari Shravana dark 8th day
- 12 years forest life (Vanavāsa) ended: Sarvari Shravana dark 7th day
- 13th year of Ajnatavās started: next day (Sarvari Shravana dark 8th day)
- Fight with Susharma: Plava Shravana dark 7th day
- 13th year of Ajnatavāsa ended: on the same day (Plava Shravana dark 7th day)
- Virata Yudh: Plava Shravana dark 8th day (next day)
- Pandavas declared their identity: Plava Shravana dark 9th day

Since these are lunar years, there would be one Adhik Māsa (extra month) every 3 years, and in 12 years there were 4 Adhik Māsas making solar and lunar years almost equal. But in the 13th year there would be an extra 10 to 12 days in the solar year when compared with the lunar year. But these extra days were merged in the coming lunar years as Adhik Māsas, just the way it had happened in the previous 12 years. Tithiway as and likewise, Bhishma and Yudhishthira followed this calculation as correct, but Duryodhana insisted on anonymity solar years, which was unacceptable in those days. Since the Ajnatavāsa period was over the previous day, Arjuna declared his identity during the fight with the Kauravas. Arjuna was carrying his bow Gandiva for 30 years from Pramodoota to Sarvari, and he told Uttara that he will carry it for another 35 years (from Plava to Pramadi).

7) The recitation of the Śrīmad-Bhāgavatam

prauṣṭhapadyāṁ paurṇamāsyāṁ hema-siṁha-samanvitam |
dadāti yo bhāgavataṁ sa yāti paramāṁ gatim ||

If on the full moon day of the month of Bhādra one places *Śrīmad-Bhāgavatam* on a golden throne and gives it as a gift, he will attain the supreme destination. (SB 12.13.13)

Śrīla Visvanātha Cakravarti Thākura *says* "*Prauṣṭhapadyām* means related to Bhādra month. In that month, Śukadeva gosvāmi completed the recitation of the *Śrīmad Bhāgavatam.* One should see the *Uttara-khaṇḍa* of *Padma Purāṇa* for details. The king of all scriptures, the sun among the Purāṇas, the *Śrīmad-Bhāgavatam,* should be seated on a throne and treated

royally. The king of all planets, the sun, enters into Siṁha or Leo zodiac sign during Bhādra month. Thus, it is like a king seated on his throne (*simhāsana*). This is the metaphor."

- *Śrāvana-kṛṣṇa-aṣthami* is the appearance tithi of Lord Śrī Kṛṣṇa.
- *Bhādrapada-śukla-aṣthami* is the appearence tithi of Śrīmati Rādhārāni.
- *Bhādrapada-śukla-dvādaśī* is the appearence tithi of Lord Vāmanadeva.
- *Bhādrapada-śukla-paurnamī* is the tithi under consideration.
- That means the Recitation of the Śrīmad-Bhāgavatam began on Rādhāṣthami tithi (15 days after Lord Śrī Kṛṣṇa's janma tithi) or on the next day, going through Vāmana-dvādaśī (*Bhādrapada-śukla-dvādaśī*), continued for the next six days and completed on the full moon day of the Bhādrapada month (*Bhādrapada-śukla-paurnamī*).

5

Seasons and Eclipses

Vertical Motion of the Sun, Moon and Rahu

After describing the sun's movement through the zodiac signs, its rising, setting and its speed along with the movements of other planets in the zodiac, Śukadeva Gosvāmī describes the vertical movements of the sun. These are the northern course (*uttarāyana*, from winter equinox to summer equinox) and southern course (*daksināyana*, from summer to winter equinox) of the sun.

As stated in SB 5.20.30: "In the middle of Puṣkaradvīpa is a great mountain named Mānasottara, which forms the boundary between the inner side and the outer side of the island. Its breadth and height are 10,000 *yojanas*. On that mountain, in the four directions, are the residential quarters of *devatās* such as Indra. The wheel of the chariot of the sun-god (*sūrya-rathasya*), making up a year (*saṁvatsarātmakaṁ*), going around Meru (*meruṁ paribhramataḥ*), travels on the top of the mountain (yad-upariṣṭāt) with northern and southern courses, called the day and night of the *devatās* (*devānām aho-rātrābhyāṁ paribhramati*)."

Because Mānasottra Mountain is circular, there are two *varṣas,* one either side of it, also of circular shape. "By days and nights of the *devatās*" means with its northern (from winter to summer equinox) and southern courses (from summer to winter equinox), the sun causes day and night for the *devatās.*

Uttarāyana and Daksināyana Movements of the Sun

yad adas taraṇer maṇḍalaṁ pratapatas tad vistarato yojanāyutam
ācakṣate dvādaśa-sahasraṁ somasya trayodaśa-sahasraṁ rāhor

The sun globe, which is a source of heat, extends for 10,000 yojanas. The

moon extends for 20,000 yojanas, and Rāhu extends for 30,000 yojanas. (SB 5.24.2)

Note1: Commentary by Śrīla Vīrarāghava ācārya - The sun disc is 10,000 yojanas (*yojanāyutam*) broad, the moon disc is 12,000 yojanas (*dvādaśa-sahasraṁ*) and that of Rāhu is 13,000 yojanas (*trayodaśa-sahasraṁ*). Because of his greater size, Rāhu is able to cover both the sun and the moon sometimes. Thus, this verse explains both about the eclipses and about the Uttarāyana and Daksināyana Movements of the sun.

Note2: Commentary by Śrīla Vijayadhvaja Tīrtha - The sun disc is 10,000 yojanas (*yojanāyutam*) broad. The moon disc is 20,000 yojanas (*dvādaśa-sahasraṁ*) broad. *Dvidaśa* means twice ten. Rāhu's disc is 40,000 yojanas (*trayodaśa-sahasraṁ*) broad. *Trayodaśa* means three tens plus ten. This is not imagination - it is according to the words of Śrīla Madhvācārya "Except the breadths of the discs Ravi, Chandra and Rāhu (and also the interchange of Milk Ocean and Emulsified Yogurt or Curds Ocean), all other things (narrated here) have to be accepted here as perfectly correct." Narrated here refers to narrated in Śrīmad-Bhāgavatam. Śrīla Madhvācārya says - "Śrīmad-Bhāgavatam is the authoritative text, which tells us about Cosmography; and information gathered from any other source should be in accordance with it."

In this regard, Madhvācārya quotes the following verses from the Brahmāṇḍa Purāṇa,

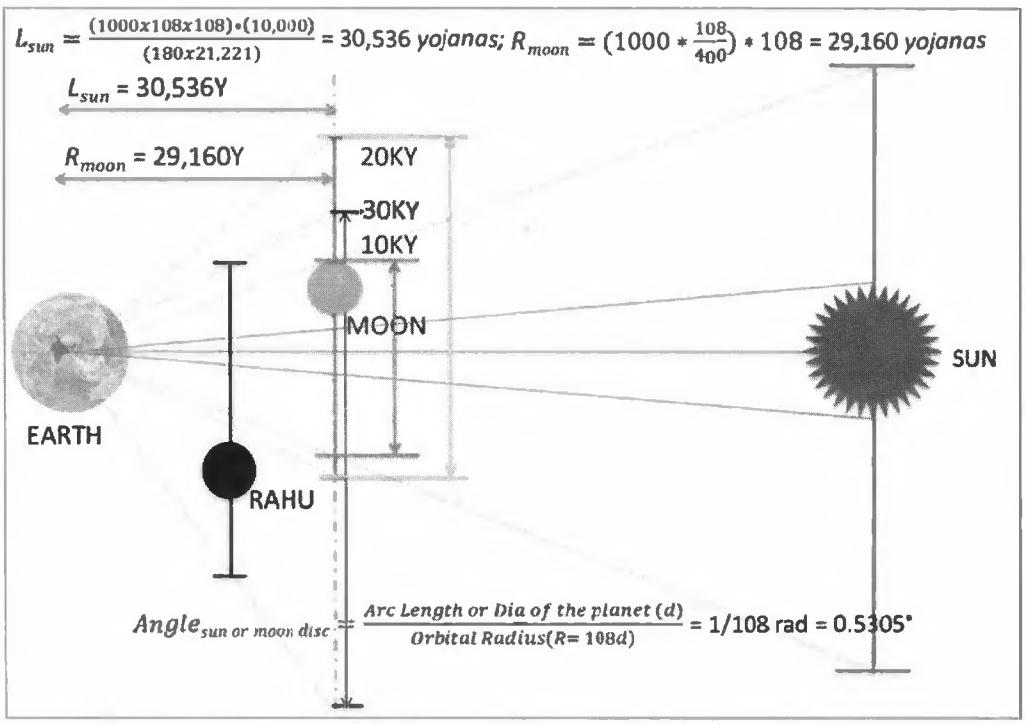
yathā bhāgavate tūktaṁ bhauvanaṁ kośa-lakṣaṇam
tasyāvirodhato yojyam anya-granthāntare sthitam

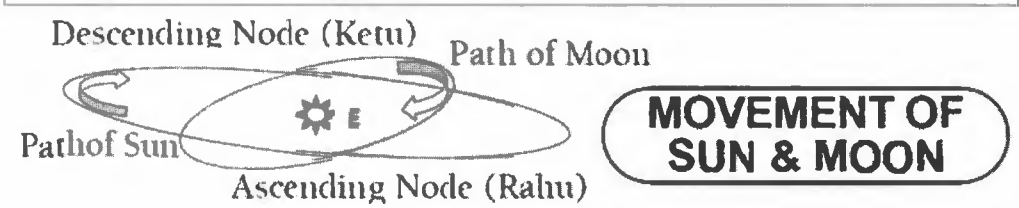
Note3: Commentary by Yādavācārya - *Dvādaśa-sahasram* means *dvidaśa-sahasraṁ,* that is two times ten thousand, meaning 20,000 yojanas, Thus, it is twice as broad as the sun's disc. *Trayodaśa* should be interpreted as *tri-daśa* and *daśa* i.e. three tens and ten meaning forty in all. Thus, Rāhu's disc is 40,000 yojanas broad - twice that of the moon. That is why it is said by Śrīla Madvācārya, "Everything should be interpreted without contradiction to Bhāgavata. The only exception being the position of the Milk and Curds (Emulsified Yogurt) Ocean and the width of Rāhu, the moon and the sun. The words should be interpreted differently so that they do not contradict other *pramanas.*"
In this regard, Madhvācārya quotes the following verses from the

Brahmāṇḍa Purāṇa,

maṇḍode puraṇaṁ caiva vyatyāsaṁ kṣīra-sāgare
rāhu-soma-ravīṇāṁ ca maṇḍalād dvi-guṇoktitām
vinaiva sarvam unneyaṁ yojanābhedato 'tra tu

As per Srila Madhvācārya, these are the two corrections in the Śrīmad-Bhāgavatam.





This extension of sun globe by 10,000 yojanas causes *uttarāyana*, *daksināyana* and *visuvāyana* (line of equator). [125]

[125] These are the scaled down values of the actual southern and northern movements of the planets Sun, Moon and Rahu. An overview of actual southern and northern movements of Sun is presented in the next chapter.

Bhugola Varnanam says[126] "Lord Sisumara, with head down and tail up, turns round and round like a gymnast right above the Meru axis and, projecting shaft-like power lines from His body and with power lines from polestar Point, controls the movements of all celestial bodies. Lord Simsumara gradually lifts the Kala Cakra up during a period of six months and lowers it during the next six months making the sun's chariot cross the Equator northward and southward, as also making the moon and the Rahu planet to keep pace with the sun. The seasons occur for this reason; and the earth need not go around the sun with inclined axis as the too clever Copernicus suggested.

Śrīmad-Bhāgavatam clearly points out that how the Time-Wheel is lifted and lowered when it speaks of "Udagāyana, Daksināyana & Vaisuvadāyana." Vishuvad rekha means the Equator. This also suggests that Bhārata-khanda is indeed globular in shape. The same Śrīmad-Bhāgavatam Canto 5 tells us about the extensions of 'Candra mandala' and 'Rahu mandala' saying that the extension in the case of the sun, moon and rahu are 10,000, 20,000 & 30,000 yojanas respectively. The extension of the sun's course is 5000 yojanas to the north of the celestial Equator and 5000 yojanas to the south of it. Hence, the extension of 'Surya mandala' is said to be or considered as 10,000 yojanas. Similarly, to keep pace with the sun in the northern and the southern courses, the extension of the moon's course is 10,000 yojanas up and 10,000 yojanas down, and the extension of Rahu's course is 15,000 yojanas in north and 15,000 yojanas in south. Some say it is 20,000 yojanas in north and 20,000 yojanas in south – *'dviguna'*. Apart from the annual northern and southern courses, the moon has also northern and southern courses every fortnight. This may be seen by any one by observing the moon's position in the sky night after night."

References on Uttarāyana and Daksināyana fom Śrīmad-Bhāgavatam[127]

As the shining sun rotated around Meru lighting up half the earth up to Lokāloka Mountain as daytime, while covering half the surface of the earth with darkness, caused them experience night. Discontent with half the earth's surface being covered by darkness, Priyavrata followed the movement of the

[126] Refer to the section "Kala Cakra - The Time Wheel (242 - 256)"

[127] SB 5.1.30-31: Maharaja Priyavrata creates seven oceans and seven dvīpas

sun (*taruṇim*), so that when the sunset, he would light up that area to give happiness to the inhabitants of earth. By his worship of the Lord, he surpassed all humans in influence. This means that by the power of *yoga* of sages like Saubhari, Priyavrata had two bodies, one as the King and one as the sun.
The movement of the wheel of his chariot produced troughs, which became seven oceans, from which seven continents were formed. This means that he had seven chariots each with one wheel, by the power of his *yoga*. Their movement was higher than the sun's, and they circled around the plane of the earth. The chariots were successively larger and larger in size, since the outer circles were larger and larger. He made larger and larger chariots in succession, each one twice the size of the previous one, traveling higher and higher so that the inhabitants of his city in Āryāvarta could see his chariot even at a great distance. He rode in each of these seven chariots for 25 days and 45.5 *ghaṭikas*[128] each (totaling half a year, 25.76x7 = 180.3 days), starting with the summer equinox, with the sun gradually descending until Pauṣa month (January).

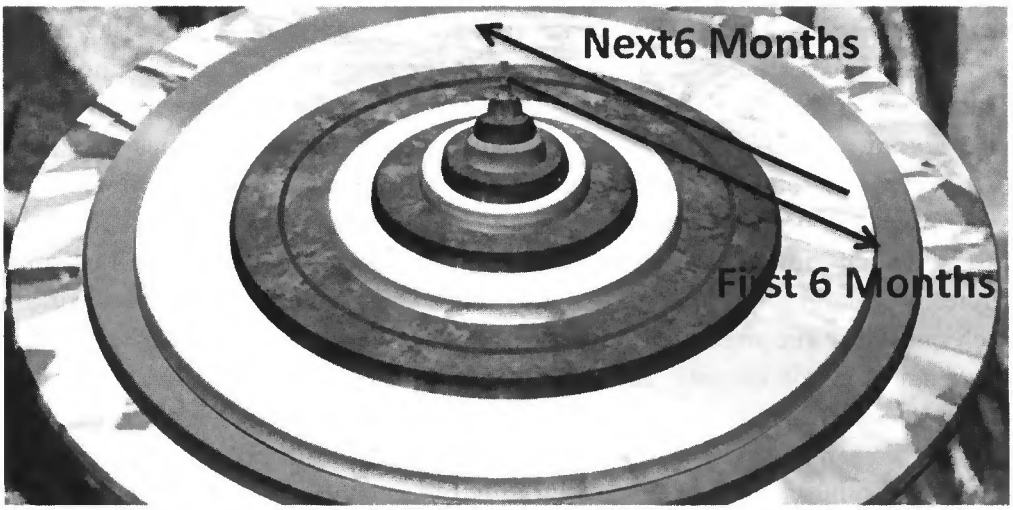
Considering the above two statements "Priyavrata followed the movement of the sun" and "starting with the summer equinox, with the sun gradually descending until Pauṣa month" indicates that the sun gradually descends from summer equinox (*Karka sankranti*, 17th July) to Pauṣa month, which generally coincides with winter equinox. That means from summer equinox to winter equinox the sun gradually comes down and goes outward (descending) as Priyavrata did. This indicates the Daksināyana movement of the sun.

Then again starting from the winter equinox (*Makara sankranti*, 14th Jan), he would travel until the cycle was completed, with the same number of days fixed, but with the opposite order of chariots, with the sun gradually ascending until Aṣaḍha month (July).

Considering the above two statements "Priyavrata followed the movement of the sun" and "starting from the winter equinox, with the sun gradually ascending until Aṣaḍha month" indicates that the sun gradually ascends from winter equinox to Aṣaḍha month, which generally coincides with summer equinox. That means from winter equinox to summer equinox the sun gradually goes up and goes inward (ascending) as Priyavrata did. This

[128] There are 60 *ghaṭikas* in a full day, with each *ghaṭika* = 24 minutes

indicates the Uttarāyana movement of the sun.



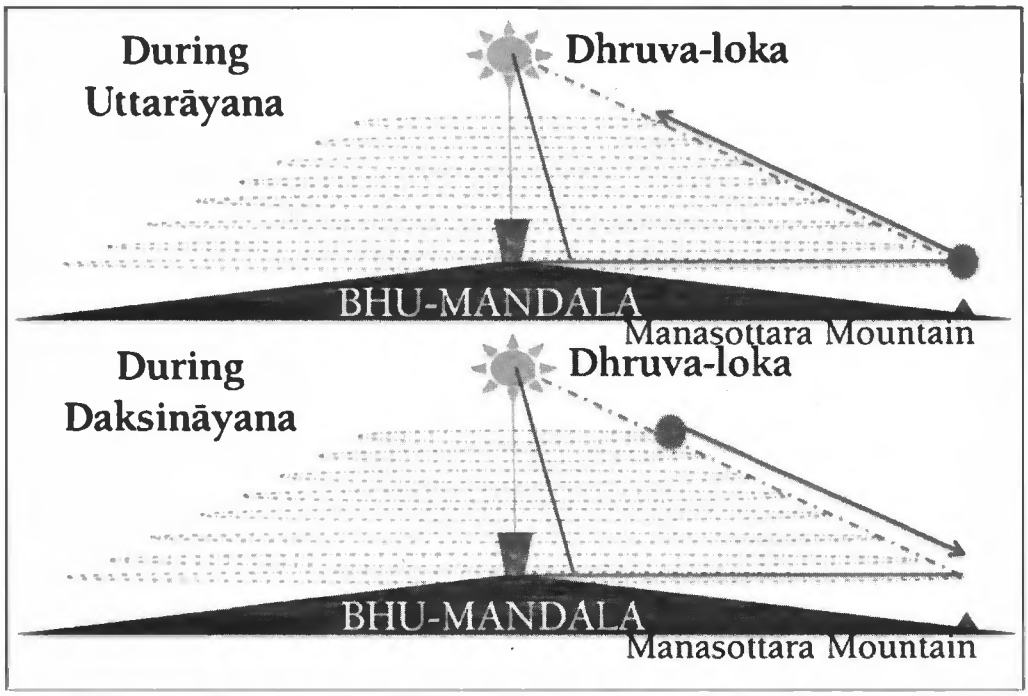
As the sun traveled around Meru clockwise along with the dependent zodiac, it gradually increased in speed, and thus, the days after the summer equinox become shorter each month, and from the winter equinox the sun gradually slowed down and the days become longer.

References from Other Puranās

Linga Purāna - Adhyāya 54, text 12-14

As the sun's chariot moves in orbits, during Daksināyana, the two ropes held by Dhruva are loosened and they become longer. Thus, the chariot goes outward into the outer tracks. During Uttarāyana, Dhruva pulls the ropes up and they become shorter. Thus, the chariot is dragged into the inner tracks. There are 180 circles between the two extreme ends (between north & south).

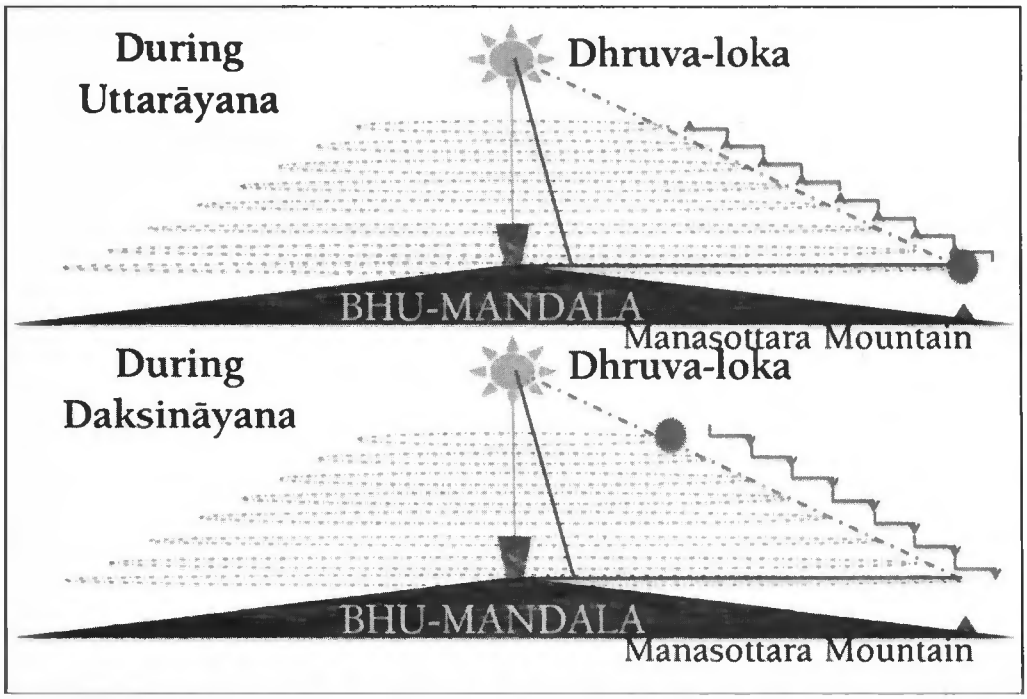
Day after day for 180 days	Uttarāyana	Daksināyana
Radious of the Orbit (r)	Decreases	Increases
Circumference of the Orbit ($C = 2\pi r$)	Decreases	Increases
Daily Speed of the Sun ($S = \frac{2\pi r}{24\ hours}$)	Decreases	Increases



Vishnu Purāna – Amsa 2; Adhyaya 10, text 1 (purport by Sridhara Svami)
There are 180 circular paths, the sun traverses in a year between the two directions. There are 180 orbits between the north and the south which the sun covers during a year by his elevated and inclined rate of movement. When Dhruva pulls the two ropes tied to the end of the yoke during Uttarāyana, every day the sun moves toward the inner orbit which is elevated more than the previous one. During Daksināyana, when Dhruva loosens the two ropes, every day the sun slips into the outer orbit which is inclined to the previous orbit. This is also stated in the Matsya Purāna.

During Uttarāyana: Track radius gradually decreases daily -> chariot's speed decreases daily	
Radius of the Orbit (r)	Decreases
Circumference of the Orbit ($C = 2\pi r$)	Decreases
Daily Speed of the Sun ($S = \frac{2\pi r}{24\ hours}$)	Decreases
During Daksināyana: *Track radius gradually increases daily -> chariot's speed increases daily*	
Radius of the Orbit (r)	Increases

Circumference of the Orbit ($C = 2\pi r$)	Increases
Daily Speed of the Sun ($S = \frac{2\pi r}{24 \text{ hours}}$)	Increases



Vāyu Purāna – Adhyaya 50, text 122-140 & 189-204

As the sun reaches the end of southern course (far end of Daksināyana) by the month of Māgha, he moves in the middle of Puṣkaradvīpa, above Mānasottara Mountain. When he starts his northern course and comes to Vishuva (the equinox points), he moves in the north side of the milk ocean. When he reaches the maximum point in the north during his stay in Sravana, he will be moving in the northern regions of the sixth island Sākadvīpa. Surya always moves in the inner circles during Uttarāyana and outer circles during Daksināyana. The sun will be on the Vishuvata peak on the day of equinoxes, thus rendering both day and night equal. On the day of equinoxes, at the end of Mesa and Tula months, both the day and night will be 15 muhurtas each.

The Sun always moves in the inner circles during Uttarāyana and outer circles during Daksināyana

Far end of Daksināyana	Sun moves in the middle of Puṣkaradvīpa (15.75MY) – 7th dvīpa
During Vishuva (the equinox points)	Sun moves in the north side of the milk ocean – 6th Ocean
Far end of Uttarāyana	Sun will be moving in the northern regions of the Sākadvīpa (11.9MY) – 6th dvīpa
The sun will be on the Vishuvata peak on the day of equinoxes, thus, rendering both day and night equal	

References from Śrīmad-Bhāgavatam - The Śiśumāra Planetary Systems[129]

Some describe the great machine or the zodiac belt *(jyotir-anīkaṁ)*, consisting of the stars and planets, resembles the form of a *śiśumāra* [dolphin] in the water. It is sometimes considered an incarnation of Kṛṣṇa, Vāsudeva. Therefore, some yogīs contemplate this imaginary *śiśumāra* to be swimming in the sky the way a dolphin swims in water. They meditate upon it as the *virāṭ-rūpa,* the gigantic form of the Supreme Lord, because it is actually visible.

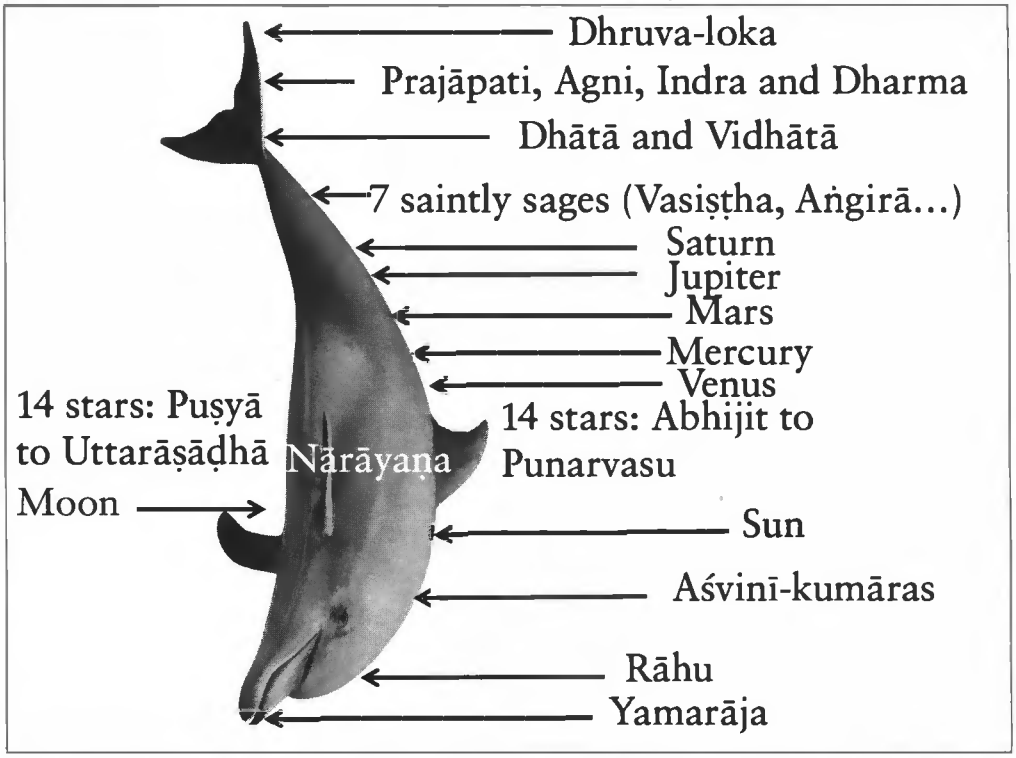
Thus, they describe this zodiac belt (*jyotir-anīkam*) to be the form of a dolphin, and worship it as Vāsudeva in meditation. This form of the *śiśumāra* has its head downward and its body coiled. The coiled body of the dolphin turns toward the right side. Its body is balanced because an equal number of stars, i.e. 14 on either side, occupy its two sides. The body of the *śiśumāra* should be considered as the external form of Lord Viṣṇu, the Supreme Lord, and not the eternal form of the Lord. This Śiśumāra-cakra is the resting place of all the demigods and all the stars and planets. Śrīla Madhvācārya quotes the following verse from the Brahmāṇḍa Purāṇa.

jñānānandātmano viṣṇuḥ śiśumāra-vapuṣy atha
ūrdhva-lokeṣu sa vyāpta ādityādyās tad-āśritā

"Lord Viṣṇu, who is the source of knowledge and transcendental bliss, has assumed the form of Śiśumāra in the seventh heaven, which is situated in the topmost level of the universe. All the other planets, beginning with the sun, exist under the shelter of this Śiśumāra planetary system."

[129] SB 5.23.4-7: Description of body of Śiśumāra (Machine of stars, planets ~ Dolphin)

Position of Planets andCconstellations on the body of Śiśumāra-cakra



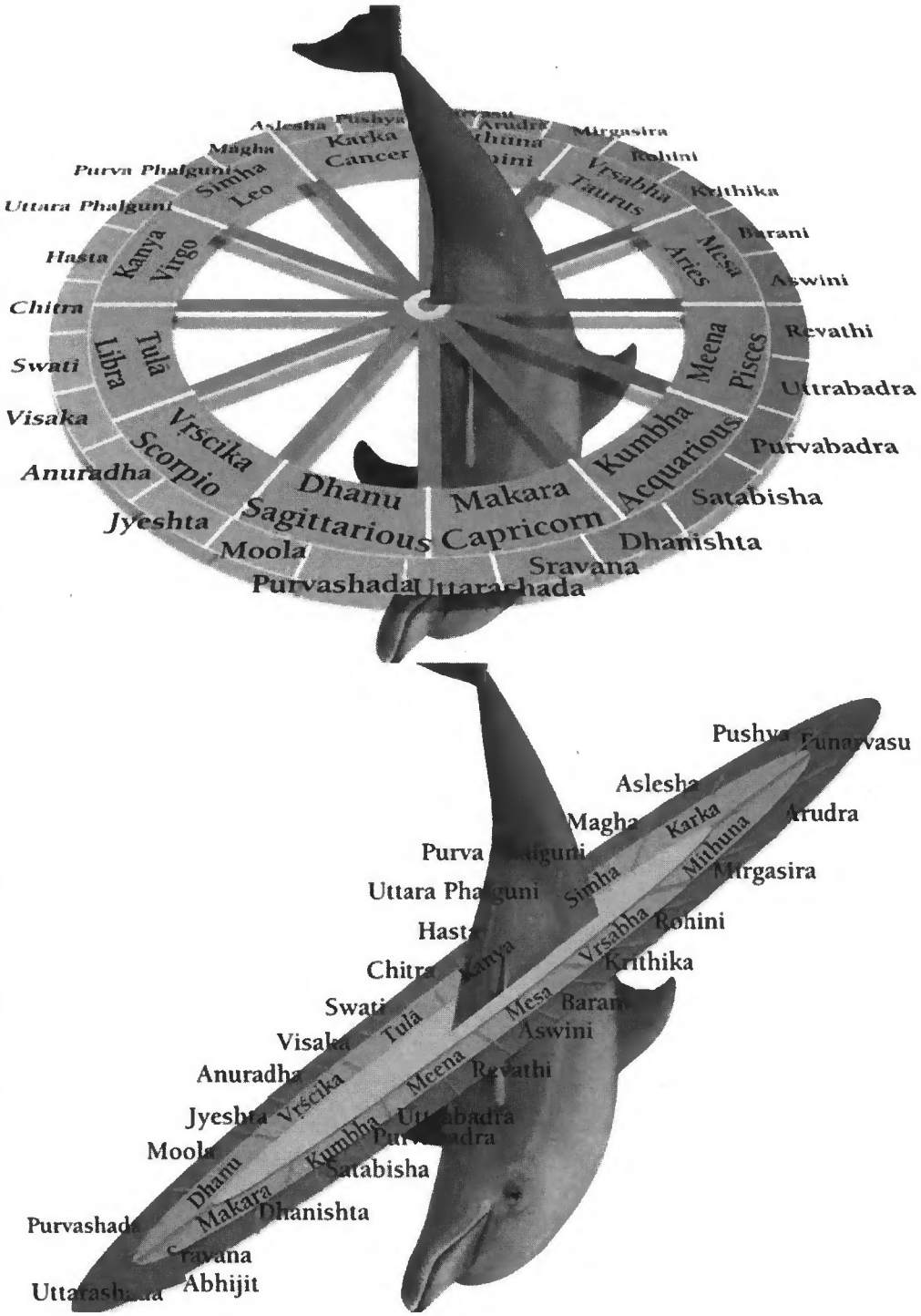
Bodily limb of Śiśumāra	The Abode of Demigods, Stars & Planets
On the end of its tail	Dhruva-loka
On the body of its tail	The demigods Prajāpati, Agni, Indra & Dharma
At the base of its tail	The planets of the demigods Dhātā and Vidhātā
The hips on the *śiśumāra*	The seven saintly sages like Vasiṣṭha and Aṅgirā
On its genitals	Saturn
On the back of its neck	Jupiter
On its mouth	Mars
Within life air (prāṇāpāna)	Mercury
On its navel	Venus
Right side of coiled body	14 stars from Abhijit to Punarvasu (*Udgayanāni*)
Left side of the coiled body	14 stars from Puṣyā-Uttarāṣāḍhā (*Dakṣiṇāyanāni*)
On the back of śiśumāra	Ajavīthī stars (1st part of southern constellations)
On its abdomen	Ākasa-Gangā (the MilkyWay)
Within its mind	The moon

On its chest	The Sun
Within the core of its heart	Nārāyaṇa
On its breasts	The Aśvinī-kumāras
On its neck	Rāhu
On its upper chin	Agasti
On its lower chin	Yamarāja
All over its body	Comets
In its pores	the numerous stars

The position of the 28 Constellations on the body of Śiśumāra-cakra

Bodily limb of the Śiśumāra-cakra	**The abode of devatās, all the stars & planets**
On the right side of the coiled body of the Śiśumāra-cakra, which turns toward its right	The 14 constellations from Abhijit to Punarvasu (*Udgayanāni*)
On its left side (body is balanced with 14 stars either side)	The 14 stars from Puṣyā to Uttarāṣāḍhā (*Dakṣiṇāyanāni*)
On the right & left sides of the loins (in between hips and genitals)	stars named Punarvasu and Puṣyā (in between sapta-rsis & Saturn)
On its right and left feet (in between hips and genitals)	Ārdrā and Aśleṣā (in between sapta-rsis & Saturn)
On its right and left nostrils (below the neck and mouth)	Abhijit and Uttarāṣāḍhā (below Rāhu, Agasti and Yamarāja)
At its right and left eyes (below neck and above the mouth)	Śravaṇā and Pūrvāṣāḍhā (below Rāhu, Agasti & above Yamarāja)
On its right and left ears (above neck)	Dhaniṣṭhā and Mūlā (above Rāhu)
On the right and left shoulders (above neck and below chest)	Śatabhiṣā and Jyeṣṭhā (above Rāhu and below Sun)
On the ribs of the left of its body (between genitals to ears)	Southern course: Maghā to Anurādhā[130] (below Saturn and above Rāhu)
On the ribs on the right side (between genitals to ears)	Northern course: Mṛgaśīrṣā-Pūrvabhādra (below Saturn and above Rāhu)

[130] The 8 constellations Mṛgaśiras, Rohiṇī, Kṛttika, Bhāraṇī, Āśvinī, Revatī, Uttarabhadrā and Pūrvabhadrā situated on the right side are in reverse order.


Aslesha
Pushya
Karka
Cancer
Magha
Purva Phalguni
Simha
Leo
Uttara Phalguni
Hasta
Kanya
Virgo
Chitra
Tula
Libra
Swati
Visaka
Vrscika
Scorpio
Anuradha
Jyeshta
Moola
Purvashada
Dhanu
Sagittarious
Makara
Capricorn
Uttarashada
Sravana
Dhanishta
Satabisha
Purvabadra
Kumbha
Acquarious
Uttrabadra
Meena
Pisces
Revathi
Aswini
Mesa
Aries
Barani
Krithika
Vrsabha
Taurus
Rohini
Mrigasira
Arudra
Punarvasu
Mithuna
Pushya
Punarvasu
Aslesha
Magha
Karka
Arudra
Purva
Uttara Phalguni
Simha
Mithuna
Mrigasira
Hasta
Rohini
Chitra
Kanya
Vrsabha
Krithika
Swati
Tula
Mesa
Barani
Visaka
Aswini
Anuradha
Meena
Revathi
Jyeshta
Vrscika
Moola
Kumbha
Dhanu
Satabisha
Makara
Dhanishta
Purvashada
Sravana
Abhijit

From the above description it is clear, that the 27 Constellation group is not on a horizontal plane, instead from Abhijit all the way up to Punarvasu, they are on an inclined plane. Even the star group indicating the Zodiac Belt also has similar inclined plane, instead of a horizontal plane. This indicates that both the sun and the moon during their transition from Abhijit to Punarvasu they travel in an elevated orbit than the previous one, and during their transition from Pusya to Uttarasada they travel in an inclined orbit than the previous one. This explains their Uttarayana and Daksinayana motions clearly, which the sun completes in one year or *samvastara* and the moon completes in one *vastara* (27 days approximately).

Description of Uttarayana-Daksinayana from Śrīmad-Bhāgavatam[131]

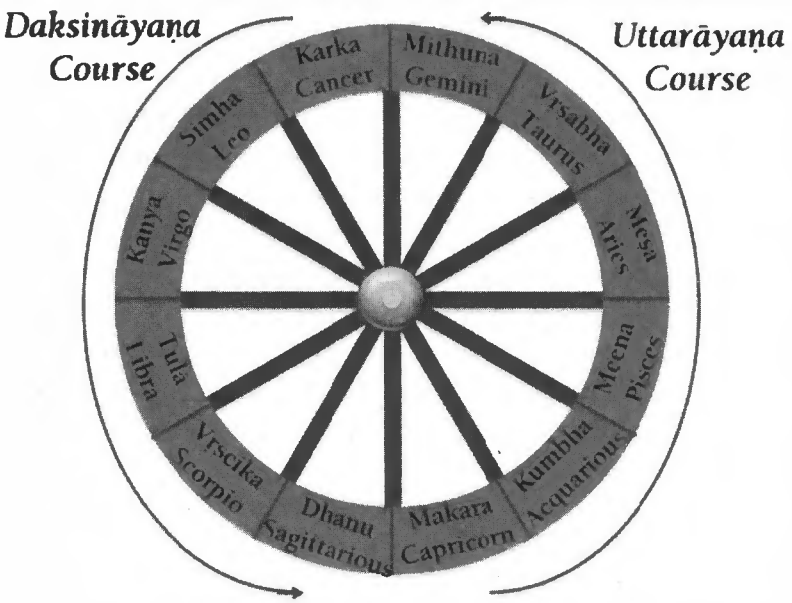
Situated in the middle of the *antarīkṣa,* the sun, master of heat, heats up and lights up the three worlds by its light. During *uttarāyaṇa,* by slow speed while ascending in the sky and passing time, the sun moves through six zodiac signs starting from Capricorn, making the days gradually longer and the nights shorter. During *dakṣiṇāyana,* moving with quick speed and descending, the sun moves through six zodiac signs starting from Cancer, making the days shorter and the nights longer. By moving at medium speed during the equinox periods, neither between ascending and descending, with days and nights equal, the sun moves through Aries and Libra.

sa eṣa udagayana-dakṣiṇāyana-vaiṣuvata-saṁjñābhir māndya-śaighrya-samānābhir gatibhir ārohaṇāvarohaṇa-samāna-sthāneṣu

During Uttarāyana: By slow speed, while ascending in the sky, the sun makes the days gradually longer and the nights shorter	
Capricorn (Makara) to Gemini (Mithuna)	duration of the days increases every month
Radios of the Orbit (r)	Decreases
Circumference of the Orbit ($C = 2\pi r$)	Decreases
Daily Speed of the Sun ($S = \frac{2\pi r}{24\ hours}$)	Decreases
During Daksināyana: moving with quick speed and descending, the sun makes the days shorter and the nights longer	

[131] SB 5.21.3-6: Movement of sun through *uttarāyaṇa* and *dakṣiṇāyana*

Cancer (Karkataka) to Sagittarius (Dhanur)	the duration of the days decreases every month
Radios of the Orbit (r)	Increases
Circumference of the Orbit ($C = 2\pi r$)	Increases
Daily Speed of the Sun ($S = \frac{2\pi r}{24\ hours}$)	Increases
During Vishuvāyana: moving at medium speed, sun makes days and nights equal	
Aries and Libra	days and nights almost equal



When the sun is in Aries and Libra, the days and nights are equal (*yadā meṣa-tulayor vartate tadāho-rātrāṇi samānāni*). Because there is no extreme difference in the day and night during this time, it is called equal. Complete equality will only take place on one day and night of the month. When the sun moves in Taurus and the five following signs, the days increase in length and then decrease every month by a *ghaṭikā* or 24 minutes (*vardhante hrasati ca māsi māsy ekaikā ghaṭikā rātriṣu*).

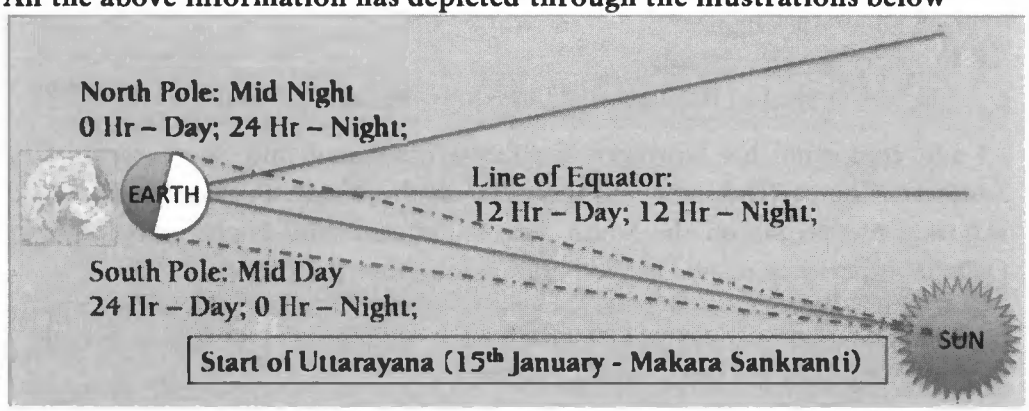
When the sun is in Taurus and Gemini (northern course) the days increase in duration, and when the sun is in Cancer and the following months Leo and Virgo (southern course), the days become shorter, whereas the nights become longer. When the nights increase in duration, the days become shorter. Actually, starting from Capricorn the days become longer and

starting from Cancer, the days become shorter. The statement that the days increase by *ghaṭikā* (24 minutes) each month is a general statement, since the increase and decrease is different each month.

When the sun passes through the five signs beginning with Scorpio, the duration of the days decreases until Capricorn (Scorpio, Sagittarius and Capricorn – southern course), and then gradually it increases month after month (*tadāho-rātrāṇi viparyayāṇi bhavanti*), until day and night become equal in Aries (Aquarius, Pisces, & Aries– northern course).

Until the sun travels on the southern course, the days grow longer (*yāvad dakṣiṇāyanam ahāni vardhante*). That means during the northern course[132] (Capricorn, Aquarius, Pisces, Aries, Taurus, Gemini) the days grow longer and nights grow shorter every month until summer equinox. Until the sun travels to the northern course, the nights grow longer (*yāvad udagayanaṁ rātrayaḥ vardhante*). That means during the southern course[133] (Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius) the nights grow longer and days grow shorter every month until winter equinox.

All the above information has depicted through the illustrations below

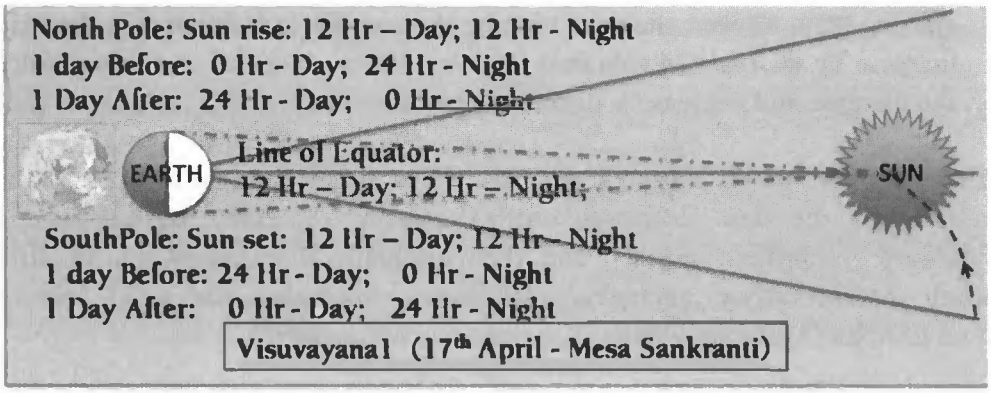


As sun is beginning his Northern Course (Uttarayana), people on the South Pole experience Mid-Day and the people on the North Pole experience Mid-Night. In addition, people on the Equator experience almost 12 hours of Day and 12 hours of Night, throughout the year.

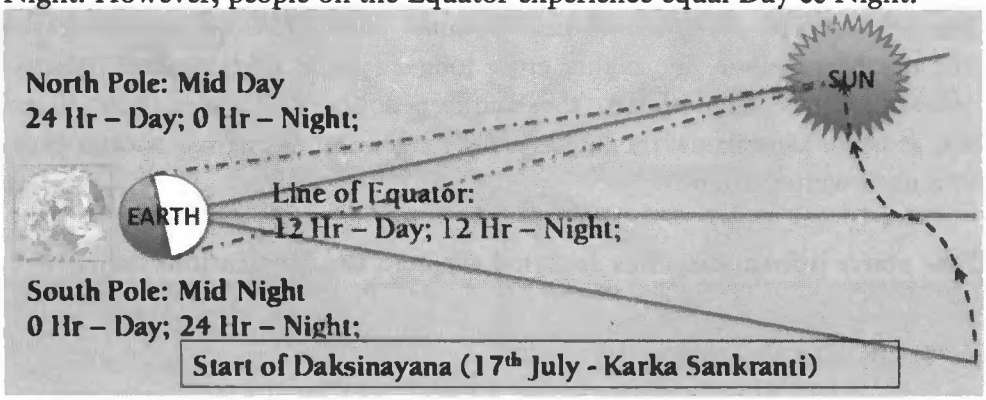
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[132] *Yāvad dakṣiṇāyanam* means from *uttarāyaṇa* until *dakṣiṇāyana*

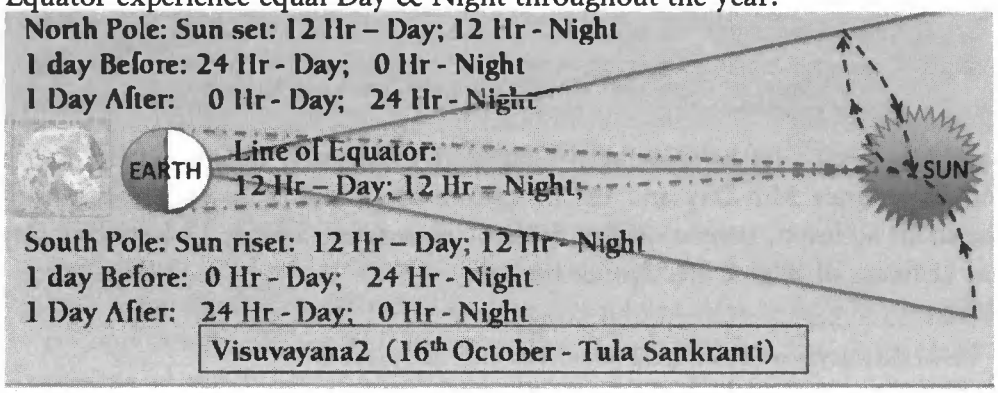
[133] *yāvad udagayanaṁ* means from *dakṣiṇāyana* until *uttarāyaṇa*



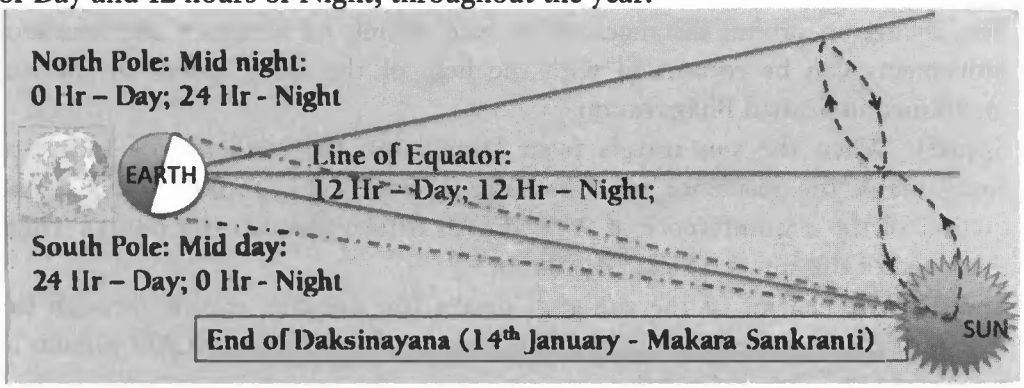
As sun is entering into Northern Hemisphere, the people on the North Pole begins to experience Day and the people on the South Pole begins to experience Night. However, people on the Equator experience equal Day & Night.



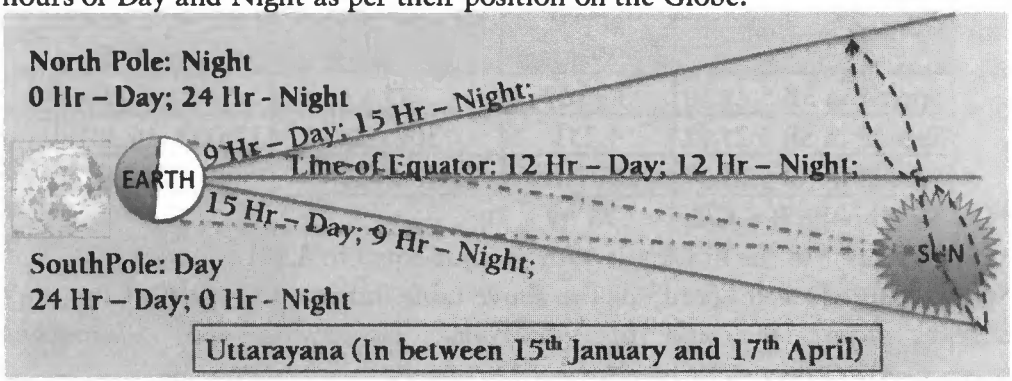
As sun completed his Northern Course (Uttarayana) and about to start his Southern Course (Dakshinayana), people on the North Pole experience Mid-Day and the people on the South Pole experience Mid-Night. People on the Equator experience equal Day & Night throughout the year.



As sun is entering into Southern Hemisphere, the people on the North Pole begins to experience Night and the people on the South Pole begins to experience Day. In addition, people on the Equator experience almost 12 hours of Day and 12 hours of Night, throughout the year.



As sun completed his Southern Course (Dakshinayana) and about to start his Northern Course (Uttarayana), people on the North Pole experience Mid-Night and the people on the South Pole experience Mid-Day. And people on the Equator experience almost 12 hours of Day and 12 hours of Night, throughout the year. People staying at in-between areas experience in-between hours of Day and Night as per their position on the Globe.



This image shows how the people staying at in-between areas experience in-between hours of Day and Night as per their position on the Globe. This picture showed above depicts the complete movement of the sun on the Zodiac for one complete year. On 15th January, sun enters into Capricorn (Makara), and in three months he enters into Aries (Mesa) on 17th April. At the interval of three months, he enters into Cancer (Karka) on 17th July, Libra (Tula) on 16th Oct, and then again into Capricorn (Makara) on 14th Jan, Thus, the yearly path of the sun is depicted. During the Northern course sun goes above and inside

increasing daytime in the north and decreasing the daytime in the south. Similarly, during Southern course the sun comes below and outside increasing daytime in the south and decreasing the daytime in the north.

The change in orbital distances of the sun during its northern and southern movements can be reconciled with the help of the three speeds of the sun mentioned in Śrīmad Bhāgavatam.

Spped1: When the sun travels from Devadhānī, the residence of Indra, to Saṁyamanī, the residence of Yamarāja, it travels 23,775,000 *yojanas* (one quarter of the circumference of 95.1 MY) in fifteen *ghaṭikās* (six hours). Thus, the speed of the sun is 3.9625 MY / hour.

Speed2: The chariot of the sun-god, meant for worship, travels through the four residences above mānasottara mountain at a speed of 3,400,800 *yojanas* in a *muhūrta* (4.251 MY / hour).

Speed3: In his orbit through a circumference of 95.1 MY on the circle of the earth, the sun travels 2,000 *yojanas* and two *krośas* (2,000.5 Yojanas) in a moment (1 kṣaṇa = 8/5 sec). Thus, the sun travels at a speed of 4.501125 MY / hour (= 2000.5*60*60/1.6 = 4.5 MY / hour)

Speed of the Sun	Speed (MY/hour)	Circumference (MY/day)	Radius of Sun's orbit (MY)
Speed1 in SB 5.21.10	3.9625	95.1	15.85 (15.136)
Speed2 in SB 5.21.12	4.251	102.024	17.004 (16.381)
Speed3 in SB 5.21.19	4.501125	108.027	18.0045 (17.193)

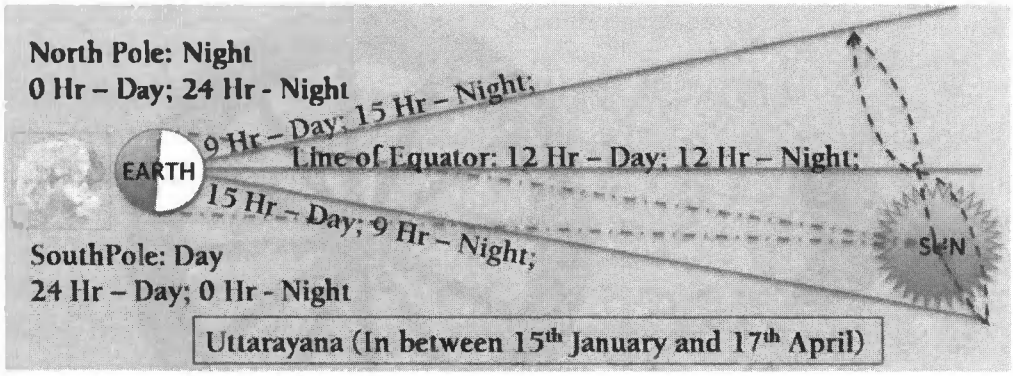
*C = Sx24hours; R = C/2pi = C/6 (pi = 3);

**Radius value in the bracket is with pi value equal to 3.141

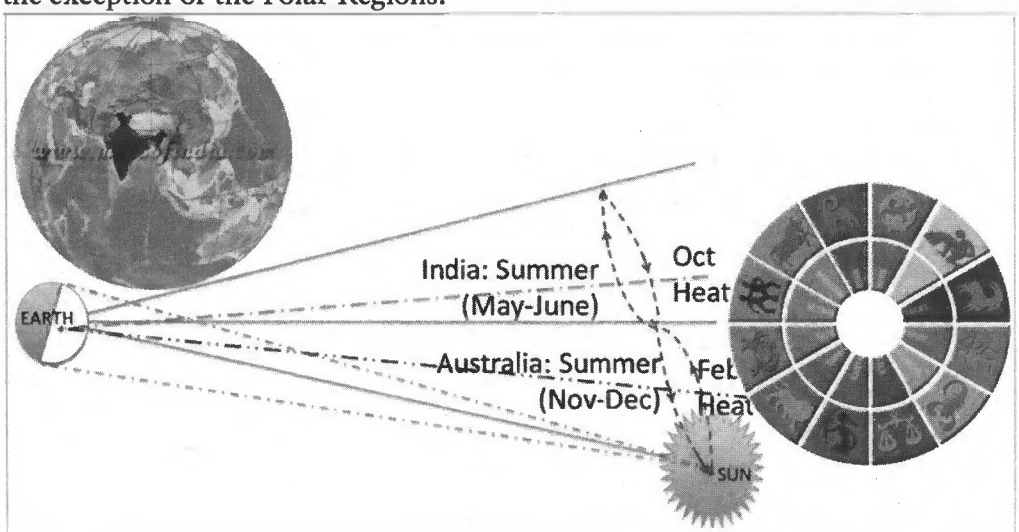
Speed 1, Speed2 and Speed3 of the above table indicate the speed of the sun's chariot during the end of *uttarāyana*, *visuvāyana* and *daksināyana* respectively.

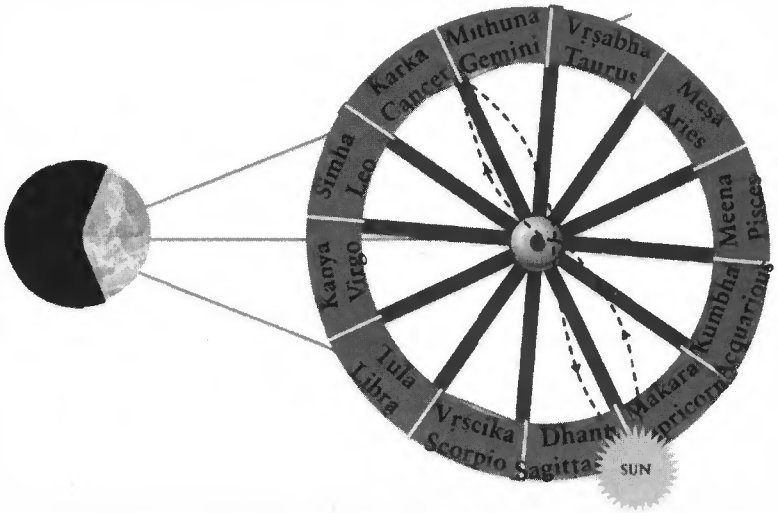
Change in Seasons and Day-Night Distribution

As the sun travels in the Northern Hemisphere starting from Aries, people on the south experience winter and on the north experience summer. As the sun travels in the Southern Hemisphere staring with Libra, people on the north experience winter and the people on the south experience summer.



The change in seasons for India is as follows: When the sun begins his northern course by entering into Capricorn (Makara) on 15th January, the temperature in India slightly increases daily. In next three months slowly passing, the sun enters into Aries (Mesa) on 17th April, during which India experiences spring with gradual increase in temperature. As sun enters northern hemisphere and moves towards Cancer (Karka), India experiences intense summer heat during April, May and June. As sun changes his course on 17th July by entering Cancer (Karka), India is experiencing Rainy season, which follows the summer. As sun enters Libra (Tula) on 16th Oct and as he goes further, India's temperature gradually goes down and they experience intense winter in January as sun again enters into Capricorn (Makara) on 14th Jan. Thus, the four seasons (spring, summer, Rainy and winter) change continuously every year. The change in seasons is somewhat similar to most of the places on the earth, with the exception of the Polar Regions.





The upper dotted line projecting from the earth Globe, representing India, intersects the sun's path twice. First in May indicating summer and next in Oct also indicating intense heat, because both the instances the sun is situated at the overhead of India. Similarly the lower dotted line projecting from the earth Globe, representing Australia, intersects the sun's path twice. First in Nov-Dec indicating summer and next in Feb indicating hot weather, because both the instances the sun is situated at the overhead of Australia.

Day-Night Variations of various tropical locations across the year

	Jun-30		Sep-30		Dec-30		Mar-30	
	Sunrise	Sunset	Sunrise	Sunset	Sunrise	Sunset	Sunrise	Sunset
Greenland (75x45)	Full day		08:00	18:50	Full night		06:10	21:10
	24 Hrs		10:50 Hrs		0 Hrs		15 Hrs	
Mumbai (20x70)	06:04	19:20	06:28	18:28	07:11	18:11	06:30	18:50
	13:16 Hrs		12 Hrs		11 Hrs		12:20 Hrs	
Singapore (LOE:1x110)	07:00	19:10	06:52	18:58	07:06	19:08	07:06	19:13
	12:10 Hrs		12:06 Hrs		12:02 hrs		12:07 Hrs	
Cape town (SA: -35, 25)	07:51	17:51	06:20	18:50	05:40	20:00	07:00	18:45
	10 hrs		12:30 Hrs		14:20 Hrs		11:45 Hrs	
Bahia/Blanca Argentina (-40x70)	08:28	17:58	06:45	19:15	05:45	20:35	07:25	19:00
	9:30 Hrs		12:30 Hrs		14:50 Hrs		11:35 Hrs	
Cape Horn (SA: -55x68)	09:00	15:55	07:00	19:40	04:40	22:20	07:48	17:18
	6:55 Hrs		12:40 Hrs		17:40 Hrs		9:30 Hrs	

Note: In Greenland (75x45) on Sept-25 there will be equal day and equal

night (12Hrs day & 12hrs night) as seen in the last year.

Notes from Bhugola Varnanam on Vertical Motion of Sun[134]

When the sun is in conjunction with the Tula (Libra) and the Mesa (Aries) constellations, almost equal division of day and night of 30 ghatikas each is experienced. These two months do not clearly witness the increase or decrease in the durations of day or night. During the months Tula (Libra) and Mesa (Aries), a day consists of 12 muhurtas; and twelve muhurtas make a night. When the sun's position goes on changing towards the upper five or the lower five constellations, the durations of days and nights also vary.

For those in the northern hemisphere, duration of day time goes on increasing as the sun migrates northward from one rasi to the other, commencing from Vrsabha (Taurus); and the duration of nights increases as the sun goes southward, commencing from Vriscika (Scorpio).

Here we have the clearest indication to say that our ancients knew our part of the earth to be a globe; and that was unlike the Greeks of Ptolemy's time who considered it to be flat and round like a rupee coin.

However they were not entirely wrong because the wide earth up to the sweet-water ocean '*Suddhodaka Samudra*' a stretch of 5 crore yojanas, is certainly round like a rupee coin although not fully flat. The Greeks and the other learned men of the Middle East had forgotten all about this wide invisible expanse of the earth and they mistook our part of the globular earth to be flat. Why for that matter, even most of the oriental panditas of our own times have such wrong notions. They must revise their opinions by learning to read between lines. How this should be done? It will be illustrated with two examples.

1. Jambūdvīpa should be considered as being basket like in shape because, it is said that the sun, the moon and the stars do not shed their lights in the Ilavrta Varṣa, the middle most region.

2. The moon's chariot should be located not too far from our globe as in the case of the sun just because it is said that its course is one lakh yojanas above that of sun. Śrīmad-Bhāgavatam clearly states that the moon's chariot is above the rays of the sun, not above the sun (*arka gabhastibhyah uparistat*, meaning, above the rays of the sun.)

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[134] Refer to the section "Ghatika as Muhurta (277 - 288)"

The Solar and Lunar Eclipses

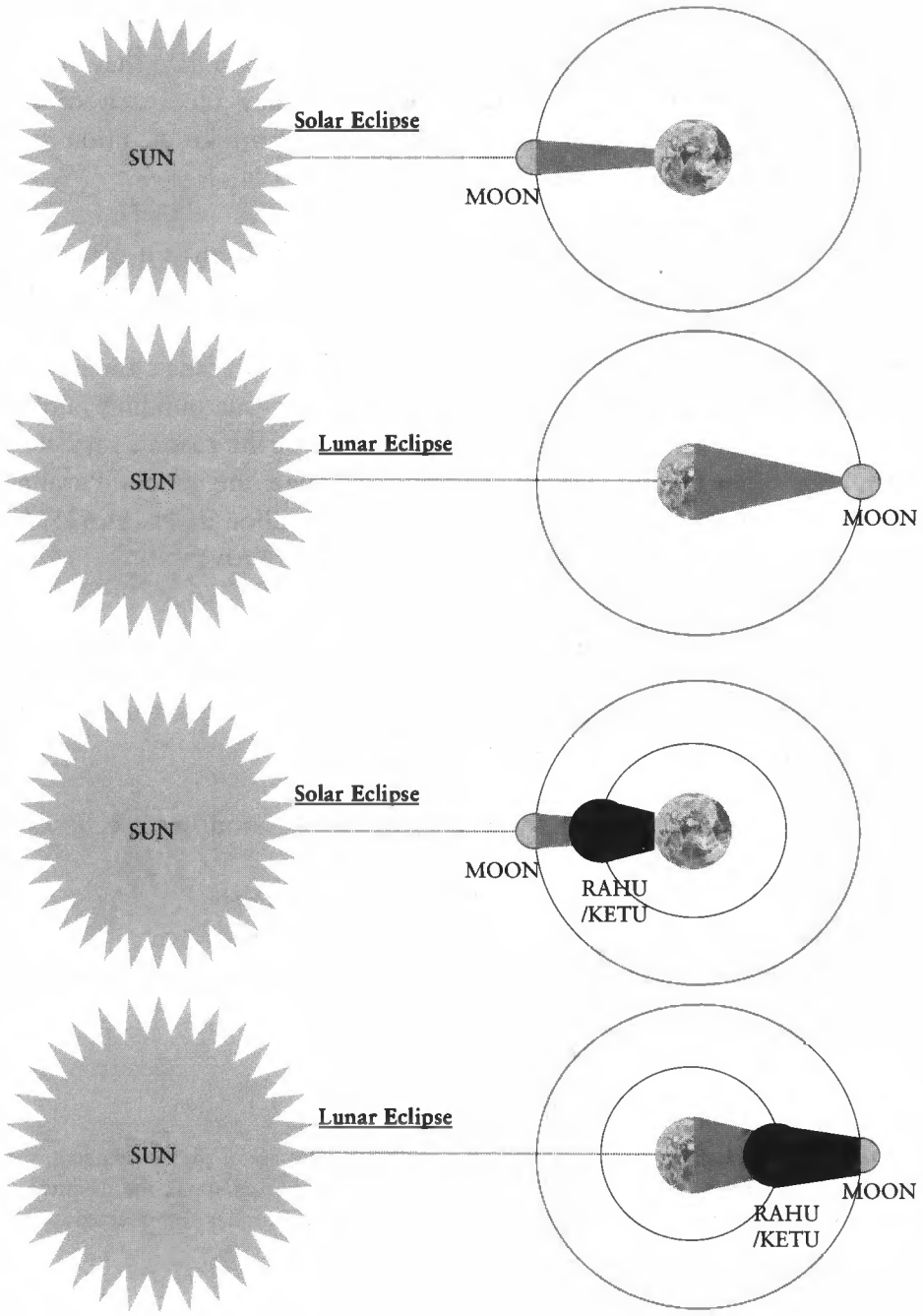
The modern science says that, "it just happens that the moon and the sun appear the same size in the sky as viewed from the earth. And since the moon orbits the earth in approximately the same plane as the earth's orbit around the sun sometimes the moon comes directly between the earth and the sun. This is called a solar eclipse; if the alignment is slightly imperfect then the moon covers only part of the sun's disk and the event is called a partial eclipse. When it lines up perfectly the entire solar disk is blocked and it is called a total eclipse of the sun. Partial eclipses are visible over a wide area of the earth but the region from which a total eclipse is visible, called the path of totality, is very narrow, just a few kilometers (though it is usually thousands of kilometers long). Eclipses of the sun happen once or twice a year. For a few precious minutes, it gets dark in the middle of the day. The stars come out. The animals and birds think it is time to sleep. And you can see the solar corona."

However, the Vedic viewpoint about the eclipse is that, "Formerly, when nectar was being distributed, Rahu tried to create dissension between the sun and moon by interposing himself between them. Thus, Rahu is inimical toward both the sun and the moon, and therefore he always tries to cover the sunshine and moonshine on the dark-moon day and full-moon night. This is called an Eclipse. Obstructing the sun and moon previously by coming between them to drink nectar, he created enmity with them when they understood what he was doing. Thus, on the full moon and dark moon days he creates an obstacle to the full moon and bright sun (dark moon) for those looking from the earth. He attacks them on the full and dark moon days (*rāhor yaḥ parvaṇi tad-vyavadhāna-kṛd vairānubandhaḥ sūryā-candramasāv abhidhāvati*)."

The speakers of the Purāṇas say that 10,000 *yojanas* below the sun is the planet known as Rāhu, which moves like one of the constellations (*adhastāt savitur yojanāyute svarbhānur nakṣatravac*). The presiding deity of that planet, who is the son of Siṁhikā, the most abominable of all *asuras,* has attained the position of a *devatā* by the mercy of the Lord, though he is not qualified to be a *devatā.*

Hearing of Rāhu's attack, the Lord engages His dear Sudarśana weapon to protect the sun and the moon. Seeing the intolerable heat of the weapon rotating in front of him for forty-eight minutes (*muhūrtam udvijamānaś*),

Rāhu, frightened and worried, flees to a distant place. People call this situation as an eclipse (*uparāgam iti vadanti lokāḥ*). Because of either precise or imprecise alignment, there are full and partial eclipses. Some say Rāhu does not actually devour the sun or moon, since he is situated 10,000 *yojanas* away from the sun."



Notes from Bugola Varnanam on Solar and Lunar Eclipses[135]

The sun's route lies on the wheel of the Kala Cakra, one lakh yojanas above the earth. He is there to bless the world with his bright auspicious rays. On another wheel, one lakh yojanas still higher, the moon moves upon its own path. On the occasions of eclipses, the sun and the moon face each other, moving as they do on their own paths. On the new-moon day, they are towards the same constellation; and on full-moon day, they face each other being exactly towards opposite constellations. If Rahu also takes position at similar longitudes, then an eclipse takes place.

Rahu is a black planet below the level of the earth. Ketu is a planet of dark-brown in color. Their path is 10,000 yojanas lower than the path of the sun; not exactly down below the sun. Similarly, the moon is one lakh yojanas higher; but not right above the sun. According to the Puranas, the moon's path lies above the 'Lavana Samudra or salt water Ocean' surrounding Jambudvlpa. Since Rahu and Ketu pester the moon by cutting off the moon's rays coming to earth, their planet should be nearer to earth than the moon. Puranas have located the sun and the moon at distances of 93 million miles (11.625MY) and 2.5 lack miles (31,250 yojanas) away from earth respectively.[136]

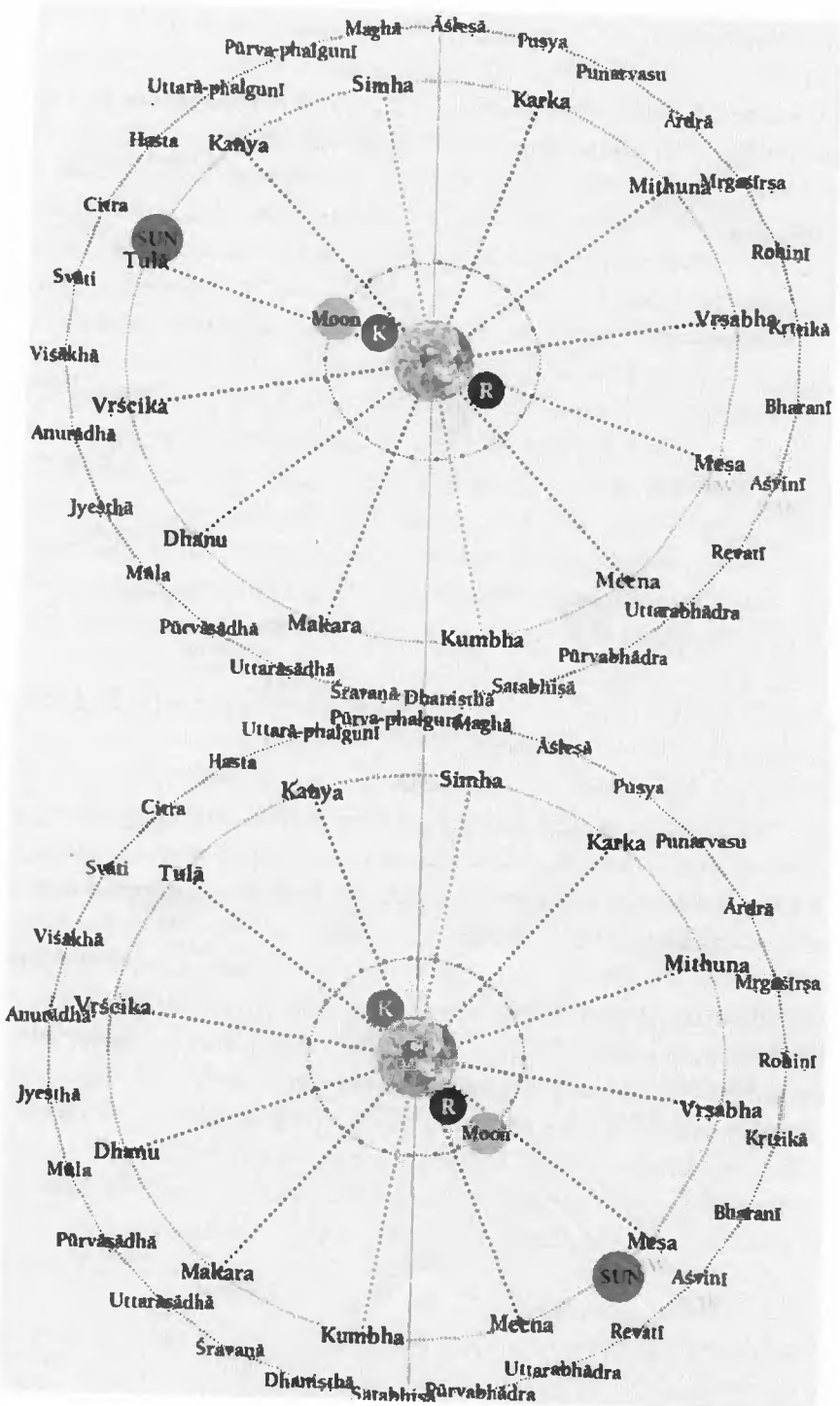
Coming to the occurrence of eclipses, we have to disagree entirely with the views of modern astronomers. Rahu and Ketu are the slowest in motion taking about 1.5 years to cover each constellation by retrogression while moving also forward along with the Time Wheel.

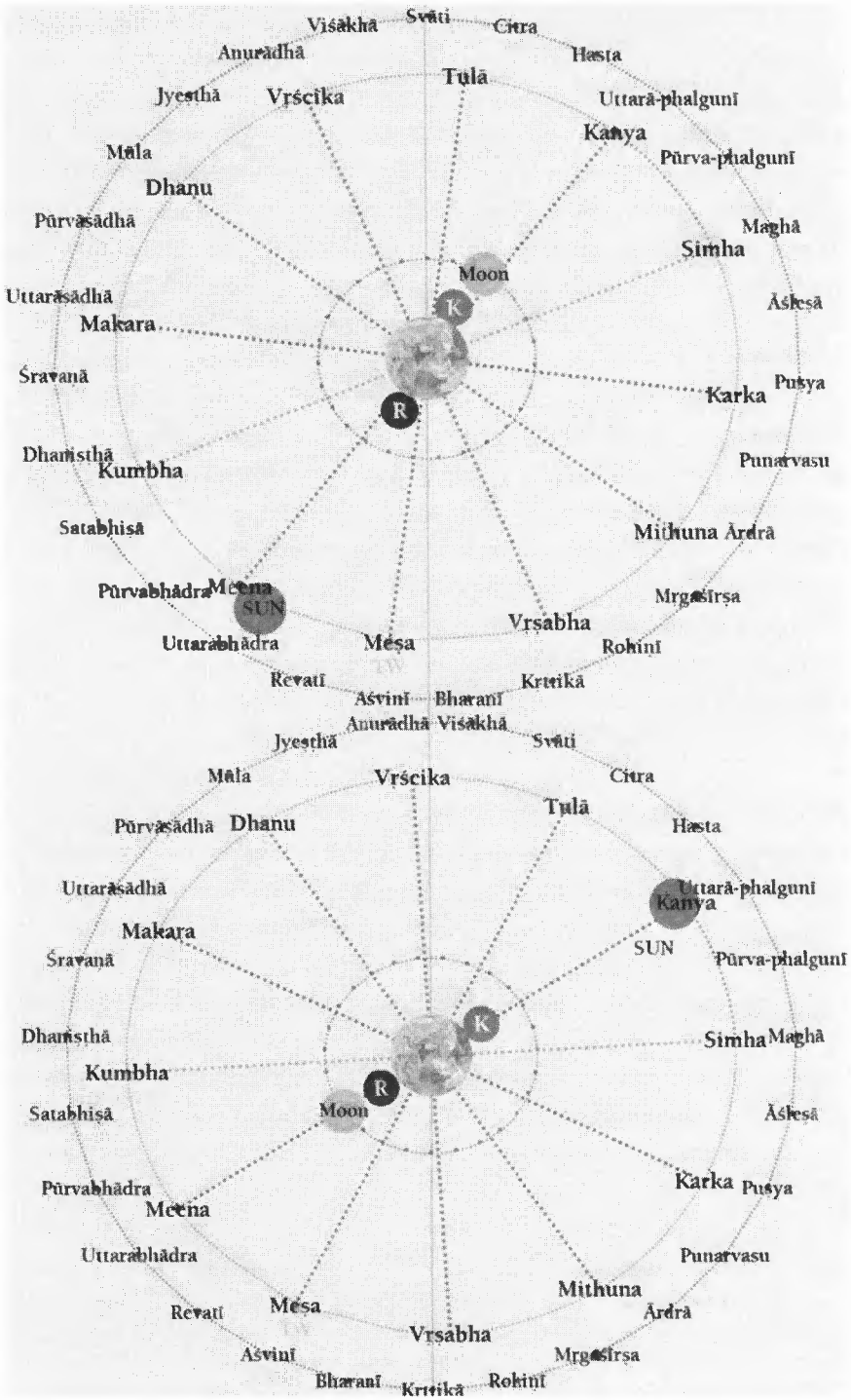
Below images show the occurances of solar eclipses and lunar eclipses on the indicated time intervals:
Solar eclipse: 177.5676 days & 355.1351 days respectively
Lunar eclipse: 340.3378 days & 517.9054 days respectively

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[135] Refer to section "Planets Rahu and Ketu (309 – 320)

[136] Although all scientific guesses may not be correct as in the case of stellar distances, we should attempt to harmonise rational findings with puranic faces. For example, the distance between earth and sun is said to be 950 lakh miles (approx) both in the puranas and in geography texts.


Maghā
Āśleṣā
Pūrva-phalgunī
Puṣya
Punarvasu
Uttarā-phalgunī
Simha
Karka
Ārdrā
Hasta
Kanya
Mithuna
Mrgaśīrṣa
Citra
SUN
Tulā
Rohiṇī
Svāti
Moon
K
Vṛṣabha
Kṛtikā
Viśākhā
R
Vṛścika
Bharaṇī
Anurādhā
Meṣa
Aśvinī
Jyeṣṭhā
Dhanu
Revatī
Mūla
Meena
Uttarabhādra
Pūrvāṣādhā
Makara
Kumbha
Pūrvabhādra
Uttarāṣādhā
Śravaṇā
Dhaniṣṭhā
Śatabhiṣā
Pūrva-phalgunī
Maghā
Uttarā-phalgunī
Āśleṣā
Hasta
Kanya
Simha
Puṣya
Citra
Karka
Punarvasu
Svāti
Tulā
Ārdrā
Viśākhā
Mithuna
Mrgaśīrṣa
Anurādhā
Vṛścika
K
Rohiṇī
Jyeṣṭhā
R
Vṛṣabha
Moon
Kṛtikā
Dhanu
Mūla
Bharaṇī
Pūrvāṣādhā
Meṣa
Makara
SUN
Aśvinī
Uttarāṣādhā
Kumbha
Meena
Revatī
Śravaṇā
Uttarabhādra
Dhaniṣṭhā
Śatabhiṣā
Pūrvabhādra


Viśākhā
Svāti
Citra
Anurādhā
Hasta
Tulā
Jyesthā
Vṛścika
Uttarā-phalgunī
Kanya
Mūla
Pūrva-phalgunī
Dhanu
Pūrvāsādhā
Maghā
Simha
Moon
K
Uttarāsādhā
Āśleṣā
Makara
Śravaṇā
Karka
Puṣya
R
Dhaniṣṭhā
Punarvasu
Kumbha
Satabhiṣā
Mithuna
Ārdrā
Pūrvabhādra
Meena
SUN
Mṛgaśīrṣa
Vṛṣabha
Uttarabhādra
Meṣa
Rohiṇī
Revatī
Kṛttikā
Aśvinī
Bharaṇī
Anurādhā
Viśākhā
Jyesthā
Svāti
Mūla
Vṛścika
Citra
Tulā
Pūrvāsādhā
Dhanu
Hasta
Uttarāsādhā
Uttarā-phalgunī
Kanya
Makara
SUN
Śravaṇā
Pūrva-phalgunī
K
Dhaniṣṭhā
Simha
Maghā
Kumbha
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Satabhiṣā
Moon
Āśleṣā
Karka
Pūrvabhādra
Puṣya
Meena
Uttarabhādra
Punarvasu
Mithuna
Revatī
Meṣa
Ārdrā
Vṛṣabha
Aśvinī
Mṛgaśīrṣa
Bharaṇī
Kṛttikā
Rohiṇī

Therefore, they often meet the sun or the moon at the same longitudes. They are not merely nodes or crossing points of the paths of the sun and the moon. The planet of Rahu and Ketu is actually there. Modern astronomers have not turned their telescopes towards that planet; or perhaps, it is not recognizable, being dark in color. To say that the earth covers the moon or that the moon covers the sun during eclipses is but a convenient guess. The three are never in the same straight line. If this argument is put forth, the scientists readily take shelter under parallax effect. That is no excuse. The fact is different. The planets being at different altitudes, there is no question of their coming together in a straight line. Rahu, when at the same longitudes, sends out a black jet and cuts off the rays of the sun or the moon as the case may be. This is known as *grasa* or swallowing. If it is the turn of Ketu, he sends up dark brown rays and consumes the rays of the sun or the moon. Anyone can verify this statement during lunar eclipses, finding black or dark brown shadow upon the moon's orb, black when it is the turn of Rahu and brown when it is the turn of Ketu. Actually, there will be no shadow upon the sun or the moon. To us it looks like that since the rays are cutoff from our view partially or fully.

Frequency of Eclipses per year

Both the sun and moon coincides on the same line, coming on one side on every new-moon day and by coming on opposite side on every full-moon night. Though both sun and moon comes on the same line once in every month, an eclipse is not seen every month. As described previously, because of the intervention of Rahu or Ketu, coming on the same line one of those days, they cause an eclipse or grahana, as is seen from earth. Generally, this happens ones in every six months, during which Rahu or Ketu coincides with both the sun and moon. A partial or full solar eclipse generally happens on a new-moon day after every six months (177 to 178 days approximately, 29.5x6), similarly a partial or full lunar eclipse generally happens on a full-moon night after every six months (177 to 178 days approximately, 29.5x6). Sometimes there can be two consecutive solar or lunar eclipses in a month. And there could be one solar and one lunar eclipse within one fortnight. Thus, in total maximum of five eclipses can happen in a year.

Between two and five solar eclipses, occur every year, with at least one eclipse per season. Since the Gregorian or western calendar was instituted in 1582,

years that have had five solar eclipses were 1693, 1758, 1805, 1823, 1870, and 1935. The next occurrence will be in 2206.

Lunar and Solar eclipse series sets from 2000–2023

Lunar Eclipse	Nature	Solar Eclipse	Nature	Lunar Eclipse	Nature	Solar Eclipse	Nature
21-Jan-00	Total	5-Feb-00	Partial	4 June 12	Partial	20-May 12	Annular
16-Jul-00	Total	1-Jul-00	Partial	28 Nov 12	Penumbral	13-Nove 12	Total
9-Jan-01	Total	31-Jul-00	Partial	25/26 Apr 13	Partial	10-May 13	Annular
5-Jul-01	Partial	25-Dec-00	Partial	24/25 May 13	Penumbral	3-Nov 13	Hybrid
30-Dec-01	Penumbral	21-Jun-01	Total	18/19 Oct 13	Penumbral	29-Apr-14	Annular
26-May-02	Penumbral	14-Dec-01	Annular	15-Apr-14	Total	23-Oct-14	Partial
24-Jun-02	Penumbral	10 / 11 Jun 02	Annular	8-Oct-14	Total	20-Mar-15	Total
19/20 Nov 02	Penumbral	4-Dec-02	Total	4-Apr-15	Partial	13-Sep-15	Partial
16-May-03	Total	31-May-03	Annular	28-Sep-15	Total	8 / 9 Mar 16	Total
8 / 9 Nov 03	Total	23/24 Nov 03	Total	23-Mar-16	Penumbral	1-Sep-16	Annular
4-May-04	Total	19-Apr-04	Partial	16-Sep-16	Penumbral	26-Feb-17	Annular
28-Oct-04	Total	14-Oct-04	Partial	10 / 11 Feb 17	Penumbral	21-Aug-17	Total
24-Apr-05	Penumbral	8-Apr-05	Total	7-Aug-17	Partial	15-Feb-18	Partial
17-Oct-05	Partial	3-Oct-05	Annular	31-Jan-18	Total	13-Jul-18	Partial
14/ 15 Mar06	Penumbral	29-Mar-06	Total	27-Jul-18	Total	11-Aug-18	Partial
7-Sep-06	Partial	22-Sep-06	Annular	21-Jan-19	Total	5 / 6 Jan 19	Partial
3 / 4 Mar 07	Total	19-Mar-07	Partial	16 / 17 Jul 19	Partial	2-Jul-19	Total
28-Aug-07	Total	11-Sep-07	Partial	10-Jan-20	Penumbral	26-Dec-19	Annular
21-Feb-08	Total	7-Feb-08	Annular	5-Jun-20	Penumbral	21-Jun-20	Annular
16-Aug-08	Partial	1-Aug-08	Total	5-Jul-20	Penumbral	14-Dec-20	Total
9-Feb-09	Penumbral	26-Jan-09	Annular	30-Nov-20	Penumbral	10-Jun-21	Annular
7-Jul-09	Penumbral	21 / 22 Jul 09	Total	26-May-21	Partial	4-Dec-21	Total
5 / 6 Aug 09	Penumbral	15 January 10	Total	19-Nov-21	Partial	30-Apr-22	Partial
31-Dec-09	Partial	11 July 10	Annular	16-May-22	Total	25-Oct-22	Partial
26-Jan 10	Partial	4 January 11	Total	8-Nov-22	Total	20-Apr-23	Total
20/21 Dec 10	Total	1 June, 2011	Partial	5-May-23	Penumbral	14-Oct-23	Annular
15/ 16 June 11	Total	1 July 11	Partial	28-Oct-23	Partial		
10 / 11 Dec 11	Total	25-Nov 11	Partial				

6

Reconciliation

Comparative Study between Modern Science and Vedic Science

Transition from Geocentric Theory to Heliocentric Theory

This section describes about the changing view of the universe. How the human conception of the universe has changed fundamentally over the past twenty-five hundred years. Let's begin with a simple observation. If you were to go outside tonight and look at the nights sky, you would observe stars and planets and the moon, and during the day the sun as well moving together across the sky. This motion is known as apparent daily-motion. Apparent daily-motion is simply when celestial objects including stars planets the moon and the sun appear to move across the sky each day. This is nothing new. People have been observing this for thousands of years. It wasn't until about 2,500 years ago when astronomers, philosophers and mathematicians began to formulate a model for the universe based on this observation of apparent daily-motion.

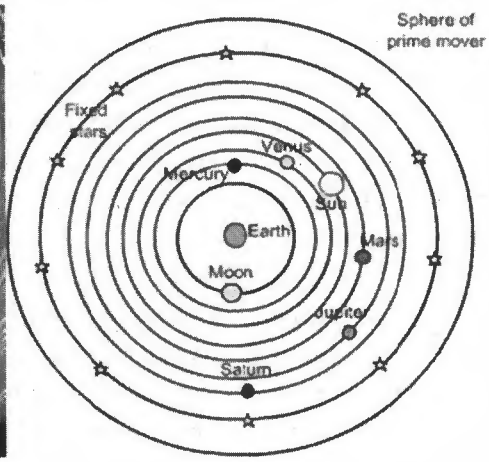
Plato (423-347 BC) & Aristotle (384-322 BC)

Plato was a Greek philosopher, mathematician and one of the first astronomers, who lived between 423 and 347 BC. Another Greek philosopher Aristotle and then a couple hundred years later Greek astronomer Claudius Ptolemy followed him. Together these three astronomers were able to formulate a fairly good functioning model of the universe.

The ancient Greeks observed the sky and noticed that the moon, sun, and stars seemed to move in a circle around the earth. It seemed that the earth was not moving and everything in the heavens revolved around the earth. As it turned out, it was very difficult to prove that the planets did not revolve around the earth without leaving the planet. Ancient Greeks such as Aristotle believed that the universe was perfect and finite, with the earth at the exact center. This is the geocentric theory, which stated, the planets, moon, sun, and stars revolve around the earth.

One interesting thing about this model in the center of everything is the earth. As a result, this became known as a Geo-centric or earth-centered model, or a model of the universe in which there is a stationary earth. The earth is sitting still in the center with other celestial objects orbiting it. This makes sense if you think of the observation of the apparent daily-motion. It certainly looks as if the earth is sitting still and everything else is moving around us and this was the basis for this model.



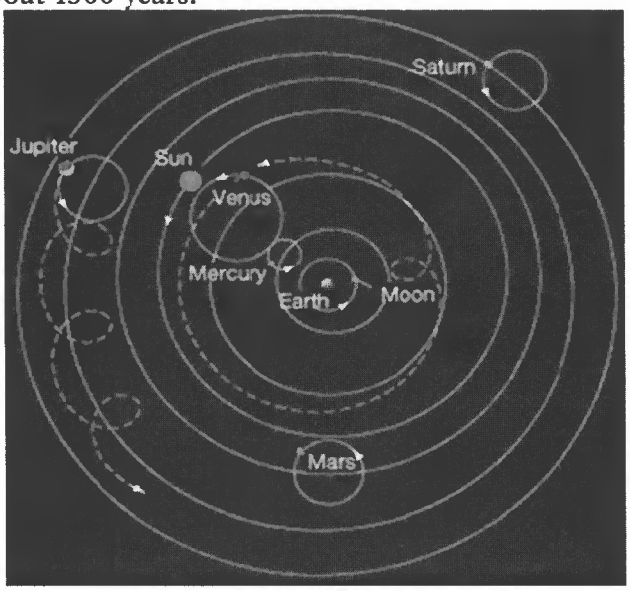


Claudius Ptolemy (85-165 AD)

In 140 AD, the Greek astronomer Claudius Ptolemy revised the geocentric model to explain all the planetary motions. His model had the planets move in little circles that also moved in bigger circles.

In the center, the earth is sitting still, not spinning or not moving at all. Around earth orbits the moon, followed by other planets Mercury and Venus. Then there is the sun orbiting around them. This was a necessary addition to the

model in order to explain their intricate motion. This includes the additional planets Mars Jupiter and Saturn but not Uranus and Neptune, which were not yet been discovered at that point of time. This model was largely accepted for more than a thousand years after Claudius Ptolemy formulated it. This belief persisted for about 1500 years.



Copernicus (1473-1543)

Everyone took for granted that the earth was sitting still in the center. Additionally they felt that the earth was a perfect sphere created by God and with no imperfections whatsoever. As time went on however, issues arose and approximately a thousand - 1200 years later came astronomer Nicolaus Copernicus. Copernicus was a Polish astronomer born in 1473 and who died in 1543. Ironically, Copernicus worked for the church and in his spare time, he made observations of the night sky and formulated a radical new view of the universe.

He created a simpler, more accurate model that could be used to predict future locations of planets. He also presented other claims that earth is spinning on axis and revolving around the sun. Copernicus was told by many scholars that he should make his new findings accessible to others by publishing it. In 1543 the book called "On the revolutions of the heavenly bodies" was released. Copernicus's book had a great impact that angered the Catholic and Protestant Church. The Church became so angry – the geocentric theory made human

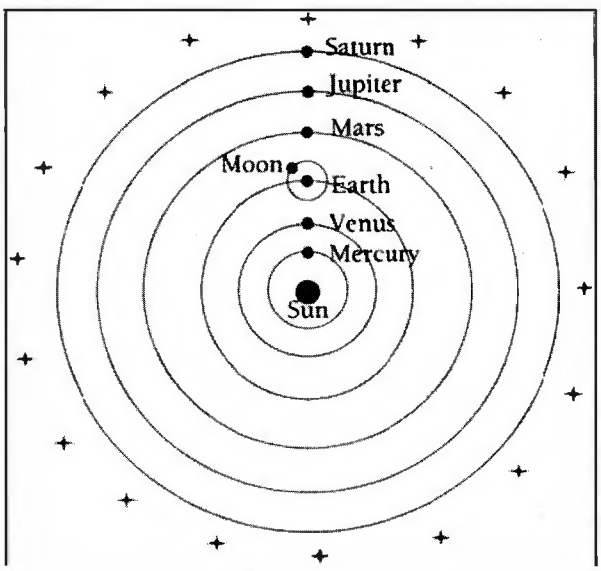
beings seem closer to God and since earth was in the center that meant humans were more special. The heliocentric theory changed that perspective completely, making humans lose that position in the universe.

Brahe (1546-1601 AD)

Later on Brahe presented that moon's orbit isn't a perfect circle, it's an ellipse through careful observations with diligent records. Brahe actually supported the Geocentric Theory.

Galileo Galilei (1564–1642)

Working in a similar time was Galileo Galilei (1564–1642, Italian physicist and astronomer), famous for his work with the telescope. Together Galileo and Copernicus formulated new radical model of the universe that went directly against the accepted geocentric model. Their model was a heliocentric model. In a heliocentric model the sun, not the earth, is in the center and all other objects including the earth orbit the sun. In this model, the earth is moving in two ways, it revolves around the sun and it also rotates on an axis around itself.



The heliocentric model proposal was quite controversial at the time, because it has suggested that the earth was not perfect at all and in fact, the earth was not special as it was orbiting another object just like the other planets did. It could not be ignored as it perfectly explains the observations that were seen.

His claims were, "our moon has craters and the sun has spots (thus they're not perfect!); Jupiter has its own moons and they don't orbit the earth. He made his observations through his carefully crafted telescope. Galileo first saw evidence that objects revolved around something besides the earth when he found Jupiter's moons. Galileo also saw that Mercury and Venus went through phases, just like the moon. The outer planets Mars, Jupiter and Saturn did not. Galileo supported Copernicus' theory, which clashed with the religious views of the time.

Galileo published his discoveries and support for the Copernican model in two books published in 1616 and 1632. Galileo was unusual for the time because he wrote in Italian rather than Latin like most scholars. He also took great pains to make his books interesting often writing them in the form of dialogues rather than dry, boring dissertations. After his first book in 1616, "*Starry Messenger*", was published he was warned by the Church not to publicly support Copernicism again.

Galileo abided by this edict until 1632 when he published "*A Dialogue on the Two Chief World Systems*". This book's outright support for the Copernican model and its ridiculing of the Ptolemaic model earned Galileo a trial before the Inquisition. Galileo was accused of heresy and sentenced to house arrest for life. However, he got off easily compared to fellow Italian Giordano Bruno who was burned at the stake in 1600 for teaching Copernican ideas.

Though he underwent many trials and tribulations, and even sentenced to house arrest for his remaining years. His view has withstood the test of time. Today we talk about our Solar system, not our earth system

When both the geocentric model and heliocentric models compared next to each other, the geocentric model has the earth sitting still in the center with the planets making complex movements around it. However, the heliocentric model is much neater with the sun in the center and the planets orbiting simply around it. The heliocentric model eventually accepted because it better predicted our observations. We could finally for the first time predict exactly where a planet would be on a given day and given time.

Johannes Kepler (1571-1630)

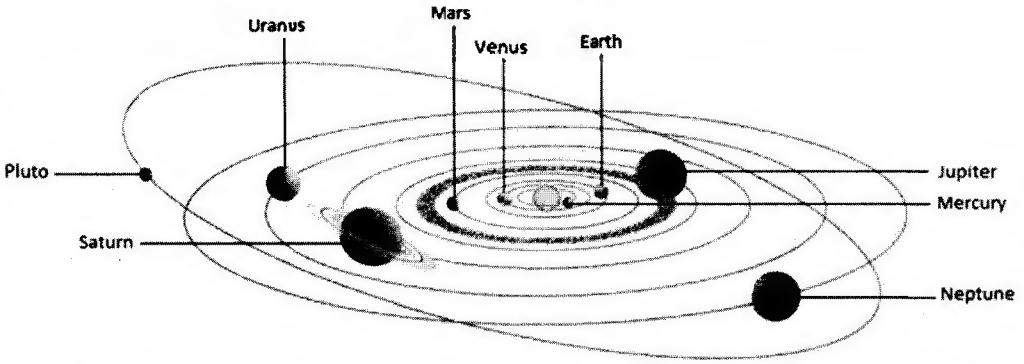
Then Johannes Kepler, a German mathematician, who lived from 1571-1630, updated this model. He presented that planets do not orbit in perfect circles but instead orbit in *ellipses*. To present this claim he used records collected by

Brahe. This discovery was the last piece needed to perfect Copernicus's model (which was not widely accepted).
Soon after Kepler did his work, Sir Isaac Newton came along, an English physicist who lived in the late sixteen hundreds. Kepler and Newton both tried to solve the problems that remained with the heliocentric model. Kepler's observations were very important. He was the first astronomer to look at the shape of planets orbits as they were going around the sun. To this day astronomers and the public felt that planets moved in the circular paths, but Kepler made a rash prediction which was that planets didn't move along perfectly circular path but rather long oval paths. These oval paths are known as ellipses. In addition, Kepler noted that when an object orbits a star, its speed changes – when it is far away it slows down and when it is closer it speeds up.

Sir Isaac Newton (1642-1727)
These observations would be further explained by Isaac's observations of universal Gravitation. Ultimately, the conclusion is in fact nothing is perfectly circular additionally it was noted that objects would speed up when closer to what they are orbiting due to the pull of gravity and so speeds change.
Everyone's conception of the universe was changing with each observation, which leads us to what we have today a heliocentric model of the solar system in which the sun is in the center, and the planets all orbit along elliptical paths speeding up and slowing down as they orbit the sun. One more thing to note, that the early astronomers had no clue how that's the universe truly was and so when they formulated a model of the universe, what they were really talking about was simply a model of our solar system. Of course, today we know that this is just a tiny piece of the entire universe. Regardless this heliocentric model has been proven true by modern technology and we owe this to all the hard work of the early astronomers.

An Overview of the Solar System

Orbits: The solar system consists of the sun; the eight official planets, at least three "dwarf planets", more than 130 satellites of the planets, a large number of small bodies (the comets and asteroids), and the interplanetary medium. (There are probably also many more planetary satellites that have not yet been discovered.)



The inner solar system contains the Sun, Mercury, Venus, Earth and Mars. The main asteroid belt (not shown) lies between the orbits of Mars and Jupiter. The planets of the outer solar system are Jupiter, Saturn, Uranus, and Neptune (Pluto is now classified as a dwarf planet).

The orbits of the planets are ellipses with the sun at one focus, though all except Mercury are very nearly circular. The orbits of the planets are all more or less in the same plane (called the ecliptic and defined by the plane of the earth's orbit). The ecliptic is inclined only 7 degrees from the plane of the sun's equator. The above diagrams show the relative sizes of the orbits of the eight planets (plus Pluto) from a perspective somewhat above the ecliptic (hence their non-circular appearance). They all orbit in the same direction (counter-clockwise looking down from above the sun's North Pole); all but Venus, Uranus and Pluto also rotate in that same sense.

Classification of Planets

The eight bodies officially categorized as planets are often further classified in several ways.

By size

- Small planets: Mercury, Venus, Earth, Mars - with diameters less than 13000 km.
- Giant planets: Jupiter, Saturn, Uranus and Neptune - with diameters greater than 48000 km.
- The giant planets are sometimes also referred to as gas giants.

By position relative to the sun

- Inner planets: Mercury, Venus, Earth and Mars.
- Outer planets: Jupiter, Saturn, Uranus, Neptune.
- The asteroid belt between Mars and Jupiter forms the boundary between

the inner solar system and the outer solar system.

By composition

- Terrestrial or rocky planets: Mercury, Venus, Earth, and Mars - composed primarily of rock and metal and have relatively high densities, slow rotation, solid surfaces, no rings and few satellites.
- Jovian or Gas planets: Jupiter, Saturn, Uranus, and Neptune - composed primarily of hydrogen and helium and generally have low densities, rapid rotation, deep atmospheres, rings and many satellites.

Planets - Data Table

	Sun	Mercury	Venus	Earth	Moon	Mars	Jupiter	Saturn	Uranus	Neptune
diameter (Earth=1)	108.968	0.382	0.949	1	0.272	0.532	11.209	9.44	4.007	3.883
diameter (km)	1,390,000	4,878	12,104	12,756	3476	6,787	142,800	120,000	51,118	49,528
mass (Earth=1)*	3.33e5	0.055	0.815	1	0.00123	0.107	318	95	15	17
mean distance from Sun (AU)	0	0.39	0.72	1	384,400 km	1.52	5.20	9.54	19.18	30.06
orbital period (Earth years)	-	0.24	0.62	1	27d 7h 43.7m	1.88	11.86	29.46	84.01	164.8
Orbital speed (AU / year)	-	12.57	7.33	6.28	-	4.96	2.62	1.99	1.42	1.14
orbital eccentricity	-	0.2056	0.0068	0.0167	-	0.0934	0.0483	0.0560	0.0461	0.0097
mean orbital velocity (km/sec)	-	47.89	35.03	29.79	-	24.13	13.06	9.64	6.81	5.43
rotation period (in Earth days)	-	58.65	-243*	1	-	1.03	0.41	0.44	-0.72*	0.72
inclination of axis (degrees)	-	0.0	177.4	23.45	-	23.98	3.08	26.73	97.92	28.8
mean temperature at surface (C)	5800 K (15.6e6K @ core)	-180 to 430	465	-89 to 58	-	-82 to 0	-150	-170	-200	-210
gravity at equator (Earth=1)	-	0.38	0.9	1	-	0.38	2.64	0.93	0.89	1.12
escape velocity (km/sec)	-	4.25	10.36	11.18	-	5.02	59.54	35.49	21.29	23.71
mean density (water=1)	-	5.43	5.25	5.52	-	3.93	1.33	0.71	1.24	1.67
atmospheric composition	-	none	CO_2	N_2 +O_2	-	CO_2	H_2+He	H_2+He	H_2+He	H_2+He
number of moons	8 planets	0	0	1	-	2	63	62	27	13
rings?	-	no	no	no	-	no	Yes	yes	yes	Yes

*Earth's mass: 5.98E24 kg

* Negative values of rotation period indicate that the planet rotates in the direction opposite to that in which it orbits the sun. This is called retrograde rotation.

The eccentricity (e) is a number which measures how elliptical orbits are. If e = 0, the orbit is a circle. All the planets have eccentricities close to 0, so their orbits are nearly circular.

Planetary Years, Distances, and Diameters, according to Modern Western Astronomy

Planet	Length of year (days)	Mean Distance from Sun (AU/MY)		Diameter (miles)
Sun	-	0.0	0	865,110
Mercury	87.969	0.39	4.528875	3,100
Venus	224.701	0.72	8.361	7,560
Earth	365.257	1.00	11.6125	7,928
Mars	686.980	1.52	17.651	4,191
Jupiter	4,332.587	5.20	60.385	86,850
Saturn	10,759.202	9.55	110.8994	72,000
Uranus	30,685.206	19.2	222.96	30,000
Neptune	60,189.522	30.1	349.5363	28,000
Pluto	90,465.38	39.5	458.6938	?
Moon	27.32166 days	0.002572	29,863.6Y	2,160

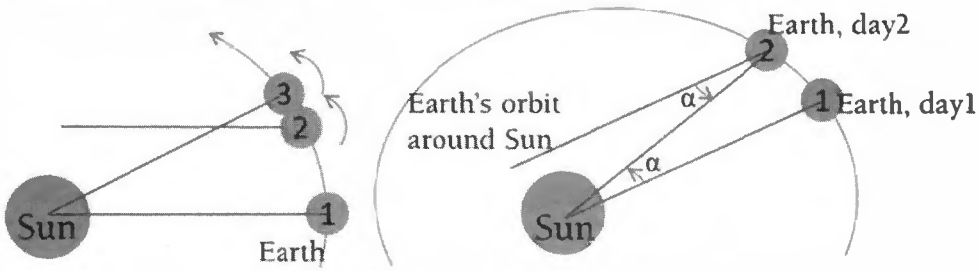
*1 AU is equal to 92.9 million miles = 11.6125 MY

The Rotation of the Earth

It is the rotation of the earth around its own axis. The earth rotates from the west towards east. As viewed from North Star or polestar, the earth turns counter-clockwise. The North Pole, also known as the Geographic North Pole or Terrestrial North Pole, is the point in the Northern Hemisphere where the earth's axis of rotation meets its surface. This point is distinct from the earth's North Magnetic Pole. The South Pole is the other point where the earth's axis of rotation intersects its surface, in Antarctica.

The earth rotates once in about 24 hours with respect to the sun and once every 23 hours, 56 minutes and 4 seconds with respect to the stars. It takes 23 hours, 56 minutes and 4 seconds for the earth to make one complete rotation on its axis, if we watch earth spin from out in space. Because earth is moving around the sun at the same time that it is rotating, earth has to turn

just a little bit more to reach the same place relative to the sun, so we experience each day on earth as 24 hours. At the equator, the earth rotates at a speed of about 1,700 kilometers per hour. Thankfully, we do not notice this movement, because it would certainly make us dizzy.



Angular speed

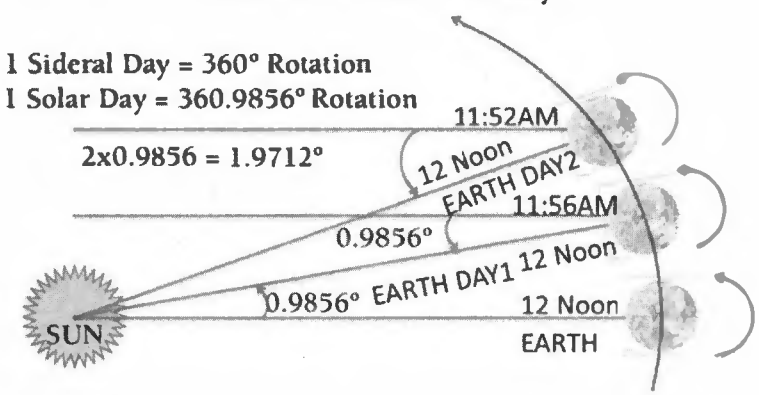
The angular speed of earth's rotation in inertial space is

(7.2921150 ± 0.0000001) $\times 10^{-5}$ radians per second (mean solar second).

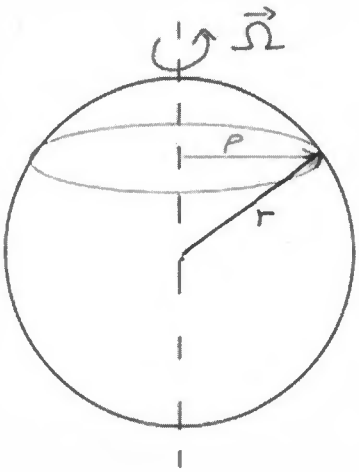
The angular speed of earth's rotation per day

= (7.2921150 ± 0.0000001) $\times 10^{-5}$ x (180°/π radians)×(86,400 seconds/mean solar day)

= 360.9856°/mean solar day, indicating that earth rotates more than 360° relative to the fixed stars in one solar day.



Earth's movement along its nearly circular orbit while it is rotating once around its axis requires that earth rotate slightly more than once relative to the fixed stars before the mean sun can pass overhead again, even though it rotates only once (360°) relative to the mean sun.

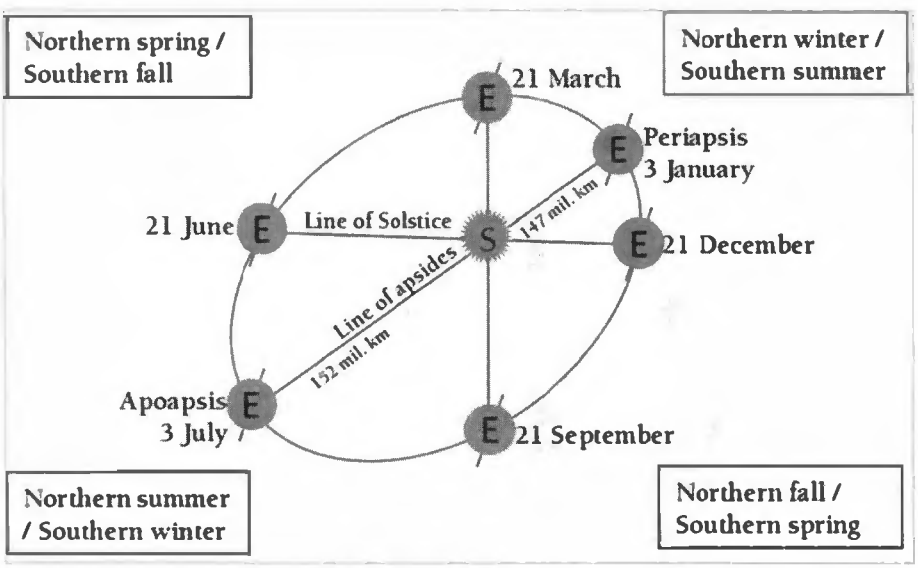


Earth's equatorial speed

Earth's equatorial circumferance: 2πx6,378,137m (radius)
= 40075.017 km/day
= 1,674.4 km/hour or 1,040.4 mile/hour
= 465.1 meter/ses

Some sources state that earth's equatorial speed is slightly less, or 1,669.8 km/h.
The tangential speed of earth's rotation at a point on earth can be approximated by multiplying the speed at the equator by the cosine of the latitude.

The Revolution of the Earth

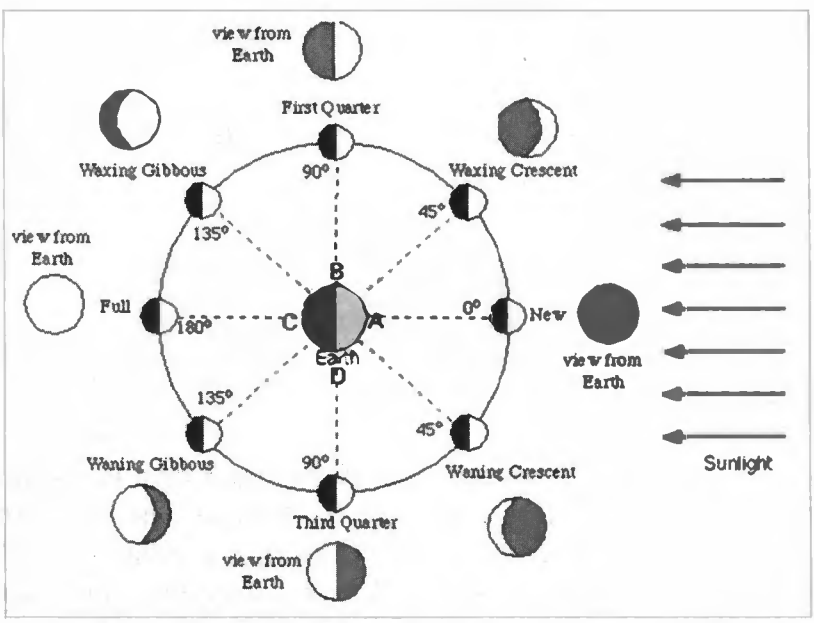


Earth's revolution around the sun takes much longer than its rotation on its axis. One complete revolution takes 365.26 days, or one year. The earth revolves around the sun because gravity keeps it in a roughly circular orbit around the sun. The earth's orbital path is not a perfect circle, but rather an ellipse, which means that it is like a slight oval in shape. This creates areas

where the earth is sometimes farther away from the sun than at other times. We are closer to the sun at perihelion (147 million kilometers) on about January 3rd and a little further from the sun (152 million kilometers) at aphelion on July 4th. Students sometimes think our elliptical orbit causes earth's seasons, but this is not the case. If it were, then the Northern Hemisphere would experience summer in January!

The Rotation of the Moon

The moon has a nearly circular orbit (e=0.05) which is tilted about 5° to the plane of the earth's orbit. Its average distance from the earth is 384,400 km. The combination of the moon's size and its distance from the earth causes the moon to appear the same size in the sky as the sun, which is one reason we can have total solar eclipses. It takes the moon 27.322 days to go around the earth once. Because of this motion, the moon appears to move about 13° against the stars each day, or about one-half degree per hour. By watching the moon over the course of several hours one night, one will notice that its position among the stars will change by a few degrees. The changing position of the moon with respect to the sun leads to lunar phases.



The lunar phase or phase of the moon is the shape of the illuminated

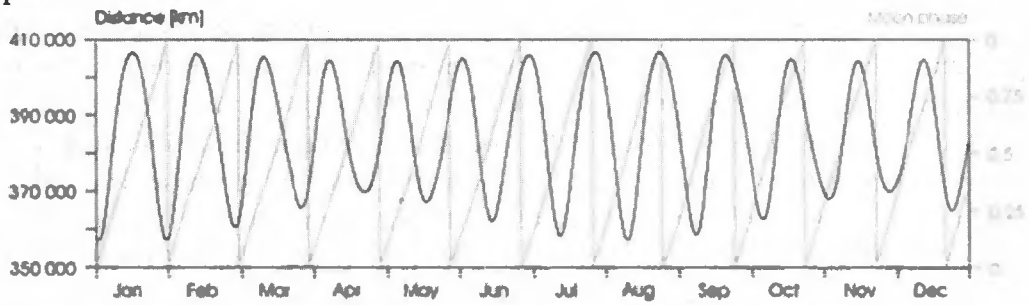
(sunlit) portion of the moon as seen by an observer on earth. The lunar phases change cyclically as the moon orbits the earth, according to the changing positions of the moon and sun relative to the earth. The moon's rotation is tidally locked by the earth's gravity, therefore the same lunar surface always faces earth. This face is variously sunlit depending on the position of the moon in its orbit. Therefore, the portion of this hemisphere that is visible to an observer on earth can vary from about 100% (full moon) to 0% (new moon). The lunar terminator is the boundary between the illuminated and darkened hemispheres. Each of the four "intermediate" lunar phases is roughly seven days (~7.4 days) but this varies slightly due to the elliptical shape of the moon's orbit. Aside from some craters near the lunar poles such as Shoemaker, all parts of the moon see around 14.77 days of sunlight, followed by 14.77 days of "night". The side of the moon facing away from the earth is sometimes called the "dark side or far-side".

When the sun and moon are aligned on the same side of the earth, the moon is "new", and the side of the moon facing earth is not illuminated by the sun. As the moon waxes (the amount of illuminated surface as seen from earth is increasing), the lunar phases progress through new moon, crescent moon, first-quarter moon, gibbous moon, and full moon. The moon is then said to wane as it passes through the gibbous moon, third-quarter moon, crescent moon and back to new moon.

Length of the lunar month

The length of each lunar cycle varies slightly from the average value. In addition, observations are subject to uncertainty and weather conditions. The average length of the synodic month is 29.530589 days. Thus, it is convenient if months are in general alternately 29 and 30 days. The average calendar month, which is $\frac{1}{12}$ of a tropical year, is about 30.44 days, while the moon's phase (synodic) cycle repeats on average every 29.53 days. Therefore, the timing of the moon's phases shifts by an average of almost one day for each successive month. Thus, in a Lunar year there are 354 days. The length of a sidereal month is 27.321662 days, with respect to the distant stars (13.36874634 passes per solar orbit = 365/27.32). And the length of Synodic month is 29.530589 days, with respect to the sun (phases of the moon,

12.36874634 passes per solar orbit = 365/29.53). The average length of a calendar month (a twelfth of a year) is about 30.4 days. This is not a lunar period, though the calendar month is historically related to the visible lunar phase.



The moon's distance from earth and moon phases in 2014. Moon phases: 0 or 1 – new moon, 0.25 - first quarter, 0.5 – full moon, 0.75 - last quarter

The Model presented in Śrīmad Bhāgavatam

Considering the details mentioned in both Śrīmad Bhāgavatam and Surya Siddhanta, we can arrive to an appreciable Geo-Observable model of the planetary system. The table below presents the brief descriptions given in Śrīmad Bhāgavatam about the annual movements of various planets around the Zodiac belt (Kala-chakra).

Sun **SB 5.21.3 & 7**	Situated in the middle of the antarīkṣa, the sun travels over all sides of Mānasottara Mountain in a circle whose length is 95.1 million yojanas every day. The time it takes for the sun to pass completely through the *antarikṣa* along with earth plane and heavenly planes, with slow, quick and medium speeds, is called *saṁvatsara* or a year.
Moon **SB 5.22.8**	The moon, moving with quicker speed, covers the sun's year in two fortnights, its month in two and half days and its fortnight in one day.
Constellations **SB 5.22.11**	Situated above moon, the constellations, twenty-eight in number with Abhijit, move clockwise around Meru on the wheel of time.
Venus **SB 5.22.12**	The Venus moves like the sun, with fast, slow and medium speed, sometimes behind and sometimes in front of the sun.

Mercury **SB 5.22.13**	Mercury, the son of moon, situated above Venus, moves in a similar fashion of that of Venus.
Mars **SB 5.22.14**	Situated above Mercury, Mars passes through one of the twelve zodiac sign every three fortnights, if not traveling retrograde.
Jupitor **SB 5.22.15**	Jupiter, situated above Mars, moves through each sign in one *prativatsara* (~360 days), if not traveling retrograde.
Saturn **SB 5.22.16**	Saturn, situated above Jupiter, travels slowly through one zodiac sign in thirty months. In this way, it travels through all twelve signs in thirty *anuvatsaras* (or ~30 years).
Sapta-ṛṣis **SB 5.22.17**	Situated above Saturn, the sages thinking of the welfare of the people rotate clockwise around Dhruvaloka.
Dhruvaloka **SB 5.23.1**	Situated above Sapta-ṛṣis, acts like pivot of all the planets, stars and constellations (luminaries).

The table below presents the brief descriptions given in Surya Siddhanta about the mean eastward revolutions of various planets around earth.

SS 1.29	In an age (*yuga*), the number of mean eastward revolutions of the sun, mercury (*Budha*) and Venus (*Śukra*) and the number of eastward revolutions of the node of conjunctions (*sīgra*) for Mars (*Mangala*), Jupitor (*Brihaspati*) and Saturn (*Śani*) is 4,320,000
SS 1.30	For the Moon, the number of revolutions is 57,753,336 and for the Mars is 2,296,832
SS 1.31	For Mercury, the number of eastward revolutions of the node of conjunction (*sīgrabhagana*) is 17,937,060 and for Jupiter the number of revolutions is 364,220
SS 1.32	For Venus, the number of revolutions of the node of conjunction is 7,022,376 and for Saturn the number of revolution is 146,568
SS 1.33	For the Moon, the number of revolutions of the node of apsis (*ucca*) is 488,203 and the number of revolutions of the node (*pāta*) on the eastward direction (left-side) is 232,238
SS 1.34	The number of revolutions (*bhagana*) for the stars is 1,582,237,828. If we subtract the number of revolutions of each planet from the number of revolutions of the stars, we get the number of risings of the respective planet in an age.

The number of mean eastward revolutions in a catur-yuga of various planets:

Planet	Eastward Revolutions in a Catur-yuga	Number of risings in a Catur-yuga	Number of Days to cover the Zodiac $(\frac{Number\ of\ risings\ of\ sun}{Eastward\ revolutions})$
Sun	4,320,000	1,577,917,828	365.258756 (1 year)
Mercury	4,320,000	1,577,917,828	365.258756 (1 year)
Venus	4,320,000	1,577,917,828	365.258756 (1 year)
Moon	57,753,336	1,524,484,492	27.321674 (days)
Mars	2,296,832	1,579,940,996	686.997494 (1.881 year)
Jupiter	364,220	1,581,873,608	4332.320652 (11.86 year)
Saturn	146,568	1,582,091,260	10765.773075 (29.474 year)
Rāhu / Pāta/Node	232,238	1,582,470,066	6794.399831 (18.602 year)
Stars	1,582,237,828	0	0

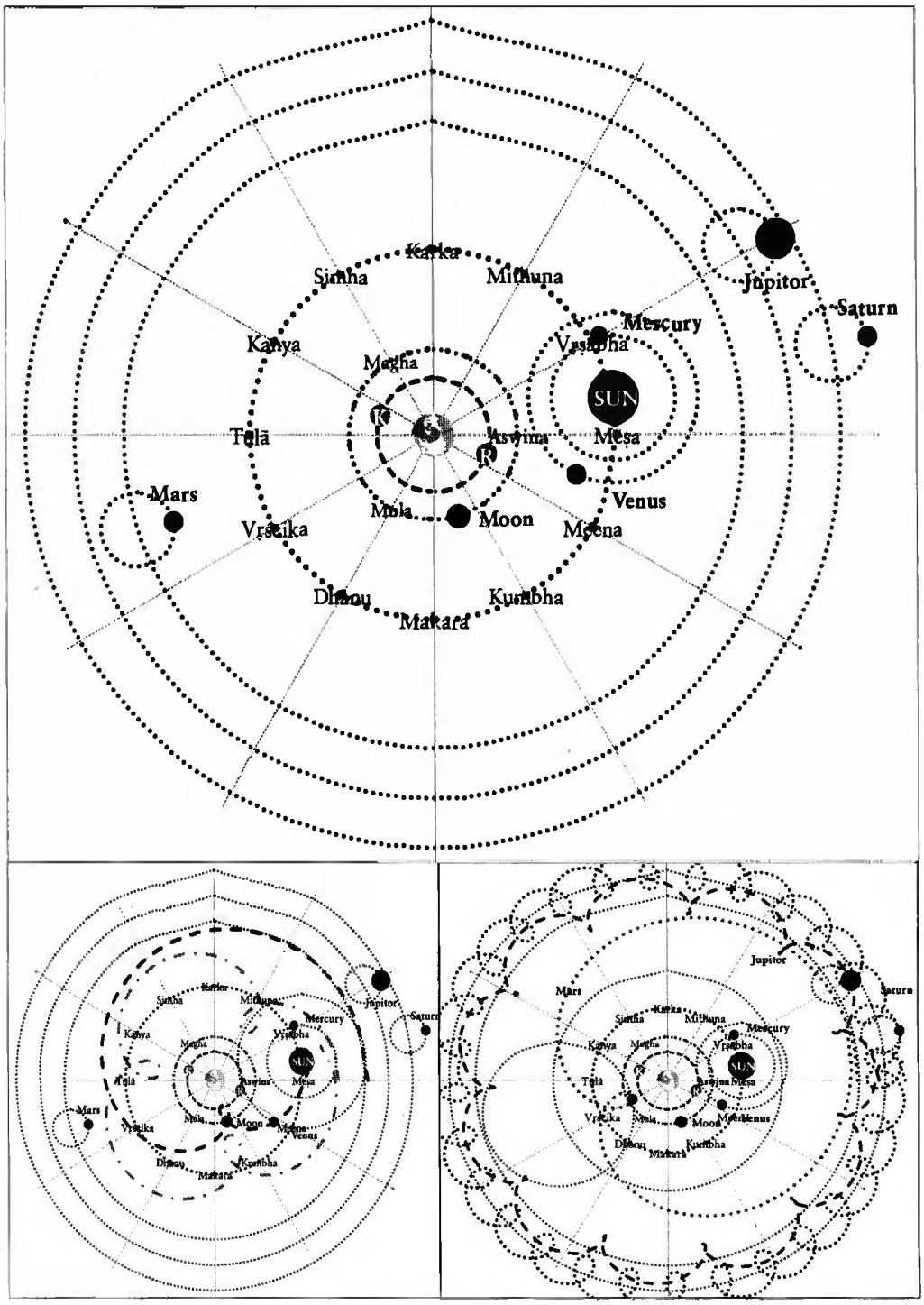
The number of eastward revolutions of the node of conjunctions (*sīgra*) in a catur-yuga:

Planet	Eastward Revolutions in a Catur-yuga	Number of risings in a Catur-yuga	Number of Days to cover the Sun / Node: $(\frac{Number\ of\ risings}{Eastward\ revolutions})$
Mercury	17,937,060	1,564,300,768	87.969702 (0.241 years)
Venus	7,022,376	1,575,215,452	224.69857 (0.615 years)
Mars	4,320,000	1,577,917,828	365.258756 (1 year)
Jupiter	4,320,000	1,577,917,828	365.258756 (1 year)
Saturn	4,320,000	1,577,917,828	365.258756 (1 year)
Ucca of Moon	488,203 (node of apsis)	1,581,749,625	3232.093674 (8.8488 year)

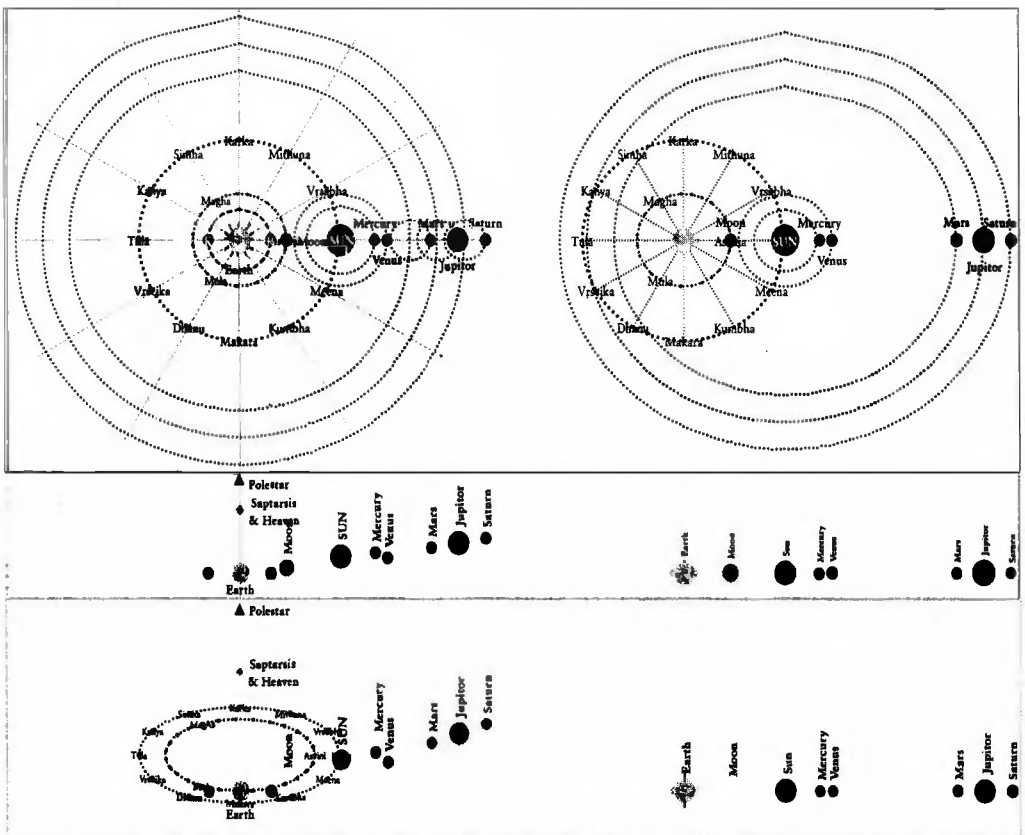
The comparison between the orbital duration of the planets as per the modern Solar system w.r.t the Vedic descriptions tabulated below

Planet	Solar System (w.r.t Sun)	Vedic Descriptions*	
		(w.r.t Zodiac)	(w.r.t Mean)
Sun	0	365.2587565days (1 year)	NA
Mercury	87.66 days (0.24 years)	365.2587565 days (1 year)	87.969702 (0.241 years)
Venus	226.459 days (0.62 years)	365.2587565 days (1 year)	224.69857 (0.615 years)
Earth	365.256 days (1 year) 365.2425 (Gregorian calendar)	0	0
Moon	27.321662 days (0.0748 years)	27.321674 (1 *idavatsara* / 1 month)	29.531 days (1 month)
Mars	687 days (1.88 years)	686.9975 days (1.88085 years)	365.2587565 days (1 year)
Jupiter	4,354 days (11.92 years)	4,332.32 days (11.86096 years)	365.2587565 days (1 year)
Saturn	10,822.5 days (29.63 years)	10,765.77 days (29.47437 years)	365.2587565 days (1 year)
Rahu & Ketu	6585.3211 days (18 years 11 days 8 hours)	6794.39983 days (18.6016 years)	NA

*Śrīmad Bhāgavatam describes that, just like the sun, both the Mercury and Venus moves through the Zodiac (or Ecliptic), sometimes going behind the sun and some other times going ahead of sun, completes the Zodiac in one year. According to Surya Siddhanta, both Mercury and Venus moves around sun in 88 and 225 days respectively. Hence, both Mercury and Venus revolve around sun with the orbital periods as indicated above (88 and 225 days respectively), and along with sun they cover the entire Zodiac plane in one year. Similarly, the outside planets (Mars, Jupiter and Saturn) cover the Zodiac plane in 1.88, 11.86 and 29.47 years respectively, while simultaneously moving in their respective epicycles with an orbital period of one year (equal to that of sun). Rahu and Ketu do not have any additional epicyclic motion, instead they simply move around the earth, in their own orbit, covering the entire Zodiac plane in 18.6 years following their left side motion. The orbits of the planets along with their epicycles presented below for better understanding.

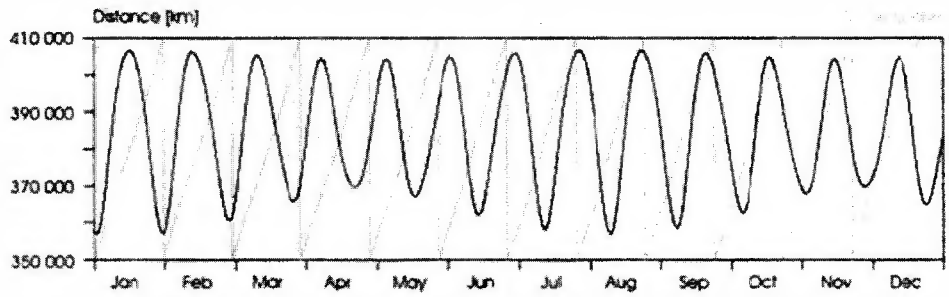

Karka
Simha
Mithuna
Jupitor
Saturn
Mercury
Kanya
Vrsabha
Magha
SUN
Tulā
Asvina
Mesa
Mars
Mula
Venus
Moon
Vrscika
Mena
Dhanu
Kumbha
Makara

Note: These orbital movements help us to understand the retrograde movements of the respective planets. In addition, these retrograde movements are similar to that are tabulated in Chapter 4, at the end of the section "The movements of the sun, moon and other planets." Thus, this explanation of Geo-Observable Model is as per the scriptural descriptions and its corresponding sun centric model is similar to that of Modern Heliocentric model. Both Vedic Geo-Observable and modern Heliocentric systems depicted below for comparison

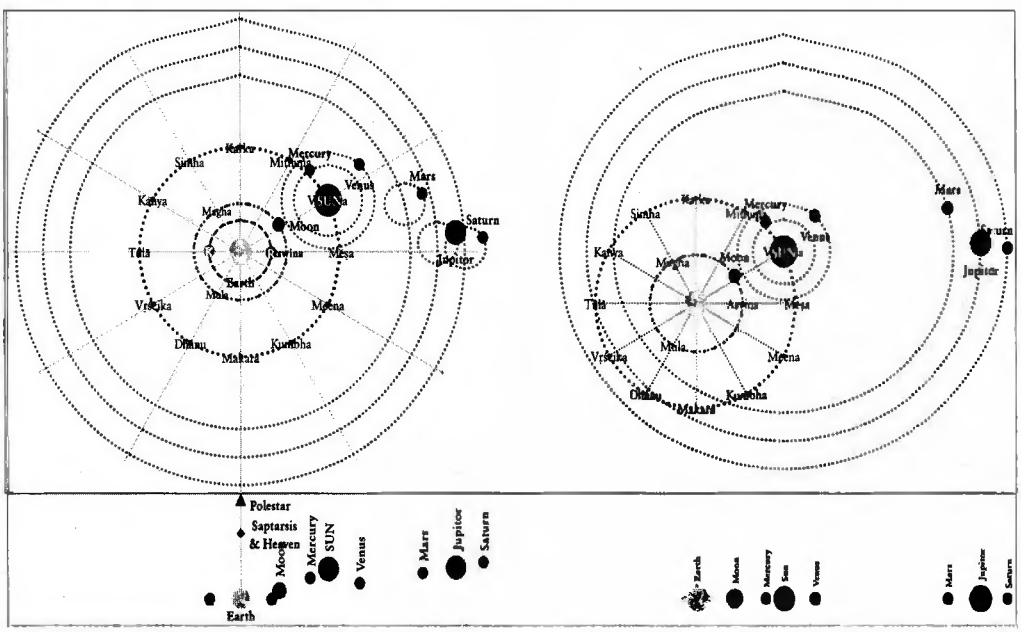


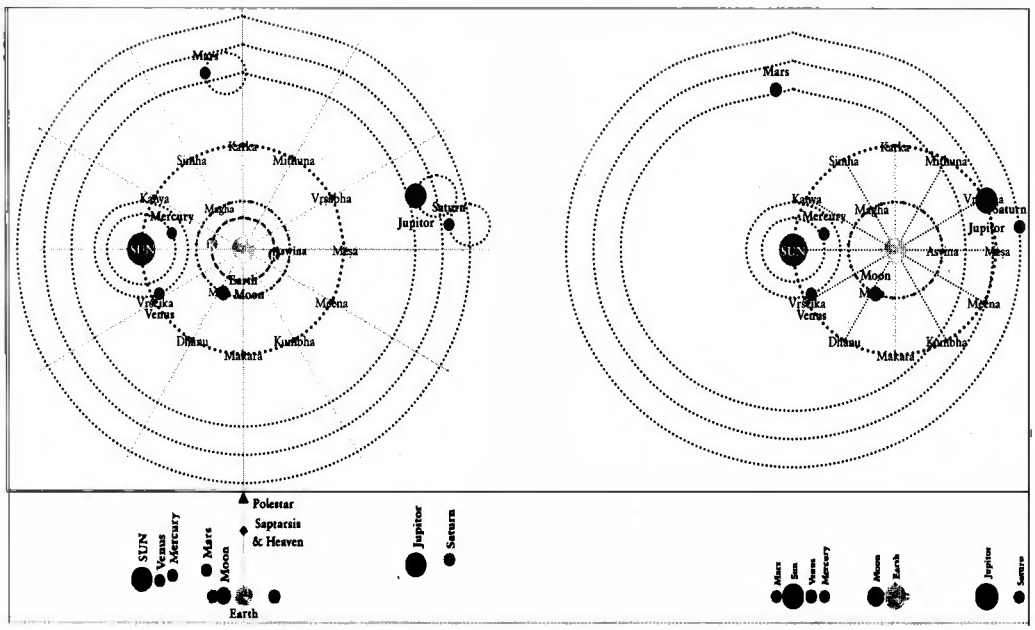
Note: As seen in the above two images, it is clear that the moon comes close to earth when it is passing through Makara rasi, gradually it goes away from earth and eventually reaches a far distant point while passing through Karka rasi. And then gradually comes close to earth while passing through subsequent rasis, until it reaches makara rasi. These points are not inline with fullmoon and new moon days. As it is explained in the previous section, that moon

covers 12 Zodiac signs and 27 stars in every 27.32167 days (sidereal month), while it reaches new moon day after every 29.531 days (Synodic month). Thus, there are 13.3688278 passes per solar orbit w.r.t stars (=365.25876/27.32167) and 12.3688278 passes per solar orbit w.r.t sun (=365.25876/29.530588). These values are same as the values mentioned in the section "Length of the lunar month" and its depiction shown below.



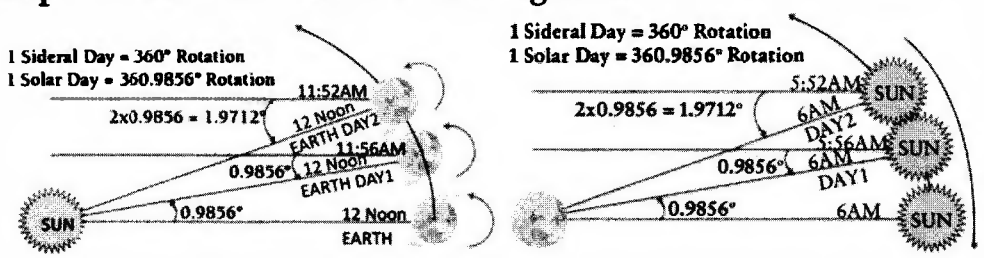
The moon's distance from earth and moon phases in 2014 (Moon phases: 0 or 1 – new moon, 0.25 - first quarter, 0.5 – full moon, 0.75 - last quarter)





Note: The above drawings are not to the scale. These images are drawn as such to show the principle of working of the Geo-Observable model with reference to the modern scientific heliocentric model.

Explanation from the Śrīmad Bhāgavatam



Earth: Daily and Yearly motion	Sun: Daily and Yearly motion
Counter clock wise movement around itself	Clock wise moevment around the earth
Counter clock wise movement around sun	Counter clock wise movement on the Time-wheel and around the earth

The orbital movements of the moon around earth, and the spinning and orbital

movement of the earth around sun of the Modern Science can be reconciled through the clockwise movements of sun and moon around earth and on the Time wheel, as it is explained in Śrīmad Bhāgavatam

The actual motion of Sun and Moon w.r.t Time-wheel: when the Time-wheel completes 366 rounds in clock wise direction, the sun completes 365 rounds and moon completes 352.667 rounds, making 13.333 reverse revolutions, indicating 12.333 months as per lunar calendar year. Thus, after the completion of every round the sun falls behind by 0.9836° w.r.t Kala-chakra, and the moon falls behind by 13.11475° respectively.

$$Sun\ daily\ lag = 360 - \frac{365\ rounds\ x\ 360°}{366\ rounds} = \frac{1\ round\ x\ 360°}{366\ rounds} = 0.9836°$$

$$Moon\ daily\ lag = 360 - \frac{352.667\ rounds\ x\ 360°}{366\ rounds} = \frac{13.333\ rounds\ x\ 360°}{366\ rounds} = 13.11475°$$

These calculations are based on the consideration of 365 days in a year, which is a close approximation to the actual value. Values that are more precise can be calculated by following the details mentioned in Surya Siddhanta. As per Surya Siddhanta, sun completes 365.2588 rounds and moon completes 352.8899 rounds (indicating 365.2588 - 352.8899 = 12.3688 lunar months in a calendar year), when Time wheel completes 366.2588 rounds.

With these actual values the daily lag of sun and moon will be 0.9856° ($\frac{360}{365.2588} = 0.9856°$) and 13.1404° ($\frac{13.3688*360}{366.2588} = 13.14038°$) respectively. With these calculated values, extra month appearance approximately after every 32nd or 33rd month approximately, instead of once in every three years.

$$Extra\ Month: \frac{1}{12.3688 - 12} = 2.7113 years = 990.3 days = \frac{990.3}{29.53} = 33.5355\ months$$

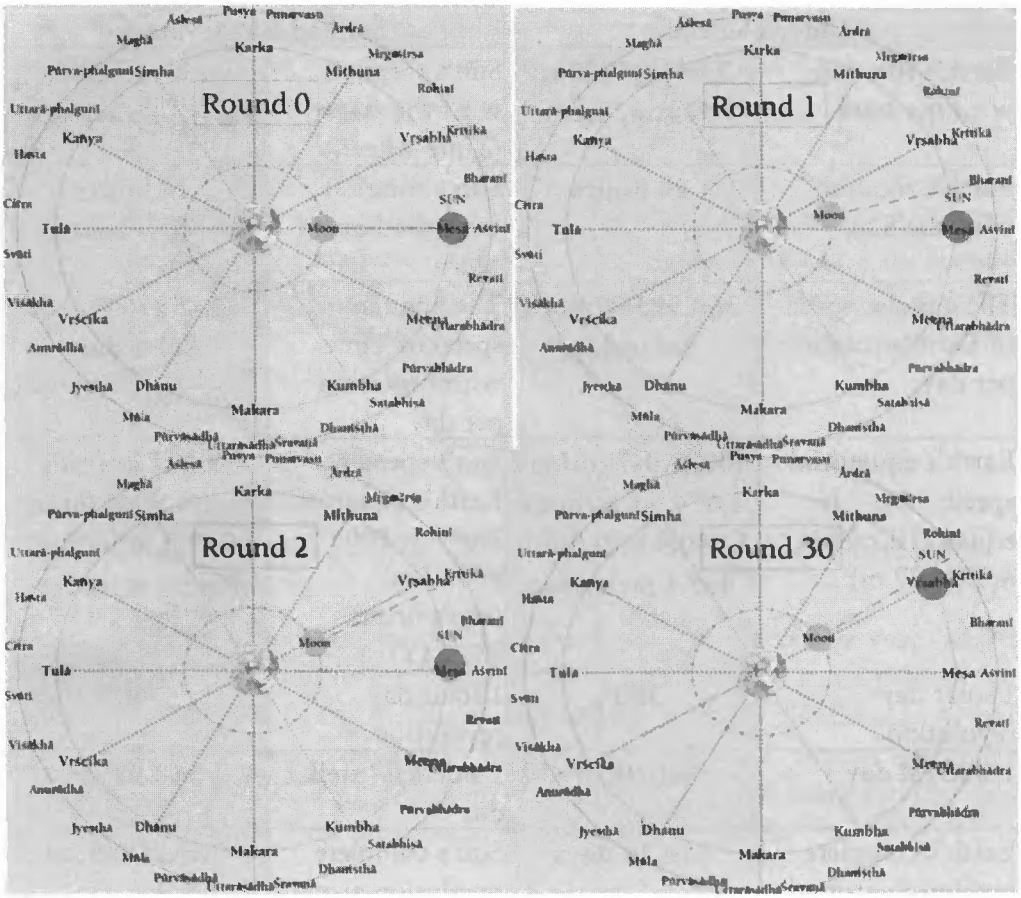
This is equal to seven extra months in nineteen (19) years:

= 32.5355*7 + 7 = 234.75 months; 234.75/12.3688 = 18.98 years

Considering the above calculated values, the comparison of various parameters w.r.t, modern observations tabulated below:

Modern Science		Śrīmad Bhāgavatam	
Earth's rotation w.r.t the stars	**23:56:04 (23hr, 56min, 4sec)**	Sun's rotation w.r.t the stars (Time-wheel)	**23:56:04** $\frac{365.2588}{366.2588} * 24 hours$
Earth's rotation w.r.t the Sun	**24 hours**	Sun's rotation w.r.t the Earth / Meru	**24 hours** $\frac{360-0}{360} * 24 hours$
The angular speed of Earth's rotation per day	**360.9856°/mean solar day**	The angular speed of Time wheel rotation per day	**360.9856°/mean solar day** $\frac{360 * 366.2588}{365.2588} = 360.9856$
Earth's equatorial speed: $2\pi r$ (r: equatorial radius, 6,378,137 m)	**40075.017km/day 1,674.4 km/hour 1,040.4 mile/hour 465.1 meter/sec**	Sun's speed at Earth's equator: $2\pi r$ (r: 500Y*, 6436 km) (sun's orbit: 95.1MY)	**40439 km/day 1,684.9 km/hour 1047.2 mile/hour 468.04 meter/sec** $(2 * \pi * 6436 = 40,438.6)$
1 solar day revolution	**360°**	1 solar day revolution	**360°**
1 sidereal day	**360.9856°**	1 sidereal / stellar day	**360.9856°**
Earth's complete revolution around Sun	**356.26 days**	Sun's complete revolution around Zodiac	**365.2588days**
Synodic month w.r.t the Sun (365.256 days / 13.3687 passes)	**29.530589 days**	Lunar Month (Moon's complete revolution w.r.t Sun)	**29.531 days** $\frac{360 * \left(\frac{365.2588}{366.2588}\right)}{13.1404 - 0.9829} = 29.531$
Sidereal month w.r.t the distant stars (365.256 days / 12.3687 passes)	**27.321662 days**	Moon's revolution around Zodiac (w.r.t stars)	**27.32167 days** $\frac{360 * \left(\frac{365.2588}{366.2588}\right)}{13.14038} = 27.3217$
Moon's daily rotation against the stars	**13° per day**	Moon's daily rotation against the stars	**13.176° per day** $\frac{13.3688 * 360}{365.2588} = 13.17635$

*1 yojana = 8 miles = 8 x 1.609 km = 12.872 km



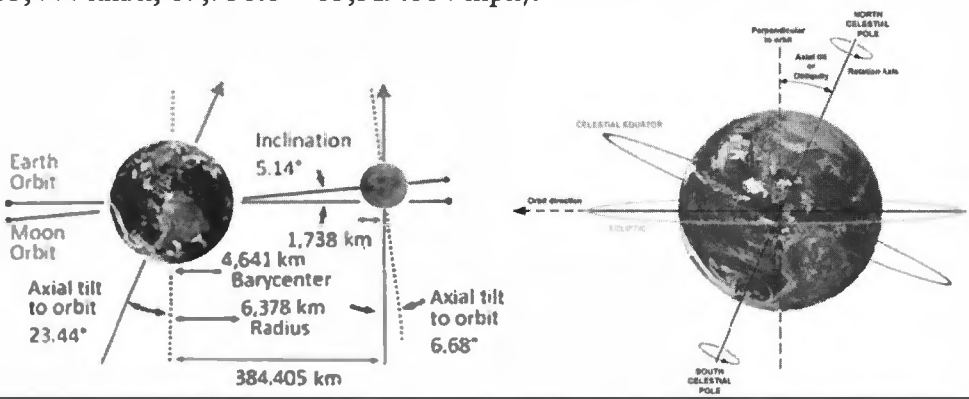
As per the Surya Siddhanta values, the length of a sidereal month is 27.32167 days, with respect to the distant stars (13.3688278 passes per solar orbit = 365.25876/27.32167). In addition, the length of Synodic month is 29.530588 days, with respect to the Sun (phases of the moon, 12.3688278 passes per solar orbit = 365.25876/29.530588). These values are same as the values mentioned in the section "Length of the lunar month."

The Orbit of the earth and the Axial Tilt

As Johannes Kepler stated in his *Astronomia Nova* (1609), the earth and Solar planets do not rotate about the sun in perfect circles. This is known as Kepler's First Law, which states that "the orbit of a planet about the sun is an ellipse with the sun's center of mass at one focus". At perihelion (i.e. its

closest) it is 147,095,000 km (91,401,000 mile) from the sun; whereas at aphelion, it is 152,100,000 km (94,500,000 mile).

This change in distance means that the earth's orbital speed increases when it is closest to the sun. While its speed averages out to about 29.8 km/s (18.5 mps) or 107,000 km/h (66487 mph), it actually ranges by a full km per second during the course of the year – between 30.29 km/s and 29.29 km/s (109,044 – 105,444 km/h; 67,756.8 – 65,519.864 mph).

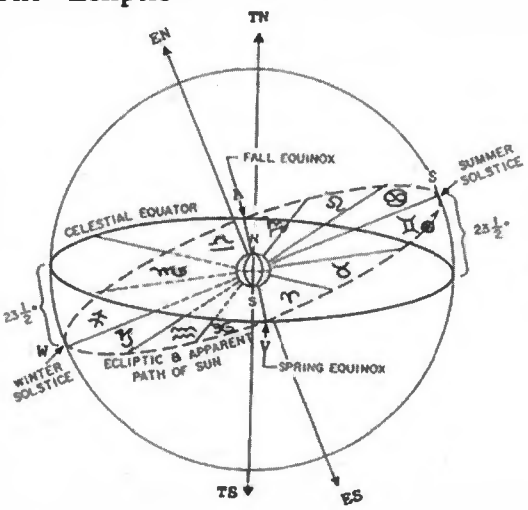


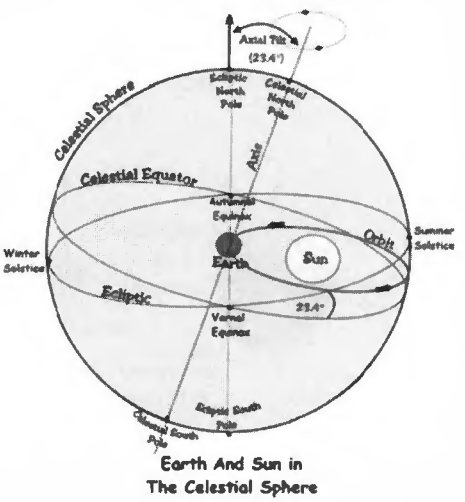
	Moon		Sun	
	At perigee (nearest)	**At apogee (farthest)**	**At perihelion (nearest)**	**At aphelion (farthest)**
Mean radius	1,737.10 kilometers (1,079.38 miles)		696,000 kilometers (432,000 miles)	
Distance	363,104 km (225,622 mi)	405,696 km (252,088 mi)	147,098,070 km (91,402,500 mi)	152,097,700 km (94,509,100 mi)
Angular diameter	33' 30" (0.5583°)	29' 26" (0.4905°)	32' 42" (0.5450°)	31' 36" (0.5267°)

Earth's axis is also inclined at approximately 23.439° towards the ecliptic. This means that when the sun crosses the equator at both equinoxes, it's daily shift relative to the background stars is at an angle to the equator. In June and December, when the sun is farthest from the celestial equator, a given shift along the ecliptic corresponds to a large shift at the equator. Because of the axial tilt of the earth in its orbit, the maximum intensity of sun rays hits

the earth 23.4 degrees north of equator at the June Solstice (at the Tropic of Cancer), and 23.4 degrees south of equator at the December Solstice (at theTropic of Capricorn).

The Ecliptic





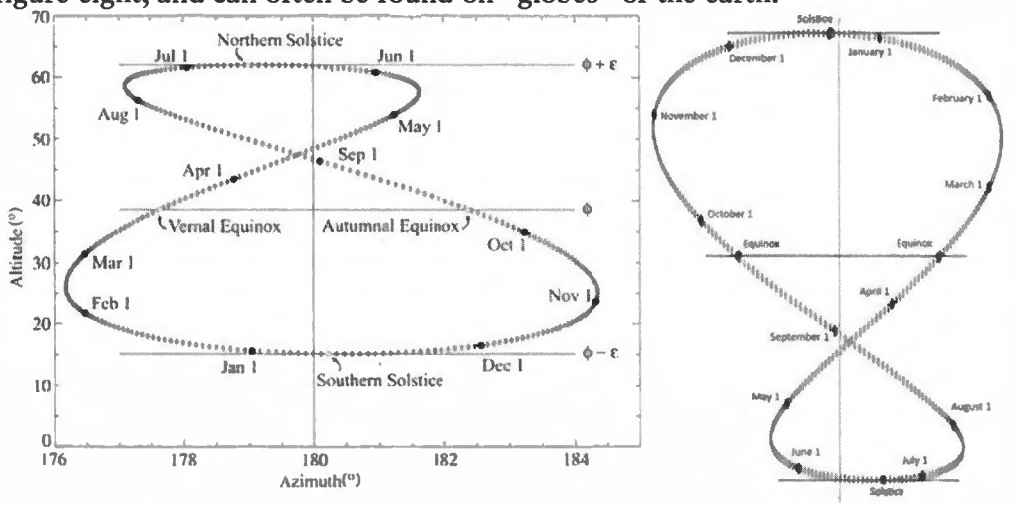
Earth And Sun in The Celestial Sphere

The ecliptic is the apparent path of the sun on the celestial sphere, and is the basis for the ecliptic coordinate system. It also refers to the plane of this path, which is coplanar with the orbit of earth around the sun (and hence the apparent orbit of the sun around earth). The ecliptic plane is tilted 23.5° with respect to the plane of the celestial equator since the earth's spin axis is tilted 23.5° with respect to its orbit around the sun. The ecliptic plane intersects the celestial equatorial plane along the line between the equinoxes. The earth's spin axis is tilted with respect to the plane of the earth's solar orbit by 23.5°. Observations show that the other planets, with the exception of Pluto, also orbit the sun in essentially the same plane. The ecliptic plane then contains most of the objects, which are orbiting the sun. This suggests that the formation process of the solar system resulted in a disk of material out of which formed the sun and the planets. The 23.5° tilt of the earth's spin axis gives the seasonal variations for sunlight received at the surface.

The Analemma

In modern astronomy, an analemma is a diagram showing the deviation of the sun from its mean motion in the sky, as viewed from a fixed location on

the earth. Due to the earth's axial tilt and orbital eccentricity, the sun will not be in the same position in the sky at the same time every day. The north–south component of the analemma is the sun's declination, and the east–west component is the equation of time. This diagram has the form of a slender figure-eight, and can often be found on globes of the earth.



There are three parameters that affect the size and shape of the analemma - obliquity, eccentricity, and the angle between the apse line and the line of solstices. Viewed from an object with a perfectly circular orbit and no axial tilt, the sun would always appear at the same point in the sky at the same time of day throughout the year and the analemma would be a dot. For an object with a circular orbit but significant axial tilt, the analemma would be a figure of eight with northern and southern lobes equal in size. For an object with an eccentric orbit but no axial tilt, the analemma would be a straight east–west line along the celestial equator. The north–south component of the analemma shows the sun's declination, its latitude on the celestial sphere, or the latitude on the earth at which the sun is directly overhead. The east–west component shows the equation of time, or the difference between solar time and local mean time. The width of the analemma is approximately 7.7°, and the length of the figure is more than six times its width.

The figure above is an example of an analemma as seen from the earth's northern and southern hemispheres. It is a plot of the position of the sun at 12:00 noon at Royal Observatory, Greenwich England (latitude 51.48°N, longitude 0.0015°W) during the year 2006. The horizontal axis is the azimuth angle in degrees (180° is facing south). The vertical axis is

the altitude in degrees above the horizon. The first day of each month is shown in black, and the solstices and equinoxes are shown in green. It can be seen that the equinoxes occur at altitude $\varphi = 90° - 51.48° = 38.52°$, and the solstices occur at altitudes $\varphi \pm \varepsilon$ where ε is the axial tilt of the earth, 23.439°.

Earth's Rotational Axis

Seasons occur because of the 23.5° tilt of the earth's rotational axis. Around the June Solstice, the Northern Hemisphere tilted towards the sun. This causes summer there. The Southern Hemisphere on the other hand, tilted away from the sun and therefore, experiences winter. The opposite occurs around the December Solstice, when the Southern Hemisphere tilted towards the sun, while the Northern Hemisphere tilted away.
There are two equinoxes every year – in March and September– when the sun shines directly on the equator and the length of night and day are nearly equal. The March equinox (March 19, 20 or 21) marks the moment the sun crosses the celestial equator – the imaginary line in the sky above the earth's equator – from south to north. Why is it Called "Equinox"? On the equinox, night and day is nearly exactly the same length – 12 hours – all over the world. This is the reason it's called an "equinox", derived from Latin, meaning "equal night". However, in reality equinoxes don't have exactly 12 hours of daylight.

What is an Equinox? Earth spins around its own axis approximately every 24 hours. It orbits the sun in around 365 days. Earth actually tilted at an angle of around 23.4 degrees toward the celestial pole, which is at a certain point in the sky. As the earth makes its yearly orbit, one hemisphere faces the sun more than the other does, the side that has summer.
What Happens on the Equinox? On any other day of the year, either the southern hemisphere or the northern hemisphere tilts a little towards the sun. However, on the two equinoxes, the tilt of the earth's axis is perpendicular to the sun's rays. However, the axis tilt of around 23.4 degrees toward the celestial pole remains the same.

Seasons: Meteorological Versus Astronomical
Most modern day calendars around the world divide the year in 4 seasons: spring, summer, fall (autumn) and winter.

People in Australia & New Zealand (South)	Consider September 1 as the beginning of spring
People in Irish and Greenland (North)	Consider spring begins on February 1 (St Brigid's Day)
People in South Asia	Have calendars that divide the year in 6 seasons

This is because the seasons in the Southern Hemisphere are opposite to the seasons in the Northern Hemisphere.

Astronomical Seasons: Astronomers and scientists use the dates of equinoxes and solstices to mark the beginning and end of seasons in a year. There are four astronomical seasons in a year in the Northern Hemisphere

Spring	March Equinox (20-21 March) to June Solstice (21 June)
Summer	June Solstice (21 June) to September Equinox (22-23 September)
Fall / autumn	September Equinox (22-23 September) to December Solstice (21-22 December)
Winter	December Solstice (21-22 December) to March Equinox (20-21 March)

Because the dates of the equinoxes and solstices can change, the length of astronomical seasons within a year and between years can vary, making it difficult to properly compare and study seasons in different years.

Meteorological Seasons: In order to be consistent and to make forecasting easier, meteorologists divide the year into 4 meteorological seasons of 3 months each,

Spring	Starting March 1 and ending May 31
Summer	Starting June 1 and ending August 31
Fall / autumn	Starting September 1 and ending November 30
Winter	Starting December 1 and ending February 28 (February 29 in a Leap year)

Summer in the North, winter in the South

Northern Hemisphere	June Solstice marks the start of summer (astronomical seasons)

Southern Hemisphere	June Solstice marks the start of winter (astronomical seasons)

The same rule applies for the other three seasons. Similarly, the meteorological seasons in the Southern Hemisphere are opposite to those in the Northern Hemisphere.

Spring in the North, fall in the South: Equinoxes and solstices are opposite on either side of the equator, and the March equinox is also known as the "spring (vernal) equinox" in the northern hemisphere and as the "autumnal (fall) equinox" in the southern hemisphere.

The Polestar and its Declination

A pole star is a visible star, preferably a prominent one, that is approximately aligned with the earth's axis of rotation; that is, a star whose apparent position is close to one of the celestial poles, and which lies approximately directly overhead when viewed from the earth's North Pole. In practice, the term pole star usually refers to Polaris, which is the current northern pole star, also known as the North Star.

The closest bright star to the north celestial pole is Polaris. At magnitude 1.97 (variable), it is the brightest star in the Ursa Minor constellation (at the end of the "handle" of the "Little Dipper" asterism). As of October 2012 its declination is +89°19'8" (at epoch J2000 it was +89°15'51.2"). Therefore, it always appears due north in the sky to a precision better than one degree, and the angle it makes with respect to the true horizon (after correcting for refraction and other factors) is equal to the latitude of the observer to better than one degree. It is consequently known as Polaris (from Latin Stella Polaris "polestar").

Orbital characteristics

Aphelion	152.1×10^6 km (94.51×10^6 miles) 1.0167 AU
Perihelion	147.10×10^6 km (91.40×10^6 miles) 0.98329 AU
Semi major axis	149.60×10^6 km (92.96×10^6 miles) 1.000001018 AU
Eccentricity (e)	0.0167086 ($= \frac{\text{(Aphelion– Perihelion)}}{\text{Semimajor axis*2}}$)

Inclination	7.155° to sun's equator 1.578690° to invariable plane
longitude of the ascending node	174.9°
longitude of perihelion[137]	102.9°
argument of periapsis[138]	288.1°
period	365.256363004 days
average speed	29.78 km/sec (18.5 miles/s) 107,200 km/hour (66,600 mph)

Sun Fact Sheet: Rotational and Orbital parameters

	Sun	Earth	Ratio (Sun/Earth)
Sidereal Rotation Period (hrs)*	609.12	23.9345	25.449
Obliquity to ecliptic (deg.)	7.25°	23.45°	0.309
Obliquity to galactic (deg.)	67.23°		
Speed relative to nearby stars (km/s)	19.4		

*This is the adopted period at 16 deg. latitude - the actual rotation rate varies with latitude L as: (14.37 - 2.33 $\sin^2$ L - 1.56 $\sin^4$ L) deg/day

North Pole of Rotation:

Right Ascension* of North Pole	286.13° (19hrs 4min 30sec)
Declination of North Pole	63.87° (63° 52' north)

* Right ascension is the angular distance measured eastward along the celestial equator from the vernal equinox to the hour circle of the point in question.

Sun Observational Parameters:

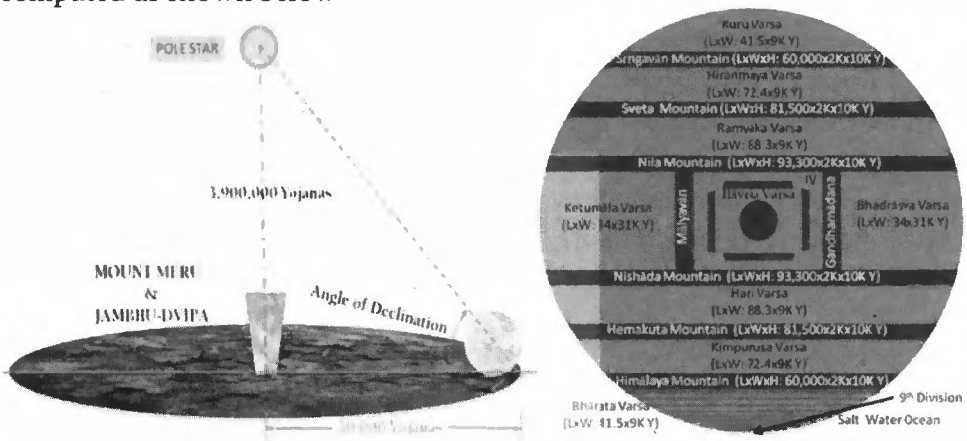
Apparent diameter from Earth	At 1 A.U.(seconds of arc) Maximum (seconds of arc) Minimum (seconds of arc)	1919 (31' 59", 0.533°) 1952 (32' 32", 0.542°) 1887 (31' 27", 0.524°)
Distance from Earth	Mean (10^6 km) Minimum (10^6 km) Maximum (10^6 km)	149.6 147.1 152.1

[137] The periapsis of an object orbiting Earth is its perigee, & the periapsis of an object orbiting Sun is its perihelion

[138] The point at which an orbiting object is closest to the body it is orbiting.

Explanation as per Śrīmad Bhāgavatam

As described in Śrīmad Bhāgavatam[139], earth sphere is situated at 50,000 yojanas from the centre of Meru on the southern direction and the polestar situated at a height of 39 lakh yojanas from the base of Mount Meru. It is said that the earth is inclined towards the polestar, as if always looking at it in the upward direction. The angle made by the earth's vertical axis connecting the polestar with the horizontal plane is called as the Declination of polestar. It is computed as shown below



Angle of Declination: $\tan^{-1}\frac{3{,}900{,}000}{50{,}000} = 89°15'56''$;

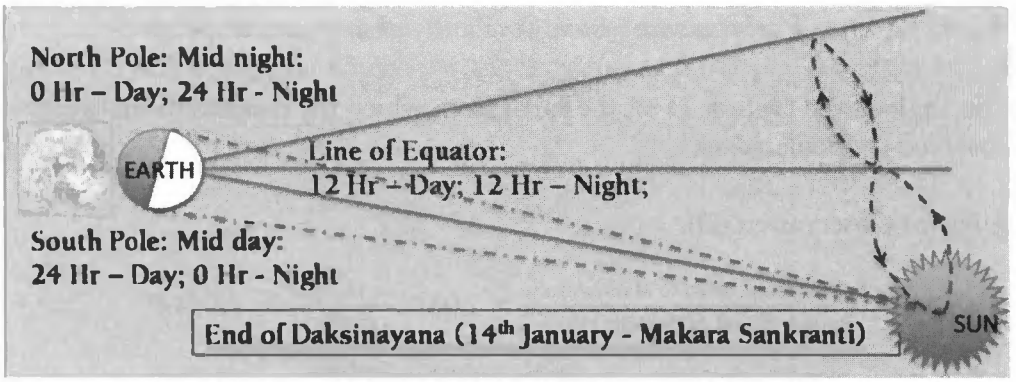
*Considering both the sun and the base of Mount Meru are on the same level as it is described in Bhugola Varnanam, the Polestar is situated at a height of 3.8MY and the earth's radius will be 500 yojanas, then the angle of declinations is calclulated to the center of the earth as:

Angle of Declination: $\tan^{-1}\frac{3{,}800{,}000}{49{,}500} = 89°15'13''$

Śrīmad Bhāgavatam further explains about the vertical movement of sun in the northern and southern courses, which are called *Uttarāyana* and *Daksināyana* movements of the sun.

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[139] Some of these calculations for comparisons are inspired by Prem Gauranga Prabhu and some of the calculations are directly taken from his research work on this subject matter.



The picture showed above depicts the complete movement of the sun on the Zodiac for one complete year. On 15th January, sun enters into Capricorn (Makara), and in three months he enters into Aries (Mesa) on 17th April. At the interval of three months, he enters into Cancer (Karka) on 17th July, Libra (Tula) on 16th Oct, and then again into Capricorn (Makara) on 14th Jan, Thus, the yearly path of the sun is depicted. During the Northern course sun goes above and inside increasing daytime in the north and decreasing the daytime in the south. Similarly, during Southern course the sun comes below and outside increasing daytime in the south and decreasing the daytime in the north. The path of the sun shown above is similar to the analemma observed by the modern scientific findings.

The change in orbital distances of the sun during its northern and southern movements can be reconciled with the help of the three speeds of the sun mentioned in Śrīmad Bhāgavatam.

Reference	Speed (MY/hour)	Circumference (MY/day)	Radius of Sun's orbit (MY)	Corresponding to
Speed1 in SB 5.21.10	3.9625	95.100	15.135635	*Uttarāyana*
Speed2 in SB 5.21.12	4.251	102.024	16.237624	*Visuvāyana*
Speed3 in SB 5.21.19	4.501125	108.027	17.193031	*Daksināyana*

*C = Sx24 hours; R = C/2pi = C/6.2632 (pi = 3.1416);

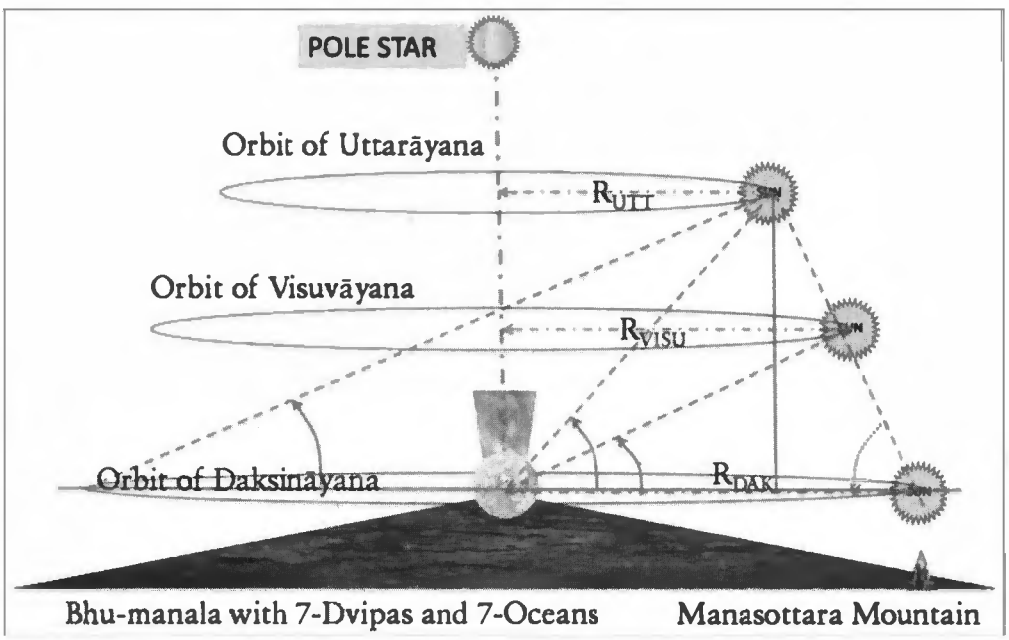
Speed 1, Speed2 and Speed3 of the above table indicate the speed of the sun

during the end of *uttarāyana, visuvāyana* and *daksināyana* respectively.

The angle of the latitude Ø on the earth from which the measurements were observed is calculated as:

Angle of Observation (Ø):

$$Ø = COS^{-1}\frac{\text{Radius from Meru to Manasottara}}{\text{Radius at end of Dakshinayana}} = COS^{-1}\frac{15.75}{17.193031} = 23.642°$$

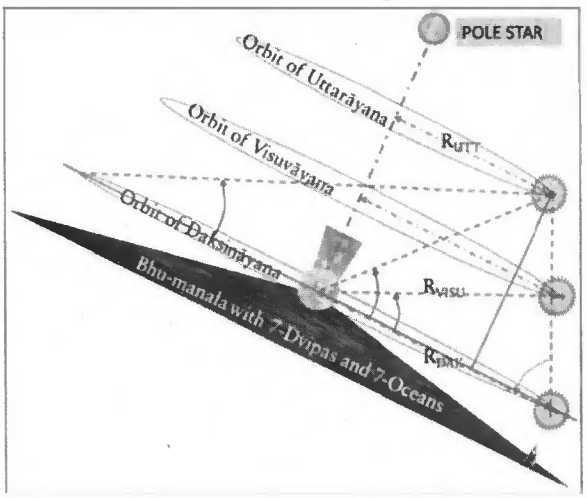


Considering the above values of Śrīmad Bhāgavatam for further calculations

	MY	Multiplying with Cos(23.642°)*
Radius during uttarāyana (R_{UTT})	15.135635	13.865284
Radius during vishuvāyana (R_{VISU})	16.237624	14.874781
Radius during daksināyana (R_{DAK})	17.193031	15.75
Difference between two (R_{DAK} - R_{UTT})	02.057	01.88472
Sum of the above two (R_{DAK} + R_{UTT})	32.329	29.615284

*Normalizing the radius values to the Line of Equator, since the values mentioned in Śrīmad Bhāgavatam or any other literature such as other Puranas

and Surya Siddhanta are generally considered to be the observed values at Ujjain[140], which approximately corresponds to the Line of Capricorn at an altitude of 23.2756°.



Vayu Purana, Adhyaya 50, text 122-140 further mentions that

- As the sun reaches the end of southern course (far end of Daksinayana) by the month of Magha, he covers 94.5MY during a Day. Then the diameter of his orbit is 31.5MY
- When he starts his northern course and comes to Vishuva (the equinox points), he moves in the north side of the milk ocean (6th Ocean), in a circle of diameter equal to 38.1MY. This value is corresponding to semi-circular diameter, i.e. the distance measured across half of the circle periphery. Then the actual diameter is 24.255MY

$$Calculated\ diameter = \frac{2 * 38.1}{\pi} = 24.255MY$$

- When he reaches the maximum point in the north during his stay in Sravana, he will be moving in the northern regions of Sakadvīpa (Sixth Island). Then the diameter of his orbit is 23.8MY[141]
- The distance b/w the tracks (180) to the extreme north and extreme south is 0.343MY
- The distance b/w the extreme points and lekhas both in the south and north is 7.175MY

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[140] Sometimes these values are w.r.t the place of discussion, either Naimiśāranya or Śukatāl

[141] These values are different from the values of Śrīmad Bhāgavatam – accounts to *kalpa-beda*

- Surya always moves in the inner circles during Uttarayana and outer circles during Daksinayana, with a magnitude of 21,221 yojanas each in diameter.
- The sun will be on the Vishuvata peak on the day of equinoxes. Thus, rendering both day and night equal. On the day of equinoxes, at the end of Mesa and Tula months, both the day and night will be 15 muhurtas each. The Vishuva is common to all the places, islands and varṣas.
- Uttarayana: Magha – Ashada; Dakshinayana: Shravana – Pusha months.

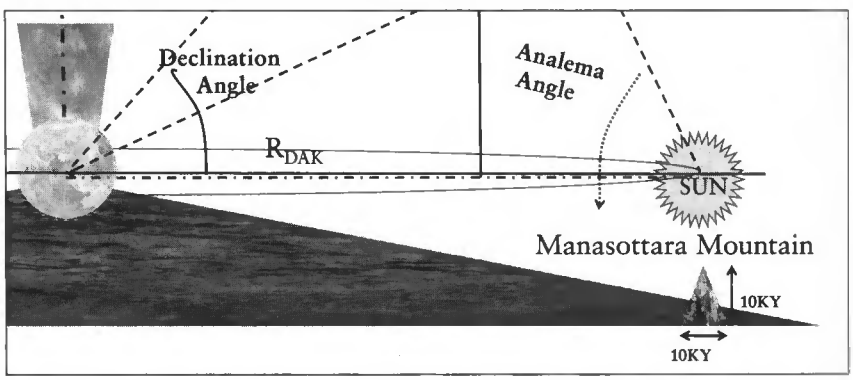
Summary of the above mentioned details of Vayu Purana

Time	Month	Circumference (MY/day)	Dia of Sun's orbit (MY)	Radius of Sun's orbit (MY)
Far end of *Daksināyana*	Magha	94.500	31.5 (30.08)	15.75 (15.04)
At the crossing of *Vishuvāyana*	Mesa, Tula	102.024	24.255 (23.162)	12.1275 (11.58)
Far end of *Uttarāyana*	Sravana	108.027	23.8 (22.727)	11.9 (11.36)

**Values in the bracket are with pi value equal to 3.1416 instead of 3*

Slope of Manasottara Mountain, whose height and width are 10,000 yojanas:

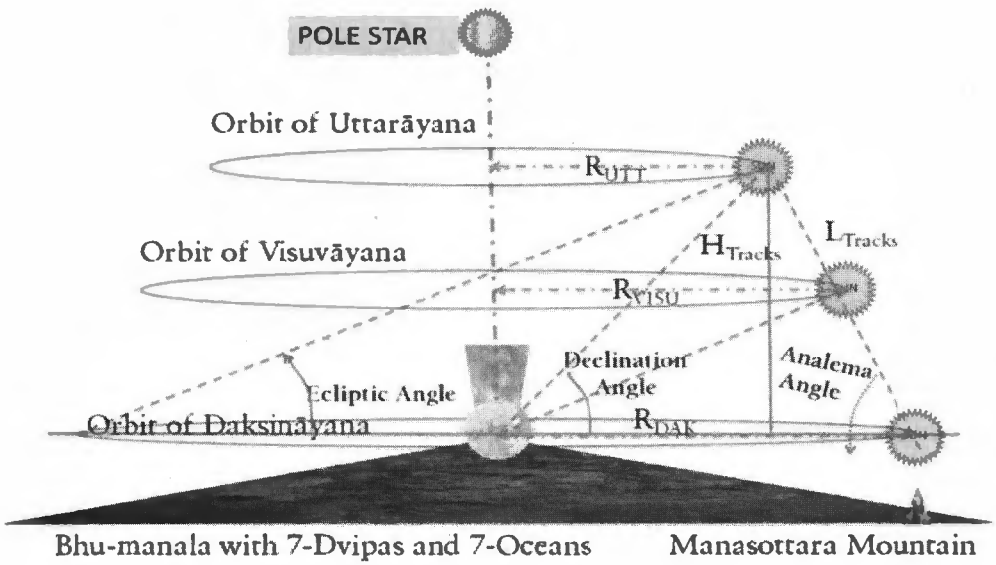
$$\tan^{-1}\frac{height}{half\ of\ the\ width} = \tan^{-1}\frac{10{,}000}{5000} = 63.435°$$



Slope of Manasottara Mountain is almost equal to the Angle of Analemma, because the chariot of the sun is said to be moving above the Manasottara Mountain and its yearly inclined path might be parallel to the slope of the Manasottara Mountain.

According to Vayu Purana, the distance between the extreme points or lekhas both in the south and north is 7.175MY, and the orbit of the sun, when he reaches the maximum point in the north during his stay in Sravana, will be in the northern regions of the sixth island Sakadvīpa. Then the diameter of his orbit measures 23.8MY. Thus, the radius of the sun is around 11.9MY.

$$\sin^{-1}\frac{\textit{distance between extreme points in south \& north}}{\textit{Radius of the orbit of the Sun at uttarayana}} = \sin^{-1}\frac{7.175}{11.9} = 37.081°$$



*The total width of 180 Tracks: 180 * 21,221 yojanas = 3,819,780 yojanas

Two possible values for H_{Track} leads to two set of values for various parameters

$H_{Track} (= \sqrt{L_{TRACK}^2 - (R_{DAK} - R_{UTT})^2})$	$L_{Track} (= \sqrt{H_{TRACK}^2 + (R_{DAK} - R_{UTT})^2})$
3,819,780 yojanas	4,261,753 yojanas
3,319,473 yojanas	3,819,780 yojanas

	Formula	**H_{Tracks} (3,819,780Y)**	**H_{Tracks} (3,319,473Y)**
Angle of Analemma (Path of Sun)	$\tan^{-1}\frac{H_{Tracks}}{(R_{DAK} - R_{UTT})}$	63.675°	60.35°

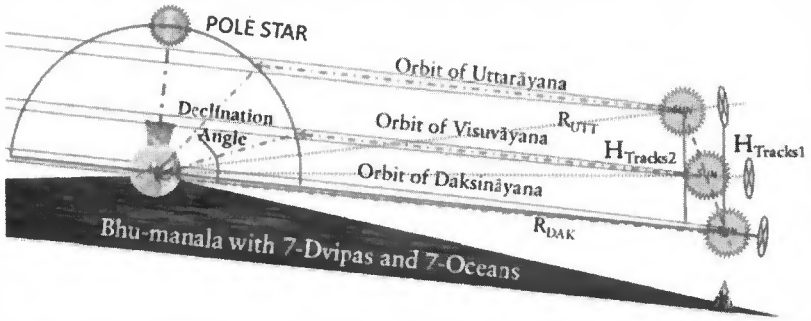
Angle of Ecliptic (Yearly Orbit)	$\tan^{-1}\frac{H_{Tracks}}{(R_{DAK} + R_{UTT})}$	7.329°	6.378°
Declination Angle at Meru Base	$\tan^{-1}\frac{H_{Tracks}}{R_{UTT}}$	15.4025°	13.4753°
Half of the Declination Angle at Meru Base	$\tan^{-1}\frac{H_{Tracks}/2}{R_{UTT}}$	7.483°	6.832°
Eccentricity	$=\frac{(R_{DAK} - R_{UTT})}{(R_{DAK} + R_{UTT})}$	0.06364	0.06364

Half of the Declination Angle at Meru Base is in line with the apparent Line of Equator of the earth. Thus, at the Line of Equator, the Declination of the sun observed is only 7.483° on both side, and 15.4025° from uttarāyana to daksināyana[142]. But as per the modern day observations, at the Line of Cancer (Karka sankrānti) and at the Line of Capricorn (Makara sankrānti), the Declination of the sun observed on either side will be 23.45°, for it is seen practically. How to understand this?

The below calculations are simply a proposal to see how the above value of the Declination of Sun (7.483°) can be seen as 23.45° to a person observing from a particular place over a year. This is explained with the help of Refraction Theory, which says that the change in direction of a light or water or sound wave passing from one medium to another or from a gradual change in the medium. This is just a proposal.

Draw a hemisphere with the Radius equal to the distance between the Base of Mount Meru and the polestar (39 lakh yojanas). Considering that the rays of the sun, during his northern and southern course, falling straight on the hemisphere drawn across the pole star with radius equal to its height (3.9MY). The rays of the sun decline due to refraction and then fall on the earth globe by changing its direction through this hemispherical surface.

[142] At the Line of Cancer (Karka sankrānti) the Declination of the Sun observed on either side will be 23.45°, for it is seen practically. The similar thing is observed at the Line of Capricorn (Makara sankrānti).



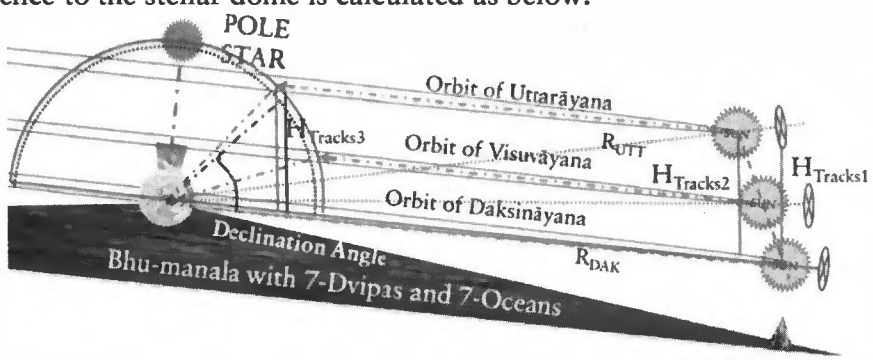
Considering the current position of the sun at the far end of *Uttarāyana:*

Description	Position of the chariot wheel	Position of the sun
Radius of the chariot wheel and sun at the far end of *Uttarāyana:*	R_{UTT}: 13.86528 MY	R_{UTT}: 15.75-(3.6+0.9/2) = 11.7MY 3.6MY: chariot carriage length 0.9MY: chariot yoke width
Vertical displacement of the chariot wheel & Sun at *Uttarāyana:*	$H_{Tracks1}$: 21,221*180 = 3.81978 MY	$H_{Tracks2}$: $= \frac{11.7}{13.865284} * 3.81978 = 3.22326\ MY$

The declination angle made by the sunrays at the far end of uttarayana:

$$Declination\ Angle = \sin^{-1}\frac{(H_{Tracks2}/2)}{\cancel{Polestar\ Height}} = \sin^{-1}\frac{(\frac{3.22326}{2})}{(3.9)} = 24.4084°$$

According to various acaryas the the polestar itself has a local orbit, which is located at a distance of 1.4 lakh yojanas below the polestar. The dome drawn at this radius ($R_{stellar}$: 3.9−0.14 = 3.76MY) is considered to be stellar dome. The declination angle made by the sunrays at the far end of uttarayana with reference to the stellar dome is calculated as below:



$$H_{\text{Tracks3}} = \frac{3.9-0.14}{3.9} * 3.22326 = 3.10755\ MY$$

$$Declination\ Angle = \sin^{-1}\frac{(\text{H}_{\text{Tracks3}}/2)}{Polestar\ Height} = \sin^{-1}\frac{\left(\frac{3.10755}{2}\right)}{(3.9)} = 23.4785°$$

Perihelion (to sun's centre): 15.75 - (3.6+0.9/2) + 0.05 = 11.75 MY
Aphelion: $sqrt(11.75^2 + 3.10755^2) = 12.15399\ MY$

Comparison of the parameters:

Modern Science		Śrīmad Bhāgavatam	
Perihelion	147.1 M km	Perihelion	11.75 MY (147.465M km)
Aphelion	152.1 M km	Aphelion	12.15399 MY (152.535 M km)
Difference	5 M km	Difference	0.40399MY (5.07167 M km)
Eccentricity of Earth's Orbit	0.0167086	Eccentricity of Sun's Orbit	0.0169006
Polestar Declination at epoch J2000 in Oct 2012	+89°19'8" +89°15'51.2"	Polestar Declination with the earth	+89°15'56" +89°15'13"
Earth's Obliquity to Ecliptic	23.45°	Earth's Obliquity to ecliptic Earth's Obliquity to ecliptic at Equator	23.4785° 7.681°
Sun's Obliquity to ecliptic Width of Analema	7.25° 7.65°	Angle of the Wheel of Time (Sun's Obliquity to ecliptic)	7.329°
Declination of North Pole	63.87°	Angle of the Path of Sun	63.675°

Conclusion

All the above comparisons goes to show that the information given in sacred texts such as Śrīmad Bhāgavatam, along with Vāyu Purana and Surya Siddhanta can stand the test of time and can be reconciled with many of the modern findings.

At the conclusion, let us come back to the observation made by Śukadeva Gosvāmī in the beginning of his description of the universe: "My dear King, it is not possible even for a person with the lifespan of a *devatā* to understand by words or mind the end of the transformations of Lord's material energy. Therefore by giving a general description I will explain the particulars of Bhū-loka, giving characteristics of name, shape and dimension (SB 5.16.4)." The history has witnessed two greatest conversations, which were held to honour the passing away of two great souls, Śrīla Bhīṣmadev and Parīkṣit maharāj. These conversations were witnessed by the greatest scholars of the Universe, such as Atri, Vasiṣṭha, Bhṛgu, Aṅgirās, Parāśara, Viśvamitra, Paraśurāma, Bhāradvāja, Gautama, Maitreya, Agastya, headed by Vyāsa and Nārada. In the first conversation, Śrīla Bhīṣmadev instructed Yudhiṣṭhira maharāj on various dharmas suitable for men according to their natures, according to varṇa and āśrama, which have qualities of renunciation and enjoyment, according to a person's detachment or attachment, while the sages were listening being headed by Lord Kṛṣṇa Himself. Śukadeva Gosvāmī is the personal witness to this conversation, whereas in the second conversation, he himself spoke to Parīkṣit maharāj for seven days about the complete perfection and the sādhana to be performed at all times for that perfection. During that seven-day discussion, in the presence of all the great sages, Śukadeva Gosvāmī briefly summarised the mysteries of the structure and the workings of this gigantic Universe. Due to lack of time and not because of prime subject matter of that discussion, he just presented the essence, within few chapters without missing any of the crucial details. Proper understanding and expansion of these few essential descriptions, reveals the complete mystery of the systematic Universal functioning. A glimpse of this we have presented in this book.

Whatever we have tried to present here is as per our limited understanding. There is lot of scope for further refinement and improvements in our understanding and presentation. This is our humble attempt to describe the details given in our esteemed Vedic Texts, primarily Śrīmad Bhāgavatam, Bhugola Varnanam and Surya Siddhanta in a reader friendly manner with suitable pictorial diagrams for easy explanation. This also tried to compare the model of Śrīmad Bhāgavatam with the modern day solar system, with lot of supporting computations and explanations. To properly answer the many questions that arise, much further research is needed. However, for further details and clarifications in this regard, we request the reader to refer to the books written by earlier authors and scholars of the subject. As it was stated in the beginning, that one of the objectives of this book is to enhance our perception and appreciation towards the wonderful creation of the Supreme Lord and the words of the Vedic literatures. So that we begin to see the events of this world in the light of Scriptural authority.

Appendix 1

Śrīla Visvanātha Chakravarti Thākura (1638 AD to 1708 AD)

Family Lineage, Birth & Studies: Śrīla Visvanātha Chakravarti Thākura took birth in a family of brahmanas found in the Radha-desa area of Nadia District, West Bengal. His father's name was Ramanarayana Cakravarti. He had two older brothers named Ramabhadra and Raghunatha, and he stayed in Devagrama during his childhood. Upon the completion of his studies in vyakarana, he moved to Saiyadabad-grama in Murasidabad District, where he studied the bhakti-sastras in the home of his guru Sri Radha-ramana Cakravarti. This Radha-ramana Cakravarti was the disciple of Sri Kṛṣṇa-carana Cakravarti, who was in turn the disciple of Sri Ganga-narayana Cakravarti (one of the chief disciples of Thakura Narottama).
From his childhood, he was of a detached temperament. At the command of his father, his brother Ramabhadra arranged for his marriage at an early age. However, through studying Śrīmad-Bhāgavatam, he developed a deep spirit of renunciation. After completing his studies, he took spiritual initiation and gradually developed an intense love for Kṛṣṇa. Finally, one day, he renounced home, took the vow of a renunciate and went to Vrndavana. After visiting several holy places, he finally sought the shelter of Mukundadasa, a disciple of Kṛṣṇadasa Kaviraja gosvāmi on the bank of the Radhākunda. At the command of his guru, he went home for one night to meet his wife. His wife, however, heard nothing other than Kṛṣṇa katha from her husband throughout the night. Early the next morning he left home and took shelter of his guru. As directed by his guru, he began copying *Śrīmad-Bhāgavatam.*

His Activities in Vrndavana: When he arrived in Vrndavana, he noticed that with the disappearance of the six gosvamis the beauty of that holy place was no longer visible and a large number of Mathas had been destroyed by the Muslims. Priests migrated from Vrndavana taking away the deities, which were in their charge. A number of deities in Vrndavana left standing alone and received no service. In addition, the devotees were in a state of constant fear. People in general were in no mood to devote attention to the study of the scriptures. During his stay in Vrndavana, many loyal workers and scholars such as Baladeva Vidyabhusana were deeply impressed upon seeing his

devotion, strength of mind and hardworking nature. he became determined to bring back the lost glory of Vrndavana. The following are some of his achievements.

- Śrīla Visvanātha Chakravarti Thākura himself installed the Deity of Gokulananda and took charge of serving Govardhana sila. He reinstated different priests to begin the service of the Deities in various places.
- Through his initiative the Sri Vardhana Matha of Kongala and some new Mathas at several other places were set up. He also arranged to renovate a large number of temples.
- At that time there was little access by the common people to the works of the Gosvamis. This was due to the fact that, there were no proper analysis and interpretation of these theological treatises. What he did was to prepare simple and lucid commentaries for these works, as well as presenting abridged forms of the original works. This enabled devotees of all types to understand and appreciate the essence of the Gosvamis works.
- Śrīla Visvanātha Chakravarti Thākura also arranged for the wide distribution of books, which Vaisnavas needed for daily study and spiritual practices. He also organized classes to be held on them to impart instructions.
- Another remarkable achievement was to establish Gaudiya Vaisnavism and its theology through Baladeva Vidyabhusana, his close follower, at the meeting of Vaisnavas held at Galta, Jaipur in 1706 AD.

During that time, at Sri Govindadeva's temple at Galta-grama (just outside Jaipur), the acaryas of the Sri Ramananda-sampradaya (Ramanandis) issued a challenge against the Gaudiya Vaisnavas. The King of Jaipur consequently invited the most prominent Gaudiya Vaisnavas of Sri Vrindavana to attend. Knowing them to be followers of Srila Rupa Gosvami, he called them to council with the followers of Sri Ramananda-sampradaya (Ramanandis). This happened in 1628 Saka (1706 A.D.), when Śrīla Visvanātha Chakravarti Thākura was very old (about 68 years). So he consulted his foremost student, Sripada Baladeva Vidya-bhusana. Thereafter, Sri Baladeva Vidya-bhusana left Vrndavana to join the assembly in Jaipur, accompanied by his own student (and disciple of Srila Cakravarti Thakura), Sri Kṛṣṇadeva Sarvabhauma. There Sri Baladeva Vidya-bhusana was obliged to write a separate commentary on the Vedanta-sutra, according to the philosophy of the Gaudiya-sampradaya.

This was done just to refute their false conclusions. Srila Cakravarti Thakura gave his full sanction and approval to this task of counteracting the challenge, which simultaneously resulted in allowing the Gaudiya Vaisnava parampara to continue preaching freely.

His Residence in Sri Vraja Mandala: By the mercy of his spiritual master, Śrīla Visvanātha Chakravarti Thākura lived in many different places within Vraja-dhama, and composed various transcendental literatures there. Most of these books are very difficult to find nowadays; however, a few of them are well known, and are considered to be the supremely honorable wealth of the Gaudiya Vaisnavas. Sometimes, Srila Cakravarti Thakura lived at Sri Govardhana, sometimes on the bank of Sri Radha-kunda, sometimes at Sri Yavata and sometimes in Sri Vrindavana within the compound of Sri Gokulananda's temple. His movements, here and there, made very clear by the statements found at the end of his books. He served the deity of Sri Gokulananda with great pleasure and devotion.

In Vrndavana, he sought the refuge of Mukundadasa. This Mukundadasa was a poet and disciple of Kṛṣṇadasa Kaviraja gosvami. Mukundadasa had some books to be completed. Noting his devotion and erudition, Mukundadasa requested him to complete those books. In Vrndavana his literary talent, blossomed and beautiful compositions began to flow like many streams of nectar. His complete works can be classified under four groups: Commentary works, abridged works of original texts, original works and compilation of Padavali. He had an extraordinary command of Vrajboli, Hindi and Sanskrit. In all three languages, he composed a total of seventy padas of which the ones in Sanskrit are the best.

Judged from the standpoint of the preceding Vaisnava acaryas and the quality of rasa, Śrīla Visvanātha Chakravarti Thākura undoubtedly contributed immensely in leading Gaudiya Vaisnavism and sadhana bhakti forward. Most of the difficult treatises of Rupa Gosvami were presented by him to devotees sometimes by adding simple commentaries and sometimes by preparing abridged editions. Devotees hailed him as `the second svarupa of Rupa' or as `avatara of Rupa'. Baladeva Vidyabhusana, a disciple of Radhadamodara, revered Śrīla Visvanātha Chakravarti Thākura deeply as his guru. Śrīla Visvanātha Chakravarti Thākura stands as a remarkable outcome of Bengali

intellect in 17th-18th century as poet, musician, thinker, theologian, and scholar and above all a devotee and preacher. His place in the Vaisnava world remains unsurpassed as far as erudition, theological knowledge, poetic talent and appreciation of rasa. He was worshiped by his contemporaries as an example of unblemished ascetic life and an ideal follower of Ragamarga. He is glorified among Vaisnava community as *rasikācārya*.

Śrīla Vādirāja Tīrtha swāmi (1480-1600)

Śrīla Vādirāja Tīrtha swāmi traditionally was a Shivalli Tulu Brahmin and native of the village of Hoovinakere, Udupi District in Karnataka state. His parents were Ramacharya and Gauri. He is considered amongst the highest saints in the Madhva hierarchy, next only to Srimad Ananda Tīrtha (1238–1317) and Sri Jayateertha (1348 – 1388). He also studied under Vyasadeva. He is regarded as an incarnation of *Latavya* and hence popularly known as '*bhāvi-samīra*' which means he will become Vayudeva in future. He was not only a great poet but also very effective in the administration of the Udupi Matha system. He brought about many changes to the operational system of the Matha which by itself showed his high placing in the Madhva hierarchy.

He became a Sannyasi at the young age of eight. The pre-sanyasa name given to Śrīla Vādirāja Tīrtha swāmi was Bhuvaraha. He worshipped Lord Vishnu in the form of Hayagriva. Śrīla Vādirāja Tīrtha swāmi used to offer daily prasadam to Lord Hayagriva by holding the prasadam on his head. Lord Hayagriva is said to appear in the form of a horse and eat the prasadam by kneeling on Śrīla Vādirāja Tīrtha swāmi's shoulders.

It was Śrīla Vādirāja Tīrtha swāmi who changed the Paryaya system of Udupi to two years from the earlier practice of two months. This extension of each individual Paryaya enabled the Swamis to travel far and wide and spread the message of Madhva tradition. Another of Śrīla Vādirāja Tīrtha swāmi's achievements is that he lived a life of 120 year and performed Lord Krishna's Paryaya at Udupi five times. He is the first Sannyasi who entered the Vrindavana (his own samadhi) alive after worshipping that for nearly 3 years. The only other saint to have followed similar way of leaving this materialistic world is Sri Rāghavendra svāmi of Mantrālaya.
In an age that knew great Sanskrit scholars and intellects, Śrīla Vādirāja

Tīrtha swāmi who himself was a great scholar, was able to accommodate the needs of the less scholarly, taking the Vaishnava tradition to the masses by translating many important works into the Kannada language. He was able to explain sophisticated concepts in the form of simple stotras.

Identity as described in Bhugola Varnanam: Among souls of divine inclination, there are several grades such as the best of humans, sages, demigods, Sankara, Garuda and Śeṣa. There is a class of souls, higher than Garuda and Śeṣa; they only are eligible to occupy the post of Caturmukha Brahma. During every Brahma Kalpa period of creation and sustenance, there will be two hundred such souls, the one who has completed his training during 199 previous Brahma Kalpas, now occupying the seat of Brahma. Next in the order of gradation, downward, are Vayu, Latavya, Gavya, Vaktavya, Jnatavya, and so on. (These details are found in Vamana Purana). Saint Śrīla Vādirāja Tīrtha swāmi is the 198th, 'Latavya', in the upward scale of gradation. He is almost as good as the 199th, Vayu, and during this (probation period), he engages himself in the activities of Brahma and Vayu, with their co-operation. The activities of these three souls are universal. They are everywhere and in everything living or non-living, doing their jobs for them. Lord Sri Lakshmi-Narayana, the Supreme Being, is their Master.

We are more likely to believe in autobiographies than in biographies penned by adorers. Saint Śrīla Vādirāja Tīrtha swāmi has given us his autobiography in his text "*Vrndavana Akhyanam*", containing 22 chapters. The wondrous part of it is that he was just known generally as a great saint and the greatest scholar of Madhva sastras during his life time; but, sometime after he entered his Samadhi at the ripe age of 120, he entered the body of a born-dumb brahmana with his spiritual part, and dictated his autobiography such that people might believe his words. This is a history of the period of the famous Krishnadevaraya of Vijaya-nagar. There is monumental evidence for all this if one cares to inspect.

The following are the salient facts in "*Vrndavana Akhyanam*" bearing relevance to "*Bhugola Varnanam*". Saint Śrīla Vādirāja Tīrtha swāmi had the ability to go anywhere in the fourteen worlds and that he was honoured by the gods to whichever upper world he went and that he was fortunate enough to go to the abode of Sriman Narayana in "Swetadvīpa", where he was blessed by

the fond embrace of the Supreme Lord. These details go to show that his "*Bhugola Varnanam*" is not just a concept formed by the study of puranas. He has given us first-hand knowledge of all the regions of the universe, which he visited and actually saw.

Knowledge of truth published by the sāstras is tested with reference to visual perception, logic and the Vedic literature (*Pratyaksa*, *Anumāna* and *Āgama*). To these is added what is known as *'Āpta Vākya'* words spoken by reliable persons. Who else could be a greater *'Āpta'* or more reliable person than Saint Sri Vādiraja, saint of the highest order, the most profound scholar, and one who had full mastery over spiritual powers? *'Āpta Vākya'*, the word of a reliable person, one who is well informed, discerning and has no intention to cheat others. Therefore, Saint Śrīla Vādirāja Tīrtha swāmi's "*Bhugola Varnanam*" is '*Apta Vākya*' for us. It is not only reliable; it can stand the test of modern science by all counts.